The Continuum Limit of the Correlation Functions of Wilson Loops in Four-Dimensional Lattice Yang–Mills Theory: Supplement on Bałaban's Papers

Abstract

This supplement contains the statements of Bałaban's papers on the renormalisation group for four-dimensional lattice gauge theories which the paper uses outside his printed hypotheses, or with a proof which he did not print, each with its proof. A result which Bałaban proved in his own setting is cited in the paper from his paper and does not appear here.

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1 Introduction

The paper runs Bałaban's construction with three changes: its parameters are evaluated at numbers fixed by the target instead of the running couplings, its test functions are continuous, and the depth of its **R**-operation is chosen by a formula of its own. It inserts a Wilson loop ensemble into the construction, and it compares two runs which differ by one step on the finest lattice. Each of these changes moves the use of some of Bałaban's results outside the hypotheses under which he proved them, and each of the comparisons needs a result about his objects which he did not state. This supplement collects those results. They are numbered as in the paper, which cites them as "Supplement, Proposition S.n", and they are grouped by the part of the paper which uses them.

We use the notation of the paper throughout, and we refer to its statements as "Paper, Lemma 3.2" or "Paper, (3.4)". Bałaban's papers are cited by page and equation, as in the paper, with the abbreviations of its bibliography: [9] and [11] for the small-field renormalisation group, [10] for the inductive description of the densities, [12] and [13] for the **R**-operation and the large-field factors, [5] for the propagators, [7] for the variational problems, [4] for the averaging operations, and [2, 3] for the random walk expansions. Every constant is one of the paper's, with its place in the order of choices of Paper, Appendix A.

2 The Construction of the Paper

The construction of Paper, Definition 3.11, is Bałaban's with the parameters of a tube, with soft test functions, and with the depth of Paper, Definition 3.9. Paper, Proposition 3.12, asserts that Bałaban's description of the densities and his bounds hold for it, and its proof consists of the three propositions of this section, one for each change.

2.1 The Parameters

In [10, 12, 13] every parameter of the scale j is a function of the running coupling g_j: the cube size R_j, the thresholds $\varepsilon_j = g_j p_0(g_j)$, $\delta_j = (A_1/A_0)\varepsilon_j$, $\delta'_j = g_j A_1 p_1(g_j)$, the radii $\alpha_{0,j}$, $\alpha_{1,j}$ of [10, (2.28), p. 259], and every threshold, radius, and support of a cutoff function derived from them [10, (2.4), (2.5), p. 255], [12, p. 183]. In the construction of the paper each of them is the same function evaluated at $\hat{g}_j$ (Paper, Definitions 3.3 and 3.11), while the couplings which enter the densities are those of a history of the tube. For the coupling flow of Paper, Section 4, the construction is also run along histories which are not flows, with the factor (4.9) of the paper at every step, and then the Wilson action of a term of the density of the step k has the coefficient function (4.10) of the paper,

$$y_k(x) := x_k + \sum_{i=1}^{k} \beta_i (1 - \phi_i(x)) \;, \tag{2.1}$$

where $\beta_i = \beta_i(x_0, \dots, x_{i-1})$, and $\phi_i \in C_0^\infty(\Lambda_i)$ equals one on $\Lambda_i^{\sim -1}$ [10, (2.24), p. 259]. Bałaban does not restrict the functions ϕ_i further, and we take them with values in $[0,1]$. Along a flow, y_k is his coefficient $1/g_k^2(x)$, the solution of the recursion (2.24) which starts from $1/g_0^2(x) = 1/g_0^2$. We put

$$\bar{\rho} := \frac{15}{8} \;, \tag{2.2}$$

which exceeds the ratio ρ of every tube, by Paper, (3.6), and we write C'_W for the constant of Paper, Theorem 4.4. The *tube of a target* is a tube whose constants are those of Paper, Theorem 4.1: $b = 2c \ln L - \delta'$, $b' = 2c \ln L + \delta'$, and $C_W = C'_W$.

When the Wilson action gives a small factor on a field which has failed a test function, the coefficient of the action enters through a lower bound by the inverse square g_m^{-2} of the running coupling of the scale m of that field. The first step is the corresponding bound for the construction, at every point at which a field of the scale m can be an integration variable.

Lemma 2.1 (The Coefficient of the Action)**.** *Let γ satisfy Paper, (3.7), the first condition of Paper, (4.12), and $C'_W\gamma^2 \le 11/40$. Let a tube be given and let $h = (x_0, \dots, x_k) \in H_k$. Consider a term of the density of the step k along h, an integer $m \in \{0, \dots, k\}$, and a point x outside Λ_{m+1}, where $\Lambda_{k+1} := \emptyset$.*

- *(i) If h is a flow, then $y_k(x) \ge \hat{g}_m^{-2}/\rho$.*
- *(ii) If $|\beta_i(x_0, \dots, x_{i-1}) - b_i| \le \delta'$ for $1 \le i \le k$, with the number δ' of Paper, Proposition 4.3(i), then $y_k(x) \ge \hat{g}_m^{-2}/\bar{\rho}$, and $y_k(x) \ge \hat{g}_m^{-2}/\rho$ if the tube is the tube of a target.*

The hypothesis of (ii) is Paper, Proposition 4.3(i), at the steps before the step k, so that the lemma is available at the step k once the densities up to that step have been constructed. The points outside Λ_{m+1} carry the fields of the scale m: the block field V_m lives on $\Gamma_m \subset \Omega^c_{m+1}$, and the fluctuation field A_m on $\Omega_{m+1} \setminus \Lambda_{m+1}$ (Paper, (3.21) and (3.24)). For $m = k$ the lemma holds at every point.

Proof. We first show that the domains of a term are nested as

$$\Lambda_{i+1} \subset \Lambda_i^{\sim -1}\ , \qquad 1 \le i < k\ , \tag{2.3}$$

or, in terms of the large-field regions, $Z_i^{\sim} \subset Z_{i+1}$. Consider first a term which the step $i+1$ has produced from a term of the step i. Let Z'_i be the union of the cubes of side LMR_{i+1} which meet Z_i, the operation $\sim$ being taken for these cubes [10, p. 264]. By [10, (3.5), p. 265] the complement of Ω_{i+1} contains $(B^{i+1}(P^1_{i+1})^c)^{\sim 3}$, and $B^{i+1}(P^1_{i+1})^c$ contains $Z_i'^{\sim 5}$ by its definition on the same page, so that $\Omega_{i+1} \cap Z_i'^{\sim 8} = \emptyset$. The partition into the cubes of side LMR_{i+1} is compatible with the earlier ones [10, p. 264], and $LMR_{i+1} \ge MR_i$ by Paper, Lemma 3.4(ii). Let $x \in \Omega_{i+1}$, let $\square$ be a cube of side MR_i which contains x, and let $\square'$ be the cube of side LMR_{i+1} which contains $\square$. Since $\square'$ is not in $Z_i'^{\sim}$, neither $\square'$ nor any cube of side LMR_{i+1} which has a point in common with it meets Z_i. Every cube of side MR_i which has a point in common with $\square$ lies in one of these cubes, and therefore in Λ_i. As Z_i is a union of cubes of side MR_i, no point of Z_i lies in $\square$ or in a cube which touches it, that is, $x \in \Lambda_i^{\sim -1}$. Since $\Lambda_{i+1} \subset \Omega_{i+1}$ [10, (2.1), p. 254], this proves (2.3) for the domains which a step produces.

The $\mathbf{R}$-operation of the step k acts on a union Z of components X of Z_k, and the operations on the large-field regions factorise as $\mathbb{T}_k(Z_k) = \mathbb{T}_k(Z_k \cap Z^c) \prod_i \mathbb{T}_k(X_i)$ [12, (1.1), p. 177]. A component X which the operation removes is removed together with $\mathbb{T}_k(X)$, which carries the regions $Z_j \cap X$, $j < k$, by Paper, (3.24), and the new term is the term $\rho(Z'', V)$ with the remaining large-field regions [12, (0.3), p. 176]. This preserves the inclusions $Z_i^{\sim} \subset Z_{i+1}$. Indeed, let Y be a component of Z_i which does not lie in X. Then $Y^{\sim}$ is a connected subset of $Z_{i+1} \subset Z_k$ which contains a point outside X, so it lies in a component of Z_k other than X, and $Y^{\sim} \subset Z_{i+1} \setminus X$.

Since ϕ_{i+1} vanishes outside $\Lambda_{i+1} \subset \Lambda_i^{\sim -1}$, while $\phi_i = 1$ on $\Lambda_i^{\sim -1}$ and both take values in $[0, 1]$, we have $\phi_{i+1} \le \phi_i$ at every point. The weights $w_i := 1 - \phi_i(x)$ therefore do not decrease in i, and we put $w_0 := 0$.

We now sum by parts. For $i > m$ we have $\Lambda_i \subset \Lambda_{m+1}$ by Paper, (3.19), so that $x \notin \Lambda_i$ and $w_i = 1$. With $S_i := \sum_{l=i}^{k} \beta_l$ for $1 \le i \le k$ and $S_{k+1} := 0$,

$$y_k(x) = x_k + S_{m+1} + \sum_{i=1}^{m} \beta_i w_i = (1 - w_m)(x_k + S_{m+1}) + \sum_{i=1}^{m} (w_i - w_{i-1})(x_k + S_i)\ . \tag{2.4}$$

The first identity is (2.1) with $w_i = 1$ for $i > m$. The second follows from the summation by parts $\sum_{i=1}^{m} \beta_i w_i = \sum_{i=1}^{m} (w_i - w_{i-1})(S_i - S_{m+1})$, which holds because $\beta_i = (S_i - S_{m+1}) - (S_{i+1} - S_{m+1})$ and $w_0 = 0$, together with $\sum_{i=1}^{m}(w_i - w_{i-1}) = w_m$. The coefficients $1 - w_m$ and $w_i - w_{i-1}$ are nonnegative and add up to one, so that $y_k(x)$ is a convex combination of the numbers $x_k + S_i$, $1 \le i \le m+1$. Since $u_{i-1} \ge u_m = \hat{g}_m^{-2}$ for $i - 1 \le m$, it suffices to show that

$$x_k + S_i \ge \frac{u_{i-1}}{\bar{\rho}} , \qquad 1 \le i \le k+1 , \tag{2.5}$$

under the hypothesis of (ii), and the same with ρ in place of $\bar{\rho}$ in the other two cases.

(i) If h is a flow, then $x_{l-1} = x_l + \beta_l$ for every $l \le k$, so that $x_k + S_i = x_{i-1}$, and $x_{i-1} \ge \ell_{i-1} \ge u_{i-1}/\rho$ by Paper, (3.11).

(ii) For $i = k+1$ the inequality (2.5) is $x_k \ge \ell_k \ge u_k/\rho$, by Paper, (3.11), and $\rho \le \bar{\rho}$. For $i \le k$ we put $n := i - 1 < k$. By hypothesis each β_l differs from b_l by at most $\delta' \le \frac{2}{5} c \ln L$, the first condition of Paper, (4.12), and $\sum_{l=n+1}^{k} b_l \ge 2c(k-n)\ln L - C'_W$ by Paper, Theorem 4.4. With $x_k \ge \ell_k = t + b(K-k) - C_W$ this gives

$$x_k + S_i \ge (t - C_W - C'_W) + b(K-k) + \tfrac{8}{5} c \ln L \,(k-n) , \tag{2.6}$$

whereas $u_n = (t + C_W) + b'(K-k) + b'(k-n)$. We compare the corresponding terms. Since $C_W \le \beta_0 t/4 < t/8$ by Paper, (3.2), and $t \ge \gamma^{-2}$,

$$\frac{t + C_W}{t - C_W - C'_W} \le \frac{9/8}{7/8 - C'_W \gamma^2} \le \frac{15}{8} , \tag{2.7}$$

the last step being $C'_W \gamma^2 \le 11/40$. Moreover $b'/b \le \frac{3}{2}$ by Paper, (3.2), and $b'/(\frac{8}{5} c \ln L) \le \frac{15}{8}$ because $b' \le b^* = 3c \ln L$ by Paper, (4.4). A quotient of two sums of positive numbers is at most the largest quotient of corresponding terms, as in Paper, (3.11), so that $u_n \le \bar{\rho}(x_k + S_i)$, which is (2.5).

If the tube is the tube of a target, then $C_W = C'_W$ and $b = 2c \ln L - \delta'$, so that the same two bounds give $S_i \ge b(k-n) - C_W$ and $x_k + S_i \ge (t - 2C_W) + b(K-n)$, while $u_n = (t + C_W) + b'(K-n)$. The quotient of the two is at most $\max((t + C_W)/(t - 2C_W),\, b'/b) = \rho$, by the definition (3.5) of the paper. *q.e.d.*

We can now state the proposition. In the displays of [13, pp. 380–383] which it concerns, the symbol $p_0(g)$ stands for $(\log g^{-2})^{p_0}$, the factor A_0 of [10, (2.4), p. 255] being written out, as in $p_1(g) = (\log g^{-2})^{p_1}$ [12, p. 183]. Only in this reading does the printed bound of $\exp(-\frac{1}{4} g_m^{-2} B_3^{-2} \frac{1}{2} \varepsilon_m^2)$ by $\exp(-\frac{1}{8} B_3^{-2} (1+\beta_0)^{-2} A_0^2 p_0^2(g_j))$, for $j - N \le m \le j$ [13, p. 381], follow from $g_m^{-2} \varepsilon_m^2 = A_0^2 (\log g_m^{-2})^{2p_0}$ and (2.7), and we read all of them so (Paper, Section 14).

Proposition S.1 (The Parameters of a Tube)**.** *Let γ satisfy the hypotheses of Lemma 2.1, let $\beta_0 \le \sqrt{2} - 1$, and let A_1 and A_0 satisfy*

$$A_1 \ge 48 \bar{\rho}^{1/2} , \qquad A_0^2 \ge \bar{\rho} A_1^2 \max(8(1+\beta_0)^2 B_3^2,\ 4O(1) B_3 M^2) , \tag{2.8}$$

with the constant $O(1)$ of [13, p. 381]. Let a tube be given, and let $h \in H_k$ be a history which is a flow or whose coefficients satisfy the hypothesis of Lemma 2.1(ii). Every statement of [10, 12, 13], and of the parts of the papers which they import, in which a parameter of a scale enters holds for the construction of the paper along h, with the following changes:

(a) a relation among the parameters alone holds as printed;

(b) a lower bound drawn from the action, in which the Wilson action gives a factor $\exp(-cF(\log g_j^{-2}))$ *on a field above a threshold, with* F *nondecreasing, holds with* c *replaced by* $c/\bar\rho$*, and by* c/ρ *along a flow and on the tube of a target;*

(c) an upper bound relative to the action holds with the printed constant, or with a constant which differs from it by a factor at most 2*;*

(d) a conversion between a scaled and an unscaled field holds with the coupling of the history as the factor and the parameter of the tube as the threshold;

(e) the **R***-terms produced by the step* k *satisfy [10, (2.31), p. 260] with the coupling* g_k *of the history.*

In particular, given Propositions S.2 and S.3 for the test functions and the depth, the conclusions of Paper, Theorem 3.6, and the bounds (3.31) *and* (3.32) *of the paper hold for the densities of the construction along* h*, the first with a constant which does not depend on the tube, on* h*, on* k *and* K*, or on the torus, the second with constants* $E_\pm$ *which depend on these only through the coupling* g_k*, as [10, Corollary 3, p. 264] allows.*

The conditions (2.8) are of the kind which Bałaban imposes himself, a lower bound on A_1 and then on A_0 in terms of earlier constants [10, p. 246, p. 265], and they belong to the stage 6 of Paper, Appendix A, as does $C'_W\gamma^2 \le 11/40$.

Proof. Theorem 1 of [10, p. 262] assumes the inequalities (0.31) of [9, p. 259], which it cites as (I.0.33) (Paper, Section 14), and its proof uses them through the inequalities (2.6)–(2.9) of [10, pp. 255–256], which Bałaban derives from the renormalisation group equations and uses in their place [10, p. 255]. A history of the tube satisfies (2.6) and (2.7) by Paper, Lemma 3.2(ii), with real exponents, and so does the sequence $(\hat g_j)$, by part (iii) of that lemma; the parameters of the tube satisfy (2.8) by Paper, Lemma 3.5(ii), and the cube sizes satisfy (2.5) and (2.9) by Paper, Lemma 3.4. It remains to go through the places at which a parameter enters, and we sort them into the kinds (a)–(e) and two further kinds.

Relations among the parameters. These are the compatibility of the partitions and the choice of the cube sizes [10, pp. 245, 264], [12, p. 177]; the definitions of k_1 [10, p. 270] and of N_0 [12, p. 179]; the inequalities (2.6)–(2.9) and their consequences, such as $L^{N_0-1} = R_{k-N_0+1} \le (L+1)(N_0+1)^{\beta_0}R_k$ and $\delta'_j \le (1+\beta_0)^2(k-j)^{\beta_0}\delta'_k$ [12, p. 192]; and the margins between powers of logarithms, such as that of $2(1+\beta_0)^{-1}p_0(g_{j+1})$ against $O(1)3(100M)^d R_{j+1}^{d+2}$ [13, (1.88), p. 387]. In the construction all of them are evaluated on the sequence $(\hat g_j)$, which lies in $(0,\gamma]$ and satisfies Bałaban's hypotheses, as we have just seen. So they hold as printed.

Lower bounds from the action. Here the Wilson action gives a small factor on a field of the scale m which exceeds a threshold. Bałaban bounds the coefficient of the action below by g_m^{-2}, the threshold being $\tau(g_n)$ for a scale n and a function $\tau(g) = cg(\log g^{-2})^p$, and then bounds $g_m^{-2}\tau(g_n)^2$ below by a nondecreasing function F of $\log g_j^{-2}$, by means of (2.6)–(2.8). In the construction the coefficient at the points which carry the field is at least $\hat g_m^{-2}/\bar\rho$, by Lemma 2.1, and the threshold is $\tau(\hat g_n)$. The same derivation, applied to the sequence $(\hat g_j)$, which satisfies (2.6)–(2.8), therefore gives

$$y_k\tau(\hat g_n)^2 \ge \frac{1}{\bar\rho}\,\hat g_m^{-2}\tau(\hat g_n)^2 \ge \frac{1}{\bar\rho}F(\log \hat g_j^{-2}) \ge \frac{1}{\bar\rho}F(\log g_j^{-2})\,, \tag{2.9}$$

the last step because $\hat g_j \le g_j$ and F does not decrease. So the factor is Bałaban's with the constant in the exponent divided by $\bar\rho$, and by ρ along a flow and on the tube of a target.

Examples are the factors $\exp(-p_0(g_0))$ and $\exp(-\frac{1}{2}p_0^2(g_0))$ of a failed plaquette test function [12, p. 175], [10, p. 244], the large-field factors of [13, pp. 380–385], and the Gaussian tails of the fluctuation integrals. A change of the constant in the exponent is harmless wherever the bound has a margin of a positive power of $\log g_j^{-2}$, and it remains to check the places where two such factors are compared at the same order. They are the following, and each of them holds with the factor $\bar\rho^{-1}$ in the exponent on the side of the action.

(i) The creation factor: [13, p. 381] bounds $\exp(-\gamma_0 \min\{\frac{1}{2}B_3^{-2}A_0^2, 2A_1^2, A_1^2\}p_0^2(g_j)(d'_j+1))$ by $\exp(-\frac{1}{2}\gamma_0 A_1^2 p_0^2(g_j)(d'_j+1) - 2p_0(g_j))$, where $d'_j \geq 0$ is the linear size of a component in terms of the cubes of side MR_j [13, p. 380]. By (2.8) the minimum is A_1^2. With $\gamma_0/\bar\rho = \frac{1}{2}\gamma_0 + \frac{1}{30}\gamma_0$ in place of γ_0 the left side is at most $\exp(-\frac{1}{2}\gamma_0 A_1^2 p_0^2(g_j)(d'_j+1) - \frac{1}{30}\gamma_0 A_1^2 p_0^2(g_j))$, because $d'_j + 1 \geq 1$, and $\frac{1}{30}\gamma_0 A_1^2 p_0^2(g_j) \geq 2p_0(g_j)$ for $g_j \leq \gamma$ small.

(ii) [13, p. 381] bounds $\exp(-\frac{1}{8}B_3^{-2}(1+\beta_0)^{-2}A_0^2 p_0^2(g_j))$ by $\exp(-A_1^2 p_0^2(g_j))$. With $\bar\rho^{-1}$ in the first exponent this is the first part of the second condition of (2.8).

(iii) [13, p. 381] bounds $\exp(-\gamma_0 g_m^{-2}(2\delta'_m)^2)$ by $\exp(-\gamma_0 4(1+\beta_0)^{-2}A_1^2 p_1^2(g_j))$ and then by $\exp(-\gamma_0 A_1^2 p_1^2(g_j))$. With $\bar\rho^{-1}$ the second step needs $4(1+\beta_0)^{-2} \geq \bar\rho$, which holds because $\beta_0 \leq \sqrt{2}-1$ gives $4(1+\beta_0)^{-2} \geq 2$.

(iv) [13, p. 381] bounds $\exp(-\frac{1}{4}g_j^{-2}(O(1)B_3M^2)^{-1}\varepsilon_j^2)$ by $\exp(-A_1^2 p_0^2(g_j))$. Since $g_j^{-2}\varepsilon_j^2 = A_0^2 p_0^2(g_j)$ in the reading above, with $\bar\rho^{-1}$ this is the second part of the second condition of (2.8).

(v) [13, p. 382] obtains the factor $\exp(-\frac{1}{4}g_j^{-2}(24R_j^4)^{-2}\delta_j'^2) = \exp(-A_1^2 48^{-2} R_j^{-8} p_1^2(g_j))$ and bounds it by $\exp(-R_j^{-8}p_1^2(g_j))$. With $\bar\rho^{-1}$ the last step is the first condition of (2.8).

(vi) The bound $\frac{1}{4}\gamma_0(14)^{-d}A_1^2 p_0^2(g_1) \geq O(1)2(64)^d M^d L^{d+1} R_1^{d+2}$ [13, p. 385] concerns the exponent obtained from (i), which holds as printed, and it has the margin of the factor $(\log g_1^{-2})^{2p_0}$ against $R_1^{d+2} < (2L)^{d+2}(\log g_1^{-2})^{(d+2)r}$, by Paper, Lemma 3.4(i), with $2p_0 \geq 10r > (d+2)r$ at $d=4$.

Upper bounds relative to the action. Here a field below a threshold enters an upper bound, often multiplied by the coefficient of the action, as in the bound $O(1)|\mathbf{B}_0|\delta_k'^3 \leq g_k^3 O(1) M^d R_k^{d+2} A_1^3 p_1^3(g_k)$ [13, p. 361] and in the bound $O(1)g_k^{-2}(\alpha_{0,k}+\alpha_{1,k})\varepsilon_k L^{-N} N^{1+\beta_0}$ [13, p. 368]. The thresholds of a tube are the smaller ones, $\tau_j \leq \tau(g_j)$ by Paper, (3.17). The coefficient of the action exceeds $x_k = g_k^{-2}$ by at most the sum of the numbers $|\beta_i|$ over the scales i involved, and each of them is at most $2c\ln L + C'_W + \delta'$, by Paper, Theorem 4.4 with $n = m+1$ and Proposition 4.3(i); this is the form $1/g_k^2 + O(k-j)$ in which the coefficient enters [13, p. 368]. So a bound of this kind holds as printed when it uses only upper bounds on the thresholds. Where it uses a quotient of two parameters of the same scale, the quotient at $\hat g_j$ differs from the quotient at g_j by a factor $(\log \hat g_j^{-2}/\log g_j^{-2})^{\pm p}$ with $p \leq p^*$, which lies between $1/2$ and 2 by Paper, Lemma 3.2(i), and the third condition of Paper, (3.7).

Conversions between scaled and unscaled fields. These are the conditions $|A(b)| < g_0^{-1}\delta_0$ [10, p. 250], $|A_j(b)| < g_j^{-1}\delta_j$ [12, (1.9), p. 178], the condition on B_j of [12, (1.55), p. 188], and $|B(b)| \geq g_j^{-1}\delta'_j$ [13, p. 383]. In the construction the factor g_j is the coupling of the history and the threshold is the parameter of the tube, by Paper, Definition 3.11(ii), so that each of them is a condition on the unscaled field with the threshold of the tube. The domains of analyticity, such as the domain (3.43) of [10, p. 276], in which $|A| < C_1 p_1(g_k)$, are domains on which bounds are asserted and not thresholds under integrals, and they are evaluated at $\hat g_j$ like every other parameter.

The **R***-terms.* The terms which the **R**-operation produces at the step k satisfy $|\mathbf{R}'^{(k)}(X,(\mathbf{U},\mathbf{J}))| \le \exp(-p_0(\hat{g}_k))\exp(-\kappa d_k(X))$ [13, p. 390], with the parameter of the tube. Since $p_0(g) = A_0(\log g^{-2})^{p_0} \ge \frac{\kappa_0}{2}\log g^{-2}$ for $g \le \gamma$ small, $\exp(-p_0(\hat{g}_k)) \le \hat{g}_k^{\kappa_0} \le g_k^{\kappa_0}$, which is [10, (2.31), p. 260] with the coupling of the history.

Quantities of the history and the geometry. Two further uses of the parameters are not comparisons with the action. Quantities of the history which are bounded by a parameter, such as the bound $O(N) \le O(R_k)$ on $1/(g_k''(\cdot))^2 - 1/g_k^2$ [13, p. 366] and the terms "bounded by any positive power of g" [10, pp. 252, 269], [12, p. 188], have the history on one side, which is bounded through the coefficients β_i and the couplings $g_j \ge \hat{g}_j$, and a parameter of the tube on the other, which differs from Bałaban's by the factors of Paper, (3.17). The domains, the partitions, and the cube sizes are fixed by the target and the configuration, and the combinatorial bounds on the numbers of polymers, cubes, and domains hold as printed with the cube sizes of the tube. The lower bounds on R_j, such as the smallness of $\exp(-R_k)$ [10, pp. 266, 268, 269] and $\log g_j^{-2} \le R_j$ [13, p. 380], hold by Paper, Lemma 3.4(i).

The imported papers. The papers [4, 7, 6] are statements about lattice gauge fields under a smallness condition on the field, and they contain no coupling; they apply to the thresholds of the tube, which are smaller than Bałaban's. The propagator and random walk estimates [5, 2, 3] contain no coupling either. The small-field steps of [9, 11], which [10] and [13] run on their own domains, use the parameters as smallness conditions on thresholds, as the threshold of the fluctuation field relative to the radii $\alpha_{0,j}$, $\alpha_{1,j}$, or in relations among the parameters, so that they fall under the kinds above.

Every place at which a parameter enters the three papers and the parts of the imported papers which they use is of one of the kinds (a)–(e) or of the two further kinds, and in each case the statement holds for the construction with constants which depend on the constants of the construction and not on the tube, on h, on k and K, or on the torus. Bałaban's proof therefore reproduces the description of the densities under these changes, given the corresponding statements for the test functions and the depth, which are Propositions S.2 and S.3. *q.e.d.*

2.2 Soft Test Functions

A soft test function differs from a characteristic function in two ways, and each of them has to be followed through Bałaban's papers. Its support is larger, so that a configuration on which a passed test function does not vanish obeys the bound $f_\square < (1+\kappa_s)t$ instead of $f_\square < t$; and it is not idempotent, so that an argument which multiplies a test function by itself, or removes it because it equals one, has to be checked. The first point is a question of margins: wherever Bałaban draws a bound from a passed test function and then uses the bound to show that another test function equals one, or that a field lies in a domain, his inequality must have room for the factor $1+\kappa_s$. The second point concerns the cascade of test functions of the **R**-operation [12, (1.22)–(1.24), p. 181], in which each test function is removed on the support of the next one, and there the widths have to be nested.

We first say which functions are soft. The cascade of the **R**-operation of the step k consists of the test functions $\chi_k^{(n)}$, $n = 0, \dots, N$, where $N = k - h$ is the number of integrations which the operation performs and h the scale at which it starts [12, pp. 181–182]. The test function $\chi_k^{(n)}$ bounds the plaquettes of the configuration $U_{k,Z}^{(n)}$ by the thresholds $T_n(p)$ of [12, (1.24), p. 182], which depend on the layer of the plaquette p.

Definition 2.2 (The Soft Functions of the Construction)**.** In the construction of the paper the following functions are soft, with the width $\kappa_s = \beta/30$ of Paper, (3.37), unless stated otherwise.

(i) The test functions of the plaquettes and of the reference configurations, [10, (1.1), (1.4), (1.8), (2.17), (3.2), (3.3)], [12, (1.3), (1.4), (1.7), (1.8), (1.27), (1.75), (1.88)], and their restatements in [13, (1.70), p. 380]: each characteristic function $\chi(\{f_\Box < t\})$ is replaced by $\bar\chi(f_\Box/t)$, and its complement by $1-\bar\chi(f_\Box/t)$, with $f_\Box := \infty$ where the configuration which defines it does not exist.

(ii) The cascade: for a cube $\Box$ let $f_\Box^{(n)}$ be the supremum of $|U_{k,Z}^{(n)}(\partial p)-1|/T_n(p)$ over the plaquettes p of $\chi_k^{(n)}$ assigned to $\Box$, each plaquette being assigned to one cube of side LM_2R_i which contains it, i the scale of $T_n(p)$. Then $\chi_k^{(n)}$ is replaced by $\prod_\Box \bar\chi_c(f_\Box^{(n)}/(1+\kappa_c)^{N-n})$, where $\bar\chi_c$ is the function (3.34) of the paper with the width

$$\kappa_c := \frac{\kappa_s}{N+1} \tag{2.10}$$

in place of κ_s, and its complement accordingly.

(iii) The cutoff $\chi(\Lambda_i)$ of [12, (1.101), p. 201] is replaced by $\prod_b \bar\chi(|(1/i)\log V'(b)|/(M_0\varepsilon_k))$, the product over the bonds b of Λ_i, with $M_0 := 2(1+\kappa_s)O(1)B_5M^6$, where $O(1)B_5M^6$ is the bound of [13, p. 378].

The characteristic functions of the fluctuation fields and of the axial gauge fixing, [10, (1.5), (1.18), (3.16)] and [12, (1.5), (1.9), (1.55), (1.82)], are sharp.

The number M_0 is left free by Bałaban, who sets it equal to the bound $O(1)B_5M^6$ on $|V_k'-1|$ which the passed test functions give [13, p. 378]; the factor $1+\kappa_s$ accounts for the soft test functions and the factor 2 for the logarithm, since $|(1/i)\log v| \le \frac{\pi}{2}|v-1|$ for a unitary v. The nesting of the cascade is the content of the next lemma.

Lemma 2.3 (The Soft Cascade)**.** *The soft test function $\chi_k^{(n)}$ equals one where $f_\Box^{(n)} \le (1+\kappa_c)^{N-n}$ for every $\Box$, and it vanishes unless $f_\Box^{(n)} < (1+\kappa_c)^{N-n+1}$ for every $\Box$. Every threshold of a soft cascade test function lies between $T_n(p)$ and $(1+\kappa_c)^{N+1}T_n(p) \le (1+2\kappa_s)T_n(p)$, and $x \mapsto \bar\chi_c(x/(1+\kappa_c)^{N-n})$ is Lipschitz with the constant $(N+1)/\kappa_s$.*

Proof. The first two statements are those of $\bar\chi_c$, which equals one on $[0,1]$ and vanishes on $[1+\kappa_c,\infty)$, at the argument $x/(1+\kappa_c)^{N-n}$. For the third, $(1+\kappa_c)^{N+1} = (1+\kappa_s/(N+1))^{N+1} \le e^{\kappa_s} \le 1+2\kappa_s$, the last step because $e^x \le 1+(e-1)x$ for $0 \le x \le 1$, by convexity. The Lipschitz constant of $\bar\chi_c$ is $1/\kappa_c = (N+1)/\kappa_s$, and the division of the argument by $(1+\kappa_c)^{N-n} \ge 1$ does not increase it. *q.e.d.*

Proposition S.2 (Soft Test Functions)**.** *Let $\beta = 1/8$, $\kappa_s = \beta/30$, $\beta_0 \le 1/7$, $L \ge 3$, and let γ be small. For every tube and every history as in Proposition S.1, the statements of [10, 12, 13] hold for the construction with the soft functions of Definition 2.2, with the following changes.*

(a) The decompositions of unity are exact, and every removal of a test function which Bałaban proves on the support of the other test functions holds on the plateau of the removed one, given the others at the thresholds of their supports.

(b) Every bound drawn from a passed test function holds with the threshold t replaced by $(1+\kappa_s)t$, and by $(1+2\kappa_s)t$ for a cascade test function; every bound drawn from a failed test function holds at the threshold t.

(c) *The changes of variables [10, (1.22), p. 251], its repetition at [10, p. 269], and [12, (1.59), p. 188] carry a soft test function into the soft test function of the new variables.*

(d) *The normalisation [12, (1.102), p. 201] holds, and the denominator of [12, (1.100)] with the soft cutoff lies within the factors* $\mathrm{e}^{\pm 1/2}$ *of the denominator with the sharp cutoff at the threshold* $M_0\varepsilon_k$.

(e) *In the term of the density of the step* k *which has no large-field region, every test function other than* χ_k *of Paper,* (3.22), *equals one on the support of* χ_k.

Proof. (a) The identity (3.35) of the paper holds for any functions with values in $[0,1]$, by Paper, Lemma 3.8(i), and so the lattice integrals are unchanged whichever decomposition expands them. Where test functions on overlapping families of cubes are combined, as $\chi_k(X_i^{\sim-6})$ with $\chi_k(\Omega_k^{\sim4})$ [13, p. 378], the soft product keeps both factors, with values in $[0,1]$ and with the support of each, and no identity $\bar{\chi}^2 = \bar{\chi}$ is used. A removal is a statement of the form: on the support of the remaining test functions the quantity f of the removed one satisfies $f < t$. With soft test functions the supports are those of Paper, Lemma 3.8(ii), and the conclusion needed is $f \le t$, the plateau, so that it is a bound drawn from passed test functions and falls under (b), with one exception, the cascade, which uses no margin at all. Indeed, Bałaban removes $\chi_k^{(n)}$ on the support of $\chi_k^{(n+1)}$ by inequalities of the form $f \le af' + y$ at each plaquette, with f the quantity of $\chi_k^{(n)}$, f' that of $\chi_k^{(n+1)}$, and numbers $a, y \ge 0$ which do not depend on the threshold of $\chi_k^{(n+1)}$: this is [12, (1.48), p. 186], with $a = 1 + \alpha\beta 2^{-(j-i)}$ and $y = \alpha\beta 2^{-(j-i)}\varepsilon_i(L^{k-i}\eta)^2$, used in [12, (1.50)–(1.52), p. 187], where he shows $aT_{n+1}(p) + y \le T_n(p)$. On the support of the soft $\chi_k^{(n+1)}$ we have $f' < (1+\kappa_c)^{N-n}T_{n+1}(p)$ by Lemma 2.3, and therefore

$$f \le (1+\kappa_c)^{N-n}(aT_{n+1}(p) + y) \le (1+\kappa_c)^{N-n}T_n(p)\,, \tag{2.11}$$

because $y \le (1+\kappa_c)^{N-n}y$. This is the plateau of the soft $\chi_k^{(n)}$, so the removal holds exactly. The coefficients a and y come from numerical factors which Bałaban makes at most $\alpha\beta$ by taking γ or A_1/A_0 small [12, p. 186], and the factor $1+\kappa_s$ of the soft test function [12, (1.27)], which enters through the field $(1/i)\log[V_j(V_Z^{(j)})^{-1}]$, is absorbed in the same way.

(b) A failed test function enters as a lower bound $f > t$ on its quantity, from which the Wilson action gives a small factor. Since $1 - \bar{\chi}(f/t)$ vanishes for $f \le t$, such a bound holds at the threshold t, as printed. This applies to the gains of [13, pp. 380–381], which are monotone in the threshold, and to the failed test functions of [10, 12].

A passed test function enters as an upper bound $f < (1+\kappa_s)t$, or $f < (1+2\kappa_s)t$ for the cascade by Lemma 2.3, and its consequences are of two kinds. Most of them carry a constant, a factor $\exp(-\delta M R_j)$ or $\exp(-R_j)$, a factor L^{-N}, or a negative power of $\log \hat{g}_j^{-2}$ such as A_1/A_0 or δ_j'/ε_j, and are then used for γ small, against such a factor, or as a regularity condition with an unspecified constant, and the factor $1 + 2\kappa_s$ is absorbed. Examples are [10, (3.7), p. 266], where the boundary-layer term $30d^2L^2B_3^2(1+\beta_0)\exp(-R_k)\varepsilon_k$ and the term $4B_3(A_1/A_0)\varepsilon_k$ scale by the factor; [10, (3.9), p. 266], $O(1)(A_1/A_0)\varepsilon_k < \varepsilon_k$; [10, (3.13), p. 267], $(4dL)^2(1+\beta_0)B_3\varepsilon_k$; the regularity $|\partial U_{k+1} - 1| < 2B_3\varepsilon_{k+1}(L^{-1}\eta)^2$ and the bound $B_3\exp(-\delta L M_2 R_{k+1})44d^2B_3\varepsilon_{k+1}$ of [10, p. 268]; $O(1)\delta_j' < \varepsilon_j$ [12, p. 184]; $O(1)B_3\exp(-\delta L M_2 R_{j+1})22d^2(1+\beta_0)(A_0/A_1)\delta_j \le \frac{1}{2}\delta_j$ [12, p. 185]; the regularity $|\partial V_k - 1| < 2L_0^{2N_0}\varepsilon_k$, absorbed in $O(1)M^2L^2R_k\varepsilon_k$ [12, p. 193]; and in [13, pp. 356–377] the bounds linear or quadratic in ε_k with a constant, such as $44d^2\varepsilon_k$ [13, p. 365], used for γ small or against L^{-N} [13, p. 363].

The others compare a quantity with a threshold at the same order, and there the margin has to be checked. They are the following.

(i) [12, (1.31), p. 184], the smallest margin. The plaquettes of $U_{j,\square}$ are bounded by $(1-\beta/2+(2\beta/10)L^{-2}+4\beta/10)\varepsilon_j\xi^2 = (1-\beta/10+\beta/(5L^2))\varepsilon_j\xi^2$, from the last inequality of (1.24) for the cascade and the bound $B_3\exp(-2\delta M_2 R_j)22d^2\varepsilon_j < (\beta/10)\varepsilon_j$ of the boundary term. With the cascade threshold enlarged by $1+2\kappa_s = 1+\beta/15$, and the boundary term by the same factor, which is absorbed for γ small, the bound becomes

$$\Big(1+\frac{\beta}{15}\Big)\Big(1-\frac{\beta}{10}+\frac{\beta}{5L^2}\Big) \le \Big(1+\frac{\beta}{15}\Big)\Big(1-\frac{7\beta}{90}\Big) = 1-\frac{\beta}{90}-\frac{7\beta^2}{1350} < 1 \qquad (2.12)$$

times $\varepsilon_j\xi^2$, where the first step is $L \ge 3$. So the test functions (1.3), and (1.4) for $j+1<k$, are on their plateaus, as on [12, p. 184].

(ii) [10, (3.8), p. 266]: the bound $(2(1+\beta_0)L^{-2}+4/10)\varepsilon_k\eta^2$ becomes $(1+\kappa_s)(2(1+\beta_0)L^{-2}+4/10)\varepsilon_k\eta^2 \le 0.66\,\varepsilon_k\eta^2$ for $L\ge 3$ and $\beta_0 \le 1/7$, which is below the plateau $\varepsilon_k\eta^2$.

(iii) [12, (1.96), p. 199]: with $\alpha = 1/12$ [12, p. 199], the bound of the plaquettes of $U''_{k,Z}$ drawn from the passed $\chi_{h,1/2}$ at the threshold $\frac{1}{2}(1+\kappa_s)\varepsilon_h$ is $(\frac{1}{2}(1+\kappa_s)+2\alpha+O(\varepsilon_h))\varepsilon_h(L^{k-h}\eta)^2 = (\frac{2}{3}+\frac{\kappa_s}{2}+O(\varepsilon_h))\varepsilon_h(L^{k-h}\eta)^2$, which is below Bałaban's $\frac{3}{4}\varepsilon_h(L^{k-h}\eta)^2$ for γ small, since $\kappa_s < 1/6$.

(iv) [12, p. 199]: the bound $22d^2\varepsilon_h+2\delta'_k < 23d^2\varepsilon_h$ becomes $22d^2(1+2\kappa_s)\varepsilon_h+2\delta'_k < 23d^2\varepsilon_h$, which holds for γ small because $44d^2\kappa_s < d^2$.

(v) [12, (1.98), p. 200]: at $j=k$ the leading term $2\varepsilon_k\eta^2$, drawn from the passed $\chi_{k,\Lambda}$, becomes $2(1+\kappa_s)\varepsilon_k\eta^2$, and the bound $(2(1+\kappa_s)+5\alpha+O(\varepsilon_k))\varepsilon_k\eta^2 \le 2.43\,\varepsilon_k\eta^2$ lies below the threshold $3(1-\beta/2)\varepsilon_k\eta^2 = 2.8125\,\varepsilon_k\eta^2$ of (1.24) at $c=3$; at $j<k$ the bound has the room of the factor $L^{-(k-j)}$.

(vi) [13, p. 378, l. 10]: the removal of $\chi_{k,\Lambda}$ on the first class compares the plaquettes of $U_{k,Z}$ with the threshold $2\varepsilon_k\eta^2$ through the passed test functions at $(1+\kappa_s)\varepsilon_k\eta^2$, and Lemma 17.21 gives the bound $(1+\kappa_s+2\tau+9\tau^2)\varepsilon_k\eta^2 < 2\varepsilon_k\eta^2$ for $\kappa_s \le \frac{1}{4}$.

(c) The change of variables [10, (1.22), p. 251] contains a smooth function g with $g=1$ on a neighbourhood of the set where the tested quantity of the reference test function reaches the threshold $2\delta_0$ [10, p. 252]. A soft test function and its complement differ from 1 and from 0 only where the tested quantity is at least the threshold, and there $g=1$, so that the two arguments coincide; where $g<1$ the tested quantity is below the threshold, and both the soft and the sharp test function equal one. The change of variables therefore carries the soft test function into the soft test function of the new variables. The same holds at [10, p. 269] and for [12, (1.59), p. 188], which are changes of variables of the same form.

(d) The cutoff of Definition 2.2(iii) is the same function in the numerator of [12, (1.100)], at $V' = V'_k$, and in its denominator, an integral over V'. At fixed V_k outside Λ the function V_Λ does not depend on $V_k\lceil_\Lambda$, so that $V_k\lceil_\Lambda \mapsto V'_k = V_k(V_\Lambda)^{-1}$ is a translation of the Haar measure, and the integral of the quotient over $V_k\lceil_\Lambda$ is one, which gives (1.102). Since $\bar{\chi}(y) = \kappa_s^{-1}\int_1^{1+\kappa_s}\mathbf{1}\{y<s\}\,\mathrm{d}s$, the denominator with the soft cutoff is the average, over independent thresholds $s_b \in [1, 1+\kappa_s]$ of the bonds, of the denominators with the sharp cutoffs $|(1/i)\log V'(b)| < s_b M_0\varepsilon_k$, and at real configurations it lies between the sharp denominators at $s_b = 1$ and $s_b = 1+\kappa_s$. In the variable B' of [13, (1.1), (1.2), pp. 356–357] the sharp cutoff is $|B'| < s\,t$ with $t := M_0 g_k^{-1}\varepsilon_k$. The difference of the two denominators is an integral over the set where $t \le |B'(b)| < (1+\kappa_s)t$ for some bond b; there the real part of the quadratic form is at least $c_\Lambda \|B'\|^2$ with a constant $c_\Lambda > 0$, by [13, (1.7)–(1.9), p. 358], and the remaining factors of the integrand and the denominator at $s = 1$, bounded below by [13, (1.10), (1.11), p. 358], contribute at most a factor $e^{C(M)}$, since Λ has at most $O(M^4)$ bonds. So the two denominators differ relatively by at most $O(M^4)e^{C(M)}\exp(-\frac{1}{2}c_\Lambda t^2)$. Since $t^2 = M_0^2\varepsilon_k^2 g_k^{-2} \ge \rho^{-1}M_0^2A_0^2(\log g_k^{-2})^{2p_0}$ by Paper, Lemma 3.2(i), this is at most $\frac{1}{4}$ for γ small, and the logarithms of the denominators differ by at most $\frac{1}{2}$.

(e) In the term without large-field regions every domain Ω_j, Λ_j is the whole torus, every test function of an earlier step has passed, and Bałaban removes each of them on the support of the test functions of the next step, [10, (3.6)–(3.9), pp. 265–266], "on $\Omega_{k+1}^{\sim}$ the functions χ_{Ax} and χ_k are equal to 1" [10, p. 265]. By (a) these removals hold for the soft test functions, and by induction on the step only χ_k remains. No **R**-operation acts on this term, since it has no large-field component. *q.e.d.*

The proposition applies verbatim to the first step of the paper, [1], which is the step of [10, Sect. 1] with the loop inserted: its bounds drawn from the passed test functions (1.4) and (1.8) of [10] hold at $(1+\kappa_s)\varepsilon_1 L^{-2}$ and $2(1+\kappa_s)\delta_0$, and its uses of failed test functions, which exclude a label by a bound below the threshold and take a witness from a nonvanishing complement, hold as printed, since $1-\bar{\chi}(f/t)$ is nonzero only for $f > t$.

2.3 The Depth

Bałaban does not fix the depth N of the **R**-operation: "conditions on N will be formulated in constructions of this section" [12, p. 177]. In the large-field induction of [13] he takes $N = R_j$ [13, p. 384], and with this choice L^N is not bounded by a power of $\log g_k^{-2}$, which the inductive step of the paper needs (Paper, Section 3.4). The paper takes $N = N_k$ of Paper, Definition 3.9, at the **R**-operation of the step k, and Paper, Lemma 3.10, gives the bounds

$$L^{-N_k} \le (\log g_k^{-2})^{-\nu_L}\ , \qquad L^{N_k} \le C_{\mathrm{dep}}(\log g_k^{-2})^{\nu_L}\ , \qquad N_k \le R_k\ , \qquad N_{k+1} \le N_k\ , \quad (2.13)$$

with $\nu_L = p_0 + q_1 + 7r + 1$, for every history of a tube. It remains to see that these are all that the two papers need.

Proposition S.3 (The Depth)**.** *Let γ satisfy the hypotheses of Paper, Lemma 3.10, and be small. For every tube and every history as in Proposition S.1, every statement of [12, 13] in which the depth enters holds for the construction of the paper, in which the* **R***-operation of the step k has the depth N_k. In the large-field induction [13, (1.80), pp. 384–387], stated for $N = R_j$, the number K of steps which a domain needs before the* **R***-operation of the step $j+K$ removes it is at most its value for $N = R_j$, and the bound (1.80) holds as printed.*

Proof. We go through the places at which N enters, and we sort them into four kinds.

The conditions which Bałaban states. They are (N1)–(N4) of Paper, Section 3.4: $N > N_0$ [12, p. 179]; the two restrictions of [12, (1.94), p. 198], $N \le O(1)(\log g_k^{-2})^{\nu}$ with $\frac{1}{2}\nu \le p_0 - p_1 - 1$, and N so large that the constant of the second term of (1.94) is at most $\frac{1}{2}\alpha$; $N \le R_k$ [13, p. 361]; and $L^{-N} \le (\log g_k^{-2})^{-\nu}$ for a sufficiently large ν [13, p. 363], which recurs as "the conditions discussed above" and "the usual conditions on N" [13, p. 364, (1.47), p. 368] and as "our assumptions on N" [13, p. 382]. Paper, Section 3.4, shows that N_k satisfies each of them, by (2.13): in particular the product $O(1)A_0C_1B_3^3B_5M^{10}p_0(g_k)q_1(g_k)R_k^7L^{-N}$ of [13, (1.47)] is at most a constant times $(\log g_k^{-2})^{p_0+q_1+7r-\nu_L} = (\log g_k^{-2})^{-1}$, and the number

$O(1)(A_0/A_1)B_3^2B_5M^7R_j^{1+\beta_0}(p_0(g_j)/p_1(g_j))L^{-N}$ of [13, p. 382], which has to be below $\frac{1}{4}$, is at most a constant times $(\log g_j^{-2})^{r(1+\beta_0)+p_0-p_1-\nu_L}$, whose exponent is negative. In the last bound every quantity, R_j, $p_0(g_j)$, $p_1(g_j)$, and the depth, belongs to the step j at which the operation is performed [13, p. 382], and in the construction of the paper the depth of that operation is N_j; the substitution is therefore direct, and no comparison between the depths or the couplings of two different scales enters.

Upper bounds. The depth enters as an upper bound, through $N \le R_k$, in $11d^2\varepsilon_h \le 11d^2(1+\beta_0)^2N^{\beta_0}\varepsilon_k \le 11d^2(1+\beta_0)^2R_k^{\beta_0}\varepsilon_k$ [13, p. 362], in $O(N) \le O(R_k)$ and $N^{2+4\beta_0}(100MR_k)^4 \le O(R_k^7)$ [13, p. 366], in $R_{j-N+1} \le (L+1)N^{\beta_0}R_j$ [13, pp. 382, 383], and in the range $j - N \le m \le j$ of the scales of a factor [13, p. 381]. Each of these expressions increases with N, and each holds for $N_k \le R_k$.

Step counts. In the proof of (1.80) the number K counts the steps after which the domain $S^K(Z)$ satisfies the conditions (i), (ii) of [12, p. 177] with $N = R_j$: $K \le n_0 - j + R_j$ [13, p. 385], $K = R_{j+1}$ for a small domain [13, p. 386], and $K \le K_2 + n_1 + R_{j+1}$ [13, p. 387]. The condition (ii) asks that no new large-field region has been created in the domain in the preceding N steps, and it is weaker for a smaller N; the **R**-operation which removes $S^K(Z)$ is that of the step $j+K$, with the depth $N_{j+K} \le N_j \le R_j$, by (2.13). So the number of steps after which the domain is removed is at most the number K of Bałaban's counts, and since the right side of (1.80) increases with K, the bound holds as printed.

The other uses. In [12] the depth enters, besides (N1) and (1.94), the proof of the removal of χ_k'': the bound $22d^2\max\{1, L^{-2N}N^{1/2}O(1)B_3B_5M^5(1+\beta_0)\}\varepsilon_h + 2\delta_k' < 23d^2\varepsilon_h$ [12, p. 199], which needs $L^{-2N}N^{1/2}O(1)B_3B_5M^5(1+\beta_0) \le 1$ and $N^{1/2}p_1(g_h)/p_0(g_h)$ small, and the bound $B_3(1+\beta_0)(1+(k-j)^{1/2})(A_1/A_0)(p_1(g_j)/p_0(g_j)) < \alpha$ for $h \le j \le k$, where $k - j \le N$ [12, p. 199]. Since $p_1(g)/p_0(g) = A_0^{-1}(\log g^{-2})^{p_1-p_0}$ with $p_0 - p_1 > 1$, and $(\log g_j^{-2})^{p_1-p_0} \le (1+\beta_0)(\log g_k^{-2})^{p_1-p_0}$ for $j \le k$ by Bałaban's (2.7), and since $N_k \le \log_L C_{\mathsf{dep}} + \nu_L \log_L \log g_k^{-2}$ by (2.13), both products are at most a constant times $(\log\log g_k^{-2})^{1/2}(\log g_k^{-2})^{p_1-p_0}$, which tends to zero; and $L^{-2N_k}N_k^{1/2} \to 0$ as well. The depth also enters definitions: the powers $L_0^{2N_0}$ of [12, p. 192], the width of a boundary layer [13, p. 363], the number of the new boundary terms [13, p. 369], and the spaces of the analytic extensions at $n = N$ [13, p. 375]; these hold at every N.

Every place at which the depth enters the two papers is of one of these kinds, and in each case the statement holds for N_k. *q.e.d.*

3 The Coefficients of the Flow

Paper, Proposition 4.3, asserts three properties of the coefficient $\beta_{k+1}(h)$ of the step $k+1$ along a history h of a tube: it differs from the coefficient b_{k+1} at zero coupling by at most δ', it differs from it by a multiple of g_k^2, and it is continuous in h. By Paper, (4.8), the difference is the polarisation coefficient Λ of the fluctuation integral of the step, and two things stand between this and the bounds. Bałaban bounds the terms of the fluctuation integral by a multiple of the threshold of the fluctuation field [11, (2.41), p. 21], which gives (i) once Λ is shown to be bounded on local terms, but a bound by g_k^2 needs a symmetry which he does not use: replaced by a complex number z in the exponent, the coupling enters only through z times the field, and the integral is an even function of z. The threshold breaks this picture, since the domain of analyticity in z shrinks as the threshold grows, and the second idea is to telescope over thresholds which double, the differences being Gaussian tails. The **R**-terms of the earlier steps enter the fluctuation integral as well, and they are separated by a family of densities analytic in a factor multiplying them. This section proves (i) and (ii), as Proposition S.4; the continuity (iii) is Proposition S.5, and the coefficients at zero coupling are the subject of Proposition S.6.

3.1 The Fluctuation Integral as a Function of the Coupling

Fix a tube, an integer $k < K$, and a history $h = (x_0, \dots, x_k) \in H_k$ with $g_k = x_k^{-1/2}$. We use the units of the step $k+1$ and the notation of [9, Sect. 2] and [11]. The fluctuation integral of the step is the logarithm of the integral [9, (2.12), (2.13), p. 268] of the exponent of [9, (2.12)] against the Gaussian measure $\mathrm{d}\mu_{C^{(k)}}(B)$, with the cutoff of the fluctuation field. In the construction of the paper the cutoff restricts the unscaled field by the threshold ϑ_k (Paper, Definition 3.11(ii)); over the bonds $b \notin \{b_0(c)\}$ on which [9, (2.9), p. 266] acts it depends on the Gaussian variable only through $|B(b)| < p$, where

$$p := \frac{\vartheta_k}{g_k} = \frac{\hat{g}_k}{g_k} A_1 (\log \hat{g}_k^{-2})^{p_0} \geq \rho^{-1/2} A_1 (\log g_k^{-2})^{p_0} \ , \tag{3.1}$$

by Paper, Definition 3.3(ii) and Lemma 3.2(i). For a threshold $q > 0$ let χ_q be the product over the same bonds of the characteristic functions of $\{|B(b)| < q\}$. For $z \in \mathbb{C}$ let $W(z, B) := \sum_Y V_k(Y; z, B)$ be the localised exponent [11, (1.41), p. 11] with g_k replaced by z, and put

$$E(z; q) := \log \int \chi_q \exp W(z, B) \, \mathrm{d}\mu_{C^{(k)}}(B) \ , \tag{3.2}$$

so that the fluctuation part of the regular term of the step is $\mathbf{E}_{\mathrm{fl}}^{(k+1)} = E(g_k; p)$.

Parity. Every term of the exponent of [9, (2.12), p. 268] is a function of $g_k CB$, or g_k^{-2} times such a function of order at least three: the terms $\operatorname{Tr} \log(I - h(\delta \tilde{D}/\delta B)(g_k CB))$ and $\log \sigma(g_k CB - h\tilde{D}(g_k CB))$, and the terms $g_k^{-2} \mathbf{G}_3(g_k CB)$ and $g_k^{-2} \langle H_1 g_k CB, \Delta_1 H_1 h \tilde{D}(g_k CB) \rangle$, $\tilde{D}$ starting at order two. The localisation of [11, (1.33)–(1.41), pp. 9–11] keeps this form when g_k is replaced by z: its pieces are functions of zCB, with the prefactor z^{-2} or 1, and the bracket of the effective action enters through $(z/g_k)B$ at the real coupling (Lemma 3.2). Hence $W(-z, B) = W(z, -B)$. The Gaussian measure is centred and χ_q is even, so that

$$E(-z; q) = E(z; q) \tag{3.3}$$

as functions of the background field and the current.

Lemma 3.1 (The Polarisation Is Bounded)**.** *Let $F = \sum_X F(X)$, with $F(X)$ analytic on the space $U_{k+1}^c(X, \alpha_0, \alpha_1)$ of [9, (1.10)–(1.16), p. 262], depending on the configuration restricted to X, and with $|F(X)| \leq A e^{-\kappa d_{k+1}(X)}$. Let $r \in (0, 1]$ be such that the polydisc of the configuration of radius r in the norms of the space, about the trivial configuration, lies in it. Then the polarisation coefficient* (4.5) *of the paper satisfies $|\Lambda(F)| \leq C_\Lambda(1) r^{-2} A$, where $C_\Lambda(1)$ depends on κ, δ_0, d, M, and L only. The bound holds for each component of the second moment of F separately and uses no invariance of F; that the second moment has a single isotropic coefficient is a property of the families to which the lemma is applied, the covariant families of [10, (2.29)], and is not part of the lemma.*

Proof. Write $u(B)$ for the configuration $(U_{k+1}(e^{iB}), J_{k+1}(e^{iB}))$ restricted to X. The map $B \mapsto U_{k+1}(e^{iB})$ is analytic for small complex B, and its derivative decays exponentially with the rate δ_0, by [7, Proposition 9, p. 309] and [7, (190), p. 308], so that $\|\partial u / \partial B_\nu(y)\| \leq C e^{-\delta_0 \operatorname{dist}(y, X)}$; the bound (189) from which [7, p. 308] derives (190) is stated there without the calculation, and Lemma 17.3 proves it. Cauchy's estimate in a single variable $B_{\nu'}(y')$ on a disc of fixed radius, applied to this derivative and symmetrised, gives

$$\left\| \frac{\partial^2 u}{\partial B_\nu(y) \partial B_{\nu'}(y')} \right\| \le C e^{-\delta_0(\mathrm{dist}(y,X)+\mathrm{dist}(y',X))/2} . \tag{3.4}$$

Cauchy's estimate on the lines $u + tv$ and $u + tv + sw$ in the configuration gives $|DF(X)[v]| \le A e^{-\kappa d_{k+1}(X)} \|v\|/r$ and $|D^2F(X)[v,w]| \le 4Ae^{-\kappa d_{k+1}(X)}\|v\|\|w\|/r^2$. By the chain rule the second derivative $\Pi(x)$ of Paper, (4.5), is the sum over X of $D^2F(X)[\partial_{B_\mu(0)}u, \partial_{B_\nu(x)}u] + DF(X)[\partial^2_{B_\mu(0)B_\nu(x)}u]$, and with $r \le 1$ the term of X is at most $CAr^{-2}e^{-\kappa d_{k+1}(X)}e^{-\delta_0(\mathrm{dist}(0,X)+\mathrm{dist}(x,X))/2}$. Finally $\sum_x |x|^2 e^{-\delta_0 \,\mathrm{dist}(x,X)/2} \le C(1 + \operatorname{diam} X + \mathrm{dist}(0,X))^{d+2}$, and $\operatorname{diam} X \le M(2d_{k+1}(X) + 4)$, by Lemma 17.1; summing first over the X which contain a given cube, the cube of X nearest to 0, by [11, (1.26), p. 8], and then over the cube, the series converges for κ large, and its sum is linear in A and proportional to r^{-2}. *q.e.d.*

Lemma 3.2 (Complex Coupling). *Let ε^* be such that $A_\chi\varepsilon^*$ satisfies every condition which [9, 11] place on their cutoff ε_1, where $A_\chi := (1 + 2^{12}(M+1)^4)(1 + 1/(\delta_{\mathrm{RG}}\kappa))$ and $\delta_{\mathrm{RG}} := (1 - 2L^{-1})/12$, and let q_{th} be the number of Lemma 3.3. Let $q \ge q_{\mathrm{th}}$ and $|z|q < \varepsilon^*$. Then the expansion [11, (2.1)–(2.13), pp. 12–14] applies to* (3.2) *and gives $E(z;q) = \sum_X E(X;z,q)$, where each $E(X;z,q)$ is analytic in z on $|z| < \varepsilon^*/q$ and in the configuration on $U^c_{k+1}(X,\alpha_0,\alpha_1)$, satisfies*

$$|E(X;z,q)| \le O(1)C_3\varepsilon^* \exp(-\kappa d_{k+1}(X)) , \tag{3.5}$$

with the constant C_3 of [11, p. 20], and equals Bałaban's term at $z = g_k$, $q = p$.

The spaces $U^c_{k+1}(X,\alpha_0,\alpha_1)$ of the lemma are those of [9, 11], with numbers α_0, α_1 which are small and do not depend on the coupling; they are not the spaces of the tube, whose radii $\alpha_{i,k}$ shrink with the coupling. The lemma concerns the exponent [11, (1.41)] of the small-field step, which is analytic on the fixed spaces, and it is on these that Lemma 3.1 is applied with a radius r independent of the coupling. The terms of the earlier steps, which the construction controls on the shrinking spaces only, do not enter through this lemma: the **R**-terms are removed first and are treated in Lemma 3.4, with the Cauchy cost of their own radii, which is the factor g_k^{-2} there.

The conditions on ε_1 are all upper bounds on it, so that ε^* exists; they are listed in Lemma 17.2. The number δ_{RG} replaces Bałaban's $\delta = (1 - 2L^{-1})/10$ [11, p. 21], which he uses only to bound $\exp(-(1 - 10\delta)\frac{1}{2}L\kappa d_{k+1})$ by $\exp(-\kappa d_{k+1})$ [11, (2.41), p. 21]; with the smaller value, $(1 - 12\delta_{\mathrm{RG}})\frac{1}{2}L = 1$ leaves room for one more loss of δ_{RG} in the exponent, which Lemma 3.3 uses, and his other conditions on δ are lower bounds on κ.

Proof. The bracket of the effective action in the exponent of [9, (2.12)] depends on (g_k, B) only through g_kCB, and [11, (1.33)] localises it at real B as $\sum_Y V'_k(Y,B)$. The bracket term of W is $\sum_Y V'_k(Y,(z/g_k)B)$, the same sum at the real coupling g_k and a complex field; V'_k is holomorphic on the polydisc $|(z/g_k)B| < \varepsilon^*/g_k$ and satisfies [11, (1.36), p. 9] there, by Lemma 17.2, so that $V'_k(Y,(z/g_k)B)$ is analytic in z for $|z||B| < \varepsilon^*$ on Y. The other terms of the exponent are built in [11, (1.38)–(1.40), p. 10] from the unscaled fluctuation field $B' = g_kCB - h\tilde{D}(g_kCB)$ [11, (1.19), p. 6], and "it can be done in the same way for all terms" [11, p. 11]; with z in place of g_k each of them is the same analytic function of zCB and of $B' = zCB - h\tilde{D}(zCB)$. The prefactor z^{-2} cancels: the second-order expansion [11, (1.40)] writes the pieces as $z^{-2}\langle \hat{Q}B', B'\rangle = \langle \hat{Q}B'/z, B'/z\rangle$, and $B'/z = CB - z^{-1}h\tilde{D}(zCB)$ is analytic also at $z = 0$, since $\tilde{D}(zCB)$ is holomorphic in z and of order $|z|^2|CB|^2$; the other terms with the prefactor z^{-2} are of order at least three in zCB. By [11, (1.20), p. 6] with $|z|$ in place of g_k, $|B'| \leq C'_1|z||B| < C'_1\varepsilon^*$, so that [11, (1.43), (1.36)] hold with ε^* in place of ε_1 and each $V_k(Y; z, B)$ is analytic in z for $|z||B| < \varepsilon^*$ on Y; Lemma 17.2 carries Bałaban's Lemmata 1 and 2, stated for real g_k, to this setting.

The expansion of [11, Sect. 2] uses the potentials only through [11, (1.42), (1.43), (1.36)] and the count of the cubes, in [11, (2.18)–(2.20), p. 16], and through their analyticity in the configuration [11, (1.34), p. 9], which does not involve z. It uses the cutoff only through a factor $\chi_{k,Y_0}\chi^c_{k,P}$ with values in $[0, 1]$ and the tail bound [11, (2.22), p. 16]: the potentials are evaluated on the support of χ_{k,Y_0} [11, (2.3), p. 12], where $|z||B| < |z|q < \varepsilon^*$, and the threshold enters the conditions of the expansion only as ε_1/g_k in [11, (2.22), (2.31), (2.33), (2.34)], which with q in its place are lower bounds on q and hold for $q \geq q_{\text{th}}$. So [11, (2.1)–(2.41)] hold with ε^* in place of ε_1 and q in place of ε_1/g_k, which gives (3.5). Each term [11, (2.14), p. 15] is an integral of a function analytic in z, dominated by [11, (2.26), p. 17] uniformly on compact subsets of $|z| < \varepsilon^*/q$, and the sums [11, (2.35)–(2.41)] converge uniformly there, so that $E(X; z, q)$ is analytic in z. *q.e.d.*

Lemma 3.3 (Threshold Differences). *There is a number q_{th}, depending on ε^*, κ, M, and the constants of [11], such that the following holds. Let $q_{\text{th}} \leq q' < q'' \leq 2q'$ and $|z|q'' < \varepsilon^*$. Then $E(z; q'') - E(z; q') = \sum_X \Delta(X; z)$, where each $\Delta(X; z)$ is analytic in z and in the configuration on $U^c_{k+1}(X, \alpha_0, \alpha_1)$, and, for every $q \geq q_{\text{th}}$,*

$$\begin{aligned} |\Delta(X; z)| &\leq C_4 \exp(-\tfrac{1}{4}\gamma_2 q'^2) \exp(-\kappa d_{k+1}(X)) \, , \\ |E(X; 0, q)| &\leq C_5 \exp(-\tfrac{1}{20}\gamma_2 q^2) \exp(-\kappa d_{k+1}(X)) \, , \end{aligned} \tag{3.6}$$

with constants C_4, C_5 which do not depend on k, the history, q', q'', q, or z, and with the constant γ_2 of [11, (2.22)].

Proof. The families $\mathbf{D}$, the sets $P \subset Y_0^{c*}$, and the domains Z_0 and Z of [11, (2.1)–(2.13)] do not depend on the threshold [11, (2.1)–(2.3), p. 12], so that the activities at q'' and at q' differ term by term only in the factor $\chi = \chi_{Y_0}\chi^c_P$. With χ'' and χ' this factor at the two thresholds,

$$\chi'' - \chi' = (\chi''_{Y_0} - \chi'_{Y_0})\chi''^c_P + \chi'_{Y_0}(\chi''^c_P - \chi'^c_P) \, . \tag{3.7}$$

The first term is nonzero only if some bond b of χ_{Y_0} has $|B(b)| \geq q'$, and the second only if $P \neq \emptyset$ and every $b \in P$ has $|B(b)| \geq q'$. Bounding each indicator as in [11, (2.22)] by $\mathbf{1}\{|B(b)| \geq q'\} \leq \exp(-\frac{1}{2}\gamma_2 q'^2 + \frac{1}{2}\gamma_2|B(b)|^2)$, the first term by a sum over its bonds, and using $\frac{1}{2}(1 + |P|) \geq \frac{1}{4} + \frac{1}{4}|P|$ and $\frac{1}{2}|P| \geq \frac{1}{4} + \frac{1}{4}|P|$ for $P \neq \emptyset$, we obtain

$$|\chi'' - \chi'| \leq (N_\chi(Y_0) + 1) \exp(-\tfrac{1}{4}\gamma_2 q'^2) \exp(-\tfrac{1}{4}\gamma_2 q'^2|P| + \tfrac{1}{2}\gamma_2\|Z_0 B\|^2) \, , \tag{3.8}$$

where $N_\chi(Y_0)$ is the number of the bonds of χ_{Y_0}. The factor $\exp(\frac{1}{2}\gamma_2\|Z_0 B\|^2)$ is absorbed into the Gaussian measure as in [11, (2.23), p. 16], and the factor $\exp(-\frac{1}{4}\gamma_2 q'^2|P|)$ runs the sums [11, (2.31)–(2.34)] with $\gamma_2/2$ in place of γ_2, which hold for $q' \geq q_{\text{th}}$ once q_{th} is large. Each bond of χ_{Y_0} has an endpoint in $Y_0 = \bigcup_{Y \in \mathbf{D}} Y$, a closed cube of side M contains at most $(M+1)^4$ lattice points, each of which is an endpoint of 8 bonds, and $M^{-4}|Y| \leq 2^9(1 + d_k(Y))$ by Lemma 17.1, so that

$$N_\chi(Y_0) + 1 \leq 1 + \sum_{Y \in \mathbf{D}} 2^{12}(M+1)^4(1 + d_k(Y)) \leq \prod_{Y \in \mathbf{D}} A_\chi e^{\delta_{\text{RG}}\kappa d_k(Y)} \, , \tag{3.9}$$

by $1 + \sum_Y a_Y \leq \prod_Y(1 + a_Y)$ and $1 + s \leq (1 + 1/(\delta_{\text{RG}}\kappa))e^{\delta_{\text{RG}}\kappa s}$ for $s \geq 0$. Each factor joins the factor of Y in [11, (2.26), p. 17], which becomes $2E_0 A_\chi \varepsilon^* C_1 \alpha_4^{-1} M^q e^{C_2\kappa_1} \exp(-(1 -$

$4\delta_{\mathrm{RG}})\kappa d_k(Y))$; since $A_\chi\varepsilon^*$ satisfies the condition of [11, p. 18], the chain [11, (2.27)–(2.38)] runs with $A_\chi\varepsilon^*$ in place of ε_1 and one more δ_{RG} in every exponent, and bounds the difference of the activities by its bound times $O(1)\exp(-\frac{1}{4}\gamma_2 q'^2)$. In the polymer series [11, (2.13)] the difference of two products is a sum of products with one factor of the difference, and [11, (2.39)–(2.41)] give the first bound of (3.6), the extra δ_{RG} being absorbed by $(1-12\delta_{\mathrm{RG}})\frac{1}{2}L = 1$.

At $z = 0$ every localised potential vanishes: in Bałaban's Lemma 1 the function $H_k(s, 0) = 0$ [11, (1.18), p. 6], so that the Cauchy integral which defines the potential vanishes; in the action terms $B' = 0$ and the coefficients of [11, (1.40)] are of order $|B'|$; and the terms without the prefactor, $\mathrm{Tr}\log(I - h(\delta\tilde{D}/\delta B))$ and $\log\sigma$, vanish at $B' = 0$, because $\tilde{D}$ starts at order two and σ there is normalised to one at $B' = 0$. So only $\mathbf{D} = \emptyset$ contributes to [11, (2.1)], every nonzero term has $P \neq \emptyset$, and [11, (2.33)] gives the factor $\exp(-\frac{1}{20}\gamma_2 q^2)$ per component in place of $\max\{\varepsilon_2, \dots\}$, which with [11, (2.37)–(2.41)] is the second bound of (3.6). *q.e.d.*

Lemma 3.4 (The **R**-Terms)**.** *Let γ be small, let $\beta_{k+1}(h)$ be the coefficient of the construction of the paper along h, and let $\beta^0_{k+1}(h)$ be the coefficient of the same recursion along h in which no* **R***-term enters the regular part. Then*

$$|\beta_{k+1}(h) - \beta^0_{k+1}(h)| \leq C_R g_k^{\kappa_0 - 4} , \tag{3.10}$$

with a constant C_R which does not depend on k, K, the history, or the tube, and with the exponent κ_0 of [10, (2.31), p. 260], which Bałaban takes ≥ 7 [10, (2.46), p. 263].

Proof. The **R**-terms satisfy $|\mathbf{R}^{(j)}(X, (\mathbf{U}, \mathbf{J}))| \leq g_j^{\kappa_0}\exp(-\kappa d_j(X))$ by Proposition S.1(e), and Bałaban includes their part R'_k into the regular term of the step [10, p. 271], while the part which stays an **R**-term needs no counterterm [10, pp. 260, 276]. For $s \in \mathbb{C}$ we run the regular recursion of the construction along h with sR' in place of R' at every step, and with the coefficients $\beta_j(s)$ computed from its own regular terms by [10, (3.57), (3.61), pp. 281–282]. At $s = 1$ this is the construction, and at $s = 0$ no **R**-term enters, so that $\beta_{k+1}(1) = \beta_{k+1}(h)$ and $\beta_{k+1}(0) = \beta^0_{k+1}(h)$. The recursion at complex s is a recursion for the family of the regular terms and not for a density: the functions R'_j are those of the construction, at $s = 1$, and only the number by which they are multiplied changes. At each step the new regular terms are obtained from the old ones and from sR'_j by the exponentiated cluster expansion of the small-field fluctuation integral [10, (3.26)–(3.47), pp. 271–278], which uses of the interaction only its localisation, its analyticity, and its bound, as in Proposition S.16(a), and no positivity; so the recursion is defined, and its terms are analytic in s and satisfy the inductive bounds, as long as $|s|$ times the bound of R'_j stays within the bound of the regular terms at every step $j \leq k$, which is what the disc of radius Θ_k below ensures.

The size of the **R***-part.* By (3.4) of the paper, $x_j \geq x_k + b(k-j) - C_W \geq \frac{1}{2}x_k + (k-j)c\ln L$ for $j \leq k$, since $b \geq c\ln L$ and $C_W \leq t/8 \leq x_k/7$. So

$$\sum_{j\leq k} g_j^{\kappa_0} \leq \sum_{m\geq 0}\left(\tfrac{1}{2}x_k + mc\ln L\right)^{-\kappa_0/2} \leq C_\Sigma g_k^{\kappa_0-2} , \qquad C_\Sigma := 2^{\kappa_0/2}\Big(1 + \frac{2}{(\kappa_0 - 2)c\ln L}\Big) . \tag{3.11}$$

The **R**-part of the input of a step enters the fluctuation integral through the potential V' of Bałaban's Lemma 1, which is built linearly from the bracket of the effective action [11, (1.33)], the sums over the domains and the scales being controlled by the exponential factor and by $g_j^{\kappa_0}$ [10, p. 260]. So there is a constant K_V, depending on the constants of the step and on κ only, such that an input whose **R**-terms are bounded with $a_i g_i^{\kappa_0}$ in place of $g_i^{\kappa_0}$ enters with $E_0 + K_V\sum_{i\leq j} a_i g_i^{\kappa_0}$ in place of E_0 in [11, (1.36), p. 9]. Here $a_i = |s|$. With $\theta := E_0/(K_V C_\Sigma)$ and $\Theta_k := \theta g_k^{2-\kappa_0}$, (3.11) gives $K_V|s|\sum_{i\leq j} g_i^{\kappa_0} \leq E_0$ for $|s| \leq \Theta_k$ and every $j \leq k$, so that the input of every step is bounded with $2E_0$ in place of E_0.

The step with $2E_0$ *and complex* s. The step accepts such inputs. Its conditions which involve E_0 are upper bounds on the cutoff relative to the radii of the domains, [11, pp. 18, 20, 21], which are imposed for $2E_0$ when the constants of the radii are chosen, and two lower bounds on the threshold which only get weaker as E_0 grows [11, pp. 16, 19]. The expansion of the step [11, Sect. 2], which [10, p. 277] uses, is analytic in complex multipliers of the interaction: [11, (2.14), p. 15] carries complex coefficients $\tau(Y)$ and is bounded through the modulus of the interaction, the only positivity used being that of the Gaussian measure at the real configuration with zero current [11, p. 15]. A complex factor s on the **R**-part enters in the same way, so that by induction over the steps $j \le k$ every regular term of the family satisfies [9, (1.18), p. 263] with E_0 and is analytic in s on $|s| < \Theta_k$.

Polarisation and Cauchy's estimate. The fluctuation term of the step $k+1$ is analytic on $U^c_{k+1}(X, \alpha_{0,k+1}, \alpha_{1,k+1})$, with the radii of the tube, and bounded by $E_0 e^{-\kappa d_{k+1}(X)}$. The polydisc about the trivial configuration of radius $r := \min(\alpha_{0,k+1}, \alpha_{1,k+1})/C_{\text{cen}}$ lies in this space, with a constant C_{cen} depending on L, M, and the averaging operations only, and $r \le 1$ for γ small. Lemma 3.1 gives, with $\hat{x} := \hat{g}_{k+1}^{-2} \le \hat{g}_k^{-2} \le \rho x_k \le 2x_k$ and $q_* := \min(q_0, q_1) \ge 2$,

$$|\beta_{k+1}(s) - b_{k+1}| \le C_\Lambda(1) r^{-2} E_0 \le C' E_0 \hat{x} (\log \hat{x})^{-2q_*} \le 2C' E_0 g_k^{-2} \, , \tag{3.12}$$

with $C' := C_\Lambda(1) C_{\text{cen}}^2 / \min(C_0^\alpha, C_1^\alpha)^2$, and $s \mapsto \beta_{k+1}(s)$ is analytic on $|s| < \Theta_k$, the infinite-volume limit of Λ being uniform in s by the domination in Lemma 3.1. For γ small, $\Theta_k \ge \theta\gamma^{2-\kappa_0} > 2$, because $\kappa_0 > 2$ and $g_k \le \gamma$, so that for $s_0 \in [0,1]$ the closed disc of radius $\Theta_k/2$ about s_0 lies in $|s| < \Theta_k$, and Cauchy's estimate gives $|\beta'_{k+1}(s_0)| \le 4C' E_0 g_k^{-2}/\Theta_k$. Integrating over $[0,1]$,

$$|\beta_{k+1}(1) - \beta_{k+1}(0)| \le \frac{4C' E_0 g_k^{-2}}{\Theta_k} = 4C' K_V C_\Sigma g_k^{\kappa_0 - 4} =: C_R g_k^{\kappa_0 - 4} \, . \tag{3.13}$$

The constants C_Σ, K_V, and C' do not depend on the tube, so neither does C_R. *q.e.d.*

The bound (3.12) shows what a second derivative costs on the spaces of the tube: the inverse square of the radius, which is of the order g_k^{-2}. The coefficients are nevertheless bounded, because the regular terms consist of a part which is analytic on spaces whose radii do not depend on the coupling and of a part which is small enough to pay this cost. The comparison of two cutoffs uses this structure for a difference of two families, and we record it.

Lemma 3.5 (The Two Parts of the Regular Terms). *Let* γ *be small, fix a tube,* $k < K$, *and a history* $h \in H_k$, *let* $\mathbf{E}^{(j)}(s)$, $j \le k+1$, $|s| < \Theta_k$, *be the family of the regular terms of the proof of Lemma 3.4, in which* s *multiplies the* **R***-terms which enter the regular part, so that* $\mathbf{E}^{(j)}(1)$ *are the regular terms of the construction, and put* $\varrho_k := 1 + \hat{g}_k^{-2}$.

- *(a) Each term* $\mathbf{E}^{(j)}(X;0)$ *of the family at* $s = 0$ *extends to an analytic function on the space* $U^c_j(X, \alpha_0, \alpha_1)$ *of [9, (1.10)–(1.16), p. 262] with the numbers* α_0, α_1 *of Lemma 3.2, which do not depend on the coupling, and it is bounded there by* $E_0 e^{-\kappa d_j(X)}$.

- *(b)* $\varrho_k \le \frac{1}{2}\Theta_k$, *and for* $|s| \le \varrho_k$ *and* $j \le k+1$ *the term* $\mathbf{E}^{(j)}(X;s)$ *is analytic on the space* $U^c_j(X, \alpha_{0,j}, \alpha_{1,j})$ *of the tube, with*

$$|\mathbf{E}^{(j)}(X;s)| \le E_0 e^{-\kappa d_j(X)} \, , \qquad |\mathbf{E}^{(j)}(X;s) - \mathbf{E}^{(j)}(X;0)| \le C_\rho g_k^{\kappa_0 - 4} E_0 e^{-\kappa d_j(X)} \, , \tag{3.14}$$

where $C_\rho := 5/\theta$ *with the number* θ *of the proof of Lemma 3.4.*

(c) For $|s| \leq \varrho_k$ *the second derivative of* $\mathbf{E}^{(k+1)}(X;s)$ *at the trivial configuration satisfies* $|D^2\mathbf{E}^{(k+1)}(X;s)[v,w]| \leq C_2 E_0 e^{-\kappa d_{k+1}(X)}\|v\|\|w\|$ *in the norms of the space, with a constant* C_2 *which does not depend on* k*,* K*, the history, or the tube.*

Proof. (a). At $s = 0$ no $\mathbf{R}$-term enters, and the recursion of the family is the small-field recursion of [9, 11]: the terms of the step $j+1$ are obtained from those of the steps up to j by the fluctuation integral [9, (2.12), (2.13), p. 268] at the coupling g_j of the history, with the counterterm computed from its own terms. Bałaban's Theorem 3 [9, p. 264] states that the terms so obtained satisfy the inductive assumptions [9, (1.1)–(1.22)], among them the analyticity on the spaces $U^c_j(X, \alpha_0, \alpha_1)$ with fixed numbers α_0, α_1 and the bound [9, (1.18), p. 263], whenever $0 < g_j \leq \gamma$ for every j. Its proof is an induction over the steps, and we check where the couplings enter it. The action of the step j is [9, (1.6), p. 261], the sum of $-g_j^{-2}A$ and of the terms $\mathbf{E}^{(i+1)}(g_i, \cdot) - \mathbf{E}^{(i+1)}(g_i, 1) - \beta_{i+1}(g_i)A$, $i < j$, each of which contains one coupling as a parameter and its own counterterm. The step produces the term [9, (2.13), p. 268], an integral which contains the coupling g_j and the terms of the earlier steps and no other coupling. The equation $g_j^{-2} = g_{j+1}^{-2} + \beta_{j+1}(g_j)$ enters once, in [9, (2.15), p. 268], where it serves to write the sum of $-g_j^{-2}A$ and of the new terms in the form [9, (1.3)] at $j+1$; this is a regrouping of the terms and concerns no bound. After it, "the proof of Theorem 3 is reduced to proving the remaining properties of (2.13)" [9, p. 269], the analyticity and the bound [9, (1.18)] of its localised terms, and the hypothesis on the couplings is $0 < g_j \leq \gamma$ for every j [9, Theorem 3]. The proof of these properties uses the earlier terms through the inductive assumptions, among which is that each counterterm is the coefficient [9, (1.22), p. 264] of its own term, and this holds for the family by its definition. Hence the theorem applies to the couplings of the history, which are small but do not satisfy the equation of [9]. The construction of the paper differs from that of [9] in the cutoff of the fluctuation field, which is ϑ_j/g_j in place of ε_1/g_j, and Lemmata 3.2 and 17.2 show that the expansion of the step holds with this cutoff on the same spaces. The localisation [10, (2.26), (2.27), p. 259] of the terms to the points z is obtained by the same expansions [10, pp. 272–278], which read the earlier terms on the spaces on which they are given. For γ small the spaces of the tube are contained in these, because $\alpha_{i,j} = \hat{g}_j C_i(\log \hat{g}_j^{-2})^{q_i} \leq \alpha_i$, and on them the terms are those of the family at $s = 0$.

(b). By Paper, Lemma 9.1(iii), in which only the tube enters, $\hat{g}_k^{-2} \leq \frac{3}{2}g_k^{-2}$, so that $\varrho_k \leq \frac{5}{2}g_k^{-2}$, and $\frac{5}{2}g_k^{-2} \leq \frac{1}{2}\theta g_k^{2-\kappa_0} = \frac{1}{2}\Theta_k$ as soon as $g_k^{\kappa_0-4} \leq \theta/5$, which holds for γ small because $g_k \leq \gamma$. The analyticity and the first bound of (3.14) on $|s| < \Theta_k$ are in the proof of Lemma 3.4. Cauchy's estimate on the disc of radius Θ_k gives $|\partial_s \mathbf{E}^{(j)}(X;s)| \leq 2E_0 e^{-\kappa d_j(X)}/\Theta_k$ for $|s| \leq \frac{1}{2}\Theta_k$, and integrating from 0 to s gives the second bound, because $\varrho_k \cdot 2/\Theta_k \leq 5g_k^{-2} \cdot g_k^{\kappa_0-2}/\theta$.

(c). We write $\mathbf{E}^{(k+1)}(s) = \mathbf{E}^{(k+1)}(0) + [\mathbf{E}^{(k+1)}(s) - \mathbf{E}^{(k+1)}(0)]$. By (a) the first term is analytic and bounded by $E_0 e^{-\kappa d_{k+1}(X)}$ on the polydisc of the radius $r^\bullet := \min(\alpha_0, \alpha_1)/C_{\text{cen}}$ about the trivial configuration, and Cauchy's estimate on the lines, as in the proof of Lemma 3.1, bounds its second derivative there by $4(r^\bullet)^{-2}E_0 e^{-\kappa d_{k+1}(X)}\|v\|\|w\|$. The second term is analytic on the polydisc of the radius r of (3.12), where $r^{-2} \leq 2(C_{\text{cen}}/\min(C_0^\alpha, C_1^\alpha))^2 g_k^{-2}$, and it is bounded there by (3.14); its second derivative is at most $8C_\rho(C_{\text{cen}}/\min(C_0^\alpha, C_1^\alpha))^2 g_k^{\kappa_0-6}E_0 e^{-\kappa d_{k+1}(X)}\|v\|\|w\|$, and $g_k^{\kappa_0-6} \leq 1$. So (c) holds with $C_2 := 4(r^\bullet)^{-2} + 8C_\rho(C_{\text{cen}}/\min(C_0^\alpha, C_1^\alpha))^2$. *q.e.d.*

Proposition S.4 (Bounds on the Coefficients)**.** *There are constants* C_Λ*,* C_3*,* C_{rem}*,* C_R*, and* $\gamma_1 > 0$*, which do not depend on the tube, on* k*, or on the history, such that for every*

$\gamma \in (0, \gamma_1]$, *every tube, every* $k < K$, *and every* $h \in H_k$,

$$|\beta_{k+1}(h) - b_{k+1}| \le C_\Lambda C_3 \gamma p_0(\gamma) + C_R \gamma^{\kappa_0 - 4} \ , \qquad |\beta_{k+1}(h) - b_{k+1}| \le (C_{\text{rem}} + C_R \gamma^{\kappa_0 - 6}) g_k^2 \ . \tag{3.15}$$

Proof. By Lemma 3.4 it suffices to prove the bounds $C_\Lambda C_3 \gamma p_0(\gamma)$ and $C_{\text{rem}} g_k^2$ for $\beta^0_{k+1}(h)$, since $C_R g_k^{\kappa_0 - 4} \le C_R \gamma^{\kappa_0 - 4}$ and $C_R g_k^{\kappa_0 - 4} \le C_R \gamma^{\kappa_0 - 6} g_k^2$, as $g_k \le \gamma$ by Paper, Lemma 3.2(i). By Paper, (4.8), and the linearity of Λ, $\beta^0_{k+1}(h) - b_{k+1} = \Lambda(\mathbf{E}_{\text{fl}}^{(k+1)})$, the part of the regular term of order zero being $\log Z^{(k)}$, whose coefficient is b_{k+1}. We write $R(z; q) := \Lambda(E(\cdot; z, q))$, so that $\beta^0_{k+1}(h) - b_{k+1} = R(g_k; p)$.

The first bound. The cutoff of the step is that of [11, Sect. 2] with ϑ_k in place of ε_1, since $p = \vartheta_k / g_k$. The conditions which [11] places on ε_1 are upper bounds, which hold for $\vartheta_k \le \gamma p_0(\gamma)$ when γ is small, and lower bounds on ε_1 / g_k, which hold for p by (3.1). So [11, (2.41), p. 21] bounds the terms of $\mathbf{E}_{\text{fl}}^{(k+1)}$ by $O(1) C_3 \vartheta_k \exp(-\kappa d_{k+1}(X))$; this is the "additional factor $O(\varepsilon_k)$ arising from the bound of the fluctuation field" of [10, p. 277]. Lemma 3.1, with the radius of the polydisc fixed by the constants α_0, α_1 of [9, (1.10)–(1.16), p. 262], gives $|R(g_k; p)| \le C_\Lambda O(1) C_3 \vartheta_k$, where $C_\Lambda := C_\Lambda(1) r^{-2}$, and we include the $O(1)$ in C_3. Finally $\vartheta_k = (A_1/A_0) \hat{g}_k p_0(\hat{g}_k) \le \gamma p_0(\gamma)$, because $A_1 < A_0$, $\hat{g}_k \le \gamma$, and $g \mapsto g p_0(g)$ increases on $(0, \gamma]$, its logarithmic derivative being $g^{-1}(1 - 2p_0 / \log g^{-2}) > 0$ once $\log \gamma^{-2} > 2p_0$.

The second bound. The function $R(\cdot; q)$ is analytic on $|z| < \varepsilon^* / q$ and bounded there by $M_0 := C_\Lambda O(1) C_3 \varepsilon^*$, by Lemmata 3.2 and 3.1, and it is even, by (3.3) and the linearity of Λ. An even function f analytic on $|z| < r_f$ with $|f| \le M_f$ has $(f(z) - f(0))/z^2$ analytic, and the maximum principle gives $|f(g) - f(0)| \le 2 M_f g^2 / r_f^2$ for $0 \le g < r_f$. Let J be the least integer with $2^J q_{\text{th}} \ge p$, put $p_j := 2^j q_{\text{th}}$ for $j < J$ and $p_J := p$, and $D_j(z) := R(z; p_{j+1}) - R(z; p_j)$. Then

$$R(g_k; p) = R(0; p) + [R(g_k; q_{\text{th}}) - R(0; q_{\text{th}})] + \sum_{j<J} [D_j(g_k) - D_j(0)] \ . \tag{3.16}$$

The middle term is at most $2 M_0 q_{\text{th}}^2 g_k^2 / \varepsilon^{*2}$. Each D_j is even and analytic on $|z| < \varepsilon^* / p_{j+1}$, which contains g_k since $g_k p_{j+1} \le g_k p = \vartheta_k < \varepsilon^*$, and bounded by $C_\Lambda C_4 \exp(-\frac{1}{4} \gamma_2 p_j^2)$, by Lemmata 3.1 and 3.3; with $p_{j+1} \le 2 p_j$ this gives $|D_j(g_k) - D_j(0)| \le 8 C_\Lambda C_4 p_j^2 \exp(-\frac{1}{4} \gamma_2 p_j^2) g_k^2 / \varepsilon^{*2}$, and the sum over j is at most $8 C_\Lambda C_4 S g_k^2 / \varepsilon^{*2}$ with $S := \sum_{j \ge 0} 4^j q_{\text{th}}^2 \exp(-\frac{1}{4} \gamma_2 4^j q_{\text{th}}^2) < \infty$. The first term is Λ of the logarithm of the Gaussian measure of the cutoff, and Lemma 3.3 at $z = 0$ gives $|R(0; p)| \le C_\Lambda C_5 \exp(-\frac{1}{20} \gamma_2 p^2) \le C_\Lambda C_5 g_k^2$, because $\frac{1}{20} \gamma_2 p^2 \ge \frac{1}{20} \gamma_2 \rho^{-1} A_1^2 (\log g_k^{-2})^{2p_0} \ge \log g_k^{-2}$ for $g_k \le \gamma$ small, by (3.1). So the second bound holds with

$$C_{\text{rem}} := \frac{2 M_0 q_{\text{th}}^2}{\varepsilon^{*2}} + \frac{8 C_\Lambda C_4 S}{\varepsilon^{*2}} + C_\Lambda C_5 \ . \tag{3.17}$$

The constants are chosen in the order ε^*, q_{th}, γ_1, and γ_1 is so small that for $g \le \gamma_1$ the number p of (3.1) is at least q_{th}, $gp < \varepsilon^*$, and the last inequality above holds. None of them depends on k, the history, or the tube. *q.e.d.*

3.2 Continuity in the History

The coefficient $\beta_{k+1}(h)$ is the end product of $k+1$ steps, each of which consists of integrals, quotients, cluster and localisation expansions, and the **R**-operation, and the history enters all of them. Continuity in h is therefore not a matter of inspecting a formula. Two features of the construction make it hold. The first is that no set on which a characteristic function, a test function, or a bump function is supported moves with the history, so that h enters only through densities which are continuous in it and through the terms of the earlier steps. The second is analyticity: the localisation expansions read the earlier terms at complex arguments and through their derivatives, and the uniform bounds of the construction turn convergence at the real points into locally uniform convergence of all derivatives, by the theorems of Montel and Vitali. The coefficient is, finally, a limit as the torus grows, and we need this limit to be uniform in h, which Bałaban does not state; it comes from the random walk expansions which give the limit its existence.

One choice in the changes of variables has to be fixed for the first feature to hold. Bałaban specifies the bump function g of [10, (1.22), p. 251] only on two sets [10, p. 252], and a choice which depends on the coupling in any other way would move with the history.

Definition 3.6 (The Bump Functions)**.** Let φ_1 be a fixed C^∞ function on $[0,\infty)$ with $0 \le \varphi_1 \le 1$, $\varphi_1 = 0$ on $[0, 4/3]$, and $\varphi_1 = 1$ on $[5/3, \infty)$. In the change of variables [10, (1.22), p. 251] of the step 1, in its repetition at the step $k+1$ [10, p. 269], and in the change of variables [12, p. 188] of the **R**-operation, the bump function is

$$g(A') := \varphi_1\Big(\frac{|g_j A'|}{\vartheta}\Big)\,, \tag{3.18}$$

where g_j is the coupling of the scale of the change, ϑ its threshold (ϑ_0, ϑ_k, and δ'_j respectively), and $|\cdot|$ the Hilbert–Schmidt norm [5, p. 390].

This choice has the properties which [10, p. 252] requires. It is C^∞ on the Lie algebra, because φ_1 vanishes near 0. It vanishes where $|g_j A'| \le \frac{4}{3}\vartheta$. It equals one where $|\exp i g_j A' - 1| \ge \frac{5}{3}\vartheta$, because $|\exp iB' - 1| \le |B'|$ for B' in the Lie algebra: both matrices are diagonal in one orthonormal basis, with the entries $e^{i\lambda} - 1$ and λ, λ real, and $|e^{i\lambda} - 1| \le |\lambda|$. In the unscaled field $B' = g_j A'$ its support is fixed by the target.

Proposition S.5 (Continuity of the Coefficients)**.** *Let $\gamma \in (0, \gamma_1]$, with γ_1 as in Proposition S.4. For every tube and every $k < K$ the function $h \mapsto \beta_{k+1}(h)$ is continuous on H_k.*

Proof. The claim of the induction. A term $F(X)$ which the step i produces is an analytic function of the configuration $(\mathbf{U}, \mathbf{J})$ of the scale $i+1$ on a domain D_F: the space $U^c_{i+1}(X, \alpha_{0,i+1}, \alpha_{1,i+1})$ for the regular terms [10, (2.26)–(2.28), p. 259] and for the **R**-terms [10, (2.31), p. 260], and the space [10, (3.43), p. 276], which contains a fluctuation field A with $|A| < C_1 p_1(\hat{g}_i)$, for the boundary terms. A *real point* of D_F is a point at which $\mathbf{U}$ is G-valued and $\mathbf{J}$ is real, and A is real for a boundary term. The radii of these domains are those of the tube, fixed by the target, except the bound of the boundary terms on the unscaled field $g_i A$, which is $g_i C_1 p_1(\hat{g}_i)$ and moves continuously with h; a real point at which this bound is not attained at h^0 is a real point for all h near h^0. For B real put $u_i(B) := (U_{i+1}(e^{iB}), J_{i+1}(e^{iB}))$, call B^0 *admissible* for F if $u_i(B^0)$ is a real point of $U^c_{i+1}(X, \frac{1}{2}\alpha_{0,i+1}, \frac{1}{2}\alpha_{1,i+1})$, and let $P_i(B^0)$ be the complex polydisc $\{|B - B^0| < c_{\mathsf{pd}} \min(\alpha_{0,i+1}, \alpha_{1,i+1})\}$. By [9, (1.15), (1.16), p. 262] and the bound on the derivative of the background field in the proof of Lemma 3.1, there is $c_{\mathsf{pd}} > 0$ such that u_i maps every $P_i(B^0)$ into D_F. The claim S_j, for $-1 \le j \le k$, is that the coefficients β_i, $i \le j+1$, are continuous on H_j, and that every term F produced by the steps $0, \dots, j$ has the following two properties at every $h^0 \in H_j$:

(a) $F(X, \mathbf{U}, \mathbf{J}; h) \to F(X, \mathbf{U}, \mathbf{J}; h^0)$ as $h \to h^0$, at every real point of D_F;

(b) if F is a regular term or an **R**-term, then $F(X, u_i(B); h) \to F(X, u_i(B); h^0)$ uniformly on the compact subsets of $P_i(B^0)$, for every admissible B^0.

A term depends on h only through the entries of the history up to the step which produces it, and the later steps read it without changing it, so that (a) and (b), once proved at that step, hold at every later stage. The density of the step 0 is $\exp[-x_0 A - E]$, continuous in x_0 at fixed U, with no earlier terms, which is S_{-1}. We assume S_{j-1} and prove S_j in five steps.

Step 1: what depends on the history. In the construction of the paper every threshold, cube size, depth, and partition, and every radius $\alpha_{i,j}$, is fixed by the target (Paper, Definition 3.11(i)). Every characteristic function is a function of the unscaled fields with a threshold fixed by the target: the functions $\chi_k(\Omega_k)$ of [10, (2.17), p. 257] on the plaquettes of the background fields [10, (2.16)], which are defined from the fields alone; the function $\chi_{k,\Lambda}$ of [12, (1.75), p. 193]; and the functions of the fluctuation fields, by Paper, Definition 3.11(ii), for example $|\exp i g_j B_j(b) - 1| < 2\delta'_j$ in [12, p. 188]. The soft test functions $\bar{\chi}(f/t)$ of Paper, Definition 3.11(iii), and those of Definition 2.2, have f a function of the unscaled fields and t fixed by the target. The changes of variables [10, (1.22)–(1.23), pp. 251–252], their repetition at [10, p. 269], and that of [12, p. 188] read, in the unscaled fields, $\exp iB = \exp iB' \exp(-ig(B')H)$, with the bump functions of Definition 3.6 and with H an analytic function of the background fields which contains no coupling ($\mathbf{H}_{1,\mathrm{Ax}}$ of [10, (1.21), p. 251], $\mathbf{H}^{(k)}$ of [10, (3.22), p. 269], and $\mathbb{H}_Z^{(j)}$ of [12, (1.58), p. 188]). These maps, their Jacobians [10, (1.24), p. 252], the constraints of the block averages, and the Haar measures do not depend on h. The history therefore enters only through the Wilson action with the coefficient function (4.10) of the paper, the Gaussian densities of the unscaled fluctuation fields, whose covariances are g_j^2 times operators which do not depend on h, the weight $\exp[-g_k^{-2} G]$ of the axial gauge and its normalisation, the Jacobians of the scalings, the densities and normalisations of the $\mathbf{R}$-operation [12, (1.100), p. 201], the counterterms β_i, and the terms of the earlier steps.

Step 2: where the step reads the earlier terms. The terms of the step j are defined by Bałaban's formulae as functions of the variables $(\mathbf{U}, \mathbf{J})$ which replace U_{j+1} and J_{j+1} [9, p. 272], and they read the earlier terms at three kinds of arguments:

(i) in [9, (3.13), p. 272], at $\exp i\xi \mathbf{A}\mathbf{U}$ and $(L^{j-1}\eta)^3 \mathbf{J} + \Delta^{\xi}_{\mathbf{U}} \mathbf{A} - P_1 \mathbf{A} + \mathbf{F}_1(\mathbf{U}, \mathbf{A})$, for $\mathbf{A}$ in the class [9, (3.14)], built from the localised functions of [9, p. 273] with the variables s of [11, p. 8];

(ii) in [9, (3.33)–(3.36), p. 277] and [9, (3.53), p. 280], at the background field and current of $\exp iw$, $w = \tau B + B'$ with $\tau \in [0, 1]$ and B' small, where B is built from the localised function of [11, (1.8), p. 3]; [10, p. 276] applies the same formulae to the boundary terms;

(iii) in the large-field steps, in the two localisations of [13, (1.59)–(1.60), p. 372] and [13, pp. 373–379], at the configurations of the passes and at the points of the Taylor remainder of [9, (3.34)], and in the terms $V(Y)$ and $V(X^{\sim})$ of [13, (1.70)–(1.72), pp. 377–379], at the configurations [13, (1.61)–(1.65), pp. 373–375], which contain the gauge transformations "given by the explicit formula (106) [12]" [13, p. 373], that is [4, (106), p. 33].

At real $(\mathbf{U}, \mathbf{J})$ and real values of the auxiliary variables each of these arguments is a real point of the domain of the earlier term. Indeed, the localised operators are sums over walks whose terms are multiplied by the real numbers $s(\Delta)$ [11, p. 3], so that the localised functions are real at real arguments, and so are those of [13, (1.54), (1.57), (1.58), pp. 370–371], built in the same way from the operators of [5, 7]. The gauge transformations of [9, (3.26)–(3.27), p. 275] are "explicitly given analytic functions" of $\mathbf{U}$ [9, p. 276], G-valued at G-valued $\mathbf{U}$. In [4, (106), p. 33] the transformation is a product of conjugations $R(X)Y = XYX^{-1}$ [4, (56),

p. 27] by contour products and of inverses of averages [4, (105)], which are exponentials of real averages of logarithms, so that it is G-valued at G-valued configurations. Every term read at these arguments is continuous in h there, by (a) of S_{j-1} for a term of an earlier step and by Step 3 for a term produced earlier in the step j.

Derivatives at real points. Let F satisfy (a) and let p be a real point of D_F at h^0. There is a complex polydisc $P \ni p$ contained in D_F for all h near h^0, because D_F does not move with h except for the bound of the boundary terms, which is not attained at p. On P the term is bounded uniformly for h near h^0 by the bound of its class, [9, (1.18), p. 263], [10, (2.31), p. 260], [10, (2.42), p. 261], [12, (1.70), p. 192], or [13, (1.11), p. 358, (1.47)–(1.48), p. 368], with constants which do not depend on the history (Paper, Proposition 3.12). By Montel's theorem every sequence $h_n \to h^0$ has a subsequence along which F_{h_n} converges locally uniformly on P; every limit equals F_{h^0} at the real points of P, which form a nonempty open subset of the real space, and therefore on P, by the identity theorem. So $F_h \to F_{h^0}$ locally uniformly on P, with all derivatives in the configuration, and finitely many such polydiscs cover a compact set of real points.

The auxiliary variables. The localisations of [11, (1.9)–(1.10), p. 4] and [9, (3.15), p. 273] differentiate the integrand $\varphi_h(t_\Box, s)$ at $t_\Box = 0$ and $s \in [0,1]^{Y_0}$, representing the derivatives by Cauchy integrals over circles [11, p. 7]. On the open polydisc D of these circles, in which $|s(\Delta)| < e^{\kappa_1}$, the field $\mathbf{A}$ stays strictly inside the class [9, (3.14)], or [9, (3.31), p. 276], so that φ_h is analytic on D and bounded there by [9, (3.17), p. 273, (3.54), p. 280], uniformly in h. In case (ii), φ_h is the remainder of [9, (3.34)], an integral over $\tau \in [0,1]$ of fifth derivatives, or a Taylor term of that formula; the function $w \mapsto \mathbf{E}^{(j)}(X, U_j(\Box_0, \exp iw))$ is analytic and bounded on a connected open set of the w above [9, (3.53)–(3.54)], and it converges at the real points of that set by the preceding paragraph, so that by Vitali's theorem it converges uniformly with all derivatives on $\{\tau B + B' \,|\, \tau \in [0,1],\ |B'| \leq \frac{1}{2}\alpha_3\}$. The transformations of [9, Sect. 4] act only on the Taylor terms at $B = 0$ ("such expressions only appear in the sum (4.6)", [9, p. 284]), and by [9, (4.2)–(4.3), p. 281] and [9, (4.9)–(4.15), pp. 283–284] these are sums of derivatives of order at most four at $w = 0$ contracted with fields and cutoffs which do not depend on h. So in cases (i) and (ii) φ_h converges at the real points of D, and by Vitali's theorem uniformly on the compact subsets of D, with all derivatives; the set $\{t_\Box = 0\} \times [0,1]^{Y_0}$ is compact in D because $e^{\kappa_1} > 1$, and the sums over Y_0 converge uniformly in h with [11, (1.24)–(1.29), pp. 7–8] as majorants. In case (iii) the expansions "(6.9), (6.10) [I]" which [13, pp. 372–374] uses are [11, (1.9)–(1.10), p. 4], obtained by the fundamental theorem of calculus in each $s(\Delta)$, so that a term is the finite alternating sum $\sum_{Z \subset \sigma} (-1)^{|\sigma \setminus Z|}$ of the expression at $s = \mathbf{1}_Z$, and no complex s is used. At $s = \mathbf{1}_Z$ the localised functions contain no coupling: the configurations of [13, (1.50), p. 370] at $B = 0$ and of [13, (1.53), (1.56)] are background fields of G-valued fields, the localised functions come from [7, Sects. D, E] with the operators of [5], and the function of [13, (1.51)] is read at the unscaled field $g_k B$. In the second localisation the expression at $s = \mathbf{1}_Z$ is a term read at a real point of its domain, which converges by (a); in the first it is, as in [9, (3.34)], a sum of Taylor terms of order at most four at a real configuration and an integral over τ of fifth derivatives at the real configurations between two members, contracted with real fields, and these converge by the preceding paragraph. The sums and the resummations converge uniformly in h, with the bounds of [11, (1.24)–(1.29), pp. 7–8], those named on [13, p. 372], and the factors [13, (1.66), p. 376] as majorants.

Step 3: the terms of the step at real points. Each term of the step j is continuous in h at every real point:

- *Fluctuation integrals.* The integrands are the densities of Step 1 at real unscaled fields, restricted by fixed characteristic functions to fields which lie in the real part of the domains: the thresholds ε_j and ϑ_j are of order $(\log \hat{g}_j^{-2})^{p_0-q_i}$ relative to $\alpha_{i,j}$, and $q_0 = q_1 > p_0 + 3r$ (Paper, Appendix A, the stage 0). The densities are continuous in h, by the continuity of β_i, $i \leq j$, in the coefficient function (4.10), by Step 2, and by S_{j-1}, and they are bounded uniformly for h near h^0; dominated convergence gives the continuity of each integral.

- *Small-field terms.* The term $\mathbf{E}^{(j+1)}(X)$ is the series [11, (2.13), p. 14] in the activities [11, (2.14), p. 15], which are integrals over $s(\Delta), t(Y) \in [0,1]$ of Cauchy integrals of Gaussian integrals in which "the operators in it are not symmetric, and the second measure is complex" [11, p. 15]. In the unscaled fields, at a real point and at fixed values on the circles, h enters the integrand only through g_j in the Gaussian densities $\exp(-Q/(2g_j^2))$, with a form Q which does not depend on h, and through the potentials, which converge by Step 2 and S_{j-1}. [11, (2.15)–(2.25), pp. 15–17] bound the modulus of the integrand by a constant times a positive Gaussian density in which g_j^{-2} multiplies a nonnegative form, and for $\frac{1}{2}g_j^0 \leq g_j \leq 2g_j^0$ this is at most 4^n times its value at $2g_j^0$, n the number of the variables. Dominated convergence gives the continuity of the activities, and the series converges uniformly in h by [11, (2.38), p. 20] and "the standard way" of [11, p. 20].

- *The* **R***-operation.* The quotients of [12, (0.3), (0.5), p. 176], with the densities and normalisations of [12, (1.100), p. 201], have positive denominators [12, p. 176], which are continuous in h by the first item; for the soft cutoff of Definition 2.2 the denominators are positive by Proposition S.2(d). The new terms $\mathbf{R}'^{(k)}(X)$ of [13, (1.98)–(1.100), p. 390] are "given by the convergent series (7.13) [I]" [13, p. 390], which is [11, (2.13)], in the activities [13, (1.91), p. 388], which are built from the operation T'_k of [13, (1.71), p. 379]. In T'_k "only the quadratic form in B and the term $V(X^{\sim})$ depend on $(\mathbf{U}, \mathbf{J})$" [13, p. 379], the integration variables are real at a real point, the Haar density of the bond, read at the unscaled field, is bounded on the small fields which the characteristic function admits, and the real part of the exponent is bounded above uniformly for h near h^0: the quadratic form and the Wilson action are nonnegative, $V(X^{\sim})$ is bounded by [13, (1.69), p. 377], and the counterterms and normalisations by $O(\log g_j^{-2})$ per site [13, p. 380]. With the Gaussian integrals dominated as in the previous item and the gauge fields in compact groups, dominated convergence gives the continuity of T'_k, and the series converge uniformly with [13, (1.92)–(1.97), (1.99)–(1.100), pp. 388–390] as majorants.

- *The terms* $\mathbf{C}_k$. They are "the sum of all these terms, including the term $I(U_0)$ from the denominator" [13, p. 369], obtained "by expanding some Wilson actions" [13, p. 369]. A Wilson action is linear in its coefficient function, and the expanded configurations contain no coupling (Step 1), so that at a real point these terms are finite sums of plaquette functions at G-valued configurations with coefficients which are continuous in h. The term $I(U_0)$ of [13, (1.10), p. 358] is continuous by dominated convergence, since its integral I_Λ is over a product of copies of G of a nonnegative integrand bounded by one, and is positive by the lower bound of [13, p. 358].

The step produces its terms in the order: the N_j preliminary integrations of [12, pp. 188–189], each a fluctuation integral "of the same type as the integral studied in Sect. 3 [III]" [12, p. 189] read as in cases (i) and (ii); the terms $\mathbf{C}_j$; the first localisation and its repetitions, and the second localisation; the **R**-operation; and the renormalisation transformation of [10, Sect. 3]. Each operation reads only terms of the earlier steps and terms produced before it, so that (a) holds for every term produced by the step j.

Step 4: polydiscs. Let F be a regular term or an **R**-term of the step j, B^0 admissible, and $f_h(B) := F(X, u_j(B); h)$ on $P_j(B^0)$. At real B the point $u_j(B)$ is a real point of D_F, so that $f_h(B) \to f_{h^0}(B)$ by Step 3. On $P_j(B^0)$ the functions f_h are analytic and bounded uniformly for h near h^0, by [9, (1.18), p. 263] or [10, (2.31), p. 260]. As in Step 2, Montel's theorem and the identity theorem on the connected set $P_j(B^0)$ give $f_h \to f_{h^0}$ locally uniformly, which is (b).

Step 5: the coefficient. The coefficient is Λ of the regular part on $\mathbb{Z}^4$, obtained from the tori $T^{(j+1)} \nearrow \mathbb{Z}^4$ ("This limit exists by the localized representation (1.7)", [9, p. 264, after (1.21)]), and the finite-volume values are continuous in h by Steps 1–4 and (b) at $B^0 = 0$. It remains to show that the limit is uniform for h near h^0. Within one step the localised terms are built from the random walk expansions of the propagators [11, (1.6)–(1.7), p. 3], from [5, (3.105)–(3.108), pp. 414–416], whose walks ω are products of local operators, each depending on the configuration on its own domain, with the bound [11, (1.7), p. 3], which decays exponentially in the length $|\omega|$ of the walk. For a fixed X the torus of side λ_T enters only through the walks of length at least a fixed multiple of λ_T, so that the term on the torus differs from the term on $\mathbb{Z}^4$, to which [10, p. 281] extends the sums, by at most $C_j e^{-c_{\mathrm{tor}}\lambda_T}$ times its bound. The inputs of the step come from the torus as well, and an induction over the steps $0, \ldots, j$, with Cauchy's estimates on the domains of [10, p. 259], carries this bound to every term with $\operatorname{diam} X \le \lambda_T/3$; the constant C_j may grow with j, but only $k+1$ steps enter. On the other hand the proof of Lemma 3.1 gives the majorant

$$|\Pi_X(x;h)||x|^2 \le m(X,x) := C E_0 r^{-2} e^{-\kappa d_{j+1}(X)} e^{-\delta_0(\mathrm{dist}(0,X)+\mathrm{dist}(x,X))/2} |x|^2 , \tag{3.19}$$

uniform in h and in the torus, with $\sum_{X,x} m(X,x) < \infty$ on $\mathbb{Z}^4$. Given $\varepsilon > 0$, a finite set of pairs (X,x) near 0 carries all but ε of this sum; for λ_T large these pairs are pairs of the torus with the same distances, and on them the torus terms converge uniformly in h. The pairs of the torus outside this finite set are of two classes, which are bounded by two different arguments. Those with $\operatorname{diam} X$, $\mathrm{dist}(0,X)$, and $|x|$ below $\lambda_T/8$ and with $d_{j+1}(X) < \lambda_T/(4M)$ lift to pairs of $\mathbb{Z}^4$ with the same distances which lie outside the finite set, and the sum of their majorants is at most ε, whatever λ_T is. The others, among them every pair whose domain wraps around the torus, have $\operatorname{diam} X$, $\mathrm{dist}(0,X)$, or $|x|$ at least $\lambda_T/8$, or $d_{j+1}(X) \ge \lambda_T/(4M)$, and then one of the factors of $m(X,x)$ is at most $e^{-c\lambda_T}$, by $\operatorname{diam} X \le M(2d_{j+1}(X)+4)$ (Lemma 17.1); the sum of the majorants of these pairs is at most a polynomial in λ_T times $e^{-c\lambda_T}$. Hence $\sum_x \Pi_{j+1}(x;h)x_\mu x_\nu$ converges uniformly for h near h^0, and β_{j+1}, a uniform limit of continuous functions, is continuous at h^0. Since $E^{(j+1)}$ contains $-\beta_{j+1}A(\phi_{j+1},\cdot)$ [10, (2.25), p. 259], it satisfies (a) and (b) as well, which completes S_j. The claim S_k contains the continuity of β_{k+1} on H_k. *q.e.d.*

4 The Coefficients at Zero Coupling

Paper, Theorem 4.4, states that the coefficients $b_j = \Lambda(\log Z^{(j-1)})$ of (4.8) add up, over any number of consecutive steps, to $2c\ln L$ per step up to a bounded error. Bałaban states the corresponding result for the running coupling as [9, Theorem 2, p. 259], for $G = SU(2)$, and writes: "A proof of this theorem, based on perturbative calculations, will be given in a separate paper". This section gives the proof, for a simple group. A single factor $Z^{(j)}$ is the Gaussian normalisation of one step, and its polarisation has no simple form, because the quadratic form of the step contains the averaging of all the earlier steps. The difficulty disappears in the sum: we show that $\sum_{j<k} \log Z^{(j)}$ is, up to terms which do not contribute to the coefficient, the difference $F_k - F_0$ of two functionals, each of which is the logarithm of the determinant of a gauge-invariant form, of the action seen at the level k and at the level

0. The polarisation of F_k, measured on the lattice of the level k, is that of a theory with a cutoff at the unit scale on both sides and is bounded uniformly in k, while that of F_0 is the one-loop polarisation of the bare lattice theory seen from the lattice of the level k, and grows like $2ck \ln L$ times the action of the background. The terms left over come from the current J of [9, (1.8), p. 261], and gauge covariance makes their polarisation vanish to the order which matters.

4.1 An Exact Decomposition

We first prove the identity, in the setting of finite-dimensional linear algebra in which it is exact.

The setting. Let X, Y, Z, and $\bar{Y}$ be finite-dimensional real inner-product spaces, and let

$$D\colon Y \to X\ , \qquad h\colon X \to X\ , \qquad Q\colon X \to Z\ , \qquad P\colon Y \to \bar{Y}\ , \qquad \bar{D}\colon \bar{Y} \to Z \tag{4.1}$$

be linear maps such that D and $\bar{D}$ are injective, h is symmetric and nonnegative with $hD = 0$ and positive definite on $(\operatorname{ran} D)^{\perp}$, Q is onto, $PP^{\mathrm{T}} = 1$, and

$$QD = \bar{D}P\ . \tag{4.2}$$

In the application X is a space of fine Lie-algebra-valued 1-forms, Y of fine 0-forms orthogonal to $\ker D$, $D = D_U$ the linearised action of the gauge group at a background U, Q the linearised k-fold average with values in the coarse 1-forms Z of the level k, P the restriction to the sites of the level k, which are fine sites because L is odd [9, p. 251], and $\bar{D}$ the coarse gauge action. The relation (4.2) is the linearisation of the gauge covariance of the average: the averages of [9, (0.4), (0.12), pp. 253–254] are built from contours based at the sites of the coarse lattice, which transform as $U(y,x) \mapsto u(y)U(y,x)u(x)^{-1}$ [9, (0.6)], so that the average of U^u is $u(c_-)\bar{U}(c)u(c_+)^{-1}$, and the linearisation of this, iterated over the k steps, is (4.2). Put $Y_0 := \ker P$, the gauge transformations which are trivial at the coarse sites, $D_0 := D|_{Y_0}$, and $W := \ker Q \ominus D(Y_0)$, where $D(Y_0) \subset \ker Q$ by (4.2).

Definition 4.1 (Forms and Functionals of the Levels)**.** In the setting (4.1), the *one-loop factor* of the k-fold average is

$$Z_k[h] := \det(QQ^{\mathrm{T}})^{-1/2} \det(h|_W)^{-1/2} \det(D_0^{\mathrm{T}} D_0)^{1/2}\ . \tag{4.3}$$

For a level j, with the average Q_j, the restriction P_j, and the coarse gauge action $\bar{D}_j$, the form Δ_j on Z_j is

$$\langle z, \Delta_j z\rangle := \min\{\langle A, hA\rangle \mid Q_j A \in z + \operatorname{ran} \bar{D}_j\}\ , \tag{4.4}$$

which depends on z modulo $\operatorname{ran} \bar{D}_j$, and

$$F_j := \tfrac{1}{2} \log \det{}_{\perp}(\Delta_j) - \tfrac{1}{2} \log \det(\bar{D}_j^{\mathrm{T}} \bar{D}_j)\ , \tag{4.5}$$

with $\det_{\perp}$ the determinant on $Z_j \ominus \operatorname{ran} \bar{D}_j$. At the level 0, where $Q_0 = 1$ and $\bar{D}_0 = D$, $\Delta_0 = h$ because $hD = 0$, and $F_0 = \frac{1}{2} \log \det_{\perp}(h) - \frac{1}{2} \log \det(D^{\mathrm{T}} D)$.

The minimum in (4.4) is attained, because a convex quadratic function which is bounded below on a finite-dimensional affine space attains its infimum. The factor (4.3) is the constrained Gaussian integral of $e^{-\frac{1}{2}\langle A, hA\rangle}\delta(QA)$ over a slice of the residual gauge orbits, in the axial gauge, in which the Faddeev–Popov determinant is one: the map $(A_s, \omega) \mapsto A_s + D\omega$ from the slice times Y_0 has the block-triangular Jacobian $\det D_{\text{tree}}$, where D_{tree} is $\omega \mapsto D\omega$

on the bonds of the trees of the axial gauge, and ordering these bonds from the coarse sites outwards makes D_{tree} triangular with orthogonal diagonal blocks, so that $|\det D_{\text{tree}}| = 1$. The integral of the δ-function is $\det(QQ^{\mathrm{T}})^{-1/2}$ times the integral over $\ker Q$, and the splitting $\ker Q = W \oplus D(Y_0)$, with the orbit measure $\det(D_0^{\mathrm{T}} D_0)^{1/2}\,\mathrm{d}\omega$, gives (4.3) up to a constant which does not depend on the background.

Proposition 4.2 (The Exact Decomposition)**.** *In the setting* (4.1)*,* $\log Z_k[h] = F_k - F_0$*.*

Proof. Let $K := \ker Q$ and $\Delta_F := h + DD^{\mathrm{T}}$. With respect to $X = \operatorname{ran} D \oplus (\operatorname{ran} D)^{\perp}$, which h preserves because it is symmetric with $hD = 0$, Δ_F is DD^{T} on $\operatorname{ran} D$, positive there because D is injective, and h on the complement, positive there by hypothesis; so Δ_F is invertible, with

$$\det \Delta_F = \det(D^{\mathrm{T}} D) \det{}_{\perp}(h)\ , \tag{4.6}$$

the nonzero eigenvalues of DD^{T} and $D^{\mathrm{T}}D$ being the same. Moreover $\Delta_F D = D(D^{\mathrm{T}} D)$, hence $\Delta_F^{-1} D = D(D^{\mathrm{T}} D)^{-1}$. We write $S := Q\Delta_F^{-1} Q^{\mathrm{T}}$ and $R := P(D^{\mathrm{T}} D)^{-1} P^{\mathrm{T}}$.

The form with the residual gauge directions. Since $PP^{\mathrm{T}} = 1$, $P^{\mathrm{T}} P$ is the orthogonal projection onto $(\ker P)^{\perp}$, so that $D_0 D_0^{\mathrm{T}} = D(1 - P^{\mathrm{T}} P)D^{\mathrm{T}}$. Let h_F be the compression of $h + D_0 D_0^{\mathrm{T}}$ to K. For $w \in W$ and $\omega \in Y_0$ we have $hD\omega = 0$, $\langle D\omega, hw\rangle = 0$, and $D_0^{\mathrm{T}} w = 0$, the last because $w \perp D(Y_0)$. So h_F is the direct sum of $h|_W$ and of $D_0 D_0^{\mathrm{T}}$ on $D(Y_0)$, whose determinant is $\det(D_0^{\mathrm{T}} D_0)$, and $\det h_F = \det(h|_W) \det(D_0^{\mathrm{T}} D_0)$. With (4.3),

$$\log Z_k[h] = -\tfrac{1}{2} \log \det(QQ^{\mathrm{T}}) - \tfrac{1}{2} \log \det h_F + \log \det(D_0^{\mathrm{T}} D_0)\ . \tag{4.7}$$

The ghost determinant. With respect to $Y = Y_0 \oplus \operatorname{ran} P^{\mathrm{T}}$, the Schur complement of the block $D_0^{\mathrm{T}} D_0$ of $D^{\mathrm{T}} D$ is the inverse of the block of $(D^{\mathrm{T}} D)^{-1}$ on $\operatorname{ran} P^{\mathrm{T}}$, which P^{T} identifies isometrically with R. Hence

$$\det(D_0^{\mathrm{T}} D_0) = \det(D^{\mathrm{T}} D) \det R\ . \tag{4.8}$$

The constraint. We have $h + D_0 D_0^{\mathrm{T}} = \Delta_F - DP^{\mathrm{T}} PD^{\mathrm{T}}$. The compression of Δ_F to K has the determinant $\det \Delta_F \det S/\det(QQ^{\mathrm{T}})$, and its inverse, extended by zero, is $C := \Delta_F^{-1} - \Delta_F^{-1} Q^{\mathrm{T}} S^{-1} Q \Delta_F^{-1}$; these are the two formulae of the constrained Gaussian integral. The determinant of a rank perturbation, $\det(1 - UV) = \det(1 - VU)$, gives

$$\det h_F = \frac{\det \Delta_F \det S}{\det(QQ^{\mathrm{T}})} \det{}_{\bar{Y}}(1 - PD^{\mathrm{T}} CDP^{\mathrm{T}})\ . \tag{4.9}$$

By $\Delta_F^{-1} D = D(D^{\mathrm{T}} D)^{-1}$ and (4.2), $D^{\mathrm{T}} \Delta_F^{-1} D = 1$ and $Q\Delta_F^{-1} D = \bar{D} P(D^{\mathrm{T}} D)^{-1}$, so that

$$1 - PD^{\mathrm{T}} CDP^{\mathrm{T}} = R\bar{D}^{\mathrm{T}} S^{-1} \bar{D} R\ , \qquad \det{}_{\bar{Y}}(1 - PD^{\mathrm{T}} CDP^{\mathrm{T}}) = (\det R)^2 \det(\bar{D}^{\mathrm{T}} S^{-1} \bar{D})\ . \tag{4.10}$$

Collecting. Inserting (4.8), (4.9), and (4.10) into (4.7), the factors $\det(QQ^{\mathrm{T}})$ cancel, and so do the factors $\det R$, which enter with $-\tfrac{1}{2} \cdot 2 + 1 = 0$:

$$\log Z_k[h] = \Big[-\tfrac{1}{2} \log \det \Delta_F + \log \det(D^{\mathrm{T}} D)\Big] + \Big[-\tfrac{1}{2} \log \det S - \tfrac{1}{2} \log \det(\bar{D}^{\mathrm{T}} S^{-1} \bar{D})\Big]\ . \tag{4.11}$$

By (4.6) the first bracket is $-\tfrac{1}{2} \log \det_{\perp}(h) + \tfrac{1}{2} \log \det(D^{\mathrm{T}} D) = -F_0$. For the second, let $M := S^{-1}$, so that $\langle z, Mz\rangle = \min\{\langle A, \Delta_F A\rangle \,|\, QA = z\}$. With respect to $Z = \operatorname{ran} \bar{D} \oplus (\operatorname{ran} \bar{D})^{\perp}$, $\det M$ is the product of the determinant of the block $M_{\|}$ on $\operatorname{ran} \bar{D}$, which is

$\det(\bar D^{\mathrm T} M\bar D)/\det(\bar D^{\mathrm T}\bar D)$, and of the determinant of the Schur complement $M/M_{\|}$. So the second bracket is $\frac12\log\det(M/M_{\|})-\frac12\log\det(\bar D^{\mathrm T}\bar D)$. The Schur complement is the form

$$\langle z,(M/M_{\|})z\rangle=\min\{\langle A,(h+DD^{\mathrm T})A\rangle\,|\,QA\in z+\operatorname{ran}\bar D\}\ , \tag{4.12}$$

and the set of the A is invariant under $A\mapsto A+D\omega$, because $QD\omega=\bar DP\omega$. As h does not change under this map, and $\omega=-(D^{\mathrm T}D)^{-1}D^{\mathrm T}A$ makes $D^{\mathrm T}(A+D\omega)=0$, the minimum (4.12) is $\langle z,\Delta_k z\rangle$, and the second bracket is F_k. *q.e.d.*

At $k=0$ the constraint fixes everything and both sides vanish. The steps of the construction average one level at a time, and the proposition applies to each of them.

Corollary 4.3 (One Step). *Let $\tilde Q$ be the linearised average from the level k to the level $k+1$ and $\tilde P$ the restriction from the sites of the level k to those of the level $k+1$, so that $Q_{k+1}=\tilde QQ_k$ and $P_{k+1}=\tilde PP_k$. Let $Z^{(k)}[h]$ be the factor* (4.3) *of the setting in which X, h, Y, D, Q, P, $\bar D$ are replaced by Z_k, Δ_k, $\bar Y_k$, $\bar D_k$, $\tilde Q$, $\tilde P$, $\bar D_{k+1}$. Then* $\log Z^{(k)}[h]=F_{k+1}-F_k$, *and* $\sum_{j<k}\log Z^{(j)}[h]=\log Z_k[h]$.

Proof. The replacements satisfy the hypotheses of the setting. Indeed, Δ_k vanishes on $\operatorname{ran}\bar D_k$ and is positive on its orthogonal complement, by Lemma 4.4 below, whose proof does not use the corollary, so that $\Delta_k\bar D_k=0$; $\tilde Q$ is onto because $Q_{k+1}=\tilde QQ_k$ is; $\tilde P\tilde P^{\mathrm T}=1$; $\bar D_k$ and $\bar D_{k+1}$ are injective; and $\tilde Q\bar D_kP_k=\tilde QQ_kD=Q_{k+1}D=\bar D_{k+1}\tilde PP_k$, with P_k onto, gives $\tilde Q\bar D_k=\bar D_{k+1}\tilde P$. The functional of the level 0 of this setting is F_k. Its form of the level 1 is $\min\{\langle z',\Delta_k z'\rangle\,|\,\tilde Qz'\in z+\operatorname{ran}\bar D_{k+1}\}=\min\{\langle A,hA\rangle\,|\,Q_{k+1}A\in z+\operatorname{ran}\bar D_{k+1}\}$, since $\tilde Q(\operatorname{ran}\bar D_k)=\bar D_{k+1}\tilde P(\bar Y_k)\subset\operatorname{ran}\bar D_{k+1}$, and $z'=Q_kA$ gives the converse inequality; this is $\langle z,\Delta_{k+1}z\rangle$, and the functional of the level 1 is F_{k+1}. Proposition 4.2 with one step gives the first identity, and the sum over $j<k$, compared with the proposition, gives the second. *q.e.d.*

Lemma 4.4 (The Form on the Whole Space). *The form Δ_k vanishes on $\operatorname{ran}\bar D_k$ and is positive definite on its orthogonal complement. Consequently, with $\mathcal D_k:=\Delta_k+\bar D_k\bar D_k^{\mathrm T}$,*

$$F_k=\tfrac12\log\det\mathcal D_k-\log\det(\bar D_k^{\mathrm T}\bar D_k)\ . \tag{4.13}$$

Proof. The form $\Delta_k\ge0$ depends on z only modulo $\operatorname{ran}\bar D_k$, so by polarisation it vanishes on $\operatorname{ran}\bar D_k$ and maps the orthogonal complement into itself. If $\langle z,\Delta_kz\rangle=0$, a minimiser A has $\langle A,hA\rangle=0$, hence $A=D\omega$, because h is positive definite on $(\operatorname{ran}D)^{\perp}$ and $hD=0$, and then $z\in Q_kD\omega+\operatorname{ran}\bar D_k=\bar D_kP_k\omega+\operatorname{ran}\bar D_k=\operatorname{ran}\bar D_k$. Since $\bar D_k\bar D_k^{\mathrm T}$ vanishes on $(\operatorname{ran}\bar D_k)^{\perp}$ and has the determinant $\det(\bar D_k^{\mathrm T}\bar D_k)$ on $\operatorname{ran}\bar D_k$, $\log\det_{\perp}\Delta_k=\log\det\mathcal D_k-\log\det(\bar D_k^{\mathrm T}\bar D_k)$, which with (4.5) gives (4.13). *q.e.d.*

4.2 Symbols and Their Integrals

The identities of Section 4.1 hold on finite-dimensional spaces on which the hypotheses (4.1) hold, whereas the coefficients are read off from symbols, which are integrals over momenta in infinite volume. This subsection connects the two and collects the operators and the elementary bounds which the next subsections use.

The operators. We work at a level k of the construction in the units of that level, in which the fine spacing is $\eta=L^{-k}$, and in the coordinates $e^{i\eta A}U$ of [5]; the coordinates Ue^{iX} of Section 4.1 are carried into these by the bondwise linear map $X=\eta\operatorname{Ad}(U)^{-1}A$, which carries

Hessians to Hessians. The Hessian $\mathbf{H} \coloneqq \operatorname{Hess} A(U)$ of the Wilson action is the form Δ of [5, (3.10), p. 392], its gradient $J \coloneqq \nabla A(U)$ the current of [5, (3.11)], $D = D_U$ the covariant derivative, $\Pi_D \coloneqq D(D^{\mathrm{T}}D)^{-1}D^{\mathrm{T}}$, and

$$h \coloneqq (1 - \Pi_D)\mathbf{H}(1 - \Pi_D) \,, \tag{4.14}$$

which satisfies the hypotheses on h in (4.1) wherever D is injective and h is positive on $(\operatorname{ran} D)^{\perp}$. Comparing the terms of first order in λ in [5, (3.117), p. 419] gives

$$\langle A, \mathbf{H}D\lambda\rangle = T(J, \lambda, A) \coloneqq \tfrac{1}{2}\langle i[\lambda(b_-), A(b)] - i[A(b), \operatorname{Ad}(U_b)\lambda(b_+)],\ J\rangle \,, \tag{4.15}$$

which is linear in J, local, and contains no derivative, so that, with an absolute constant $c_{\mathcal{N}}$,

$$|T(J, \lambda, A)| \leq c_{\mathcal{N}} \sum_b \eta^4 |J(b)||A(b)|(|\lambda(b_-)| + |\lambda(b_+)|) \,.$$

With $\varphi_A \coloneqq (D^{\mathrm{T}}D)^{-1}D^{\mathrm{T}}A$ the difference $K \coloneqq \mathbf{H} - h$, which equals $\Pi_D\mathbf{H} + \mathbf{H}\Pi_D - \Pi_D\mathbf{H}\Pi_D$, is therefore

$$\langle A', KA\rangle = T(J, \varphi_{A'}, A) + T(J, \varphi_A, A') - T(J, \varphi_A, D\varphi_{A'}) \,, \tag{4.16}$$

linear in J and nonlocal through φ. The average $\tilde{Q}$ of Section 4.1 is the linearised k-fold average of the construction. Bałaban's estimates of the propagators use a dressed average instead [5, p. 392], [4, (89)–(92), p. 31], and so do we: $Q = Q(U)$ is the linear part of the one-step average of [5, p. 418], with the dressing factors [5, (3.113)] and with the contour variables of [9, (0.11), p. 253] in place of the contours, composed over the k steps as in [5, (3.15), p. 393], and Q' is [5, (3.19)] with the same replacement, $Q'\lambda(y) = \sum_{x \in B(y)} L^{-d} \operatorname{Ad}(U(y, x))\lambda(x)$, composed over the steps. Then:

(i) $QD = \bar{D}Q'$, by the derivation of [5, (3.113)–(3.115), p. 418], which uses only the covariance of the average at the coarse sites and that of the contour variables [9, (0.6)];

(ii) $Q - \tilde{Q}$ takes values in $\operatorname{ran} \bar{D}$: the dressing factors are 1 at the trivial configuration, so the linear part of one step is $\tilde{Q}^{(1)}A + \bar{D}\omega_d(A)$, with $\omega_d(A)$ the linear part of the logarithm of the dressing factor, and over the steps $Q_{\leq j} - \tilde{Q}_{\leq j} = Q^{(j)}(Q_{\leq j-1} - \tilde{Q}_{\leq j-1}) + (Q^{(j)} - \tilde{Q}^{(j)})\tilde{Q}_{\leq j-1}$ with $Q^{(j)}\bar{D} = \bar{D}Q'^{(j)}$. Hence $\{QA \in z + \operatorname{ran} \bar{D}\} = \{\tilde{Q}A \in z + \operatorname{ran} \bar{D}\}$, and Δ_k and F_k are the same for Q and for $\tilde{Q}$;

(iii) at $U = 1$, Q is the composite Q_k of the straight averages of [2, (1.16)–(1.18), p. 20], with centred blocks, and Q' is the centred block average [2, (1.20)]: at $U = 1$ the contour factors contribute $\omega_d(A)(c_-) - \omega_d(A)(c_+)$ to the linear part, by [9, (0.8)], and the dressing contributes the opposite.

Further, $N(Q')$, the projection R onto $\Delta_U^{\eta}N(Q')$, and $G' = (\Delta_U^{\eta} + Q'^{*}aQ')^{-1}$ are those of [5, (3.21)–(3.25), p. 394], Δ_π is the extension [5, (3.119), p. 419], $\Delta'_\pi \coloneqq \mathbf{H} - \Delta_\pi$ is given by [5, (3.120)], every term of which contains $\lambda_A \coloneqq G'RD^{*}A$, and

$$G \coloneqq (\Delta_\pi + DRD^{*} + aQ^{*}Q)^{-1} \,, \qquad H \coloneqq GQ^{*}(QGQ^{*})^{-1} \,, \qquad C_G \coloneqq G - GQ^{*}(QGQ^{*})^{-1}QG \tag{4.17}$$

[5, (3.122), (3.126), p. 420], with $RD^{*}H = 0$ [5, (3.124)]. By (iii), at $U = 1$ these are Bałaban's own operators at the trivial configuration, conjugated by the fixed translation from the blocks of [4, (2), p. 17] to the centred blocks of [9], since Bałaban's average has the same linear part at $U = 1$ [4, (124), p. 36].

Cosets. The hypotheses (4.1) fail on a whole torus near $U = 1$. Indeed, at $U = 1$ the kernel of D consists of the constant 0-forms, while for a background $U = U_k(e^{itB})$ with a plane wave $B = ae^{ip\cdot y} + \text{c.c.}$, $a = e_1 w$, $0 \neq w$ in the Lie algebra, and $p_2 \not\equiv 0$, some holonomies are $\exp(sw)$ with $s \neq 0$ small, so that the parallel sections take values in the centraliser of w, a proper subspace; $D_U^{\mathrm{T}} D_U$ then has eigenvalues of order t^2, and $\log\det(D_U^{\mathrm{T}} D_U)$ is unbounded as $t \to 0$. The remedy is to work on one family of momenta at a time. Let T be a torus of side N, $p \in (2\pi/N)\mathbb{Z}^4$ with $p \not\equiv 0 \bmod 2\pi$, and $T_p := \{y \in T \mid e^{ip\cdot y} = 1\}$. Every operator above is a translation-covariant function of the background and so commutes with T_p; decomposing the fields into Bloch fibres of quasi-momentum $p' \bmod 2\pi$ under the translations of T, the characters which are trivial on T_p are the multiples of p, so that every such operator is block-diagonal in the cosets $\mathcal{O} := p' + \mathbb{Z}p$. We write $X_{\mathcal{O}}$, $Y_{\mathcal{O}}$, and so on, for the real fields with quasi-momenta in $\mathcal{O} \cup (-\mathcal{O})$. Let $\mathcal{O} \neq \mathbb{Z}p$. On a nonzero fibre at $U = 1$, D and $\bar{D}$ are injective and h is positive on $(\operatorname{ran} D)^{\perp}$, since its kernel there consists of harmonic 1-forms, which are constant. The spaces are finite-dimensional, a T_p-invariant background preserves them, and the operators are real-analytic functions of the link variables; so for T_p-invariant U near 1 these properties persist, the hypotheses (4.1) hold on $\mathcal{O}$ with $Y = Y_{\mathcal{O}}$, and Proposition 4.2, Corollary 4.3, and Lemma 4.4 hold there.

Symbols. At $U = 1$ every operator commutes with the translations of the coarse lattice, and its Bloch fibre at quasi-momentum q is its *symbol*; the symbol of a quadratic form is that of its kernel. The second-order Taylor coefficient in the field B of the coarse lattice of a function of B is a quadratic form whose symbol $\Pi(p)$ is the *polarisation*, and the coefficient Λ of Paper, (4.5), is read off from its part of order $|p|^2$ [9, (1.22), p. 264]. On the coset $\mathcal{O} \neq \mathbb{Z}p$ the second-order coefficient in t of each quantity of this section, restricted to $X_{\mathcal{O}}$, is $\sum_{p' \in \mathcal{O}\cup(-\mathcal{O})} \mathcal{M}(p', p)$, where $\mathcal{M}$ is the one-loop integrand of its Bloch form, with internal lines at the quasi-momenta p' and $p' + p$. For N larger than the ranges of the local operators the fibre of each operator at $U = 1$ is its infinite-volume symbol, because inverses, projections, and Schur complements are taken fibrewise. We put $\Pi^N(p) := N^{-4} \sum_{p' \notin \mathbb{Z}p} \mathcal{M}(p', p)$. The sets $\mathcal{O} \cup (-\mathcal{O})$ with $\mathcal{O} \neq \mathbb{Z}p$ partition the $p' \notin \mathbb{Z}p$, so the identities of Section 4.1, which hold on each of them, hold exactly for Π^N.

Lemma 4.5 (Infinite Volume)**.** *Suppose that $\mathcal{M}(p', p)$ is jointly continuous in (p', p) on the set where $p \not\equiv 0$ and $p' \notin \{0, -p\} \bmod 2\pi$, and that it has singularities of degree at most 2 at $p' \in \{0, -p\} \bmod 2\pi$, uniformly for p in compact subsets of $\mathbb{R}^4 \setminus 2\pi\mathbb{Z}^4$. Then $\Pi^N(p) \to \int_{[-\pi,\pi]^4} \mathcal{M}(p', p)\, \mathrm{d}^4 p'/(2\pi)^4$ as $N \to \infty$ through the N with $p \in (2\pi/N)\mathbb{Z}^4$, and the limit is continuous in $p \not\equiv 0$. Consequently every identity between such limits which holds for $p \in \bigcup_N (2\pi/N)\mathbb{Z}^4$ holds for every $p \not\equiv 0$.*

Proof. In dimension 4 the grid points within distance r of a singular point contribute at most Cr^2 to Π^N, uniformly in N, because $N^{-4}\#\{p' \in (2\pi/N)\mathbb{Z}^4 \mid |p'| < r\}|p'|^{-2}$ is a Riemann sum of $\int_{|p'|<r} |p'|^{-2}\, \mathrm{d}^4 p' = \pi^2 r^2$. The point $p \in 2\pi\mathbb{Q}^4$ is fixed along the sequence, so $\mathbb{Z}p$ consists of a fixed number m_0 of points, which apart from 0 and $-p$ are not singular, and their omission changes the Riemann sum by at most $C_p m_0 N^{-4}$. Away from the r-neighbourhoods of the singular points the Riemann sums converge to the integral, and r is arbitrary. For the continuity let $p_n \to p \not\equiv 0$. The integrals over the r-neighbourhoods of the singular points are $O(r^2)$ uniformly in n, by the uniform bound; on the complement of the $2r$-neighbourhoods of 0 and $-p$, which for large n avoids the r-neighbourhoods of the singular points of $\mathcal{M}(\cdot, p_n)$, the functions $\mathcal{M}(\cdot, p_n)$ converge to $\mathcal{M}(\cdot, p)$ uniformly, by the joint continuity on a compact set; so the integrals converge, r being arbitrary. Continuity in p' for each p would not suffice here: a function which does not depend on p' and jumps in p satisfies every other hypothesis. The symbols to which the lemma is applied are products of the singular and the regular

factors defined below, which are jointly continuous away from the singular points. Finally $\bigcup_N (2\pi/N)\mathbb{Z}^4$ is dense. *q.e.d.*

Bloch norms. A field X is a Bloch wave of quasi-momentum p if $X(x+y) = e^{ip\cdot y}X(x)$ for the coarse translations y, and matrix elements between Bloch waves are taken per coarse unit cell. We put $\|X\|_\infty := \sup|X|$ and $\|X\|_2 := (\sum_{x\in\text{cell}} \eta^4|X(x)|^2)^{1/2}$, so that $\|X\|_2 \le \|X\|_\infty$; if three Bloch waves have quasi-momenta adding to zero, then $\sum_{\text{cell}} \eta^4|fgh| \le \|f\|_\infty\|g\|_\infty\|h\|_2$. We write $v_\nu(p) := e^{ip_\nu} - 1$, so that the pure gauge fields of the coarse lattice are $B = v(p)\hat{\omega}e^{ip\cdot y}$, and $|\hat{p}| := |v(p)|$, so that $(2/\pi)|p| \le |\hat{p}| \le |p|$ for $|p_\mu| \le \pi$. At $U = 1$ the symbol of the fine Laplacian satisfies $\Delta(q) \ge (4/\pi^2)|q|^2$ at every alias q of a coarse momentum p, and $|q| \ge |p|$, so that $\|(D^{\mathrm{T}}D)^{-1}\| \le (\pi^2/4)|p|^{-2}$, $\|(D^{\mathrm{T}}D)^{-1}D^{\mathrm{T}}\| \le (\pi/2)|p|^{-1}$, and $\|\Pi_D\| \le 1$ on quasi-momentum p. We call these the *singular factors*, of the degrees 2, 1, and 0; a factor of degree s has fibre norm at most $Cd(q)^{-s}$ and q-derivative at most $Cd(q)^{-s}(1 + d(q)^{-1})$, uniformly in η, where $d(q)$ is the distance from q to $2\pi\mathbb{Z}^4$. A *regular factor* is one whose symbol is bounded with a bounded q-derivative, uniformly in k.

Lemma 4.6 (Second-Order Vanishing)**.** *Let $T(p)$ be 4×4 matrices in the vector index, tensored with the identity on the Lie algebra, with $T(p)v(p) = 0$ for all p.*

(a) If T is C^2 on a ball about $p = 0$, with $\|D^2T(q)[u,u]\| \le C|u|^2$ there, and covariant under the signed permutations R, $T(Rp) = \rho(R,p)T(p)\rho(R,p)^{-1}$ with $\rho(R,\cdot)$ unitary and differentiable and $\rho(R,0)$ the signed permutation matrix of R, then $\|T(p)\| \le (C/2)|p|^2$ on that ball.

(b) If $\|T(p) - T(0)\| \le M|p|^\gamma$ for all p, with $\gamma > 0$, then $T(0) = 0$ and $\|T(p)\| \le M|p|^\gamma$.

Proof. (a) At $p = 0$ covariance says that $T(0)$ commutes with every signed permutation matrix; commuting with the sign changes makes it diagonal, and commuting with the permutations makes it $t \cdot 1$. Differentiating $T(p)v(p) = 0$ at 0 in the direction e_μ, with $v(0) = 0$ and $\partial_\mu v(0) = ie_\mu$, gives $ite_\mu = 0$, so $t = 0$. Let $a_\varrho := \partial_\varrho T(0)$. Differentiating the covariance under the reflection R_λ of the λ-th coordinate at $p = 0$, where $T(0) = 0$ makes the derivatives of $\rho(R_\lambda,\cdot)$ drop out, gives $(a_\varrho)_{\mu\nu} = (-1)^{[\varrho=\lambda]+[\mu=\lambda]+[\nu=\lambda]}(a_\varrho)_{\mu\nu}$ for every λ. The exponents add up to 3 over $\lambda = 1,\dots,4$, so one of them is odd and the entry vanishes; hence $\nabla T(0) = 0$, and Taylor's theorem gives the bound. (b) For $0 < s < 1$, $v(se_\mu) = (e^{is} - 1)e_\mu$ with $e^{is} \ne 1$, so $T(se_\mu)e_\mu = 0$ and $|T(0)e_\mu| = |(T(0) - T(se_\mu))e_\mu| \le Ms^\gamma$; letting $s \to 0$ gives $T(0) = 0$. *q.e.d.*

Lemma 4.7 (A One-Loop Lipschitz Bound)**.** *Let $a, b \ge 0$ with $a + b \le 2$, and let $f(q,p)$ be defined where $d(q) > 0$ and $d(q+p) > 0$, 2π-periodic in q and continuously differentiable in p, with*

$$|f(q,p)| \le Md(q)^{-a}d(q+p)^{-b}\ , \qquad |\partial_p f(q,p)| \le Md(q)^{-a}d(q+p)^{-b}(1 + d(q+p)^{-1})\ . \tag{4.18}$$

Then $I(p) := \int_{[-\pi,\pi]^4} f(q,p)\,\mathrm{d}^4q$ satisfies $|I(p)| \le C_I M$ for all p and $|I(p) - I(0)| \le C_I M|p|$ for $|p| \le \pi$, with an absolute constant C_I. For a sum of n such functions, with (a,b) varying, the bounds hold with nC_I in place of C_I.

Proof. On $[-\pi,\pi]^4$, $d(q) = |q|$, and $\int_{|q|<\varrho} |q|^{-s}\,\mathrm{d}^4q = 2\pi^2\varrho^{4-s}/(4-s)$ for $s < 4$; by periodicity the integral of $d(q+p)^{-s}$ over a set on which $d(q+p) < \varrho$ is at most the same. Where $d(q+p) \ge |q|$ the integrand is at most $M|q|^{-a-b}$, and elsewhere at most $Md(q+p)^{-a-b}$, which gives $|I(p)| \le CM$ since $a + b < 4$. On the region $|q| \ge 2|p|$ we have $d(q+tp) \ge |q|/2$ for $t \in [0,1]$, so that the mean value theorem gives $|f(q,p) - f(q,0)| \le 2^b(2\pi+2)M|p||q|^{-a-b-1}$, integrable since $a+b+1 \le 3$. On the region $|q| < 2|p|$, $f(q,0)$ and $f(q,p)$ are bounded as in the first bound, on sets on which $|q|$ or $d(q+p)$ is less than $2|p|$, with integrals at most $CM|p|^{4-a-b} \le CM\pi^{3-a-b}|p|$, because $4-a-b \ge 2$ and $|p| \le \pi$. *q.e.d.*

4.3 The Terms of the Current

Bałaban's Gaussian of the step $k+1$ is taken at the constrained minimiser $U_{k+1}(e^{iB})$, at which ∇A does not vanish, and its quadratic form contains a term with the current J of [9, (1.8), p. 261]. The decomposition of Section 4.1 is stated for the gauge-invariant form h of (4.14), which involves no Lagrange multiplier, and we define the *term of the current* of the step $k+1$ as the difference

$$\mathcal{J}^{(k)}(U) := \log Z^{(k)}(U) - (F_{k+1} - F_k)(U)\ , \tag{4.19}$$

with $Z^{(k)}$ the normalisation (4.7) of the paper. We show that it does not contribute to the coefficient. On the coarse lattice $T^{(k+1)}$ of the step, with $p \in [-\pi,\pi]^4$, let $\Pi^{(k)}_{\mathcal{J}}(p)$ be the polarisation of $B \mapsto \mathcal{J}^{(k)}(U_{k+1}(e^{iB}))$.

Proposition 4.8 (The Terms of the Current)**.** *There is a constant C, depending on d, L, G, and the constants of [5] only, such that $\|\Pi^{(k)}_{\mathcal{J}}(p)\| \le C|p|^3$ for $|p| \le \pi$ and $\|\Pi^{(k)}_{\mathcal{J}}(p)\| \le C$ for all p, for every k. In particular $\Lambda(\mathcal{J}^{(k)}) = 0$.*

Every symbol at $U=1$ which occurs in the proof commutes with the adjoint action on the complexified Lie algebra, by the covariance under constant gauge transformations (step (b) below). The complexification of a compact simple Lie algebra is simple, so its adjoint representation is irreducible, and Schur's lemma gives every such symbol the form $T_0(p) \otimes 1$, as in Lemma 4.6.

Proof. (a) Bałaban's form of the step. In the variable B' of the step, by [9, (1.5), p. 261] with $j=k$ and [9, (2.11), p. 267], the quadratic form of $Z^{(k)}$ is

$$\langle B', \Delta^{(k)} B'\rangle = \langle H_1 B', \Delta_1 H_1 B'\rangle - 2\langle H_1 h_B \tilde{C}^{(2)}(B'), J\rangle + \mathbf{G}^{(2)}(B')\ , \tag{4.20}$$

where h_B is the operator h of [9, (2.4), p. 266], $\tilde{C}^{(2)}$ the quadratic part of the average of the step, and $\mathbf{G}^{(2)}$ twice the quadratic part of the axial gauge-fixing term $\mathbf{G}$ of [9, (2.5), p. 266]. Eliminating the variables $B'(b_0(c))$ by $\delta(\tilde{Q}B')$ writes $B' = \mathcal{P}B''$ (Bałaban's $B' = CB$), $Z^{(k)}(U) = \mathrm{j}(U)\det(\mathcal{P}^*\Delta^{(k)}(U)\mathcal{P})^{-1/2}$, and $C^{(k)} = (\mathcal{P}^*\Delta^{(k)}\mathcal{P})^{-1}$ [9, pp. 267–268], with $\mathrm{j}(U)$ the Jacobian of the elimination, which depends on the background only through $\tilde{Q}$. Here H_1 and Δ_1 are [5, (3.127)–(3.129), p. 421], so that $\langle H_1B', \Delta_1 H_1 B'\rangle = \min\{\langle A, G_1^{-1}A\rangle \mid QA = B'\} - a|B'|^2$, and by [5, (3.128), (3.120), (3.134)–(3.135)], with Bałaban's $\Delta = \mathbf{H} = h + K$,

$$G_1^{-1} = h + DRD^* + aQ^*Q + X(U,J)\ , \qquad X(U,j) := K(U,j) - \Delta'_\pi(U,j) - \Delta^{(2)}_\pi(U,j)\ , \tag{4.21}$$

where $K(U,j)$, $\Delta'_\pi(U,j)$, and $\Delta^{(2)}_\pi(U,j)$ are (4.16), [5, (3.120)], and [5, (3.134)–(3.135)] with an independent fine 1-form j in place of J, each linear in j.

(b) The term as a determinant. For a fine 1-form j let $\Delta^{(k)}(U,j)$ be the form (4.20) with G_1^{-1} replaced by $h + DRD^* + aQ^*Q + X(U,j)$ and J by j, and put $\mathcal{F}(U,j) := -\frac{1}{2}\log\det(\mathcal{P}^*\Delta^{(k)}(U,j)\mathcal{P}) + \frac{1}{2}\log\det(\mathcal{P}^*\Delta^{(k)}(U,0)\mathcal{P})$. By (a), $\Delta^{(k)}(U, J(U)) = \Delta^{(k)}(U)$. At $j=0$ the argument of Lemma 4.19 below gives $\langle B', \Delta^{(k)}(U,0)B'\rangle = \langle B', \Delta_k B'\rangle + \mathbf{G}^{(2)}(B')$. On $\{\tilde{Q}B' = 0\}$ the gauge directions of Δ_k are the ω which vanish at the sites of $T^{(k+1)}$, by (4.2), and they shift the linear part $\tilde{B}'(y,x)$ of $(V'V^{(k)})(y,x)V^{(k)}(y,x)^{-1}$ [9, p. 266]

by $-i\,\mathrm{Ad}(V^{(k)}(y,x))\omega(x)$, which is orthogonal at real $V^{(k)}$; so the Gaussian integral over these directions against $e^{-\frac{1}{2}\mathbf{G}^{(2)}}$ is a constant, the axial slice has the Faddeev–Popov determinant one, and the integral $\int \delta(\tilde{Q}B')\exp[-\frac{1}{2}\langle B', \Delta^{(k)}(U,0)B'\rangle]\,\mathrm{d}B'$, which carries the same Jacobian $\mathfrak{j}(U)$, is $Z^{(k)}[h]$ up to a constant. By Corollary 4.3, on each coset $\mathcal{O} \neq \mathbb{Z}p$,

$$\mathcal{J}^{(k)}(U) = \mathcal{F}(U, J(U)) + \mathrm{const}\;, \qquad \mathcal{F}(U,0) = 0\;. \tag{4.22}$$

Every operator in X and in the term of J in (4.20) is gauge covariant, the averages by (4.2) and (i) above, the step variables transform by the orthogonal maps $\mathrm{Ad}(g)$ at the sites of $T^{(k)}$, $\mathbf{G}^{(2)}$ is invariant, and $J(U^g) = \mathrm{Ad}(g)J(U)$; hence $\mathcal{F}(U^g, \mathrm{Ad}(g)j) = \mathcal{F}(U,j)$ for every fine gauge transformation g.

(c) The second order in B. On $X_{\mathcal{O}}$, $\mathcal{F}$ is real-analytic in (t,j) near $(0,0)$, because the minimum in (b) is a Schur complement of a form which is positive on $\ker Q$ at $(1,0)$, where it is $G(1)^{-1}$, and $\mathcal{P}^*\Delta^{(k)}(1,0)\mathcal{P} = C^{(k)}(1)^{-1}$ is invertible. Write $\partial_j\mathcal{F}(U,0)[j] =: \langle j, \Theta(U)\rangle$ and $\mathcal{F}_{jj} := \partial_j^2\mathcal{F}(1,0)$; by gauge covariance $\Theta(U^g) = \mathrm{Ad}(g)\Theta(U)$. Expand $J(U_{k+1}(e^{iB})) = j_1(B) + j_2(B) + \dots$. Since $\mathcal{F}(U,0) = 0$ and $J(1) = 0$, the part of second order of $\mathcal{J}^{(k)}$ is $\langle j_2, \Theta(1)\rangle + \langle j_1, \Theta_1 B\rangle + \frac{1}{2}\mathcal{F}_{jj}[j_1, j_1]$, with Θ_1 the linearisation of $B \mapsto \Theta(U_{k+1}(e^{iB}))$. First, $\Theta(1) = \mathrm{Ad}(u)\Theta(1)$ for every constant u, so each $\Theta(1)(b)$ is an invariant vector of the simple Lie algebra, hence zero. Secondly, the linearisation A_1 of $U_{k+1}(e^{iB})$ minimises $\|\partial A\|^2$ on $\{\tilde{Q}_{k+1}A = B\}$, so that it has the curvature of $H_{k+1}B$, with H_{k+1} the operator of [2, (1.65), p. 29]; on a quasi-momentum $p \not\equiv 0$ a curl-free Bloch wave is a gradient, so $A_1 = H(1)B + \partial\omega'$, and Θ_1 annihilates gradients, being the derivative of $\Theta(1^{e^{it\omega'}}) = \mathrm{Ad}(e^{it\omega'})\Theta(1) = 0$. Hence $\Theta_1 B$ is the derivative of Θ at 1 in the direction $A_1 := H(1)B$, whose kernel decays exponentially [5, (3.133), p. 422]. Thirdly, $j_1 = \mathbf{H}(1)A_1 = \partial^{\mathrm{T}}\partial H_{k+1}B = Q_{k+1}^{\mathrm{T}}\lambda_B$ for a Lagrange multiplier λ_B, because $H_{k+1}B$ minimises $\|\partial A\|^2$ subject to $Q_{k+1}A = B$; then $\langle B, \Delta_{k+1}B\rangle = \|\partial H_{k+1}B\|^2 = \langle B, \lambda_B\rangle$ [2, p. 29], and polarisation gives $\lambda_B = \Delta_{k+1}B$, so that

$$j_1 = Q_{k+1}^{\mathrm{T}}\Delta_{k+1}B\;, \qquad \|\Delta_{k+1}(p)\| \leq C\gamma_1|p|^2\;, \tag{4.23}$$

by $0 \leq \langle B, \Delta_{k+1}B\rangle \leq \gamma_1\langle\partial^1 B, \partial^1 B\rangle$ [2, (1.67), p. 29] and $|\partial^1_\mu(p)| \leq |p_\mu|$. Hence the part of second order is $\langle\Delta_{k+1}B, \tau B\rangle + \frac{1}{2}\langle\Delta_{k+1}B, \tilde{R}\Delta_{k+1}B\rangle$, with $\tau := Q_{k+1}\Theta_1$ and $\tilde{R} := Q_{k+1}\mathcal{F}_{jj}Q_{k+1}^{\mathrm{T}}$, and

$$\|\Pi^{(k)}_{\mathcal{J}}(p)\| \leq \|\Delta_{k+1}(p)\|\|\tau(p)\| + \tfrac{1}{2}\|\Delta_{k+1}(p)\|^2\|\tilde{R}(p)\|\;. \tag{4.24}$$

(d) The loop integrals at $U = 1$. Let $\mathcal{C}(U) := (\mathcal{P}^*\Delta^{(k)}(U,0)\mathcal{P})^{-1}$ and $H_h(U)$ the minimiser map of the minimum at $j = 0$; at $U = 1$, $\mathcal{C}(1) = C^{(k)}(1)$ and $H_h(1) = H(1)$. The minimum is stationary at its minimiser, so the part of $\Delta^{(k)}(U,j)$ linear in j is $V(U,j)(B') := \langle H_hB', X(U,j)H_hB'\rangle - 2\langle H_hh_B\tilde{C}^{(2)}(B'), j\rangle$, and

$$\langle j, \Theta(U)\rangle = -\tfrac{1}{2}\,\mathrm{Tr}(\mathcal{C}(U)\mathcal{P}^*V(U,j)\mathcal{P})\;, \qquad \tfrac{1}{2}\mathcal{F}_{jj}[j,j] = -\tfrac{1}{2}\,\mathrm{Tr}(\mathcal{C}\mathcal{P}^*V_2(j)\mathcal{P}) + \tfrac{1}{4}\,\mathrm{Tr}((\mathcal{C}\mathcal{P}^*V(1,j)\mathcal{P})^2)\;, \tag{4.25}$$

at $U = 1$ for the second, with $V_2(j)$ the part of $\Delta^{(k)}(1,j)$ quadratic in j, which the constrained Gaussian gives as $-\langle H(1)B', X(1,j)C_GX(1,j)H(1)B'\rangle$ plus the quadratic part $-2\langle\partial_jH_1[j]h_B\tilde{C}^{(2)}(B'), j\rangle$ of the term of j, with $\partial_jH_1[j] = -C_GX(1,j)H(1)$. In the direction A_1 the derivative in U falls on $\mathcal{C}$, $\mathcal{P}$, H_h, X, h_B, or $\tilde{C}^{(2)}$, and $\partial_Uh(1)[A_1] = \partial\mathbf{H}(1)[A_1] - K(1, \mathbf{H}(1)A_1)$, since $h = \mathbf{H} - K(U, J(U))$ and $\partial J(1)[A_1] = \mathbf{H}(1)A_1$. At $U = 1$ every operator commutes with the translations of $T^{(k+1)}$, and for B of momentum p each trace is $\int_{[-\pi,\pi]^4}\mathrm{Tr}\,\mathcal{M}(q,p)\,\mathrm{d}^4q/(2\pi)^4$ by Lemma 4.5, with $\mathcal{M}(q,p)$ the product of the symbols of its factors, a linear map on the $dL^d\dim G$ step variables of one cell, so that $|\mathrm{Tr}\,\mathcal{M}| \leq dL^d\dim G\,\|\mathcal{M}(q,p)\|$. The factors are of two kinds.

- *Regular factors:* $\mathcal{C}(1)$, $\mathcal{P}$, H, G, G', C_G, R, Q, Q^{T}, A_1, Δ'_π, $\Delta^{(2)}_\pi$, h_B, $\tilde{C}^{(2)}$, $\mathbf{G}^{(2)}$, the local factors of T, and the derivatives at 1 in the direction A_1 of $\mathcal{P}$, R, Q, Q^{T}, D, $\mathrm{Ad}(U_b)$, h_B, $\tilde{C}^{(2)}$, $\mathbf{G}^{(2)}$, G', H, and $\partial\mathbf{H}(1)[A_1]$, once the fine derivatives in $\partial\mathbf{H}(1)$, Δ'_π, D, and D^* have been moved onto the adjacent G', H, G, or C_G. Their symbols are bounded with bounded q-derivatives uniformly in k, by the exponential decay of their kernels and the Schur test: [5, Theorems 3.1–3.3, 3.12, (3.131)–(3.133), pp. 397–422] at $U = 1$, with the block-to-block bound [5, (3.46), p. 398] for $\nabla G'\nabla^*$, whose kernel [5, p. 423] does not bound pointwise; the explicit one-step objects h_B, $\tilde{C}^{(2)}$, $\mathcal{P}$, $\mathbf{G}^{(2)}$, which are fixed analytic functions of the $n(d,L)$ bond variables of $V^{(k)}$ on two blocks, with finite range and with Cauchy estimates on a polydisc of radius $r_1(d,L,G)$ about 1 for their derivatives; and Lemma 4.9 below for $\mathcal{C}(1)$.
- *Singular factors:* Π_D, $\varphi = (\partial^{\mathrm{T}}\partial)^{-1}\partial^{\mathrm{T}}$, $\partial(\partial^{\mathrm{T}}\partial)^{-1}$, and $(\partial^{\mathrm{T}}\partial)^{-1}$, of the degrees 0, 1, 1, and 2.

The derivatives of $\mathcal{C}$ and of H_h in the direction A_1 are not regular factors, because these two operators are built from the compressed form h and not from $\mathbf{H}$. Differentiating the inverse and the Schur complement which define them,

$$\begin{aligned} \partial_U\mathcal{C}[A_1] &= -\mathcal{C}(\partial_U(\mathcal{P}^*\Delta^{(k)}\mathcal{P})[A_1])\mathcal{C}\ , \\ \partial_U H_h[A_1] &= -C_G(\partial_U G_1^{-1}[A_1])H(1) + (\text{terms with } \partial_U Q[A_1])\ , \end{aligned}$$

and in both the derivative of h occurs, $\partial_U h(1)[A_1] = \partial\mathbf{H}(1)[A_1] - K(1,\mathbf{H}(1)A_1)$; the first term is regular and the second contains one factor φ. So each of these two derivatives is a sum of products of regular factors with at most one singular factor of degree 1, and it is counted in this way. Each insertion of $X(1,j)$ has degree at most 1, since each term of (4.16) contains one φ, the third also $D\varphi$ of degree 0, and Δ'_π and $\Delta^{(2)}_\pi$ are regular. The derivative in U raises the degree by at most 1: $K(1,\mathbf{H}(1)A_1)$ has degree at most 1, whether it enters directly or through $\partial_U\mathcal{C}$ and $\partial_U H_h$, and replacing φ_A by its derivative $(\partial^{\mathrm{T}}\partial)^{-1}[D_1^{\mathrm{T}}(1-\Pi_D)A - \partial^{\mathrm{T}}D_1\varphi_A]$, with $D_1\lambda(b) = i[A_1(b),\lambda(b_+)]$, raises the degree of its term by at most 1. The quantity Θ_1 has one j and one derivative in U, and $\mathcal{F}_{jj}$ has two j's and none, so every integrand has degrees (a,b) at $(q,q+p)$ with $a+b \le 2$, and its p-derivative raises b by at most one or falls on a regular factor. Lemma 4.7 gives $\|\tau(p)-\tau(0)\| \le C|p|$ for $|p| \le \pi$ and $\|\tau(p)\|, \|\tilde{R}(p)\| \le C$ for all p.

(e) Pure gauge. For $B = \bar{D}\omega$ the minimum of $\|\partial A\|^2$ on $\{\tilde{Q}_{k+1}A = \bar{D}\omega\}$ is zero, attained at a gradient by (4.2), so A_1 is a gradient for $p \ne 0$, $\Theta_1 B = 0$ by (c), and $\tau(p)v(p) = 0$; at $p = 0$, $v(0) = 0$.

(f) Conclusion. τ is 2π-periodic and bounded, so $\|\tau(p)-\tau(0)\| \le C|p|$ for all p, and by (e) and Lemma 4.6(b) with $\gamma = 1$, $\|\tau(p)\| \le C|p|$. With (4.23) and (4.24), $\|\Pi^{(k)}_{\mathcal{J}}(p)\| \le C|p|^3 + C|p|^4$ for $p \not\equiv 0$, and $\Pi^{(k)}_{\mathcal{J}}$ vanishes at $p \equiv 0$ by continuity; so $\|\Pi^{(k)}_{\mathcal{J}}(p)\| \le C|p|^3$ for $|p| \le \pi$ and $\le C$ for all p. The coefficient is determined by the part of order $|p|^2$ [9, (1.22)], which vanishes. *q.e.d.*

Lemma 4.9 (Coercivity of the Step)**.** *The symbol of $C^{(k)}(1)$ is bounded with a bounded q-derivative, uniformly in k.*

Proof. At $(U,j) = (1,0)$, $G_1 = G(1)$ with $G(1)^{-1} = \mathbf{H}(1) + \partial R\partial^* + aQ_k^*Q_k$, so that $\langle B', \Delta^{(k)}(1,0)B'\rangle = \min\{\langle A, (\mathbf{H}(1)+\partial R\partial^*)A\rangle \mid Q_kA = B'\} + \mathbf{G}^{(2)}(B')$. As R is a projection, $\partial R\partial^* \ge 0$; at $J = 0$, $\mathbf{H}(1) = h(1)$, and $\{Q_kA = B'\} \subset \{QA \in B' + \operatorname{ran}\bar{D}\}$, so the minimum is at least $\langle B', \Delta_k B'\rangle \ge \gamma_0'\|\partial^1 B'\|^2$, by [2, (1.65), (1.67), p. 29], where $\partial^1 B'$ is the curvature of B' on the plaquettes of $T^{(k)}$ and $\gamma_0' > 0$ depends on d and L only. Let c be a bond of $T^{(k+1)}$ and P the set of the bonds of $T^{(k)}$ with both ends in $B(c_-) \cup B(c_+)$. The linear map T_c which sends $B'|_{\mathsf{P}}$ to its curvature on the plaquettes of P, to $\tilde{B}'(y,x)$ for $y = c_\pm$, $x \in B(y)$, and to $\tilde{Q}B'(c)$, at $V^{(k)} = 1$, is injective. Indeed, if $T_c B' = 0$, then in each block B' has zero curvature, so $B' = \partial\omega$ there, a cube being simply connected, and since the contours of $\mathbf{G}(y,x)$ stay in $B(y)$, $\tilde{B}'(y,x) = i(\omega(x) - \omega(y)) = 0$, so that $B' = 0$ on the bonds inside the blocks; two crossing bonds at neighbouring sites of the common face bound a plaquette whose other two bonds lie inside the blocks, so all L^{d-1} crossing bonds carry the same value β, each with the coefficient L^{-d} in $\tilde{Q}B'(c)$, so that $\tilde{Q}B'(c) = L^{-1}\beta = 0$. The map T_c is the same for every c and k, on a space of dimension at most $n(d,L)\dim G$, so $|B'|_{\mathsf{P}}|^2 \le C_{d,L}|T_c B'|^2$, and summing over c, with each bond in at least one P and each plaquette and block in at most $2d$, gives $\|B'\|^2 \le 2dC_{d,L}(\|\partial^1 B'\|^2 + \mathbf{G}^{(2)}(B') + \|\tilde{Q}B'\|^2)$. The field $\mathcal{P}B''$ lies in $\{\tilde{Q}B' = 0\}$ and equals B'' on the bonds other than the $b_0(c)$, so $\|\mathcal{P}B''\| \ge \|B''\|$, and $C^{(k)}(1)^{-1} = \mathcal{P}^*\Delta^{(k)}(1,0)\mathcal{P} \ge c_{\mathcal{C}} := \min(\gamma_0', 1)/(2dC_{d,L})$. The symbol of $\mathcal{P}^*\Delta^{(k)}(1,0)\mathcal{P}$ has a bounded q-derivative, because $(QG(1)Q^*)^{-1}$ has an exponentially decaying kernel [5, (3.132), p. 422] and $\mathcal{P}$ and $\mathbf{G}^{(2)}$ have finite range, and $\partial_q \hat{C} = -\hat{C}(\partial_q \hat{C}^{-1})\hat{C}$ is bounded by $C/c_{\mathcal{C}}^2$. *q.e.d.*

4.4 The Total of k Steps

We now pass to the lattice $T^{(k)}$ of the level k, on which the coefficient $\Phi_k := \sum_{j=1}^k b_j$ is to be read off, and use its units, in which the bare spacing is $\eta_k = L^{-k}$. For a functional $\mathfrak{F}$ of the field B of this lattice we write $\mathfrak{F}(e^{iB}) = \mathfrak{F}(1) + \frac{1}{2}\langle B, \Pi_{\mathfrak{F}} B\rangle + O(B^3)$ and call $\Pi_{\mathfrak{F}}(p)$ its polarisation. The Wilson action of the background satisfies $A(U_k(e^{iB})) = \frac{1}{2}\langle B, \Delta_k B\rangle + O(B^3)$, where Δ_k is Bałaban's operator of [2, (1.65)–(1.67), p. 29] (we prove this in Proposition 4.13, step 1): its symbol is analytic, it annihilates the pure gauge fields, and

$$\gamma_0 \langle \partial^1 B, \partial^1 B\rangle \le \langle B, \Delta_k B\rangle \le \gamma_1 \langle \partial^1 B, \partial^1 B\rangle \ , \tag{4.26}$$

with ∂^1 the curl on the unit lattice and $\gamma_0, \gamma_1 > 0$ "dependent on d only" [2, p. 29]. Let $\mathfrak{B}_r$ be the complex ball of the fields B of $T^{(k)}$ with $\sup|B| \le r$ and $\sup|\bar{\partial}B| \le r$, where $\bar{\partial}B := (\partial_\mu B_\nu)_{\mu,\nu}$. For r small, $U_k(e^{iB})$ with $B \in \mathfrak{B}_r$ lies in the spaces $U_j^c(X, \alpha_0, \alpha_1)$ of [9, (1.11)–(1.16), p. 262] for every $j \le k$, by [7, Theorem 1, (181), p. 307], so that every functional which Bałaban proves analytic on those spaces is analytic on $\mathfrak{B}_r$.

Lemma 4.10 (Cauchy's Estimate for Polarisations). *Let $\mathfrak{F}$ be analytic on $\mathfrak{B}_r$, translation covariant, and bounded by $M_{\mathfrak{F}}$ per unit cube of $T^{(k)}$. Then $\|\Pi_{\mathfrak{F}}(p)\| \le 16M_{\mathfrak{F}}/r^2$ for all p.*

Proof. For unit vectors v, w put $B := ve^{ip\cdot Y}$ and $B' := we^{-ip\cdot Y}$. For $|s|, |t| \le r/4$ the field $sB + tB'$ has $\sup|\cdot| \le r/2$ and $\sup|\bar{\partial}\,\cdot| \le r$, since $|e^{i\theta} - 1| \le 2$, so it lies in $\mathfrak{B}_r$, where $|\mathfrak{F}| \le M_{\mathfrak{F}}|T|$, with $|T|$ the number of unit cubes. The coefficient of st in $\mathfrak{F}(sB + tB')$ is $\langle B', \Pi_{\mathfrak{F}} B\rangle = |T| w^{\mathrm{T}} \Pi_{\mathfrak{F}}(p) v$, by translation covariance, and Cauchy's estimate on $|s| = |t| = r/4$ bounds it by $16M_{\mathfrak{F}}|T|/r^2$. *q.e.d.*

Proposition 4.11 (The Total of the Gaussian Factors). *Let $\mathcal{L}_k(B) := \sum_{j=1}^k \mathbf{E}^{(j)}(0, U_k(e^{iB}))$ be the sum of the terms of the steps at zero coupling [9, (1.6), p. 261], evaluated at the background of the level k. Then*

$$\Pi_{\mathcal{L}_k}(p) = \Phi_k \Delta_k(p) + \mathcal{E}_k(p) \ , \qquad \sup_p \|\mathcal{E}_k(p)\| \le C \ , \tag{4.27}$$

with C independent of k.

Proof. By [9, (0.28), p. 258], $\mathbf{E}^{(j)}(U_k) = \beta_j A(U_k) + \sum_{X \in \mathbf{D}_j} \mathbf{V}^{(j)}(X, U_k)$, where each $\mathbf{V}^{(j)}(X, \cdot)$ is analytic on U_j^c and $|\mathbf{V}^{(j)}(X, U)| \leq O(1)(L^j \eta)^{4+\alpha} e^{-\kappa d_j(X)}$ with $\alpha > 0$ [9, (0.29)]; these are the results of [9, Sects. 3–5] in the form summarised on [9, p. 258]. As in [9, (0.30)], the sum over the X which meet a fixed unit cube of $T^{(k)}$ is at most $C(L^j \eta)^\alpha = C(L^{j-k})^\alpha$, so by Lemma 4.10 the polarisation of $\sum_X \mathbf{V}^{(j)}$ is at most $C(L^{j-k})^\alpha$, and the sum over $j \leq k$ is at most $C(1 - L^{-\alpha})^{-1}$. At zero coupling $\beta_j = b_j$, and the polarisation of $A(U_k(e^{iB}))$ is Δ_k. *q.e.d.*

The terms of the current, on the other hand, are bounded on the lattice of the step at which they arise, and we have to see them from the lattice $T^{(k)}$. The fine field $H_k B$ which carries a coarse plane wave is a sum over the aliases of its momentum, and the next lemma is the summability we need.

Lemma 4.12 (Alias Summability). *Fix $\alpha \in (0, 1)$. At $U = 1$, for every k and every coarse momentum p,*

$$\sum_l |(d_\eta H_k)^\wedge(p + l)| \, |p + l|^\alpha \leq C(d, \alpha) \,, \tag{4.28}$$

the sum over the aliases $p + l$ of p in the fine Brillouin zone, where $(d_\eta H_k)^\wedge$ is the symbol of $B \mapsto d_\eta(H_k B)$ and d_η the lattice curl at the spacing $\eta = L^{-k}$.

A bound of this kind is stated after [2, (1.63), p. 28], with an exponent which is not defined there; we derive it from [2, (1.63)] for every $\alpha \in (0, 1)$, together with bounds on the functions $\Delta_0 \varphi_\mu$ of [2, (1.62)] with constants of d only.

Proof. We use the units of [2]: the coarse lattice has the spacing 1, $p \in [-\pi, \pi)^d$, and $l_\mu = 2\pi m_\mu$ with $|m_\mu| \leq (L^k - 1)/2$ [2, p. 23]. Write $q := p + l$, $\Delta(q) = \sum_\mu |\partial_\mu(q)|^2$ with $\partial_\mu(q) = (e^{i\eta q_\mu} - 1)/\eta$, $\Delta_0(p) = \sum_\mu |\partial^1_\mu(p)|^2$ with $\partial^1_\mu(p) = e^{ip_\mu} - 1$, $v_\mu(q) = \partial^1_\mu(p)/\partial_\mu(q)$, and $u(q) = \prod_\mu v_\mu(q)$ [2, (1.31), (1.61), p. 24].

Elementary bounds. Since $|e^{ix} - 1| = 2|\sin(x/2)|$ lies between $(2/\pi)|x|$ and $|x|$ for $|x| \leq \pi$, and $|\eta q_\mu| \leq \pi$, we have $(2/\pi)|q_\mu| \leq |\partial_\mu(q)| \leq |q_\mu|$, hence $(4/\pi^2)|q|^2 \leq \Delta(q) \leq |q|^2$, and likewise for $\Delta_0(p)$. Moreover $|\partial^1_\mu(p)| \leq |p_\mu| \leq |q_\mu|$, because $|p_\mu| \leq \pi \leq |p_\mu + l_\mu|$ when $l_\mu \neq 0$; so $|v_\mu(q)| \leq \pi/2$ and $|\partial_\mu(q)||v_\mu(q)| = |\partial^1_\mu(p)| \leq 2$. Also $|q_\mu| \geq |l_\mu|/2$. If $l_\mu \neq 0$, then $|l_\mu| \geq 2\pi$ and $|v_\mu(q)| \leq (\pi/2)|p_\mu|/|q_\mu| \leq \pi^2/|l_\mu|$, so that in all cases

$$|v_\mu(q)| \leq c_v(1 + |l_\mu|)^{-1} \,, \qquad |u(q)| \leq c_v^d \prod_\mu (1 + |l_\mu|)^{-1} \,, \qquad c_v := \pi^2 + \pi/2 \,, \tag{4.29}$$

which is [2, (1.36), p. 24] with a constant depending on d only; at $q = p$, $|v_\mu(p)| \geq 2/\pi$ and $|u(p)| \geq (2/\pi)^d$. With $\varphi_\mu(p) = \sum_{l'} |u(p + l')|^2 |v_\mu(p + l')|^2 / \Delta(p + l')$ of [2, (1.62)], the term $l' = 0$ of $\Delta_0(p)\varphi_\mu(p)$ lies between $c_0 := (2/\pi)^{2d+4}$ and $(\pi^2/4)(\pi/2)^{2d+2}$, and for $l' \neq 0$, $\Delta_0(p)/\Delta(p + l') \leq d\pi^4/|l'|^2 \leq d\pi^2/4$, so that

$$c_0 \leq \Delta_0(p)\varphi_\mu(p) \leq c_1 := \tfrac{\pi^2}{4}\left(\tfrac{\pi}{2}\right)^{2d+2} + \tfrac{d\pi^2}{4} c_v^{2d+2} \Big(\sum_{m \in \mathbb{Z}} (1 + 2\pi|m|)^{-2} \Big)^d \,. \tag{4.30}$$

These are bounds of the form $\gamma_0 \leq \Delta_0 \varphi_\mu \leq \gamma_1$ of [2, p. 28], with constants depending on d only. Finally $N(p) := \sum_{l''} |u(p + l'')|^2 \Delta_0(p)^2 / \Delta(p + l'')^2 \geq c_N := (2/\pi)^{2d}(4/\pi^2)^2$, from its term $l'' = 0$.

The symbol of H_k. By [2, (1.63), p. 28], $(H_k B)^\wedge_\mu(q) = T_1 + T_2$, both linear in $\hat{B}(p)$. The first is $T_1 = (\Delta_0(p)/\Delta(q))\bar{u}(q)\bar{v}_\mu(q)(\Delta_0(p)\varphi_\mu(p))^{-1}\hat{B}_\mu(p)$, so $|T_1| \leq$

$(\pi/(2c_0))\Delta_0(p)|u(q)|\Delta(q)^{-1}|\hat{B}(p)|$. The second is $\partial_\mu(q)\bar{u}(q)\Delta(q)^{-1}$ times $\sum_{l'\neq l}|u(p+l')|^2\Delta_0(p)^2\Delta(p+l')^{-1}[|v_\mu(p+l')|^2\Delta(q)^{-1}-|v_\mu(q)|^2\Delta(p+l')^{-1}]$, times $N(p)^{-1}(\Delta_0(p)\varphi_\mu(p))^{-1}$ and a factor $b(p)$ with $|b(p)|\le(c_1\sqrt{d}/c_0)|\partial^1(p)||\hat{B}(p)|$. Extending the two sums of the bracket to all l', they are $\Delta_0(p)^2\varphi_\mu(p)\Delta(q)^{-1}\le c_1\Delta_0(p)\Delta(q)^{-1}$ and $|v_\mu(q)|^2N(p)$, so that

$$|T_2|\le\frac{c_1\sqrt{d}}{c_0^2}|\partial_\mu(q)||u(q)|\Delta(q)^{-1}\Big[\frac{c_1\Delta_0(p)}{c_N\Delta(q)}+|v_\mu(q)|^2\Big]|\partial^1(p)||\hat{B}(p)|\,. \tag{4.31}$$

The alias $l=0$. Here $q=p$, and the factors $\Delta_0(p)/\Delta(p)$, $|\partial_\mu(p)||\partial^1(p)|/\Delta(p)$, and $|v_\mu(p)|$ are bounded by constants of d, so that $|(H_kB)^\wedge(p)|\le C_1|\hat{B}(p)|$ and $|(d_\eta H_k)^\wedge(p)||p|^\alpha\le 2|\partial(p)|C_1|p|^\alpha\le C_2$, with $|p|\le\pi\sqrt{d}$.

The aliases $l\neq 0$. Here $|q|\ge\pi$, and the elementary bounds give $|T_1|,|T_2|\le C|u(q)||q|^{-2}|\hat{B}(p)|$. The curvature has the symbol $\partial_\mu(q)\hat{A}_\nu(q)-\partial_\nu(q)\hat{A}_\mu(q)$, and $\sum_{\mu,\nu}|a_\mu b_\nu-a_\nu b_\mu|^2\le 4|a|^2|b|^2$, so $|(d_\eta H_k)^\wedge(q)|\le C|u(q)||q|^{-1}$ and

$$\sum_{l\neq 0}|(d_\eta H_k)^\wedge(q)||q|^\alpha\le C\sum_{l\neq 0}|l|^{\alpha-1}\prod_\mu(1+|l_\mu|)^{-1}\le C2^{1-\alpha}\Big(\sum_{m\in\mathbb{Z}}(1+2\pi|m|)^{-1-(1-\alpha)/d}\Big)^d\,, \tag{4.32}$$

by $\prod_\mu(1+|l_\mu|)\le(2|l|)^d$, which gives $|l|^{\alpha-1}\le 2^{1-\alpha}\prod_\mu(1+|l_\mu|)^{(\alpha-1)/d}$ since $\alpha<1$. The sum is finite and depends on d and α only. *q.e.d.*

Proposition 4.13 (The Total of the Terms of the Current). *Let* $\mathcal{J}_k(B):=\sum_{j=0}^{k-1}\mathcal{J}^{(j)}(U_k(e^{iB}))$. *Then* $\sup_p\|\Pi_{\mathcal{J}_k}(p)\|\le C$, *uniformly in* k.

Proof. Fix j and put $m:=j+1\le k$.

Step 1: reduction to the level m. Let $V^*(B):=(U_k(e^{iB}))^{(m)}$ be the average of the level m. The configurations $U_k(e^{iB})$ and $U_m(V^*(B))$ agree up to a gauge transformation which is 1 on $T^{(m)}$: M^k is M^{k-m} after M^m, so a minimiser on $\{M^k(U)=e^{iB}\}$ also minimises on $\{M^m(U)=V^*(B)\}$, where the minimal orbit is unique [9, (0.21)–(0.22), p. 256]. The term $\mathcal{J}^{(j)}$ is invariant under every fine gauge transformation, by (4.22) and the covariance of $\mathcal{F}$ in step (b) of Proposition 4.8, so $\mathcal{J}^{(j)}(U_k(e^{iB}))=\mathcal{J}^{(j)}(U_m(V^*(B)))$. Moreover $\mathcal{J}^{(j)}\circ U_m$ has a vanishing coefficient of first order at the trivial background, by step (c) of that proof. We show below that the first-order part of $V^*(B)$ is $iL^{m-k}Q_mH_kB$ up to a field of the level m of zero curvature, with the straight average Q_m of (4.34). A field of zero curvature is a gradient plus a constant, and the symbol $\Pi_{\mathcal{J}}^{(j)}$ of Proposition 4.8 annihilates it: it is transverse, $\Pi_{\mathcal{J}}^{(j)}(p')v(p')=0$, because $\Delta_m(p')v(p')=0$ and $\tau(p')v(p')=0$, and it is $O(|p'|^{2+2\alpha})$ for every $\alpha\in(0,\frac{1}{2})$, with the constant $C_\alpha:=C\max(\pi^{1-2\alpha},\pi^{-2-2\alpha})$, since $|p'|^3\le\pi^{1-2\alpha}|p'|^{2+2\alpha}$ for $|p'|\le\pi$ and $1\le\pi^{-2-2\alpha}|p'|^{2+2\alpha}$ for $|p'|\ge\pi$. So the polarisation of $\mathcal{J}^{(j)}(U_k(e^{iB}))$ on $T^{(k)}$ is $\langle V_1,\Pi_{\mathcal{J}}^{(j)}V_1\rangle$ with $V_1:=Q_mH_kB$.

The first-order background. The configuration $U_k(V)$ lies in the unique regular orbit of minimisers of the Wilson action on $\{M^k(U)=V\}$ [9, (0.21)–(0.22), p. 256]; at $V=1$ the configuration 1 is such a minimiser, so we may take $U_k(1)=1$, and then $U_k(e^{itB})=\exp(it\eta A_1+O(t^2))$ bondwise, analytic in t [7, Sect. G]. Let $\tilde{Q}_k$ be the first-order part of M^k at $U=1$; by the computation below with $m=k$, $\tilde{Q}_kA=Q_kA+\partial\omega_k(A)$ with ω_k linear, and $\tilde{Q}_kA_1=B$. Three facts give $d_\eta A_1=d_\eta H_kB$. First, A_1 minimises $\|d_\eta A\|^2$ on $\{\tilde{Q}_kA=B\}$: for each such A the implicit function theorem gives regular $U_t=\exp(it\eta A+O(t^2))$ with $M^k(U_t)=e^{itB}$, the quadratic part of the Wilson action at $U=1$ is $\frac{1}{2}\|d_\eta X\|^2$ with $U=e^{i\eta X}$, because $1-\operatorname{Re}\operatorname{tr}e^{iY}=\frac{1}{2}\operatorname{tr}Y^2+O(Y^4)$, and minimality gives $t^2\|d_\eta A_1\|^2+O(t^3)\le t^2\|d_\eta A\|^2+O(t^3)$.

Secondly, the sets $\{d_\eta A \,|\, \tilde{Q}_k A = B\}$ and $\{d_\eta A \,|\, Q_k A = B\}$ coincide: a gauge transformation $A \mapsto A - \partial^\eta \lambda$ changes $Q_k A$ by $-\partial Q_k' \lambda$ [2, (1.20), p. 20] and $\tilde{Q}_k A$ by the gradient of $-\lambda$ on $T^{(k)}$, either change can cancel $\partial \omega_k(A)$, and neither changes $d_\eta A$; the common set is affine, so $\|\cdot\|^2$ has exactly one minimiser on it. Thirdly, $H_k B$ attains that minimum: [2, p. 29] gives $H_k B$ as a minimiser of $\frac{1}{2}\langle \partial A, \partial A\rangle$ on $\{Q_k A = B,\ R\partial^* A = 0\}$, a slice which meets every orbit of the gauge transformations with $Q_k'\lambda = 0$, since R is the projection onto $\Delta N(Q_k')$ [2, p. 25]; the blocks of [2] are corner blocks [2, p. 18] and those of [9] centred, which conjugates Q_k, H_k, and R by a fixed translation, and minimality and curvature norms are translation invariant. So $A_1 - H_k B$ has zero curvature, and the same three facts give $A(U_k(e^{iB})) = \frac{1}{2}\langle B, \Delta_k B\rangle + O(B^3)$, since [2, p. 29] defines $\langle B, \Delta_k B\rangle$ as $\langle \partial H_k B, \partial H_k B\rangle$.

The first-order average of the level m. The average of the level m is the m-fold iterate of the one-step average [9, (0.17), (0.19), p. 255]. At $U = 1$, on a coarse bond c with $c_+ = c_- + Le_\mu$, the first-order part of the logarithm in [9, (0.4), p. 253], and in [9, (0.12), p. 254], whose linear part is the same, is

$$\sum_{x \in B(c_-)} L^{-d} A([x, x']) + \omega(c_-) - \omega(c_+)\ , \qquad \omega(y) := \sum_{x \in B(y)} L^{-d} |\mathbf{G}(y,x)|^{-1} \sum_{\Gamma \in \mathbf{G}(y,x)} A(\Gamma)\ , \quad (4.33)$$

with $A(\Gamma)$ the sum of the bond logarithms along Γ: the logarithm of the loop is $A(\Gamma) + A([x, x']) - A(\Gamma') - A(c)$ to first order, the term $-A(c)$ cancels against the factor $U(c)$, and $x \mapsto x' = x + Le_\mu$ maps $B(c_-)$ onto $B(c_+)$. The first term is L times the straight average [2, (1.11), p. 19], and the endpoint terms form a coarse gradient. A straight average of a gradient is the gradient of the block means, and the composition of m straight averages over centred blocks is the straight average over the centred block of side L^m ([2, (1.16)–(1.18), p. 20] states this for corner blocks; for corner and centred blocks alike the offsets $sL + t$ run once over $\{0, \ldots, L^2 - 1\}$, and for centred blocks the offsets $Lr + o$ run once over the centred block of side L^2). So the first-order bond logarithm of the level m is $L^{m-k} Q_m A$ plus a gradient of that level, where

$$(Q_m A)_\mu(Y) := |B_m|^{-1} \sum_{r \in B_m} L^{-m} \sum_{t=0}^{L^m - 1} A_\mu(\eta(L^m Y + r + te_\mu))\ , \qquad B_m := \{-\tfrac{L^m - 1}{2}, \ldots, \tfrac{L^m - 1}{2}\}^4\ . \quad (4.34)$$

Step 2: the change of units. In the units of $T^{(k)}$ the modes of V_1 are q with $|q_\mu| \le \pi L^{k-m}$, with the momentum $p' = L^{-(k-m)} q$ of the level m; a bond of the level m carries $L^{-(k-m)}$ times the amplitude of $T^{(k)}$, and there are $L^{4(k-m)}$ sites of the level m per site of $T^{(k)}$. So, with $|p'|^{2+2\alpha} = L^{-(2+2\alpha)(k-m)} |q|^{2+2\alpha}$,

$$|\langle V_1, \Pi^{(j)}_{\mathcal{J}} V_1\rangle| \le C_\alpha L^{2(k-m)} \sum_q |p'|^{2+2\alpha} |\hat{V}_1(q)|^2 = C_\alpha L^{-2\alpha(k-m)} \sum_q |q|^{2\alpha} |q|^2 |\hat{V}_1(q)|^2\ . \quad (4.35)$$

Step 3: transversality. Let $\tilde{V}_1(q)$ be the component of $\hat{V}_1(q)$ orthogonal to $v(p')$. As $\Pi^{(j)}_{\mathcal{J}}(p')$ is Hermitian and annihilates $v(p')$, (4.35) holds with $\tilde{V}_1$ in place of $\hat{V}_1$. The symbol of the difference of the level m in the direction μ is $L^{k-m} v_\mu(p')$, so Lagrange's identity gives $|(\bar{d} V_1)^\wedge|^2 = L^{2(k-m)} |v(p')|^2 |\tilde{V}_1|^2$, with $\bar{d} := d_{L^{m-k}}$, and since $L^{k-m} |v_\mu(p')| \ge (2/\pi)|q_\mu|$,

$$|\langle V_1, \Pi^{(j)}_{\mathcal{J}} V_1\rangle| \le \tfrac{\pi^2}{4} C_\alpha L^{-2\alpha(k-m)} \sum_q |q|^{2\alpha} |(\bar{d} V_1)^\wedge(q)|^2\ . \quad (4.36)$$

Step 4: alias summability. The curl of a straight-path average is the sum of the fine plaquettes in the $L^m \times L^m$ square which the path sweeps out, so that $\bar{d} Q_m A = P_m d_\eta A$, where

$$(P_m F)_{\mu\nu}(Y) := |B_m|^{-1} L^{-2m} \sum_{r\in B_m} \sum_{a,b=0}^{L^m-1} F_{\mu\nu}(\eta(L^m Y + r + a e_\mu + b e_\nu)) \tag{4.37}$$

has nonnegative weights which add up to one, so that on a fine plane wave of momentum p it acts by a symbol $\hat{P}_m(p)$ with $|\hat{P}_m(p)| \le 1$, times the character of p on the level m. The polarisation is Hermitian, so it suffices to take a single mode $B = \hat{B}(p)e^{ip\cdot Y}$. At $U = 1$ the field $H_k B$ is then a finite sum of plane waves with the momenta $p_a = p + l_a$ over the aliases of Lemma 4.12; let $\hat{c}_{\mu\nu}(a)$ be the coefficient of $(d_\eta H_k B)_{\mu\nu}$ at p_a, and let $q(a)$ be p_a reduced componentwise by multiples of $2\pi L^{k-m}$ into $(-\pi L^{k-m}, \pi L^{k-m}]$, which has the same character on the level m and $|q(a)| \le |p_a|$. Distinct representatives give distinct characters, so the coefficient of $(\bar{d}V_1)_{\mu\nu}$ at q is $\sum_{a:q(a)=q} \hat{P}_m(p_a)\hat{c}_{\mu\nu}(a)$, and by Lemma 4.12

$$\sum_q |q|^{2\alpha} \|(\bar{d}V_1)^\wedge_{\mu\nu}(q)\|^2 \le \Big(\sum_a |p_a|^\alpha \|\hat{c}_{\mu\nu}(a)\|\Big)^2 \le C(d,\alpha)^2 \|\hat{B}(p)\|^2 \,, \tag{4.38}$$

uniformly in k and m; the norms are the invariant norms on the Lie algebra, on whose components H_k acts in the same way at $U = 1$. Summing over the six planes gives $6C(d,\alpha)^2\|\hat{B}(p)\|^2$.

Step 5: the sum. With (4.36), $\|\Pi_{\mathcal{J}_k}(p)\| \le \frac{3\pi^2}{2} C_\alpha C(d,\alpha)^2 \sum_{m\le k} L^{-2\alpha(k-m)} \le \frac{3\pi^2}{2} C_\alpha C(d,\alpha)^2 (1 - L^{-2\alpha})^{-1}$. *q.e.d.*

4.5 The Polarisation of the Bare Lattice

The functional $-F_0$ is built from the bare lattice alone, and its polarisation, seen from the lattice $T^{(k)}$, produces the coefficient $2c \ln L$ per step. On the bare lattice it is a one-loop integral with two propagators, which in four dimensions produces a logarithm of the external momentum; the coefficient of the logarithm is that of the continuum, because the logarithm comes from small loop momenta, where the lattice vertices are close to the continuum ones. In this subsection we use the units of the bare lattice, the external momentum ξ and the loop momentum ℓ in $[-\pi,\pi]^4$, and $s(\ell) := \sum_\mu 4\sin^2(\ell_\mu/2) = |\ell|^2 + O(|\ell|^4)$; as in Section 4.2, the symbols are the limits of the momentum sums of the finite tori, written $\int \mathrm{d}^4\ell/(2\pi)^4$.

Lemma 4.14 (The Infrared Lemma). *Let $N(\ell,\xi)$ be a matrix-valued trigonometric polynomial in $(\ell/2, \xi/2)$ with $N(-\ell,-\xi) = N(\ell,\xi)$ and $N(0,0) = 0$, and write $N = N_2 + N_r$ with N_2 homogeneous of degree 2 in (ℓ,ξ) and $|N_r| \le C(|\ell| + |\xi|)^3$ near 0. Put*

$$I(\xi) := \int_{[-\pi,\pi]^4} \frac{N(\ell,\xi)}{s(\ell)s(\ell+\xi)} \frac{\mathrm{d}^4\ell}{(2\pi)^4} \,. \tag{4.39}$$

Then, as $\xi \to 0$,

$$I(\xi) = I(0) + \Sigma(\xi) \ln\frac{1}{|\xi|} + O(|\xi|^2) \,, \qquad \Sigma(\xi) := (2\pi)^{-4} \int_{S^3} f_2(\omega,\xi)\, \mathrm{d}\sigma(\omega) \,, \tag{4.40}$$

where $f_2(\ell,\xi)$ is the part quadratic in ξ of the Taylor expansion at $\xi = 0$ of $f(\ell,\xi) := N_2(\ell,\xi)/(|\ell|^2|\ell+\xi|^2)$, $\ell \in \mathbb{R}^4 \setminus \{0\}$. An integral $\int \tilde{N}(\ell,\xi)/s(\ell)\, \mathrm{d}^4\ell$ with a single propagator and a trigonometric polynomial $\tilde{N}$ is C^∞ in ξ.

Proof. Let $G(\ell,\xi) := N(\ell,\xi)/(s(\ell)s(\ell+\xi))$ and $\Delta^2 G(\ell) := G(\ell,\xi) + G(\ell,-\xi) - 2G(\ell,0)$. The substitution $\ell \mapsto -\ell$, with the parity of N and of s, gives $I(-\xi) = I(\xi)$, so that $I(\xi) - I(0) = \frac{1}{2}\int \Delta^2 G(\ell)\, \mathrm{d}^4\ell/(2\pi)^4$. Let $0 < |\xi| \le \pi/16$; we work entrywise, so that N is scalar, and use the norm $|(\ell,\xi)| := \max(|\ell|, |\xi|)$.

Bounds on N. Evenness gives $DN(0) = 0$, so $N_2 = \frac{1}{2}D^2N(0)$. Let M_N bound $|D^2N|$ and $|D^3N|$ on the ball $|(\ell,\xi)| \le 2\pi + 1$, which contains every (ℓ,ξ) with $\ell \in [-\pi,\pi]^4$ and $|\xi| \le \pi/8$. The mean value theorem, applied to DN, N, D^2N_r, DN_r, and N_r, each of which vanishes at 0, gives $|DN(q)| \le M_N|q|$, $|N(q)| \le M_N|q|^2$, and $|D^tN_r(q)| \le M_N|q|^{3-t}$ for $t = 0,1,2$. For $F \in \{N, N_r\}$ the mean value theorem along $v \mapsto (\ell,v)$, $|v| \le |\xi|$, gives $|F(\ell,\xi) - F(\ell,0)| \le |\xi|\sup|DF|$, and applied to $v \mapsto F(\ell,v) - DF(\ell,0)(0,v)$ it gives $|\Delta^2F| \le 2|\xi|^2\sup|D^2F|$, the linear terms cancelling in Δ^2.

A calculus of second differences. On a region $\{0 < |\ell| \le R,\ |\xi| \le |\ell|/2\}$ we say that $F \in \mathrm{DC}(m)$ if $|F| \le C|\ell|^m$, $|F(\ell,\xi) - F(\ell,0)| \le C|\xi||\ell|^{m-1}$, and $|\Delta^2F| \le C|\xi|^2|\ell|^{m-2}$. If $|\xi| \le |\ell|/2$, every point of the segments above has norm $|\ell|$, so that $N \in \mathrm{DC}(2)$ and $N_r \in \mathrm{DC}(3)$ on $\{0 < |\ell| \le \pi/2\}$, and $N \in \mathrm{DC}(0)$ on the outer region $\{|\ell| \ge \pi/2\}$. The classes are stable under sums; under products, by

$$\begin{gathered}\Delta^2(F_1F_2) = \Delta^2F_1 \cdot F_2(\ell,0) + F_1(\ell,0)\cdot\Delta^2F_2 + \Delta^+F_1\Delta^+F_2 + \Delta^-F_1\Delta^-F_2\ ,\\ \Delta^\pm F := F(\ell,\pm\xi) - F(\ell,0)\ ;\end{gathered} \tag{4.41}$$

and under reciprocals of functions with $|F| \ge c|\ell|^m$, which go into $\mathrm{DC}(-m)$, by $1/x_+ + 1/x_- - 2/x_0 = (-x_0(x_+ + x_- - 2x_0) - 2(x_+ - x_0)(x_- - x_0))/(x_+x_-x_0)$ with $x_\pm := F(\ell,\pm\xi)$ and $x_0 := F(\ell,0)$. On bounded regions $\mathrm{DC}(m) \subset \mathrm{DC}(m')$ for $m' \le m$. From $\cos(\ell\pm\xi) = \cos\ell\cos\xi \mp \sin\ell\sin\xi$, $|x - \sin x| \le 2|x|^3$, and $|2 - 2\cos y - y^2| \le 5y^4$ one obtains $|\ell+\xi|^2, s(\ell+\xi) \in \mathrm{DC}(2)$ and $r_s(\ell+\xi) \in \mathrm{DC}(4)$, with $r_s(p) := |p|^2 - s(p)$; since $s(p) \ge (36/25\pi^2)|p|^2$ when all $|p_\mu| \le 5\pi/4$, $h_s(p) := 1/s(p) - 1/|p|^2 = r_s(p)/(s(p)|p|^2)$ satisfies $h_s(\ell+\xi) \in \mathrm{DC}(0)$.

The outer region $|\ell| \ge \pi/2$. There $s(\ell)$ and $s(\ell+\xi)$ are bounded below, since $\ell+\xi$ stays at distance at least $\pi/2 - \pi/16$ from $2\pi\mathbb{Z}^4$, so that $G \in \mathrm{DC}(0)$, $|\Delta^2G| \le C|\xi|^2$, and the region contributes $O(|\xi|^2)$.

The inner ball $|\ell| < 2|\xi|$. From $|N(\ell,\eta)| \le M_N(|\ell| + |\eta|)^2$, $|\eta| \le |\ell| + |\ell+\eta|$, and the lower bound on s, $|G(\ell,\eta)| \le C(|\ell|^{-2} + |\ell+\eta|^{-2})$ for $\eta \in \{\xi, -\xi, 0\}$, and since $\int_{|\ell|<r}|\ell|^{-2}\,\mathrm{d}^4\ell = \pi^2r^2$, the region contributes $O(|\xi|^2)$.

The annulus $2|\xi| < |\ell| < \pi/2$. Exactly,

$$G = f + F_r\ , \qquad F_r = \frac{N_r}{|\ell|^2|\ell+\xi|^2} + N\Big[\frac{h_s(\ell)}{|\ell+\xi|^2} + \frac{h_s(\ell+\xi)}{|\ell|^2} + h_s(\ell)h_s(\ell+\xi)\Big]\ , \tag{4.42}$$

because $1/s = 1/|\cdot|^2 + h_s$. By the calculus $F_r \in \mathrm{DC}(-1)$, so that $|\Delta^2F_r| \le C|\xi|^2|\ell|^{-3}$, whose integral over the annulus is $O(|\xi|^2)$. For the homogeneous part write $a = |\ell|^2$, $u = \langle\ell,\xi\rangle$, $w = |\xi|^2$, and $N_2 = n_0 + n_1 + n_2$, with n_0 quadratic in ℓ, n_1 bilinear, and n_2 quadratic in ξ. Expanding $f(\ell,t\xi) = a^{-2}(n_0 + tn_1 + t^2n_2)(1 + (2tu + t^2w)/a)^{-1}$ to second order in t gives

$$f_2 = \frac{n_2}{a^2} - \frac{2n_1u}{a^3} - \frac{n_0w}{a^3} + \frac{4n_0u^2}{a^4}\ , \tag{4.43}$$

and, with $|\ell\pm\xi|^2 = a \pm 2u + w$, a direct computation, collecting the coefficients of n_0, n_1, and n_2 separately, gives the identity

$$\Delta^2 f - 2f_2 = \frac{2\left[\begin{gathered} n_0(a^2w^2 + aw^3 - 12awu^2 - 4w^2u^2 + 16u^4)\\ + 2n_1ua(2aw + w^2 - 4u^2) + n_2(-a^3w - a^2w^2 + 4a^2u^2)\end{gathered}\right]}{a^4|\ell+\xi|^2|\ell-\xi|^2}\ , \tag{4.44}$$

so that $|\Delta^2f - 2f_2| \le C|\xi|^4|\ell|^{-6}$, with an integral $O(|\xi|^2)$ over the annulus. Finally $f_2(\cdot,\xi)$ is homogeneous of degree -4, so that in polar coordinates $\int_{2|\xi|<|\ell|<\pi/2} f_2\,\mathrm{d}^4\ell =$

$\ln(\pi/(4|\xi|))\int_{S^3} f_2(\omega,\xi)\,\mathrm{d}\sigma(\omega)$, and $\Sigma(\xi) = O(|\xi|^2)$. The annulus thus contributes $\Sigma(\xi)\ln(1/|\xi|) + O(|\xi|^2)$, and the three regions together give (4.40). The last assertion holds because $1/s$ is integrable in four dimensions and $\tilde{N}$ is smooth in ξ. *q.e.d.*

The functional in the Feynman gauge. Let

$$B_{\mathrm{F}}(U) := -\tfrac{1}{2}\log\det(\mathbf{H} + D_U D_U^{\mathrm{T}}) + \log\det(D_U^{\mathrm{T}} D_U)\ , \tag{4.45}$$

with $\mathbf{H} = \operatorname{Hess} A(U)$ in the coordinates $X \mapsto Ue^{iX}$, the one-loop functional of the background field in the Feynman gauge, for the Wilson action with the normalised trace. It is gauge invariant, because the coordinates transform linearly under gauge transformations, so that the Hessian is covariant even where $\nabla A \neq 0$. Let $\Psi_{\mathrm{F}}(\xi)$ be its polarisation in the direction of a fine background b. At $U = 1$ both $\mathbf{H} + D_U D_U^{\mathrm{T}}$ and $D_U^{\mathrm{T}} D_U$ are the lattice Laplacian, with the symbol $s(\ell)$ on each component, so that the propagator is $s(\ell)^{-1}$. We write V^1 and V^2 for the vertices of first and second order in b of $\tilde{M} := \tilde{\mathbf{H}} + \check{D}\check{D}^{\mathrm{T}}$, where $\tilde{\mathbf{H}}$ is the Hessian of $\tilde{X} \mapsto A(e^{i(b+\tilde{X})})$ at $\tilde{X} = 0$ and $\check{D}$ the linearised gauge action in the chart $\tilde{X} \mapsto e^{i(b+\tilde{X})}$, and of the ghost operator $D_U^{\mathrm{T}} D_U$ at $U = e^{ib}$. On a torus Λ of the bare lattice we put $e_\ell(x) := |\Lambda|^{-1/2} e^{i\langle \ell, x\rangle}$ on 0-forms and $(e_\ell)_\mu(x) := |\Lambda|^{-1/2} e^{i\langle \ell, x+e_\mu/2\rangle}$ on the μ-component of 1-forms, with the phase at the midpoint of the bond, and $\hat{b}_\mu(\xi) := \sum_x e^{-i\langle \xi, x+e_\mu/2\rangle} b_\mu(x)$. With $(m_\xi\beta)_\mu(x) := |\Lambda|^{-1}\beta_\mu e^{i\langle \xi, x+e_\mu/2\rangle}$, so that $b = \sum_\xi m_\xi \hat{b}(\xi)$, the symbols of the vertices are

$$v^1(\ell,\xi)[\beta] := |\Lambda|\langle e_{\ell+\xi}, V^1[m_\xi\beta]e_\ell\rangle\ , \qquad v^2(\ell,\xi)[\beta',\beta] := |\Lambda|^2\langle e_\ell, V^2[m_{-\xi}\beta', m_\xi\beta]e_\ell\rangle\ , \tag{4.46}$$

matrices on the fibre, which is the complexified Lie algebra tensored with $\mathbb{C}^4$ for 1-forms and the complexified Lie algebra for 0-forms; by translation covariance $\langle e_{\ell'}, V^1[m_\xi\beta]e_\ell\rangle = 0$ unless $\ell' = \ell + \xi$. We write tr_f for the unnormalised trace over the fibre.

Lemma 4.15 (The Naive Continuum Limit)**.** *The symbols v^1 and v^2 are trigonometric polynomials in $(\ell/2, \xi/2)$, and $v^1(-\ell,-\xi) = -v^1(\ell,\xi)$. The part of v^1 linear in (ℓ,ξ) is the symbol of the part of first order in b of the continuum operators $-D_b^2 \cdot 1 - 2i \operatorname{ad} F_b$ for the vector field and $-D_b^2$ for the ghost, where $D_b := \partial + i\operatorname{ad} b$, $F(b)_{\mu\nu} := \partial_\mu b_\nu - \partial_\nu b_\mu + i[b_\mu, b_\nu]$, and $(\operatorname{ad} F\, X)_\nu := \sum_\mu [F_{\nu\mu}, X_\mu]$, in the Hermitian convention of [9] for the Lie algebra.*

Proof. Trigonometric polynomials. $A(e^{i(b+\tilde{X})})$ is a sum over plaquettes of functions of their four bond variables, and $\check{D}$ and D_U map the values of a 0-form at the two ends of a bond to that bond, with coefficients analytic in b at the bond. So $\tilde{M}$ and $D_U^{\mathrm{T}} D_U$ are finite sums of shifts with coefficients analytic in the values of b within distance 2, and their parts of order $n = 1, 2$ in b are finite sums of terms which multiply $\psi(\cdot + a_0)$ by a coefficient multilinear in $b(\cdot + a_1), \ldots, b(\cdot + a_n)$; with the phases at sites and midpoints, the symbol of each term is its coefficient times a product of exponentials $e^{i\langle \ell, a\rangle}$ and $e^{i\langle \xi, a'\rangle}$ with $a, a' \in \frac{1}{2}\mathbb{Z}^4$.

Parity. Let $\mathfrak{i}$ be the point inversion: $(\mathfrak{i}U)_\mu(x) := U_\mu(-x-e_\mu)^{-1}$, $(\mathfrak{i}_1 X)_\mu(x) := -X_\mu(-x-e_\mu)$ on 1-forms, and $(\mathfrak{i}_0\omega)(x) := \omega(-x)$ on 0-forms, the last two orthogonal involutions. The holonomy of $\mathfrak{i}U$ around the plaquette at $-x - e_\mu - e_\nu$ in the $\mu\nu$-plane is conjugate to that of U around the plaquette at x, so $A \circ \mathfrak{i} = A$; moreover $\mathfrak{i}(e^{i(b+\tilde{X})}) = e^{i(\mathfrak{i}_1 b + \mathfrak{i}_1\tilde{X})}$, and $\mathfrak{i}$ carries the gauge transform of U by u to that of $\mathfrak{i}U$ by $\mathfrak{i}_0 u$. Hence $\tilde{\mathbf{H}}(\mathfrak{i}_1 b) = \mathfrak{i}_1\tilde{\mathbf{H}}(b)\mathfrak{i}_1$, $\check{D}(\mathfrak{i}_1 b) = \mathfrak{i}_1\check{D}(b)\mathfrak{i}_0$, and $\tilde{M}(\mathfrak{i}_1 b) = \mathfrak{i}_1\tilde{M}(b)\mathfrak{i}_1$; in the chart of B_{F}, $\mathfrak{i}(Ue^{iX}) = (\mathfrak{i}U)e^{iX'}$ with an isometry $X \mapsto X'$ of the 1-forms, so that $D_{\mathfrak{i}U}^{\mathrm{T}} D_{\mathfrak{i}U} = \mathfrak{i}_0 D_U^{\mathrm{T}} D_U \mathfrak{i}_0$. Comparing the parts of first order, and using $\mathfrak{i}_0 e_\ell = e_{-\ell}$, $\mathfrak{i}_1 e_\ell = -e_{-\ell}$, and $(\mathfrak{i}_1 b)^\wedge(\xi) = -\hat{b}(-\xi)$, we obtain $v^1(-\ell,-\xi) = -v^1(\ell,\xi)$.

Linear parts. With $\Xi := b + \tilde{X}$ and, for the plaquette at x in the $\mu\nu$-plane, $\Xi_1 := \Xi_\mu(x)$, $\Xi_2 := \Xi_\nu(x+e_\mu)$, $\Xi_3 := \Xi_\mu(x+e_\nu)$, $\Xi_4 := \Xi_\nu(x)$, we have $U_p = e^{i\Xi_1}e^{i\Xi_2}e^{-i\Xi_3}e^{-i\Xi_4} = e^{i\Xi_p}$, and the Baker–Campbell–Hausdorff formula gives $\Xi_p = (d\Xi)_p + \frac{i}{2}[\Xi,\Xi]_p + O(\Xi^3)$, with $(d\Xi)_p := \Xi_1 + \Xi_2 - \Xi_3 - \Xi_4$ and $[\Xi,\Xi]_p := [\Xi_1,\Xi_2] - [\Xi_1,\Xi_3] - [\Xi_1,\Xi_4] - [\Xi_2,\Xi_3] - [\Xi_2,\Xi_4] + [\Xi_3,\Xi_4]$. Since $\operatorname{tr}\Xi_p = 0$ and $\operatorname{tr}\Xi_p^3$ is real, $1 - \operatorname{Re}\operatorname{tr} e^{i\Xi_p} = \frac{1}{2}\operatorname{tr}\Xi_p^2 + O(\Xi_p^4)$, so the part of $A(e^{i\Xi})$ of order two in $\tilde{X}$ and one in b is that of $\sum_p \frac{i}{2}\operatorname{tr}((d\Xi)_p[\Xi,\Xi]_p)$. Its symbol is a sum of products of the symbol of one difference, which vanishes at $(0,0)$, and of the symbol of a commutator; the linear part keeps the linear part of the first and the value at $(0,0)$ of the second, where all four Ξ_a sit at one point and $[\Xi,\Xi]_p = 2[\Xi_\mu,\Xi_\nu]$. The linear part of the symbol of $(d\Xi)_p$ is that of $\partial_\mu\Xi_\nu - \partial_\nu\Xi_\mu$, so the linear part of the symbol of the first-order part of $\tilde{\mathbf{H}}$ is that of the continuum form $\frac{1}{4}\sum_{\mu\nu}\int \operatorname{tr} F(\Xi)_{\mu\nu}^2$ at the same order. By the same formula $e^{i\omega(x)}e^{ib_\mu(x)}e^{-i\omega(x+e_\mu)} = e^{i(b_\mu(x)+(\check{D}\omega)_\mu(x)+O(\omega^2))}$ with $(\check{D}\omega)_\mu(x) = \omega(x) - \omega(x+e_\mu) - \frac{i}{2}[b_\mu(x), \omega(x) + \omega(x+e_\mu)] + O(b^2\omega)$, whose symbol has the linear part of that of $-D_b\omega$. With $D_U = T\check{D}$, T the derivative of the change of chart of the next paragraph, and $T^{\mathrm{T}}T = 1 + O(b^2)$, the first-order parts of $\check{D}\check{D}^{\mathrm{T}}$ and $D_U^{\mathrm{T}}D_U$ have the linear parts of those of $D_bD_b^{\mathrm{T}}$ and $D_b^{\mathrm{T}}D_b = -D_b^2$. Finally, the continuum Hessian of $\frac{1}{4}\sum_{\mu\nu}\int \operatorname{tr} F(b+X)_{\mu\nu}^2 + \frac{1}{2}\int \operatorname{tr}(D_b^{\mathrm{T}}X)^2$ is $-D_b^2 \cdot 1 - 2i\operatorname{ad} F_b$, by integration by parts from $[D_\mu, D_\nu] = i\operatorname{ad} F_{\mu\nu}$, $\langle F, [X_\mu, X_\nu]\rangle = \langle [F, X_\mu], X_\nu\rangle$, and $F(b+X)_{\mu\nu} - F(b)_{\mu\nu} = D_\mu X_\nu - D_\nu X_\mu + i[X_\mu, X_\nu]$. *q.e.d.*

The chart of B_{F}. At $U = e^{ib}$ the functional B_{F} uses the chart $X \mapsto e^{ib}e^{iX}$. Define $X = \phi(\tilde{X})$ bondwise by $e^{ib}e^{i\phi(\tilde{X})} = e^{i(b+\tilde{X})}$; the Baker–Campbell–Hausdorff formula gives $\phi(\tilde{X}) = \tilde{X} - \frac{i}{2}[b,\tilde{X}] + O(b^2\tilde{X}, b\tilde{X}^2)$, so $T := D\phi(0) = 1 - \frac{i}{2}\operatorname{ad} b + O(b^2)$ and, since $i\operatorname{ad} b$ is antisymmetric for tr, $T^{\mathrm{T}}T = 1 + O(b^2)$. With $\tilde{A}(\tilde{X}) := A(e^{i(b+\tilde{X})})$, $\mathbf{H} = T^{-\mathrm{T}}\tilde{\mathbf{H}}T^{-1} + \mathbf{Gr}$ with $\mathbf{Gr} := \langle \nabla\tilde{A}(0), D^2\phi^{-1}(0)\rangle = O(b^2)$, which contains no differences, and $D_U = T\check{D}$, both charts parametrising the same gauge orbit. Hence $\mathbf{H} + D_UD_U^{\mathrm{T}} = T^{-\mathrm{T}}(\tilde{M} + E)T^{-1}$ with $E := T^{\mathrm{T}}T\check{D}\check{D}^{\mathrm{T}}T^{\mathrm{T}}T - \check{D}\check{D}^{\mathrm{T}} + T^{\mathrm{T}}\mathbf{Gr}T = O(b^2)$, and

$$-\tfrac{1}{2}\log\det(\mathbf{H} + D_UD_U^{\mathrm{T}}) = -\tfrac{1}{2}\log\det(\tilde{M} + E) + \log\det T\ . \tag{4.47}$$

Here $\log\det T$ is a sum over the bonds of functions of the value of b at the bond alone; its first-order part, $-\frac{i}{2}\operatorname{Tr}(\operatorname{ad} b)$ summed over the bonds, vanishes, since $\operatorname{Tr}\operatorname{ad} X = 0$, and its second-order part has a symbol independent of ξ, so it carries no logarithm. Since $\tilde{M}^{-1} = s^{-1} + O(b)$, E enters the second order once, as $\operatorname{Tr}(s^{-1}E_2)$, a term with one propagator and a trigonometric polynomial as numerator. By Lemma 4.14 the terms with one propagator, $\operatorname{Tr}(s^{-1}V^2)$ and $\operatorname{Tr}(s^{-1}E_2)$, are C^∞ in ξ and carry no logarithm, while the terms with two propagators,

$$\operatorname{Tr}(s^{-1}V^1s^{-1}V^1) = |\Lambda|^{-2}\sum_{\ell,\xi} s(\ell)^{-1}s(\ell+\xi)^{-1}\operatorname{tr}_f[v^1(\ell+\xi,-\xi)[\hat{b}(-\xi)]\,v^1(\ell,\xi)[\hat{b}(\xi)]]\ , \tag{4.48}$$

have numerators $N(\ell,\xi)$ which satisfy the hypotheses of Lemma 4.14: trigonometric polynomials, even because v^1 is odd, and zero at 0. Their N_2 is the product of the linear parts of the two vertices, the numerator of the continuum bubble in the Feynman gauge. The logarithm in Ψ_{F} is therefore fixed by the continuum vertices.

Lemma 4.16 (The Coefficient of the Logarithm)**.** *Fix a mode $\hat{b}_\mu(\xi)$ of the background, put $\langle X, Y\rangle := \operatorname{tr}(\overline{X}Y)$, positive definite on the Lie algebra in the Hermitian convention, and $B_{\mu\nu}(\xi) := \langle \hat{b}_\mu(\xi), \hat{b}_\nu(\xi)\rangle$, the bar denoting the conjugation of the complexification. The part of second order of B_{F} in this mode has the logarithmic part*

$$c\ln\frac{1}{|\xi|^2}\cdot\tfrac{1}{2}(|\xi|^2\delta_{\mu\nu} - \xi_\mu\xi_\nu)B_{\mu\nu}(\xi)\ , \qquad c = \frac{11}{48\pi^2}\kappa\ , \tag{4.49}$$

up to the factor $|\Lambda|^{-1}$ *common to all modes, with the constant* κ *of Paper, (4.2). The same mode of the quadratic part of the Wilson action is* $\frac{1}{2}(|\xi|^2\delta_{\mu\nu}-\xi_\mu\xi_\nu)B_{\mu\nu}$ *up to* $O(|\xi|^3)\sum_\mu B_{\mu\mu}$*, so that the logarithmic part of* Ψ_{F} *is* $c\ln(1/|\xi|^2)$ *times the leading part of the polarisation of the Wilson action.*

Proof. By Paper, Lemma 4.2, $\mathrm{Tr}(\mathrm{ad}\,X\,\mathrm{ad}\,Y)=\kappa\langle X,Y\rangle$ on the Lie algebra in the Hermitian convention as well, since $X\mapsto iX$ changes the signs of both sides, and we extend it bilinearly to the complexification. Since b is real, $\hat b(-\xi)=\overline{\hat b(\xi)}$.

Expansion. If $\Delta=\Delta^0+V^1+V^2+O(b^3)$ and $G=(\Delta^0)^{-1}$, then $\log\det\Delta=\log\det\Delta^0+\mathrm{Tr}\,GV^1+\mathrm{Tr}\,GV^2-\frac{1}{2}\mathrm{Tr}(GV^1GV^1)+O(b^3)$. So, by (4.47) and the preceding paragraph, the logarithmic part of the second order of B_{F} is that of $\frac{1}{4}\mathrm{Tr}(GV_v^1GV_v^1)-\frac{1}{2}\mathrm{Tr}(GV_{\mathrm{gh}}^1GV_{\mathrm{gh}}^1)$, with the first-order vertices V_v^1 of $\tilde M+E$ and V_{gh}^1 of $D_U^{\mathrm{T}}D_U$.

Continuum vertices. By Lemma 4.15 the linear parts of the vertices are those of $-D_b^2\cdot 1-2i\,\mathrm{ad}\,F_b$ and $-D_b^2$. Their parts linear in b have, with $\xi=\ell'-\ell$ and $\hat b=\hat b(\xi)$, the matrix elements $(\ell+\ell')_\mu\,\mathrm{ad}\,\hat b_\mu/|\Lambda|$ (convective), on each of the four components for the vector field and on the ghost, and, for the vector field only, $W_{\nu\mu}=2\,\mathrm{ad}(\xi_\nu\hat b_\mu-\xi_\mu\hat b_\nu)/|\Lambda|$ (magnetic), since $\hat F_{\nu\mu}=i(\xi_\nu\hat b_\mu-\xi_\mu\hat b_\nu)$ at first order.

Traces. With $\ell'=\ell+\xi$, each trace is $\kappa|\Lambda|^{-1}\sum_\xi$ times a lattice bubble (4.39) whose N_2 is: for the ghost, $(2\ell+\xi)_\mu(2\ell+\xi)_\nu B_{\mu\nu}$, the vertex factors being real and $\mathrm{Tr}(\mathrm{ad}\,\hat b_\mu(-\xi)\,\mathrm{ad}\,\hat b_\nu(\xi))=\kappa B_{\mu\nu}$; for the convective part of the vector field, four times that of the ghost; for the mixed part, zero, because the trace of W over the vector index is $\sum_\nu W_{\nu\nu}=0$; and for the magnetic part, $4\sum_{\mu\nu}|\xi_\mu\hat b_\nu-\xi_\nu\hat b_\mu|^2=8(|\xi|^2\delta_{\mu\nu}-\xi_\mu\xi_\nu)B_{\mu\nu}$.

Angular averages. By Lemma 4.14 each bubble has the logarithmic part $\Sigma(\xi)\ln(1/|\xi|)=(16\pi^2)^{-1}\langle f_2(\cdot,\xi)\rangle\ln(1/|\xi|^2)$, where $\langle\cdot\rangle$ is the average over S^3, since $(2\pi)^{-4}|S^3|=1/(8\pi^2)$. The moments are $\langle\omega_\mu\omega_\nu\rangle=\delta_{\mu\nu}/4$ and $\langle\omega_\mu\omega_\nu\omega_\varrho\omega_\sigma\rangle=(\delta_{\mu\nu}\delta_{\varrho\sigma}+\delta_{\mu\varrho}\delta_{\nu\sigma}+\delta_{\mu\sigma}\delta_{\nu\varrho})/24$: the signed permutations of the coordinates leave only $\langle\omega_1^2\rangle$, $\langle\omega_1^4\rangle$, and $\langle\omega_1^2\omega_2^2\rangle$; $\sum_\mu\omega_\mu^2=1$ gives $\langle\omega_1^2\rangle=1/4$ and $4\langle\omega_1^4\rangle+12\langle\omega_1^2\omega_2^2\rangle=1$; and the rotation by $\pi/4$ in the $(1,2)$-plane gives $\langle\omega_1^4\rangle=3\langle\omega_1^2\omega_2^2\rangle$. By (4.43) at $|\ell|=1$, with $u=\langle\omega,\xi\rangle$ and $w=|\xi|^2$, the convective numerator gives $f_2=\xi_\mu\xi_\nu-4u(\omega_\mu\xi_\nu+\xi_\mu\omega_\nu)-4w\omega_\mu\omega_\nu+16u^2\omega_\mu\omega_\nu$, so that

$$\langle f_2\rangle=\xi_\mu\xi_\nu-2\xi_\mu\xi_\nu-w\delta_{\mu\nu}+\tfrac{2}{3}(w\delta_{\mu\nu}+2\xi_\mu\xi_\nu)=-\tfrac{1}{3}(|\xi|^2\delta_{\mu\nu}-\xi_\mu\xi_\nu)\,, \tag{4.50}$$

while the magnetic numerator is already quadratic in ξ, so that $f_2=N_2$ on S^3 and $\langle f_2\rangle=8(|\xi|^2\delta-\xi\xi)B$.

Sum. In units of $\kappa|\Lambda|^{-1}(16\pi^2)^{-1}\ln(1/|\xi|^2)(|\xi|^2\delta-\xi\xi)B$ the logarithmic part is $\frac{1}{4}[4\cdot(-\frac{1}{3})+8]-\frac{1}{2}\cdot(-\frac{1}{3})=\frac{5}{3}+\frac{1}{6}=\frac{11}{6}$, which is $(11\kappa/96\pi^2)\ln(1/|\xi|^2)(|\xi|^2\delta-\xi\xi)B=c\ln(1/|\xi|^2)\cdot\frac{1}{2}(|\xi|^2\delta-\xi\xi)B$. The statement on the Wilson action is the expansion of the plaquette above at $\tilde X=0$. *q.e.d.*

The positivity of c is asymptotic freedom, and the rate $2c\ln L=(11\kappa/24\pi^2)\ln L$ per step is the universal running at one loop. The sign agrees with Bałaban's conventions: the effective action is $-g_k^{-2}A+\mathbf{E}_k$ and $g_{k+1}^{-2}=g_k^{-2}-\beta_{k+1}$ [9, (0.20), (0.22), p. 256], so that a positive multiple of A in the regular part, which is what $-F_0$ contributes to $\log Z_k=F_k-F_0$, lowers g^{-2}. The affine part of Ψ_{F} vanishes. Indeed, with $\check v_\mu(\xi):=2i\sin(\xi_\mu/2)$, the symbol of the gradient in the phases above, $\Psi_{\mathrm{F}}(\xi)\check v(\xi)=0$: the first-order term of $B_{\mathrm{F}}(e^{ib})$ is an invariant linear form in $\hat b(0)$, hence zero, the Lie algebra being simple, and the part of order one in b and one in ω of $B_{\mathrm{F}}(u\cdot e^{ib})=B_{\mathrm{F}}(e^{ib})$, $u=e^{i\omega}$, is then $-\langle b,\Psi_{\mathrm{F}}\partial_1\omega\rangle=0$, because

$\check{D}\omega = -\partial_1\omega + O(b\omega)$. Moreover $\Psi_{\mathrm{F}}(R\xi) = R\Psi_{\mathrm{F}}(\xi)R^{-1}$ for every signed permutation R, which maps midpoints of bonds to midpoints. The steps of the proof of Lemma 4.6(a) which determine $T(0)$ and $\nabla T(0)$ use v only through $v(0) = 0$ and $\partial_\mu v(0) = ie_\mu$, which $\check{v}$ has, and applied to the affine part $a_0 + a_1\xi$ of Ψ_{F}, the logarithmic part being covariant and annihilating $\check{v}$ up to $O(|\xi|^5 \ln)$ and the rest being $O(|\xi|^2)$, they give $a_0 = 0$ and $a_1 = 0$.

Proposition 4.17 (The Bare Polarisation)**.** *Let F_0 be the functional of the level* 0 *of Definition 4.1, built from the form* (4.14)*. The polarisation of $-F_0$ is $\Psi_0 = \hat{d}^*\tilde{\Psi}_0\hat{d}$, with $\hat{d}$ the symbol of the lattice curl on* 1*-forms, $(\hat{d}X)_{\mu\nu} = \check{v}_\mu X_\nu - \check{v}_\nu X_\mu$, the phase of a plane wave of* 2*-forms being taken at the centre of the plaquette, and with $\tilde{\Psi}_0$ acting on* $\operatorname{ran}\hat{d}$*, and*

$$\tilde{\Psi}_0(\xi) = c\ln\frac{1}{|\xi|^2}\cdot 1 + R_0(\xi)\ , \qquad \sup_{\xi\in[-\pi,\pi]^4\setminus 0} \|R_0(\xi)\| < \infty\ . \tag{4.51}$$

Proof. We write $\mathbf{H} = \operatorname{Hess} A(U)$, $D = D_U$, $\Pi_D = D(D^{\mathrm{T}}D)^{-1}D^{\mathrm{T}}$, $\Pi := 1 - \Pi_D$, $h_\perp$ for h on $(\operatorname{ran} D)^\perp$, $S_D := (D^{\mathrm{T}}D)^{-2}$, and $J = \nabla A(U)$; differentiating $\langle\nabla A(U), D_U\omega\rangle \equiv 0$ gives $\mathbf{H}D = \mathcal{N}_U(J)$, where $\mathcal{N}_U(j)$ is linear in j, local, contains no difference, depends on U only through the transports, and is gauge covariant; in the coordinates of [5] it is (4.15). All operators are restricted to a coset $\mathcal{O} \neq \mathbb{Z}\xi$. The loop momentum ℓ is that of the line of 0-forms, $a := |\ell|$ and $a' := |\ell + \xi|$, both modulo 2π, and at $U = e^{ib}$ a subscript n marks the part of order n in b.

Step 1: one determinant. By Lemma 4.4 at the level 0 and (4.6), $-F_0 - B_{\mathrm{F}} = -\frac{1}{2}[\log\det(h + DD^{\mathrm{T}}) - \log\det(\mathbf{H} + DD^{\mathrm{T}})]$. With respect to $(\operatorname{ran} D)^\perp \oplus \operatorname{ran} D$, $h + DD^{\mathrm{T}} = h_\perp \oplus DD^{\mathrm{T}}$, and $\mathbf{H} + DD^{\mathrm{T}}$ has the diagonal blocks $h_\perp$ and $\Pi_D\mathbf{H}\Pi_D + DD^{\mathrm{T}}$, with the Schur complement $DD^{\mathrm{T}} + \Pi_D\mathbf{H}_S\Pi_D$ of $h_\perp$, where $\mathbf{H}_S := \mathbf{H} - \mathbf{H}\Pi h_\perp^{-1}\Pi\mathbf{H}$. On $\operatorname{ran} D$, $(DD^{\mathrm{T}})^{-1} = DS_DD^{\mathrm{T}}$, and Sylvester's identity $\det(1 + XY) = \det(1 + YX)$ gives

$$-F_0 - B_{\mathrm{F}} = \tfrac{1}{2}\log\det{}_Y(1 + S_D\mathcal{W})\ , \qquad \mathcal{W} := D^{\mathrm{T}}\mathbf{H}_SD = D^{\mathrm{T}}\mathcal{N}_U(J) - \mathcal{N}_U(J)^{\mathrm{T}}\Pi h_\perp^{-1}\Pi\mathcal{N}_U(J)\ . \tag{4.52}$$

Step 2: the integrand. Since $\mathcal{W}_0 = 0$, the part of second order is $\frac{1}{2}\operatorname{Tr}(S_D\mathcal{W}_2) + \frac{1}{2}\operatorname{Tr}(S_{D,1}\mathcal{W}_1) - \frac{1}{4}\operatorname{Tr}(S_D\mathcal{W}_1S_D\mathcal{W}_1)$, with an integrand $I(\ell,\xi)$, the fibre trace of its symbol. The form $\mathbf{H}_S$ vanishes at $U = 1$, since $\mathbf{H}_0 = \Pi\mathbf{H}_0\Pi$, and its first-order part is $\Pi_D\mathbf{H}_1\Pi_D$, the derivatives of the projections dropping out because $\Pi(\Pi)_1\Pi = 0$ and $\mathbf{H}_0(\Pi)_1\Pi = 0$; hence $\mathcal{W}_1 = D_0^{\mathrm{T}}\mathbf{H}_1D_0$ and $\mathcal{W}_2 = D_0^{\mathrm{T}}(\mathbf{H}_2 - \mathbf{H}_1\Pi h_\perp^{-1}\Pi\mathbf{H}_1)D_0 + D_1^{\mathrm{T}}\Pi_D\mathbf{H}_1\Pi_DD_0 + D_0^{\mathrm{T}}\Pi_D\mathbf{H}_1\Pi_DD_1$, since $\mathbf{H}_0D_0 = 0$. The symbols satisfy $|D_0| \le a$, $s \ge (4/\pi^2)a^2$, $h_0 = s$ on coclosed 1-forms, $|\mathbf{H}_2| \le C$, $|(D^{\mathrm{T}}D)_1| \le C(a + a')$, and $|\mathbf{H}_1| \le C(a + a')$, because $\mathbf{H}_1$ is a trigonometric polynomial which vanishes at zero momenta, the action being quartic in constant fields; D_1 contains no difference. The five terms are therefore bounded by Ca^{-2}, $C(a+a')^2a^{-2}a'^{-2}$, $C(a+a')a^{-3}$, $C(a+a')^2(a^{-1}a'^{-3}+a^{-3}a'^{-1})$, and $C(a+a')^2a^{-2}a'^{-2}$, of degree at most 3 at $\ell = 0$ and $\ell = -\xi$, and as in Lemma 4.5, with Cr in place of Cr^2, $\Psi_0 - \Psi_{\mathrm{F}} = \int I(\ell,\xi)\,\mathrm{d}^4\ell/(2\pi)^4$. Since $J_0 = 0$, I is the integrand of $\frac{1}{2}\operatorname{Tr}(S_DD_0^{\mathrm{T}}\mathcal{N}_0(J_2))$ plus $I_{\text{lin}} + I_{\text{quad}}$, where $I_{\text{lin}} := \theta_1(\ell,\xi)[\hat{b},\hat{J}_1]$, θ_1 being the part of first order in b at fixed j of the integrand of $\frac{1}{2}\operatorname{Tr}(S_DD_U^{\mathrm{T}}\mathcal{N}_U(j))$, and I_{quad} is the integrand of $-\frac{1}{2}\operatorname{Tr}(S_D\mathcal{N}_0(J_1)^{\mathrm{T}}\Pi h_\perp^{-1}\Pi\mathcal{N}_0(J_1)) - \frac{1}{4}\operatorname{Tr}(S_DD_0^{\mathrm{T}}\mathcal{N}_0(J_1)S_DD_0^{\mathrm{T}}\mathcal{N}_0(J_1))$. Only the part of J_2 of zero momentum reaches the trace, and for constant j, $\mathcal{N}_0(j) = \sum_\nu A_\nu(j)\otimes n_\nu(\ell)$ with $A_\nu(j)$ a multiple of $\operatorname{ad} j_\nu$ and every other factor scalar on the Lie algebra; the fibre trace of $A\otimes P$ is $\operatorname{Tr} A$ times the trace of P, and $\operatorname{Tr}\operatorname{ad} = 0$, so the term of J_2 vanishes for every ℓ. Finally $J_1 = \mathbf{H}_0b$ has the symbol $O(|\xi|^2)$, so that $|\hat{J}_1|, |\mathcal{N}_0(J_1)| \le C|\xi|^2$.

Step 3: the size of the difference. Let $|\xi| \le \xi_0$ with ξ_0 small. In the inner region $|\ell| < 2|\xi|$, where $a + a' \le 5|\xi|$, the bounds of Step 2 apply: $\int a^{-2} \le C|\xi|^2$, $|\xi|\int a^{-3} \le C|\xi|^2$, and for

$p, q \leq 3$ with $p + q = 4$ the substitution $\ell = |\xi|\lambda$ gives $|\xi|^2 \int a^{-p} a'^{-q} = |\xi|^2 \int_{|\lambda|<2} |\lambda|^{-p} |\lambda + \xi/|\xi||^{-q} \, \mathrm{d}^4\lambda \leq C|\xi|^2$, with C independent of the direction. In the outer region $a' \geq a/2$, so $|I_{\text{quad}}| \leq C|\xi|^4 a^{-6}$, with an integral at most $C|\xi|^2$; and $|\theta_1(\ell, t\xi)| \leq Ca^{-4}$, $|\partial_\xi \theta_1(\ell, t\xi)| \leq Ca^{-5}$ for $t \in [0,1]$, because the three pieces $S_D D_1^{\mathrm{T}} \mathcal{N}_0$, $S_D D_0^{\mathrm{T}} \mathcal{N}_1$, and $S_{D,1} D_0^{\mathrm{T}} \mathcal{N}_0$ are bounded by Ca^{-4}, Ca^{-3}, and $C(a + a')a^{-6}a'$ and a derivative in ξ costs at most one factor a^{-1}, so that $\theta_1(\ell, \xi) - \theta_1(\ell, 0)$ contributes at most $C|\xi|^3 \int_{a \geq 2|\xi|} a^{-5} \leq C|\xi|^2$. It remains to bound the contribution of $\theta_1(\ell, 0)[\beta, j]$, the integrand for a constant background β and a constant j. To first order in β, $U_\beta := (e^{i\beta_\mu})$ is the gauge transform of 1 by $g(x) := e^{-i\beta \cdot x}$, so that $D_{U_\beta} = g D_0 g^{-1}$ and $\mathcal{N}_{U_\beta}(j) = g \mathcal{N}_0(\mathrm{Ad}(g)^{-1} j) g^{-1}$; the conjugation by g and the dependence of $\mathrm{Ad}(g)^{-1} j$ on x enter through commutators with the coordinates x_μ, and $[x_\mu, P]$ has the symbol $i\partial_{\ell_\mu} P$ for translation-invariant P, the terms multiplied by x_μ cancelling because U_β is translation invariant. What remains is a sum of terms in which one factor P of the product at $U = 1$ is replaced by a fixed multiple of $\mathrm{ad}\, \beta_\mu \, \partial_{\ell_\mu} P$, and of terms whose only factor which is not scalar on the Lie algebra is $\mathrm{ad}[\beta_\mu, j]$; the latter vanish after the fibre trace, by the factorisation above, and by Leibniz the former give $\theta_1(\ell, 0)[\beta, j] = \sum_{\mu\nu} C_{\mu\nu} \partial_{\ell_\mu} \Phi_\nu(\ell)$, with $C_{\mu\nu}$ a fixed multiple of $\mathrm{Tr}(\mathrm{ad}\, \beta_\mu A_\nu(j))$ and $\Phi_\nu(\ell) := \mathrm{tr}_f[S_D(\ell) D_0(\ell)^* n_\nu(\ell)]$. The function Φ_ν is smooth away from 0 with $|\Phi_\nu| \leq Ca^{-3}$, and the faces of $[-\pi, \pi]^4$ cancel by periodicity, so that $|\int_{a \geq 2|\xi|} \partial_{\ell_\mu} \Phi_\nu| = |\int_{a = 2|\xi|} \Phi_\nu \omega_\mu \, \mathrm{d}\sigma| \leq C$, and with $|\hat{J}_1| \leq C|\xi|^2$ this part is $O(|\xi|^2)$. So $\Psi_0 - \Psi_{\mathrm{F}} = O(|\xi|^2)$ for $|\xi| \leq \xi_0$, and for $|\xi| \geq \xi_0$ the singularities of $I(\cdot, \xi)$ have degree at most 3, locally uniformly in ξ, so that $\Psi_0 - \Psi_{\mathrm{F}}$ is continuous and bounded there.

Step 4: transversality. On each coset F_0 is invariant under the gauge transformations which commute with T_ξ, because these preserve the cosets and every operator of F_0 is gauge covariant. Its first-order coefficient at $U = 1$ vanishes on T_ξ-invariant backgrounds: a mode of momentum $\xi' \in \mathbb{Z}\xi$, $\xi' \not\equiv 0$, couples different fibres and has zero trace, and a constant mode gives an invariant element of the simple Lie algebra, which is zero. Differentiating $\langle \nabla F_0(U), D_U \lambda \rangle \equiv 0$ at $U = 1$ therefore gives $\Psi_0(\xi) \check{v}(\xi) = 0$ on each coset and in the limit. As $\Psi_0(\xi)$ is Hermitian, it acts on the 1-forms orthogonal to $\check{v}(\xi)$, on which $\hat{d}$ is injective for $\xi \not\equiv 0$, since $\hat{d}^* \hat{d} = |\check{v}|^2 \cdot 1$ there; so $\Psi_0 = \hat{d}^* \tilde{\Psi}_0 \hat{d}$.

Step 5: conclusion. By Lemma 4.16, the vanishing of the affine part, and Lemma 4.14, $\Psi_{\mathrm{F}}(\xi) = c \ln(1/|\xi|^2)(|\xi|^2 \delta - \xi\xi) + O(|\xi|^2)$ near 0. Also $\hat{d}^* \hat{d}(\xi) = |\check{v}|^2 \delta - \overline{\check{v}} \check{v}^{\mathrm{T}} = |\xi|^2 \delta - \xi\xi + O(|\xi|^4)$, and $c \ln(1/|\xi|^2) O(|\xi|^4) = O(|\xi|^2)$, so that $\hat{d}^* R_0 \hat{d} = \Psi_0 - c \ln(1/|\xi|^2) \hat{d}^* \hat{d} = O(|\xi|^2)$ by Step 3. On the 1-forms orthogonal to $\check{v}$, $\hat{d}^* \hat{d} = |\check{v}|^2 \geq (4/\pi^2)|\xi|^2$, since $4 \sin^2(\xi_\mu/2) \geq 4\xi_\mu^2/\pi^2$ for $|\xi_\mu| \leq \pi$, and R_0 is Hermitian, so that $\|R_0(\xi)\| \leq C$ near 0. Away from 0, Ψ_0 is continuous and bounded. *q.e.d.*

We now look at $-F_0$ from the lattice $T^{(k)}$, on which it is a functional of the field B through the fine field $H_k B$.

Proposition 4.18 (The Bare Functional Seen from the Level k)**.** *For a fixed momentum* $p_0 \neq 0$ *of* $T^{(k)}$, $\Pi_{-F_0}(p_0) = c \ln(1/\eta_k^2) \Delta_k(p_0) + O_{p_0}(1)$, *uniformly in* k, *where* $\eta_k = L^{-k}$.

Proof. By the paragraph on the first-order background in the proof of Proposition 4.13, $U_k(e^{itB}) = \exp(it\eta A_1 + O(t^2))$ bondwise, with $d_\eta A_1 = d_\eta H_k B$. The coefficient of first order of F_0 at $U = 1$ is a translation-invariant linear form, that is, an invariant element of the Lie algebra paired with the constant mode of each component, and it vanishes as in Step 4 of the proof of Proposition 4.17; so the part of second order in t of the configuration does not contribute, and since $\Psi_0 = \hat{d}^* \tilde{\Psi}_0 \hat{d}$, the polarisation depends on A_1 only through $d_\eta A_1$. Since $X = \eta A$ and the curl of the bare lattice is $\eta^2 d_\eta$, a sum $\sum_p |(dX)_p|^2$ over the bare plaquettes equals $\eta^4 \sum_p |(d_\eta A)_p|^2$, the square of the norm of $d_\eta A$ in the units of $T^{(k)}$. The polarisation of

$-F_0$ is therefore $\sum_l \langle (d_\eta H_k B)^\wedge(q), \tilde{\Psi}_0(\eta_k q)(d_\eta H_k B)^\wedge(q)\rangle$, summed over the aliases $q = p_0 + l$, with the bare momentum $\xi = \eta_k q$; here $(d_\eta H_k B)^\wedge$ has the phases of [2] at the corners, and passing from the phases at the centres of Proposition 4.17 conjugates $\tilde{\Psi}_0$ by a diagonal unitary, which keeps $c \ln(1/|\xi|^2) \cdot 1$ and $\|R_0\|$. By Proposition 4.17 it is

$$c \ln \frac{1}{\eta_k^2} \sum_l |(d_\eta H_k B)^\wedge(q)|^2 + c \sum_l \ln \frac{1}{|q|^2} |(d_\eta H_k B)^\wedge(q)|^2 + \sum_l \langle (d_\eta H_k B)^\wedge(q), R_0(\eta_k q)(d_\eta H_k B)^\wedge(q)\rangle \ . \tag{4.53}$$

The first sum is $\|d_\eta H_k B\|^2 = \langle B, \Delta_k B\rangle$, by [2, (1.65), p. 29]. For $l \neq 0$, $|q| \geq \pi$, so that for every l, $|\ln(1/|q|^2)| \leq |\ln(1/|p_0|^2)| + \alpha^{-1}|q|^{2\alpha}$, with the exponent α of Lemma 4.12, because $\ln t < t$ at $t = |q|^{2\alpha}$. Since $|q| \geq \min(|p_0|, \pi)$ for every alias, the right side is at most $C(p_0)|q|^{2\alpha}$, and as $\sum_l a_l^2 \leq (\sum_l a_l)^2$ for $a_l \geq 0$, the second sum is at most $C(p_0)(\sum_l |q|^\alpha |(d_\eta H_k)^\wedge(q)|)^2 |\hat{B}|^2$, which is $O_{p_0}(1)|\hat{B}|^2$ by that lemma. The third is at most $\sup\|R_0\| \cdot \|d_\eta H_k B\|^2$, which is bounded by (4.26). *q.e.d.*

4.6 The Functional of the Top Level

The functional F_k remains. Seen from the lattice $T^{(k)}$, it consists of the logarithm of the determinant of the form Δ_k, which is a minimum over the fine fields, and of a ghost determinant of the coarse lattice, and unlike $-F_0$ it has no reason to produce a logarithm: at the trivial background the minimum is attained through Bałaban's propagators, whose kernels decay exponentially on the unit scale of $T^{(k)}$, uniformly in k. Two features make the bound less direct than this suggests. The form h which defines Δ_k is the gauge-invariant compression of the Hessian, and it differs from the forms of [5] by terms of the current J which contain the factor φ of (4.16), singular at the zero coarse momentum; and the coarse ghost determinant has singular factors as well. We show that in every vertex of the expansion of $\log\det\Delta_k$ in the background these singular factors sit at the momenta $p \pm p_0$ of the internal lines adjacent to the background, so that in four dimensions they are integrable once $p_0 \neq 0$ separates them from the singularity of the propagator of Δ_k at $p = 0$.

Throughout this subsection we use the operators and the notation of Section 4.2 at the level k, in the units of $T^{(k)}$, so that $L^k \eta = 1$. All the domains Ω_j of [5] are the whole torus, so that only the level k occurs in [5, (3.24), p. 394]; we take Bałaban's constant $a = 1$, the value of [5, Theorem 3.3, p. 399], and write a for the coefficient a_k of the level k in (3.24), which satisfies $1 - L^{-2} \leq a \leq 1$ by the recursion $a_{j+1} = a a_j/(aL^{-2} + a_j)$, $a_1 = a_0 = a$, printed after (3.24), so that $\langle A, aQ^*QA\rangle = a|QA|^2$. We write $c_\mathfrak{g}$ for a constant with $|[X,Y]| \leq c_\mathfrak{g}|X||Y|$ on the complexified Lie algebra, so that $|\mathrm{ad}\, X| \leq c_\mathfrak{g}|X|$, and $\|\cdot\|$ for the ℓ^2-norm on the whole torus, with the weight η^4 on fine fields and 1 on coarse ones, and for the operator norm which it defines.

Lemma 4.19 (The Form of the Top Level through Bałaban's Propagators)**.** *Let U be a background at which the hypotheses* (4.1) *hold, on the whole torus or on the fields of a coset, at which Q' is onto, and at which G', R, and G exist. Then for every coarse field z*

$$\langle z, \Delta_k z\rangle = \min\{\langle A, (h + DRD^* + aQ^*Q)A\rangle \,|\, QA = z\} - a|z|^2 \ . \tag{4.54}$$

Proof. By (4.4) and (ii) of Section 4.2, $\langle z, \Delta_k z\rangle$ is the minimum of $\langle A, hA\rangle$ over $\{QA \in z + \operatorname{ran}\bar{D}\}$. If $QA = z + \bar{D}\bar{\omega}$, we choose λ with $Q'\lambda = \bar{\omega}$; then $Q(A - D\lambda) = z$ by (i) of Section 4.2, and $\langle A - D\lambda, h(A - D\lambda)\rangle = \langle A, hA\rangle$ because $hD = 0$. So the minimum may be taken over $\{QA = z\}$, on which $\langle A, (h + DRD^* + aQ^*Q)A\rangle - a|z|^2 = \langle A, hA\rangle + \|RD^*A\|^2 \geq \langle A, hA\rangle$, R being an orthogonal projection. Conversely, let $QA = z$. Since RD^*A lies in the range

$\Delta^\eta_U N(Q')$ of R [5, (3.21), p. 394], there is $\lambda \in N(Q')$ with $D^*D\lambda = RD^*A$, and $A' := A - D\lambda$ satisfies $QA' = z - \bar{D}Q'\lambda = z$, $\langle A', hA' \rangle = \langle A, hA \rangle$, and $RD^*A' = RD^*A - R^2D^*A = 0$; for it the two sides of the inequality agree. The minimum on the right of (4.54) is attained, because the form is bounded below by $\langle A, hA \rangle \geq 0$ on an affine space. *q.e.d.*

The expansion of Δ_k in the background needs the derivatives of the dressed average $Q(U)$ and of $Q'(U)$ at the trivial background. We use the steps $j = 1, \ldots, k$ of the composite, the step j mapping fields of the level $j-1$, of the spacing $s_j := L^{j-1-k}$, to fields of the level j, of the spacing $\varepsilon_j := Ls_j$, and on a lattice of spacing s we put $\|X\|_p^p := \sum_b s^4 |X(b)|^p$ for $p \in \{1, 2\}$, summed over the bonds or sites of one cell of $T^{(k)}$ for Bloch waves and over the whole torus otherwise. A field of a level is normalised by the factor s^{-1} in its logarithm, $U_b = e^{isa(b)}$.

Lemma 4.20 (The Dressed Average near the Trivial Background)**.** *Let* $U(t) = \exp(i\eta(tA_1 + t^2A_2 + O(t^3)))$ *bondwise, with fine* 1*-forms* A_1 *and* A_2*, and let* $Q^{[n]}$ *and* $Q'^{[n]}$ *be the coefficients of* t^n *in* $Q(U(t))$ *and* $Q'(U(t))$*. There is a constant* C*, depending only on* d*,* L*, and* G*, such that for* $p \in \{1, 2\}$*, for fine* 1*-forms* A*, and for fine* 0*-forms* λ*,*

$$\|Q^{[1]}A\|_p \leq C\|A_1\|_\infty \|A\|_p\,, \qquad \|Q^{[2]}A\|_p \leq C(\|A_2\|_\infty + \|A_1\|_\infty^2)\|A\|_p\,, \tag{4.55}$$

and the same bounds hold for $Q'^{[1]}\lambda$ *and* $Q'^{[2]}\lambda$*. Moreover, at* $U = 1$ *the composite* $\tilde{Q}_{(i\to j)}$ *of the linearised averages [9, (0.4)] of the steps* $i, \ldots, j$ *satisfies* $\|\tilde{Q}_{(i\to j)}X\|_\infty \leq (1 + d)\|X\|_\infty$ *for normalised fields.*

Proof. One step. The dressed one-step average at a bond c [5, p. 418, l. 6, (3.113)], with the contour variables [9, (0.11), p. 253] in place of the contours, is built from the bond variables of the level $j-1$ of the background and of the fluctuation on the set $\mathcal{R}_c$ of the bonds of $B(c_-) \cup B(c_+)$ and of the lines $[x, x']$, by products, inverses, conjugations, the group average [9, (0.10)], and averages $\exp(i \sum w \log)$. These are analytic near 1 on the complexified group [9, p. 252, l. 37; (0.8), p. 253], and no length enters them. So there is one analytic map $\Psi\colon (\mathfrak{g}^c)^n \times (\mathfrak{g}^c)^n \to \mathfrak{g}^c$ on a complex polydisc of radius $r_0(d, L, G)$, with $n = |\mathcal{R}_c|$ depending on d and L, such that the logarithm of the dressed average is $\Psi(s_ja, s_jB)$ for every j, k, and c, where a and B are the normalised background and fluctuation on $\mathcal{R}_c$; the normalised step is $\varepsilon_j^{-1}\Psi(s_ja, s_jB)$, with the linear part $L^{-1}\partial_B\Psi(s_ja, 0)$. Moreover $\Psi(a, 0) = 0$: at the trivial fluctuation the dressing factors (3.113) are 1, and the formula of l. 6 gives $\bar{U}_c(\bar{U}_c)^{-1}\bar{R}_c 1 = 1$, the transport $\bar{R}_c = R(\bar{U}_c)$ acting on the factor to its right, as the computation on l. 13 of the same page shows. The raw one-step average [9, (0.4)] is likewise $\varepsilon_j^{-1}\tilde{\Psi}(s_ja)$ with an analytic $\tilde{\Psi}$ of the same kind, whose normalised linear part at $U = 1$ is the linearised raw average $\tilde{Q}^{(j)}$.

Cauchy's estimates and norms. The Taylor coefficients of order N of Ψ and $\tilde{\Psi}$ are at most $K_N := 2\max(\sup|\Psi|, \sup|\tilde{\Psi}|)r_0^{-N}$, the suprema over the polydisc. Each bond lies in at most $m(d, L)$ of the sets $\mathcal{R}_c$, so that a linear map from the level $j-1$ to the level j whose kernel has entries at most K on $\mathcal{R}_c$ has the ℓ^1-norm at most L^dKm and the ℓ^2-norm at most $L^{d/2}K(nm)^{1/2}$, by $\sum_c (Ls)^d |\sum_{b\in\mathcal{R}_c} K|X(b)||^p \leq L^dK^pn^{p-1}m\sum_b s^d|X(b)|^p$.

The trivial background. At $U = 1$ the linear part of each step is the straight average $Q^{(j)}$, by (iii) of Section 4.2, and it contracts $\|\cdot\|_1$, $\|\cdot\|_2$, and $\|\cdot\|_\infty$. Indeed $(Q^{(j)}X)(c)$ is the mean of the L^{d+1} values of X on the lines $[x, x']$, $x \in B(c_-)$, and each bond of the level $j-1$ lies on exactly L of the lines which occur, for all c together, so that Jensen's inequality gives $\|Q^{(j)}X\|_p^p \leq \sum_b \varepsilon_j^d L \cdot L^{-d-1}|X(b)|^p = \|X\|_p^p$. The same holds for the block average $Q'^{(j)}$ at $U = 1$. Next, $\tilde{Q}^{(i)}X = Q^{(i)}X - \bar{D}\omega^{(i)}(X)$ at $U = 1$, by (iii) of Section 4.2, where

$\omega^{(i)}(X)(y)$ is the mean over $x \in B(y)$ of the contour means of X over $\mathbf{G}(y,x)$ [9, (0.8)]; the points of $B(y)$ are $x = y + \sum_\mu s_i n_\mu e_\mu$ with $|n_\mu| \le (L-1)/2$, since L is odd [9, p. 251] and $B(y)$ is the cube of side ε_i centred at y [9, (0.3), p. 252], so the contours have at most $d(L-1)/2$ bonds and $|\omega^{(i)}(X)| \le d(L-1)s_i\|X\|_\infty/2$. By $\tilde{Q}^{(i)}\bar{D} = \bar{D}P$, the composite satisfies $\tilde{Q}_{(1\to i)}X = Q_{(1\to i)}X - \bar{D}\omega_i$ with $\omega_i := P\omega_{i-1} + \omega^{(i)}(Q_{(1\to i-1)}X)$, so that $\sup|\omega_{j-1}| \le (d/2)(L-1)\sum_{i<j} s_i\|X\|_\infty \le (d/2)\varepsilon_{j-1}\|X\|_\infty$, because $\sum_{i<j} s_i \le \varepsilon_{j-1}/(L-1)$. The normalised gradient of ω_{j-1} is therefore at most $d\|X\|_\infty$, and $\|\tilde{Q}_{(1\to j-1)}X\|_\infty \le (1+d)\|X\|_\infty$; the same argument applies to every composite $\tilde{Q}_{(i\to j)}$.

The backgrounds of the steps. The background of the step j is the raw average of the order $j-1$ of $U(t)$ [5, (3.15), p. 393], [4, (91), p. 31]. Let $a^{(j)}(t)$ be its normalised logarithm, with the coefficients $a_1^{(j)}$ and $a_2^{(j)}$ of t and t^2, so that $a^{(0)} = tA_1 + t^2A_2 + O(t^3)$. Then $a_1^{(j)} = \tilde{Q}_{(1\to j)}A_1$ and $a_2^{(j)} = \tilde{Q}^{(j)}a_2^{(j-1)} + (s_j/L)\tilde{\Psi}_2(a_1^{(j-1)})$, with $|\tilde{\Psi}_2(x)| \le K_2 n^2\|x\|_\infty^2$, the part of second order of $\tilde{\Psi}$. Iterating, and using the bound on the composites and $\sum_i s_i \le 1/(L-1)$, we obtain $\|a_1^{(j)}\|_\infty \le (1+d)\|A_1\|_\infty$ and $\|a_2^{(j)}\|_\infty \le C(\|A_2\|_\infty + \|A_1\|_\infty^2)$.

The coefficients of Q. The kernel of the linear part $L^{-1}\partial_B\Psi(s_j a^{(j-1)}(t), 0)$ of the step j has a coefficient of t with entries at most $L^{-1}K_2 n s_j\|a_1^{(j-1)}\|_\infty$ and a coefficient of t^2 with entries at most $L^{-1}(K_2 n s_j\|a_2^{(j-1)}\|_\infty + K_3 n^2 s_j^2\|a_1^{(j-1)}\|_\infty^2)$, by Cauchy's estimates. The coefficient $Q^{[1]}$ is the sum over j of the composites in which the step j is replaced by its coefficient of t and all others are straight averages, and $Q^{[2]}$ the sum of the composites with the coefficient of t^2 of one step or the coefficients of t of two steps. The straight averages are contractions, so the bounds on the kernels, the norms of linear maps, and $\sum_j s_j \le 1/(L-1)$ give (4.55).

The coefficients of Q'. The step j of Q' is $\lambda \mapsto \sum_{x\in B(y)} L^{-d}R(e^{i\theta_j(y,x)})\lambda(x)$ [5, (3.19), p. 393], with $\theta_j(y,x)$ the logarithm of the contour variable of the background of the step, an analytic function of the logarithms on the contours of $\mathbf{G}(y,x)$ whose linear part is the contour mean [9, (0.8)]. Its coefficients satisfy $|\theta_{j,1}| \le s_j d(1+d)(L-1)\|A_1\|_\infty/2$ and $|\theta_{j,2}| \le s_j d(L-1)\|a_2^{(j-1)}\|_\infty/2 + Cs_j^2\|a_1^{(j-1)}\|_\infty^2$, and $R(e^{i\theta}) = \exp(i\,\mathrm{ad}\,\theta)$ has the coefficients $i\,\mathrm{ad}\,\theta_{j,1}$ and $i\,\mathrm{ad}\,\theta_{j,2} - \frac{1}{2}(\mathrm{ad}\,\theta_{j,1})^2$, of the norms at most $c_{\mathfrak{g}}|\theta_{j,1}|$ and $c_{\mathfrak{g}}|\theta_{j,2}| + c_{\mathfrak{g}}^2|\theta_{j,1}|^2/2$. Summing over one or two steps as for Q gives the bounds for $Q'^{[1]}$ and $Q'^{[2]}$. All the counts are local, so the bounds hold over one cell and over the whole torus alike. *q.e.d.*

The gauge parameter of the minimiser at the trivial background is bounded, and not only square summable, and for this reason the factor φ of (4.16) does no harm when it acts on the field Hz carried by a coarse plane wave. We use the notation of the proof of Lemma 4.12.

Lemma 4.21 (The Gauge Parameter of the Minimiser). *Let $z = \hat{z}e^{ip\cdot y}$ be a coarse plane wave with $p \ne 0$, $|p_\mu| \le \pi$, which is transverse, $\sum_\mu \overline{\partial_\mu^1(p)}\hat{z}_\mu = 0$, and let $H = H(1)$. Then $\varphi := (\partial^*\partial)^{-1}\partial^* Hz$ satisfies $\|\varphi\|_\infty + \|\partial\varphi\|_\infty \le C|\hat{z}|$, with C depending only on d and L.*

Proof. The minimiser is Bałaban's H_k. At $U = 1$, $J = 0$, $Q = Q_k$ with centred blocks by (iii) of Section 4.2, and $\Delta_\pi = \mathbf{H}(1)$ by [5, (3.119), p. 419], since $\mathbf{H}(1)\partial = 0$; moreover $\langle A, \mathbf{H}(1)A\rangle = \|\partial A\|^2$, the curl. The form $G^{-1} = \mathbf{H}(1) + \partial R\partial^* + aQ_k^*Q_k$ is positive definite, G being bounded [5, Theorem 3.3, p. 399], so Hz is the unique minimiser of $\langle A, G^{-1}A\rangle$ on $\{Q_kA = z\}$ [5, (3.122), p. 420]. On that set $\langle A, G^{-1}A\rangle = \|\partial A\|^2 + \|R\partial^*A\|^2 + a|z|^2 \ge \|\partial A\|^2 + a|z|^2$. The field H_kz minimises $\|\partial A\|^2$ on $\{Q_kA = z\}$, by the paragraph on the first-order background in the proof of Proposition 4.13, and $R\partial^*H_kz = 0$ [2, p. 29], so H_kz attains the lower bound and $Hz = H_kz$, up to the fixed translation between the blocks of [2] and the centred blocks, which multiplies the Fourier coefficients by phases of modulus one. Hence $\varphi = (\partial^*\partial)^{-1}\partial^*H_kz$.

The symbol. Put $S(p) := \sum_\nu |\partial^1_\nu(p)|^2/(\Delta_0(p)^2\varphi_\nu(p))$ and $\beta(p) := \sum_\lambda \overline{\partial^1_\lambda(p)}(\Delta_0(p)\varphi_\lambda(p))^{-1}\hat{z}_\lambda$. Applying ∂^*, with the symbol $\sum_\mu \overline{\partial_\mu(q)}$, to the first form of [2, (1.63), p. 28], and using $\overline{\partial_\mu(q)}\,\overline{v_\mu(q)} = \overline{\partial^1_\mu(p)}$ and $\sum_\mu |\partial_\mu(q)|^2 = \Delta(q)$, the term in $\hat{z}_\mu$ gives $\bar{u}(q)\Delta_0(p)\beta(p)/\Delta(q)$, the term in $\partial^1_\mu(p)$ gives $-\bar{u}(q)\Delta_0(p)\beta(p)/\Delta(q)$, because $\sum_\mu |\partial^1_\mu|^2/(\Delta_0\varphi_\mu) = \Delta_0 S$, and the two cancel, so that

$$(\partial^* H_k z)^\wedge(q) = \frac{\overline{u(q)}\Delta_0(p)\beta(p)}{N(p)S(p)\Delta(q)}\,, \qquad \hat{\varphi}(q) = \frac{\overline{u(q)}\Delta_0(p)\beta(p)}{N(p)S(p)\Delta(q)^2}\,. \tag{4.56}$$

With $N \geq c_N$, $S \geq 1/c_1$ by (4.30), $\Delta_0(p)/\Delta(p)^2 \leq (\pi^4/16)|p|^{-2}$, and, for $l \neq 0$, $|q| \geq \pi$ and (4.29), this gives $\sum_l (1+|\partial(q)|)|\hat{\varphi}(q)| \leq C|\beta(p)||p|^{-2}$, and $\|\varphi\|_\infty + \|\partial\varphi\|_\infty$ is at most this sum.

The size of β. We claim $|\Delta_0(p)\varphi_\lambda(p) - 1| \leq C|p|^2$. With $\mathrm{s}(x) := \sin x/x$ we have $|\partial^1_\mu(p)|^2 = p_\mu^2 \mathrm{s}(p_\mu/2)^2$ and $|\partial_\mu(p)|^2 = p_\mu^2 \mathrm{s}(\eta p_\mu/2)^2$, and since $\mathrm{s} \in [2/\pi, 1]$ and $|\mathrm{s}(x)^2 - 1| \leq x^2/3$ for $|x| \leq \pi/2$, the term $l' = 0$ of $\Delta_0\varphi_\lambda$, which is $|u(p)|^2|v_\lambda(p)|^2\Delta_0(p)/\Delta(p)$, is a product of factors $1 + O(|p|^2)$. For $l' \neq 0$ some $l'_\nu \neq 0$, so that $|v_\nu(p + l')|^2 \leq C|p|^2(1 + |l'_\nu|)^{-2}$ and $\Delta_0(p)/\Delta(p + l') \leq \pi^2|p|^2/|l'|^2$, and with (4.29) for the other factors these terms add up to at most $C|p|^4$. For transverse z, $\beta = \sum_\lambda \overline{\partial^1_\lambda}((\Delta_0\varphi_\lambda)^{-1} - 1)\hat{z}_\lambda$, so that $|\beta| \leq C|p|^3|\hat{z}|$ by the claim and (4.30). Hence $\|\varphi\|_\infty + \|\partial\varphi\|_\infty \leq C|p||\hat{z}| \leq C\pi\sqrt{d}|\hat{z}|$. *q.e.d.*

We now fix the background. Let $p_0 \not\equiv 0$ and $B = be^{ip_0\cdot y} + \text{c.c.}$, a real plane wave on $T^{(k)}$. By [7, p. 281, ll. 2–3] there is exactly one gauge transformation u with $\overline{R_0 u}^k = 1$ such that $U_k(e^{iB})^{u^{-1}}$ satisfies the Landau gauge conditions [7, (19)–(21), p. 281], and by [7, Proposition 9, p. 309] this configuration is $\exp(i\eta\mathscr{H}(B))$, with $\mathscr{H}$ analytic for small complex B and satisfying (19)–(21) with $\varepsilon_2 = B_5\varepsilon_1$ [7, (173), p. 305]. At $U_k(1) = 1$, with every Ω_j the whole torus, these conditions are invariant under the translations of $T^{(k)}$, and so is the axial gauge of $U_k(e^{iB})$, whose orbit is unique [7, p. 279]; by the uniqueness of u, the transformation u and the configuration $\mathscr{H}(B)$ are covariant under the translations of $T^{(k)}$, in particular invariant under those which leave B invariant. For real τ near 0 we put $A(\tau) := \mathscr{H}(\tau B) = \tau A_1 + \tau^2 A_2 + O(\tau^3)$ and $U(\tau) := \exp(i\eta A(\tau))$, and for an operator $X(U)$ we write $X^{[n]}$ for the coefficient of τ^n in $X(U(\tau))$, so that $X^{[0]} = X(1)$; we put $\beta_n := |b|^n$, and π denotes the orthogonal projection of the matrices onto the Lie algebra for the form tr.

Lemma 4.22 (Taylor Coefficients at the Trivial Background). *There is a constant C, depending on d, L, G, and the constants of [5] and [7], and not on k, such that for $n = 1, 2$:*

- *(i) $\|A_n\|_\infty, \|\nabla A_n\|_\infty, \|d^*_\eta d_\eta A_n\|_\infty \leq C\beta_n$, $J^{[0]} = 0$, and $\|J^{[n]}\|_\infty \leq C\beta_n$;*
- *(ii) $\|(S^{-1})^{[n]}\| \leq C\beta_n$ for $S := QGQ^*$, and $|\langle A', M^{[n]}A\rangle| \leq C\beta_n\|A'\|_E\|A\|_E$ for $M := G^{-1}$, where $\|A\|_E := \langle A, G(1)^{-1}A\rangle^{1/2}$;*
- *(iii) for a coarse plane wave $z = \hat{z}e^{ip\cdot y}$, $H^{[1]}z$ is the sum of two Bloch waves of the quasi-momenta $p \pm p_0$, whose sum has $\|\cdot\|_2 \leq C|b||\hat{z}|$, and $\|H(1)z\|_\infty \leq C|\hat{z}|$; if $2p_0 \not\equiv 0 \bmod 2\pi$, as at the momentum $p_0 = (\pi/2, 0, 0, 0)$ of Proposition S.6, the two quasi-momenta are different, the two waves are orthogonal, and each of them satisfies the same bound. For $2p_0 \equiv 0$ the two waves have the same quasi-momentum and only their sum is bounded.*

Proof. (i). For complex τ with $|\tau| \leq \rho := r/(4|b|)$ the field τB lies in $\mathfrak{B}_r$, since $\sup|B| \leq 2|b|$ and $\sup|\bar{\partial}B| \leq 4|b|$, and for r small $V' = e^{i\tau B}$ and $V_0 = 1$ satisfy the assumptions of [7, Sect. G]: V_0 satisfies [7, (7), p. 278] trivially, and $|V' - 1| \leq re^r < C_1\varepsilon_1$, which is

[7, (172), p. 305]. Proposition 9 of [7] states that $\mathscr{H}$ satisfies (19) there, which at $j = k$, $U_0 = 1$, and $L^k\eta = 1$ reads $|\mathscr{H}|, |\nabla\mathscr{H}|, |d_\eta^* d_\eta \mathscr{H}| < B_5\varepsilon_1$ on the torus; we use this as printed. Cauchy's estimate on $|\tau| = \rho$ gives the first three bounds. For the current, $J = D_U^*\pi\Phi$ with $\Phi(p) := \eta^{-2}\operatorname{Im} U(\partial p)$: indeed $J = D_U^*\Phi$ [5, (3.11), p. 392], and for Lie-algebra-valued A, $\langle A, D_U^*\Phi\rangle = \langle D_U A, \Phi\rangle = \langle D_U A, \pi\Phi\rangle$. Write $U(\tau)(\partial p) = \exp(i\eta^2 F_p)$; the Baker–Campbell–Hausdorff formula gives $F_p^{[1]} = (d_\eta A_1)(p)$ and $F_p^{[2]} = (d_\eta A_2)(p) + F_p^{\mathrm{com}}$, $F_p^{\mathrm{com}} := \frac{i}{2}\sum_{j<l}[x_j, x_l]$, with $x_1, \ldots, x_4$ the signed values of A_1 on the bonds of ∂p in order, and $\eta^{-2}\operatorname{Im} U(\partial p) = \eta^{-2}\sin(\eta^2 F_p)$ has the coefficients $F_p^{[1]}$ and $F_p^{[2]}$, which lie in the Lie algebra. So $J^{[0]} = 0$, $J^{[1]} = d_\eta^* d_\eta A_1$, and $J^{[2]} = d_\eta^* d_\eta A_2 + d_\eta^* F^{\mathrm{com}} + (D^*)^{[1]} F^{[1]}$, with d_η^* the adjoint [5, (3.9), p. 392] at $U = 1$. By (3.8)–(3.9) of [5], $(d_\eta^* F^{\mathrm{com}})(b)$ is a sum of at most d terms $\eta^{-1}(F_{p'}^{\mathrm{com}} - F_p^{\mathrm{com}})$, p' the translate of p by $-\eta e_\nu$; the values of A_1 on $\partial p'$ and ∂p differ by at most $\eta\|\nabla A_1\|_\infty$, so each term is at most $6c_{\mathfrak{g}}\|A_1\|_\infty\|\nabla A_1\|_\infty \leq C\beta_2$. Finally $(D^*)^{[1]}$ replaces one transport $R(U(x, x - \eta e_\nu))$ in (3.8)–(3.9) by its coefficient $i\eta \operatorname{ad} A_1$, so that $|(D^*)^{[1]}F^{[1]}| \leq dc_{\mathfrak{g}}\|A_1\|_\infty\|F^{[1]}\|_\infty \leq C\beta_2$, as $|F^{[1]}| = |d_\eta A_1| \leq 2\|\nabla A_1\|_\infty$.

Energy norms. On the finite torus all operators below are real-analytic in τ near 0, since the inverses in them exist at $\tau = 0$, and $X^{[n]}$ is homogeneous of degree n in $(b, \bar{b})$. With $G = G(1)$ and $G' = G'(1)$ we put $\|\lambda\|_{E'} := \langle\lambda, G'^{-1}\lambda\rangle^{1/2}$ and $\|f\|_{E^*} := \sup_A|\langle f, A\rangle|/\|A\|_E = \|G^{1/2}f\|$. Then $\|A\| \leq \|G\|^{1/2}\|A\|_E$ and $a^{1/2}\|QA\| \leq \|A\|_E$, because $G^{-1} = \mathbf{H}(1) + \partial R\partial^* + aQ^*Q$ with $\mathbf{H}(1), \partial R\partial^* \geq 0$; the block-to-block bound [5, (3.46), p. 398] for G [5, Theorem 3.3, p. 399] at $U = 1$, summed over the cubes by the Schur test, gives $\|\nabla G\nabla^*\| \leq C$, that is $\nabla^*\nabla \leq CG^{-1}$, so that $\|\nabla A\| \leq C\|A\|_E$ and $\|\partial^* A\| \leq \sqrt{d}\|\nabla A\|$. Likewise $\|\lambda\| \leq \|G'\|^{1/2}\|\lambda\|_{E'}$ and $\|\partial\lambda\| \leq \|\lambda\|_{E'}$, since $G'^{-1} = \partial^*\partial + Q'^* aQ'$ at $U = 1$ [5, (3.23)–(3.24), p. 394]. Moreover $\|Gf\|_E = \|f\|_{E^*}$, $\|Q^* w\|_{E^*} \leq a^{-1/2}\|w\|$, and $\|X^* w\|_{E^*} \leq \|G\|^{1/2}\|X\|\|w\|$ for bounded X, and the same holds for E' and G'.

Factors at $U = 1$. $\|S\| \leq a^{-1}$, since $aQ^*Q \leq G^{-1}$, and $\|S^{-1}\| \leq C$ by the exponential decay of the kernel of S^{-1} [5, (3.132), p. 422] and the Schur test; $S' := Q'G'^2Q'^*$ has $\|S'^{-1}\| \leq C$ by [5, Theorem 3.2, (3.48), p. 398] with $L^k\eta = 1$. Further, $H = GQ^*S^{-1}$ has $\|Hw\|_E^2 = \langle w, S^{-1}w\rangle \leq \|S^{-1}\|\|w\|^2$, and $C_G = G^{1/2}(1 - P)G^{1/2}$ with the orthogonal projection $P := G^{1/2}Q^*S^{-1}QG^{1/2}$, so that $C_G \geq 0$ and $\|C_G f\|_E \leq \|f\|_{E^*}$. The kernel of H is bounded by $Ce^{-\delta_1 d(y,y')/2}$ [5, (3.133), p. 422], which gives $\|Hz\|_\infty \leq C|\hat{z}|$.

Local factors. $R(U_b(\tau)) = \exp(i\eta \operatorname{ad} A(\tau)(b))$ has the coefficients $i\eta \operatorname{ad} A_1(b)$ and $i\eta \operatorname{ad} A_2(b) - \frac{\eta^2}{2}(\operatorname{ad} A_1(b))^2$. So the coefficients $D^{[n]}$, $n = 1, 2$, of $D_U\lambda(b) = \eta^{-1}(R(U_b)\lambda(b_+) - \lambda(b_-))$, of D_U^*, and of the plaquette derivative [5, (3.4), p. 391] contain no difference quotient, have range 1, and have norms at most $C\beta_n$. By Lemma 4.20 and (i), $\|Q^{[n]}\|, \|Q'^{[n]}\| \leq C\beta_n$.

The Hessian. By [5, (3.10), p. 392], $\mathbf{H} = D^*D + \Delta'$ with D the plaquette derivative, and Δ' has two terms: $\operatorname{tr}((\eta DA)(p))^2\eta^{-2}(\operatorname{Re} U(\partial p) - 1)$, and the commutators of the transported bond values of (3.6) against $\eta^{-2}\operatorname{Im} U(\partial p)$. Here $\operatorname{Re} U(\partial p) - 1 = \cos(\eta^2 F_p) - 1$ has the coefficients 0 and $-\eta^4(F_p^{[1]})^2/2$, and the transports in the transported bond values have coefficients at most $C\eta\beta_n$. Hence $|\langle A', \mathbf{H}^{[n]}A\rangle| \leq \sum_{i+j=n}\|D^{[i]}A'\|\|D^{[j]}A\| + C\beta_n(\|A'\|\|A\| + \|d_\eta A'\|\|d_\eta A\|) \leq C\beta_n\|A'\|_E\|A\|_E$, with $D^{[0]} = d_\eta$ and $\|d_\eta A\| \leq 2\|\nabla A\|$.

G' and R. $M' := G'(U)^{-1} = D^*D + Q'^* aQ'$ has $|\langle\lambda', M'^{[n]}\lambda\rangle| \leq C\beta_n\|\lambda'\|_{E'}\|\lambda\|_{E'}$, by the bounds on $D^{[i]}$ and $Q'^{[i]}$ and by $\|\partial\lambda\| \leq \|\lambda\|_{E'}$, $\|\lambda\| \leq \|G'\|^{1/2}\|\lambda\|_{E'}$. From $G'^{[1]} = -G'M'^{[1]}G'$ and $G'^{[2]} = G'M'^{[1]}G'M'^{[1]}G' - G'M'^{[2]}G'$ we get $\|G'^{[n]}f\|_{E'} \leq C\beta_n\|G'^{1/2}f\|$, hence $\|G'^{[n]}\|, \|\partial G'^{[n]}\| \leq C\beta_n$ for $n = 0, 1, 2$. By [5, (3.25), p. 394], $R = 1 - G'Q'^*S'^{-1}Q'G'$ at every U, and expanding the products, with $(S'^{-1})^{[1]} = -S'^{-1}S'^{[1]}S'^{-1}$ and $(S'^{-1})^{[2]} = S'^{-1}S'^{[1]}S'^{-1}S'^{[1]}S'^{-1} - S'^{-1}S'^{[2]}S'^{-1}$, gives $\|S'^{[n]}\|, \|R^{[n]}\| \leq C\beta_n$. So RD^* and the map

$\Lambda\colon A \mapsto \lambda_A = G'RD^*A$ satisfy $\|(RD^*)^{[n]}A\|, \|\Lambda^{[n]}A\|, \|(D\Lambda)^{[n]}A\| \leq C\beta_n\|A\|_E$ for $n = 0, 1, 2$.

(ii). By [5, (3.120), (3.122), pp. 419–420], $M = \mathbf{H} - \Delta'_\pi + (RD^*)^*(RD^*) + aQ^*Q$ at every U, R being an orthogonal projection. The form Δ'_π is linear in J and built from Λ, $D\Lambda$, and the transports R_b; since $J^{[0]} = 0$, $(\Delta'_\pi)^{[1]}$ is (3.120) at $U = 1$ with $J^{[1]}$, and $(\Delta'_\pi)^{[2]}$ is the same with $J^{[2]}$ plus the coefficient of the dependence on U at $J^{[1]}$. With $\sum_b \eta^4|J(b)||X(b)||Y(b')| \leq C\|J\|_\infty\|X\|\|Y\|$ for neighbouring b, b', this gives $|\langle A', (\Delta'_\pi)^{[n]}A\rangle| \leq C\beta_n\|A'\|_E\|A\|_E$, and with the Hessian, RD^*, and $Q^{[n]}$ the bound on $M^{[n]}$. Then $G^{[1]} = -GM^{[1]}G$ and $G^{[2]} = GM^{[1]}GM^{[1]}G - GM^{[2]}G$ satisfy $\|G^{[n]}f\|_E \leq C\beta_n\|f\|_{E^*}$; $S^{[n]}$ is the sum of the $Q^{[i]}G^{[j]}Q^{[l]*}$ with $i+j+l = n$, and $|\langle w', Q^{[i]}G^{[j]}Q^{[l]*}w\rangle| \leq \|Q^{[i]*}w'\|_{E^*}\|G^{[j]}Q^{[l]*}w\|_E \leq C\beta_n\|w'\|\|w\|$; and the expansion of S^{-1} as for S'^{-1} gives $\|(S^{-1})^{[n]}\| \leq C\beta_n$.

(iii). Differentiating $H = GQ^*S^{-1}$ gives $H^{[1]} = -C_G M^{[1]}H + C_G Q^{[1]*}S^{-1} - HQ^{[1]}H$, so that $\|H^{[1]}w\|_E \leq C\beta_1\|Hw\|_E + \|Q^{[1]*}S^{-1}w\|_{E^*} + \|S^{-1}\|^{1/2}\|Q^{[1]}Hw\| \leq C\beta_1\|w\|$, and $\|H^{[1]}w\| \leq \|G\|^{1/2}\|H^{[1]}w\|_E$. The coefficient $H^{[1]}$ is linear in A_1, which is the sum of its Bloch waves of the quasi-momenta $\pm p_0$ by the covariance of $\mathscr{H}$, so that $H^{[1]}z$ is the sum of two Bloch waves of the quasi-momenta $p \pm p_0$. These are orthogonal when $2p_0 \not\equiv 0 \bmod 2\pi$, which holds at the momentum $p_0 = (\pi/2, 0, 0, 0)$ of Proposition S.6, the only one at which the lemma is used, and then each of them has norm at most that of the sum; for Bloch waves the norm over the torus is $|T|^{1/2}$ times the norm over one cell. *q.e.d.*

Proposition 4.23 (The Functional of the Top Level). *Let* $p_0 \in [-\pi, \pi]^4 \setminus \{0\}$, *and let* $\Pi_{F_k}(p_0)$ *be the polarisation of* $B \mapsto F_k(U_k(e^{iB}))$ *at* p_0 *on* $T^{(k)}$, *the limit of the coset sums of Section 4.2. Then* $\sup_k\|\Pi_{F_k}(p_0)\| < \infty$.

Proof. The gauge. Let g be a fine gauge transformation which commutes with the translations of $T^{(k)}$ fixing B, and $\hat{g}$ its restriction to the sites of $T^{(k)}$. The coordinates $X \mapsto Ue^{iX}$ at U and at U^g are related by $X \mapsto \operatorname{Ad}(g)X$, bondwise $\operatorname{Ad}(g(b_+))$, so that the covariance of the average [9, (0.6)] gives $\tilde{Q}(U^g)\operatorname{Ad}(g) = \operatorname{Ad}(\hat{g})\tilde{Q}(U)$, the gauge invariance of the action gives $h(U^g) = \operatorname{Ad}(g)h(U)\operatorname{Ad}(g)^{-1}$, and $\bar{D}_{V^{\hat{g}}} = \operatorname{Ad}(\hat{g})\bar{D}_V\operatorname{Ad}(\hat{g})^{-1}$. By (4.4), Δ_k and $\bar{D}^{\mathrm{T}}\bar{D}$ are therefore conjugated by $\operatorname{Ad}(\hat{g})$, which is orthogonal for real B and maps the fields of each coset onto themselves, because g commutes with the translations which fix B; so F_k does not change. For the background τB we apply this with $g = u^{-1}$, the transformation of [7, p. 281] for τB: the background becomes $U(\tau)$, and $e^{i\tau B}$ becomes its transform by $\hat{g}$, which is the raw average of $U(\tau)$ by [9, (0.6)]. Below, U and V denote these configurations, except in the term $\log\det(\bar{D}_V^{\mathrm{T}}\bar{D}_V)$, which the transformation does not change and which we take at $V = e^{i\tau B}$; we write B for τB where only the order in τ matters.

(a) A fixed transverse space. Let $T_0 := (\operatorname{ran}\bar{D}_0)^\perp$, with $\bar{D}_0$ the coarse gauge action at $V = 1$ and the orthogonal projection P_T, and let Π_V be the orthogonal projection onto $(\operatorname{ran}\bar{D}_V)^\perp$. We work on the fields of one coset $\mathcal{O} \neq \mathbb{Z}p_0$, on which $\bar{D}_V$ is injective for small real B, so that $(\operatorname{ran}\bar{D}_V)^\perp$ has constant dimension, $\Pi_V \to P_T$ as $B \to 0$, and Π_V maps T_0 isomorphically onto $(\operatorname{ran}\bar{D}_V)^\perp$. By Lemma 4.4, $\Delta_k = \Pi_V\Delta_k\Pi_V$, so that $\det_{T_0}(P_T\Delta_kP_T) = \det_\perp(\Delta_k)\det_{T_0}(P_T\Pi_VP_T)$, and

$$F_k = \tfrac{1}{2}\log\det{}_{T_0}(P_T\Delta_kP_T) - \tfrac{1}{2}\log\det{}_{T_0}(P_T\Pi_VP_T) - \tfrac{1}{2}\log\det(\bar{D}_V^{\mathrm{T}}\bar{D}_V)\,. \tag{4.57}$$

(b) The expansion of the form. At U near 1 the hypotheses of Lemma 4.19 hold on the coset, since they hold at $U = 1$ and are open conditions on finite-dimensional spaces. Since $\mathbf{H} = \Delta_\pi + \Delta'_\pi$ and $h = \mathbf{H} - K$, we have $h + DRD^* + aQ^*Q = G^{-1} - K'$ with $K' := K - \Delta'_\pi$, which is linear in J and symmetric. For real t let $m_t(z) := \min\{\langle A, (G^{-1} - tK')A\rangle \mid QA = z\} - a|z|^2$, so that $\Delta_k = m_1$ by (4.54). The minimum is $\langle z, (QG_tQ^*)^{-1}z\rangle$, and for complex t we define

m_t by this expression, which is analytic in t; no minimum of complex numbers is meant. Here $G_t = (G^{-1} - tK')^{-1} = G + tGK'G + t^2GK'GK'G + \dots$, and expanding the inverse gives the coefficient $(QGQ^*)^{-1} - a$ of t^0 and $-H^*K'(C_GK')^{n-1}H$ of t^n, $n \geq 1$. On the coset $K' = O(B)$, since $J = O(B)$, so for small B the map $t \mapsto m_t$ is analytic on $|t| \leq 2$ and its coefficient of t^n is of order n in B. Moreover $H^*\Delta'_\pi H = 0$ at every U: polarising [5, (3.120)], every term contains λ_{Hz} or $\lambda_{Hz'}$, which vanish because $RD^*H = 0$ [5, (3.124), p. 420]. So, up to terms of order three in B,

$$\Delta_k = [(QGQ^*)^{-1} - a] - H^*KH - H^*K'C_GK'H \ . \tag{4.58}$$

(c) The vertices. On T_0 write $P_T\Delta_kP_T = \Delta^0 + V^1[B] + V^2[B,B] + \dots$. Here Δ^0 is the operator Δ_k of [2, (1.65), p. 29] on transverse fields, because at $U = 1$ the gauge moves of the proof of Lemma 4.19 reduce the minimum (4.4) to that of $\|\partial A\|^2$ on $\{Q_kA = z\}$, which H_kz attains, so that $\gamma_0|\hat{p}|^2 \leq \Delta^0(p) \leq \gamma_1|\hat{p}|^2$ by [2, (1.66)–(1.67), p. 29], the curl of a transverse plane wave having the squared symbol $|\hat{p}|^2|\hat{z}|^2$ by Lagrange's identity. Let $z = \hat{z}e^{ip\cdot y}$ and z' be transverse plane waves. We claim

$$|\langle z', V^1z\rangle| \leq C|b||\hat{z}||\hat{z}'| \ , \qquad |\langle z', V^2[b,\bar{b}]z\rangle| \leq C(1 + |(p+p_0)^\wedge|^{-2} + |(p-p_0)^\wedge|^{-2})|b|^2|\hat{z}||\hat{z}'| \ , \tag{4.59}$$

with z' of the momentum $p + p_0$ in the first and p in the second, and we bound the terms of (4.58) one by one. Throughout we use $\|Hz\|_\infty \leq C|\hat{z}|$ and $\|\varphi_{Hz}\|_\infty + \|D\varphi_{Hz}\|_\infty \leq C|\hat{z}|$, by Lemmata 4.22(iii) and 4.21, and the bound on (4.15) for Bloch waves: since a cell has the volume 1 and each site is an end of $2d$ bonds, Cauchy–Schwarz gives $\sum_b \eta^4|A(b)| \leq \sqrt{d}\|A\|_2$ and $\sum_b \eta^4(|\lambda(b_-)| + |\lambda(b_+)|) \leq 2d\|\lambda\|_2$ over a cell, so that

$$|T(J,\lambda,A)| \leq 2dc_{\mathcal{N}}\|J\|_\infty \min(\|\lambda\|_\infty\|A\|_2, \ \|A\|_\infty\|\lambda\|_2) \ . \tag{4.60}$$

- *The analytic term.* $(QGQ^*)^{-1} - a$ contributes at most $C|b||\hat{z}||\hat{z}'|$ and $C|b|^2|\hat{z}||\hat{z}'|$ to V^1 and V^2, by Lemma 4.22(ii).

- *First order in J at $U = 1$.* The part of V^1 which contains J is $-\langle Hz', K(1,j_1)Hz\rangle$, with $j_1 := J^{[1]}$. By (4.16) and (4.60) it is at most $2dc_{\mathcal{N}}\|j_1\|_\infty(\|\varphi_{Hz'}\|_\infty\|Hz\|_2 + \|\varphi_{Hz}\|_\infty\|Hz'\|_2 + \|\varphi_{Hz}\|_\infty\|D\varphi_{Hz'}\|_2) \leq C|b||\hat{z}||\hat{z}'|$, and in the same way $-\langle Hz', K(1,J^{[2]})Hz\rangle$ is at most $C|b|^2|\hat{z}||\hat{z}'|$.

- *The term $H^*K'C_GK'H$ at $U = 1$.* Let $X := K'(1,j)Hz$ with j one of the two Bloch parts of j_1, of the quasi-momentum $p' = p \pm p_0$. Since $C_G \geq 0$ is the inverse of the compression of G^{-1} to $\ker Q$, extended by zero, $\langle X, C_GX\rangle = \sup\{2\operatorname{Re}\langle X,Y\rangle - \langle Y, G^{-1}Y\rangle \,|\, Y \in \ker Q\}$, over Bloch waves Y of the quasi-momentum p'. Here $\langle X,Y\rangle = \langle Hz, KY\rangle - \langle Hz, \Delta'_\pi Y\rangle$, and (4.16), (4.60), and $\|\varphi_Y\|_2 \leq (\pi/2)|p'|^{-1}\|Y\|_2$ give $|\langle Hz, KY\rangle| \leq C(1 + |\hat{p}'|^{-1})|b||\hat{z}|\|Y\|_2$. Polarising (3.120) with $\lambda_{Hz} = 0$ gives $\langle Hz, \Delta'_\pi Y\rangle = T(j, \lambda_Y, Hz)$, at most $2dc_{\mathcal{N}}\|j\|_\infty\|Hz\|_\infty\|\lambda_Y\|_2$. At $U = 1$, $\Delta_\pi = \mathbf{H}(1) \geq 0$, so that $\langle Y, G^{-1}Y\rangle \geq \frac{1}{2}\|RD^*Y\|_2^2 + \frac{1}{2}\|G\|^{-1}\|Y\|_2^2$, and $\|\lambda_Y\|_2 \leq \|G'\|\|RD^*Y\|_2$. Hence $\langle X, C_GX\rangle \leq C(1 + |\hat{p}'|^{-1})^2|b|^2|\hat{z}|^2$, and by the Cauchy–Schwarz inequality for $C_G \geq 0$ these terms contribute at most $C(1 + |(p+p_0)^\wedge|^{-2} + |(p-p_0)^\wedge|^{-2})|b|^2|\hat{z}||\hat{z}'|$ to V^2.

- *First order in J and in U.* The remaining part of V^2 is the part of first order in U of $-\langle H(U)z', K(U,j_1)H(U)z\rangle$, the term of Δ'_π having vanished at every U. Where the derivative falls on H, (4.16) gives terms such as $T(j_1, \varphi_{H^{[1]}z'}, Hz)$, $T(j_1, \varphi_{Hz}, H^{[1]}z')$, and $T(j_1, \varphi_{Hz}, D\varphi_{H^{[1]}z'})$, and the same with z and z' exchanged, and with $\|\varphi_{H^{[1]}z'}\|_2 \leq (\pi/2)|p'|^{-1}\|H^{[1]}z'\|_2$ at the quasi-momentum p' of each part of $H^{[1]}z'$, $\|D\varphi_{H^{[1]}z'}\|_2 \leq$

$\|H^{[1]}z'\|_2$, and Lemma 4.22(iii), they are at most $C(1+|\hat{p}'|^{-1})|b|^2|\hat{z}||\hat{z}'|$. Where it falls on D_U, $D^{[1]}\lambda(b) = i[A_1(b), \lambda(b_+)]$ contains no difference and the coefficient of the transports in T carries a factor η; at fixed A the coefficient of φ_A is $(D^{\mathrm{T}}D)^{-1}[D^{[1]\mathrm{T}}(1-\Pi_D)A - D^{\mathrm{T}}D^{[1]}\varphi_A]$, of the norm at most $C\|A_1\|_\infty(|p'|^{-2}\|A\|_2 + |p'|^{-1}\|\varphi_A\|_\infty)$ on the quasi-momentum p', by the bounds on the singular factors, and with $A = Hz, Hz'$ these terms are at most $C(1+|\hat{p}'|^{-2})|b|^2|\hat{z}||\hat{z}'|$.

The four kinds of term give (4.59), with constants which do not depend on k.

(d) The coarse terms. The coefficient of τ in the normalised logarithm of the configuration V is $B_1 := \tilde{Q}(1)A_1$, a Bloch wave of the quasi-momenta $\pm p_0$ with $\|B_1\|_\infty \le (1+d)\|A_1\|_\infty \le C|b|$, by Lemmata 4.20 and 4.22(i). Since $P_T\bar{D}_0 = 0$, $P_T\bar{D}_V = P_T\bar{D}^{[1]}[B_1] + O(B^2)$, where $\bar{D}^{[1]}[B_1]\omega(c) = i[B_1(c), \omega(c_+)]$ contains no difference, and $P_T\Pi_V P_T = P_T - P_T\bar{D}_V(\bar{D}_V^{\mathrm{T}}\bar{D}_V)^{-1}\bar{D}_V^{\mathrm{T}}P_T$ gives

$$\log\det{}_{T_0}(P_T\Pi_V P_T) = -\operatorname{Tr}(P_T\bar{D}^{[1]}(\bar{D}_0^{\mathrm{T}}\bar{D}_0)^{-1}\bar{D}^{[1]\mathrm{T}}P_T) + O(B^3)\,, \tag{4.61}$$

whose symbol at p_0 is at most $C|b|^2\int|\hat{p}|^{-2}\,\mathrm{d}^4p$. For $\log\det(\bar{D}_V^{\mathrm{T}}\bar{D}_V)$ at $V = e^{iB}$, the vertices of first and second order of $\bar{D}_V^{\mathrm{T}}\bar{D}_V$ have the matrix elements at most $C|b|(|\hat{p}| + |(p+p_0)^\wedge|)$ and $C|b|^2(1+|\hat{p}|)$, so its symbol at p_0 is at most $C|b|^2\int(|\hat{p}|^{-2}(1+|\hat{p}|) + |\hat{p}|^{-2}|(p+p_0)^\wedge|^{-2}(|\hat{p}| + |(p+p_0)^\wedge|)^2)\,\mathrm{d}^4p$.

(e) Integration. The coefficient of second order of $\frac{1}{2}\log\det_{T_0}(\Delta^0 + V^1 + V^2)$ is $\frac{1}{2}\operatorname{Tr}((\Delta^0)^{-1}V^2) - \frac{1}{4}\operatorname{Tr}((\Delta^0)^{-1}V^1(\Delta^0)^{-1}V^1)$, the traces being sums over the quasi-momenta of the coset of fibre traces over the transverse fields, of dimension $3\dim G$. By (4.59) and $\Delta^0 \ge \gamma_0|\hat{p}|^2$, and with (d), the coset sums are bounded by the Riemann sums of

$$C\Big[|\hat{p}|^{-2}(1 + |(p+p_0)^\wedge|^{-2} + |(p-p_0)^\wedge|^{-2}) + |\hat{p}|^{-2}|(p+p_0)^\wedge|^{-2}\Big] \tag{4.62}$$

times $|b|^2$, plus the terms of (d). The singular points 0 and $\mp p_0$ of these functions lie in $\mathbb{Z}p_0$, which the coset sums omit, and they are separated because $p_0 \ne 0$; at each the singularity has the degree 2, integrable in four dimensions. So the proof of Lemma 4.5, which applies verbatim to the three singular points, shows that the coset sums converge and that $\|\Pi_{F_k}(p_0)\|$ is at most the integral of (4.62) over $[-\pi,\pi]^4$ plus that of the bounds of (d), which is finite and independent of k. *q.e.d.*

4.7 The Window

The four estimates of this section combine into the statement of Paper, Theorem 4.4, once we know that the polarisations which they compute, each as the limit of sums over cosets or of alias sums of symbols of a lower level, are the pieces of one identity at a fixed momentum.

Proposition S.6 (The Coefficients at Zero Coupling)**.** *There is a constant C_W, depending on L, G, and the averaging operations only, such that for all integers $0 \le m < n$*

$$\Big|\sum_{j=m+1}^{n} b_j - 2c(n-m)\ln L\Big| \le C_W\,. \tag{4.63}$$

Proof. Since $\sum_{j=m+1}^{n} b_j = \Phi_n - \Phi_m$ with $\Phi_0 = 0$, it suffices to show $|\Phi_k - 2ck\ln L| \le C_W/2$ for every $k \ge 1$. Fix $p_0 := (\pi/2, 0, 0, 0)$.

Step 1: the identity. By [9, (1.3), p. 260, and (1.6), p. 261], the functional $\mathcal{L}_k$ of Proposition 4.11 is $\sum_{j<k}[\log Z^{(j)}(U_k) - \log Z^{(j)}(1)]$, with $U_k = U_k(e^{iB})$. For B of the momentum p_0

every operator in these factors commutes with the translations of $T^{(k)}$ which fix B, so that each $\log Z^{(j)}$ is the sum of its restrictions to the fields of the cosets. On a coset $\mathcal{O} \neq \mathbb{Z}p_0$, where the functionals F_j are defined, the definition (4.19) of $\mathcal{J}^{(j)}$ makes the sum telescope, $\mathcal{L}_k = F_k - F_0 + \mathcal{J}_k + \text{const}$, and hence

$$\Pi_{\mathcal{L}_k}(p_0) = \Pi_{F_k}(p_0) + \Pi_{-F_0}(p_0) + \Pi_{\mathcal{J}_k}(p_0) \ , \tag{4.64}$$

each term being the limit of the sums over the cosets $\mathcal{O} \neq \mathbb{Z}p_0$. These limits are the symbols of Propositions 4.11, 4.23, 4.18, and 4.13, for the following reasons. The polarisation of F_k is computed on the same cosets in Proposition 4.23. The polarisation of $\mathcal{L}_k$ in Proposition 4.11 is that of Bałaban's step factors on the whole torus; their quadratic forms are built from the operators G_1, H_1, Δ_1, G', R, h_B, $\tilde{C}^{(2)}$, and $\mathbf{G}^{(2)}$ of (4.20) and [5, (3.128)–(3.129), p. 421], which contain no factor $(D^{\mathrm{T}}D)^{-1}$, the forms $C^{(j)}(1)^{-1}$ are invertible on the whole torus by Lemma 4.9, and the Jacobian $\mathfrak{j}(U)$ is a product of local factors; so the block of the cosets $\mathbb{Z}p_0$ is regular, its fibres are the m_0 points of $\mathbb{Z}p_0$, and it adds $O(N^{-4})$ to the coefficient on the torus of side N. Finally, $\mathcal{J}^{(j)}$ and $-F_0$ are functionals of the fields of the level $m = j+1$ and $m = 0$, with a vanishing coefficient of first order at $U = 1$ (step (c) of the proof of Proposition 4.8, and Step 4 of the proof of Proposition 4.17), so that the coefficient of second order on $T^{(k)}$ is the form of second order of the level m evaluated on the first-order lift of B to the level m, whose quasi-momenta are the aliases $q = p_0 + l$ of p_0. The fields of the level m decompose under the translations of $T^{(k)}$, so that each coset of $T^{(k)}$ is a union of fibres of the level m; at $U = 1$ every operator of the level m conserves the quasi-momentum of that level, so that the terms which pair different aliases vanish fibre by fibre, and the sum over the cosets of $T^{(k)}$ is the sum over l of the sums of the level m at the external momentum q, over the quasi-momenta of the level m which lie above the quasi-momenta of $T^{(k)}$ outside $\mathbb{Z}p_0$. Besides $\mathbb{Z}q$ these sums omit the points of $(\mathbb{Z}p_0 + 2\pi\mathbb{Z}^4) \setminus \mathbb{Z}q$ modulo $2\pi L^{k-m}$, in the units of $T^{(k)}$, a set of at most $m_0 L^{4(k-m)}$ points which does not depend on N and avoids the singular points 0 and $\mp q$ of the integrands of the level m, where these integrands are therefore bounded. At fixed k the limits are thus the alias sums of the symbols of the level m of Propositions 4.8 and 4.17, as used in Propositions 4.13 and 4.18.

Step 2: the estimates. Let e be a unit vector with $\langle v(p_0), e\rangle = 0$. Inserting Proposition 4.11 on the left of (4.64), and Propositions 4.23, 4.18, and 4.13 on the right, we obtain

$$\Phi_k \langle e, \Delta_k(p_0)e\rangle = c \ln \frac{1}{\eta_k^2} \langle e, \Delta_k(p_0)e\rangle + O(1) \ , \tag{4.65}$$

with a bound on the error term which does not depend on k.

Step 3: the division. The curl of $e e^{ip_0 \cdot y}$ has the squared symbol $|\hat{p}_0|^2|e|^2 - |\langle v(p_0), e\rangle|^2 = |\hat{p}_0|^2 = |e^{i\pi/2} - 1|^2 = 2$, by Lagrange's identity, so that [2, (1.67), p. 29] gives $2\gamma_0 \leq \langle e, \Delta_k(p_0)e\rangle \leq 2\gamma_1$, with $\gamma_0 > 0$ independent of k. Dividing (4.65) by this number, and using $\ln(1/\eta_k^2) = 2k \ln L$, we obtain $|\Phi_k - 2ck \ln L| \leq C_W/2$ with a constant C_W independent of k. The constants of the four propositions depend on $d = 4$, L, G, the averaging operations, and the constants of [2, 5, 7], which depend on these alone. *q.e.d.*

5 The First Step at the Retreated Region

The paper carries a loop ensemble through the first step on labelled terms of a refined form [1, Section 9.4]: the conditional integral over the fluctuation field, which Bałaban performs on his domain Λ_1, is performed on a smaller region Λ, from which the cubes near every large value of the fluctuation field on the boundary of the integration have been removed. The retreat keeps the mean of the conditional Gaussian below a fixed fraction of the cutoff radius on the whole region of integration [1, Remark 9.11], and the analysis of the loop needs this margin. Bałaban's description of the first step, however, is written for Λ_1, a domain built from the test functions alone [10, (1.20), p. 251], and his inductive assumption and the large-field bounds of [13, Section 1] are stated for the domains which his construction produces. We show in this section that his construction runs at Λ, that its output has the form of his inductive assumption with Λ in the place of Λ_1, and that the large-field factors of [13] still pay for the region Λ^c, which is larger than his and depends on the fluctuation field. Only the last point changes a constant.

5.1 The Retreat

Throughout this section lengths are measured in fine units, the cubes are those of side $\ell := LMR_1$ of which Ω_1 and Λ_1 are unions, and the operation $\sim$ adds one layer of these cubes. We fix a term of the decomposition of the first step with the domains Ω_1 and Λ_1 of [10, (1.10), p. 248; (1.20), p. 251]. Let Ω_1^* be the set of the free bonds of Ω_1, and $I \subset \Omega_1^*$ the set of the bonds over which Bałaban's conditional integral at Λ_1 runs, both in the variables of [1, Section 3]. In these variables the quadratic form $\langle A, C^*\Delta_1 CA\rangle$ of [10, (1.15), p. 249] is $\langle A, HA\rangle$ with a symmetric operator H on Ω_1^*, whose entries vanish between bonds at a distance greater than $\rho_L := 4L$, and whose principal restrictions to subsets of I satisfy $\gamma_L/2 \le H \le m_L$, with constants γ_L and m_L which depend on L and G [1, Section 9.1, (I1)–(I3)]. For a union $\Lambda \subset \Lambda_1$ of cubes, the *carrier*, the *exterior*, and the *collar* of Λ are

$$\begin{aligned} D(\Lambda) &:= \{b \in I \,|\, b \text{ has an endpoint in } \Lambda\} \,, \\ J(\Lambda) &:= \Omega_1^* \setminus D(\Lambda) \,, \\ J_{\mathrm{col}}(\Lambda) &:= \{e \in J(\Lambda) \,|\, \mathrm{dist}(e, D(\Lambda)) \le \rho_L\} \,, \end{aligned} \tag{5.1}$$

the distance of two bonds being that of their initial sites [1, Section 9.3.2]. The cutoff radius of the fluctuation field in the construction of the paper is $t_0 := \vartheta_0/g_0$ (Paper, Definition 3.11(ii)). A collar bond e is *bad* if $|A(e)|_\sigma \ge \vartheta_V t_0$ and *good* otherwise, where $|\cdot|_\sigma$ is the norm on $\mathfrak{g}$ of [1, Section 9.1] and $\vartheta_V \le 1/16$ is the constant of [1, Remark 9.3].

Definition 5.1 (The Retreat)**.** Let Fill be the iterated filling of [10, p. 251], which replaces each component of the complement which lies in a cube of side 100ℓ by the smallest rectangular parallelepiped of cubes containing it, until no component changes [1, Section 9.4.1]. Put $\Lambda^{(0)} := \mathsf{Fill}(\Lambda_1)$. Given $\Lambda^{(k)}$, the bonds of $J_{\mathrm{col}}(\Lambda^{(k)})$ which have not been tested before are tested, and $B^{(k)}$ is the set of those found bad. If $B^{(k)} = \emptyset$, we put $\Lambda := \Lambda^{(k)}$; otherwise

$$\Lambda^{(k+1)} := \mathsf{Fill}\Big(\Lambda^{(k)} \setminus \bigcup_{e\in B^{(k)}} \mathcal{Q}^+(e)\Big) \,, \tag{5.2}$$

where $\mathcal{Q}^+(e)$ is the union of the cubes which meet the closed ℓ^∞-ball of radius $\ell + \rho_L + 1$ about the initial site x_e of e. The *refined labels* of the term are $(\Omega_1, \Lambda_1, B^{(0)}, B^{(1)}, \dots)$, and Λ is its *retreated region.*

The tests enter the integrand through the decomposition $1 = \prod_e(\mathbf{1}\{|A(e)|_\sigma < \vartheta_V t_0\} + \mathbf{1}\{|A(e)|_\sigma \ge \vartheta_V t_0\})$ over the tested bonds, so that the terms with Bałaban's labels (Ω_1, Λ_1) are the sums of the terms with the refined labels [1, Section 9.4.1]. In [1] the retreat is the union $\mathcal{Q}(e)$ of the cubes which meet the ball of radius ρ_L+1. We use the larger ball because it gives two placements which the smaller one does not: every bad bond lies outside $\Lambda^\sim$, where Bałaban's construction admits restrictions of small fields only, and at a distance at least ℓ from Λ, so that the quadratic form produces the small factor of a bad bond.

Lemma 5.2 (The Retreated Region)**.** *On every term the following hold.*

(i) $\Lambda \subset \Lambda^{(0)} \subset \Lambda_1 \subset \Omega_1^{\sim -2} \cap (R_1'^{\sim})^c$, *where* R_1' *is the union of the cubes which meet* R_1*;* Λ *is a union of cubes, and every component of* Λ^c *which lies in a cube of side* 100ℓ *is a rectangular parallelepiped.*

(ii) Every point of $\Lambda^{(0)} \backslash \Lambda$ *lies within* 103ℓ *of* x_e *for a bad bond* e*, and at most* $N_{\mathrm{ret}}^{+} := 4(209\ell)^4$ *bonds have an endpoint within* 103ℓ *of a given site.*

(iii) Every bond of $J_{\mathrm{col}}(\Lambda)$ *is good.*

(iv) Every bad bond e *has* $\mathrm{dist}(e, \Lambda) \geq \ell$*, and no cube which contains a point of* e *lies in* $\Lambda^{\sim}$*.*

Proof. (i) This is [1, Lemma 9.12(i)], whose proof uses of the retreat only that each round removes cubes and that every region $\Lambda^{(k)}$ is a fixed point of **Fill** [1, Lemma 9.10(c)]; the inclusion $\Lambda_1 \subset \Omega_1^{\sim -2} \cap (R_1'^{\sim})^c$ is [10, (1.20), p. 251], which the filling preserves.

(ii) The proof of [1, Lemma 9.10(c), (d)] uses of the retreat only that every point of it lies within 2ℓ of x_e, and gives the distance 100ℓ plus that bound. A point of a cube which meets the ball of radius $\ell + \rho_L + 1$ about x_e lies within $2\ell + \rho_L + 1 \leq 3\ell$ of x_e, because $2(\rho_L + 1) + 1 < \ell$ by [1, (3.1)], and so the distance is 103ℓ. The sites within 103ℓ of a site number $(206\ell + 1)^4$, the bonds with an endpoint among them have their initial sites within $103\ell + 1$, and each site is the initial site of four bonds, which gives at most $4(206\ell + 3)^4 \leq N_{\mathrm{ret}}^{+}$ bonds.

(iii) This is [1, Lemma 9.10(b)], whose proof uses only that $\Lambda \subset \Lambda^{(j)} \setminus \mathcal{Q}(e)$ for the round j in which e was found bad, and $\mathcal{Q}(e) \subset \mathcal{Q}^{+}(e)$.

(iv) Let e be found bad in the round j. Since the rounds and the filling only remove cubes, $\Lambda \subset \Lambda^{(j)} \setminus \mathcal{Q}^{+}(e)$, so that no site of Λ lies in the closed ball of radius $\ell + \rho_L + 1$ about x_e. As the other endpoint of e is a neighbour of x_e, $\mathrm{dist}(e, \Lambda) \geq \ell + \rho_L \geq \ell$. Let Q be a cube which contains a point of e, and Q' a cube which has a point in common with Q. Then Q' has a point within $\ell + 1$ of x_e, so that Q' meets the ball and is not a cube of Λ. Hence Q has no point in common with Λ, that is, $Q \not\subset \Lambda^{\sim}$. *q.e.d.*

5.2 The Construction at the Retreated Region

We now state what the paper uses. Part (i) is Bałaban's conditional integral [10, (1.26), p. 253] with its expansion, part (ii) is the form [10, (1.28)–(1.30), pp. 253–254] of the output, part (iii) is the large-field factor which [13, (1.79), p. 383] attributes to the first step, and part (iv) is the passage from the thresholds of [1] to those of a tube. We write $\mathbf{D}_1$ for the localisation domains of the scale 1, $d_1(X)$ for the linear size of $X \in \mathbf{D}_1$ [9, p. 257], and d_1' for the linear size measured in cubes of side MR_1 of $T^{(1)}$ [13, p. 380]; the cubes of side MR_1 of $T^{(1)}$ are the cubes of side ℓ in fine units. The constants B_0, κ, and $\beta = 1/8$ are those of [10, (2.42), p. 261] (Paper, Appendix A), and c_4 is the constant

$$c_4 := \tfrac{1}{2} \min\Big\{ \gamma_0 m_A, \frac{c_b}{4 \cdot 10^d} \Big\} , \qquad m_A := \tfrac{1}{4} \min\Big\{ \tfrac{1}{2} B_3^{-2} A_0^2 A_1^{-2}, 2 \Big\} , \qquad c_b := \frac{\gamma_L}{16 c_h^2} , \tag{5.3}$$

with the absolute constant γ_0 of [13, p. 380], the constant B_3 of [7, Theorem 1, p. 279], and the constant c_h of the two norms (Paper, Appendix B).

Proposition S.7 (The First Step at the Retreated Region)**.** *Let* γ *be sufficiently small. Consider the first step of the construction for a tube, with a loop ensemble inserted, and a term with refined labels and retreated region* Λ*.*

(i) *The integral over the fluctuation field on $D(\Lambda)$, at fixed values of the other variables, satisfies the two equalities of [10, (1.26), p. 253] with Λ_1 replaced by Λ and with the covariance $C^{(0)}(\Lambda) := (H_{D(\Lambda)D(\Lambda)})^{-1}$. The logarithm of this integral, plus $V^{(0)}(S_1)$, is the sum of a constant, of the quadratic form $\frac{1}{2}\langle \ell_\Lambda, C^{(0)}(\Lambda)\ell_\Lambda\rangle$ with $\ell_\Lambda := H_{D(\Lambda)J(\Lambda)}\Lambda^c A$, of*

$$\mathbf{E}^{(1)}(\Lambda) := \sum_{z\in\Lambda^0} \sum_{X\in\mathbf{D}_1,\, z\in X\subset\Lambda} \mathbf{E}^{(1)}(X,z)\ , \tag{5.4}$$

where $\mathbf{E}^{(1)}(X,z)$ are the terms of [10, (2.26)–(2.27), p. 259] which Bałaban's construction produces on the whole lattice and Λ^0 is Λ with one layer of cubes removed [10, p. 259], and of a boundary term $\mathbf{B}^{(1)} = \sum_X \mathbf{B}^{(1)}(X)$ over the $X \in \mathbf{D}_1$ with $X\cap\Omega_1 \neq \emptyset$ and $X \cap \Lambda^{c\sim} \neq \emptyset$, whose terms have the properties (i)–(iii) of [10, p. 261] on the spaces $\tilde{U}_1^c(X)$ with the radii enlarged by the factor $1+\beta$, and the bound

$$|\mathbf{B}^{(1)}(X)| < B_0 e^{-(1+4\beta)\kappa d_1(X)}\ , \qquad \Big|\sum_X \mathbf{B}^{(1)}(X)\Big| \le 2B_1|\Omega_1^{(1)}\setminus\Lambda^0| \le 2B_1 3^d|\Lambda^c\cap\Omega_1|\ , \tag{5.5}$$

with the constant B_1 of [10, (2.47), p. 264], the volumes counted in points of $T^{(1)}$.

(ii) *The sum of the terms with given $(\Omega_1, \Lambda_1, \Lambda)$ is $\chi_1(\Omega_1)\mathbf{T}_1^{\Lambda_1,\Lambda}\exp A_1^\Lambda$, where A_1^Λ is [10, (1.30), p. 254] with Λ_1 replaced by Λ, with $\phi_1 \in C_0^\infty(\Lambda)$ equal to one on $\Lambda^{\sim -1}$ and $\mathbf{E}^{(1)}(\Lambda)$ of (i), and $\mathbf{T}_1^{\Lambda_1,\Lambda}$ is [10, (1.29), p. 254] with $C^{(0)}(\Lambda)$ in its quadratic form, with the integral over the fluctuation field on $J(\Lambda)\cap\Omega_1$, and with three further nonnegative factors: the bounds $\mathbf{1}\{|A(b)|_\sigma < t_0\}$ on $I\setminus D(\Lambda)$, the test functions of the retreat, and the factors of the rows outside Λ of the function $z^{(0)}(U_1)$ of [1, (3.5)]. The domains $\Omega_1 \supset \Lambda$ satisfy [10, (2.1), p. 254] and the condition after [10, (2.3), p. 255] at $j=1$, Λ_1 enters $\mathbf{T}_1^{\Lambda_1,\Lambda}$ and E_1 as a parameter, and $\mathbf{T}_1^{\Lambda_1,\Lambda}$ factorises over the components of Λ^c as in [10, (2.19), (2.21), p. 258].*

(iii) *Let $\mathcal{X}$ be a component of Λ^c, k a step at which the basic step of the $\mathbf{R}$-operation acts on a component X_k of the large-field region with $\mathcal{X}\subset X_k$, and $\mathbf{T}_1'(\mathcal{X})$ the part of the operation $\mathbf{T}_k'(X_k)$ of [13, (1.71), p. 379] which integrates the variables of the first step in $\mathcal{X}$. Let G run over the components in $\mathcal{X}$ of the cover by cubes of side ℓ of $P_1\cup Q_1\cup R_1\cup W_1$, where W_1 is the union of the cubes of side $L^2M_2R_1$ of [10, (1.18), p. 250] which contain the initial site of a bad bond. Then $\mathcal{X}\subset\bigcup_{G\subset\mathcal{X}} G^{\sim 107}$, and, with $a := A_1^2(\log g_1^{-2})^{2p_0}$,*

$$\mathbf{T}_1'(\mathcal{X})1 \le \exp(O(1)M^d R_1^{d+1}(d_1'(\mathcal{X})+1)) \sup \prod_{G\subset\mathcal{X}} \exp(-\tfrac{1}{2}c_4\vartheta_V^2 a(d_1'(G)+1) - 2p_0(g_1))\ , \tag{5.6}$$

the supremum over the domains P_1, Q_1, R_1, S_1 in $\mathcal{X}$, where $p_0(g) := (\log g^{-2})^{p_0}$, the reading of the display on [13, p. 381], whose term $\frac{1}{2}B_3^{-2}A_0^2$ is $\frac{1}{2}(B_3^{-1}\varepsilon_j/g_j)^2$ with $\varepsilon_j/g_j = A_0(\log g_j^{-2})^{p_0}$. Consequently [13, (1.82), p. 385] holds with $\frac{1}{4}c_4 d^{-1}(434)^{-d}\vartheta_V^2$ in place of $\frac{1}{4}\gamma_0(14)^{-d}$, and [13, (1.80), p. 384] holds at $j=1$.

(iv) *The conditions on the coupling under which [1] proves its results hold at the radius t_0 for γ small, except the condition behind $e^{-c_{\mathrm{ph}}r^2} \le g_0$ of [1, (9.29)(iii)], $c_{\mathrm{ph}} = 9c_n\alpha_G/128$, which at the radius t_0 reads*

$$\frac{9c_n\alpha_G}{64\rho}\, A_1^2 \log g_0^{-2} \ge 1 \tag{5.7}$$

and holds for γ small. Every statement of [1] holds at the radius t_0 with the constants of its exponents which come from lower bounds on the radius divided by ρ.

The proof occupies the rest of this section. Bałaban's proof of the first step, in [10, pp. 247–254], rests on [11, Sections 1–2] and on the propagators of [5, Sections C–E], and the place where it is generalised to the later steps, [10, pp. 277–278], concerns his own domains Λ_{k+1}. We therefore go through the properties of Λ_1 which these pages use and check each at Λ; where a property fails, or where the retreat adds an integrand factor, we give the argument which replaces it.

Proof of (i). Step 1: the properties of the domain. Bałaban's construction of the first step uses the following properties of Λ_1:

(P1) Λ_1 is a union of cubes of side ℓ;

(P2) $\Lambda_1 \subset \Omega_1^{\sim -2}$;

(P3) $\Lambda_1 \cap R_1'^{\sim} = \emptyset$;

(P4) every component of Λ_1^c which lies in a cube of side 100ℓ is a rectangular parallelepiped [10, p. 251, and after (2.3), p. 255];

and the following, which are stated in the text:

(α) "On the set $\Lambda_1^{\sim}$ we have only the small fluctuation field characteristic functions from (1.18)" [10, p. 251], used again at the general step [10, p. 269];

(β) the boundary term "can be bounded by $O(1)|\Lambda_1 \cap \Lambda_1^c|$; therefore it is controlled by the large field estimates" [10, p. 253];

(γ) "If $\Omega_j \cap \Lambda_j^c$ is nonempty, then S_j is nonempty too", and the characteristic function $\chi(\Omega_j \cap \Lambda_j^c, S_j)$ includes a large-fluctuation function "associated with the set R_j" [10, p. 255];

(δ) the bound on $\mathbf{H}_{1,Ax}$ uses that the distance between a cube $\square' \subset S_1$ and the support of its argument near $\partial\Lambda_1$ is greater than MR_1 [10, p. 251].

By Lemma 5.2(i), Λ has (P1)–(P4). For (α) at Λ we locate a test function of [10, (1.18), p. 250] by its index cube $\square'$, as on [10, p. 251], and a test function of a single bond by its bond; with this convention (α) holds for Λ_1 as printed, since the index cubes of the large fluctuations lie in $R_1 \subset R_1'$, outside $\Lambda_1^{\sim}$. On $\Lambda^{\sim} \subset \Lambda_1^{\sim}$ the restrictions of the integrand are then Bałaban's small-field functions of (1.18), the bounds $|A(b)|_\sigma < t_0$ on $I \setminus D(\Lambda)$, which [1, Lemma 9.7] retains, and the test functions of the retreat at the good bonds, since every bad bond lies outside $\Lambda^{\sim}$ by Lemma 5.2(iv); all of them are restrictions $\{|A(b)| < t\}$ with $t \le t_0$, which is the form in which (α) is used. Property (δ) holds because $\Lambda \subset \Lambda_1$ while the background $U_1(\Omega_1 \cap \Lambda_1^c, V)$, the sets S_1, R_1, and the change of variables [10, (1.22), p. 251] are those of Λ_1 [1, Section 9.4.1], so that the distance in question is Bałaban's. Property (β) is proved at Λ in Step 4 below, and (γ), which fails at Λ when Ω_1 is the whole torus, is replaced by the factor of part (iii).

Step 2: completing the square. Write $D := D(\Lambda)$ and $J := J(\Lambda)$. Since $D \subset I$, H_{DD} is a principal restriction of H_{II}, and $\gamma_L/2 \le H_{DD} \le m_L$ by [1, (I3)], which passes to principal restrictions [1, Remark 9.11]. Hence $C^{(0)}(\Lambda) = H_{DD}^{-1}$ exists, and by [1, Lemma 9.1]

$$\|C^{(0)}(\Lambda)(b,b')\| \le \frac{2}{\gamma_L} e^{-\mu_H \operatorname{dist}(b,b')}\,, \qquad \mu_H := \frac{\gamma_L}{8Lm_L}\,. \tag{5.8}$$

The integrand of [10, (1.26)] at Λ is the integrand of [10, (1.15), p. 249] with the variables on D integrated and those on J held fixed; the cross term is $-\langle \ell_\Lambda, A_D\rangle$ with $\ell_\Lambda = H_{DJ}A_J$, and the cutoff on D is the box $\prod_{b\in D} \mathbf{1}\{|A(b)|_\sigma < t_0\}$ wherever the fibre of the constraint is nonempty [1, Lemma 9.12(iv)]. Completing the square in the Gaussian measure of covariance $C^{(0)}(\Lambda)$ gives the two equalities of (1.26) at Λ, with the term $\frac{1}{2}\langle \ell_\Lambda, C^{(0)}(\Lambda)\ell_\Lambda\rangle$, and the quadratic form which remains on J is

$$-\tfrac{1}{2}\langle A_J, H_{JJ}A_J\rangle + \tfrac{1}{2}\langle A_J, H_{JD}C^{(0)}(\Lambda)H_{DJ}A_J\rangle\ , \tag{5.9}$$

the form of [10, (1.29)] at Λ. The change of variables [10, (1.22)] differs from the identity only on S_1^*, and $\operatorname{dist}(I, S_1^*) \geq \ell - 2 > \rho_L$ [1, (3.39)], so the blocks of H which meet D are the same in both sets of variables; the change on S_1^* enters $V^{(0)}(S_1)$ as for Λ_1.

Step 3: the expansion and the whole-lattice terms. The logarithm of the normalised Gaussian integral is expanded by the cluster expansion of [11, Sections 1–2], as on [10, p. 253]. The inputs of this expansion are: the decoupling of the covariance in the cubes of side $L^{-1}\ell$ [11, (1.3)–(1.4), p. 2], which needs Λ to be a union of such cubes, true by (P1); the random walk expansions of the propagators with Dirichlet conditions outside Λ [5, Theorems 3.7, 3.9, 3.10, Corollary 3.8, pp. 409–416], and the unit-lattice covariance [5, Theorem 3.15, p. 432], whose hypotheses on [5, p. 427] are that Λ be a union of big blocks at a distance greater than RM from the complement of the domain of the step; and the polymer bounds of [11, (2.38), p. 20]. The cubes of side ℓ are unions of big blocks, and $\operatorname{dist}(\Lambda, \Omega_1^c) \geq 2\ell > RM$ by (P2), so all of these hold at Λ with the constants of Λ_1. The result is Bałaban's representation of $\log Z^{(0)}(\Lambda)$ and of the expectation of $e^{\mathbf{V}^{(0)}}$ as sums over $X \in \mathbf{D}_1$ of terms which depend on the configuration on X only, with $|\tilde{\mathbf{E}}(\Lambda, X)| \leq O(1)C_3\varepsilon_1 e^{-(1-10\delta)\frac{1}{2}L\kappa d_1(X)}$ [11, (2.41), p. 21], analytic on $\tilde{U}_1^c(X)$ by the domain inclusion of [9, pp. 262–263].

A term whose domain X satisfies $\tilde{X}^5 \subset \Lambda$ coincides with the term which the same construction gives on the whole lattice. Indeed, by [5, Theorem 3.14, p. 427] two operators constructed for two sequences of domains differ only through the walks whose localisations meet the complement of the common domain, and the same holds for the unit-lattice covariances of [5, Section E], which are built from these operators; the terms of the expansion are functions of the walks localised in $\tilde{X}^5$. Each operation $\sim$ on a localisation domain adds a layer of big blocks of side LM_1 [5, p. 408], so that $\tilde{X}^5$ lies within $5LM_1 \leq \frac{1}{2}\ell$ of X. The terms with $X \subset \Lambda$ and $\operatorname{dist}(X, \Lambda^c) > \ell$, in particular all $X \ni z$ with $z \in \Lambda^0$ and $X \subset \Lambda$ which do not meet the layer removed from Λ, are therefore the whole-lattice terms; we collect the terms with $z \in \Lambda^0$ in (5.4), as on [10, p. 259], and the others, which meet $\Lambda^{c\sim}$, in $\mathbf{B}^{(1)}$. Since the terms of (5.4) do not depend on Λ except through the range of summation, $\mathbf{E}^{(1)}(\Lambda, g_0, U_1) - \mathbf{E}^{(1)}(\Lambda, g_0, 1)$ is renormalised by the counterterm $\beta_1(g_0)A(\phi_1, U_1)$ of [10, p. 253] with ϕ_1 localised in Λ, exactly as for Λ_1: the coincidence with the terms of [9] is the reason Bałaban gives on [10, p. 254], and it holds at Λ for the same reason.

Step 4: the boundary terms. The terms of $\mathbf{B}^{(1)}$ are of three kinds: the logarithms of the determinants and the cluster terms of the expansion whose domains meet $\Lambda^{c\sim}$; the quotients of the cutoff functions, which Bałaban expands in [10, (3.32), p. 272] at $k = 0$; and the expanded $V^{(0)}(S_1)$, which is Bałaban's, since S_1 and $\mathbf{H}_{1,Ax}$ are those of Λ_1. Each term depends on the configuration on its domain X only, it is gauge invariant because the expansion is, and it extends analytically to $\tilde{U}_1^c(X)$ with the radii enlarged by $1+\beta$, because the expansion of Step 3 is analytic on that domain. For the bound, the terms of the first two kinds whose domains X meet $\Lambda^{c\sim}$ differ from the whole-lattice terms only through walks which reach Λ^c; by (5.8) and [5, (3.108), p. 416] the bond terms are at most $C_{\text{bond}}e^{-(2\kappa+1)d_1(X)}$ and the terms attached to a point z at most $2^{22}L^4e^{4\kappa+2}C_{\text{bond}}e^{-(2\kappa+1)d_1(X)}$, with C_{bond} depending on the constants of the determinants and of the expansion of the cutoffs. The rate of the cutoff quotients is the rate of [10, (3.32)], which the polymer bounds of [11, (2.38)] carry with $B'_V = O(p_0^3(g_0))$ by Lemma 17.15. Since $(2\kappa+1)d_1(X) \geq (1+4\beta)\kappa d_1(X) + d_1(X)$ for $\beta = 1/8$, the first bound of (5.5) holds once B_0 exceeds $2^{10}M^4$ times these constants plus $4O(1)C_3\varepsilon_1 + 1$, a choice of the stage at which B_0 is fixed. For the second bound we follow [10, (2.47)–(2.48), p. 264] at $k = j = n = 1$: the sum of the terms with $X \ni z$ is at most B_1 for each z, and the terms are attached to the points of $\Gamma_1 \setminus \Lambda^0 = \Omega_1^{(1)} \setminus \Lambda^0$, which gives $2B_1|\Omega_1^{(1)} \setminus \Lambda^0|$. Every cube of $\Lambda \setminus \Lambda^0$ has a point in common with a cube of Λ^c, and that cube lies in Ω_1 because $\Lambda \subset \Omega_1^{\sim -2}$; as a cube has at most $3^d - 1$ neighbours, $|\Omega_1^{(1)} \setminus \Lambda^0| \leq 3^d|\Lambda^c \cap \Omega_1|$. This is (β) at Λ: a volume term of the form which [13, p. 380, l. 27–29] places in the first exponential of [13, (1.79)], and which (iii) pays.

Step 5: the outer field. The fixed field A_J of the conditional integral lives on $\Omega_1 \cap \Lambda^c = (\Omega_1 \cap \Lambda_1^c) \cup (\Lambda_1 \setminus \Lambda)$. On $\Omega_1 \cap \Lambda_1^c$ it carries Bałaban's restrictions [10, (1.25), p. 252], and on $\Lambda_1 \setminus \Lambda$ the small-field functions of Λ_1, the bounds on $I \setminus D(\Lambda)$, and the test functions of the retreat. Since ℓ_Λ and every term of $\mathbf{V}^{(0)}$ which depends on both A_D and A_J are supported within $4L+3$ of Λ^c [1, Section 3.3], and $\mathrm{dist}(\Lambda, S_1) \geq \ell > 4L+4$, the conditional integral sees the outer field only on bonds of J within $4L+3$ of D, where it is bounded by t_0 and, on the collar, by $\vartheta_V t_0$ (Lemma 5.2(iii)). The mean of the conditional Gaussian, $m = -C^{(0)}(\Lambda)\ell_\Lambda$, therefore satisfies $|m(b)|_\sigma \leq t_0/4$ on D [1, (9.11), Remark 9.11], and the shift of the box cutoff by m is the cutoff Θ^t of [1, (9.13)]. The term $\frac{1}{2}\langle \ell_\Lambda, C^{(0)}(\Lambda)\ell_\Lambda\rangle$ is not local, and Bałaban moves it into the form (5.9) of $\mathbf{T}_1$, which couples only fields in the same component [10, p. 258]; what is left is the difference

$$\mathbf{B}_{\text{out}} := \log \tilde{I}_1 - \log \tilde{I}_0 - \tfrac{1}{2}\langle \ell_\Lambda, C^{(0)}(\Lambda)\ell_\Lambda\rangle \ ,$$

where $\tilde{I}_1$ is the normalised integral with the outer field and $\tilde{I}_0$ the same at $A_J = 0$; this replaces the first expression of [10, (3.28)] at $k = 0$. To write it as a sum of local terms we replace the shift m by the local shift $m^{\text{loc}} := -\sum_{|\omega| \leq \rho_2} C_\omega \ell_\Lambda$, the walk expansion of $C^{(0)}(\Lambda)$ [5, (3.107), p. 416] truncated at $\rho_2 := (2/\delta_{\text{tl}})\log(K_{\text{tl}} m_L t_0^3)$ steps, K_{tl} and δ_{tl} being the constants of [5, (3.108)]. Then $|m - m^{\text{loc}}| \leq 1/(m_L t_0^2)$, the function m^{loc} is supported in the collar $\mathcal{K}_3 := \{b \in D \mid \mathrm{dist}(b, J) \leq R_U\}$ with $R_U = O(\log t_0)$, and $m^{\text{loc}}(b)$ depends on the configuration within R_U of b only. At a real configuration, [1, Lemma B.5] on a rectangle of side $2R_U + 3$ gives a gauge in which the configuration is close to the identity, and in it the bound $|m(b)|_\sigma \leq t_0/4$ holds with the local data alone; so $|m^{\text{loc}}(b)|_\sigma \leq t_0/2$.

With $A = A'' + m^{\text{loc}}$ the integral becomes the expectation, in the Gaussian measure of covariance $C^{(0)}(\Lambda)$, of $e^{-\langle A'', w\rangle}\chi^{(0)}(A'' + m^{\text{loc}})e^{\mathbf{V}^{(0)}(A_J, A'' + m^{\text{loc}})}$, times $\exp(\frac{1}{2}\langle \ell_\Lambda, C^{(0)}(\Lambda)\ell_\Lambda\rangle - \frac{1}{2}\langle m^{\text{loc}} - m, H_{DD}(m^{\text{loc}} - m)\rangle)$, where $w := H_{DD}(m^{\text{loc}} - m)$ satisfies $|w| \leq 2/t_0^2$. The construction of [11, (2.1)–(2.13), pp. 12–14] is run on this expectation with three changes. The first concerns the linear factor $e^{-\langle A'', w\rangle}$. The function w is local in A'', but not in the configuration: since $H_{DD}m = -\ell_\Lambda$, it is the sum of the walks which the truncation has removed,

$$w = H_{DD}(m^{\text{loc}} - m) = \sum_{|\omega| > \rho_2} w_\omega \ , \qquad w_\omega := H_{DD}C_\omega \ell_\Lambda \ , \tag{5.10}$$

and the term of a walk depends on the configuration on the whole domain of the walk [5, Theorem 3.10, p. 416]. A potential $-\sum_{b \in \Delta}\langle A''(b), w(b)\rangle$ of a single cube Δ of side LM would therefore depend on the configuration in a neighbourhood of Δ of the radius of the walks, and its size, of the order $1/t_0$, could not pay for such a domain. We keep the decomposition (5.10) instead. For a cube Δ and a walk ω with $|\omega| > \rho_2$ whose term w_ω does not vanish on Δ let

$$W(\Delta, \omega) := -\sum_{b \in \Delta}\langle A''(b), w_\omega(b)\rangle \ , \tag{5.11}$$

a function of A'' on Δ and of the configuration and the outer field on the domain $Y(\Delta, \omega)$, the union of Δ, of the cubes of the walk, and of their neighbours within the range $4L + 3$ of H and of ℓ_Λ, a connected union of at most $3^d(|\omega| + 2)$ cubes. By [5, (3.108)] and $|\ell_\Lambda| \leq m_L t_0$, $|w_\omega(b)| \leq K_{\mathrm{tl}} m_L^2 t_0 e^{-\delta_{\mathrm{tl}}|\omega|}$, and for $|\omega| > \rho_2$ the choice of ρ_2 gives $e^{-\delta_{\mathrm{tl}}|\omega|} \leq e^{-\delta_{\mathrm{tl}}|\omega|/2}(K_{\mathrm{tl}} m_L t_0^3)^{-1}$, so that on the support of the cutoff, where $|A''(b)|_\sigma \leq 2t_0$,

$$|W(\Delta, \omega)| \leq \frac{2c_h m_L n_\Delta}{t_0}\, e^{-\delta_{\mathrm{tl}}|\omega|/2} \,, \tag{5.12}$$

with n_Δ the number of the bonds of Δ. These are the potentials which enter the Mayer expansion of [11, (2.1)], each with its own domain $Y(\Delta, \omega)$. Their bounds decay exponentially in the number of the cubes of the domain, at the rate $\frac{1}{2}\delta_{\mathrm{tl}}$ per step of the walk, and $e^{-\delta_{\mathrm{tl}}}$ is of the order M^{-1} [5, (3.108)], so that for M large, as [9, Theorem 3, p. 264] requires in terms of κ, the rate exceeds the rate which [11, (1.36), p. 9] asks of a potential, with room for the number of the walks of a given length from a cube, which is at most a constant to the power of the length [5, p. 413]. No domain is enlarged for them. The second change concerns the potentials of $\mathbf{V}^{(0)}$ whose domains meet $\mathcal{K}_3$. They are read at $A'' + m^{\mathrm{loc}}$, and $m^{\mathrm{loc}} = -\sum_{|\omega| \leq \rho_2} C_\omega \ell_\Lambda$ depends on the configuration along its walks. This dependence is localised in the way in which Bałaban localises the dependence of the potentials on the background field: the decoupling parameters $s(\Delta)$ of [11, (1.6)–(1.10), pp. 3–4] are introduced in the walks of m^{loc} as in those of the other propagators, and the expansion in these parameters produces potentials whose domains contain the walks which they involve and whose bounds carry the factors [11, (1.7), p. 3] of these walks, so that every cube which a walk adds to a domain is paid by a step of the walk, as for the linear potentials. The third change concerns the cutoff. On the bonds of $\mathcal{K}_3$ which the expansion keeps, the shifted cutoff is written $\chi(A''(b) + m^{\mathrm{loc}}(b)) = \mathbf{1}\{|A''(b)|_\sigma < t_0/2\} + \Delta_b$, where Δ_b vanishes unless $|A''(b)|_\sigma \geq t_0/2$, since $|m^{\mathrm{loc}}(b)|_\sigma \leq t_0/2$. On the bonds where the cutoff fails, the penalty

$$\mathbf{1}\{|x + a|_\sigma \geq t_0\} \leq e^{-\frac{1}{2}\gamma_2 \nu_p^2 t_0^2} e^{\frac{1}{2}\gamma_2 |x|_\sigma^2} \,, \qquad |a|_\sigma \leq (1 - \nu_p) t_0 \,,$$

with $\nu_p = 1/2$, takes the place of [11, (2.22), p. 16], and [11, (2.22)–(2.37), pp. 16–20] go through, since they use only lower bounds on the coefficient of the penalty, which hold for small g_0. A piece of the expansion depends on the configuration on its decoupled domain, which contains the domains of its potentials, those of the linear potentials and of the potentials at the shifted field included, and on the neighbourhood of radius R_U of each bond of $\mathcal{K}_3$ at which a cutoff is expanded; its domain X' is the domain of [11] enlarged by these neighbourhoods. At a complex configuration of $\tilde{U}_1^c(X')$ the operators are evaluated at the complex point, $H = H_R + iH_I$ with entries of $H - H(U)$ of size $\epsilon_H = O(\alpha_{0,1} + \alpha_{1,1})$, and the shift acquires an imaginary part $|\mathrm{Im}\, m^{\mathrm{loc}}| \leq \epsilon_m t_0$ with $\epsilon_m = O(\rho_2 \epsilon_H) \to 0$; the determinant ratio, the outer forms, and the cutoff differences are then bounded by Cauchy's estimate as in [11, (2.15)–(2.17), p. 16], each expanded cutoff contributing at most $2e^2 C \epsilon_m t_0^2 \leq g_0^{1/2}$ or the penalty $e^{-\gamma_2 t_0^2/8}$. The enlargement of X to X' costs at most $g_0^{-1/8}$ per enlarged cube against the decay in $d_1(X')$, since $(2R_U/(LM) + 3)^4 = O((\log\log g_0^{-2})^4)$, and one sixth of the penalty, $e^{-\gamma_2 t_0^2/48} \leq g_0$, pays it, because $t_0^2 \geq \rho^{-1} A_1^2 (\log g_0^{-2})^{2p_0}$ by Paper, Lemma 3.5; there is one such penalty for every expanded cutoff, and no other part of the expansion enlarges a domain. The pieces are therefore terms of $\mathbf{B}^{(1)}$ with the bounds of Step 4, and the pairs of long walks in $-\frac{1}{2}\langle m^{\mathrm{loc}} - m, H_{DD}(m^{\mathrm{loc}} - m)\rangle$, whose sizes carry the factor $e^{-\delta_{\mathrm{tl}}\rho_2/2} = (K_{\mathrm{tl}} m_L t_0^3)^{-1}$, are terms of $\mathbf{B}^{(1)}$ as well. *q.e.d.*

Proof of (ii). The sum of the refined terms with given (Ω_1, Λ_1) is Bałaban's term with these labels, as said after Definition 5.1. Given the field and (Ω_1, Λ_1), exactly one sequence of bad sets has all its test functions equal to one, so that $\chi_\Lambda := \sum \mathbf{1}_{\mathrm{lab}}$ over the refined labels with retreated region Λ is the indicator of $\{\Lambda^{\mathrm{fin}} = \Lambda\}$, and it vanishes unless $\Lambda \subset \Lambda^{(0)}$. By (i) the action and the operation of a refined term depend on its labels only through Λ, besides Ω_1, Λ_1, and S_1, the last inside $\mathbf{T}_1$ as in [10, (1.29)]. Summing over the refined labels with given Λ therefore gives $\chi_1(\Omega_1)\mathbf{T}_1^{\Lambda_1,\Lambda} \exp A_1^{\Lambda}$, with the operation and the action described in (ii). The label Λ_1 stays an outer label, because the kernel of $\mathbf{T}_1$ depends on it through $\chi(\Omega_1 \cap \Lambda_1^c, S_1)$, $U_1(\Omega_1 \cap \Lambda_1^c, V)$, the change of variables, and $\mathbf{H}_{1,Ax}$, which has only "the exponential decay property" [10, p. 251], so that moving Λ_1 inside $\mathbf{T}_1$ would destroy the factorisation [10, (2.19)]. The domains satisfy [10, (2.1)] and the condition after [10, (2.3)] by Lemma 5.2(i).

For the factorisation we take the regions $\mathcal{X}$ of [10, (2.19)] to be the components of Λ^c for the relation of cubes having a point in common; distinct regions are at distance at least ℓ. The form (5.9) couples bonds of different components of $\Omega_1 \cap \Lambda^c$, taken for the relation of bonds at distance less than $\ell/2$, only through its second term, since the range $4L$ of H is less than $\ell/2$. For bonds x, y of different components that term is at most $C' e^{-\mu_H(\mathrm{dist}(x,y)-8L)}$ by (5.8), with C' depending on L and G, and the fields there are bounded by t_0, so the cross-component part at a boundary bond is at most $C' t_0^2 \sum_{\mathrm{dist}(x,y)\geq \ell/2} e^{-\mu_H(\mathrm{dist}(x,y)-8L)} \leq C'' t_0^2 e^{-\mu_H \ell/4} \leq 1$ for small g_0, since $\mu_H \ell$ is at least a constant times $(\log g_1^{-2})^r$ while $\log t_0^2$ is of the order of $\log\log g_0^{-2}$; these blocks are moved into $\mathbf{B}^{(1)}$ as on [10, p. 258], with its format.

It remains to show that the cutoff factorises over the regions, that is, $\chi_\Lambda = \prod_{\mathcal{X}} \chi_{\Lambda,\mathcal{X}}$ on the range $\Lambda \subset \Lambda^{(0)}$, where $\chi_{\Lambda,\mathcal{X}}$ is the indicator that the retreat started from $(\Lambda^{(0)})^c$, with every bond which has an endpoint outside $\mathcal{X}$ declared good, ends with the complement $\mathcal{X} \cup ((\Lambda^{(0)})^c \setminus \mathcal{X})$. Every change which the retreat makes near $\mathcal{X}$ is a face-connected set which meets $\mathcal{X}$: a collar bond has both endpoints in one region, since parts of the complement in different regions are at distance at least $\ell > 2\rho_L + 2$; $\mathcal{Q}^+(e)$ is face-connected and contains the cube of x_e; and a filling acts on one component at a time. If $\Lambda^{\mathrm{fin}} = \Lambda$, every intermediate complement lies in $\Lambda^c = \bigcup \mathcal{X}$, a face-connected set inside $\bigcup \mathcal{X}$ lies in one region, and by induction over the rounds the global retreat agrees near each $\mathcal{X}$ with the local one, which therefore ends with the complement stated. Conversely, if each local retreat ends so, the same induction shows that the global one never adds a point outside $\bigcup \mathcal{X}$, and it stops at the last local stop. The other factors of the cutoff are bounds on single bonds, each in one region, and Bałaban's $\chi(\Omega_1 \cap \Lambda_1^c, S_1)$ and $\zeta(\Omega_1^c)$, which factorise over the components of Λ_1^c by [10, (2.19)] for his own $\mathbf{T}_1$; each component of Λ_1^c lies in one region. Hence $\mathbf{T}_1^{\Lambda_1,\Lambda}$ has the form [10, (2.19)–(2.21)] at $j = 0$, with one integrand factor which Bałaban's has not, the functions of the rows of the constraint off Λ, which are positive, gauge invariant, and local, like the coefficients of [10, (1.29)]. The later steps leave $\mathbf{T}_1^{\Lambda_1,\Lambda}$ untouched [10, (2.22), p. 258], and [13] uses its integrand only through positivity and gauge invariance [13, p. 379]. *q.e.d.*

Proof of (iii). The geometry. Each G lies in one component of Λ^c: the cubes of the cover of $P_1 \cup Q_1$ lie in Ω_1^c, which is a union of cubes, and $\Lambda \subset \Omega_1^{\sim -2}$; the cover of R_1 is R_1', which lies in Λ_1^c by [10, (1.20)]; and the cube of side ℓ which contains a cube of W_1 contains the initial site x_e of a bad bond, so that it meets the ball which defines $\mathcal{Q}^+(e)$ and lies in Λ^c. Let p be a point of a component $\mathcal{X}$ of Λ^c. If $p \notin \Lambda^{(0)}$, then p lies within 100ℓ of a point q of the unfilled region $Q_1'^{\sim 4} \cup (B(P_1^1)^c)^{\sim 5} \cup R_1'^{\sim}$, in the same component [1, Lemma 9.9(b)]; this region lies in Λ_1^c by [10, (1.10), (1.20)]. By Bałaban's form of [13, (1.76), p. 381] at $j = 1$, used on [13, p. 385, l. 15–16], the component of Λ_1^c which contains q lies in $\bigcup_i (Z_1^{(i)})^{\sim 2}$ over the components $Z_1^{(i)}$ of the cover of $P_1 \cup Q_1 \cup R_1$ which it contains, so that $p \in G^{\sim 102}$ for the component G which contains such a $Z_1^{(i)}$. If $p \in \Lambda^{(0)} \setminus \Lambda$, then p lies within 100ℓ of a point of $\mathcal{Q}^+(e)$ for a bad bond e, in the same component (the proof of Lemma 5.2(ii)), and every point of $\mathcal{Q}^+(e)$ lies within 3ℓ of x_e, which lies in a cube of W_1; so $p \in G^{\sim 103}$. In both cases

$p \in G^{\sim 107}$ for some $G \subset \mathcal{X}$.

The small factor. Let W consist of one bad bond with its initial site in each cube of W_1, and of one bond $e(\square')$ of $(\square'^{\sim 2})^*$ with $|A(e(\square'))|_\sigma \geq t_0$ for each cube $\square'$ of R_1, which exists on the support of $\chi^{(0)c}(R_1)$ [10, (1.18), p. 250]; a bond lies in $(\square'^{\sim 2})^*$ for at most $2 \cdot 5^d$ cubes, so $|W| \geq |W_1|_\# + |R_1|_\#/(2 \cdot 5^d)$, where $|\cdot|_\#$ counts cubes. For a component X of $\Omega_1 \cap \Lambda^c$ let $Q_X := \langle A_J, (H_{JJ} - H_{JD}C^{(0)}(\Lambda)H_{DJ})A_J \rangle$, restricted to the fields in X. It is nonnegative, the form (5.9) of the exponent is $-\frac{1}{2}Q_X$, and since $H_{JJ} - H_{JD}H_{DD}^{-1}H_{DJ}$ is the Schur complement of H, Q_X is the minimum over the fields A_D of $\langle z, Hz \rangle$, $z = (A_D, A_J)$; a lower bound of $\langle z, Hz \rangle$ which does not involve A_D is therefore a lower bound of Q_X. Let P_X be the bonds of Ω_1^* within a distance R_ϑ of $W \cap X$, where R_ϑ is the least integer with $\sigma(R_\vartheta - \rho_L)^2 m_L^2 (2\pi + 1)^2 N(R_\vartheta) \leq \vartheta_V^2/(4c_h^2)$, $\sigma(T) := (2/\gamma_L)\sum_{t \geq T} 32(2t+1)^3 e^{-\mu_H t}$ and $N(R) := 4(2R + 2\rho_L + 1)^4$; it depends on L and G only. A bad bond lies within ρ_L of I and, by Lemma 5.2(iv), at distance at least ℓ from Λ; a bond $e(\square')$ lies at distance at least $\frac{3}{4}\ell - 2$ from $D(\Lambda)$, because R_1' is at distance at least ℓ from Λ_1 [1, Lemma 3.11(i)]. Hence P_X and its boundary layer ∂P_X of width ρ_L miss $D(\Lambda)$ once $R_\vartheta + 2\rho_L \leq \ell/2$. We complete the square in the variables of P_X: with $u := -H_{P_X P_X}^{-1} H_{P_X, P_X^c} z$,

$$\langle z, Hz \rangle \geq \frac{\gamma_L}{2}|z_{P_X} - u|^2 \geq \frac{\gamma_L}{4}|z_W|^2 - \frac{\gamma_L}{2}\sigma(R_\vartheta - \rho_L)^2 m_L^2 |z_{\partial P_X}|^2 ,$$

since the Schur complement is nonnegative, $H_{P_X, P_X^c} z$ depends on $z_{\partial P_X}$ only and is at most $m_L|z_{\partial P_X}|$, and the block of $H_{P_X P_X}^{-1}$ between W and the bonds at distance at least $R_\vartheta - \rho_L$ has row and column sums at most $\sigma(R_\vartheta - \rho_L)$ by (5.8). The local gap $\gamma_L/2 \leq H_{P_X P_X}$ holds near the bad bonds by [1, Lemma 3.4], whose proof uses the position of its set only through the plaquette bounds of [1, Lemma B.4], valid there since P_X lies within $\frac{5}{8}\ell + O(1)$ of Λ_1, the $O(1)$ depending on L and G; near the cubes of R_1 it holds by the same proof, with the plaquette bound $\|U(\partial p) - 1\| \leq c_h \varepsilon_1 L^{-2} + 4c_h O(L^2)\varepsilon_0$ which the test functions of Ω_1 give on every cube $\square''$ with $\square''^{\sim 4} \subset \Omega_1$ [10, (1.8), p. 247; (1.11), p. 248]. On the support of the integrand $|z_{\partial P_X}| \leq (2\pi + 1)t_0 N(R_\vartheta)^{1/2}|W \cap X|^{1/2}$, by the bounds of the small-field functions and of the change of variables near R_1, which moves the field by at most $t_0/2$. Every bond e of W has $|z(e)|_\sigma \geq \vartheta_V t_0$, for the bonds $e(\square')$ after that change of variables because $t_0 - t_0/2 \geq \vartheta_V t_0$, and the norm h in which H is bounded satisfies $h \geq |\cdot|_\sigma/c_h$. With the choice of R_ϑ this gives

$$Q_X \geq 2c_b \vartheta_V^2 t_0^2 |W \cap X| ,$$

so that the factor $e^{-\frac{1}{2}Q_X}$ of the integrand is at most $e^{-c_b \vartheta_V^2 t_0^2 |W \cap X|}$. Since the integrand of $\mathbf{T}_1'(\mathcal{X})$ is nonnegative [13, (1.73), p. 380], its other factors are bounded as on [13, pp. 380–383]: the factors of P_1 and Q_1 of [13, p. 380], which give at least $2\gamma_0 m_A a$ per cube because $\varepsilon_1/(A_0 g_1) = \vartheta_1/(A_1 g_1) \geq \rho^{-1/2}(\log g_1^{-2})^{p_0}$ by Paper, Lemma 3.5; the volume terms of [13, p. 380, l. 25–29], among them the boundary terms of (i); the normalisations, at most $O(\log g_1^{-2})$ per site; the integrals over the fluctuation field, bounded through the Jacobian $g_0^{\dim G}$ per bond and the range $|A(b)| \leq C/g_0$ of the chart; the factors of the rows of the constraint off Λ, at most e^{2Cd} per site; and the number of refined labels with retreated region Λ, at most $2^{|S|}$, where S is the set of the bonds with both endpoints in Λ^c and initial site within $\rho_L + 1$ of $\Lambda^{(0)}$, because the labels are determined by the union of the bad sets, which lies in S [1, Lemma 9.12]. All of these are at most $\exp(O(1)M^d R_1^{d+1}(d_1'(\mathcal{X}) + 1))$, since $|\mathcal{X}|$ is at most $(MR_1)^d 3 \cdot 2^{d-1}(d_1'(\mathcal{X}) + 1)$ points of $T^{(1)}$ and $\log g_1^{-2} \leq R_1$. The factor $e^{-c_b \vartheta_V^2 t_0^2 |W|}$ is split among the components G: a bond of W serving cubes of R_1 is within $2L^2 M_2 R_1 + 1$ of each of them, so they meet at most 2^d cubes of side ℓ and at most 2^d components G, each of which receives the share 2^{-d}. With $t_0^2 \geq a/2$, each G receives the exponent $\frac{1}{2}\gamma_0 m_A a$ per cube of $P_1 \cup Q_1$ and at least $\frac{1}{2}c_b \vartheta_V^2 a |W_1 \cap G|_\# + (c_b \vartheta_V^2 a/(4 \cdot 10^d))|R_1 \cap G|_\#$, which is at least $2c_4 \vartheta_V^2 a$

times the number of generating cubes of G, by (5.3) and $\vartheta_V^2 \le \frac{1}{2}$. The generating cubes are disjoint, and $d_1'(G)+1 \le 2$ times the number of cubes of side ℓ in G, which is at most the number of generating cubes, since LM_2 divides M. So G receives at least $c_4\vartheta_V^2 a(d_1'(G)+1)$, and this is at least $\frac{1}{2}c_4\vartheta_V^2 a(d_1'(G)+1)+2p_0(g_1)$ under $\frac{1}{2}c_4\vartheta_V^2 A_1^2(\log g_1^{-2})^{p_0} \ge 2$, a condition on γ, since $d_1'(G)+1 \ge 1$. This is (5.6).

The inequality (1.82). Since $\mathcal{X}$ is connected and $\mathcal{X} \subset \bigcup_{G\subset\mathcal{X}} G^{\sim 107}$, the sets $\hat{G} := G^{\sim 108}$ have a connected union which contains $\mathcal{X}$, and the argument of [13, p. 385], with $\hat{G}$ in place of $(Z_1^{(i)})^{\sim 3}$, gives $d_1'(\mathcal{X})+1 \le \sum_G (d_1'(\hat{G})+1)$. The set $\hat{G}$ has at most 217^d times as many cubes as G, hence at most $217^d \cdot 3 \cdot 2^{d-1}(d_1'(G)+1)$ by the count of [13, p. 384], and a tree through the centres of n cubes which have points in common along a spanning tree has length at most $\sqrt{d}(n-1)$, so that $d_1'(\hat{G})+1 \le 2\cdot 217^d \cdot 3\cdot 2^{d-1}(d_1'(G)+1) \le 2d(434)^d(d_1'(G)+1)$ at $d=4$. Hence $d_1'(\mathcal{X})+1 \le 2d(434)^d \sum_{G\subset\mathcal{X}}(d_1'(G)+1)$, and $\prod_G \exp(-\frac{1}{2}c_4\vartheta_V^2 a(d_1'(G)+1)) \le \exp(-\frac{1}{4}c_4 d^{-1}(434)^{-d}\vartheta_V^2 a(d_1'(\mathcal{X})+1))$, which is [13, (1.82)] with the constant stated. The condition of [13, p. 385] for [13, (1.80)] at $j=1$ becomes $\frac{1}{4}c_4 d^{-1}(434)^{-d}\vartheta_V^2 A_1^2(\log g_1^{-2})^{2p_0} \ge O(1)2(64)^d M^d L^{d+1} R_1^{d+2}$, which holds for γ small, since $R_1^{d+2} \le (2L)^{d+2}(\log g_1^{-2})^{6r}$ and $2p_0 > 6r$. The modified form of [13, (1.76)] enters only here: on [13, p. 386] it relates the components of Z_{j+1}, $j \ge 1$, to those of Z_j and to the new regions, which the later steps form as printed from $Z_1 = \Lambda^c$, and no further retreat occurs after the first step. *q.e.d.*

Proof of (iv). Put $\lambda := \log g_0^{-2}$ and $r_0 := A_1\lambda^{p_0}$, the radius of [1, (D.1)]. By Paper, Lemma 3.5(i), $\rho^{-1/2}r_0 \le t_0 \le r_0$, and the plaquette thresholds of the tube are at most those of [1] and at least $\rho^{-1/2}$ times them; the cube size satisfies $(\log g_1^{-2})^r \le R_1 < L(\log g_1^{-2}+\log\rho)^r$. The conditions on the coupling of [1] are listed in [1, Appendix B.8], and each is of one of three kinds. An upper bound on a quantity increasing in the radius or in a threshold, such as $C_0 g_0(1+3r)^3$ small, ε_0 small, or $\delta_0 \le \delta_G$, holds a fortiori at t_0. A lower bound on the radius or on R_1, such as $r \ge 1$ or LMR_1 large, holds for small coupling, because $t_0^2 \ge r_0^2/\rho \to \infty$ and $R_1 \to \infty$. A relation among the thresholds holds as printed, because the sequence $(\hat{g}_j)$ satisfies [10, (2.6)–(2.8)] (Paper, Lemmata 3.2 and 3.5). The lines of [1] in which the functional form of the radius enters are of the same kinds, with one exception, the bound $e^{-c_{\text{ph}}r^2} \le g_0$ of [1, (9.29)(iii)], derived there from $r \ge A_1\lambda$. At the radius t_0 we have $t_0^2 \ge \rho^{-1}A_1^2\lambda^2$, so that

$$e^{-c_{\text{ph}}t_0^2} \le \exp\Big(-\frac{c_{\text{ph}}}{\rho}A_1^2\lambda^2\Big) = g_0^{(2c_{\text{ph}}/\rho)A_1^2\lambda} \le g_0$$

once $(2c_{\text{ph}}/\rho)A_1^2\lambda \ge 1$, which is (5.7) with $c_{\text{ph}} = 9c_n\alpha_G/128$. Since $g_0 \le \rho^{1/2}\hat{g}_0 \le \rho^{1/2}\gamma$ by Paper, Lemma 3.2, it holds for γ small. Where [1] uses the radius from below in an exponent, as in the parameter $\tau = C[g_0(1+r)^3 + e^{-cr^2}]$ or the rates of its tail bounds, the bound at t_0 is the bound at r_0 with the constant c replaced by c/ρ, and since $\rho \le 3/2$ the comparisons between the powers $2p_0 > 4r$ of λ in which these rates are used are not affected. In (i)–(iii) the radius t_0 has been used throughout. *q.e.d.*

6 The Test Functions at the Averages of the Background

At the step $k+1$ every factor of a dressed loop is expanded around the block fields $V_j^* := M^j U_{k+1}$, $j \le k$, of the new background U_{k+1}, and the expansion is useful only if these block fields lie in the support of the test functions of the step j: otherwise the bounds which the paper draws from the passed test functions are not available at the point of expansion. For $j = k$ and the cubes of the domain $\Omega_{k+1}^{\sim -1}$ this is Bałaban's computation [10, (3.16)–(3.19), pp. 268–269], which compares $V^{(k)} = M^k(U_{k+1})$ with the reference fields $V_{\square'}^{(k)}$ of the cubes

$\square'$ of side $L^2M_2R_{k+1}$ of [10, (3.16)]. Two cases are not covered by it. The cubes of the band $\Omega^{\sim}_{k+1} \setminus \Omega^{\sim -1}_{k+1}$ lie in no cube of [10, (3.16)], and for a cube $\square'$ whose neighbourhood $\square'^{\sim 4}$ crosses the boundary of Ω_{k+1} the constraints of U_{k+1} on $\square'^{\sim 4}$ are of two scales, so that the identity [10, (3.17)] does not hold as printed. Bałaban treats a set of this kind on [10, p. 269], the set $\Lambda^c_{k+1} \cap \Lambda_k$, whose boundary does not cross that of Ω_{k+1}; here the crossing has to be dealt with.

Throughout, $1 \leq i \leq k$, the cubes $\square$ are those of side LM_2R_i of [10, (2.17), p. 257], and $\square'$ is the cube of side $L^2M_2R_{i+1}$ with $\square \subset \square'$ [10, p. 266]. The *support of the step* j is the set of the block fields V_j at which the test functions $\chi_j(\square)$ of [10, (2.17)], $\square \subset \Omega_j$, and the axial factors χ_{Ax} of [10, (3.9), p. 266] of the term do not vanish. We write $\mathbf{B}_j(\Omega)$ for the determining set of [10, p. 256] localised in Ω, $U(\mathbf{B}, \cdot)$ for the minimal configuration with the constraints $\mathbf{B}$ [10, (2.13), p. 257], and B_3 for the constant of [7, Theorem 1, p. 279]; c_{12} is the constant of the bounds [4, (159)–(163), p. 42] on the averages, and

$$C_9 := 4 + 4c_{12}B_3 \ . \tag{6.1}$$

Proposition S.8 (The Test Functions at the Averages of the Background)**.** *Let γ be sufficiently small and $A_1/A_0 < 1/(20B_3C_9)$. Consider a labelled term of the step $k+1$ of the construction, with domains Ω_j and background U_{k+1}, and let $1 \leq i \leq k$. If V^*_{i+1} lies in the support of the step $i+1$, then V^*_i lies in the support of the step i. More precisely:*

(a) for every cube $\square \subset \Omega^{\sim}_{i+1}$,

$$\sup_{p \subset \square^{\sim}} |U_{i,\square}(V^*_i, \partial p) - 1| < \varepsilon_i \eta^2 \ , \tag{6.2}$$

so that the test function of $\square$, sharp or soft, equals one at V^*_i;

(b) for every cube $\square \subset \Omega_i \setminus \Omega^{\sim}_{i+1}$, the test function of $\square$ at V^*_i is its value at V_i, which is a factor of the term and does not vanish on its support, so that

$$\sup_{p \subset \square^{\sim}} |U_{i,\square}(V^*_i, \partial p) - 1| < (1+\kappa_s)\varepsilon_i \eta^2 \ , \tag{6.3}$$

with κ_s the width of the soft test functions (Paper, Definition 3.7) and $\kappa_s = 0$ for the sharp ones;

(c) $|V^*_i(y,x) - 1| < \varepsilon_i$ for $y \in (\Omega^{\sim}_{i+1})^{(i+1)}$ and $x \in B(y)$.

The distinction between (a) and (b) matters for the soft test functions only. A sharp test function which does not vanish equals one, but a soft one which does not vanish may take any value in $(0,1]$, and where the block field is unchanged, on the cubes of (b), nothing improves the value which the term already has. On the cubes of (a) the block field is the average of the new background, and the proof gives the strict bound (6.2) with a margin of its own; these are the cubes on which the paper uses the proposition (Paper, Section 6.1), while on the cubes of (b) it uses only that the factor of the term is the same function of the same field.

Since $V^*_{k+1} = V_{k+1}$ lies in the support of the step $k+1$ on every term, the proposition gives the statement for every $i \leq k$ by a downward induction. The proof uses Bałaban's estimates [10, (3.6)–(3.9), pp. 266–267] at $V_i = V^*_i$, which hold for any block field once two inputs are available: the reference configuration $U_{i+1,\square'}$ lies in the support of the test function [10, (3.2), p. 265], and the argument of the function $\mathbf{H}_{i,\square}$ in [10, (3.6)] is small. The second input is the localisation of the minimiser.

Proof. Step 1: the block fields of the background. The average M^jU is the j-fold iterate of the one-step average [9, (0.21), p. 256], and the average on a bond c depends only on the bonds of the two blocks $B(c_-) \cup B(c_+)$ [4, (42)–(43), p. 24]. On Γ_{i+1} the field V_i^* is the minimiser $V^{(i)}(V_{i+1}^*) = M^i(U_{i+1})$ of [10, (3.10), p. 266], by the hypothesis on V_{i+1}^* and the uniqueness of the minimal orbit [10, p. 256]; on $\Gamma_i = \Omega_i^{(i)} \setminus \Omega_{i+1}^{(i)}$ it is V_i, since the constraints of U_{k+1} there are those of V_i [10, (2.10)–(2.12), p. 256]; and on Ω_i^c it is $V_i = \bar{V}_{i-1}$ by the same constraints at the scale $i-1$, the δ-function of the step i, and the locality of the average.

Step 2: the cubes away from Ω_{i+1}. Let $\Box \subset \Omega_i \setminus \Omega_{i+1}^{\sim}$. Its distance from Ω_{i+1} is at least LMR_{i+1}, and $\Box^{\sim 4}$ extends at most $4LM_2R_i \leq 4L^2M_2R_{i+1}$ beyond $\Box$, because $R_i \leq LR_{i+1}$ (Paper, Lemma 3.4(ii)). Hence $\Box^{\sim 4} \cap \Omega_{i+1} = \emptyset$ for $M \geq 4LM_2$, and by Step 1 $V_i^* = V_i$ on $\Box^{\sim 4}$. The function $U_{i,\Box}$ depends on the block field on $\Box^{\sim 4}$ only [10, (2.16), p. 257], so the test function of $\Box$ at V_i^* is the test function at V_i. This function is a factor of the term: the small-field factors of these cubes are present by the resummation of [10, p. 267] into $\zeta(\Omega_{i+1}^c)$, and no $\mathbf{R}$-operation has removed them [12, p. 177, (ii)]. On the support of the term it does not vanish, and a factor $\bar{\chi}(f_\Box/t)$ of Paper, (3.35), which does not vanish has $f_\Box < (1+\kappa_s)t$ (Paper, Lemma 3.8(ii)), which is (6.3). This is (b); it does not give the value one for a soft test function, and nothing in what follows uses more.

Step 3: the reference configuration lies in the support of its test function. Let $\Box \subset \Omega_{i+1}^{\sim}$, so that $\Box \subset \Box' \subset \Omega_{i+1}^{\sim}$ [10, p. 266]. "On the domain $\Omega_{k+1}^{\sim 2}$ there are only the small field characteristic functions from the above decompositions of unity" [10, p. 265], in every term of the resummation over the large-field regions, by [10, (3.5), p. 265]; for $\Box' \not\subset \Omega_{i+1}$, that is $\Box' \subset \Omega_{i+1}^c$ since $\Box'$ lies in one cube of side LMR_{i+1}, they are common factors of $\zeta(\Omega_{i+1}^c)$ [10, p. 267]. So $U_{i+1,\Box'}$ passes [10, (3.2)] on the support.

Step 4: the cubes with $\Box'^{\sim 4} \subset \Omega_{i+1}$. These include all $\Box' \subset \Omega_{i+1}^{\sim -1}$, because $4L^2M_2R_{i+1} \leq LMR_{i+1}$. On $\Box'^{\sim 4}$ the constraints of U_{i+1} are those of $\mathbf{B}_{i+1}(\Box'^{\sim 4})$, so that [10, (3.17), p. 268] is an identity, and Bałaban's derivation of [10, (3.19)] applies: its inputs are the uniqueness of the minimal orbit [7, Theorem 1, p. 279], the decay [7, (190), p. 308], and the bounds [4, (106)–(108), p. 33; (159)–(163), p. 42], none of which refers to the position of $\Box'$ inside Ω_{i+1}. Bałaban states the result for $\Box' \subset R_{k+1}^c \subset \Omega_{k+1}^{\sim -1}$, and the argument gives it for every $\Box'$ with $\Box'^{\sim 4} \subset \Omega_{i+1}$. Then the argument of $\mathbf{H}_{i,\Box}$ is at most $4\delta_i$ off the boundary layer and [10, (3.7)–(3.8)] hold as printed.

Step 5: the crossing cubes. Let $\Box'^{\sim 4}$ meet Ω_{i+1}^c. There the constraints of U_{i+1} are of the scale i, and we compare U_{i+1} with the minimiser of a determining set of two scales,

$$\mathbf{B}_{\mathrm{mix}} := \mathbf{B}_{i+1}(\Box'^{\sim 4})|_{\Omega_{i+1}} \cup \mathbf{B}_i(\Box'^{\sim 4})|_{\Omega_{i+1}^c}\ , \qquad \hat{U}_{\Box'} := U(\mathbf{B}_{\mathrm{mix}}, \cdot)\ , \tag{6.4}$$

the set of [10, (2.14), p. 257] for a boundary in several layers, as on [10, p. 269], with all sets on the lattice of U_{i+1} and the constraints of the scale i being those of the same blocks. Let $\Omega_n(\Box'^{\sim 4})$ be the domains of $\mathbf{B}_{i+1}(\Box'^{\sim 4})$ [10, p. 256] and put $\Omega_n' := \Omega_n \cap \Omega_n(\Box'^{\sim 4})$.

(1) $\mathbf{B}_{\mathrm{mix}}$ *is a determining set of the class of [7, (1), p. 277].* The sets Ω_n' are unions of blocks of the scale n, as intersections of two such unions, and they are nested. Since $(\Omega_n')^c = \Omega_n^c \cup \Omega_n(\Box'^{\sim 4})^c$ and Ω_{n+1}' lies in Ω_{n+1} and in $\Omega_{n+1}(\Box'^{\sim 4})$, the distance from $(\Omega_n')^c$ to Ω_{n+1}' is at least the smaller of the two distances which the global sequence and the local one already satisfy; near a crossing of the two boundaries the condition involves $(\Omega_n')^c$ only, so the crossing imposes nothing further.

(2) *The localisation identity.* Let W be a configuration on $\Box'^{\sim 4}$ with the same $\mathbf{B}_{\mathrm{mix}}$-averages as U_{i+1}, glued to U_{i+1} outside. We claim that for every bond c of the scale n with no endpoint in Ω_{n+1}' the glued field and U_{i+1} have the same average of the scale n. For

$n = 0$, such a bond either lies in $\Omega'_0 \setminus \Omega'_1$, where $\mathbf{B}_{\text{mix}}$ fixes the field itself [7, (3), p. 277], or outside $\Box'^{\sim 4}$, where the two fields agree. Assume the claim at $n-1$. If c has an endpoint in Ω'_n, it is a constraint of $\mathbf{B}_{\text{mix}}$, and both of its blocks lie in $\Box'^{\sim 4}$, because the block of that endpoint lies in $\Omega_n(\Box'^{\sim 4})$, whose distance from the complement of $\Box'^{\sim 4}$ is at least the side of a block of the scale n [10, p. 256]; there the glued field is W, which has the required average. Otherwise $B(c_-) \cup B(c_+)$ misses Ω'_n, every bond of the scale $n-1$ with both endpoints in these blocks has no endpoint in Ω'_n, and the claim at $n-1$ with the formula [9, (0.4), p. 253] for the one-step average gives it at n. A global constraint of the scale m is a bond of Γ_m, which has no endpoint in $\Omega_{m+1} \supset \Omega'_{m+1}$, so the glued field satisfies all constraints of U_{i+1}. A plaquette with a bond outside $\Box'^{\sim 4}$ has its other bonds in $\Omega'_0 \setminus \Omega'_1$, so the difference of the actions of the glued field and of U_{i+1} is the difference of the local functionals [7, (5), p. 277] on $\Box'^{\sim 4}$. Since U_{i+1} is a minimiser of the global problem, its restriction is a critical point of the problem with the constraints $\mathbf{B}_{\text{mix}}$, and it lies in the space [7, (6), p. 278] of that problem, because its regularity at a point of the global scale m is the bound of the scale m, stronger than that of the local scale $n \le m$. By [7, Theorem 1, p. 279] the critical orbit is unique, so that $U_{i+1} = \hat{U}_{\Box'}(M^{\cdot}(U_{i+1}))^{u^{-1}}$ on $\Box'^{\sim 4}$ with $u = 1$ on $\mathbf{B}_{\text{mix}}$ [7, (4), p. 277]. The same induction, with the domains $\Omega_n(\Box'^{\sim 4})$ in place of the global ones, shows that the constraints of $\mathbf{B}_{\text{mix}}$ imply those of $\mathbf{B}_{i+1}(\Box'^{\sim 4})$, so that $U_{i+1,\Box'}$, which minimises under the coarser set, is also the minimiser for $\mathbf{B}_{\text{mix}}$ with its own averages as data.

(3) *The argument of* $\mathbf{H}_{i,\Box}$. Following [10, (3.6)] with the reference $U_{i+1,\Box'}$, we split the argument $(1/i)\log[V_i^*(V_{\Box'}^{(i)})^{-1}]$ on the blocks of Γ_{i+1} in $\Box^{\sim 4}$, off the boundary layer of width $2M_1$, at $M^i\hat{U}_{\Box'}$. The first part compares $V^{(i)} = M^i(U_{i+1})$ with $M^i\hat{U}_{\Box'}$. By (2) and the representation [7, Proposition 9, p. 309], $U_{i+1} = (\exp(iL^{-1}\eta\hat{\mathbf{H}}(\cdot))\hat{U}_{\Box'})^{\hat{u}^{-1}}$, where the argument of $\hat{\mathbf{H}}$ vanishes off the boundary layer of $\Box'^{\sim 4}$ and is at most $44d^2B_3\varepsilon_{i+1}$ on its blocks in Ω_{i+1} [10, p. 268], at most $30d^2L^2B_3(1+\beta_0)\varepsilon_i$ on those in Ω^c_{i+1} [10, p. 266], and $O(\varepsilon_i)$ on the constraints of the scale i which straddle $\partial\Omega_{i+1}$. The decay [7, (190), p. 308] over the distance LM_2R_{i+1} and the bounds [4, (106)–(108), (159)–(163)] on the averages then give, as [10, (3.19)],

$$|V^{(i)}(M^i\hat{U}_{\Box'})^{-1} - 1| < O(1)44d^2B_3^2(1+\beta_0)\varepsilon_i e^{-R_i} \quad \text{on } \Box'^{\sim 2} .$$

The second part compares $M^i\hat{U}_{\Box'}$ with $V_{\Box'}^{(i)} = M^i(U_{i+1,\Box'})$. Both $\hat{U}_{\Box'}$ and $U_{i+1,\Box'}$ are minimisers for $\mathbf{B}_{\text{mix}}$, with data which agree on the blocks in Ω_{i+1} and differ on those in Ω^c_{i+1} by $X := (1/i)\log[V_i(V_{\Box'}^{(i)})^{-1}]$. By the representation [7, Proposition 9] for $\mathbf{B}_{\text{mix}}$, $\hat{U}_{\Box'} = (\exp(i\eta\mathbf{H}_{\text{mix}}(X))U_{i+1,\Box'})^{u^{-1}}$. On $(\Box'^{\sim 2})^{(i)*}$ we have $|X| < 4\delta_i$, by the small-field test function [10, (3.3)] of $\Box'$, which is a factor of the term on $\Omega^{\sim 2}_{i+1}$, and $|(1/i)\log x| \le 2|x-1|$. On $\Box'^{\sim 4} \setminus \Box'^{\sim 2}$ we have $|X| < O(1)\varepsilon_i$, because V_i and $V_{\Box'}^{(i)}$ are regular there and have the same averages of the scale $i+1$: $V_{i+1} = \bar{V}_i$ on Ω^c_{i+1}, and $M^{i+1}U_{i+1,\Box'} = V_{i+1}$ on every bond of the scale $i+1$ with both blocks in $\Box'^{\sim 4}$, by the induction of (2) with the local domains and the identity $M^{i+1}(Q^{s*}_{i+1}V_{i+1}) = V_{i+1}$; the bound is then the reasoning of [6, (1.65), p. 87], that is, [6, Lemma 1, (1.24)–(1.25), p. 79] applied block by block, which needs only the bounds on $B(c_-) \cup B(c_+)$ [6, p. 80], with the axial gauge of [10, (3.9)] for $V_{\Box'}^{(i)}$ and of χ_{Ax} for V_i. On the boundary layer $|X| < O(1)L^2B_3(1+\beta_0)\varepsilon_i$ by the same lemma. These data lie in the domain of [7, Proposition 9] for γ small, since every bound is $O(\varepsilon_i)$.

(4) *The estimate.* The argument of $\mathbf{H}_{i,\Box}$ is, up to quadratic corrections, $Y_1 + Y_2 + Y_3$ with $Y_1 = X$ on the blocks in Ω^c_{i+1} of $\Box'^{\sim 2}$, $|Y_1| < 4\delta_i$; Y_2 the first part above, $|Y_2| < O(1)\varepsilon_i e^{-R_i}$; and Y_3 the average of the scale i of $\eta\mathbf{H}_{\text{mix}}$ on the blocks of Γ_{i+1}, $|Y_3(b)| \le c_{12}B_3(4\delta_i + O(1)\varepsilon_i \sum_{b'} e^{-\delta\,\mathrm{dist}(b,b')})$ by [7, (190)] and [4, (159)–(163)], the sum over the blocks b' of $\Box'^{\sim 4} \setminus \Box'^{\sim 2}$. Every point of $\Box^{\sim}$ is at distance at least $LM_2R_{i+1} \ge M_2R_i$ from $\Box'^{\sim 4} \setminus \Box'^{\sim 2}$, because $\Box^{\sim}$ extends at most M_2R_i beyond $\Box'$ while $\Box'^{\sim 2}$ extends at least $2LM_2R_{i+1}$ beyond it

[10, p. 245]; hence the far data contribute to $|\mathbf{H}_{i,\Box}|$ on $\Box^\sim$ at most $C_{\mathrm{vol}} B_3^2 O(1) \varepsilon_i e^{-(\delta/2) M_2 R_i} \leq \varepsilon_i/40$ for γ small, C_{vol} counting the blocks, since $R_i \geq (\log g_i^{-2})^r$. By [7, (190)] for $\mathbf{H}_{i,\Box}$ and its gradient,

$$|\mathbf{H}_{i,\Box}|,\ |\nabla^\eta \mathbf{H}_{i,\Box}| \leq B_3 C_9 \delta_i + O(1) e^{-R_i} \varepsilon_i + \tfrac{1}{40}\varepsilon_i < \tfrac{1}{10}\varepsilon_i \quad \text{on } \Box^\sim \ ,$$

since $\delta_i = (A_1/A_0)\varepsilon_i$ and $B_3 C_9 A_1/A_0 < 1/20$; the quadratic corrections are of the order of ε_i^2. This is [10, (3.7)] with $4B_3$ replaced by $B_3 C_9$.

Step 6: conclusion. By Steps 4 and 5, the bound [10, (3.7)] holds for every $\Box \subset \Omega_{i+1}^\sim$, and with the identity [10, (3.6)] and [6, (1.43)] it gives [10, (3.8)], which is (6.2); this is (a), and with Step 2 every $\Box \subset \Omega_i$ is covered by (a) or (b), so that V_i^* lies in the support of the step i. For the soft test functions the regularity of $U_{i+1,\Box'}$ which enters the first term of [10, (3.8)] is that of a soft test function which does not vanish, at most $(1+\kappa_s)\varepsilon_{i+1}$ times the plaquette area (Paper, Lemma 3.8(ii)), and $2(1+\kappa_s)(1+\beta_0)L^{-2} + 4/10 < 1$, so (6.2) still holds and the soft test function is one. The axial factor holds on Γ_{i+1} because $V_i^* = V^{(i)}$ satisfies the constraint $V^{(i)}(y,x) = 1$ of [10, (3.10)], on the other points of $(\Omega_{i+1}^\sim)^{(i+1)}$ by [10, (3.9)] at $V_i^* = V_i$, and farther out by the resummation of [10, p. 267], which contains χ_{Ax}. *q.e.d.*

7 Attached Insertions and the Large-Field Factors

The insertions of a dressed loop which are attached to a large-field region carry no deviation of the background, and nothing makes them smaller from one step to the next: a segment of the step n which runs along the large-field data of the step $j < n$ contains L^{n-j} traversals of the step j, each with an attached part of the size of $\hat\rho_j$. Paper, Section 6.4, therefore pays their total over each component Z of the large-field region of the step j once, from the factor which Bałaban's large-field estimates attach to Z [13, pp. 381, 384]. For this to work three things must be shown: that the attached parts do not grow during the lifetime of Z, although they are evaluated at new backgrounds, partly integrated by the later steps, and merged with other insertions; that the scalar terms near the large-field data are local and of the size of $\hat\tau^2$ per traversal; and that Bałaban's induction for the factors $\exp(-\kappa_j(Z))$ [13, (1.80)–(1.89), pp. 384–387] survives the additional costs, with a reserve left over.

We use the notation of Paper, Section 6. For a group W made at the step j and attached to a component Z of Z_j, let $\mathbf{Coll}_j \supset Z$ be the union of Z and of its boundary layer, and let $\mathbf{ext}^{(n)}(W)$ be the largest distance, in sides of the cubes of π_n, from a cube meeting the domain or the dependence set of W to $\mathbf{Coll}_j$; it decreases by the factor L^{-1} per step. The *weighted mass* of W is

$$w(W) := e^{\nu\sigma_j(W)} e^{\frac{3}{8}\nu_e \mathrm{ext}^{(n)}(W)} e^{\theta(W)} \|Y_W\| \ , \tag{7.1}$$

with the size $\sigma_j(W)$, the reserve ν, the extent rate $\nu_e = \nu/(2C_{\mathrm{box}})$, and the interaction exponent $\theta(W)$ of Paper, Definition 5.2, and the norm the supremum over the polydiscs of Paper, Definition 5.2(iii), restricted to the subdomain on which the attached bounds are stated. The insertions attached at the step j to Z, together with those attached to its boundary layer through the marks $\mathfrak{m}_j$, form the *species* of (j, Z); a group made later which contains a part of an insertion of this species belongs to it as well. Let c_W be the constant part of the shares which an old member pays in a new group, for its root, its positions, and its transports; it depends on ν, L, and M only, and the factor e^{c_W} is taken once for every insertion. Let $A_j(Z)$ be the total of $e^{c_W} w(D)$ over the insertions D of the species at its production, each counted once at its first traversal, and put

$$\delta_n := 2\bar{c} e^{-\frac{25}{8}\frac{\nu_e}{L} R_{n+1}} \ , \qquad S_R := \sum_{m=1}^{K'-1} e^{-\frac{25}{8}\frac{\nu_e}{L} R_{m+1}} \ , \tag{7.2}$$

the sum over the steps $m \to m+1$ at which the loops are carried, which end at the stopping step K'. Here $\bar{c} := e^{\frac{3}{8}\nu_e} \max(2K_\beta, C'') \sup_{m\leq K'} b_m/\bar{\alpha}_m$, with b_m the size of the fresh activities of the step m, $\bar{\alpha}_m = g_m \min(C_0^\alpha, C_1^\alpha)(\log g_m^{-2})^{q_0}$ the radius of the polydiscs, and constants K_β and C'' of the class, which depend on L, M, κ, ν, and the constants of the size bound and not on the step or on K; since $q_0 > p_0$, $b_m/\bar{\alpha}_m$ is bounded by a negative power of $\log g_m^{-2}$.

Proposition S.9 (Attached Insertions). *Let $L \geq 16$, and let γ_q be sufficiently small, depending on $c_{\mathcal{C}}$. Fix a labelled term and a component Z of the large-field region Z_j of a step $j \leq K'$.*

(i) Production. *$A_j(Z) \leq e^{c_W} C_A(j)(d'_j(Z)+1)$, where $C_A(j)$ is g_j or g_j^2 times a polynomial in R_j and $\log g_j^{-2}$ for $j \geq 2$, and a polynomial in $\log g_0^{-2}$ for $j = 1$.*

(ii) Carrying. *$2\bar{c}S_R \leq \log 2$. At every later step $n+1$ of the lifetime of the component which contains Z, the products of the insertions of the species of (j, Z) which are present at the step $n+1$, summed over the terms of the multilinear expansion of the words, are at most*

$$\prod_D \Big(1 + e^{c_W} w(D) \prod_{m=j}^{n} (1+\delta_m)\Big) \leq \exp(2A_j(Z)) \ , \tag{7.3}$$

the product over the insertions D of the species made at the step j. In particular an attached insertion is not increased by its evaluation at the background U_{n+1}, and its parts which the step $n+1$ integrates or merges into new groups have weighted mass at most δ_n times its own.

(iii) Exposed segments. *The difference $E^{\partial}_{k+1}(\ell) = \log S^\circ(\ell) - \log S^\circ(\ell_0)$ of Paper, Section 7, is a sum $\sum_X e_X^{(k+1)}$ over the clusters X of the reference gas which meet the large-field region Z_{k+1} or its boundary layer, each $e_X^{(k+1)}$ a function of the data in X attached to one component of Z_{k+1}, and*

$$\sum_{X \ni \mathfrak{n}} |e_X^{(k+1)}| e^{\theta(X)} \leq C\hat{\tau}_{k+1}^2 \ , \qquad \sum_{X \cap Q \neq \emptyset} |e_X^{(k+1)}| e^{\frac{\nu}{64}\#_{k+1}(X)} \leq C'_e M R_{k+1} \hat{\tau}_{k+1}^2 \ , \tag{7.4}$$

for every traversal $\mathfrak{n}$ and every cube Q of side MR_{k+1}, with $\hat{\tau}_{k+1}^2 = \max(\tau_{k+1}^2, C\hat{\rho}_{k+1})$ and constants of the class.

(iv) The payment. *Let ε_A be the coefficient of $d'_j(Z)+1$ in the lifetime total of the costs (i)–(iii), Paper, (6.10) and (6.11). Then Bałaban's induction [13, pp. 384–389] holds for $\hat{\kappa}_j(Z) - \mathrm{Res}(Z) - \mathrm{Pol}(Z)$, with $\hat{\kappa}_j(Z)$ the exponent of Proposition S.11, in which half of the gain of every new region is set aside, and with the summand of [13, (1.80)] enlarged by these costs, where $\mathrm{Res}(Z) = \frac{1}{8}K_j(d'_j(Z)+1)$ and $\mathrm{Pol}(Z) \geq \frac{1}{2}K_j(d'_j(Z)+1)$ at the creation of Z, with $K_1 = \frac{1}{8}c_4 d^{-1}(434)^{-d}\vartheta_V^2 A_1^2(\log g_1^{-2})^{2p_0}$ (half of the coefficient of Proposition S.7(iii)) and $K_{j+1} = \frac{1}{12}\gamma_0(14)^{-d}A_1^2 p_0^2(g_{j+1})$; and the reserve satisfies $4\,\mathrm{mass}(Z) \leq \mathrm{Res}(Z)$ through the induction, where $\mathrm{mass}(Z)$ is the total of the left side of the second bound of (7.4) over the cubes of the regions which descend from Z, multiplied by 13^d.*

(v) The cross species. *$2\bar{c} \leq 1$. For an insertion D of the cross species of Paper, Definition 5.2(ii), let $\|D\|$ be the supremum over the same subdomain of the projective norm of Paper, (5.5), and let its weighted mass after the step n be $w(D) := e^{\lambda_n \nu \sigma(D)} e^{\theta(D)} \|D\|$, with the rate $\lambda_n := 1 - \sum_m (e^{-p_0(g_m)/4} + e^{-31R_m/512})$, the sum over the $\mathbf{R}$-steps $m \leq n$; then $\lambda_n \geq \frac{31}{32}$. At a step $n \to n+1$, before the $\mathbf{R}$-operation of the step $n+1$, the products*

of the insertions of the cross species made at a step $j \leq n$ which are present at the step $n+1$, summed over the terms of the multilinear expansion of the words, are at most

$$\prod_{D} \Big(1 + e^{c_W} w(D)(1+2\bar{c})\Big)\,, \tag{7.5}$$

the product over the insertions of that species present at the step n. In particular such an insertion is not increased by its evaluation at U_{n+1}, and its parts which the step integrates or merges into new groups have weighted mass at most $2\bar{c}$ times its own, so that the total weighted mass of the species grows by at most the factor $1+2\bar{c} \leq 2$.

Proof. (ii), the coupling sum. By [10, (2.5), p. 255], $R_m \geq (\log g_m^{-2})^r$ with $r \geq 2$, and by Paper, Theorem 4.1, $u_m := g_m^{-2}$ satisfies $u_m \geq u_{m'} + b(m'-m) - C_W$ for $m \leq m'$. The sum S_R runs over the indices $2 \leq m+1 \leq K'$, at which the radii R_{m+1} and the couplings are those of the steps before the stopping step. Two such indices $m < m'$ with $u_m, u_{m'}$ in one interval $[X + sb, X + (s+1)b)$, $X := \min_{m \leq K'} u_m$, satisfy $b(m'-m) - C_W < b$, so that such an interval contains at most $2 + C_W/b$ of the numbers u_m. With the decreasing function $f(u) := e^{-\frac{25}{8}\frac{\nu_e}{L}(\log u)^r}$,

$$S_R \leq \Big(2 + \frac{C_W}{b}\Big)\Big(f(X) + \frac{1}{b}\int_X^\infty f(u)\,\mathrm{d}u\Big) \leq \Big(2 + \frac{C_W}{b}\Big)\Big(f(X) + \frac{X^{-1}}{b}\Big)\,,$$

because $f(u) \leq u^{-2}$ for $u \geq X$ once $\frac{25}{8}\frac{\nu_e}{L}(\log X)^{r-1} \geq 2$. Since $X \geq \gamma_q^{-2}$ on the steps $m \leq K'$, $2\bar{c}S_R \leq \log 2$ for γ_q small.

(ii), evaluation. At the step $n \to n+1$ every carried insertion D is evaluated at the points $e^{i\eta(\mathbf{H}_n(s,tB')+a'')}U_{n+1}$ of the expansion of the step, and these lie in the subdomain of the step n around the real point $U_n(V_n)$ on which the attached bound of D is stated: the analyticity domains shrink at each step by the powers of $\frac{1}{2}$ in [10, (2.36)–(2.39), p. 261], and "the exponential decay of the minimizing function $\mathbf{H}_k$ imply that this difference is still much greater than bounds on this function" [10, p. 277], with the bound [11, (1.18), p. 6] on all the norms of the fluctuation field. The attached bound is a supremum over this subdomain, and an attached insertion carries no deviation, so its norm at every point reached, in particular at U_{n+1}, is at most its bound at production. Its transports do not grow either: a transport along an arc of $\ell_j(W)$ traversals of the step j has, on the subdomain of the step $n+1$, norm at most $e^{c'_\pi \ell_j(W)\alpha_{1,n+1}L^{-(n+1-j)}} \leq e^{c'_\pi \ell_j(W)\alpha_{1,j}}$, because $\alpha_{1,n+1} \leq (1+\beta_0)^2(n+1-j)^{1/2}\alpha_{1,j}$ by Paper, Lemma 3.5(ii), and $m^{1/2}L^{-m} \leq L^{-1} \leq (2(1+\beta_0)^2)^{-1}$ for $m \geq 1$.

(ii), the touched parts. The part of D which the step $n+1$ integrates is $\int_0^1 \partial_t D(e^{i\eta\mathbf{H}_n(s,tB')}U_{n+1})\,\mathrm{d}t$, with the localisation in the parameters s of [11, (1.9)–(1.10), p. 4]. With $s = 0$ off a polymer, the propagators vanish across its components [11, p. 4], so D depends on B' only through the components which meet its dependence set, and B' is supported in Ω_{n+1}; a nonzero touched term therefore contains a connected chain of cubes of π_n, the *bridge*, from the dependence set of D to Ω_{n+1}, each cube of which carries the factor $e^{-\nu_e}$ set aside from the rate of the polymers. By Cauchy's estimate on the polydisc of radius T_Y, which is not reduced by the layers which D meets, $\|\partial_t D\| \leq 2\sup\|D\|/T_Y$, and the sum of the new groups containing the touched part, rooted at D, is at most $c_n e^{-\nu_e N}$ times the weighted mass of D, where $c_n = \bar{c}_* b_n/\bar{\alpha}_n$ and N is the number of cubes of the bridge. The bridge is long: $Z \subset \Lambda_j^c \subset \Lambda_n^c$ and $\mathbf{Coll}_j$ lies within six sides of the cubes of π_j of Λ_j^c, while Ω_{n+1} is at distance at least $6LR_{n+1}$ sides of π_n from Λ_n^c [10, (3.5), p. 265; (3.20), p. 269], so that $\mathbf{ext}^{(n)}(D) + N \geq 5R_{n+1}$. The descendant has extent at most $(\mathbf{ext}^{(n)}(D) + N)/L + E' + 2\sum_{D''}\sigma(D'')/L + 1$ in sides of π_{n+1}, where E' is the spread of the

supports of its new vertices, paid at $e^{-\nu_e}$ per unit by the extent shares of the vertices, and the last sum runs over the old insertions which it contains, each charged to the reserve of its own size. With $a = \frac{3}{8}\nu_e$ and $L \geq 2$, so that $a/L - a \leq -a/2$ and $a/L - \nu_e \leq -a/2$, its weighted mass is at most

$$e^a c_n e^{-\frac{a}{2}(\mathrm{ext}^{(n)}(D)+N)} w(D) \leq e^a c_n e^{-\frac{15}{16}\nu_e R_{n+1}} w(D) \; .$$

(ii), merged parts. An untouched old insertion W merged into a new group adds at most $(2\varepsilon_n^{\mathrm{new}}/\varepsilon_*)\tilde{w}(W)b(W)e^{b(W)/2}$, by the bound for a sum rooted at an old vertex of Paper, Section 6.2, with $b(W) = \frac{\nu}{4}\sigma(W)$ and $\varepsilon_n^{\mathrm{new}} \leq K_\beta(b_n/\bar{\alpha}_n)\varepsilon_*$, where K_β is the constant of the class in the definition of $\bar{c}$ and involves neither the step nor the number of steps; the factor $b(W)e^{b(W)/2}$ is paid from the reserve $e^{\nu\sigma(W)}$, which is spent as $\frac{5}{8}\nu\sigma$ for the activity and the root, $\frac{7}{32}\nu\sigma$ for the polynomial factor, $\frac{\nu}{16}\sigma$ for the transports, $\frac{\nu}{16}\sigma$ for the extent of the merge, and $\frac{\nu}{32}\sigma$ for the occurrence charge, in total $\nu\sigma$. The merging group depends on B', so it reaches Ω_{n+1}, and the path from W to Ω_{n+1}, of at least $5R_{n+1} - \mathrm{ext}^{(n)}(W)$ sides of π_n, is paid by the extent shares at $e^{-\nu_e}$ per side of π_{n+1} and by the merge-extent charges of the old members it crosses; for $L \geq 16$ this gives the factor $e^{-\frac{25}{8}\frac{\nu_e}{L}R_{n+1}}$, the slower of the two rates. An insertion is a non-first member of at most one merge, after which it is part of a new insertion, so the constant e^{cW} is taken once per insertion.

(ii), the product. Each new group which contains a touched or merged part of an old insertion of the species is assigned to the first such insertion, and by the hard core each old insertion has at most one group assigned to it in a term. Hence, at fixed old insertions, the sum over the new groups is at most $\prod_D(w(D)+m_D)$ times the bound of the new species of the step, with $m_D \leq \delta_n w(D)$ by the two estimates above, since $\frac{25}{8}L^{-1} \leq \frac{15}{16}$ for $L \geq 4$. Summing over the old insertions present gives $\prod_D(1 + e^{cW}w(D)(1+\delta_n))$, and the recursion is homogeneous, so it iterates over the lifetime; with $\prod_m(1+\delta_m) \leq e^{2\bar{c}S_R} \leq 2$ and $\prod(1+x_D) \leq e^{\sum x_D}$ this is (7.3). The lifetime does not enter: the species persists when Z merges into a larger component [13, (1.84)–(1.88), pp. 385–387], and it ends when the **R**-operation removes the component. For the species of the first step the extent is measured from the 2ℓ-neighbourhood $\mathcal{N}_1'$ of Λ^c, which is at distance at least $(4L-2)R_2 \geq 5R_2$ sides of π_1 from Ω_2 [10, (3.5), p. 265; (2.9), p. 256], and the same argument applies.

(i). The attached parts of the step j have, per traversal $\mathfrak{n}$ of the step j, total weighted mass at most $C\hat{\rho}_j e^{-c\,d_j^Z(\mathfrak{n})/M}$, the second term of Paper, (5.9), with the carried weights inside. A traversal with $d_j^Z(\mathfrak{n}) \in [D, D+1)$ lies within $D+1$ of Z; the loop has at most $c_C M R_j$ traversals per cube of side MR_j by Paper, Lemma 5.4(i), and the $(D+1)$-neighbourhood of Z has at most $(2(D+1)/(MR_j)+5)^d$ times as many cubes as Z. Summing over D gives $\sum_{\mathfrak{n}} e^{-cd_j^Z(\mathfrak{n})/M} \leq c_C C_c M^2 R_j \#_j(Z)$, and $\#_j(Z) \leq 3 \cdot 2^{d-1}(d_j'(Z)+1)$ [13, p. 384]. The marks of the boundary layer and, at $j=1$, the parts of the first step which decay from the bonds of the retreated region add terms of the same form, with the factor 205^d at $j=1$ for the neighbourhood of width 101ℓ of Z which contains their anchors (Lemma 5.2(ii)).

(iii). The two reference series are evaluated at the trivial data, so they differ only through the label. By the localisation of [11, (1.9)–(1.10)] each activity is a function of the fields on its polymer, and the label enters the integrand of the step only through Z_{k+1} and the cutoffs on its boundary layer; hence a cluster whose polymer meets neither has the same activity on ℓ and on ℓ_0, and the clusters which do not see the loop cancel between the numerator and the normaliser. Here we use that an attached group has the activity 0 in the reference gas (Paper, Section 7.1): the groups which contain a touched or a merged part of an attached insertion exist on ℓ alone and reach into the small-field region with it, and they would otherwise give clusters of ℓ which meet neither Z_{k+1} nor its boundary layer and have no counterpart on

ℓ_0 (Section 18). So $E^{\partial}_{k+1}(\ell) = \sum_X(\Phi_\ell(X) - \Phi_{\ell_0}(X))$ over the clusters X which see the loop and meet the large-field data, each term attached to the component whose boundary layer it meets first in a fixed order. The first bound of (7.4) is the tree bound of Kotecký and Preiss [15] with the activity sum $C\hat{\rho}_{k+1}$ per traversal, and the second follows from it with half of the reference weight $\theta(X) \geq \frac{\nu}{32}\#_{k+1}(X)$, the other half paying the sum over the roots by the count of (i).

(iv). By (ii) the carried mass of the species of (j, Z) is at most $2A_j(Z)$ at every step of the lifetime, and each later step n adds the species of (n, Z_n), $Z_n = S^{n-j}(Z)$ while no new large fields occur near Z [13, (1.83), p. 385], and the exposed segments of (iii), at most $13^d C'_e M R_n \hat{\tau}_n^2$ per cube of Z_n, since the clusters attached to Z_n meet its boundary layer, inside $Z_n^{\sim 6}$. Along Bałaban's chain [13, pp. 384–385], and with $d'_n(X) + 1 \leq d\,\#_n(X)$ for a connected union X of cubes, which holds because a union of paths of length at most dMR_n inside the cubes, joined at common corners along a spanning tree, meets every cube, the lifetime total is at most $\varepsilon_A(d'_j(Z) + 1)$, with ε_A a polynomial in R_j times $\hat{\tau}^2$ or g_* for $j \geq 2$, and a polynomial in $\log g_1^{-2}$ for $j = 1$; Paper, (6.10) and (6.11) are these bounds. At $j = 1$, Proposition S.7(iii) and Proposition S.11 give $\hat{\kappa}_1(X) \geq (K_1 - B_1)(d'_1(X) + 1)$ with $B_1 = (O(1) + \lambda_1)(64)^d M^d L^{d+1} R_1^{d+2}$, and we require $K_1 \geq 8B_1$ and $\varepsilon_A \leq K_1/8$, both of which hold for γ_q small, since K_1 is of the order of $(\log g_1^{-2})^{2p_0}$ while B_1 and ε_A are polynomials of degree at most $r(d+2) + 1 < 2p_0$ in $\log g_1^{-2}$. Then $(K_1 - B_1)(d'_1 + 1)$ splits into Bałaban's future sum of [13, (1.81)], at most $B_1(d'_1 + 1)$; the costs, at most $\frac{1}{8}K_1(d'_1 + 1)$; the reserve $\mathsf{Res} = \frac{1}{8}K_1(d'_1 + 1)$; and $\mathsf{Pol} = (K_1 - 2B_1 - \frac{1}{4}K_1)(d'_1 + 1) \geq \frac{1}{2}K_1(d'_1 + 1)$. For a component created at a step $j + 1 \geq 2$ the same split applies to the recursion of $\hat{\kappa}$, the analogue of [13, (1.85), p. 386] with the gain $\frac{1}{6}\gamma_0 A_1^2 p_0^2(g_{j+1})(d'_{j+1}(Z^{(i)}_{j+1}) + 1)$ which Proposition S.11 leaves for each new region, converted to $d'(Z)$ with the factor $2(14)^d$ of [13, p. 385], under $K_{j+1} \geq 8B_{j+1}$, which is Bałaban's condition of [13, p. 385] with a larger constant. The branches of the induction carry Res and Pol: under S they are unchanged and [13, (1.83)] is used verbatim; in the branch of a small domain with a new large field [13, pp. 385–386], $\kappa_j(Z_0)$ is kept and the new factor $p_0(g_j)$ is split into four quarters, for Res, Pol, the costs, and Bałaban's future sum; at a merge [13, (1.86)–(1.88), pp. 386–387] the costs are subadditive up to the slack $2d$ of [13, p. 386], and they and a merge reserve are paid from the surplus $2(1 + \beta_0)^{-1}p_0(g_{j+1})$ of [13, (1.88)] with its constant doubled; and [13, (1.89)] is used verbatim. The inequality $4\,\mathsf{mass}(Z) \leq \mathsf{Res}(Z)$ holds at creation, because $\mathsf{mass}(Z)$ is at most $13^d C'_e M L\, C'_d R_j(LR_j + 1)\bar{\tau}^2(d'_j(Z) + 1)$ with $C'_d = d(3(126)^d + 64^d + 2^d)$, a small multiple of $d'_j(Z) + 1$ against $\frac{1}{8}K_j$, and it survives the branches by the same subadditivity.

(v). The constant $\bar{c}$ is a multiple of $\sup_m b_m/\bar{\alpha}_m$, which is at most a negative power of $\log \gamma_q^{-2}$ on the steps $m \leq K'$, so that $2\bar{c} \leq 1$ for γ_q small; and $\sum_m(e^{-p_0(g_m)/4} + e^{-31R_m/512}) \leq \frac{1}{32}$ for γ_q small, by the argument for S_R in (ii). The proof of (7.5) is that of (ii) without the bridge. The insertion of a cross group is a sum of integrals of elementary tensors, with one factor for each interval, and every estimate of (ii) bounds an ordered product of factors along an interval by the product of their norms, interval by interval; it therefore holds in the projective norm, in which products along the intervals are submultiplicative, and the Cauchy estimates hold for analytic functions with values in a normed space of finite dimension. *Evaluation.* An insertion of the cross species carries no deviation, its bound is a supremum over the subdomain of its step, and the points of the expansion of the step $n + 1$ lie in that subdomain, as in (ii); its transports do not grow, on each interval, by the same inequality. *The touched parts.* The argument of (ii) gives, for the sum of the new groups which contain the touched part, rooted at D, at most $c_n e^{-\nu_e N}$ times the weighted mass of D, where now $N \geq 0$ is all that is known: the support of B' lies in Ω_{n+1}, which may contain the loops and the dependence set of D. With the extent weight e^a, $a = \frac{3}{8}\nu_e$, of the new group this is at most

$e^a c_n w(D) \leq \bar{c}\, w(D)$. *The merged parts.* An untouched insertion W of the species merged into a new group adds at most $(2\varepsilon_n^{\mathrm{new}}/\varepsilon_*)\tilde{w}(W) b(W) e^{b(W)/2} \leq \bar{c}\, w(W)$, with the shares of (ii) except the share $\frac{\nu}{16}\sigma$ for the extent of the merge, which is not needed here; the remaining shares add up to $\frac{15}{16}\nu\sigma(W) \leq \lambda_n \nu\sigma(W)$. *The product.* As in (ii), each new group which contains a touched or merged part of an insertion of the species is assigned to the first such insertion, the hard core allows at most one group per insertion in a term, and $m_D \leq 2\bar{c}\, w(D)$; summing over the insertions present gives (7.5). An insertion of the cross species which becomes a member of a group of an attached species keeps its own factor in this product, the group being counted in the species of its first attached parent. *q.e.d.*

8 The R-Operation with a Loop Ensemble

Bałaban constructs the **R**-operation for the density ϱ_k alone [12, Sect. 1] [13, pp. 377–391]: on the components of the large-field region which are small and old it integrates the term over the block field, replaces it by a normalised fraction, and resums the resulting operations into the exponent, as the terms $\mathbf{R}'^{(k)}(X)$ and $\mathbf{B}'^{(k)}(X)$ of [13, (1.98)–(1.101), p. 390]. In the numerator of the paper the same term carries the dressed loop, a product of matrices along the loops which depends on every integrated variable through the multiscale minimiser [10, (2.12), p. 256]. Nothing of the exponentiation survives this factor, and three questions arise: whether the operation still preserves the integral of the numerator, what becomes of the loops' words inside the components which the operation integrates, and how large the result is. We answer them by writing the part of Bałaban's polymer gas which is linked to the loops as a ratio of two gases, in which the loops' words stand inside the polymers as ordered factors, and by splitting each group of polymers along a loop into its value at flat data, a scalar by Schur's lemma, and a remainder which vanishes at flat data.

We fix an **R**-step k, that is $k > N_k$ (Paper, Section 7), and a term of [13, (1.72), p. 379], with the components $X_1, \ldots, X_n$ of the first class, $Y_1, \ldots, Y_m$ of the second class, and the components of $Z_k \cap Z^c$, which are unchanged. For a component K of either class, K^+ is the set $K^\sim$ [13, p. 377] widened by one layer of cubes of side MR_k, and the *homes* are the classes of the components under the equivalence generated by the overlap of their sets K^+. For a first-class X, the operation $\mathcal{T}_X := (\mathbf{T}'_k(X)1)^{-1}\mathbf{T}'_k(X)$ is positive [13, p. 380] and normalised [13, p. 391]. The *flat reference* of the term is the identity configuration on the small-field region and on $X^+ \setminus \Lambda$ for every first-class X, with $\mathbf{J} = 0$ and the large-field data far from the loops unchanged. On a fixed term of the multilinear expansion of its insertions, the dressed loop of a loop C is the trace of an ordered product of *letters*: the link transports on single bonds and the insertions of the term, each on its interval. An insertion whose interval meets an arc of $C \cap X^+$, for a component X of either class, belongs with its whole interval to the *slot* of that arc, and slots which meet are merged, so that every slot is a union of whole letters and its endpoints lie outside X^+, away from the sites at which the construction of $\mathbf{T}'_k(X)$ fixes a gauge [12, pp. 195–196].

Bałaban's bracket in [13, (1.72)] is the polymer gas

$$\Xi := 1 + \sum_{r \geq 1} \sum_{\{X'_1, \ldots, X'_r\}} \prod_{p=1}^{r} F(X'_p) \,, \tag{8.1}$$

over the families of polymers with disjoint localisation domains, with the activities F of [13, (1.91)] and the bounds [13, (1.92)–(1.97), pp. 388–390], $|F(X')| \leq c_1 e^{-(1+\beta)\kappa d_{k,\cup Y_i}(X')}$ with $c_1 = e^{-p_0(g_k)}$ if X' does not meet $\bigcup_i Y_i$ and $c_1 = \alpha^{1/3}$ otherwise. With the loops present, a polymer is a connected union of class components, localisation domains, and the domains of the pieces of the loops' letters given by Lemma 17.19; it *meets* the ensemble if its activity carries a slot, and then its loop-linked activity T_P is the expression [13, (1.91)] with its slots in loop order, a tensor with one pair of indices per slot. For a family $\mathfrak{F}$ of compatible polymers which meet the ensemble, $\Xi_{\neg\mathcal{C}}(\mathfrak{F})$ is the gas of the polymers which do not meet it and are compatible with every member of $\mathfrak{F}$, and $A^{\mathcal{C}}(\mathfrak{F})$ is the product over the loops of the traces of the ordered products of the tensors T_P, $P \in \mathfrak{F}$, and of the letters at the renormalised background U^R on the bonds in no slot. For a polymer P with slots on a loop C, $\mathsf{span}(P)$ is the shortest arc of C which contains the slots of P on C (the whole loop if they cover it, and otherwise the complement of a longest arc between two consecutive slots, the first in the orientation of C from a fixed base bond if several are longest); two polymers *interleave* if a slot of one shares a bond with the span of the other, and the *groups* of $\mathfrak{F}$ are the classes of the equivalence generated by interleaving, each with the span $\mathsf{span}(g) := \bigcup_{P\in g} \mathsf{span}(P)$ and the hull $\mathsf{hull}(g)$, the union of its span and of the domains of its polymers. A group whose slots lie on two loops is a *cross group.* The size $|P|$ of a polymer is the number of cubes of side MR_k which meet its domain, the domains of its pieces, or a set K^+ of one of its components, plus, for each insertion with a piece in P, the diameter of its support in units of MR_k.

Proposition S.10 (The **R**-Operation with a Loop Ensemble)**.** *Let γ_q be sufficiently small, depending on $c_{\mathcal{C}}$, let $k \leq K'$, and let the ensemble be dressed at the step k (Paper, Definition 5.2).*

(i) The depth. *If $k \leq N_k$, the* **R***-operation of the step k is the identity on every labelled term of the numerator and of the denominator.*

(ii) The integrals. *Let $k > N_k$. For every integrable function f of the variables of a term, $\int (\mathbf{R}' f)\, \mathrm{d}V_k = \int f\, \mathrm{d}V_k$; the removal of the δ-functions $\delta_{G_0 \cap X_i}(V_k')$ preserves the integral of every integrand which is invariant under the gauge transformations of V_k on X_i; and the dressed loop is such an integrand. Hence the* **R***-operation preserves the integrals of the numerator and of the denominator.*

(iii) The ratio of two gases. *On every term, with $\Xi^{\mathcal{C}}$ the bracket with the dressed loop inserted,*

$$\frac{\Xi^{\mathcal{C}}}{\Xi} = \sum_{\mathfrak{F}} e^{-\Phi(\mathfrak{F})} A^{\mathcal{C}}(\mathfrak{F})\ , \qquad \Phi(\mathfrak{F}) := \log \Xi - \log \Xi_{\neg\mathcal{C}}(\mathfrak{F})\ , \tag{8.2}$$

the sum over the compatible families of polymers which meet the ensemble, $\mathfrak{F} = \emptyset$ included. Every polymer which meets the ensemble has the amplitude $\hat{c}_1 := e^{3+a_k} e^{-\mathrm{p}_0(g_k)} \leq e^{-\mathrm{p}_0(g_k)/2}$ with $a_k \leq \frac{11}{10}$, $|\Phi(\emptyset)| \leq e^{-\mathrm{p}_0(g_k)/4} \#_k(\mathcal{C})$, with $\#_k(\mathcal{C})$ the number of cubes of side MR_k which meet the loops, and $|\Phi(\mathfrak{F}) - \Phi(\emptyset)| \leq B_Q \#_k(\bigcup \mathfrak{F})$, with $B_Q := 6^d C_\Psi ((8\alpha)^{1/3} + e^{-\mathrm{p}_0(g_k)}) + e^{-\mathrm{p}_0(g_k)/2}$ and the constant C_Ψ of Lemma 17.20(iii), which depends on d, κ, and β.

(iv) The flat part. *Call a group g* eligible *if it is not a cross group and none of its polymers meets a second-class or an unchanged component. At the flat reference the contraction of an eligible group is a multiple $\mathfrak{a}_g$ of the identity, its* scalar activity*; for the other groups put $\mathfrak{a}_g := 0$. The scalar activity is a number of its own: it is not a multiple of the product $\prod_{P\in g} F(P)$ of the activities of the polymers without the ensemble, and it need not vanish when this product does. The logarithm $E_k^{\mathbf{R}}(\mathcal{C};\ell)$ of the gas $\Xi^\flat := \sum_{\mathfrak{F}} \prod_{g\subset\mathfrak{F}} \mathfrak{a}_g\, \Xi_{\neg\mathcal{C}}(\mathfrak{F})$, divided by Ξ, is $\sum_i E_k^{\mathbf{R}}(C_i;\ell_0) + E_k^{\mathbf{R}\times} + E_k^{\mathbf{R},\partial}(\ell)$, where ℓ_0 is the term without second-class and unchanged components and:*

- *(a)* *$E^{\mathbf{R}}_k(C;\ell_0)$ has the form of Paper, Proposition 7.10, with $\mu^{\mathbf{R}}_{k,v} := \kappa^{\mathbf{R}}_v/(LMR_k)$, $\kappa^{\mathbf{R}}_v$ the sum of the densities of the cluster series of the line of orientation v over one period of LMR_k traversals of the step $k-1$, and $\zeta^{\mathbf{R}}_{k,v,w}$ the corner coefficient of the bent path of orientations v, w;*
- *(b)* *$E^{\mathbf{R},\partial}_k(\ell)$ is a sum of cluster terms, each attached to one second-class or unchanged component, whose sum rooted at a cube of side MR_k is at most $m^{\mathbf{R}}_k := C^{\mathbf{R}}_m e^{-p_0(g_k)/4} \le g_k^2$;*
- *(c)* *the cross constant $E^{\mathbf{R}\times}_k := E^{\mathbf{R}}_k(\mathcal{C};\ell_0) - \sum_i E^{\mathbf{R}}_k(C_i;\ell_0)$ is a number which does not depend on the labelled term, it vanishes for a single loop, and $|E^{\mathbf{R}\times}_k| \le C^{\mathbf{R}\times}_E P_k(\mathcal{C}) e^{-p_0(g_k)/4} e^{-\kappa a_k/(6MR_k)}$, with a constant $C^{\mathbf{R}\times}_E$ of d, κ, and β.*

(v) The deviation insertions. *With $E^{\mathbf{R}}_k(\mathcal{C};V_k)$ the logarithm of $\Xi^\flat/\Xi$ at the actual data,*

$$\frac{\Xi^{\mathcal{C}}}{\Xi} = e^{E^{\mathbf{R}}_k(\mathcal{C};V_k)} \sum_S e^{-\Psi(S)} \prod_{i=1}^{n} \mathrm{tr}[\mathcal{W}^{\mathrm{sf}}_{C_i}(U^R) \textit{ with } D_g \textit{ on } \mathsf{span}(g) \textit{ for } g \in S]\,, \tag{8.3}$$

the sum over the compatible non-interleaving sets S of groups, where $\mathcal{W}^{\mathrm{sf}}_C(U^R)$ is the small-field word of the class at U^R with the term's insertions, $|\Psi(S)| \le \sum_{g\in S} 2e^{-p_0(g_k)/4}\#_k(\mathsf{hull}(g))$ up to a part of at most a constant per second-class component, and each D_g of an eligible group is a deviation insertion of the class, whether or not its scalar activity vanishes: covariant at the ends of $\mathsf{span}(g)$, a function of U^R on $\mathsf{hull}(g)$ widened by one layer of cubes of side MR_k, analytic on the polydiscs of the class, and zero at pure-gauge data there. A group which is not eligible is an attached insertion at the component it meets, or a cross group. The cross groups which meet no second-class or unchanged component are of two kinds. Those which contain no insertion of the cross species, in their slots or among their members, form the cross species of the step k (Paper, Definition 5.2(ii)); in the projective norm of Paper, (5.5), and with each group counted once, at the first traversal of its slots, their total is at most

$$C^{\mathbf{R}}_{\mathsf{cr}} P_k(\mathcal{C})\, e^{-p_0(g_k)/4} e^{-\kappa d_k/(6MR_k)}\,, \tag{8.4}$$

with d_k the least distance between two loops in the units of the step k and a constant $C^{\mathbf{R}}_{\mathsf{cr}}$ of d, κ, β, and the representations. A cross group which contains an insertion of the cross species belongs to the species of the first such insertion W, and the groups assigned to W have total mass at most $y^{\mathbf{R}}_k$ times that of W, with $y^{\mathbf{R}}_k := C_y e^{-p_0(g_k)/4}$ and $\prod_k (1+y^{\mathbf{R}}_k) \le 2$, the product over the **R***-steps $k \le K'$. A group on one loop one of whose members wraps around the torus is of the wrapping species, it is counted with its first such member, and the groups assigned to a member add at most $y^{\mathbf{R}}_k$ times its mass. The deviation insertions which the* **R***-step makes have, per traversal, total size at most $C^{\mathbf{R}}(g_k)\tilde{\rho}_k \delta^*_{\mathsf{n}}(U^R)$, with*

$$C^{\mathbf{R}}(g) := \exp\Bigl(-\frac{1}{4}A_0(\log g^{-2})^{p_0} + \log g^{-2} + C\log\log g^{-2}\Bigr)\,, \tag{8.5}$$

and at an **R***-step the recursion of Paper, (6.7), holds with $C_{\mathsf{new}} + C^{\mathbf{R}}(g_k)$ in place of C_{new} and $a + C_a MR_k e^{-p_0(g_k)/4}$ in place of a.*

(vi) The post-**R** term. *The post-***R** *term carries a dressed loop of the step k in the sense of Paper, Definition 5.2, whose step factor contains the* **R***-part of Paper, (7.23); the attached insertions which the operation makes at a second-class component are paid from its reserve as in Proposition S.9(iv), with $m^{\mathbf{R}}_n \le m_n$ added to the cost of each* **R***-step n.*

(vii) Commensurability. *MR_kL^k divides $MR_{K'}L^{K'}$, which divides the fine length of a bond of $\mathbb{T}_q$, so that in the units of the step k the corners of the loops lie in $MR_k\mathbb{Z}^4$ and the length of every segment is a multiple of MR_k.*

Proof. (i). Every component of Z_k contains a large-field region created at a step $j \geq 1$. If $k \leq N_k$, then $j \geq 1 \geq k - N_k + 1$, so that a new large-field region was created inside the component in the preceding N_k steps, and the condition (ii) of [12, p. 177] fails. The class of the components on which the operation acts is therefore empty, and on a term with an empty class none of the rewritings of [12, Sect. 1] acts, since each acts on $\mathbb{T}_k(Z)$ and on the factors of $\exp A_k$ localised in Z [12, (1.1), p. 177]; the bracket of [13, (1.72)] has no **V**-term, because $V(Y, U_k)$ is defined only when $Y \cap Z^{\sim}$ is a nonempty union of components of $Z^{\sim}$ [13, p. 377], and the admissible domains of $\mathbf{R}'^{(k)}$ "contain at least one large field region connected with one of the $\mathbf{T}'_k$-operations" [13, p. 390]. So the operation is the identity, on the numerator as on the denominator, since the class is defined by the labels of the term alone.

(ii). By [12, (1.100), p. 201], $\mathbf{R}'f$ is the integral of f over $V_k\lceil_\Lambda$, multiplied by a fraction in a fresh variable $V_k\lceil_{\Lambda_i}$ whose integral over this variable is 1 for every value of the other variables; Fubini's theorem then gives $\int(\mathbf{R}'f)\,\mathrm{d}V_k = \int f\,\mathrm{d}V_k$ for every integrable f, which is [12, (1.102)], printed there for ϱ_k and with no property of $\exp A_k$ used. The removal of the δ-functions on the first-class components is the reversed Faddeev–Popov procedure of [13, p. 378], possible "only if we integrate the density with respect to the gauge field variables V_k, at least on the components X_i", and it preserves the integral of every integrand invariant under the gauge transformations which the δ-function fixes. The construction inserts the Faddeev–Popov δ-function δ_{T_0} [12, pp. 195–196], changes to the axial gauge [12, p. 197], and compensates by gauge transformations of V_h and of V', each of which needs the same invariance. The dressed loop has it: the averaging $M_{\mathbb{B}}$ of [10, (2.11), p. 256] is covariant, because the averaged loops of [9, (0.4), p. 253] are based at the initial site of the bond and the principal logarithm is equivariant under conjugation; regularity is a condition on plaquettes, so that $U \mapsto U^u$ maps the minimal orbit of [10, (2.12)] for V onto that for V^v when u restricts to v; the fluctuation fields transform by the adjoint [10, (2.41)(iii), p. 261]; and every insertion transforms by π at the ends of its interval (Paper, Definition 5.2(iii)), so that the traces are invariant. The rewritings of [12, pp. 190–193] are changes of variables, decompositions of unity, and Fubini's theorem on the old variables, identities of integrals of integrable functions which hold with the bounded measurable factor of the loops. Finally, the class and every exclusion of the construction are characteristic functions of the fields alone (the functions $1 - \chi'^{(j)}$, $1 - \chi_{k,\Lambda}$, $1 - \chi'$ of [12, (1.55), p. 188; (1.75), p. 193; (1.82), p. 196] and $\chi^c_k(Y_i^{\sim -6})$ of [13, (1.72)]), so that the operation has the same components and the same characteristic functions on a term of the numerator as on the corresponding term of the denominator.

(iii), the identity. Expand every letter of a loop into the pieces of Lemma 17.19, each a function of the external data and of the data of the components in its domain. A configuration of the gas with the loops is, uniquely, the compatible family $\mathfrak{F}$ of its polymers which meet the ensemble together with a configuration of $\Xi_{\neg\mathcal{C}}(\mathfrak{F})$, because the polymers which do not meet the ensemble are Bałaban's, the activity of each being a function of the renormalised configuration on the polymer [13, p. 379] ("All the expressions in the above integral operation are localized in the domain X"), and the pieces are local by Lemma 17.19(ii). Hence $\Xi = \sum_{\mathfrak{F}} \prod_{P\in\mathfrak{F}} F(P)\,\Xi_{\neg\mathcal{C}}(\mathfrak{F})$ and $\Xi^{\mathcal{C}} = \sum_{\mathfrak{F}} \Xi_{\neg\mathcal{C}}(\mathfrak{F})A^{\mathcal{C}}(\mathfrak{F})$, and dividing by Ξ gives (8.2). No matrix is exponentiated: the right side is multilinear in the ordered arc factors, and every sum over the positions, the shapes, and the internal sequences of the components stays inside the sum over $\mathfrak{F}$.

(iii), the bounds. The positivity of $\mathbf{T}'_k(X)$ and [13, (1.73), p. 380], a triangle inequality for a positive measure which does not see the parameters of the integrand, give $|T_P| \leq \bar{F}(P)\prod_{\text{slots}} \sup\|w\|$, with the bound $\bar{F}(P)$ of [13, (1.92)]; on the analyticity domain [13, (1.75), p. 380] adds $e^{3\sup|\sigma|} \leq e^3$, by the bounds on σ of [13, p. 379]. The norm of the word of a slot is at most the product of the masses $e^{2A_j(Z)}$ of the attached insertions of the old components inside the polymer, by Proposition S.9(ii), and each component pays its own from $\kappa_j(Z)$ in the induction of Proposition S.9(iv), so that these masses enter no smallness condition. The activities of the gas are produced by the operations $\mathbf{T}'_k(X)$ of the first-class components only. The letters in the slots of the second-class and of the unchanged components, and the pieces of the letters all of whose units are of these two classes, are not integrated by the operation; they stay in the words of the new term, as insertions attached to those components (Proposition S.9), and they are not part of any activity. A polymer whose activity carries a slot therefore contains a first-class component which carries a slot or a unit of a piece, so that $q \geq 1$ in [13, (1.95), p. 389], and the factor $e^{-p_0(g_k)}$ is extracted before the sum in both cases of c_1, as on [13, p. 389]; the pieces of the letters add at most e^{a_k} per first-class component, with $a_k \leq \frac{11}{10}$ by Lemma 17.19(iv). Hence $\hat{c}_1 = e^{3+a_k}e^{-p_0(g_k)} \leq e^{-p_0(g_k)/2}$ once $p_0(g_k) \geq 9$. The number $\Phi(\emptyset)$ is the sum of the cluster terms which contain a polymer meeting the ensemble. By the Kotecký–Preiss criterion [15] for the bounds [13, (1.97)], the cluster terms through a cube Q of side MR_k which contain such a polymer are at most the weighted activities of these polymers through Q; each is rooted at one of the R_k^d cubes of π_k in Q, or at one of the $3^d(101R_k)^d + 1$ cubes of the layer $Y_i^+ \setminus Y_i$ when its domain contains a second-class Y_i, its shapes on the scale M cost a constant per root, paid by a part $e^{-\frac{\beta}{8}\kappa d_k}$ of the decay in [13, (1.95)], and the root counts are less than $p_0(g_k)$ by the condition $-\frac{1}{4}p_0(g_k) + 1 + 2\kappa(100R_k)^d \leq 0$ of [13, p. 389], since $3^d 101^d < 2\kappa 100^d$ for $\kappa \geq 4^d$. So the sum through Q is at most $2p_0(g_k)e^C e^{-p_0(g_k)/2} \leq e^{-p_0(g_k)/4}$ for γ_q small, and summing over the cubes which meet the loops gives the bound on $\Phi(\emptyset)$. The difference $\Phi(\mathfrak{F}) - \Phi(\emptyset)$ is the sum of the cluster terms which are incompatible with $\mathfrak{F}$ and contain no polymer meeting the ensemble; each meets $\bigcup\mathfrak{F}$, and the Kotecký–Preiss sum of Bałaban's polymers through a cube of side MR_k is at most $6^d C_\Psi(\alpha^{1/3} + e^{-p_0(g_k)}) + e^{-p_0(g_k)/2} \leq B_Q$ by Lemma 17.20(iv), which gives the second bound.

(iv), the spans. Two polymers of $\mathfrak{F}$ whose spans on a loop share a bond interleave: if one span is the whole loop, a slot of the other lies in it; otherwise walk along the second span from the shared bond to one of its ends, which is either a bond of a slot of the second polymer lying in the first span, or, if the walk leaves the first span, an end of the first span, which is a bond of a slot of the first polymer lying in the second span. So the spans of distinct groups share no bond, and the span of a group shares no bond with a slot of another group. A polymer P has its slots in its domain and the supports of its insertions within $|P|$ cubes, so the slots lie in a set of ℓ^∞-diameter at most $|P|MR_k$, and by Paper, Lemma 5.4(ii), $\mathsf{span}(P)$ has at most $4\Lambda(C)|P|MR_k + 2$ bonds or is more than half of C; in both cases it meets at most $C_\Lambda|P|$ cubes of side MR_k, with C_Λ depending on $\Lambda(C)$ and d, because a span of more than half of C has, by Paper, Lemma 2.9, diameter at least $P_k(C)/(2\Lambda(C))$, which is at most $|P|MR_k$. Hence $\mathsf{hull}(g)$ meets at most $(C_\Lambda + 1)\sum_{P\in g}|P|$ such cubes.

(iv), the flat scalar. Let $u_0 \in G$ be a constant gauge transformation. It fixes the flat reference and leaves every factor of the construction of $\mathbf{T}'_k(X)$ invariant: the Haar measures; the Wilson action and the terms $\mathbf{E}$, $\mathbf{R}$, $\mathbf{B}$ of A'_k, which are gauge-invariant functions [10, (2.23), p. 258; (2.30), p. 260]; the Gaussian form and $\sigma(g_k B)$ of [13, (1.70)], since the inner product on $\mathfrak{g}$ is Ad-invariant; the characteristic functions, which depend on $|\partial U - 1|$, $|B|$, $|A|$; the δ-functions of the trees, since $u_0 1 u_0^{-1} = 1$; the minimiser V_Λ, covariant by [12, Proposition 1, p. 194] and fixed at flat data; and the functions $\mathbf{H}$ and the configurations

of [13, (1.61)–(1.63)]. The contraction of a group's slots in loop order, with the transports 1 of the flat reference on the bonds of its span in no slot, transforms as $\pi(u_0)\cdot\pi(u_0)^{-1}$, because every link is conjugated, every insertion is covariant at its ends, and so is every piece by Lemma 17.19(iii). It therefore commutes with $\pi(G)$, and since π is irreducible, Schur's lemma makes it a multiple of the identity. This multiple is the scalar activity $\mathfrak{a}_g$ of the group. Schur's lemma gives no more than this: the contraction is an integral of the activities of the polymers against the insertions of the slots, and it need not be a multiple of the product $\prod_{P\in g} F(P)$ of the integrals without the insertions, nor vanish when that product vanishes through a cancellation. We therefore use $\mathfrak{a}_g$ as the activity of the group and never divide by the activities of its polymers. The species of a group is fixed by its type, as in (v), and not by the value of its scalar activity: an eligible group belongs to the deviation species also when $\mathfrak{a}_g$ happens to vanish, and a cross group or a group with a polymer at a second-class or an unchanged component has $\mathfrak{a}_g = 0$ by definition, whatever its contraction is. On a term whose insertions off the slots are evaluated at the flat reference, and for a family $\mathfrak{F}$ all of whose groups are eligible, the contraction of each loop is the normalised trace of a product of scalars, so that $A^{\mathcal{C}}(\mathfrak{F}) = \prod_{g\subset\mathfrak{F}} \mathfrak{a}_g$; the traces of the loops are the normalised traces $\chi/\chi(1)$ of the paper, and no factor $\dim\pi_i$ appears. For a family which contains a group that is not eligible this equality is not asserted, and such a family contributes to the deviation, attached, or cross insertions of (v). Accordingly $\Xi^\flat$ is the gas whose polymers are those not meeting the ensemble and the groups, with the activities $\mathfrak{a}_g$, two groups being compatible when their domains are disjoint and they do not interleave. The bound of $\mathfrak{a}_g$ which the cluster expansions use is that of the contraction, the product of the amplitudes $\hat{c}_1$ of the polymers of the group and of the bounds of the insertions of its slots, and not a bound of $\prod_{P\in g} F(P)$.

(iv), the entropy of the groups. A third of the decay $e^{-\kappa|P|}$ of the activities, which a tree of [13, (1.67)] through the cubes of the scale M provides at the rate κ per cube of side MR_k, pays the entropy: a polymer whose span meets a given cube has its domain within $|P|$ layers of it, hence meets one of $(2|P|+1)^d$ cubes; the next member of a group has its span meeting one of the at most $C_\Lambda|P|$ cubes of the span of a member already placed; and the part of the polymer off the second-class components is rooted at one of at most $\frac{1}{8}p_0(g_k)$ cubes of π_k. The second third pays the Kotecký–Preiss margin, and the last third is kept for the diameters in (v). So the groups whose hulls meet a given cube of side MR_k sum, with their Kotecký–Preiss weights, to at most $e^{-p_0(g_k)/4}$, and both cluster expansions converge under the hypothesis of [13, p. 389].

(iv), the periodicity. Bałaban's construction is covariant under the translations of the lattice of the step k which preserve the partitions of the scales $j \le k$ that enter $\mathbf{T}'_k(X)$: the cubes, the trees averaged over the reflections and the permutations [12, p. 201], the characteristic functions, and the operations are defined from these partitions, and for the terms $\mathbf{R}'^{(k)}(X)$ Bałaban states it, "by their construction it is also clear that they are Euclidean covariant" [13, p. 390]. The partition of the scale j has side MR_jL^{j-k} in the units of the step k, and MR_jL^j divides MR_kL^k by (vii). So the translations by $MR_k\mathbb{Z}^4$ preserve every partition, and on ℓ_0, where no second-class or unchanged component breaks the covariance, the activities of $\Xi^\flat$ at the flat reference are periodic along each segment with the period MR_k, that is LMR_k traversals of the step $k-1$.

(iv)(a). With these three properties, locality, decay, and periodicity, the proof of Paper, Theorem 7.9, applies to the cluster series of $\Xi^\flat$ along C on ℓ_0 with the following changes. The clusters are allocated to the traversals of their spans. The sum of the densities over the traversals in one cube of side MR_k is at most a constant times $e^{-p_0(g_k)/4}$, by the entropy bound, and the clusters through that cube which reach a distance t, in units of MR_k, from it

contribute at most $C_t e^{-p_0(g_k)/4} e^{-\kappa t/6}$, with a constant C_t of d, κ, and β, by the margin; the same holds for the gas of the polymers of a subensemble, the gas Ξ of the empty subensemble included, whose polymers are among those of the bounds of (iii). A cluster with a span which contains more than half of C reaches the distance $P_k(C)/(2\Lambda(C))$, by Paper, Lemma 2.9. The corners lie in $MR_k\mathbb{Z}^4$ by (vii), so the bent-path gas at a corner is the translate of the bent-path gas at the origin, and every segment is a union of whole periods. Hence $\kappa_v^{\mathbf{R}}$, a sum over one cube, is at most a constant times $e^{-p_0(g_k)/4}$, so is $\mu_{k,v}^{\mathbf{R}} = \kappa_v^{\mathbf{R}}/(LMR_k)$, and $\zeta_{k,v,w}^{\mathbf{R}}$, a sum over the cubes at the distances $t \geq 0$ from the corner of terms at most $e^{-p_0(g_k)/4} e^{-\kappa t/6}$, is at most a constant times $e^{-p_0(g_k)/4}$. The comparison of the gas of C with the bent-path gas costs, per cube, at most a constant times $e^{-p_0(g_k)/4}(e^{-\kappa r_k(C)/(6MR_k)} + e^{-\kappa\iota_{T,k}/(6MR_k)})$, and at most $2^d(P_k(C)/(MR_k)+1)$ cubes meet C; the tails of the corner sums cost at most a constant times $e^{-p_0(g_k)/4} e^{-\kappa s_k(C)/(12MR_k)}$ per corner, and $s_k(C) \geq 2r_k(C)$. Since $P_k(C)/(MR_k)+1 \leq 2P_k(C)$, this is (7.29) with $c_E^{\mathbf{R}} = \kappa/12$, a fortiori with the factor MR_k. The coefficients are sums of activities of the gas on the trivial term, which involve neither the loops, nor the torus, nor the labels.

(iv)(b). A polymer whose domain lies in the small-field region, at distance at least six layers from Λ_k^c, depends on U_k restricted to its domain only [13, pp. 377, 379], so that it has the same activity on ℓ and on ℓ_0; the activities differ only for the polymers which meet a second-class or an unchanged component or its collar of six layers, where $c_1 = \alpha^{1/3}$ [13, p. 390], the **V**-terms depend on the sequences restricted to the components inside their domains [13, p. 379], and the domains equal to components of $Z^\sim$ are excluded [13, p. 379]. So $E_k^{\mathbf{R},\partial}(\ell)$ is the sum of the cluster terms of ℓ meeting such a collar minus those of ℓ_0 meeting it, each attached to the first component whose collar it meets in a fixed order. Every one of these clusters involves the loops, since the others cancel between $\Xi^\flat$ and Ξ, so it contains a first-class component with a slot, whose factor $e^{-2(1+\beta_0)^{-1}p_0(g_k)+1}$ of [13, (1.89), p. 387] is extracted before the sum, as on [13, p. 389]. With the root counts of (iii) the rooted sum through a cube of side MR_k is at most $2e^{-p_0(g_k)/4}$ for each of the two parts, which gives $m_k^{\mathbf{R}}$, and $m_k^{\mathbf{R}} \leq g_k^2$ for γ_q small because $p_0 > 1$.

(iv)(c). For $A \subseteq \{1, \ldots, n\}$ let $\Xi_A^\flat$ be the gas $\Xi^\flat$ on ℓ_0 of the subensemble of the loops C_i, $i \in A$, so that $\Xi_\emptyset^\flat = \Xi$ and

$$E_k^{\mathbf{R}\times} = \log \Xi_{\{1,\ldots,n\}}^\flat - \sum_{i=1}^{n} \log \Xi_{\{i\}}^\flat + (n-1)\log \Xi \,, \tag{8.6}$$

a number which involves no label of a term. We expand each logarithm into its cluster series. For a cluster c let $\mathbb{L}(c)$ be the set of the loops met by the domains of its polymers. The activity of a polymer or of a group in $\Xi_A^\flat$, and the compatibility of two of them, depend only on the loops of A which their domains meet, because the slots, the contractions, and the scalar activities $\mathfrak{a}_g$ are computed from these loops alone (Lemma 17.19(ii)), a polymer which meets no loop of A having its activity in Ξ; so the term $v_A(c)$ of c in $\log \Xi_A^\flat$ depends on A only through $A \cap \mathbb{L}(c)$. The coefficient of c in (8.6) is therefore $v_{\mathbb{L}(c)} - \sum_{i\in\mathbb{L}(c)} v_{\{i\}} + (|\mathbb{L}(c)|-1)v_\emptyset$, which vanishes if $|\mathbb{L}(c)| \leq 1$; in particular $E_k^{\mathbf{R}\times} = 0$ for one loop. A cluster with $|\mathbb{L}(c)| \geq 2$ has polymers meeting two loops $C_i \neq C_{i'}$, and since two objects of a cluster are linked by an overlap of their domains or by an interleaving, which joins two groups with slots on one loop, a shortest chain of the cluster from the polymers meeting C_i to those meeting $C_{i'}$ consists of overlapping domains. Its domains reach from a cube of side MR_k meeting C_i to the distance $a_k - 2MR_k$, at least, because the loops are at distance at least a_k in the units of the step k (Paper, Lemma 3.14). We charge $v_{\mathbb{L}(c)}$ to one loop of $\mathbb{L}(c)$, $v_{\{i\}}$ to C_i, and $(|\mathbb{L}(c)|-1)|v_\emptyset| \leq \sum_{i\in\mathbb{L}(c)} |v_\emptyset|$ to the loops of $\mathbb{L}(c)$; then each loop C_i receives three sums of cluster terms over the clusters through the at most $2^d(P_k(C_i)/(MR_k)+1) \leq 2^{d+1}P_k(C_i)$ cubes of side MR_k which meet it, each reaching the distance $a_k/(MR_k)-2$ in units of MR_k. By (iv)(a) this gives the bound of (iv)(c) with $C_E^{\mathbf{R}\times} := 3 \cdot 2^{d+1}e^{\kappa/3}C_t$.

(v), the expansion. For an eligible group g let D_g be the contraction of its tensors in loop order over $\mathsf{span}(g)$, with the transports at U^R on the bonds of the span in no slot and the insertions of the term whose intervals share a bond with the span (the *members*, which lie in the span because their intervals contain no slot bond and the ends of the span are slot bonds), minus $\mathfrak{a}_g$ times the small-field word on the span with the members on their intervals; for a group which is not eligible, $\mathfrak{a}_g = 0$ and D_g is the whole contraction. Since the spans of distinct groups are disjoint and none meets a slot of another, the word is a product over disjoint arcs, on each span the factor is $\mathfrak{a}_g\mathcal{W}^{\mathrm{sf}}_{\mathsf{span}(g)} + D_g$, and expanding the product gives a sum over the sets S of the groups at which D_g is taken. At fixed S the sum over the other groups is $\Xi^\flat(S)/\Xi$, the gas restricted to the configurations compatible with S, which is $e^{E_k^{\mathbf{R}}(\mathcal{C};V_k)}e^{-\Psi(S)}$ with $\Psi(S) := \log\Xi^\flat - \log\Xi^\flat(S)$, the sum of the cluster terms of $\Xi^\flat$ which overlap or interleave with S, each of which meets $\bigcup_{g\in S}\mathsf{hull}(g)$. This is (8.3). The clusters which contain a part of a localisation domain containing a second-class component have the amplitude $\alpha^{1/3}$ [13, pp. 389–390] and give at most C_Ψ per such component, by Lemma 17.20(iii) at its root; the others, assigned to the first group of S which they meet, give $|\Psi_g(S)| \leq 2e^{-p_0(g_k)/4}\#_k(\mathsf{hull}(g))$ by the entropy bound.

(v), the properties of D_g. Locality: each slot is a piece of Lemma 17.19, a function of the data of the components of its polymer and of U^R on its domain, and the transports are bond variables of U^R, so D_g depends on U^R on the widened hull $\mathsf{hull}(g)^+$ only. Covariance: the contraction is covariant at the ends a, b of the span, by the external covariance of the averaged arcs (the change of variables induced by an external gauge transformation u, under which the Haar and Gaussian measures are invariant and $V_\Lambda(u\cdot V) = u\cdot V_\Lambda(V)$ by [12, Proposition 1, p. 194]; at complex external data the same holds for real u, since the integration variables stay real), and the small-field word transforms in the same way. Vanishing: if U^R is a pure gauge $u\cdot 1$ on $\mathsf{hull}(g)^+$, we may evaluate D_g at a configuration which is the pure gauge $\tilde{u}\cdot 1$ on the whole small-field region with $\tilde{u} = u$ there, by locality; there the contraction is $\pi(u(a))\mathfrak{a}_g\pi(u(b))^{-1}$ by the flat scalar and the covariance, and the small-field word is $\pi(u(a))\pi(u(b))^{-1}$, so $D_g = 0$. Analyticity and size: on the polydiscs of the class, of radius $2r_k^{\mathrm{ext}}$ about a real datum, $|D_g| \leq B(g) := 2e^{2s(g)}\prod_{P\in g}\bar{F}(P)\,e^{A_k}\dim\pi$, with $s(g)$ the number of first-class components with a slot and e^{A_k} the analyticity bound of the transports of the class; since D_g vanishes at deviation zero, Cauchy's estimate on the disc of radius r_k^{ext} through the actual deviation ε, or the bound $B(g)$ itself when $\|\varepsilon\| \geq r_k^{\mathrm{ext}}$, gives $|D_g| \leq B(g)\|\varepsilon\|_{\mathsf{hull}(g)^+}/r_k^{\mathrm{ext}}$, and in an axial gauge rooted in the widened hull $\|\varepsilon\| \leq C\,\mathrm{diam}_k(\mathsf{hull}(g)^+)\delta^*$. For a group with members the members' bounds on the same polydiscs give $\|D_g\| \leq B(g)\prod_W\|W\|$ without a Cauchy step, and D_g is a deviation insertion when every member is one. A group which is not eligible and which meets a second-class or an unchanged component, or whose box meets the six layers of Λ_k^c, is attached to that component and charged to it, with amplitude $\hat{c}_1$ per component, at most $\frac{1}{2}e^{-p_0(g_k)/4}$ per cube of side MR_k; this charge is the summand $c_n^{\mathbf{R}}(d_n'+1)$, $c_n^{\mathbf{R}} = Ce^{-p_0(g_n)/4} \leq Cg_n^2$, of the induction of Proposition S.9(iv).

(v), the cross groups. Let g be a cross group which meets no second-class or unchanged component. Its flat contraction is not a scalar, $\mathfrak{a}_g = 0$ by definition, and D_g is the whole contraction, a tensor with one factor for each arc of its slots between two bonds in no slot of g. The bound of (iii) on T_P is a bound on the integral, against a positive measure, of the product of the norms of the words of the slots, and the members enter as factors along their

intervals, so that $\|D_g\| \le B(g)\prod_W \|W\|$ in the projective norm. Suppose first that g contains no insertion of the cross species, in its slots or among its members. Two polymers interleave on one loop, on which both have slots, so that, if every polymer of g had its slots on one loop, all of them would have their slots on the same loop; hence g contains a polymer P with slots on two loops, and since the insertions in its slots are on one loop each, the domain of P meets both loops. These are at distance at least $d_k \ge a_k \ge L^2 MR_k$ in the units of the step k (Paper, Lemma 3.14), so that $|P| \ge d_k/(MR_k) - 2$. The groups through a cube of side MR_k which contain a polymer of this size sum, with the bounds $B(g)e^{|\Psi_g|}$ and the masses of their members, to at most $C_t e^{-p_0(g_k)/4} e^{-\kappa(d_k/(MR_k)-2)/6}$, by the entropy bound of (iv) and half of the margin's third of $e^{-\kappa|P|}$, as for the clusters of (iv)(a); the masses of the members, which are of the deviation species here, are summed in the factor e^{x_g} of the source below. Each group is counted at the first traversal of its slots, which lies in one of the at most $2^{d+1}P_k(\mathcal{C})$ cubes of side MR_k which meet the loops, and this gives (8.4) with $C^{\mathbf{R}}_{\mathrm{cr}} := 2^{d+2}e^{\kappa/3}C_t \max_i \dim \pi_i$. Suppose now that g contains insertions of the cross species, and let W be the first one in the order of the loops and of the traversals. In a term of the expansion every insertion of the cross species is untouched or lies in the slots of exactly one polymer, hence in exactly one group, because the polymers of a family have disjoint domains; the groups which contain W are rooted at the cubes of side MR_k which meet the intervals of W, and they sum to at most $e^{-p_0(g_k)/4}$ per cube by the entropy bound. The number of these cubes is paid from the weight of W, of which the **R**-step spends the fraction $e^{-p_0(g_k)/4} + e^{-31R_k/512}$ of Proposition S.9(v), and this gives the factor $y^{\mathbf{R}}_k = C_y e^{-p_0(g_k)/4}$; the other members of the cross species in g keep their own factors. The product of the $1 + y^{\mathbf{R}}_k$ is at most $\exp(C_y \sum_k e^{-p_0(g_k)/4}) \le 2$ for γ_q small, by the argument for S_R in the proof of Proposition S.9(ii). Finally let g be a group on one loop with a member W which wraps around the torus. The contraction D_g depends on U^R on the box of W, which does not lift to the universal cover, so that D_g does not lift either: it is an insertion of the wrapping species and not a deviation insertion, whatever its other members, and its mass, at most $B(g)\prod_W \|W\|$, enters the last term of Paper, (5.9), through its first wrapping member. The groups which contain a given member are counted as in the second case above, which gives the same factor $y^{\mathbf{R}}_k$.

(v), the source. The diameter of a hull is at most $3MR_k \sum_{P\in g}|P|$, since the hull is covered by the connected sets $(\mathsf{dom}\,P)^{+|P|}$, chained through intersecting spans, so $\mathrm{diam}_k(\mathsf{hull}(g)^+) \le MR_k(2 + 3n_g)$ with $n_g := \sum_{P\in g}|P|$, and $e^{|\Psi_g|} \le e^{30e^{-p_0(g_k)/4}n_g}$. Once $30e^{-p_0(g_k)/4} \le \kappa/6$, the last third of $e^{-\kappa|P|}$ pays both, and the sum of the masses $e^{|\Psi_g|}B(g)C\,\mathrm{diam}_k(\mathsf{hull}(g)^+)/r^{\mathrm{ext}}_k$ over the groups through a traversal, together with the deviation parts of the cluster terms of $E^{\mathbf{R}}_k(\mathcal{C}; V_k)$ and of $\Psi(S)$, which are local functions equal to their numbers at the flat reference and are bounded in the same way, is at most $C'MR_k e^{-p_0(g_k)/4}/r^{\mathrm{ext}}_k$, which is $C^{\mathbf{R}}_k\tilde{\rho}_k$ with $C^{\mathbf{R}}_k := C'MR_k e^{-p_0(g_k)/4}/(r^{\mathrm{ext}}_k\tilde{\rho}_k)$. The radius r^{ext}_k is a constant times $\hat{g}_k C_0(\log \hat{g}_k^{-2})^{q_0}$ [10, (2.28), p. 259], and $\tilde{\rho}_k = \rho_k/\min(\alpha_{0,k}, \alpha_{1,k})$ is of order g_k times a power of $\log g_k^{-2}$ (Paper, (5.9)), so $1/(r^{\mathrm{ext}}_k\tilde{\rho}_k) \le g_k^{-2}$ times a power of $\log g_k^{-2}$, and MR_k is a power of $\log g_k^{-2}$ by Paper, Lemma 3.4(i). This gives $C^{\mathbf{R}}_k \le C^{\mathbf{R}}(g_k)$. The groups with members replace their members' contributions and add per traversal at most $\sum_g 2B(g)(e^{x_g} - 1)$ with $x_g \le 4MR_k n_g C_\chi \tilde{\rho}_k \delta^*$; for γ_q small $4MR_k C_\chi \tilde{\rho}_k \delta^* \le 1$, and $e^x - 1 \le xe^x$ gives at most $C_a MR_k e^{-p_0(g_k)/4} C^{(k)}_\chi \tilde{\rho}_k \delta^*$, which is the change of a. The function $C^{\mathbf{R}}$ increases for small arguments and $g_k \le \gamma_q$ for $k \le K'$ (Paper, (3.46)), so $\sup_k C^{\mathbf{R}}(g_k) \le C^{\mathbf{R}}(\gamma_q)$, which is small since $p_0 > 1$.

(vi). By (iii)–(v) the post-**R** dressed loop is $\prod_i \kappa^{\mathbf{R}}_k(C_i)\, e^{\mathcal{E}_k + \sum_i E^{\circ,\mathbf{R}}_k(C_i) + E^{\mathbf{R}\times}_k + E^{\mathbf{R},\partial}_k(\ell)}$ times the small-field words with the deviation insertions D_g and the deviation parts of the cluster terms, and with the attached insertions at the second-class and unchanged components, where $\log \kappa^{\mathbf{R}}_k(C) = \sum_v N^{(k-1)}_v(C)\mu^{\mathbf{R}}_{k,v} + \sum_{c\in\mathcal{K}(C)} \zeta^{\mathbf{R}}_{k,v_c,w_c}$ is the **R**-part of Paper, (7.23). The deviation

insertions have the form which the class requires by (v), the attached insertions are paid by κ as in (v) or, for the terms of $E_k^{\mathbf{R},\partial}(\ell)$, within the reserve, by the count $4\,\mathsf{mass}(Z) \leq \mathsf{Res}(Z)$ of Proposition S.9(iv) with $m_n + m_n^{\mathbf{R}} \leq 2m_n$ in place of m_n at the **R**-steps, which halves its margin. The post-**R** density has the form of [10, Sect. 2] with the new domains, by [13, Theorem 1, p. 355], so the next step applies to it. A first-class component is removed, and its attached insertions and the words on its arcs are slots of its polymer; their masses are paid from its own factor, so their lineage ends. A second-class component stays a large-field region with the operation [13, (1.102), p. 390] and the factor $\mathbf{T}'_k(Y_i)1 \leq e^{-2(1+\beta_0)^{-1}p_0(g_k)}$ of [13, (1.89)], and “this new action satisfies the induction hypothesis” [13, p. 390], so the induction of Proposition S.9(iv) continues through it, as the branch of a small domain with a new large field [13, pp. 385–386], whose factor $\exp(-p_0(g_k))$ Lemma 17.20(v) provides.

(vii). By Paper, Lemma 3.4(ii), $R_{i+1}L^{i+1} \in \{R_iL^{i+1}, R_iL^i\}$, and both are powers of L [10, p. 245; (2.5), p. 255], so MR_iL^i divides $MR_{i+1}L^{i+1}$, and by induction MR_jL^j divides MR_kL^k for $j \leq k$ and MR_kL^k divides $MR_{K'}L^{K'}$. By Paper, (3.44), $L^{j_q-q} = L^3MR_{K'-1} \geq L^3MR_{K'}$, so $MR_{K'}L^{K'}$ divides $L^{j_q-q+K'}$, the fine length of a bond of $\mathbb{T}_q$. In the units of the step k a bond of $\mathbb{T}_q$ has length $a_k = L^{j_q-q}L^{K'-k}$, which is therefore a multiple of MR_k, and the corners and segment lengths of the loops are multiples of a_k (Paper, Lemma 3.14 and Lemma 7.1).

The conditions. The smallness conditions used are: $p_0(g_k) \geq 9$; the root counts $R_k^d, 3^d(101R_k)^d + 1 < p_0(g_k)$, from the condition of [13, p. 389]; $2p_0(g_k)e^Ce^{-p_0(g_k)/2} \leq e^{-p_0(g_k)/4}$; $30e^{-p_0(g_k)/4} \leq \kappa/6$; $m_k^{\mathbf{R}} \leq g_k^2$; $4MR_kC_\chi\tilde{\rho}_k\delta^* \leq 1$; and $C_dL^2\lambda_\alpha(\gamma_q) + C_aMR_ke^{-p_0(g_k)/4} \leq L^{-1}$, with λ_α of Paper, (6.6), which keeps $a \leq \frac{1}{2}$ in the fixed point of the recursion (Paper, Section 6.3). Each is a condition of the kind “p_0 large, g small” which [13, pp. 385, 389] imposes, and each holds for γ_q small because $p_0 > rd$ and $g_k \leq \gamma_q$ for $k \leq K'$. *q.e.d.*

9 The Large-Field Factors with Half of the Gains Set Aside

The weight of the large-field terms at the stopping step (Paper, Proposition 8.4) is bounded by comparing whole integrals, and the comparison has a price: every point of the history of a component, at every scale, costs a factor $e^{c_n+C_P}$ with c_n of the order of $\log g_n^{-2}$, which is more than the cost $O(1)M^dR_n^{d+1}d'_n$ per step that Bałaban’s induction [13, (1.80)–(1.88), pp. 384–387] pays. We pay it by raising the cost of each step and setting aside half of the gain of every new region, and we show that the induction survives both changes. Bałaban’s numbers $\kappa_j(Z)$ are defined by [13, (1.83), (1.85)] from the factors of [13, (1.79), p. 383], with one exception: for $Z = S(Z_0)$, Z_0 satisfying the conditions (i), (ii) of [12, p. 177] and a new large field in the preparatory operations, he puts $\kappa_{j+1}(Z) := p_0(g_j) - O(1)M^dR_{j+1}^{d+1}d'_{j+1}(Z)$ [13, p. 386], discarding $\kappa_j(Z_0) \geq 0$. We let $\kappa_j^{\mathrm{tr}}(Z)$ be the exact exponent of the factor of Z, defined by the same recursion with $\kappa_j^{\mathrm{tr}}(Z_0)$ kept in this case, so that $\kappa_{j+1}^{\mathrm{tr}}(Z) = \kappa_j^{\mathrm{tr}}(Z_0) + p_0(g_j) - O(1)M^dR_{j+1}^{d+1}d'_{j+1}(Z)$ there. In the construction of the paper the gain of a new region Z' created at the step $j' \geq 2$ is at least $\frac{1}{3}\gamma_0A_1^2p_0^2(g_{j'})(d'_{j'}(Z') + 1)$, and the terms $2p_0$ and the preparatory factor $p_0(g_j)$ are $(2/\rho)p_0$ and $p_0(g_j)/\rho$ (Paper, Proposition 3.12), while a region of the first step has the gain $\frac{1}{2}c_4\vartheta_V^2A_1^2(\log g_1^{-2})^{2p_0}(d'_1(Z') + 1)$ of Proposition S.7(iii). With the numbers $\Upsilon_{j'}$ of Paper, (8.21), which are half of these gains per unit of $d' + 1$, and with

$d = 4$, put, for a component Z of Z_j,

$$\begin{aligned} G_j(Z) &:= \sum_{Z'} \Upsilon_{j'}(d'_{j'}(Z') + 1) , \\ H_j(Z) &:= \sum_{n\le j} \lambda_n \sum_{Z''} M^d R_n^{d+1} d'_n(Z'') , \\ \lambda_n &:= 1 + 3 \cdot 2^{d-1} L^d C_\lambda (1 + \log g_n^{-2}) , \end{aligned} \tag{9.1}$$

the first sum over the new regions merged into Z during its history, the second over the components Z'' of Z_n inside Z, with the constant C_λ of Paper, (8.24).

Proposition S.11 (Bałaban's Induction with Half of the Gains)**.** *Let γ be sufficiently small, and let $p_0 > r(d+2)+1$. For every term, every $j \le K'$, and every component Z of Z_j, the number $\hat\kappa_j(Z) := \kappa_j^{\mathrm{tr}}(Z) - G_j(Z) - H_j(Z)$ satisfies:*

(i) $\hat\kappa_j$ obeys the recursion of κ_j^{tr} with the gain of every new region diminished by $\Upsilon_{j'}(d'_{j'}(Z')+1)$ and the cost $O(1)M^d R_n^{d+1} d'_n(Z'')$ of every step raised to $(O(1)+\lambda_n)M^d R_n^{d+1} d'_n(Z'')$;

(ii) with K_Z the number of steps of [13, (1.80)], which Bałaban writes K, and $n_{\max}$ the smaller of $j+K_Z$ and the index of the last step for which the tube defines the parameters g_n and R_n,

$$\hat\kappa_j(Z) \ge \sum_{n=j+1}^{n_{\max}} (O(1)+\lambda_n) M^d R_n^{d+1} d'_n(S^{n-j}(Z)) \ge 0 ; \tag{9.2}$$

the sum is cut at the end of the run because a domain whose removal would fall beyond it meets no further step, and the induction, which passes from j to $j+1$ by subtracting the cost of the step $j+1$, gives the statement with the sum cut in this way;

(iii) a component X of $Z_{K'}$ which fails the condition (i) or (ii) of [12, p. 177] at the step K' satisfies $\hat\kappa_{K'}(X) \ge \hat c_0 \#_{K'}(X)$ with $\hat c_0 := M^d R_{K'+1}^{d+1} (3\cdot 2^{d-1})^{-1}(2L^2)^{-d}$;

(iv) $\kappa_j^{\mathrm{tr}}(Z) \ge \kappa_j(Z)$, and [13, (1.79)] holds with κ^{tr} in place of κ.

Proof. (iv). The number $\kappa_j^{\mathrm{tr}}(Z)$ is, by definition, the exponent of the product of the factors of [13, (1.79)] which belong to Z, so [13, (1.79)] holds with it, and it differs from $\kappa_j(Z)$ only by the discarded numbers $\kappa_j^{\mathrm{tr}}(Z_0)$ of the preparatory case, which are nonnegative by (ii) at the step j; so $\kappa_j^{\mathrm{tr}} \ge \kappa_j$ by induction on j.

(i). The recursion of κ^{tr} has three cases: $Z = S(Z_0)$ with no new large field, in which $\kappa_{j+1}^{\mathrm{tr}}(Z) = \kappa_j^{\mathrm{tr}}(Z_0) - O(1)M^d R_{j+1}^{d+1} d'_{j+1}(Z)$ [13, (1.83)]; the preparatory case above; and a merger of components $Z_j^{(n)}$ and new regions $Z_{j+1}^{(i)}$, in which [13, (1.85)] adds the numbers of the components, the gains of the new regions, the terms $2p_0$, and subtracts the cost of the step and $2p_0(g_{j(Z)})$. In each case the history of Z is the union of the histories of its parents, the new regions of the step, and Z itself as a component of Z_{j+1}. So $G_{j+1}(Z)$ is the sum of the G_j of the parents and of $\Upsilon_{j+1}(d'_{j+1}(Z_{j+1}^{(i)})+1)$ over the new regions, and $H_{j+1}(Z)$ is the sum of the H_j of the parents and of $\lambda_{j+1} M^d R_{j+1}^{d+1} d'_{j+1}(Z)$. The recursion is affine in the gains and the costs, and the terms $2p_0$ and $p_0(g_j)$ enter κ^{tr} and the recursion of (i) alike; subtracting, the number defined by the recursion of (i) differs from $\kappa_{j+1}^{\mathrm{tr}}(Z)$ by exactly $G_{j+1}(Z)+H_{j+1}(Z)$, if it does so for the parents. At $j=1$ the history of a component of Z_1 consists of its regions and of itself, and the same computation applies.

(ii). We run Bałaban's proof of [13, (1.80)] [13, pp. 384–387] for the recursion of (i), by induction on j. Its geometric parts, the bound [13, (1.81)] on the right side through the

operation S, the covering [13, (1.84)], the tree of the domains in the merger, the subadditivity $d'_{j+1}(X) + d'_{j+1}(Y) + 2d \geq d'_{j+1}(Z)$ of [13, p. 386], and the counts $K \leq n_0 - j + R_j$, $K = R_{j+1}$, $K \leq K_2 + n_1 + R_{j+1}$ of [13, pp. 385–387], do not involve the sizes of the gains and of the costs. The cost constant enters [13, (1.81), (1.83), (1.87)] and the middle line of [13, (1.88)] on both sides alike, with $O(1) + \lambda_n \leq O(1) + \lambda_{j+1}$ for $n \geq j + 1$, since g_n increases with n along a history up to the factor $1 + \beta_0$ of Paper, Lemma 3.2, which a change of the constant absorbs. The sizes enter only through three inequalities. First, at the creation of a region, the analogue of the condition of [13, p. 385]: with the gain halved, the conversion $d'(Z)+1 \leq 2(14)^d \sum_i (d'(Z^{(i)})+1)$ of [13, p. 385] gives $\frac{1}{12}\gamma_0(14)^{-d} A_1^2 p_0^2(g_j) \geq (O(1)+\lambda_j)2(64)^d M^d L^{d+1} R_j^{d+2}$ as the requirement for $j \geq 2$, and $\frac{1}{8}c_4 d^{-1}(434)^{-d}\vartheta_V^2 A_1^2(\log g_1^{-2})^{2p_0} \geq (O(1)+\lambda_1)2(64)^d M^d L^{d+1} R_1^{d+2}$ at $j = 1$, by Proposition S.7(iii). Secondly, in the preparatory case, $p_0(g_j)/\rho \geq (O(1) + \lambda_{j+1}) M^d R_{j+1}^{d+1} d'_{j+1}(Z)$ summed over the $K = R_{j+1}$ steps of a small domain, that is $p_0(g_j)/\rho \geq C_d(O(1) + \lambda_{j+1}) M^d R_{j+1}^{d+2}$ with a constant C_d of d, since a domain in a cube of side $100MR_{j+1}$ has $d' \leq d\,100^d$; here $\hat{\kappa}_j(Z_0) \geq 0$ by the induction, so that $\hat{\kappa}_{j+1}(Z) \geq p_0(g_j)/\rho - (O(1) + \lambda_{j+1}) M^d R_{j+1}^{d+1} d'_{j+1}(Z)$, the right side being Bałaban's value with the raised cost. Thirdly, the last inequality of [13, (1.88)], $2(1 + \beta_0)^{-1} p_0(g_{j+1})/\rho \geq (O(1) + \lambda_{j+1})3(100M)^d R_{j+1}^{d+2}$. The gains do not enter the merger otherwise, since [13, (1.86)] uses only the numbers of X and Y. Each of the three inequalities compares $(\log g^{-2})^{2p_0}$ or $(\log g^{-2})^{p_0}$ with $\lambda R^{d+2} \leq C(\log g^{-2})^{r(d+2)+1}$, by $R_n \leq 2L(\log g_n^{-2})^r$ (Paper, Lemma 3.4(i)), so all three hold for γ small when $p_0 > r(d + 2) + 1$, which is a form of Bałaban's "p_0 large" [13, pp. 385, 387].

(iii). For a component X which fails (i) or (ii) at the step K', the number K of [13, (1.80)] is the smallest positive integer of [13, p. 384], so $K \geq 1$ and the first term of (9.2) gives $\hat{\kappa}_{K'}(X) \geq M^d R_{K'+1}^{d+1} d'_{K'+1}(S(X))$, since $O(1) + \lambda_{K'+1} \geq 1$. The domain $S(X)$ is the cover of X by cubes of side $LMR_{K'+1}$ of the lattice of the step K', which are the cubes of side $MR_{K'+1}$ of the step $K' + 1$, with ten layers of them added [13, p. 384]; so it has at least $21^d > 2^d$ of them and $d'_{K'+1}(S(X)) \neq 0$, because a union of cubes of linear size zero has at most 2^d cubes, all containing a common vertex. Hence $\#_{K'+1}(S(X)) \leq 3 \cdot 2^{d-1} d'_{K'+1}(S(X))$ [13, p. 384]. A cube of side $LMR_{K'+1}$ of the step K' is at most $L^2 MR_{K'}$ across, because $R_{K'+1} \leq LR_{K'}$ (Paper, Lemma 3.4(ii)), so it contains at most L^{2d} cubes of side $MR_{K'}$, and $\#_{K'+1}(S(X)) \geq L^{-2d} \#_{K'}(X) \geq (2L^2)^{-d} \#_{K'}(X)$. Together, $\hat{\kappa}_{K'}(X) \geq M^d R_{K'+1}^{d+1} (3 \cdot 2^{d-1})^{-1} (2L^2)^{-d} \#_{K'}(X)$. *q.e.d.*

10 The Propagators on Two Fine Lattices

Paper, Section 9, compares the terms of one walk of Bałaban's expansions in two constructions whose fine lattices have the spacings ε and ε/L, through the Galerkin identities of Paper, Lemma 9.2, and the bound on the bilinear defect of Paper, Proposition 9.3. The paper proves the identities, the reduction over the shells, and the defect of the averaging terms; here we prove the remaining inputs. We use the notation of that section: run A and run B, the shared scales, the aligned background $\tilde{U} = MU^B$, the local operators $\mathcal{O}^X$ of a cube $\square$ at the shared scale j with their spaces $\mathfrak{F}^X$, the linearised one-step average Q_1 at U^B, the shells indexed by run A's n, run B's extra shell with the index -1, the norms $\|\cdot\|_{1,\beta;y}$ of Paper, (9.14), and ϖ_n of Paper, (9.16). The block size M in this section is the size KR_0M_0 of [5, p. 408].

Proposition S.12 (The Propagators on Two Fine Lattices)**.** *Let* $\beta \in [\frac{3}{4}, 1)$*, let* $R_0 M_0$ *satisfy*

$$\begin{gathered} \tfrac{1}{6}\delta R_0 M_0 > 2d \log c_0(\tfrac{1}{6}) + 1 \,, \qquad 4\delta R_0 M_0 \geq 3(1 + \beta) \log L \,, \\ R_0 M_0 \geq L \,, \qquad \delta_Q R_0 M_0 \geq 12 \log L \,, \end{gathered} \tag{10.1}$$

with $\delta \in \{\delta_0, \delta_0'\}$*,* $\delta_0' := \min(\delta_0, \frac{1}{4}\delta_Q, \frac{1}{4}\delta_1, \frac{1}{4}\delta_c)$*, the rates* δ_Q *of the inverse of* $Q_1 Q_1^*$*,* δ_1 *of the operator* G_1 *of [5, (3.128)],* $\delta_c := \frac{1}{16}\min(\delta_V, \delta_0, \delta_1)$*,* c_0 *the constant of [3, (2.58)], and let* $K \geq 2$ *in* $M = KR_0M_0$*. Two one-step averages occur, and we fix the notation. The average of the construction is the average* M *of [9, (0.4), p. 253], and* Q_1^{pl} *denotes its linear part at* U^B*,* $M(e^{i(\varepsilon/L)A}U^B) = e^{i\varepsilon(Q_1^{\mathrm{pl}}A + O(A^2))}MU^B$*. The operators of [5] are built with the average of the fluctuation relative to the background,* $\bar{\bar{U}}_c' = (\overline{R_{c-}U'})^{-1}(\overline{U'U})_c(\bar{U}_c)^{-1}\bar{R}_c\overline{R_{c+}U'}$ *with the block means [5, (3.113), p. 418] of the contour variables, and* $Q_1 = Q(U^B)$ *always denotes its linear part [5, (3.13), p. 392], [4, (121)–(124), p. 36]. With*

$$(\Lambda_1 A)(y) := \frac{\varepsilon}{L} \sum_{x \in B(y)} L^{-d}(R_y A)(\Gamma_{y,x}) \,, \tag{10.2}$$

where $A(\Gamma)$ *is the sum of* A *over the bonds of the contour, so that* $\Lambda_1 A$ *is the linear part of* $\frac{1}{i}\log \overline{R_y U'}$ *for* $U' = e^{i(\varepsilon/L)A}$*, a function on the sites of run A, the two linear parts are related by item (j) below. At aligned backgrounds on the complex domain of [5, (3.37), p. 396] the following hold, with constants independent of* j*,* k*,* K*,* ε*, the torus, the block, and the field.*

(a) Averages. $Q_{n+1}^B(U^B) = Q_n^A(\tilde{U})Q_1$ *and* $Q_{n+1}'^B(U^B) = Q_n'^A(\tilde{U})Q_1'$ *for* $n \geq 0$*, with* $Q_0^A = 1$*, after the gauge rotation of the coarse sites which identifies the averaged configuration of run B with* $\tilde{U}$*.*

(b) The scalar operator. *The defect of* Δ_a' *has the form of Paper,* (9.15)–(9.16)*, with the constant* $C_{\mathcal{L}}'(\beta) + 8d + a$*.*

(c) The data operator. *With run B's data space split as* $\mathfrak{D}^A \oplus \mathfrak{D}_0^B$*,* $\mathfrak{D}_0^B$ *the data of the extra shell, and* $K^X := Q'^X G_\square'^{X2} Q'^{X*}$*,*

$$[C_\square^B]_{11} - C_\square^A = C_\square^A(K^A - K_{11} + K_{12}K_{22}^{-1}K_{21})[C_\square^B]_{11} \,, \tag{10.3}$$

with the blocks K_{ik} *of* K^B *for this splitting, and* $|([C_\square^B]_{11} - C_\square^A)(z \leftarrow z')| \leq C_\square(\beta)(\varepsilon/a_j)^\beta \ell_m^{-4} e^{-\frac{1}{27}\delta_0 d(z,z')}$ *for* z *in the layer* m*.*

(d) The one-form operator. *The defect of* $\Delta_a = D^*D + \Delta' + DRD^* + Q^*aQ$ *has the form of Paper,* (9.15)–(9.16)*, the gauge-fixing part with a norm weighted by the decay of* $P = G'Q'^*(Q'G'^2Q'^*)^{-1}Q'G'$*, with the rate* $\frac{1}{3}\delta_{\mathrm{gf}}'$*,* $\delta_{\mathrm{gf}}' := \min(\frac{1}{486}\delta_0, \delta_0')$*.*

(e) The lift. *Let* $\mathcal{L}_\iota := H_1$ *be the operator of [5, (3.129), p. 421] of the one-step problem on* $Y^\sim$ *at the background* U^B*, whose constraints are* Q_1 *on the bonds of run A and the identity on the fine bonds of the collar; when* U^B *is the localised lift* $\iota_Y(\tilde{U})$ *of its own average,* $\mathcal{L}_\iota$ *is the part in the Landau gauge of the derivative of* ι_Y *at* $\tilde{U}$*. On the fields of run A which vanish within the distance* 2ε *of the collar,* $Q_1\mathcal{L}_\iota = 1$*; the cut-off* $\chi_B\mathcal{L}_\iota$ *by the characteristic function of* $\Omega_0^B(\square)$ *maps* $\mathfrak{F}^A$ *into* $\mathfrak{F}^B$ *and is an exact right inverse there,* $\chi_A Q_1 \chi_B \mathcal{L}_\iota X = X$ *for* $X \in \mathfrak{F}^A$*; and for such a field* v*, at every fine bond* z*,*

$$\begin{aligned} |\mathcal{L}_\iota v(z)| &\leq C_{\mathcal{L}'} \sum_c e^{-\delta_0' d_\varepsilon(z,c)} |v(c)| \,, \\ |\nabla \mathcal{L}_\iota v(z)| &\leq C_{\mathcal{L}'} \sum_c e^{-\delta_0' d_\varepsilon(z,c)} (|\nabla v(c)| + \bar{\vartheta}(c)\varepsilon^{-1}|v(c)|) \,, \end{aligned} \tag{10.4}$$

the sums over the bonds c *of run A, with* d_ε *the distance in the units of* ε *and*

$$\bar{\vartheta}(c) := \sup_{c'} e^{-\delta_0' d_\varepsilon(c,c')} \vartheta(c') \,, \qquad \vartheta := \varepsilon^2|F| + \varepsilon^3|J| \,, \tag{10.5}$$

where $F(p) := (\varepsilon/L)^{-2}(U^B(\partial p) - 1)$ *and* J *is the current [5, (3.11), p. 392] of* U^B*, and* $\vartheta := 1$ *at the collar and outside* $Y^\sim$*. On the shell* n *of a local sequence* $\bar\vartheta\varepsilon^{-1} \le 2(L^n\varepsilon)^{-1}$*, so that* $\chi_B\mathcal{L}_\iota$ *satisfies the hypothesis of Paper, Lemma 9.4, with the constant* $3C_{\mathcal{L}'}$ *and the rate* δ_0' *in place of the range of one block.*

(f) The explicit factors. *The multiplications by the partition functions* $h_\square$*,* $\zeta_{\tilde\square}$ *and their commutators* $K(h_\square)$*,* $P_{\square,1}(\partial h_\square)$ *with the local operators have, between the two lattices, a defect of relative size at most*

$$\Big[\frac{3(2d+1)c_h^*(1 + C_{\mathcal{L}'}m_Q)}{M} + \frac{8dc_h^*}{c_K}\Big]\frac{\varepsilon}{a_j} + \frac{MC_{[h]}(\beta)}{c_K}\Big(\frac{\varepsilon}{a_j}\Big)^\beta , \tag{10.6}$$

where $a_j = L^j\eta$ *is the spacing of the scale* j *of the cube. Every term carries a positive power of* ε/a_j*.*

(g) The class of pairs. *The pair* $(e^{it\varepsilon A}\tilde U, J' + t\delta\mathbf{J})$ *of a configuration and a current lies in the domain of analyticity of the operators of [5, Sect. D] for* $|t| \le \alpha_*'/(2\bar d(\omega))$*, with* $\alpha_*' := \min(\alpha_1, \alpha_J/\hat x_J)$*,* $\alpha_J := M\alpha_0$*, and* $\hat x_J$ *the constant of the bound* $a_{j(x)}^3|\delta\mathbf{J}(x)| \le \hat x_J\bar d(x)$*, provided* $4\bar B_J'\varepsilon_1 \le \alpha_J$*.*

(h) The nonlinear functions. *On the polydisc* $|s(\Delta)| \le e^{\kappa_1}$*,* $|(Q_1H^B(s) - H^A(s))X| \le C_{\text{tel}}\delta_w(Y)B_Te^{2\cdot 6^d\kappa_1}|X|$*, and the same for* $Q_1H_0^B(s) - H_0^A(s)$*, for* $Q_1\tilde G^B(s) - \tilde G^A(s)\mathcal{L}_\iota^*$*, and for the fine differences* $H^B(s) - \mathcal{L}_\iota H^A(s)$*,* $H_0^B(s) - \mathcal{L}_\iota H_0^A(s)$*, and* $\tilde G^B(s) - \mathcal{L}_\iota\tilde G^A(s)\mathcal{L}_\iota^*$*, in the supremum norm, where* $\delta_w(Y)$ *is the supremum of* $\delta_w(\omega)$ *over the walks in* Y*,* B_T *the constant of the sums over the composite walks, and* C_{tel} *a constant of d, L, and the constants of the lifts, under* $M \ge 4\kappa_1/(\rho\delta_g)$*. The first differences of these six differences satisfy the same bounds, with the power of the scale lowered by one and with* $\delta_w(Y)^{\beta/(1+\beta)}$ *in place of* $\delta_w(Y)$*.*

(i) The reconstruction. *Let* $\tilde C_1 := G_1 - H_1QG_1$*, with the operators and the constraint* Q *of the one-step problem of (e). Every field* u *of run* B *on* $Y^\sim$ *which vanishes within the distance* 2ε *of the collar satisfies*

$$(1 - \mathcal{L}_\iota Q_1)u = \tilde C_1(D^*D + DRD^* + \Delta' - \Delta_\pi' - \Delta_\pi^{(2)})u , \tag{10.7}$$

with the operators Δ'*,* Δ_π'*,* $\Delta_\pi^{(2)}$ *of [5, (3.10), (3.120), (3.135)]. For a scale* $a \ge \varepsilon$ *and a block* z' *of side* ε *let* $N_a(u; z') := \sup(a^{-1}|u| + |\nabla u|) + a^\beta[\nabla u]_\beta$*, the supremum over the points within the distance* a *of* z' *and the Hölder quotient of [5, (3.40)] over the pairs of such points. Then, at every fine bond* z*,*

$$\begin{aligned} |(1 - \mathcal{L}_\iota Q_1)u(z)| &\le C_{\text{rc}}\varepsilon\sum_{z'} e^{-\delta_0' d_\varepsilon(z,z')}\Big(1 + \bar\vartheta(z')\frac{a}{\varepsilon}\Big)N_a(u; z') , \\ |\nabla(1 - \mathcal{L}_\iota Q_1)u(z)| &\le C_{\text{rc}}(\beta)\sum_{z'} e^{-\delta_0' d_\varepsilon(z,z')}\Big(\Big(\frac{\varepsilon}{a}\Big)^\beta + \bar\vartheta(z')\frac{a}{\varepsilon}\Big)N_a(u; z') , \end{aligned} \tag{10.8}$$

the sums over the blocks z' *of side* ε*, with constants* C_{rc} *and* $C_{\text{rc}}(\beta)$ *of d, L,* β*, and the constants of [5, Theorems 3.1, 3.2, 3.12].*

(j) The two averages. *With* $\bar D$ *the covariant difference of the functions on the sites of run A at* $\tilde U$*,* $(\bar D\mu)(c) := \varepsilon^{-1}(R(\tilde U_c)\mu(c_+) - \mu(c_-))$ *as in [5, (3.114)],*

$$Q_1 = Q_1^{\text{pl}} + \bar D\Lambda_1 , \qquad Q_1D\lambda = \bar DQ_1'\lambda , \qquad Q_1^{\text{pl}}D\lambda = \bar D(\lambda\restriction_{\textit{sites of run A}}) , \tag{10.9}$$

for the fields A and the functions λ of run B, where Q'_1 is the block mean of [5, (3.19)]. More precisely, for every fluctuation U' the configuration $\bar{\bar{U}}'\tilde{U}$ is the gauge transform of $M(U'U^B)$ by the gauge transformation $y \mapsto (\overline{R_y U'})^{-1}$ of the sites of run A. Consequently: every gauge-invariant function of the averaged configuration, in particular A_ε, has the same value at $\bar{\bar{U}}'\tilde{U}$ and at $M(U'U^B)$; the two sets of fine configurations on $Y^\sim$ with $\bar{\bar{U}}'\tilde{U} = V$ and with $M(U'U^B) = V$, for a configuration V of run A, are mapped onto each other by gauge transformations of the fine lattice, and so are the minimisers of the action on them; and if $U^B = \iota_Y(\tilde{U})$, then for every field v of run A which vanishes within the distance 2ε of the collar there is a function λ on the fine sites with

$$d\iota_Y(\tilde{U})[v] = \mathcal{L}_\iota v + D\lambda \,. \tag{10.10}$$

The statements (a)–(i) concern Q_1. The plain linear part Q_1^{pl} is not the straight average plus terms which vanish with the curvature: at a flat background it contains the term $-\bar{D}\Lambda_1$, which does not vanish, and the form [4, (124), (125)] used in the proof of (e) belongs to Q_1.

Proof. (a). By [5, (3.15), p. 393], $Q_j(U) = Q(\bar{U}^{j-1}) \cdots Q(\bar{U})Q(U)$, a composition of one-step averages at the iterated averaged configurations, and by [5, (3.19)] the scalar averages $Q'_j(U)$ are the compositions of the one-step averages $Q'(V)\lambda(y) = \sum_{x \in B(y)} L^{-d} R(V(\Gamma_{y,x}))\lambda(x)$. Run B's first averaged configuration is $\overline{U^B} = MU^B = \tilde{U}$, and its iterated averages are run A's averages of $\tilde{U}$, since the scale $n+1$ of run B is the scale n of run A (Paper, Section 9.1). So $Q^B_{n+1}(U^B) = Q^A_n(\tilde{U})Q(U^B)$ and $Q'^B_{n+1}(U^B) = Q'^A_n(\tilde{U})Q'(U^B)$, exactly at the plain average; at the double bar of [4, (78)–(80)] the identity holds after the gauge rotation v_1 at the coarse sites which carries $\overline{U^B}$ to $\tilde{U}$, so with $Q'_1 := R(v_1)^{-1}Q'(U^B)$. The rotation is a gauge transformation, real at real backgrounds, and at complex backgrounds it changes the bounds by at most the factor 2 of **Ad** under $8dc_G\alpha_1 \le \log 2$. In both readings the one-step factor is the linear part Q_1 of the average of the fluctuation relative to the background, which is the average of [5, (3.13)], and not the linear part Q_1^{pl} of M; the background of the next factor is the average $MU^B = \tilde{U}$ of the background, because $(\bar{U})_c$ in the definition of $\bar{\bar{U}}'_c$ is the average [9, (0.4)] of U.

(j). In the definition of $\bar{\bar{U}}'_c$ the factor $(\overline{U'U})_c$ is the average $M(U'U^B)_c$, $\bar{U}_c = \tilde{U}_c$, and $\bar{R}_c$ is the conjugation by $\tilde{U}_c$, so that

$$\bar{\bar{U}}'_c\,\tilde{U}_c = g(c_-)^{-1}\, M(U'U^B)_c\, g(c_+) \,, \qquad g(y) := \overline{R_y U'} \,, \tag{10.11}$$

which is the gauge transform of $M(U'U^B)$ by $y \mapsto g(y)^{-1}$. For $U' = e^{i(\varepsilon/L)A}$ the definition [5, (3.113)] gives $g(y) = \exp(i(\Lambda_1 A)(y) + O(A^2))$, and the linear parts of the logarithms of the two sides of (10.11), taken relative to $\tilde{U}_c$, are $\varepsilon(Q_1 A)(c)$ and $\varepsilon(Q_1^{\mathrm{pl}} A)(c) - (\Lambda_1 A)(c_-) + R(\tilde{U}_c)(\Lambda_1 A)(c_+)$; this is the first identity of (10.9). The second is [5, (3.114), p. 418], and the third is the linear part of the covariance $M(W^u) = M(W)^{u\restriction}$ of the average under the gauge transformations [9, (0.6), p. 253], $u \restriction$ being the restriction of u to the sites of run A. The three agree: along a contour the sum of $D\lambda$ telescopes, so that $\Lambda_1 D\lambda = Q'_1\lambda - \lambda\restriction$.

The first consequence is the invariance of the function. For the second, let $W = U'U^B$ with $\bar{\bar{U}}'\tilde{U} = V$. By (10.11), $M(W) = V^g$, and if h is the gauge transformation of the fine sites which equals g at the sites of run A and 1 elsewhere, then $M(W^{h^{-1}}) = M(W)^{g^{-1}} = V$ by the covariance, so that $W^{h^{-1}}$ lies in the second set. Conversely, let $M(W) = V$ and $U' = WU^{B\,-1}$. For a gauge transformation h which is 1 off the sites of run A the fluctuation U'' of W^h has the contour variables $h(y)U'(\Gamma_{y,x})$, $x \neq y$, and by (10.11) and the covariance, $\bar{\bar{U}}''\tilde{U} = V$ as soon as $h(y) = \overline{R_y U''}$ at every site y of run A. In the logarithm ξ of $h(y)$ this is the equation $\xi = \sum_{x\in B(y),x\neq y} L^{-d}\frac{1}{i}\log(e^{i\xi}U'(\Gamma_{y,x}))$, whose right side has the derivative $1-L^{-d}$ at $\xi=0$, $U'=1$; for the fluctuations of the domain of [4, (121)–(123)] it is a contraction of a ball about the origin, with a unique fixed point which is analytic in U'. So the two sets are mapped onto each other by gauge transformations, which preserve the action, and their minimisers correspond; the minimisers are unique up to gauge transformations [7, Theorem 1, p. 279].

For (10.10), let $W_t := \iota_Y(e^{it\varepsilon v}\tilde{U})$, so that $W_0 = U^B$ and $M(W_t) = e^{it\varepsilon v}\tilde{U}$, and let W'_t be the minimiser on the set of the fluctuations with $\bar{\bar{U}}'\tilde{U} = e^{it\varepsilon v}\tilde{U}$, in the Landau gauge about U^B. The logarithm of the average of the fluctuation is the datum of [7, (20), p. 281], and the derivative of the minimiser in this datum, in the Landau gauge, is H_1 [7, (174), (177), p. 306]: $W'_t = \exp(i(\varepsilon/L)(t\mathcal{L}_\iota v + O(t^2)))U^B$. By what has been shown, $W'_t = W_t^{h_t}$ with gauge transformations h_t of the fine sites, analytic in t with $h_0=1$, which contain the return to the axial gauge of [7, p. 307]; differentiating at $t=0$, with the rule $A\mapsto A-D\lambda$ for the gauge transformation $e^{i\lambda}$ [5, p. 418], gives (10.10) with λ the derivative of $\frac{1}{i}\log h_t$ at $t=0$.

(b). By [5, (3.23), (3.24), p. 394], $\Delta'_a = (\Delta^\eta_U + Q'^*aQ')\restriction_{\Omega_0}$ with $\langle\lambda, Q'^*aQ'\lambda\rangle = \sum_n a_n \sum_{y\in\Lambda_n}(L^n\eta)^{d-2}|(Q'_n(U)\lambda)(y)|^2$ and $a_{n+1} = aa_n/(aL^{-2}+a_n)$, $a_1=a_0=a$. The averaging part of the defect is computed exactly in Paper, Proposition 9.3, from (a): it is $\langle\mathcal{F}, s_0^{\mathrm{av}}\rangle_B$ with $s_0^{\mathrm{av}} = \sum_{n\geq 0}(a_n - a_{n+1})(L^n\varepsilon)^{-2}Q'^{B*}_{n+1}1_{\Lambda^A_n}Q'^B_{n+1}\mathcal{A} - a(\varepsilon/L)^{-2}1_{\Lambda^B_0}\mathcal{A}$, and $|a_{n+1}-a_n|/a_n \leq L^{-2n}/(1-L^{-2})$. For the covariant Laplacian we fix a point z and the cell of fine bonds within ℓ^1-distance $4d\varepsilon$ of it; in a complex axial gauge about z, every transport along a path of at most $3dL$ fine bonds in the cell has adjoint action within $\delta_T(n) \leq 32d^2c_F(M\alpha_0+\alpha_1+\alpha_1^2)L^{-2n}$ of 1, by the holonomy around the plaquettes and the curvature bound $c_F(M\alpha_0+\alpha_1+\alpha_1^2)(L^n\varepsilon)^{-2}$ which [5, (3.35), (3.37), p. 396] give on the shell n. In this gauge run A's difference quotient of $Q'_1\mathcal{A}$ is the average over a block of run B's difference quotients of $\mathcal{A}$ along the transported paths, up to the transports, so that $\langle\nabla^A Q'_1\mathcal{F}, \nabla^A Q'_1\mathcal{A}\rangle_A - \langle\nabla^B\mathcal{F}, \nabla^B\mathcal{A}\rangle_B$ is a sum over the blocks of the covariance of $\nabla^B\mathcal{F}$ and $\nabla^B\mathcal{A}$ inside the block, plus the transport errors. The covariance is an oscillation: by the Hölder bound of $\nabla\mathcal{A}$ at the scale $L^n\varepsilon$, the deviation of $\nabla^B\mathcal{A}$ from its block mean is at most $(\varepsilon/(L^n\varepsilon))^\beta(L^n\varepsilon)^\beta\|\nabla\mathcal{A}\|_\beta$, which is $L^{-\beta n}$ times the norm $\|\mathcal{A}\|_{1,\beta;y}$. Summing by parts, the defect is $\langle\nabla\mathcal{F}, s_1\rangle_B + \langle\mathcal{F}, s_0\rangle_B$ with s_1 the deviation field and s_0 the transport errors, bounded by $(C'_{\mathcal{L}}(\beta)+8d)\varpi_n\|\mathcal{A}\|_{1,\beta;y}$, and together with the averaging part this is (b). On run A's shell 0 and on run B's extra shell nothing relates the two operators, and the same computation gives the bound with $\varpi=1$.

(c). Since run B's data space is $\mathfrak{D}^A \oplus \mathfrak{D}^B_0$, with $Q'^B_0 = 1$ on the extra shell, the inverse of the block matrix gives $[C^B_\square]_{11} = (K_{11}-K_{12}K_{22}^{-1}K_{21})^{-1}$, the inverse of the Schur complement, and (10.3) is the resolvent identity $X^{-1}-Y^{-1} = Y^{-1}(Y-X)X^{-1}$ with $X = K_{11}-K_{12}K_{22}^{-1}K_{21}$ and $Y=K^A$. By (a), $K_{11} = Q'^A_nQ'_1G'^{B2}_\square Q'^*_1Q'^{A*}_n$ on $\mathfrak{D}^A$, so $K^A - K_{11}$ is the defect of $G'^2_\square$, which is $\mathsf{def}(G')G'^B + G'^A\mathsf{def}(G')$ with $\mathsf{def}(G') := \chi_AQ'_1G'^B - G'^A\chi_AQ'_1$, bounded by Paper, Lemma 9.4, with (b) and the transported block-constant lift $\mathcal{L}'f(x) := R(U^B(\Gamma_{y,x}))^{-1}f(y)$, $x\in B(y)$, which is an exact right inverse of Q'_1 of range one block by [5, (3.19)], maps $\mathfrak{F}^A$ into $\mathfrak{F}^B$ when $R_0M_0 \geq L$, and is an isometry for the bilinear trace form. The term $K_{12}K_{22}^{-1}K_{21}$ couples to the extra shell, at weighted distance at least $2R_0M_0j$ from a block of $\square$, and is of relative size $L^{-(1+\beta)j}$ by the second condition of (10.1). The per-block bounds of the factors are [5, (3.48), Theorem 3.2, p. 398] for $C_\square$ and [5, (3.42)] for $G'_\square$, and on the data spaces transposition for the weighted pairing is exact; three compositions, each at a third of the rate by the triangle inequality and the count [3, (2.58)], give the rate $\frac{1}{27}\delta_0$, under the conditions $\frac{2}{3}\kappa R_0M_0 \geq (|p+q|+1)\log L$ for the powers ℓ^p, ℓ^q of the intermediate layers, which are of the form of (10.1); the innermost shell needs its width $ML^j\eta \geq 2R_0M_0L^j\eta$, that is $K\geq 2$.

(d). The averaging part Q^*aQ of Δ_a [5, (3.16), p. 393] is exact by (a), with only the

extra-shell term left. The action part $D^*D + \Delta'$ is the second variation of the Wilson action at the background [5, (3.10)]; differentiating twice the identity which defines the blocking artefact $b_\square = A_\varepsilon \circ \bar{M} - A_{\varepsilon/L}$ on the plaquettes of Ω_0^A and Ω_0^B, where $\bar{M}(U'U^B) := \bar{\bar{U}}'\tilde{U}$ is the average of the fluctuation relative to the background, with $d\bar{M} = Q_1$ and $d^2\bar{M} = 2C_1^{(2)}$, and where $A_\varepsilon \circ \bar{M} = A_\varepsilon \circ M$ by (j), so that $b_\square$ is the blocking artefact of the construction, gives the defect as $d^2 b_\square[\mathcal{F}, \mathcal{A}] - 2\langle J^A, C_1^{(2)}[\mathcal{F}, \mathcal{A}]\rangle_A$ plus the plaquettes cut by χ_A, all in the shells with $\varpi = 1$, where $J^A = D_A^* F^A$ is the current of $\tilde{U}$ [5, (3.11)–(3.12)] and $|C_1^{(2)}[\mathcal{F}, \mathcal{A}]| \le C_2'\varepsilon|\mathcal{F}||\mathcal{A}|$ by Cauchy's estimate in the domain of [4, (123)]. The complex background reduces to a real one by the Baker–Campbell–Hausdorff series, $e^{i\eta X}U'U^B = e^{i\eta Z}U^B$ with Z analytic, and the identity extends by the identity theorem. The second variation of $b_\square$ is a sum over the coarse plaquettes of local Taylor remainders, an oscillation of relative order L^{-n} on the shell n between fields regular at that scale, of which the Hölder exponent keeps $L^{-\beta n}$. The gauge-fixing part $DRD^* = DD^* - DPD^*$ [5, p. 414] is nonlocal through P; its defect is reduced to those of $G'_\square$ by (b) and of $C_\square$ by (c), through the telescoping of the product $G'Q'^*CQ'G'$, each factor at a third of the rate, and the source fields are bounded in the norm $\|\cdot\|_{1,\beta;y}$ weighted by the decay $e^{-\frac{1}{3}\delta'_{\mathrm{gf}}d}$ of these factors, which is the form used in Paper, Lemma 9.4.

(e). The one-step problem on $Y^\sim$ is the local problem of [5, p. 417] at $k = 1$, with $\Omega_0 = Y^\sim$, $\eta = \varepsilon/L$, the layer Λ_1 of the bonds of run A, and the collar Λ_0 of fine bonds on which $Q_0 = 1$; its hypotheses [5, (3.35), (3.36)] are bounds on U^B at the scales ε/L and ε, which are weaker than the bounds U^B satisfies on its own layers. Its operator $H_1 = G_1 Q^*(QG_1Q^*)^{-1}$ [5, (3.129), p. 421] satisfies $QH_1 = 1$ identically, and on Λ_1 the constraint Q is the linear part at U^B of the averaging [5, (3.13), p. 392], which is Q_1. Hence $Q_1\mathcal{L}_\iota v = v$ for a field v of run A which vanishes at the collar, whose data are v on Λ_1 and zero on Λ_0. Now let $U^B = \iota_Y(\tilde{U})$, where $\iota_Y(U)$ is Bałaban's local minimiser on $Y^\sim$ with the block data U and the collar data $Q_1^{s*}U$. Its constraint is that of the construction, $M(W) = U$ [7, (3), p. 278], while the constraint of H_1 is Q_1, and the two are related by (j): by (10.10) the derivative of ι_Y at $\tilde{U}$ in the direction v is $\mathcal{L}_\iota v$ plus a linear gauge transformation. This is the sense in which $\mathcal{L}_\iota$ is the part in the Landau gauge of the derivative of ι_Y at $\tilde{U}$; the terms to which the lift is applied are gauge invariant and do not distinguish the two. In general the operator H_1 decays at the rate $\frac{1}{2}\delta_1$ by [5, (3.133), p. 422], so that it does not map $\mathfrak{F}^A$ into $\mathfrak{F}^B$, and we cut it off: let χ_B be the characteristic function of $\Omega_0^B(\square)$. Then $\chi_B\mathcal{L}_\iota$ maps $\mathfrak{F}^A$ into $\mathfrak{F}^B$, and it is still an exact right inverse there, $\chi_A Q_1 \chi_B \mathcal{L}_\iota X = X$ for $X \in \mathfrak{F}^A$, for a geometric reason and not up to a tail. The kernel $Q_1(c, b)$ vanishes unless the fine bond b lies within the ℓ^∞-distance 2ε of the bond c of run A, because the paths of the average, the straight paths $[x, x']$ of the tubes and the paths $\Gamma_{c_\pm,x}$ inside the blocks $B(c_\pm)$ [4, (124), p. 36], stay within $2L$ fine spacings; the margin between $\Omega_0^A(\square) = \Omega_1^B(\square)$ and the complement of $\Omega_0^B(\square)$ is $2R_0M_0\varepsilon/L$ (Paper, Section 9.1), which is at least 2ε by the third condition of (10.1); hence $\chi_A Q_1(1 - \chi_B) = 0$ and $\chi_A Q_1 \chi_B \mathcal{L}_\iota X = \chi_A Q_1 \mathcal{L}_\iota X = X$. The proofs of the identities of Paper, Lemma 9.2, use the lift only through these two properties. What the cut-off costs is a term in the derivative of $\chi_B \mathcal{L}_\iota w$ on the bonds which cross the boundary of $\Omega_0^B(\square)$, at most $(L/\varepsilon)|\mathcal{L}_\iota w|$; these bonds lie in run B's extra shell, whose own norm carries the factor $(\varepsilon/L)^{-1}$ and which Paper, Lemma 9.4, bounds by $L^{-\beta j}$. The bounds (10.4) are proved after (i), on which the second of them rests.

(f). The functions $h_\square$ and $\zeta_{\tilde{\square}}$ are restrictions to the two fine lattices of one function on the torus each, built from one fixed $h \in C_0^\infty(]-\frac{2}{3}, \frac{2}{3}[)$ [2, (1.118), p. 36] [3, p. 229] [5, pp. 408, 414], with $|\partial^k h_\square| \le c_h^*(ML^j\eta)^{-k}$ for $k \le 3$. The multiplication by such a function commutes with the average Q_1 up to the oscillation of the function over the reach of its kernel: the kernel of Q_1 at a bond b of run A has total weight at most 3 and reads the fine bonds b' within the ℓ^1-distance $(2d + 1)\varepsilon$ of the point at which h is taken, so that

$[h,Q_1]u(b)=\sum_{b'}w(b,b')(h(b)-h(b'))u(b')$ and

$$|[h,Q_1]u|\le 3(2d+1)\varepsilon\,\sup|\partial h|\,\sup|u|\le\frac{3(2d+1)c^*_h}{M}\,\frac{\varepsilon}{a_j}\,\sup|u|\ ,$$

by $|\partial h|\le c^*_h(Ma_j)^{-1}$. This is the first term of (10.6), the factor $1+C_{\mathcal{L}'}m_Q$ coming from the same commutator with the lift on the side of the sources. For the scalar averaging commutators, $Q'^B=Q'^AQ'_1$ and $Q'_1Q'^*_1=1$ give the defect $Q'_1[Q'^*_B,h]-[Q'^*_A,h]=Q'_1[Q'^*_1,h]Q'^*_A$ exactly, which is at most $8d\varepsilon c^*_h(Ma_j)^{-1}$ times the supremum of its argument; relative to the size c_KM^{-1} of these commutators this is the second term, with no loss of M. Both terms are thus of the first order in ε/a_j. Finally the commutators $K(h)$, $P_{\Box,1}(\partial h)$ of [5, (3.88), (3.101)–(3.105), pp. 409, 414] read the local operators within one block of the support of $\partial h_\Box$, where their defect is (b)–(d), at the cost of the factor M from ∂h, which gives the third term with the constant $C_{[h]}(\beta)$.

(g). By Paper, Proposition 10.6, $U^A=e^{i\varepsilon A}\tilde{U}$ with $a_{j(x)}|A(x)|+a^2_{j(x)}|\nabla A(x)|\le\bar{d}(x)$. For $|t|\le\alpha_1/(2\bar{d}(\omega))$ the configuration $U(t)$ satisfies [5, (3.37)] layer by layer, a condition which on the finer shells of a local sequence is weaker. The operators of [5, Sect. D] carry the current, and along the path $\mathbf{J}(t)=J'+t\delta\mathbf{J}$ from the current J' paired with $\tilde{U}$ to the current $J(U^A)$ of run A, $a^3_{j(x)}|\mathbf{J}(t)(x)|\le\bar{B}'_J\varepsilon_1+|t|\hat{x}_J\bar{d}(\omega)\le\bar{B}'_J\varepsilon_1+\frac{1}{2}\alpha_J$ for $|t|\le\alpha_J/(2\hat{x}_J\bar{d}(\omega))$. The class of pairs bounds the current by $\alpha_Ja^{-3}_{j(x)}$, with $\alpha_J=M\alpha_0$ fixed with α_0, so under $4\bar{B}'_J\varepsilon_1\le\alpha_J$ the current is at most $\frac{3}{4}\alpha_Ja^{-3}_{j(x)}$ and the pair stays in the class for $|t|\le\alpha'_*/(2\bar{d}(\omega))$. The local propagators are analytic there with the bounds of [5, Theorems 3.1–3.3], by [5, Theorem 3.4, p. 400], whose proof uses only these theorems at U, and they hold for the local sequences by [5, Corollary 3.6, p. 408].

(h). The operator $\mathbf{H}_k(B')=H_0B'+\mathbf{A}_0-HD(H_0B'+\mathbf{A}_0)$ of [11, (1.2)–(1.4), p. 2] is built from the propagators through the fixed points D and $\mathbf{A}_0$, whose local functions C_j [7, (44), (49)] and $\delta V/\delta A'$ [7, (80)] are each run's own, and the decoupling replaces only the propagators [11, p. 3]. Its walk expansion is a composite walk, with vertices at which three chains meet [5, p. 427], and $H_0=G_aQ^*(QG_aQ^*)^{-1}$ [7, (129), p. 297] contains an inverse which [5] does not expand, whose local inverses $(QG_{a,\Box}Q^*)^{-1}$ we compare by the identity of (c) with $G_{a,\Box}$ in place of $G'^2_\Box$. By (a), the n-fold average of run B is the $(n-1)$-fold one of run A applied to the one-step average, so the functions C_j of the two runs differ by the one-step remainder C_1 composed with run A's, which is of order ε relative to the scale. The telescoping of Paper, (9.22), over the $N_c(|\omega|+1)$ constituents of a composite walk, with the per-constituent defects of (b)–(f) and of these functions, bounds the difference of a walk term by $C_{\mathrm{tel}}2^{|\omega|}N_c\max_c\mathfrak{d}_c$ times its bound, with $C_{\mathrm{tel}}:=\max(3,\frac{5}{2}(c_1+c_2))$ and no smallness condition on the defects: when $N_c\max_c\mathfrak{d}_c\le\log\frac{3}{2}$ the mixed terms give at most $3\cdot2^{|\omega|}N_c\max_c\mathfrak{d}_c$, and otherwise the two sides are bounded separately by c_1 and c_2 times the bound, with $c_1\le3$ the weight of Q_1 and c_2 the constant of the slot map. The factor $|\omega|+1\le2^{|\omega|}$ is absorbed by the factors $O(M_1^{-1/2})^{|\omega|}$, which "are used to control the sum over ω, and they determine the constant B_0" [11, p. 3], at the price of M_1 four times Bałaban's lower bound. Bałaban's argument for [11, (1.11), p. 5] uses $\delta_0d(\omega)\ge\delta_1mM$ for $m>2^4$ and $\delta_1M\ge\kappa_1$; at the rate of the composite walks in the global distance this is the condition $M\ge4\kappa_1/(\rho\delta_g)$, and it gives (h) on $|s(\Delta)|\le e^{\kappa_1}$, where the functions enter the Cauchy integrals of [11, (1.23), p. 7].

(i), the identity. In the one-step problem of (e) the layer Λ_1 has the scale $L\eta=\varepsilon$. By [5, (3.120), (3.128), (3.134), (3.135)], $G_1^{-1}=\Delta-\Delta'_\pi-\Delta^{(2)}_\pi+DRD^*+Q^*aQ$ with $\Delta=D^*D+\Delta'$ [5, (3.10)], which is also the content of the series [5, (3.138)]. Since $H_1=G_1Q^*(QG_1Q^*)^{-1}$, we have $\tilde{C}_1Q^*=G_1Q^*-H_1QG_1Q^*=0$ and $\tilde{C}_1G_1^{-1}=1-H_1Q$. For a field u which vanishes within the distance 2ε of the collar, Qu is Q_1u on Λ_1 and zero on Λ_0, so that $H_1Qu=\mathcal{L}_\iota Q_1u$,

and the second identity applied to u, in which the term $\tilde{C}_1 Q^* a Q u$ vanishes by the first, is (10.7).

(i), the kernel of $\tilde{C}_1$. By [5, Theorem 3.12, p. 423], G_1 satisfies the bounds [5, (3.42)–(3.44)], the one with the covariant Laplacian excepted. At the scale ε they say that for f supported in a block y', at a point of a block y, the numbers $|G_1 f|$, $|\nabla G_1 f|$, and $|G_1 \nabla^* f|$ are at most $B_0[\varepsilon^2, \varepsilon, \varepsilon] e^{-\delta_0 d(y,y')} |f|$, and that $|\nabla G_1 \nabla^* f| \leq B_0'(\frac{1}{2}) e^{-\delta_0 d(y,y')} (\|f\|_{1/2}^{\xi'} + |f|)$, where ∇ and ∇^* may be exchanged in any place [5, p. 398]. On the lattice of spacing ε/L the Hölder quotient of exponent $\frac{1}{2}$ at the scale ε of a bounded field is at most $2L^{1/2}$ times its supremum, because distinct points are at least ε/L apart, so that $|\nabla G_1 \nabla^* f| \leq (1 + 2L^{1/2}) B_0'(\frac{1}{2}) e^{-\delta_0 d(y,y')} |f|$. The kernel of H_1 satisfies $|H_1(x, y')| + \varepsilon |\nabla H_1(x, y')| \leq c_H \varepsilon^{-d} e^{-\frac{1}{2}\delta_1 d(y,y')}$ [5, (3.133)], and Q_1 has the weight at most 3 and the reach 2ε by [4, (124)]. Since $2\delta_0' \leq \frac{1}{32} \min(\delta_0, \delta_1)$, the sums over the intermediate blocks converge, with a constant N_1 of d, L, δ_0, and δ_1 [3, (2.58)], and each of

$$|\tilde{C}_1 f|, \qquad \varepsilon |\nabla \tilde{C}_1 f|, \qquad \varepsilon |\tilde{C}_1 \nabla^* f|, \qquad \varepsilon^2 |\nabla \tilde{C}_1 \nabla^* f| \tag{10.12}$$

is at most $\tilde{B} \varepsilon^2 e^{-2\delta_0' d(y,y')} |f|$ at a point of the block y, where $\tilde{B} := (1 + 3c_H N_1)(B_0 + (1 + 2L^{1/2}) B_0'(\frac{1}{2}))$. By the symmetry of $\tilde{C}_1$ the same bounds hold for the sums over a block y', with the weights of the pairing, of the kernel $\tilde{C}_1(\cdot, z)$ and of its first differences.

(i), the five terms. We bound the terms of (10.7) at a fine bond z and write $|\cdot|_{z'}$ for the supremum over a block z' of side ε. First, Du is the covariant curl of u, so that $|Du| \leq 2|\nabla u|$, and the third entry of (10.12) gives $|\tilde{C}_1 D^*(Du)(z)| \leq 2\tilde{B}\varepsilon \sum_{z'} e^{-2\delta_0' d_\varepsilon(z,z')} |\nabla u|_{z'}$. Second, $R = 1 - P$, and the kernel of P satisfies $|P(x, x')| \leq c_P \varepsilon^{-d} e^{-\frac{1}{2}\delta_0 d(y,y')}$ [5, (3.49), p. 399], so that $|RD^* u|_{z'} \leq d(1 + c_P) \sum_{z''} e^{-2\delta_0' d_\varepsilon(z',z'')} |\nabla u|_{z''}$; the third entry, with the gradient of functions in the place of ∇^*, then gives $|\tilde{C}_1 D(RD^* u)(z)| \leq d(1 + c_P) N_1 \tilde{B} \varepsilon \sum_{z'} e^{-\delta_0' d_\varepsilon(z,z')} |\nabla u|_{z'}$. Third, by [5, (3.7), (3.10)] the form $\langle A_1, \Delta' A_2 \rangle$ is the sum over the plaquettes p of $\mathrm{tr}(DA_1(p) DA_2(p))(\mathrm{Re}\, U^B(\partial p) - 1)$ and of a bilinear expression of at most six terms in the values of A_1 and A_2 on the bonds of p, multiplied by $(\varepsilon/L)^{-2} \,\mathrm{Im}\, U^B(\partial p)$; the first coefficient is at most ϑ and the second at most $\vartheta \varepsilon^{-2}$, by (10.5). With $A_1 = \tilde{C}_1(\cdot, z)$ and the first two entries, $|\tilde{C}_1 \Delta' u(z)| \leq 6\tilde{B}\varepsilon \sum_{z'} e^{-2\delta_0' d_\varepsilon(z,z')} \vartheta(z') (|\nabla u|_{z'} + \varepsilon^{-1} |u|_{z'})$. Fourth and fifth, the forms Δ'_π and $\Delta^{(2)}_\pi$ are linear in the current, which enters Δ'_π at the bond at which the form is evaluated [5, (3.120)] and $\Delta^{(2)}_\pi$ through $H^* J$ [5, (3.136)]. Bałaban's bounds [5, (3.131), (3.137)], with [5, (3.42), (3.49)] for the factors $DG'RD^*$ of [5, (3.135)], contain the current only through the factor $O(1) M \alpha_0$, his bound [5, (3.36)] of $\varepsilon^3 |J|$; with the values of $\varepsilon^3 |J|$ at the bonds in its place, weighted by the decay of the kernels, they read

$$\begin{aligned} &|\langle A_1, (\Delta'_\pi + \Delta^{(2)}_\pi) A_2 \rangle| \\ &\qquad \leq c_\pi \sum_{y,y'} (\|D^* A_1\|_{L^1(y)} + \varepsilon^{-1} \|A_1\|_{L^1(y)}) \bar{\vartheta}(y) e^{-2\delta_0' d(y,y')} (|D^* A_2|_{y'} + \varepsilon^{-1} |A_2|_{y'}) \,. \end{aligned}$$

With $A_1 = \tilde{C}_1(\cdot, z)$ this gives $|\tilde{C}_1(\Delta'_\pi + \Delta^{(2)}_\pi) u(z)| \leq 2c_\pi N_1 \tilde{B} \varepsilon \sum_{z'} e^{-\delta_0' d_\varepsilon(z,z')} \bar{\vartheta}(z') (d|\nabla u|_{z'} + \varepsilon^{-1} |u|_{z'})$. Adding the five bounds, and repeating them with the second and the fourth entry of (10.12) for the first differences,

$$|(1 - \mathcal{L}_\iota Q_1) u(z)| + \varepsilon |\nabla (1 - \mathcal{L}_\iota Q_1) u(z)| \leq C_{\mathrm{rc}} \varepsilon \sum_{z'} e^{-\delta_0' d_\varepsilon(z,z')} (|\nabla u|_{z'} + \bar{\vartheta}(z') \varepsilon^{-1} |u|_{z'}) \,, \tag{10.13}$$

with $C_{\mathrm{rc}} := 2(8 + d(1 + c_P) N_1 + 2dc_\pi N_1) \tilde{B}$. Since $|\nabla u|_{z'} \leq N_a(u; z')$ and $\varepsilon^{-1} |u|_{z'} \leq (a/\varepsilon) N_a(u; z')$, this contains the first line of (10.8).

(i), the first differences. For the first differences (10.13) gives no gain, and the gain comes from the first two terms, in which a covariantly constant field is annihilated up to the curvature. For a value λ at a point z let $\lambda_z(p) := R(U^B(\Gamma_{p,z}))\lambda$ be its transport along the shortest contours of [5, (3.40)]. The covariant difference of λ_z along a fine bond at the distance r from z is $(\varepsilon/L)^{-1}(R(h)-1)\lambda$ with h the holonomy of a closed contour through z which bounds at most $(rL/\varepsilon+1)^2$ plaquettes, so that it is at most $c_G L\varepsilon^{-1}(r/\varepsilon+1)^2|\lambda|$ times the supremum of ϑ within the distance r of z, which is at most $e^{\delta_0' r/\varepsilon}\bar{\vartheta}(z)$; at the boundary of $Y^{\sim}$, where the Dirichlet condition cuts λ_z, the difference is at most $L\varepsilon^{-1}|\lambda|$, which is the same bound with $\vartheta = 1$. With $\lambda := Du(z)$ we write $\nabla\tilde{C}_1 D^*(Du)(z) = \nabla\tilde{C}_1 D^*(Du-\lambda_z)(z) + \nabla\tilde{C}_1(D^*\lambda_z)(z)$. In the first term $|Du-\lambda_z|_{z'} \leq 2(d_\varepsilon(z,z')+2)^\beta(\varepsilon/a)^\beta N_a(u;z)$ for the blocks within the distance a of z, by the Hölder quotient, and $|Du-\lambda_z|_{z'} \leq 2N_a(u;z') + 2N_a(u;z)$ for the others, for which $e^{-\delta_0' a/\varepsilon} \leq c_\beta(\varepsilon/a)^\beta$ with $c_\beta := (\beta/(e\delta_0'))^\beta$; the fourth entry of (10.12) bounds the term by a constant times $(\varepsilon/a)^\beta \sum_{z'} e^{-\delta_0' d_\varepsilon(z,z')} N_a(u;z')$. The second term is at most $2dc_G L N_2 \tilde{B}\bar{\vartheta}(z)|\nabla u(z)|$ by the second entry, with N_2 the sum of $(d_\varepsilon+1)^2 e^{-\delta_0' d_\varepsilon}$ over the blocks. For the gauge-fixing term let $\mu := D^*u$ and x a point. Since $(\mu-\mu_x)(x) = 0$, $R(\mu-\mu_x)(x) = -P(\mu-\mu_x)(x)$, which is bounded as the first term, with c_P and the kernel of P. For the transported constant we use that $G'^{-1} = D^*D + Q'^*aQ'$ on functions [5, (3.23), (3.24), p. 394] and that $P = G'Q'^*(Q'G'^2Q'^*)^{-1}Q'G'$ fixes the range of $G'Q'^*$: from $\mu_x = G'D^*D\mu_x + G'Q'^*aQ'\mu_x$ we obtain $R\mu_x = RG'D^*(D\mu_x)$, and [5, (3.42)] for $G'\nabla^*$ with the bound of $D\mu_x$ above gives $|R\mu_x(x)| \leq c_R\bar{\vartheta}(x)|\mu(x)|$, $c_R := (1+c_P)c_G L N_2 B_0$. Hence $|RD^*u|_{z'}$ is at most a constant times $\sum_{z''} e^{-\delta_0' d_\varepsilon(z',z'')}((\varepsilon/a)^\beta + \bar{\vartheta}(z''))N_a(u;z'')$, and the fourth entry carries this to $\nabla\tilde{C}_1 D(RD^*u)(z)$. The three remaining terms are bounded as in (10.13), by $\bar{\vartheta}(z')(1+a/\varepsilon)N_a(u;z')$ per block. This is the second line of (10.8).

(e), the regularity. The first bound of (10.4) is the bound of the kernel of H_1. For the second we use an interpolation. For a fine bond b in the direction μ whose first endpoint lies in the block $B(y)$ let $c(b)$ be the bond of run A from y in the direction μ, and let $(I_0 v)(b)$ be the transport of $v(c(b))$ from y to that endpoint along the contour of the average. Two values of $I_0 v$ on neighbouring fine bonds are transports of the same value of v, or of its values on neighbouring bonds of run A, so that $|\nabla I_0 v| \leq L(|\nabla v| + c_G\vartheta\varepsilon^{-1}|v|)$, the second term from the holonomy of a contour inside two blocks. By [4, (124), (125), p. 36], which describe the linear part of the average $\bar{\bar{V}}_1$ of the fluctuation and hence Q_1, and not Q_1^{pl}, Q_1 is the average Q_0 over the straight paths $[x,x']$, $x \in B(c_-)$, plus terms whose operators vanish at vanishing curvature and are at most $O(1)L^2$ times the plaquette variables, that is at most $c_{\mathrm{nm}}\vartheta$. Each of the L bonds of a path $[x,x']$ has its first endpoint in $B(c_-)$ or in $B(c_+)$, where $I_0 v$ is the transport of $v(c)$ or of the value of v at the next bond in the direction of c, and the weights of Q_0 sum to one; hence $|v - Q_1 I_0 v| \leq \varepsilon|\nabla v| + (c_G + c_{\mathrm{nm}})\vartheta|v|$. Now $\mathcal{L}_\iota v = I_0 v - (1-\mathcal{L}_\iota Q_1)I_0 v + \mathcal{L}_\iota(v - Q_1 I_0 v)$. The first differences of the three terms are bounded by the bound of $\nabla I_0 v$, by (10.13) for $u = I_0 v$, and by the kernel bound $c_H\varepsilon^{-1}$ of ∇H_1 applied to $v - Q_1 I_0 v$, which together give the second bound of (10.4) with $C_{\mathcal{L}'} := (1 + c_G + c_{\mathrm{nm}})(c_H + L + 2LC_{\mathrm{rc}})$. On the shell n of a local sequence $\vartheta \leq c_F(M\alpha_0 + \alpha_1 + \alpha_1^2)L^{-2n} + c_J(M\alpha_0 + \alpha_1)L^{-3n}$, by the curvature bound of (b) and the bounds [5, (3.36), (3.38)] of the current. A point of a shell $m < n$ lies at the distance at least $2R_0M_0(n-m-1)\varepsilon$, and $e^{-2\delta_0' R_0 M_0(n-m-1)} \leq L^{-\frac{3}{2}(1+\beta)(n-m-1)}$ by the second condition of (10.1), which pays the ratio $L^{2(n-m)}$ of the two bounds up to the factor L^3; the collar lies outside $\Omega_0(\square)$, at a distance of at least $2R_0M_0n\varepsilon$. Since $\frac{3}{2}(1+\beta) \geq 2$, $\bar{\vartheta} \leq c_\vartheta L^{-2n}$ on the shell n, with $c_\vartheta := L^3(c_F + c_J)(M\alpha_0 + \alpha_1 + \alpha_1^2) + 1$, and therefore $\bar{\vartheta}\varepsilon^{-1} \leq 2(L^n\varepsilon)^{-1}$ for α_0 and α_1 small, which gives the hypothesis of Paper, Lemma 9.4. The same argument applies to the layers of the set on which a walk lives: a point of the layer j lies at the distance at least $M_1 a_j$ from the collar and at the distance at least $RM_1 a_{j-1}$ from every layer finer

than $j-1$ [7, (1), p. 277], and $\sup_{t\geq 1} t^2 e^{-\delta_0' M_1 t} \leq (2/(e\delta_0' M_1))^2$, so that there

$$\bar{\vartheta} \leq c_\vartheta' \Big(\frac{\varepsilon}{a_j}\Big)^2 , \qquad c_\vartheta' := L^2(c_\vartheta + 2(2/(e\delta_0' M_1))^2) . \tag{10.14}$$

(h), the fine form. Let W^B and W^A be the terms of a walk in the two runs, or the two functions of (h) at one s, and $e := (Q_1 W^B - W^A \mathcal{J}')\lambda$ their averaged difference. The fields which they produce vanish within the distance 2ε of the collar, because the kernels of the decoupled operators vanish unless both arguments lie in the interior of Y [11, p. 4], at the distance at least $M_1 a_k$ from the collar. Hence (i) applies to the field $u := W^B\lambda$ of run B and (10.4) to e, and we use the algebraic identity

$$(W^B - \mathcal{L}_\iota W^A \mathcal{J}')\lambda = (1-\mathcal{L}_\iota Q_1) W^B \lambda + \mathcal{L}_\iota e , \tag{10.15}$$

in which the product $\mathcal{L}_\iota Q_1$ acts only on u. This field satisfies the bound of its walk in the supremum norm, in the norm of the first differences, and in the Hölder norm [5, (3.107), (3.108), p. 416]; writing w for the first of these bounds at a point of the layer j, we have $N_{a_j}(u; z') \leq 3a_j^{-1} w$ there. By (10.8) and (10.14), the first term of (10.15) is at most $3(1 + c_\vartheta')C_{\mathrm{rc}}(\varepsilon/a_j)$ times the sum of $e^{-\delta_0' d_\varepsilon(z,z')} w(z')$ over the blocks, and its first differences are at most $3(1+c_\vartheta')C_{\mathrm{rc}}(\beta)(\varepsilon/a_j)^\beta a_j^{-1}$ times the same sum. By the first bound of (10.4), the second term is at most $C_{\mathcal{L}'}$ times the sum of $e^{-\delta_0' d_\varepsilon(z,c)}|e(c)|$. It remains to compare the bound w at the blocks z' with its value at z. The exponential weights are in the units of ε, while the layers of the set have the spacings $a_j \geq \varepsilon L^{m(\omega)}$: a block at the distance r from z changes the weighted distance in w by at most $r/(\varepsilon L^{m(\omega)}) + 1$ and the power of the scale by at most L^4 per layer crossed, of which there are at most $r/(\varepsilon L^{m(\omega)}) + 1$. Let m_{rc} be the least integer with $L^{m_{\mathrm{rc}}} \geq (8\log L + \delta_0)/\delta_0'$. For $m(\omega) \geq m_{\mathrm{rc}}$ half of the weight $e^{-\delta_0' r/\varepsilon}$ pays both changes, at the rates $\frac{1}{2}\delta_0$ and $\frac{1}{6}\delta_0$ of the bounds of the walk and of the averaged difference, and the sums are at most $L^4 e^{\delta_0} N_1 w(z)$; this is the reason for the condition $K' \geq 2m_{\mathrm{rc}} + 2$ of Paper, Theorem 9.5, under which every layer of the set has $j \geq K'/2 - 1 \geq m_{\mathrm{rc}}$. Together, the fine difference is bounded in the supremum norm by $L^4 e^{\delta_0} N_1[3(1 + c_\vartheta')C_{\mathrm{rc}}(\varepsilon/a_j) + C_{\mathcal{L}'}\delta_w]$ times the bound of the walk at the rate $\frac{1}{6}\delta_0$, and both terms of the bracket are of the form of Paper, (9.21).

(h), the first differences. For a field f on a lattice and a scale a, summing the first differences along a straight path of length $r \leq a$ with their transports gives $|\nabla f| \leq 2r^{-1}\sup|f| + r^\beta[\nabla f]_\beta$, and the choice of r gives

$$\sup|\nabla f| \leq 4(\sup|f|)^{\beta/(1+\beta)}([\nabla f]_\beta)^{1/(1+\beta)} + 4a^{-1}\sup|f| , \tag{10.16}$$

the Hölder quotient over the separations at most a: for $r_* := (2\sup|f|/[\nabla f]_\beta)^{1/(1+\beta)}$ between the spacing and a we take $r = r_*$ up to a factor 2, for $r_* > a$ we take $r = a$, and for r_* below the spacing η' the trivial bound $2\sup|f|/\eta'$ is at most the first term. The averaged difference e is the difference of two fields of run A each of which satisfies the Hölder bound of its walk at the scale a_j: $W^A\mathcal{J}'\lambda$ by [5, (3.108)], and $Q_1 W^B\lambda$ because run A's difference quotient of an average is the average of run B's difference quotients up to the transports, as in (b). Hence (10.16) bounds $|\nabla e|$ by $8\delta_w^{\beta/(1+\beta)}$ times the bound of the walk in the norm of the first differences. For the fine difference, the first differences of the first term of (10.15) were bounded above with the factor $(\varepsilon/a_j)^\beta$, and those of the second are bounded by the second line of (10.4) applied to e, in which $\bar{\vartheta}\varepsilon^{-1}|e| \leq c_\vartheta'(\varepsilon/a_j)a_j^{-1}|e|$. Since $(\varepsilon/a_j)^\beta \leq \delta_w^{\beta/(1+\beta)}$ once $C_w \geq 1$, this is the last statement of (h). *q.e.d.*

11 The Minimisers on Two Fine Lattices

The second input of the comparison of two cutoffs which Bałaban's analysis of a single run does not contain is the distance between the minimisers of the two runs at equal data (Paper, Proposition 10.6). Both are minimisers of [7, Theorem 1, p. 279], on the lattices of the spacings ε and ε/L, and the point is that the average MU^B of run B's minimiser satisfies run A's constraint, so that it is a competitor in run A's problem whose action exceeds the minimum only by a lattice artefact. We write $W = e^{i\eta A}$, $F_p := (i\eta^2)^{-1} \log W(\partial p)$, ∇ for the covariant lattice difference, $j(x)$ for the layer of a point x of the determining set of the data and $a_{j(x)} = \varepsilon L^{j(x)}$ for its spacing, and ε_1 for the regularity parameter of the data in [7, (7)]. On every cube $\square$ of side $2R_1M_1a$ in the layer of the spacing a, in the class of cubes described before [7, Theorem 1, p. 279], the bounds [7, (9), (10), p. 279] with $M = R_1M_1$ and $\beta = 1$ give, in a gauge on $\square$,

$$|F| \leq C\varepsilon_1 a^{-2}\,, \qquad |\nabla F| \leq C\varepsilon_1 a^{-3}\,, \qquad |A| \leq C\varepsilon_1 a^{-1}\,, \tag{11.1}$$

with $C = C_0^{\mathrm{VAR}} := 4R_1M_1(B_3 + B_4(1))$, B_3 and B_4 those of [7, Theorem 1], since the Hölder bound of exponent 1 controls every second difference of A. We set $C_0 := 3(1 + 4c_9)C_0^{\mathrm{VAR}}$, with $c_9 \geq 1$ a constant of L and d defined in the proof of (a); the factor $3(1 + 4c_9)$ is the room which the averaged fields MW need.

Proposition S.13 (The Minimisers on Two Fine Lattices)**.** *Let* $0 < \varepsilon_0 < \frac{1}{2}$*, and let* ε_1 *be small, depending on* ε_0*,* d*,* L*,* G*, and the constants of [7, 6, 4, 3, 5]. Let* U^A *and* U^B *be the minimisers of the two runs for the same data* V *on a determining set at the shared scales, let* $\tilde{U} := MU^B$*, and let* N *be the number of points of the lattice of the data.*

(a) If W *satisfies* (11.1) *with* $C = C_0^{\mathrm{VAR}}$ *on a single layer of the spacing* a*, the blocking artefact* $b(W) := A_\varepsilon(MW) - A_{\varepsilon/L}(W)$ *satisfies* $|b(W)| \leq C_b(\varepsilon/a)\varepsilon_1^2 N$*,* $C_b := 48c_9^2C_0^2$*.*

(b) In a gauge, $U^A = e^{i\varepsilon A'}\tilde{U}$ *with* $a_{j(x)}|A'(x)| + a_{j(x)}^2|\nabla A'(x)| \leq 2\bar{\rho}(x)$*,* $\bar{\rho}(x) := C_G(\varepsilon_0)(\varepsilon/a_{j(x)})^{1-\varepsilon_0}\varepsilon_1$*, which is Paper,* (10.9)*; on a single layer of the spacing* a *the gap satisfies* $|A_\varepsilon(\tilde{U}) - A_\varepsilon(U^A)| \leq C_{\mathrm{gap}}(\varepsilon_0)(\varepsilon/a)^{2-2\varepsilon_0}\varepsilon_1^2 N$*.*

(c) The test functions of the multiscale minimisers, of the cascade [12, (1.22)–(1.24), p. 181], of the reference configurations [12, (1.27), p. 182], and of [12, (1.75), p. 193] evaluate plaquette variables which differ between the runs, at equal data, by at most $3\bar{\rho}(p) + 2c_9C_0(\varepsilon/a_{j(p)})\varepsilon_1$ *in the units of the layer* $j(p)$ *of the tested plaquette* p*.*

(d) For each of the configurations $c \in \{U_k'', U_k^0, U_{k,\Lambda}, U_k\}$ *of the* **R***-operation of [13, §1] on a young history born at the scale* n*, at the data of the two runs,* $c^A = e^{i\varepsilon A_c}Mc^B$ *with* Paper, (10.10)*. On a matched pair of young terms the difference of the two classical actions at these configurations is at most* $N_q^4 C_{\mathrm{cl}}(\varepsilon_0)((g_{K'}^A)^{-2} + b_2K')L^{-(1-\varepsilon_0)K'/2}\varepsilon_1^2$*, with* N_q^4 *the number of points of the stopping lattice and* b_2 *the constant of [10, (2.6), p. 255].*

Proof. (a). The average M of [9, (0.4), p. 253] multiplies the straight transport of W along a coarse bond c by the exponential of a convex combination of the logarithms of the holonomies of W around closed loops formed by c and fine paths in the two blocks of its endpoints, each spanned by at most dL^2 fine plaquettes. By the lattice Stokes identity with transports, $(1/i)$ times the logarithm of such a holonomy is the transported sum of $(\varepsilon/L)^2F$ over a spanning surface, up to commutators of two fluxes, of order $\varepsilon^4|F|^2$. For a coarse plaquette the corrections of two opposite bonds are the same convex combinations of fluxes through surfaces translated by one coarse spacing, so their difference is at most $C\varepsilon^3|\nabla F|$, and

$$\frac{1}{i\varepsilon^2}\log(MW)(\partial p) = \bar{F}_p + O(\varepsilon|\nabla F| + \varepsilon^2|F|^2)\ , \qquad \bar{F}_p := L^{-2}\sum_{p' \text{ tiling } p} F_{p'}\ , \tag{11.2}$$

the fine field strengths transported to a corner of p; c_9 bounds the constants of the remainders. The two actions are $A_\varepsilon(MW) = \sum_p \varepsilon^4[\frac{1}{2}|\bar{F}_p|^2 + O(\varepsilon|F||\nabla F| + \varepsilon^2|F|^3 + \varepsilon^4|F|^4)]$ and $A_{\varepsilon/L}(W) = \sum_p \varepsilon^4 L^{-4}\sum_{p'\in\mathsf{cell}(p)}[\frac{1}{2}|F_{p'}|^2 + O(\varepsilon^4|F|^4)]$, with $\mathsf{cell}(p)$ the L^4 fine plaquettes parallel to p in the four-cell of p, each fine plaquette lying in one cell. All the values of F in one coarse term lie within 2ε of each other, so $||\bar{F}_p|^2 - L^{-4}\sum|F_{p'}|^2| \le 4|F|\varepsilon|\nabla F|(1 + O(\varepsilon|A|))$. Per coarse plaquette, with $Y := (\varepsilon/a)C_0^2\varepsilon_1^2 a^{-4}$ and $C_0\varepsilon_1 \le 1$, the quadratic mismatch, the cross term with the remainder of (11.2), half its square, and the quartic terms are at most $2c_9Y$, $2c_9Y$, $2c_9^2Y$, and $2Y$, so at most $8c_9^2Y$, and the $6(a/\varepsilon)^4N$ coarse plaquettes of weight ε^4 give (a).

(b), the first variation. The same expansion gives the first variation of b at W in the direction $e^{i(\varepsilon/L)\delta\mathcal{A}}W$, in weak form,

$$\begin{gathered}\delta b(W)[\delta\mathcal{A}] = \sum_{p'}(\varepsilon/L)^4\langle r_1(p'), (D\delta\mathcal{A})(p')\rangle + \sum_{b'}(\varepsilon/L)^4\langle r_0(b'), \delta\mathcal{A}(b')\rangle\ , \\ |r_1| \le C_r\frac{\varepsilon\varepsilon_1}{a^3}\ , \qquad |r_0| \le C_r\frac{\varepsilon\varepsilon_1^2}{a^4}\ ,\end{gathered} \tag{11.3}$$

with $C_r := 8c_9C_0(1 + C_0)$: the fine side is exact, the term $\langle\theta, [\theta, \delta\mathcal{A}]\rangle$ of the commutator vanishing because $\operatorname{tr} X[X,Y] = 0$; the coarse side pairs $\bar{F}_p$ with the linearised average of the curl, a weighted average of the fine curls over the plaquettes within 2ε, with nonnegative weights $w_{p,p'}$ which sum to 1 over p' and to L^{-4} over p; and the difference $\tilde{F}_{p'} - F_{p'}$, $\tilde{F}_{p'} := \sum_p L^4 w_{p,p'}\bar{F}_p$, is at most $6\varepsilon|\nabla F|(1 + O(\varepsilon|A|))$. Only ∇F enters.

(b), the current. We take [7]'s reference configuration $U_0 := \tilde{U}$ for run A's problem. By (11.2), $\tilde{U}$ satisfies [7, (2)] with the parameter $2C_0\varepsilon_1$, and $\bar{U}_0^k = V$ exactly, since $M^kM = M^{k+1}$, so that the data B of [7, (20)] vanish. By [7, Proposition 5] the minimiser is $U^A = e^{i\varepsilon A'_*}\tilde{U}$, $A'_* = A_1^* - HD(A_1^*)$, with A_1^* the fixed point of $T(A_1) := -\mathfrak{G}J - \mathfrak{G}(\delta V/\delta A')(A_1)$ [7, (116)], a contraction with constant $\frac{1}{2}$ on a ball [7, (120)–(121)], where J is the current of $\tilde{U}$ and $\mathfrak{G} = G_1\mathfrak{P}^* = \mathfrak{P}G_1$ [5, (3.153)], $\mathfrak{P}$ the projection onto $\mathcal{T} := \{QA = 0, RD^*A = 0\}$ orthogonal for $\langle A, (\Delta_1 + DRD^* + aQ^*Q)A'\rangle$ [7, p. 293]; hence $\|A_1^*\| \le 2\|\mathfrak{G}J\|$. Let $\mathcal{L}$ be a right inverse of the plain linearised average $Q_1^{\mathsf{pl}} := dM(U^B)$, $\mathcal{L}g := Ig + Q_1^{\mathsf{pl}*}(Q_1^{\mathsf{pl}}Q_1^{\mathsf{pl}*})^{-1}(g - Q_1^{\mathsf{pl}}Ig)$, I the multilinear interpolation in the gauge of the cube; $Q_1^{\mathsf{pl}}Q_1^{\mathsf{pl}*}$ has finite range, is gauge covariant, and has spectrum in $[\frac{1}{2}m_Q^{\mathsf{pl}}, \tilde{M}_Q^{\mathsf{pl}}]$ at U^B, by its lower bound at $U = 1$ through the straight-tube average and the localisation in an axial gauge, so its inverse decays at a rate δ_Q^{pl}, and $\sup|\mathcal{L}g| \le C_{\mathcal{L}}\sup|g|$, $\sup|D\mathcal{L}g| \le C_{\mathcal{L}}(\sup|\nabla g| + (\varepsilon_1/a)\sup|g|)$. For $\delta A \in \ker Q_A$, $\mathcal{L}\delta A$ lies in the tangent space $\ker Q_B = \ker(Q_AQ_1^{\mathsf{pl}})$ of run B's constraint at U^B, on which $dA_{\varepsilon/L}(U^B)$ vanishes by criticality, and for $\delta A \in \mathcal{T}$ the same follows from the representation of run B's current through the multipliers of the constraint. Therefore

$$\langle\delta A, J\rangle = dA_\varepsilon(MU^B)[Q_1^{\mathsf{pl}}\mathcal{L}\delta A] = d(A_\varepsilon\circ M)(U^B)[\mathcal{L}\delta A] = db(U^B)[\mathcal{L}\delta A] \tag{11.4}$$

for $\delta A \in \ker Q_A + \mathcal{T}$, so that $\mathfrak{G}J = \mathfrak{G}\mathcal{L}^*(D^*r_1 + r_0)$: the residual of the fixed-point equation at $\tilde{U}$ is the propagator applied to the first variation of the blocking artefact.

(b), the residual and the gap. By [5, Theorem 3.13, p. 426] the bounds [5, (3.42)–(3.47)], except the entry of [5, (3.42)] with the covariant Laplacian, hold for $\mathfrak{G}$ at $\tilde{U}$. Summing by

parts, $(\mathfrak{G}\mathcal{L}^*D^*r_1)(x)$ pairs the curl of $\mathcal{L}\mathfrak{G}_x$ with r_1, which by the structure of $\mathcal{L}$ is $\mathfrak{G}\nabla_U^*$ applied to a field of size $O(|r_1|)$. With $\delta_f := \min(\delta_0, \delta_Q^{\mathrm{pl}})$, the smaller of the rates of $\mathfrak{G}$ and of $\mathcal{L}$, let $c_{\mathcal{K}} \geq 2$ be a constant of L, d, and G which bounds $\sup_x \sum_y e^{-\delta_f d(x,y)/2}(a_{j(x)}/a_{j(y)})^2$, the factor $4d$ of the lattice divergence, the constants $C_{\mathcal{L}}$ of $\mathcal{L}$, and the constants of the remainders of (11.2), in which C_0 enters only with a factor ε_1 and $C_0\varepsilon_1 \leq 1$; the constant B_0 of [5, (3.42)] is left unchanged. Then [5, (3.42)] gives $\sup|\mathfrak{G}\mathcal{L}^*D^*r_1| \leq c_{\mathcal{K}}B_0 a \sup|r_1|$ and $\sup|\mathfrak{G}\mathcal{L}^*r_0| \leq c_{\mathcal{K}}B_0 a^2 \sup|r_0|$. For the gradient, [5, (3.44)] bounds $\nabla\mathfrak{G}\nabla^*$ through the Hölder seminorm of exponent ε_0 of the source at the scale ε/a, which for a bounded source is at most $2(a/\varepsilon)^{\varepsilon_0}$ times its supremum; this is the loss ε_0. Together $\|\mathfrak{G}J\| \leq \rho := \frac{1}{2}C_G(\varepsilon_0)(\varepsilon/a)^{1-\varepsilon_0}\varepsilon_1$ with $C_G(\varepsilon_0) := 8c_{\mathcal{K}}C_r(B_0 + B_0'(\varepsilon_0))$, in the norm $\max(a\sup|\cdot|, a^2\sup|\nabla\cdot|)$, so $\|A_1^*\| \leq 2\rho$, and [7, (57), (58), pp. 286–287] give $\|A_*'\| \leq 4\rho = 2\bar{\rho}$ once $\rho \leq 1$, $54C_2B_0\rho \leq 1$, and $6\rho \leq c_4$; since $\varepsilon/a \leq 1$ these three conditions hold once $C_G(\varepsilon_0)\varepsilon_1$ is small, which is the smallness of ε_1 in the hypothesis. The representative of [7, Proposition 5] differs from that of another gauge by a gauge transformation, which the gauge-invariant quantities do not see. By [7, (81)], $A_\varepsilon(\tilde{U}) - A_\varepsilon(U^A) = -\langle A_1^*, J\rangle - \frac{1}{2}\langle A_1^*, \Delta_1 A_1^*\rangle - V(A_1^*)$. The first term is $db(U^B)[\mathcal{L}A_1^*]$ by (11.4), at most $32C_rC_{\mathcal{L}}N(\varepsilon/a)\varepsilon_1\rho$ by (11.3); the quadratic term is at most $C_\Delta N\rho^2$, C_Δ the bound of Δ_1 in these norms, plus the term $\langle HC^{(2)}(A_1^*), J\rangle$ of [7, (79)], which is at most $512B_0C_2C_0N\varepsilon_1\rho^2$ since $|J| \leq 8C_0\varepsilon_1 a^{-3}$; and $|V(A_1^*)| \leq 256C_4N\rho^3$ by [7, Proposition 4, (97)]. Since $(\varepsilon/a)^{2-\varepsilon_0} \leq (\varepsilon/a)^{2-2\varepsilon_0}$ and $\rho \leq 1$, this is the gap bound with $C_{\mathrm{gap}}(\varepsilon_0) := 16C_rC_{\mathcal{L}}C_G(\varepsilon_0) + \frac{1}{4}(512B_0C_2C_0 + C_\Delta + 256C_4)C_G(\varepsilon_0)^2$.

(b), several layers. On a determining set with several layers, [7, Theorem 1, (9)–(10), p. 279] gives (11.1) on each cube of the class with the spacing $a = a_{j(x)}$ of its layer, so that (11.2), (11.3), and (11.4) hold pointwise with the spacing of the layer of the point: $|r_1(x)| \leq C_r\varepsilon\varepsilon_1 a_{j(x)}^{-3}$ and $|r_0(x)| \leq C_r\varepsilon\varepsilon_1^2 a_{j(x)}^{-4}$. The bounds of [5, Theorem 3.13] are stated in the weighted norms of the multiscale lattice, with kernels that decay at the rate δ_0 in the distance of [3, (2.46)], and the kernel of $\mathcal{L}$ decays at the rate δ_Q^{pl}. A path from a point of the layer j to a point of the layer $j - m$ crosses $m - 1$ layers, each of width more than RM in the units of its own scale for the domains of [3, (2.2), p. 224], and at least one layer of M-cubes of its scale for the frozen layers of [12, p. 179, ll. 21–24]; since $M \geq 2(d+1)L^2M_1 \geq LM_1$ (Paper, Appendix C), the path has length at least $(m-1)LM_1$ in that distance, and half of the decay at the rate δ_f absorbs the factors $(a_{j(x)}/a_{j(y)})^2 \leq L^{2m}$ of the weights, up to a factor $L^{7/3}$, under the condition

$$\tfrac{1}{2}\delta_f LM_1 \geq \tfrac{7}{3}\log L\ , \tag{11.5}$$

a lower bound on M_1 of stage 3 (Paper, Appendix C), met by M_1 a large power of L. The weighted sum of the kernels is then the one which $c_{\mathcal{K}}$ bounds, and the residual at x is bounded by the argument above with $a = a_{j(x)}$; this gives $a_{j(x)}|A_*'(x)| + a_{j(x)}^2|\nabla A_*'(x)| \leq 2\bar{\rho}(x)$.

(c). A test function of the multiscale minimiser at a plaquette p of the layer $j(p)$ evaluates the plaquette variable of the minimiser on the cube of p, at the scale of that layer [12, (1.22)–(1.24), (1.27), pp. 181–182; (1.75), p. 193]. The plaquette variables of U^A and of $\tilde{U}$ differ by the curl of $\varepsilon A'$ with transports, at most $a^2|\nabla A'| + C(a|A'|)^2 + C(a|A'|)C_0\varepsilon_1 \leq 3\bar{\rho}(p)$ in the units of the layer for ε_1 small, by (b), and those of $\tilde{U}$ and of U^B differ by the remainder of (11.2), at most $c_9(\varepsilon C_0\varepsilon_1 a^{-3} + \varepsilon^2C_0^2\varepsilon_1^2 a^{-4})$ times a^2, which is at most $2c_9C_0(\varepsilon/a)\varepsilon_1$ since $C_0\varepsilon_1 \leq 1$ and $\varepsilon/a \leq 1$.

(d). On a young history the data of the two runs differ themselves on the region Λ on which the $\mathbf{R}$-operation integrates, through the minimiser V_Λ of the nested problem of [12, Proposition 1, p. 194]. Its functional Φ grows quadratically away from the minimum: by [13, (1.7)–(1.9), p. 358] the second variation is at least $(2\kappa_\Lambda - O(1)(M^6R_k\varepsilon_k + e^{-R_k}))\|Y\|_2^2$ with $\kappa_\Lambda := \gamma_0/(2d(100M)^{d+1})$, so $\Phi(e^{iY}V_0') - \Phi(V_0') \geq \frac{1}{4}\kappa_\Lambda\|Y\|_2^2$ for $\|Y\|_\infty$ small. Let Δ be

the ℓ^2-distance of the minimising data of the two runs, $V_0'^A = e^{-i\mathfrak{X}} V_0'^B$. Adding the growth inequalities of the two runs at $Y = \pm\mathfrak{X}$ gives $\frac{1}{2}\kappa_\Lambda \Delta^2 \leq \delta\Phi(V_0'^B) - \delta\Phi(V_0'^A)$, $\delta\Phi := \Phi^A - \Phi^B$, and the right side is a sum over the blocks y of the smaller of $2\bar{\sigma}$, the bound of $|\delta\Phi|$ by (b) at the scale n of the history, of order $(\varepsilon/a_n)^{1-\varepsilon_0}$, and of a bound $C\varepsilon_1 K_\Lambda(y)\Delta$ linear in Δ, with $\sum_y K_\Lambda(y)^{1/2}$ bounded by the decay of the kernels. Since $\min(u,v) \leq (uv)^{1/2}$, $\Delta^{3/2} \leq C(\varepsilon/a_n)^{(1-\varepsilon_0)/2}\varepsilon_1^{3/2}$, so $\Delta \leq C(\varepsilon/a_n)^{(1-\varepsilon_0)/3}\varepsilon_1$. The configurations U_k'', U_k^0, $U_{k,\Lambda}$, U_k are built from these data by the minimisations of [13, (1.51), p. 370] and of (b), each of which propagates a data difference with the decay of its kernels, so they satisfy Paper, (10.10), with the exponent of the worse of the two sources, $(1-\varepsilon_0)/3$, and the scale $\min(j(x), n)$. Finally the classical group of a matched pair of young terms is the difference of the two actions at these configurations, weighted by the cutoff ζ_1 of [13] and the smooth coupling function $1/g^2(x)$ of [10, (2.24), p. 259], whose value lies between $g_{j(x)}^{-2}$ and $g_{j(x)-1}^{-2}$; (a) and (b) applied point by point give $C(\varepsilon_0)g_{j(x)}^{-2}(\varepsilon/a_n)^{1-\varepsilon_0}\varepsilon_1^2$ per point, and on the stopping lattice $g_{j(x)}^{-2} \leq (g_{K'}^A)^{-2} + b_2 K'$ by [10, (2.6)], with $a_n \geq \varepsilon L^{K'/2}$ on a young history, which gives the last bound. *q.e.d.*

12 The Large-Field Integrands on Two Fine Lattices

The large-field operations of [13, §1] integrate over the region Λ of a component with integrands built from minimisers, from their linearisations, and from the factors $V(Y)$ which re-sum the old terms of the effective action; at the steps at which large-field regions exist they create boundary terms, and on a term carrying a loop they dress it with words whose transports are read on the same minimisers. All of these are computed on the fine lattice of the run, and the comparison of two cutoffs needs their two-run differences, per point of the history of a component, at an amount which decays geometrically in its birth scale (Paper, Theorem 10.8). Bałaban's analysis concerns one run; the proposition below is the comparison. Two devices recur. The first is Proposition S.13, which aligns the minimisers of the two runs. The second carries a real bound to the complex arguments of the inductive class.

Lemma 12.1 (Two Constants)**.** *Let $1 < v < v_1$, let f be analytic on $\{|z| < v_1\}$ with $|f| \leq K$ there, and let $|f| \leq \kappa$ on the real segment $(-v, v)$, $0 < \kappa$. Then $|f(z)| \leq \kappa^{e(z)}K^{1-e(z)}$ for $|z| < v$ and $\kappa \leq K$, with $e(z) := 1 - (2/\pi)|\arg((v+z)/(v-z))|$. For $v = 24$, $v_1 = 25$, and every $\kappa > 0$, $|f(z)| \leq \kappa^{15/16}K^{1/16}$ for $|z| \leq 1$.*

Proof. On the half-disc $D_+ := \{|z| < v, \operatorname{Im} z > 0\}$ the map $w := (v+z)/(v-z)$ is a bijection onto the open first quadrant, which carries the segment to the positive real axis and the semicircle to the positive imaginary axis. So $h := 1 + (2i/\pi)\operatorname{Log} w$ is analytic on D_+, with $\operatorname{Re} h = e$ equal to 1 on the segment and to 0 on the semicircle, and $F := f\kappa^{-h}K^{h-1}$ is bounded on D_+, with $|F| \leq 1$ on its boundary except at $\pm v$. For $q > 0$ the function $F((v^2 - z^2)/(4v^2))^q$, with the principal power, is at most 1 on the boundary and tends to 0 at $\pm v$, so it is at most 1 on D_+ by the maximum modulus principle; letting $q \to 0$ gives the bound on D_+, and the lower half-disc follows from $z \mapsto \overline{f(\bar{z})}$, since $e(\bar{z}) = e(z)$. For $|z| \leq 1$, $|\arg(v \pm z)| \leq \arcsin(1/v)$, so $e(z) \geq 1 - (4/\pi)\arcsin(1/24) > 15/16$, because $\arcsin x \leq x(1-x^2)^{-1/2}$ and $(4/3)575^{-1/2} < 1/16$. Since $\kappa \leq K$, $\kappa^e K^{1-e} = K(\kappa/K)^e \leq \kappa^{15/16}K^{1/16}$, and for $\kappa > K$ this bound exceeds K. *q.e.d.*

At a point of the inductive class $(\mathbf{U}, \mathbf{J})$, $\mathbf{U} = e^{i\xi A'}U$ with $A' = A_R' + iA_I'$, $\mathbf{J} = \mathbf{J}_R + i\mathbf{J}_I$, the four fields real, we use the line $\mathbf{U}(z) := e^{i\xi(A_R' + zA_I')}U$, $\mathbf{J}(z) := \mathbf{J}_R + z\mathbf{J}_I$, which passes through the point at $z = i$ and is real for real z. Since $\mathfrak{g}$ consists of hermitian matrices,

$|A'_R|, |A'_I| \leq |A'|$, so that for real $|z| < 24$ the configuration $\mathbf{U}(z)$ is real with data regular with the parameter

$$\varepsilon'_1 := C(M) R_k \varepsilon'_k \ , \qquad \varepsilon'_k := \varepsilon_k + 60(\hat{\alpha}_0 + \hat{\alpha}_1) + 25 g_k b_k \ , \tag{12.1}$$

where $\hat{\alpha}_i := \max_{m \leq k} \alpha_{i,m}$ are the thresholds of [10, (2.28), p. 259], $b_k := \max(A_1 p_1(\hat{g}_k), M_0 p_0(\hat{g}_k))$ bounds the fields B and B' of the **R**-operation on the supports of its cutoffs [13, p. 356, p. 360], and $C(M) \geq 4$ is a constant of M and of Bałaban's constants; ε'_1 is at most a constant times $\hat{g}_k$ times a power of $\log \hat{g}_k^{-2}$. For $|z| < 25$ the data of all the minimisation problems of the operation lie within $C_{V'} \varepsilon'_1$ of their real data at $z = 0$, so that by [7, Sect. G, Proposition 9] the minimisers, their linearisations, the actions, and the forms are analytic in z on that disc, with each run's minimiser a complex gauge transform of the real one by a field $\mathbf{H}$ obeying [7, (19)] at $\varepsilon_2 := 6 B_1 B_3 C_{V'} \varepsilon'_1$.

Proposition S.14 (The Large-Field Integrands on Two Fine Lattices)**.** *Let* $0 < \varepsilon_0 < 1/125$, *let the constants satisfy the conditions of Paper, Appendix C, and let* $\bar{g}$ *be small. Let* X *be a component of a young matched pair, born at the scale* n, *let* $k \geq n$ *be the step of an* **R***-operation of its history, and let* ε_1 *be the parameter of Paper,* (10.11)*. There are constants* $C_A(M)$, $C_F(M)$, $C_{dc}(M)$, c_4, C_h, *and* $C_{\mathcal{B}}$, *independent of* K, k, ε, *and the torus, such that the following hold.*

- *(a)* The actions. *Per point of the data of* $\Lambda \cap \Omega''^{\sim 2}_{h+1}$ *at its own scale and per bond of* $\Lambda^{(k)}$, *the two runs' weighted actions of [13, (1.70), p. 377] and the functionals of [13, (1.3), p. 357] differ at real data by at most* $C_A(M) g_j^{-2} (\varepsilon/a_n)^{(1-\varepsilon_0)/3} \varepsilon_1^2$, *and on the inductive class by at most* $C_{dc}(M) g_j^{-2} (\varepsilon/a_n)^{5(1-\varepsilon_0)/16} \varepsilon_1'^2$.

- *(b)* The forms. *Per point of* $\mathbf{B}_0$ *and of* $\Lambda^{(k)}$, *the forms* $\frac{1}{2}\langle DH''_{1,k,Z} B, \zeta DH''_{1,k,Z} B \rangle$ *of [13, (1.70)] and* $\langle H_{1,k} B', \Delta_1(\zeta_0) H_{1,k} B' \rangle$ *of [13, (1.2), p. 356] differ at real data by at most* $C_F(M) (\varepsilon/a_n)^{(1-\varepsilon_0)/3} N_k$, *and on the class by at most* $\max(C_F(M), 2c_4)(\varepsilon/a_n)^{5(1-\varepsilon_0)/16} N_k$, *where* $N_k := 8 d^2 B_0^2 c_{\mathcal{K}}^2 (1 + 2dL)^2 b_k^2$ *bounds each form per point.*

- *(c)* The factors $V(Y)$. *The two runs' factors* $V(Y)$ *of [13, (1.68), (1.91)] differ, relative to the bound* $\alpha e^{-(1+2\beta)\kappa d_{k,Z}(Y)}$ *of [13, (1.68)], by at most* $C_h \tilde{\theta}^n$ *per point of* Y *at the layers of the history, at real points and on the class, plus the differences of the terms of the effective action which they re-sum, which enter linearly with the factor* $L^{-(\beta'/2)(m-j)}$ *of the age of a term of the generation* j *at a point of the layer* m, *and the differences of the re-summing propagators, which enter through Paper, Theorem 9.5.*

- *(d)* The boundary terms. *Let* b_i *be the smallest number with* $|\mathbf{B}^{B,(i)}(X) - \mathbf{B}^{A,(i)}(X)| \leq b_i B_0 e^{-\kappa d_i(X)}$ *on the common class of [10, (2.40)–(2.42), p. 261], over the terms whose domains have no point of a layer at most* $k_s/8$, k_s *the stopping scale. Under the third condition of Paper,* (C.2), $b_i \leq 2 s_i^\sharp$ *for every* $i > k_s/8$, *where* $s_i^\sharp := C_{\mathcal{B}}(\mathcal{D}_i^* + C_h \tilde{\theta}^{k_s/8})$ *and* $\mathcal{D}_i^*$ *is the largest, over the steps* $k_s/8 < i' \leq i$, *of the quantities of the recursion of Paper, Section 11, at the step* i'.

- *(e)* The quotients. *The bounds (a)–(c) hold for the numerators and denominators of the renormalisation quotients of [13, (1.1), (1.2), p. 356] at the complex arguments of the inductive class, relative to Bałaban's majorants of their moduli, and they enter linearly in the norms of the classes of the effective actions of the two runs.*

- *(f)* The dressings. *On a young matched pair whose components are born at scales above* $k - \lfloor k/8 \rfloor$, *at every point of the real integration region, the normalised dressed loops* *of the two runs differ by at most* $[e^{D^{\mathrm{dev}}}(\exp(\sum_X C_M s_K^W P(X) + s_K^{W,0}) - 1) + (e^{D^{\mathrm{dev}}} - 1)]$ *times the majorant of run A's loop, where* D^{dev} *is the weighted difference of the deviation insertions, bounded by the recursion of Paper, Section 11, and* s_K^W *and* $s_K^{W,0}$ *are given in* (12.4)*; the sum* s_K^W *of Paper, Proposition 11.10, which also contains the pairs whose components are born earlier, is summable in* K*. For an ensemble of several loops the same holds with the insertions of the cross species included, with the numbers* $s_K^{W,\times}$ *and* $s_K^{W,0,\times}$ *of* (12.5) *in the places of* s_K^W *and* $s_K^{W,0}$*.*

Proof. Throughout, $a_j = \varepsilon L^j$ is the spacing of the layer j in the units of run A, $j(x)$ the layer of a point, and a_n the finest spacing of the history; since the history is young, $k < 8n + j(\mathcal{C})$, with $j(\mathcal{C})$ the scale at which the loops are attached, and $n \geq k/2 - 1$.

(a). At real data, the configurations at which the actions are evaluated are, by Proposition S.13(d), aligned as $c^A = e^{i\varepsilon A_c} M c^B$ with $a_{j(x)}|A_c| + a_{j(x)}^2 |\nabla A_c| \leq \bar{d}(x) = C_U(\varepsilon/a_{\min(j(x),n)})^{(1-\varepsilon_0)/3}\varepsilon_1$, and the data of the minimisation of [13, (1.3)] differ by at most $C_\Delta(M)(\varepsilon/a_n)^{(1-\varepsilon_0)/3}\varepsilon_1$ in the norm $\|\cdot\|_\infty$, by the growth argument of its proof. At a block y, the difference of the two actions is $[a_y(c^A) - a_y(Mc^B)] + b_y(c^B)$, the first bracket at most $C_{bd}(\varepsilon_1 + \bar{d})\bar{d}$ by the expansion of the plaquette action to second order at a field of strength at most $2C_0\varepsilon_1 a_j^{-2}$, and the second the blocking artefact of Proposition S.13(a) per block, at most $C_b(\varepsilon/a_n)\varepsilon_1^2$. The weight of the action, $1/(g''(\cdot))^2$ of [13, (1.70)] with the cutoff ζ_1, is at most $2g_{j(x)}^{-2}$ by [10, (2.24), p. 259] and $2b_2\bar{g}^2 \leq 1$; together, $C_A(M) := C_{\mathrm{pr}} c_{\mathcal{K}}(\frac{\pi}{2}C'_\Delta(M) + C_\Delta(M)) + C_{bd}(1 + C_U)C_U + C_b$. On the class, each run's unweighted density is at most $C_{\mathrm{act}}^c \varepsilon_1'^2$ per point on the disc $|z| < 25$, $C_{\mathrm{act}}^c := d^2(c_{\mathrm{pl}} C_0^{\mathrm{ms}})^2$: the plaquette term $1 - \frac{1}{2}\operatorname{tr}(X + X^{-1}) = -\frac{1}{2}\operatorname{tr}((X - 1)^2 X^{-1})$ is invariant under complex conjugations $X \mapsto gXg^{-1}$, so it may be evaluated at the Landau representative, whose plaquettes satisfy $|X - 1| \leq c_{\mathrm{pl}} C_0^{\mathrm{ms}} \varepsilon_1' \leq \frac{1}{2}$. The real bound holds on the segment $|z| < 24$ with ε_1' in place of ε_1, and Lemma 12.1 at $z = i$ gives the bound on the class with the exponent $\frac{15}{16} \cdot \frac{1-\varepsilon_0}{3} = \frac{5(1-\varepsilon_0)}{16}$ and $C_{dc}(M) := \max(C_A(M), 2C_{\mathrm{act}}^c)$; the factor $(\varepsilon_1'^2)^{1/16}$ of the disc bound is at most 1.

(b). In each run the data derivative of a minimiser with a multiscale determining set is $H_1(1+\Xi)$, with Bałaban's $H_1 = G_1 Q^*(QG_1Q^*)^{-1}$ [5, (3.129)] and the interface twist Ξ, which is at most $2dL$ per point. At aligned backgrounds the constraints of run B are those of run A composed with the linearised one-step average, $Q^B = Q^A Q_1$, layer by layer, by the chain rule $d(M^{j+1})(U^B) = d(M^j)(MU^B) \circ dM(U^B)$; this is the hypothesis of Proposition S.12, which compares the Green operators G_1 of the two runs, and the identity $H_1 = G_1 Q^*(QG_1Q^*)^{-1}$ carries the comparison to the lifts. By [5, Theorem 3.12, (3.132)–(3.133), pp. 422–423] the lifts decay at the rate $\frac{1}{2}\delta_1$ between any two layers, with the Hölder bound at the scale of the receiving layer, so that $a_j|H_1\tilde{B}| + a_j^2|\nabla H_1\tilde{B}| \leq \sigma_k := 2B_0 c_{\mathcal{K}}(1 + 2dL)b_k$ per point, which gives $N_k = 2d^2\sigma_k^2$. The two runs' forms then differ by the sum of the parts of Proposition S.12 at the aligned backgrounds, of the alignment of Proposition S.13, which moves the background by $\bar{d}$ and the currents by second differences of the alignment field, bounded through the multiscale form of Proposition S.12(d), and of the cutoffs ζ_0, ζ_1, fixed smooth functions of the continuum point with $|\nabla\zeta_0| \leq c_\zeta/a_k$ and $|\nabla\zeta_1| \leq c_\zeta/a_h$, which change by at most $C_\zeta\varepsilon/a_n$, $C_\zeta := 2c_\zeta(1 + L)$, between plaquettes within 2ε; the gradients of ζ_0, ζ_1 are not bounded in [12, p. 201] and [13, p. 360], and the choice is admissible because the **R**-operation preserves the integral of the density for every such pair [12, (0.4), p. 176]. This gives the real bound with $C_F(M)$ the sum of these constants. On the disc each run's forms are at most $c_4 N_k$ per point, since the Landau representative of each run lies in the class of Proposition S.12 with the lifts obeying [5, (3.133)] there, and Lemma 12.1 gives the bound on the class.

(c). Bałaban's factor $V(Y)$ re-sums, at fixed Y, the terms of the expansions [13, (1.59)–(1.64), pp. 372–374] with their completions [13, p. 377, ll. 22–26], each evaluated at a configuration of the chain U_k'', U_k^0, U_1, U_2, $U_{k,\Lambda}$, U_k, or at a decoupled configuration of [13, (1.60)] built from the propagators of the run. We telescope the two runs' $V(Y)$ in three parts. First, each old term of run B is evaluated at the lift ι_Y of run A's configuration, with the current lifted through $Q_1^{\mathrm{pl}}(\iota_Y c^A)^*$, and compared with run A's term at c^A: this is the difference of the terms of the same shared scale, which the recursion of Paper, Section 11, bounds, and which enters $V(Y)$ linearly with the age factor of [10, (2.43), p. 263], of which half, $L^{-(\beta'/2)(m-j)}$, is kept. Second, the propagators, the fixed points, and the gauge transformations by which the expansions are formed are replaced by those of run B; their differences are those of Paper, Theorem 9.5. Third, $V^B(Y)$ is moved from the lifted configurations $\iota_Y c^A$ to run B's own c^B. The third part is a variation of an analytic function on its domain: along the path from $\iota_Y c^A$ to c^B the configuration moves, in the units of the layer m of a point, by $\sigma_c(x) \le C(\varepsilon_0)(a_k/\varepsilon)^{\varepsilon_0}\bar{d}(x)$, and the domain of $V(Y)$ has the radius $\varrho(x) = c_\rho \alpha_{0,m}/(4B_3)$ there, so that Cauchy's estimate bounds the variation by $2r(Y)$ times the bound [13, (1.68)], $r(Y) := \sup_{x\in Y} \sigma_c(x)/\varrho(x)$, once $r(Y) \le \frac{1}{2}$. On a domain at the layers $m \ge n$, $(a_k/\varepsilon)^{\varepsilon_0} = L^{k\varepsilon_0} \le L^{j(\mathcal{C})\varepsilon_0} L^{8n\varepsilon_0}$, $\varepsilon_1/\varepsilon_k \le C(M)R_k(8n)^{1/2}$, and $\varepsilon_k/\alpha_{0,m} \le (1+\beta_0)(k-m+1)^{1/2}$ by [10, (2.6), (2.28)], so $r(Y)$ is at most a polynomial in n times $L^{-(1-25\varepsilon_0)n/3}$, and since $\tilde{\theta} \ge L^{-(1-\varepsilon_0)/4}$ this is at most $C\tilde{\theta}^n L^{-(1-97\varepsilon_0)n/12}$, with $1-97\varepsilon_0 > 0$. On the class the same argument runs with the configurations of the line and Lemma 12.1, with the exponent $5(1-25\varepsilon_0)/16$ in place of $(1-25\varepsilon_0)/3$ and $1-121\varepsilon_0 > 0$. For the finitely many n below the least n_0 at which $r(Y) \le \frac{1}{2}$, each run's $V(Y)$ is at most the larger of α of [13, (1.68)] and the $O(1)$ of [13, (1.69)] per point, which is at most $C_h\tilde{\theta}^n$ once C_h contains $2\max(\alpha, O(1))\tilde{\theta}^{-n_0}$. A domain which meets a layer $m < n$ is bounded in the same way with m in place of n, or, if it contains a point of a layer at most $k_s/8$, by the deep terms of Paper, Section 11.

(d). A boundary term $\mathbf{B}^{(i)}$ of [10, (2.40)–(2.42), p. 261], after the step which creates it, is a fixed function evaluated at the later backgrounds; at a fluctuation step k it enters [10, (3.26)] only through the integrand $\mathbf{B}_k(U_k(\cdot))-\mathbf{B}_k(U_{k+1})$ of the t-integral and through the measure [10, (3.27), p. 271], and its output goes into $\mathbf{B}^{(k+1)}$ only. The two-run difference of $\mathbf{B}^{(k+1)}$ therefore has three parts: the terms in which the frozen difference of one input of the generation i is marked, bounded through the analyticity in the fluctuation by $\varepsilon_k^{*B} w_{k+1-i} L^{d(k+1-i)} b_i$, with $\varepsilon_k^{*B} = 2C_{\mathrm{pt}}C_X C_{\mathrm{KP}} C_\varepsilon \varepsilon_k$, C_ε the constant of the factor $O(\varepsilon_k)$ of [10, p. 277, ll. 7–9], and w_r the weight of the age r; the terms in which the step's own difference is marked, applied to run B's boundary terms; and the boundary parts of the other marked activities. The last two are at most $s_{k+1}^\sharp$, the variation of an old term between the backgrounds of the two runs being bounded as in (c) on the class of [10, (2.34)–(2.39), p. 261], with the room $\frac{1}{4}\beta 2^{-(i-m)_+}\min(\alpha_{0,m}, \alpha_{1,m})$ which the prefactors $1-\beta(1-2^{-(i-m)})$ leave at the layer m, against the move $C_{mv}\mathfrak{d}_c(x)$ of the path, $\mathfrak{d}_c(x) = 4(1+\hat{x}_J)\bar{d}_c(x)^{15/16}$ by Lemma 12.1. On a domain without points of the layers at most $k_s/8$, the ratio of move to room is at most a constant times $2^{j(\mathcal{C})}P(k)2^{7k_s/8}L^{-5(1-\varepsilon_0)k_s/128}$, $P(k)$ a polynomial in k and $\log \hat{g}_k^{-2}$, because $i-m < 7k_s/8 + j(\mathcal{C})$ and $\min(m,n) > k_s/8$; the condition

$$L^{5(1-\varepsilon_0)/128} \ge 4 \tag{12.2}$$

of Paper, (C.1), turns $2^{7k_s/8}L^{-5(1-\varepsilon_0)k_s/128}$ into at most $2^{-k_s}2^{-k_s/8} \le 2^{-k_s}\tilde{\theta}^{k_s/8}$, since $\tilde{\theta} \ge \frac{1}{2}$. The weight of the age satisfies $w_r \le C_{wr}e^{-\sigma(r-1)}$, $\sigma := \min(\frac{1}{2}\delta\kappa R, \frac{1}{16}\delta_0 RM)$, because a term of the generation i which is still carried has a point at a layer at most i, and the layers $\Gamma_{i+1}, \dots, \Gamma_k$ between it and the region of the fluctuation each have width more than RM in their units [3, (2.2), p. 224]; here δ is the small number of [9, p. 272, ll. 2–4], of which half is spent on the layers. A point carries the ordinary terms of a generation and at most $C_{\mathrm{ov}}N_0^+$ families of terms created by the preliminary integrations of [12, (1.60)–(1.63), p. 189], $C_{\mathrm{ov}} = 2$, because the regions $Z_{h+i} \setminus Z''_{h+i}$ in which they are localised are pairwise disjoint for $h + i \leq k_0$ and contain a common domain for the at most N_0 larger i. With $\mu_N(a) := 1 + 4(2\beta_0 + \log(a + 1))/\log L$, [10, (2.6)] gives $N_0^+(i) \leq \mu_N(k - i)N_0^+(k)$, and the recursion

$$b_{k+1} \leq \varepsilon_k^{*B} \sum_{k_s/8<i\leq k} w_{k+1-i}L^{d(k+1-i)}2C_{\mathrm{ov}}N_0^+(i)b_i + s^\sharp_{k+1} \tag{12.3}$$

gives $b_{k+1} \leq (4C_{\mathrm{ov}}\varepsilon_k^{*B}N_0^+(k)W^\sharp + 1)s^\sharp_{k+1} \leq 2s^\sharp_{k+1}$ by induction, with $W^\sharp := \sum_{r\geq 1} w_r\mu_N(r)L^{dr} \leq 4(1 + 4(2\beta_0 + 1)/\log L)C_{wr}L^d$ under $\sigma \geq (d + 1)\log L + 1$, a lower bound on RM of stage 3, and $s^\sharp$ nondecreasing. The sum is empty at the first generation above $k_s/8$, because the output domains of a step contain those of its inputs [11, p. 12], so that a term built from a deep input is deep. The terms created at a step of an **R**-operation, the logarithms of the integrals [12, (1.60)] with the coupling terms of [12, (1.62)], are compared by the same interpolation as the fluctuation integrals of a step, with the bound [12, (1.70), p. 192]; their differences are sources of (12.3) and are contained in $s^\sharp$.

(e). The renormalisation quotients of [13, (1.1), (1.2)] are formed from the actions, forms, and factors of (a)–(c), at the complex arguments of the class through the line $\mathbf{U}(z)$, and the dependence on the effective actions of the two runs enters through the factors $V(Y)$, which are linear in the old terms. The bounds on the class in (a)–(c) therefore give (e), relative to Bałaban's bounds of the moduli of the numerators and of the denominators [13, pp. 357–358], which hold on the class with ε_k replaced by $O(\alpha_{0,k} + \alpha_{1,k})$.

(f). The normalised dressed loop of a labelled term is $e^{\mathcal{E}_k}$ times its leading word plus the words $\mathrm{tr}(Y_{W_1}U_{(1)}\cdots Y_{W_m}U_{(m)})$ with the weights Q_k (Paper, Section 6). On a young pair four kinds of factors depend on the run: the transports, read on the background; the insertions of the deviation, attached, and wrapping species; the cluster weights and the local terms, which are numbers given the labels; and the parts produced by the **R**-operations of the components (Proposition S.10). For a product of factors x_i with $|x_i^A| \leq \bar{x}_i$ and $|x_i^B - x_i^A| \leq \Delta_i\bar{x}_i$, $|\prod x_i^B - \prod x_i^A| \leq (e^{\sum \Delta_i} - 1)\prod \bar{x}_i$, so it suffices to bound the relative differences. The transports along the bonds c of a loop differ by $\delta_c \leq \|d\pi\|(4C_G(\varepsilon_0)c''_K + C_Y)(\varepsilon/a_{j(c)})^{2-\varepsilon_0}\varepsilon_1$: by Proposition S.13(b), $U^A = e^{i\varepsilon A'_*}MU^B$ with $a_j|A'_*| \leq 2\bar{\rho}$, and $MU^B(c) = e^{iY_c}W(c)$ with $W(c)$ the transport of U^B along the L fine bonds of c and $|Y_c| \leq C_Y(\varepsilon/a_{j(c)})^2\varepsilon_1$, $C_Y := 8d(3d + 1)L^2C_0^{\mathrm{ms}}$, by the expansion of the holonomies of [9, (0.4)] in the axial gauge of a block. Since a layer-j block holds the initial points of at most $4a_j/\varepsilon$ bonds of the loop and $L^{-(1-\varepsilon_0)} \leq \tilde{\theta}$, the transports contribute $L^dC_W^{\mathrm{tr}}\varepsilon_1\tilde{\theta}^{k-m_W}P(X)$ per component and $s_K^{W,\mathrm{tr}} := \ell_{\mathcal{C}}C_W^{\mathrm{tr}}\varepsilon_1L^{-(1-\varepsilon_0)k}$ from the stopping layer, $C_W^{\mathrm{tr}} := 4\|d\pi\|(4C_G(\varepsilon_0)c''_K + C_Y)$, $m_W := \lfloor k/8\rfloor$, and $\ell_{\mathcal{C}}$ the length of the loops in stopping-scale bonds. Each insertion is analytic on a domain about $(U^A, J(U^A))$ whose radius in the alignment path is at least R_i with $1/R_i \leq c_{RW}\varepsilon_1((1-\beta)\min(\alpha_{0,i}, \alpha_{1,i}))^{-1}L^{3m_W-(1-\varepsilon_0)(k-m_W)}$, so by Cauchy's estimate it changes along the path by at most $\bar{r}_K := 2c_{RW}C(M)(1 + \beta_0)(1 - \beta)^{-1}R_k(m_W + 1)^{1/2}L^{-(4-7\varepsilon_0)k/8}$ times its bound; at aligned arguments the insertions generated by a step differ by at most $8\rho_K^{\mathrm{LF}}$ times their bounds at doubled amplitudes, ρ_K^{LF} the larger of the relative differences of the step's ingredients, by Cauchy's estimate in the interpolation parameter. The carry of the attached species through at most $m_W + 1$ steps, with the per-step growth $1 + 2\delta_i^{\mathrm{cl}}$ and $\prod_i(1 + 2\delta_i^{\mathrm{cl}}) \leq 4$, gives the difference mass $\bar{c}_K^W \sum_X \bar{Y}_X + 24(m_W + 1)\tau_K^w$, $\bar{c}_K^W := 4(m_W + 1)(\bar{r}_K + 8\rho_K^{\mathrm{LF}})$, with $\bar{Y}_X \leq 2\bar{\varepsilon}_AP(X)$ the masses of the attached species of the components of X; the wrapping species, which grow by up to L per step, are compared only for the ages below m_K^w, and those older differ by at most $2\tau_K^w$. At the steps of an **R**-operation the ratio of gases of Proposition S.10 is interpolated between the runs, every interpolated activity obeying the bound of both since these are

bounds of moduli, and the marked-cluster bound gives the change $2C_{\mathrm{pt}}C_X(C_{\mathrm{KP}}+C''_{\Psi})\bar{b}_K$ per point of the component, $\bar{b}_K$ the larger of the relative differences of the operations $\mathbf{T}'$ and of the factors $V(Y)$ given by (c). Collecting,

$$\begin{aligned} s_K^W &:= \frac{L^d C_W^{\mathrm{tr}}\varepsilon_1\tilde{\theta}^{k-m_W}+2(1+C_Q^W)\bar{c}_K^W\bar{\varepsilon}_A+2C_{\mathrm{pt}}C_X(C_{\mathrm{KP}}+C''_{\Psi})L^d\bar{b}_K}{C_M}\,, \\ s_K^{W,0} &:= s_K^{W,\mathrm{tr}}+2\tau_K^w+24(m_W+1)\tau_K^w+D_K^{w,\mathrm{LF}}+C_Q^W\bar{c}_K^W b_{\mathrm{wrap}}\,, \end{aligned} \tag{12.4}$$

with C_Q^W the constant of the change of the cluster weights, $D_K^{w,\mathrm{LF}}$ the difference of the young wrapping species, and b_{wrap} their weighted mass. Every term is summable in K: $\tilde{\theta}^{k-m_W}\le\tilde{\theta}^{7k/8}$, $L^{-(1-\varepsilon_0)k}$, and $\bar{r}_K$ are geometric in k up to powers of $\log(k+2)$, τ_K^w and $D_K^{w,\mathrm{LF}}$ are summable by Paper, Section 11, and $\bar{c}_K^W\bar{\varepsilon}_A$ is at most a power of $\log(K+2)$ times quantities which are summable after multiplication by any power of $\log K$. For the pairs whose oldest component is born at a scale between $k-\lfloor k/2\rfloor$ and $k-m_W$, that component meets the large-field region at every scale from its birth to k, so $P(X)\ge m_W+1$. The number $B(\mathcal{C})$ bounds the normalised expectation and not the dressed loop of a single term, whose bound carries the weights of its components (Paper, Lemma 8.6); these pairs are therefore removed by Paper, Lemma 10.2, with $p=m_W+1$, which sums the terms with their weights: removing them from both runs changes $W_{K+1}(\mathcal{C})-W_K(\mathcal{C})$ by at most $4B(\mathcal{C})e^{-(C_P-1)(m_W+1)}\bar{\delta}_{\mathrm{LF}}$, which is summable since $C_P\ge 2$.

(f), several loops. For an ensemble of several loops the cross species (Paper, Definition 5.2(ii)) is a fifth kind of factor which depends on the run, and it changes the bound in three places. First, its insertions born at least $m_K^{\times}$ steps before the stopping step, with $m_K^{\times}$ of Proposition S.16(i), are not compared. The bound Paper, (5.11), is a property of the class and holds on every labelled term, so that their weighted norm is at most $\tau_K^{\times}$ in each run, as on the trivial terms, and they contribute $2\tau_K^{\times}$; an old insertion of the cross species which an attached group takes in has an attached descendant, whose difference between the runs is at most $2\tau_K^{\times}$ at every step and which is carried as the old wraps are, with the count which gave $24(m_W+1)\tau_K^w$. Second, the younger insertions of the cross species are compared together with the younger wraps by the joint recursion of Proposition S.16(i). That recursion uses of a step only the relative differences of its constituents at their births and the carrying of Propositions S.9(v) and S.10(v), which hold on every term. On a young pair the constituents of a step are those of Proposition S.16(g)–(h), among them the operations of the young components and the factors $V(Y)$, with the relative difference $8\rho_K^{\mathrm{LF}}$ at the enlarged amplitudes after the old components have been removed at the cost $4e^{-C_Pk/4}$; these are the numbers with which the recursion was run on the trivial terms. It therefore gives the same bound D_K^{cw}, and the young insertions of the cross species and the young wraps change the logarithm of the majorant by at most $2D_K^{\mathrm{cw}}M^{\times}$, with $M^{\times}$ the total weighted mass in the exponent of $\mathfrak{b}(\mathcal{C})$. Third, an attached group born at a young step takes in the cluster occurrences of the step before, which contain wraps and insertions of the cross species; their differences add $C_\rho^w D_K^{\mathrm{cw}}$ to the relative difference of each birth, with C_ρ^w of (14.6), and the birth at the first step of the pair is compared after the removal of the old components. Carrying these terms through the at most m_W+1 steps as above replaces $\bar{c}_K^W$ by

$$\bar{c}_K^{W,\times}:=\bar{c}_K^W+4(m_W+1)C_\rho^w D_K^{\mathrm{cw}}+32\rho_K^{\mathrm{LF}}+16e^{-C_Pk/4}\,.$$

The cluster weights of the words with a member of the cross species are bounded as those of the other words, with the rate $\lambda\nu$, $\lambda\ge\frac{31}{32}$, of that species in place of ν, which costs the factor $\frac{16}{15}$ because $\sigma e^{-(\lambda-\frac{1}{2})\nu\sigma}\le 32/(15e\nu)$. Collecting, for several loops the statement holds with

$$
\begin{aligned}
s_K^{W,\times} &:= s_K^W|_{\bar c_K^W \to \bar c_K^{W,\times}} ,\\
s_K^{W,0,\times} &:= s_K^{W,0}|_{\bar c_K^W \to \bar c_K^{W,\times}} + 2\tau_K^\times + 24(m_W+1)\tau_K^\times + 2D_K^{\mathrm{cw}} M^\times + \tfrac{16}{15} C_Q^W \bar c_K^{W,\times} A^\times(\mathcal{C})
\end{aligned}
\tag{12.5}
$$

in the places of s_K^W and $s_K^{W,0}$. They are summable in K. Indeed $m_W + 1 \leq K$; $\tau_K^\times$ is at most a power of $\log(K+2)$ times $(K+2)^{-4}$; and D_K^{cw} is at most a power of $\log(K+2)$ times $\rho_K^{\mathrm{LF}} + \tau_K^w + \tau_K^\times$ (Proposition S.16(i)), where $K\rho_K^{\mathrm{LF}}$ is at most a geometric sequence plus a constant times $K^{5-\kappa_0/2}\ell_K$ by (14.6), which is summable with any power of $\log(K+2)$ because $\kappa_0 \geq 13$. *q.e.d.*

13 Theorem 2 for a Difference of Two Families

Bałaban's Theorem 2 [10, (2.43), p. 263] bounds the irrelevant part of a term of the effective action of the generation j at a regular background by a power of L^{j-n}, the ratio of the scale of the term to that of the layer at which it is evaluated. The comparison of two cutoffs needs the same bound for the difference $\Delta_j = \mathcal{G}_j - \mathbf{E}^{A,(j)}$ of run A's term and run B's term at the lift (Paper, (11.2)), with the size of the difference in place of Bałaban's constant, and it needs the difference of the irrelevant parts of one term of run B at two configurations. The first is Bałaban's proof run for a difference, and the point is to check which properties of the family the proof uses; the second is an interpolation between the bound of the age and a Cauchy estimate along a line.

We use the notation of Paper, Section 11.3. The reduced radii are $\alpha'_{i,m} := \alpha_{i,m}/(4B_3)$ and $r_{i,m} := \frac{1}{2}\alpha'_{i,m}$, $i = 0, 1$, on the layer m; $\mathcal{P}_j(X; s)$ is the set of pairs $(\mathbf{U}, \mathbf{J})$ on $X^\sim$ whose orbit under the complex gauge transformations contains a representative whose restriction to X satisfies the conditions (i)–(iii) of $\mathcal{U}_j^c(X, s_{0,j}, s_{1,j})$ [9, p. 262], with the current at most $s_{0,j}/C_{Q^*}$, $C_{Q^*} := 2(1 + (2d+1)L^{d-1})$, and whose lift ι_X has in its orbit, through a gauge transformation $\tilde v$ with $\|\mathrm{Ad}\,\tilde v^{\pm 1}\| \leq e^{2\beta_I}$, a configuration which satisfies on X the configuration conditions of run B's $\mathcal{U}_{j+1}^c(X, s_{0,j}, s_{1,j})$. The *evaluation set* $\mathcal{E}_j(X; s)$ is the set of the pairs $(\mathbf{U}, J(\mathbf{U})) \in \mathcal{P}_j(X; s)$ in which $\mathbf{U}$ is, in a complex gauge on $X^\sim$, a minimiser of real or of complex data of a scale $p \geq j$, or one of the configurations which [9, (3.13), (3.30)–(3.31)] build from such minimisers, and $J(\mathbf{U})$ is its current; these are the pairs at which the constructions below read the terms, and the terms are constant on the orbits of the complex gauge transformations. In particular $\mathcal{E}_j(X; s)$ contains the pairs $(U_j(e^{iB}), J_j(e^{iB}))$ of the minimisers of the complex data B of the scale j with $|B| < \min(s_{0,j}, s_{1,j})/C_{\mathrm{cen}}$, where C_{cen} is the constant of the proof of Lemma 3.4, enlarged by the constants of [7, Proposition 9, (190), pp. 308–309] for the lift; this is the polydisc on which Lemma 3.1 is applied to a difference. Besides the radii of the tube we use the *fixed radii* $\alpha_i^\bullet := \alpha_i/(4B_3)$, $i = 0, 1$, with the numbers α_0, α_1 of Lemma 3.2, which do not depend on the coupling or the scale, and we write $\mathcal{P}_j^\bullet(X)$ and $\mathcal{E}_j^\bullet(X)$ for the spaces $\mathcal{P}_j(X; s)$ and $\mathcal{E}_j(X; s)$ with $s_{i,j} = \alpha_i^\bullet$ and $s_{i,j} = \frac{1}{2}\alpha_i^\bullet$. For γ small these contain the spaces of the tube. For $z \in \Lambda_j^0 \cap (\Omega_n \setminus \Omega_{n+1})$, $\square \in \pi_n$ the cube of z, and $\square_0 := \tilde{\square}^5$ its near domain, the regular subclass $\mathcal{S}_z$ consists of the pairs $(\mathbf{U}, J(\mathbf{U}))$ of the class whose restriction to $\square_0$ is, in a complex gauge on $\square_0$, the local minimiser $U_n(\square_0, M^{\cdot}(\mathbf{U}))$ of its own block data at the scale of the layer [9, p. 273, ll. 39–42; p. 275, ll. 4–7], or at the scale $n-1$ where $\square_0$ meets the layer $n-1$.

Proposition S.15 (Theorem 2 for a Difference). *Let $\mathbf{E}^A$ and $\mathcal{G}$ be two families of terms of the generation j on the same lattices, $\mathbf{E}^A$ with the properties (i)–(iv) of [10, p. 259] and the covariance [10, (2.29)], and $\mathcal{G}$ with (i) on $X^\sim$ in place of X, (ii) on $\mathcal{P}_j(X; \alpha')$, and (iii), and* *let $\Delta := \mathcal{G} - \mathbf{E}^A$ satisfy $|\Delta(X,(\mathbf{U},J(\mathbf{U})),z)| \le dE_0 e^{-\kappa d_j(X)}$ on the evaluation sets of $\mathcal{P}_j(X;r)$. Let $\beta' := \frac{1}{2}\min(\beta_H, 1-\beta_E)$ as in Paper, (11.8).*

(a) For $z \in \Lambda_j^0 \cap (\Omega_n \setminus \Omega_{n+1})$ and every $U \in \mathcal{S}_z$,

$$|\Delta(\Lambda_j,U,z) - \Delta(\Lambda_j,1,z) - \Pi^\Delta A(h_z,U) - N_j(U)| \le dC_\alpha E_0 (L^{j-n})^{4+\beta'} \ , \qquad (13.1)$$

with Π^Δ the isotropic coefficient of Δ, N_j the part of the second moment of $\mathcal{G}$ off its isotropic coefficient (Paper, Section 11.1), and C_α independent of j, n, k, K, and the torus. For two covariant families (13.1) *holds with $N_j = 0$.*

(b) For two families of $\mathbf{R}$-terms with the properties of [10, p. 260, ll. 18–25], whose difference has the class norm $d^{\mathbf{R}}$, the difference $\Delta^{\mathbf{R}}(\Lambda_j,U,z) - \Delta^{\mathbf{R}}(\Lambda_j,1,z)$ is at most $d^{\mathbf{R}} C_\alpha g_j^{\kappa_0}(L^{j-n})^4$ for $U \in \mathcal{S}_z$.

(c) On the all-small-field pair at the stopping scale $k = K'$, the variation group satisfies Paper, (11.10), with $\theta_V = \beta'/(2(4+\beta'))$, $\hat{\sigma}_{k,j} := \sigma_k + C(\varepsilon_0)\rho_k + C(\varepsilon_0,\delta_c M_1)\varepsilon_1 e^{-\delta_c M_1 L^j}$ the distance of the pair line in the units of a_k, $\delta_c = \frac{1}{16}\min(\delta_V,\delta_0,\delta_1)$ the rate of Proposition S.12, and $E_V := (2C_\alpha E_0)^{1-\theta_V}(C_X(64+16C_J+2496C_0^2)E_0 + C'_A\bar{\beta})^{\theta_V}$. For the $\mathbf{R}$-terms the corresponding group is at most $(64+16C_J)(\hat{\sigma}_{k,j}/r_{0,k})(g^B_{j+1})^{\kappa_0}$ per localisation domain, without a factor of the age.

(d) The lifted family. *For a localisation domain X of the scale j and a configuration U in the domains of ι and of ι_X, in the Landau gauge about the global lift $\iota(U)$,*

$$\iota_X(U) = e^{i(\varepsilon/L)T_X}\iota(U) \quad \text{on } X^\sim \ , \qquad \tfrac{\varepsilon}{L}(|T_X(x)| + \varepsilon|\nabla T_X(x)|) \le C_\iota \varepsilon_1 e^{-2\delta_c \operatorname{dist}(x,(X^\sim)^c)/\varepsilon} \ , \qquad (13.2)$$

and $\operatorname{dist}(x,(X^\sim)^c) \ge M_1 a_j$ for $x \in X$. The second moment $\Pi^{\mathcal{G}}_{\mu\nu,\kappa\lambda}$ of the family $\mathcal{G}_j$ of Paper, (11.2)*, in the sense of [10, (3.57), p. 281], satisfies, for $\kappa < \mu$ and $\lambda < \nu$,*

$$|\Pi^{\mathcal{G}}_{\mu\nu,\kappa\lambda} - \beta^B_{j+1}\delta_{\mu\nu}\delta_{\kappa\lambda}| \le C_\Pi E_0 e^{-\delta_c M_1 L^j} \ , \qquad (13.3)$$

with C_Π independent of j, k, K, and the torus. In particular $|\Pi^{\mathcal{G}} - \beta^B_{j+1}|$ and each of the 36 *coefficients of N_j are at most $C_\Pi E_0 e^{-\delta_c M_1 L^j}$.*

(e) The fixed radii and the families. *Let $\mathbf{E}^A(s)$ and $\mathbf{E}^B(s)$ be the families of Lemma 3.5 for the two runs, $\mathcal{G}(s)$ the family of run B at the lift, and $\Delta(s) := \mathcal{G}(s) - \mathbf{E}^A(s)$, so that $\Delta(1)$ is the difference of the paper. For $|s| \le \varrho_{j-1}$ the statements (a), (c), and (d) hold for these families on the spaces of the tube, with constants which do not depend on s. At $s = 0$ they hold on the spaces $\mathcal{P}^\bullet_j(X)$ and $\mathcal{E}^\bullet_j(X)$ of the fixed radii. Moreover, if $|\Delta(X;0)| \le d^\bullet E_0 e^{-\kappa d_j(X)}$ on $\mathcal{E}^\bullet_j(X)$, and $|\Delta(X;s)| \le d^\sharp E_0 e^{-\kappa d_j(X)}$ on $\mathcal{E}_j(X;r)$ for every $|s| \le \varrho_{j-1}$, then the isotropic coefficient of $\Delta(1)$ satisfies*

$$|\Pi^\Delta| \le C^\bullet_\beta d^\bullet + C^\sharp_\beta d^\sharp \ , \qquad C^\bullet_\beta := \frac{4C_\Lambda(1)C^2_{\mathrm{cen}}E_0}{\min(\alpha^\bullet_0,\alpha^\bullet_1)^2} \ , \quad C^\sharp_\beta := \frac{64C_\Lambda(1)B_3^2C^2_{\mathrm{cen}}E_0}{\min(C^\alpha_0,C^\alpha_1)^2} \ , \quad (13.4)$$

with the constant $C_\Lambda(1)$ of Lemma 3.1 and the constants C^α_i of the radii $\alpha_{i,j} = \hat{g}_j C^\alpha_i(\log \hat{g}_j^{-2})^{q_i}$. Neither constant depends on j, K, the coupling, or the torus.

Proof. (a). Bałaban proves [10, (2.43)] on [10, pp. 279–283], from the analysis of [9, Sects. 3–4] applied with η replaced by L^{-n} in the layer $\Omega_n \setminus \Omega_{n+1}$ [10, p. 276, ll. 2–16; p. 280, ll. 14–22].

The proof is linear in the family of terms, and it reads the family through four properties only. First, the localisation, by which the sum over the domains $X \ni z$ splits into the near domains $X \subset \tilde{\square}^2$, analysed through the local minimiser on $\square_0$, and the far domains, bounded directly by [10, (3.48)] [10, p. 280, ll. 3–6]. Second, the analyticity on the polydiscs of [9, (3.13)–(3.14), Lemma 4, p. 280], on which the composites of [9, (3.30)–(3.31)] are evaluated. Third, the bound $E_0 e^{-\kappa d_j(X)}$. Fourth, the Euclidean covariance, used only at [10, (3.60)–(3.61), p. 282] to show that the second moment is a multiple of the isotropic action $A(h_z, \cdot)$. The difference Δ has the first property with $X^{\sim}$, which enlarges the near domains by one layer of cubes and changes only the counting constants; it has the second on $\mathcal{P}_j(X; \alpha')$, which contains the polydiscs at which Lemma 4 is run at the radii r, because the conditions of $\mathcal{P}_j$ are those of Bałaban's spaces at constants C_0, C_1 divided by $8B_3$ and at the current bound divided by C_{Q^*}, and his statements hold at such reduced thresholds once C_0, C_1 exceed his lower bounds by the factor $\lambda_Q := \max(8B_3, 2^{13}B_3^2L^3, 64B_3C_{Q^*}, C_{\mathrm{mem}})$, with C_{mem} the constant by which the configurations of Proposition S.14 lie in the reduced spaces, a choice which [10, (2.28), p. 259] allows because it bounds C_0 and C_1 from below only; it has the third with dE_0 in place of E_0. The fourth fails for $\mathcal{G}$, since the lift evaluates every domain at its own lift, and the part of the second moment off the isotropic coefficient is kept as N_j, a sum of the 36 plane-pair actions $\operatorname{tr} F_{\kappa\mu}F_{\lambda\nu}$, $\kappa < \mu$, $\lambda < \nu$, of [10, (3.57)]. The isotropic coefficient does not control these coefficients, since a tensor without isotropic part has the isotropic coefficient zero. Nor does the number d bound any coefficient of the second moment uniformly in the coupling: Cauchy's estimate for a second derivative of the family on the polydisc costs the inverse square of its radius, so that each coefficient is at most a constant times $dE_0 \min(r_{0,j}, r_{1,j})^{-2}$, and the function $A(h_z, \cdot)$ with such a coefficient has itself the class norm of the order d. What the proof needs is the bound of the functions $\Pi^{\Delta} A(h_z, \cdot)$ and N_j on the class, and this is a constant times dE_0, because the plane-pair actions are at most a constant times the square of the radius there. The coefficients themselves are bounded separately: those of N_j and the defect of the isotropic one by (d), and Π^{Δ} through the norms of (e). Bałaban's proof applies, therefore, to Δ, and gives (13.1) with the exponent of [10, (3.67), p. 283]. That exponent is $5 - \beta_E$ for every $\beta_E > 0$ [10, p. 283, l. 31], and the regularity of the background enters through the Hölder index β_H of [9, (3.27), (3.32), (4.18)], which Bałaban states for every index below one and which the minimisers have up to the index one [7, Theorem 1, (9), p. 279]; this is why the bound is stated on $\mathcal{S}_z$, where the configuration on $\square_0$ is a local minimiser, and not on the whole class, which bounds first derivatives only. With $\beta_H \geq 19/20$ and $\beta_E \leq 1/20$, $4 + \beta' < 5 - \beta_E$, and since $L^{j-n} \leq 1$ the bound holds with the exponent $4 + \beta'$. At $n = j$, and at a near domain which meets the layer $n - 1$ for $j \in \{n-1, n\}$, the direct bound $\sum_X |\Delta(X, U, z)| \leq O(1)dE_0$, with the same at $U = 1$ and for $\Pi^{\Delta} A(h_z, U)$ and $N_j(U)$, gives (13.1) once $C_\alpha \geq O(1)(2 + c_2^{-2})L^{4+\beta'}$. Bałaban's spaces of [9, p. 277, l. 34] carry a condition (iv) on the maximal sequence of $\square_0$ which the class does not impose; the proof of Lemma 4 uses (iv) only through [9, (3.40)], and on $\mathcal{S}_z$ this is the configuration's own condition (iii) up to the adjoint action of a gauge transformation g with $\|\mathrm{Ad}\, g\| \leq e^{\beta_I}$, so that it holds where the proposition is applied.

(b). The **R**-terms have the properties (i)–(iii) of the **E**-terms with the bound $g_j^{\kappa_0} e^{-\kappa d_j(X)}$ of [10, (2.31)] on the class, carry no coupling coefficient and hence no counterterm, and at a regular configuration each is $O(1)(L^j L^{-n})^4 g_j^{\kappa_0} e^{-\kappa d_j(X)}$ [10, p. 283, ll. 34–38]. The argument of (a), linear in the family, gives (b).

(c). The group compares $\mathbf{irr}_{j+1}^{B}(U^B)$ with $\mathbf{irr}_j^{\mathcal{G}}(U^A)$, and each of the two values has the bound (13.1) for one family with $d = 1$, so the group is at most $2C_\alpha E_0 (L^{j-k})^{4+\beta'}$ per point of the scale $j + 1$. For the second bound we join the two configurations: by Proposition S.13(b) and the tail of the localised lift, $\iota_X U^A = e^{i(\varepsilon/L)\mathbf{S}_X} U^B$ on X with $\mathbf{S}_X$

and its gradient at most $\hat{\sigma}_{k,j}L^{j-k}$ in the units of a_j. The terms are functions of pairs with the current an independent variable [9, (1.14), p. 262], so with $W(s) := e^{is(\varepsilon/L)\mathbf{S}_X}U^B$, $\Delta J := Q_1^{\mathrm{pl}}(\iota_X U^A)^* J(U^A) - J(U^B)$, and $F(s,u) := \mathbf{E}^B(W(s), J(U^B)+u\Delta J; z)$, the difference is $[F(1,0)-F(0,0)]+[F(1,1)-F(1,0)]$. In the units of a_j the line $s \mapsto W(s)$ moves the plaquettes by at most $4|s|\hat{\sigma}_{k,j}L^{j-k}$ and the field of the cube gauge with its gradient by $|s|\hat{\sigma}_{k,j}L^{j-k}$, against the room $r_{0,j} \geq r_{0,k}L^{j-k}$, which follows from $r_{0,j} \geq r_{0,k}(1+\beta_0)^{-2}(k-j)^{-1/2}$ by [10, (2.6), (2.28)]. For $|s| \leq T := r_{0,k}/(16\hat{\sigma}_{k,j})$ the line stays in the domain of $\mathbf{E}^B$, so Cauchy's estimate gives $|F(1,0) - F(0,0)| \leq 2E_0/T = 32(\hat{\sigma}_{k,j}/r_{0,k})E_0$ for $T \geq 2$, and $2E_0 < 64(\hat{\sigma}_{k,j}/r_{0,k})E_0$ for $T < 2$. For the current, both minimisers carry the lifts of multipliers, $J = \mathcal{Q}^*\lambda$, and since the averagings compose, $J(U^B) = Q_1^{\mathrm{pl}}(U^B)^* J'$ with $J' := \mathcal{Q}(MU^B)^*\lambda^B$; then $\Delta J = Q_1^{\mathrm{pl}}(\iota_X U^A)^*(J(U^A) - J') + [Q_1^{\mathrm{pl}}(\iota_X U^A) - Q_1^{\mathrm{pl}}(U^B)]^* J'$, the first term bounded through the multipliers of the two problems and the second through the transports of $Q_1^{\mathrm{pl}*}$, which gives $a_j^3 \sup|\Delta J| \leq C_J\hat{\sigma}_{k,j}L^{3(j-k)}$, $C_J := 2x_J + 1$, and the current part at most $16C_J(\hat{\sigma}_{k,j}/r_{0,k})E_0$. The coupling term $-\beta^B_{j+1}[A_{\varepsilon/L}(h_z, U^B) - A_\varepsilon(h_z, U^A)]$ splits through MU^B into the alignment, at most $C_A\rho_k\varepsilon_1 L^{4(j-k)}$, and the localised blocking artefact, at most $(16C_b + 768C_0^2)(\varepsilon/a)\varepsilon_1^2 L^{3(j-k)}$ by Proposition S.13(a) with the weight h_z, which is Lipschitz with the constant $1/a_j$; with $\bar{\beta} := \sup_j|\beta_j^B|$, finite by the window bound of Paper, Section 4, this is at most $C'_A\bar{\beta}\hat{\sigma}_{k,j}\varepsilon_1$, $C'_A := C_A + 16C_b + 768C_0^2$. The moment defect $(\Pi^{\mathcal{G}} - \beta^B_{j+1})A_\varepsilon(h_z, U^A) + N_j(U^A)$ is at most $2496C_X C_0^2 E_0 C_\Pi \varepsilon_1^2 e^{-\delta_c M_1 L^j}$, since the isotropic action and each of the 36 plane-pair actions are at most $32C_0^2\varepsilon_1^2 L^{4(j-k)}$ and their coefficients are bounded by (13.3). So the group has the second bound $(\hat{\sigma}_{k,j}/r_{0,k})(C_X(64 + 16C_J)E_0 + C'_A\bar{\beta}) + 2496C_X C_0^2 E_0 C_\Pi \varepsilon_1^2 e^{-\delta_c M_1 L^j}$, without a factor of the age. The smaller of two bounds is at most their geometric mean with the weights $1-\theta_V$ and θ_V, $(x+y)^\theta \leq x^\theta + y^\theta$, and $(4+\beta')(1-\theta_V) = 4 + \beta'/2$, which gives Paper, (11.10). For the **R**-terms the same split of the pair line gives the stated bound, with $(g^B_{j+1})^{\kappa_0}$ in place of E_0 and no moment, since they carry no coupling coefficient.

(d), the tail. The global lift $W_0 := \iota(U)$, restricted to $X^\sim$, is the minimiser of the local problem of ι_X with its own values on the collar as data. Indeed, let W be a configuration on $X^\sim$ with the averages of W_0 on the bonds of run A in $X^\sim$ and with the values of W_0 on the collar, and continue it by W_0 outside $X^\sim$. The continued configuration has the averages of W_0 everywhere, because the average of a bond of run A reads the fine bonds within the distance 2ε [4, (124), p. 36] and the collar is wider than that; and its action is the action of W on $X^\sim$ plus a number which does not depend on W, because every plaquette which is not contained in $X^\sim$ has its bonds in the collar or outside $X^\sim$. So W_0 minimises the local action, and the local minimiser is unique up to gauge transformations [7, Theorem 1, p. 279]. Hence $\iota_X(U)$ and W_0 are minimisers of one local problem, with data which agree on the bonds of run A and differ on the collar by $B := (1/i)\log[(Q_1^{s*}U)W_0^{-1}]$, and $|B| \leq C_1\varepsilon_1$ in the units of the collar, because both collar fields are regular and have the same averages [6, Lemma 1, (1.24)–(1.25), p. 79]. By [7, Proposition 9, p. 309], in the Landau gauge $\iota_X(U)W_0^{-1} = \exp(i(\varepsilon/L)\mathscr{H}(B))$, and the derivative of $\mathscr{H}$ in the datum of a block y' is bounded, with its first differences, by [7, (190), p. 308], with the decay $e^{-\frac{1}{8}\delta_V d(y,y')}$. Integrating along the segment from 0 to B and summing over the blocks of the collar gives (13.2), with $2\delta_c \leq \frac{1}{8}\delta_V$ and C_ι a multiple of C_1 by the constant of [7, (190)] and the count N_1. The distance from X to the complement of $X^\sim$ is at least $M_1 a_j$ because $X^\sim$ is X with the adjacent cubes of side $M_1 a_j$.

(d), the moments. By [10, (3.50), (3.57)] the second moment of a family is the value of the second derivative of $B \mapsto \mathbf{E}(U_j(e^{iB}), z)$ at $B = 0$, in the block data B of the scale j and through the minimiser U_j, on the pairs of the affine data $B_\mu(x) = (x-z)_\kappa$ and $B_\nu(y) = (y-z)_\lambda$; the terms are those of the whole lattice, as in [10, p. 281], and the sums converge by the

decay of the kernels. We compare the second derivatives of $B \mapsto \mathcal{G}_j(U_j^A(e^{iB}), z)$ and of $B \mapsto \mathbf{E}^{B,(j+1)}(U_{j+1}^B(e^{iB}), z)$ on these data. The second is $\beta_{j+1}^B \delta_{\mu\nu}\delta_{\kappa\lambda}$ by [10, (3.60)–(3.61), p. 282] for the covariant family of run B. The first derivative of a term $\mathbf{E}^B(X, \cdot)$ at the trivial configuration vanishes: it is a linear form $A \mapsto \sum_b \operatorname{tr}(c_b A(b))$ which the gauge invariance of the term makes invariant under $A \mapsto gAg^{-1}$ for every constant $g \in G$, so that every c_b commutes with G, and the centre of the Lie algebra of a simple group is trivial. The second derivative of the first function is therefore $\sum_{X \ni z} d^2\mathbf{E}^B(X)[\mathcal{L}_X a, \mathcal{L}_X a']$, and that of the second is $\sum_{X \ni z} d^2\mathbf{E}^B(X)[b, b']$, up to terms which contain the currents; here $\mathcal{L}_X := d\iota_X(1)$, a and a' are the linear minimisers of run A at the two data, and b, b' those of run B.

For affine data the linear minimisers are affine. Let p be an affine field on a lattice and $H = GQ^*(QGQ^*)^{-1}$ the linear minimiser at the trivial background, with $G^{-1} = D^*D + DRD^* + Q^*aQ$ there. The identity of Proposition S.12(i), which uses only the form of H, reads $(1 - HQ)p = (G - HQG)(D^*D + DRD^*)p$. Now Dp is a constant two-form, so $D^*Dp = 0$, and D^*p is a constant c, which R annihilates: $P = 1 - R$ fixes the range of $G'Q'^*$, and $c = G'Q'^*(aQ'c)$ at the trivial background, where $G'^{-1} = D^*D + Q'^*aQ'$ and $D^*Dc = 0$. Hence $HQp = p$. Applied to the minimisers of the two runs this gives $a = p^A$ and $b = p^B$, the affine fields of the two lattices with the data B; applied to the one-step problem it gives $\mathcal{L}p^A = p^B$ for the global lift $\mathcal{L} := d\iota(1)$, since $Q_1 p^B$ is affine with the data of p^A. The currents of affine fields vanish, so the terms with the currents drop out, and

$$\Pi^{\mathcal{G}}_{\mu\nu,\kappa\lambda} - \beta_{j+1}^B \delta_{\mu\nu}\delta_{\kappa\lambda} = \sum_{X \ni z} \Big(d^2\mathbf{E}^B(X)[\mathcal{L}_X p, \mathcal{L}_X p'] - d^2\mathbf{E}^B(X)[\mathcal{L}p, \mathcal{L}p'] \Big) , \tag{13.5}$$

with p, p' the affine fields of run A at the two data. It remains to bound $\mathcal{L}_X - \mathcal{L}$ on X. Let θ be a function with values in $[0, 1]$ which equals one within the distance $\operatorname{diam}(X^\sim)$ of z, vanishes beyond twice this distance, and has first differences at most $1/\operatorname{diam}(X^\sim)$. The field $(1 - \theta)p$ vanishes on $X^\sim$, so $\mathcal{L}_X$ does not see it, and $|\mathcal{L}((1-\theta)p)|$ is at most $c_H N_1 e^{-\frac{1}{4}\delta_1 \operatorname{diam}(X^\sim)/\varepsilon}$ times a polynomial in $\operatorname{diam}(X^\sim)/\varepsilon$ on X, by the decay of the kernel of $\mathcal{L}$. For θp the difference $(\mathcal{L}_X - \mathcal{L})\theta p$ is the derivative at $U = 1$ of the tail T_X in the direction θp, and since T_X is analytic in U on a polydisc of radius $c_\iota \varepsilon_1$ in the units of the scale j, on which (13.2) holds, Cauchy's estimate gives $|(\mathcal{L}_X - \mathcal{L})\theta p| \le (C_\iota/c_\iota)e^{-2\delta_c M_1 L^j} \sup|\theta p|$ on X, with the first differences. The supremum of $|\theta p|$ is at most $3 \operatorname{diam}(X^\sim) \le 3M(3 + d_j(X))a_j$ for the unit data. The second derivative of $\mathbf{E}^B(X)$ at the trivial configuration is at most $C_2 E_0 e^{-\kappa d_{j+1}(X)}$ in the norms of its space, with a constant C_2 which does not depend on the coupling: this is Lemma 3.5(c), and it is not Cauchy's estimate on the polydisc of the tube, whose radius would cost a factor of the order g_j^{-2}. With it the sum over $X \ni z$, in which half of κ pays the polynomial in $d_j(X)$, bounds (13.5) by a constant times $E_0 e^{-2\delta_c M_1 L^j}$ times a polynomial in $M_1 L^j$, which is at most $C_\Pi E_0 e^{-\delta_c M_1 L^j}$; the units a_j^2 of the two affine data are those of the second moment. This is (13.3), and its consequences for $\Pi^{\mathcal{G}}$, the coefficient with $\mu = \nu = 2$ and $\kappa = \lambda = 1$ [10, (3.61)], and for the coefficients of N_j follow.

(e), the families. For $|s| \le \varrho_{j-1}$ the families $\mathbf{E}^A(s)$ and $\mathbf{E}^B(s)$ have the first three of the four properties through which the proof of (a) reads a family, with the constants of the construction, by Lemma 3.5(b) at $k = j - 1$ and at run B's corresponding step. They have the fourth as well, in the sense in which the regular terms of the construction have it: the **R**-terms which enter the regular part are covariant under the Euclidean transformations which leave the next lattice invariant [10, p. 271, ll. 1–5], a complex multiple of a covariant function is covariant, and the recursion preserves the covariance. The proofs of (a), (c), and (d) are linear in the families and use nothing else, the proof of (d) through Lemma 3.5(c), which holds for these s. This gives the first statement.

(e), the fixed radii. At $s = 0$ the terms of both runs are analytic on the spaces $U_j^c(X, \alpha_0, \alpha_1)$ and bounded there by $E_0 e^{-\kappa d_j(X)}$, by Lemma 3.5(a). These are the spaces for which the analysis of [9, Sects. 3–4], on which the proof of (a) rests, is written, so that the proof of (a) applies with them in the place of the spaces of the tube. The reduction of the radii by the factors $4B_3$ and λ_Q of that proof is admissible: Bałaban's conditions bound the radii from below relative to the thresholds of the fields, and in the construction of the paper the thresholds ε_j and ϑ_j tend to zero with γ while α_0 and α_1 are fixed. The local lift, its tail (13.2), and the comparison of the lifts in the proof of (d) are statements on the minimisers of [7, Theorem 1, p. 279; Proposition 9, p. 309], which require of the data only a regularity with a parameter that is small independently of the coupling; they hold therefore at the data of the fixed radii, once α_0 and α_1 are below that parameter, and in the proof of (d) the second derivative of $\mathbf{E}^B(X; 0)$ is bounded by the first term of the constant C_2 of Lemma 3.5(c).

(e), the coefficient. The coefficient is linear in the family, so that $\Pi^\Delta = \Lambda(\Delta(0)) + \Lambda(\Delta(1) - \Delta(0))$, where Λ denotes the isotropic coefficient of a family, the polarisation coefficient of Lemma 3.1 taken through run A's minimisers. The family $\Delta(0)$ is analytic and bounded by $d^\bullet E_0 e^{-\kappa d_j(X)}$ on $\mathcal{E}_j^\bullet(X)$, which contains the minimisers of the complex data B with $|B| < \frac{1}{2}\min(\alpha_0^\bullet, \alpha_1^\bullet)/C_{\text{cen}}$. This is a polydisc about the trivial configuration whose radius does not depend on the coupling, and Lemma 3.1 gives $|\Lambda(\Delta(0))| \leq C_\beta^\bullet d^\bullet$. For the second term, fix a pair of $\mathcal{E}_j(X; r)$. The function $s \mapsto \Delta(X; s)$ is analytic on $|s| < \Theta_{j-1}$ and bounded by $d^\sharp E_0 e^{-\kappa d_j(X)}$ on $|s| \leq \varrho_{j-1}$, and for $s \in [0, 1]$ the disc of the radius $\varrho_{j-1} - 1 = \hat{g}_{j-1}^{-2}$ about s lies in this set. Cauchy's estimate bounds the derivative in s on $[0, 1]$ by $\hat{g}_{j-1}^2 d^\sharp E_0 e^{-\kappa d_j(X)}$, and so does it bound $|\Delta(X; 1) - \Delta(X; 0)|$. The set $\mathcal{E}_j(X; r)$ contains the minimisers of the complex data B with $|B| < r_\Delta := \min(r_{0,j}, r_{1,j})/C_{\text{cen}}$, and since $r_{i,j} = \alpha_{i,j}/(8B_3) \geq \hat{g}_j C_i^\alpha/(8B_3)$, we have $r_\Delta^{-2} \leq 64 B_3^2 C_{\text{cen}}^2 \min(C_0^\alpha, C_1^\alpha)^{-2} \hat{g}_j^{-2}$. Lemma 3.1 on this polydisc gives

$$|\Lambda(\Delta(1) - \Delta(0))| \leq C_\Lambda(1)\, r_\Delta^{-2}\, \hat{g}_{j-1}^2\, d^\sharp E_0 \leq C_\beta^\sharp \, \frac{\hat{g}_{j-1}^2}{\hat{g}_j^2}\, d^\sharp \leq C_\beta^\sharp d^\sharp \ ,$$

because $\hat{g}_{j-1} \leq \hat{g}_j$. The factor $\hat{g}_j^{-2}$ which the radius of the tube costs is paid by the radius of the disc of s, and this is possible because the $\mathbf{R}$-terms are of the order g^{κ_0}: the disc of the radius ϱ_{j-1} lies in the disc of Lemma 3.4. *q.e.d.*

14 The Recursion of the Differences

At a shared scale the two runs apply one step to inputs on the same lattice, and the difference of the outputs is, by the interpolation identity of Paper, Lemma 11.7, the expectation of a marked activity under a measure of Bałaban's kind. Bałaban's construction of the step [10, pp. 274–278] treats one insertion of this type, the localised term $\operatorname{tr} \mathbf{V}_k'(CA)(b)$ of [10, (3.37), p. 274], and the first item of the proposition below records that his argument applies to any localised insertion with the properties it uses. The others assemble the recursion of Paper, Section 11, solve it, and treat the parts of the dressed loops which the recursion carries. We write a_k for the bound of the activities of the interaction, a positive power of g_k times a power of $\log g_k^{-2}$, and μ_k for the square of the ratio of the supremum of the cutoff to the radius of analyticity of the frozen terms, a negative power of $\log g_k^{-2}$.

Proposition S.16 (The Recursion of the Differences)**.** *Let the constants satisfy the conditions of Paper, Appendix C, let $\bar{g}$ be small, and let γ satisfy Paper,* (C.2)*.*

- *(a)* The marked-cluster bound. *Let $\mathbf{P}_t$ be an admissible interaction of the step $k+1$ and let $I(Y)$, $Y \in \mathbf{D}_k$, be localised insertions, analytic on the fluctuation domain of Bała-*

ban's terms of the same kind with $|I(Y)| \leq \iota(Y)e^{-\kappa d_k(Y)}$ *there. Then* $\langle \sum_Y I(Y) \rangle_t = \sum_{X \in \mathbf{D}_{k+1}} \mathbf{I}(X)$*, with* $\mathbf{I}(X)$ *analytic on the class of the scale* $k+1$ *and*

$$|\mathbf{I}(X)| \leq C_{\mathrm{pt}} C_X C_{\mathbf{KP}} \sup_{Y \cap X \neq \emptyset} [\iota_G(Y) + a_k \iota(Y)] e^{-(1+7\beta/2)\kappa d_{k+1}(X)} , \tag{14.1}$$

where $\iota_G(Y) \leq \iota(Y)$ *bounds the terms in which the insertion is connected to no activity of* $\mathbf{P}_t$*,* $\iota_G(Y) = 0$ *for an insertion odd in the fluctuation field, and* $\iota_G(Y) \leq C''_G(|c| + |Y| \|K\|_Y)$ *for* $I(Y) = c + \langle A, KA \rangle$*,* $\|K\|_Y$ *the larger of the row and column sums of* K*.*

(b) The recursion. *Let* $d^{\sharp}_k$ *be the class norm of the differences* $\Delta_k(s)$ *of the families of Proposition S.15(e) on the evaluation sets of the tube, uniformly in* $|s| \leq \varrho_{k-1}$*, and* $d^{\bullet}_k$ *the class norm of* $\Delta_k(0)$ *on the evaluation sets of the fixed radii, with the carried sums* $D^{\sharp}_k$ *and* $D^{\bullet}_k$*; the class norm of the paper's difference is* $d_k \leq d^{\sharp}_k$*. They satisfy Paper,* (11.19) *and* (11.20)*, with* $\varepsilon^*_k = C_X C_{\mathrm{pt}} C_{\mathbf{KP}} L^4 C^{\mathbf{A}}_{\alpha} [C_T \mu_k + 4a_k]$*, and with* $C_\beta := 2\max(C^{\bullet}_{\beta}, C^{\sharp}_{\beta})$ *of* (13.4) *and* $u_k := \max(d^{\bullet}_k, d^{\sharp}_k)$*,*

$$|\beta^B_{k+2} - \beta^A_{k+1}| \leq C_\beta u_{k+1} + C_\Pi E_0 e^{-\delta_c M_1 L^{k+1}} , \qquad |\delta_j| \leq C_\beta \sum_{i>j} u_i + 2C_\Pi E_0 e^{-\delta_c M_1 L^{j+1}} . \tag{14.2}$$

The second terms, the moment defect of Proposition S.15(d), enter the recursion as sources of the form of the depth sources e_i *and are counted among them. The norm* d_k *alone does not bound the difference of the coefficients: on the evaluation sets of the tube it gives* $|\Pi^{\Delta}_k| \leq C \hat{g}_k^{-2} d_k E_0$ *and no more.*

(c) The **R**-terms. *The class norms of the differences of the* **R***-terms satisfy Paper,* (11.21)*, with* $C_S := 2\exp(2C_\Sigma)$*,* $C_\Sigma := \sup_K \sum_{j \leq K} (c_j + c^B_j + c^V_j + c'_j)$*, which is finite for* $\kappa_0 > 2$*.*

(d) The integrands. *The relative difference of the two runs' integrands of the* **R***-operation, per point of the history of a component at the scale* m*, is at most* $\mathcal{D}_m := D^{\flat}_m + S_{m-1} + C|\delta_m| + e_m + h_m$*, with* $D^{\flat}_m := \sum_{i \leq m} q^{m-i} d_i$*,* $q := L^{-1/3} e^{-1/12}$*,* $h_m := C_h \tilde{\theta}^m$*; Paper,* (11.21)*, holds with the memory-weighted maxima* $\mathcal{D}'_{[j]} := \max_{k_s/8-1 < m \leq j} e^{-C_P(j-m)/2} (\mathcal{D}_m - S_{m-1})$ *and the corresponding maxima* $\mathcal{D}^{*\prime}_j$*,* $\mathcal{D}^{V\prime}_j$ *of the boundary terms and of the old terms in the factors* $V(Y)$*.*

(e) The solution. *The constant* $C_{\mathcal{T}} := \tilde{\theta}^{-1}[(1 - L^{-\beta'}/\tilde{\theta})^{-1}(1 + C_S C_m) + (1 - L^{-1/3}/\tilde{\theta})^{-1} C_S C_m + C C_\beta C_S C'_m + C_S C_*(1 + C_\mu)(C^B_{R,1}/C_R) C^N_m + C_S C_*(1 + C^B_V)(1 + C_\mu)(C^V_R/C_R) C_m + C_g C_\beta (2(1 - \tilde{\theta})^{-2} + b^{-1})]$*, with the constants of the proof, satisfies* $C_{\mathcal{T}} \geq (1 - L^{-\beta'})^{-1}$ *and bounds the norm of the operator of the recursion by* $\varepsilon^* C_{\mathcal{T}}$ *in the weighted norm* $\sup_i u_i/m_i$*,* $m_i := i\tilde{\theta}^i + w_K$*; the depth sources satisfy* $e_i \leq C_e \tilde{\theta}^i$ *at every* i*; and Paper,* (11.24) *and* (11.25)*, hold with the constants of the proof.*

(f) The insertions. *The class norms* d^{ins}_i *of the differences of the deviation insertions satisfy* $d^{\mathrm{ins}}_{i+1} \leq (\varepsilon^* + \varepsilon^Q_i) D^{\mathrm{ins}}_i + \varepsilon^*(D_i + S_i) + C_g \varepsilon^* g_i^2 |\delta_i| + e_{i+1}$*, with* $\varepsilon^Q_i := \varepsilon^Q(g_i)$ *of order* g_i^2 *times a power of* $\log g_i^{-2}$*, and under Paper,* (11.26)*,* $d^{\mathrm{ins}}_i \leq 2C^*_{\mathrm{ins}} m_i$*.*

(g) The wraps and the weights. *The insertions which wrap around the torus and the interaction weights* $Q_{K'}(\mathfrak{B})$ *change the normalised dressed loop by at most* $(x^w_K + x^Q_K)\mathfrak{b}(\mathcal{C})$ *with* x^w_K *and* x^Q_K *summable in* K*.*

(h) The loop constant. $|E^B_{\mathrm{const}} - E^A_{\mathrm{const}}| \leq \Delta E_K \leq \min(2\tau^E_K + \Delta^E_K, 2B_E(\mathcal{C}))$*, summable in* K*.*

(i) The cross species. *For an ensemble of several loops and on the trivial terms of the two runs, the insertions of the cross species (Paper, Definition 5.2(ii)) and the wraps born at least $m_K^\times$ steps before the stopping step, with an age $m_K^\times$ which grows like* $\log\log K$*, are not compared, and their totals $\tau_K^\times$ and τ_K^w in each run are at most a power of* $\log(K+2)$ *times $(K+2)^{-4}$. At each of the last $m_K^\times$ steps i, the two runs' differences e_i of the younger insertions of the cross species and of the younger wraps in weighted mass, the differences $\delta\hat{\varepsilon}_i$ of the activities of the reference gas which contain wraps, and the part F_i of the class norm of the deviation insertions which is due to the wraps in the merged occurrences are at most*

$$D_K^{\mathrm{cw}} := 1024\,\varpi^{m_K^\times+1}\Big[B_K^H + (m_K^\times+1)(C_1 e^{c\mathrm{w}}\beta_K + 6C_{\hat{\varepsilon}}\tau_K^w)\Big]\,, \tag{14.3}$$

with the constants and the sources of the proof, and D_K^{cw} is summable in K. Consequently the cross species, the wraps, and the weights change the normalised dressed loop by at most $x_K^\times\mathfrak{b}(\mathcal{C})$ with $x_K^\times$ summable in K, and (f) holds with the further source $2c_o C_1 D_K^{\mathrm{cw}}$. The cross constants of the steps satisfy (h), and the cross groups of the attached species satisfy the statements for these species, with the counts of one loop and the total perimeter in place of the perimeter.

Proof. (a). Bałaban defines the new constant term through $\int_0^1\langle\operatorname{tr}\mathbf{V}_k'(tg_k,A,b)(CA)(b)\rangle_t\,\mathrm{d}t$ [10, (3.37), p. 274], and he expands "the last expectation value in (3.37)" [10, p. 277, ll. 22–44; p. 278, ll. 1–15] by the cluster expansion [10, (3.44)–(3.47)], in which the nominator carries one factor $H'(Z_0)$ "given by an integral with the integrand including, as one of the factors, a localized term from the expansion of the integrated function", which "although different, …satisfies the same bound as the other factors" [10, p. 277, l. 44 – p. 278, l. 2]. The argument uses of the insertion only its localisation, its analyticity on the fluctuation domain of [9, (6.34)], and its bound, which enters linearly, so the output bound is the insertion's bound times a constant of the expansion. The split into ι_G and $a_k\iota$ separates the terms of $H'(Z_0)$ whose integrand carries no activity of $\mathbf{P}_t$ from those with at least one; each activity carries the factor a_k, the vertices of order $n\geq 3$ of the action being at most $g_k O(p_0(\hat{g}_k)^3)$ on the cutoff ball and the fluctuation parts of the frozen terms being bounded "with the constant B_0 replaced by $O(p_0^3(g_k))$" [10, p. 278, ll. 32–35]. The terms without activities are Gaussian expectations of the insertion with the cutoff, which Bałaban treats for the integrand $\partial_t\chi^{(k)}(tA(b))$ [10, p. 278, ll. 37–42] by the steps [11, (2.3)–(2.13)], which use of the integrand only its localisation and its bound; so $\iota_G\leq\iota$, and $\iota_G=0$ for an odd insertion because the cutoff measure is even. For $I(Y)=c+\langle A,KA\rangle$, Schur's test gives $|\langle A,KA\rangle|\leq\|K\|_Y\sum_{b\in Y}|A(b)|^2$, and the Cauchy formula $\langle A,KA\rangle=(2\pi i)^{-1}\oint e^{\tau\langle A,KA\rangle}\tau^{-2}\,\mathrm{d}\tau$ over $|\tau|=\alpha_4^G/(\|K\|_Y|Y|)$ reduces the quadratic insertion to an exponential of a form bounded by $\alpha_4^G\sum_{b\subset Y_0}|A(b)|^2$, which the steps [11, (2.14)–(2.25), pp. 15–17] absorb into the Gaussian integral with $\alpha_5+O(1)\alpha_4^G$ in place of α_5; the Cauchy formula costs $\|K\|_Y|Y|/\alpha_4^G$, so $C_G'':=(1+1/\alpha_4^G)$ times Bałaban's constant. For the rate, the construction is applied with the input rate $(1-\beta/8)\kappa$, allowed since κ may be chosen large [10, p. 260, l. 27], and its output carries $(1+4\beta)(1-\beta/8)\kappa\geq(1+15\beta/4)\kappa$ for $\beta\leq 1/4$ [9, p. 262, ll. 25–28]. Of the remaining factor $e^{-(\beta/8)\kappa d_k(Y)}$ one half pays the factors $|Y|$, at the cost C_{pt}, and the other the sum over the Y meeting X, at most $n_k(X)C_X$, $C_X:=\sup_x\sum_{Y\ni x}e^{-(\beta/16)\kappa d_k(Y)}$, with $n_k(X)\leq C_{\mathrm{pt}}e^{(\beta/4)\kappa d_{k+1}(X)}$, which leaves the rate $(1+7\beta/2)\kappa$. For the insertions from the boundary terms the same holds with the fluctuation domain $\tilde{\mathcal{U}}_{k+1}^c(Y,\tilde{\alpha}_0,\tilde{\alpha}_1)\times\{|A|<2\vartheta_k/g_k\}$, on which they are analytic, and the output is analytic on the class of $\mathbf{B}^{(k+1)}$ [10, (2.41)–(2.42), p. 261].

(b). By (a) applied to the right side of the interpolation identity, with the marked activity $\mathbf{P}^B-\mathbf{P}^A$ and the quadratic insertion $-\frac{1}{2}\langle A,C^*\delta\Delta^{(k)}CA\rangle$, the generated difference has localised

terms bounded by (14.1). The marked activity consists of the fluctuation parts of the frozen differences of the generations $j \leq k$, which Proposition S.15 bounds at the composites with the constant $C^{\mathbf{A}}_{\alpha} := 6^5 C_{\alpha}$; of the coupling, whose two values differ relatively by at most $g_k^2|\delta_k|$; and of the depth sources. The fluctuation part of a frozen difference is not of the second order in the ratio of the cutoff to the radius, and its two Taylor parts enter (a) differently. Let ρ be the supremum of the cutoff and r the radius of analyticity in the fluctuation field, $\mu_k = (\rho/r)^2$, and write the fluctuation part as $f_1 + f_2$, with f_1 linear in the fluctuation field and f_2 the remainder. On the cutoff ball Cauchy's estimate gives $|f_1| \leq 2\mu_k^{1/2}$ and $|f_2| \leq C_T \mu_k$ times $C^{\mathbf{A}}_{\alpha}(D_k + S_k)$ per point, with C_T the constant of the Taylor remainder. The part f_1 is odd, so that in (14.1) it has $\iota_G = 0$ and contributes only through the terms connected to an activity, with the factor a_k; the part f_2 contributes through ι_G. Since $2\mu_k^{1/2} + C_T\mu_k \leq 4$ for $\bar{g}$ small, this gives the bracket $C_T\mu_k + 4a_k$ of ε^*_k.

(b), the sources. We now say what the depth sources e_{k+1} consist of and why they are bounded uniformly in the coupling. The argument uses the two runs only at backgrounds of spaces whose radii do not depend on the coupling, the domains of [5, (3.37), p. 396] and of [4, Proposition 3, p. 36], and it applies therefore to every evaluation set contained in them, those of the tube and those of the fixed radii alike. Let $(\mathbf{U}, J(\mathbf{U}))$ be a point of an evaluation set of the scale $k+1$, at which run A's terms are read, and $(\iota_X\mathbf{U}, Q_1^{\mathrm{pl}}(\iota_X\mathbf{U})^* J)$ its lift, at which run B's are. Since $M\iota_X\mathbf{U} = \mathbf{U}$, the configurations of the scale k of the two runs coincide, $V^{(k)} = M^k\mathbf{U}$ in both, and with them every object of the step which is built from $V^{(k)}$ and the fluctuation variable B alone: the map C, the functions $\tilde{C}$, $\tilde{D}$, and h, the density σ, and the gauge-fixing terms [9, pp. 266–268]. The exponent of the step is [9, (2.12), p. 268], and its terms $\operatorname{Tr}\log(I - h(\delta\tilde{D}/\delta B))$, $\log\sigma$, and $g_k^{-2}\mathbf{G}_3$ are therefore the same functions of B in the two runs. The other terms contain objects of the fine lattices, the operator H_1, the form Δ_1, the current, and the remainder V of the third order of the action, and none of them is of an order lower than three in g_kCB apart from the Gaussian form; in particular the exponent has no term linear in B in either run. We write $x_1 := g_kCB$, $x_2 := h\tilde{D}(g_kCB)$, $x_3 := h\tilde{D}_3(g_kCB)$, and $\varphi := x_1 - x_2$; on the cutoff ball $|x_n| \leq C\vartheta_k^n$ by Lemma 17.2. The differences are the following, each at a shared scale $k \geq m_{\mathrm{rc}}$, with m_{rc} the integer of the proof of Proposition S.12(h), and each for the operators decoupled by the parameters $s(\Delta)$ on the polydisc of [11, (1.23), p. 7], as the localisation of the potentials requires.

(1) *The Gaussian form.* Its difference is the quadratic insertion with the kernel bound r_k of Proposition S.17, whose proof compares the two forms at aligned backgrounds of the complex domain of [5]; by (a), with $\iota_G \leq C''_G|Y|\|K\|_Y$, it contributes at most $C'_e r_k$ per point of the scale $k+1$.

(2) *The forms $g_k^{-2}\langle H_1x_1, \Delta_1H_1x_2\rangle$ and $g_k^{-2}\langle H_1x_2, \Delta_1H_1x_2\rangle$.* We write the difference of the two runs' values of $\langle H_1x, \Delta_1H_1y\rangle$ as the sum of $\langle H_1^Bx, \Delta_1^BH_1^By\rangle - \langle Q_1H_1^Bx, \Delta_1^AQ_1H_1^By\rangle$, which is the bilinear defect at the pair (H_1^Bx, H_1^By) and is bounded by Proposition S.12(d) as in the aligned comparison of the proof of Proposition S.17, and of $\langle Q_1H_1^Bx, \Delta_1^AQ_1H_1^By\rangle - \langle H_1^Ax, \Delta_1^AH_1^Ay\rangle$, in which $Q_1H_1^B - H_1^A$ is bounded with its first differences by Proposition S.12(h). Both are at most a constant times $(\varepsilon/a_k)^{\gamma_H}$ times the common bound of the two forms, which is a constant times $g_k^{-2}\vartheta_k^3$ per point.

(3) *The term $g_k^{-2}\langle H_1x_3, J\rangle$.* Run B's current is $Q_1^{\mathrm{pl}}(\iota_X\mathbf{U})^*J$, so that the difference is $g_k^{-2}\langle(Q_1^{\mathrm{pl}}H_1^B - H_1^A)x_3, J\rangle$. By (10.9), $Q_1^{\mathrm{pl}} = Q_1 - \bar{D}\Lambda_1$, and $\langle\bar{D}\mu, J\rangle = 0$ for every function μ, because $J = J(\mathbf{U})$ is the first variation of the action, which is gauge invariant. The difference is therefore $g_k^{-2}\langle(Q_1H_1^B - H_1^A)x_3, J\rangle$, at most $C_{\mathrm{tel}}\delta_w(Y)B_Te^{2\cdot 6^d\kappa_1}$ times $g_k^{-2}\vartheta_k^3 \sup|J|$ per point by Proposition S.12(h), and $\sup|J|$ is at most the radius

of the space.

(4) *The remainder* $g_k^{-2}V(H_1\varphi)$. Let $a^X := H_1^X\varphi$, $W := \iota_X\mathbf{U}$, and let $b(W') = A_\varepsilon(\bar{M}W') - A_{\varepsilon/L}(W')$ be the blocking artefact, with $\bar{M}(e^{i(\varepsilon/L)a}W) = \exp(i\varepsilon(Q_1 a + C_1(a)))\mathbf{U}$ and C_1 the part of the second and higher orders of the average [4, (122), (123), p. 36]. Taking the parts of the third and higher orders in a of the identity $A_{\varepsilon/L}(e^{i(\varepsilon/L)a}W) = A_\varepsilon(e^{i\varepsilon(Q_1 a+C_1(a))}\mathbf{U}) - b(e^{i(\varepsilon/L)a}W)$ gives

$$\begin{aligned} V^B(a) = V^A(Q_1 a + C_1(a)) &+ \langle C_1^{(\geq 3)}(a), J\rangle + \langle Q_1 a, \Delta^A C_1(a)\rangle \\ &+ \tfrac{1}{2}\langle C_1(a), \Delta^A C_1(a)\rangle - R_3[b](a)\ , \end{aligned} \tag{14.4}$$

where Δ^A is the second variation [5, (3.10), (3.12), p. 392] of the action at $\mathbf{U}$, V^X is the remainder of the third order of the action of run X at its background, and $R_3[b]$ is the Taylor remainder of the third order of $a \mapsto b(e^{i(\varepsilon/L)a}W)$. Here $|C_1(a)| \leq C_1\varepsilon|a|^2$ by [4, (123)], which is of the relative order $\varepsilon|a| \leq CL^{-k}\vartheta_k$ against $Q_1 a$, and the function V^A and its gradient are bounded by constants times the third and the second power of their argument. The artefact b is analytic in a on a polydisc of a radius ρ_b which does not depend on the coupling, in the units of the scale k, and it is bounded there by $C_b(\varepsilon/a_k)(\varepsilon_1 + \rho_b)^2$ per point: the proof of Proposition S.13(a) uses of the configuration only the bounds (11.1) and expansions which are identities between analytic functions, so that it holds at the complex configurations of this polydisc. Cauchy's estimate gives $|R_3[b](a)| \leq 2C_b L^{-k}(\varepsilon_1+\rho_b)^2(|a|/\rho_b)^3$ per point for $|a| \leq \frac{1}{2}\rho_b$. Finally $V^B(a^B) - V^A(a^A)$ is the sum of (14.4) at $a = a^B$, less $V^A(Q_1 a^B)$, and of $V^A(Q_1 a^B) - V^A(a^A)$, which is at most a constant times $|a|^2|Q_1 a^B - a^A|$ and is bounded by Proposition S.12(h). Since $|a^X| \leq C\vartheta_k$ in these units, all of this is at most a constant times $(L^{-k} + \delta_w(Y))g_k^{-2}\vartheta_k^3$ per point.

(5) *The configurations of the frozen terms.* The term of the generation j is read by run B at its own composite, the minimiser of run B at the data $e^{i\varphi}V^{(k)}$ of the scale k, and $\mathcal{G}_j$ reads it at the lift of run A's composite. The two are configurations of run B whose distance is bounded by Proposition S.13(b) and by the tail of the lift, relative to the radius of the space, and the difference of the values of the term is the variation group of Proposition S.15(c) at the fluctuated data. At complex fluctuations the two values are bounded separately by the age bound, on the disc of the fluctuation, and their difference is bounded on the real diameter of the disc, where the data are real and regular; the two-constants theorem, with the harmonic measure of the circle at the points of the disc of a quarter of the radius at most $\frac{1}{3}$, gives on that smaller disc the bound of Proposition S.15(c) to the power $\frac{2}{3}$ times the age bound to the power $\frac{1}{3}$. The quotient of the distance and the radius is the same on the spaces of the tube and on those of the fixed radii, the data parameter and the radius being proportional.

In (2)–(4) the common bounds are at most a constant times $g_k^{-2}\vartheta_k^3 \leq g_k p_0(\hat{g}_k)^3$ per point, which tends to zero with the coupling, and no negative power of the coupling remains: the classical action enters the comparison only through its parts of the third and higher orders in the fluctuation field, and not through the difference of its values at the two minimisers, which is of the order $g_k^{-2}L^{-k}$ times the square of the regularity of the background and is not bounded uniformly in the coupling on a space of fixed radius. The relative differences are at most a constant times $L^{-\gamma_H(k-1)} + L^{-(1-\varepsilon_0)k}$ in (1)–(4), because $\delta_w(Y) \leq C_w L^{-\gamma_w k}$ and $\gamma_H < \gamma_w$, and the bound of (5) is geometric in k at the rate of Paper, (11.22). By (a) the sources are therefore at most $C_e\tilde{\theta}^{k+1}$ per point of the scale $k+1$ for $k \geq k_{\mathrm{src}} := \max(k_0(\varepsilon_0), m_{\mathrm{rc}}, k_r + 1)$, with k_r the number of Paper, Section 11.5, up to which the interpolation identity is not used, and with a constant C_e which does not depend on k, K, the coupling, or the torus. At the steps $k < k_{\mathrm{src}}$, whose number does not depend on K, the difference of two outputs of the class is at most 2 in the class norm.

(b), the two norms. The argument so far reads the regular families of the two runs through the properties of Proposition S.15(a), the admissibility of the interactions, and the bounds of the sources, and it applies in the two settings of the statement. The first is that of the families $\mathbf{E}^A(s)$ and $\mathbf{E}^B(s)$ of Lemma 3.5 on the spaces of the tube, uniformly in $|s| \leq \varrho_k$, and we go through the places at which a family enters the argument. First, the interactions: the interaction of the step of the family consists of the terms of [9, (2.12)] outside its curly bracket, which do not contain s, of the fluctuation parts of the regular terms $\mathbf{E}^{(j)}(s) - \mathbf{E}^{(j)}(s)(1) - \beta_j(s)A$, $j \leq k$, and of s times those of the **R**-terms which enter the regular part. By Lemma 3.5(b) the regular terms satisfy the bound of the class for $|s| \leq \varrho_k$, by the proof of Lemma 3.4 the multiple s of the **R**-terms keeps the interaction within the bound with $2E_0$, and the counterterms are bounded uniformly, $|\beta_j(s) - \beta_j(0)| \leq C_\Lambda(1) r^{-2} C_\rho g_{j-1}^{\kappa_0 - 4} E_0 \leq C$ by Lemma 3.1 and (3.14), as in the proof of Lemma 3.5(c). So the interpolated interactions $\mathbf{P}_t$ are convex combinations of two interactions with the bounds of the class. Secondly, the interpolation identity of Paper, Lemma 11.7: its proof uses of the interactions these bounds only, the integral Z_t does not vanish because it is the exponential of the convergent cluster series, and this series converges for complex interactions, the expansion [11, (2.14), p. 15] being bounded through the moduli of its factors, as in the proof of Lemma 3.4; the backgrounds of the class are complex in any case. Thirdly, the marked-cluster bound (a), which uses of the interaction its admissibility and is linear in the insertion. Fourthly, the frozen differences: Proposition S.15(e) bounds them, with the moment defects, uniformly in s. Finally the couplings, the cutoffs, the minimisers, and the Gaussian forms are those of the two runs and do not depend on s, so that δ_k and the sources e_{k+1} are unchanged. The one change is in the **R**-terms which enter the regular part: their differences enter the marked activity multiplied by s, hence with $|s| S_k \leq \varrho_k S_k$ in the place of S_k. The **R**-terms of the recent generations, which [10, (3.26), p. 271] carries by an interpolation of its own, contribute to the **R**-terms and to the boundary terms of the next step and not to its regular term, and they are not multiplied by s. This gives Paper, (11.19), for $d^{\sharp}_{k+1}$.

The second setting is that of the families at $s = 0$ on the spaces of the fixed radii. No **R**-term enters. The interactions satisfy the bounds of [11] on these spaces (Lemmata 3.2 and 17.2), and the expansion of the step is that of [11, Sect. 2], in which the marked factor is treated as in (a). The frozen differences are bounded by Proposition S.15(e). The step depends on the coupling analytically, on a disc of the radius ε^*/p (Lemma 3.2), and $g_k p = \vartheta_k$ is small against ε^*; since the two values of the coupling differ by at most $2g_k^3|\delta_k|$, by Paper, Lemma 9.1(iii), Cauchy's estimate on this disc shows that the two outputs differ relatively by at most a constant times $g_k^2|\delta_k|\vartheta_k/\varepsilon^* \leq g_k^2|\delta_k|$, and the difference of the two cutoffs is bounded by Lemma 3.3. The sources are those of the preceding paragraph, which were bounded at the backgrounds of the complex domain of [5, (3.37), p. 396]; this domain does not shrink with the coupling, and it contains the configurations of the fixed radii and their lifts when α_0 and α_1 are small. The data of the minimisers in (5) have a parameter of the order of the fixed radii, which is below the number that [7, Theorem 1, p. 279] and Proposition S.13 require, again for α_0 and α_1 small. The ratio μ_k of the cutoff to the radius is smaller than on the spaces of the tube, the radius being larger. This gives Paper, (11.20), for $d^{\bullet}_{k+1}$. We write ε^*_k and e_{k+1} for the larger of the two coefficients and of the two sources.

(b), the coefficients. The coefficient β_{k+1} is the marginal part of the output, and by (13.4) the quadratic coefficient Π^Δ of [10, (3.61)–(3.64), pp. 282–283] satisfies $|\Pi^\Delta_{k+1}| \leq$

$C^{\bullet}_{\beta} d^{\bullet}_{k+1} + C^{\sharp}_{\beta} d^{\sharp}_{k+1} \leq C_{\beta} u_{k+1}$. Cauchy's estimate on the evaluation sets of the tube would give $C\hat{g}^{-2}_{k+1} d_{k+1}$ instead, and with this bound the recursion does not close: the coupling differences would enter the step i with the coefficient $C\varepsilon^* g_i^2 \sum_{l>i} g_l^{-2} u_l$, whose kernel g_i^2/g_l^2 is not summable over l, and the bounds would compound by a fixed factor at every step. The difference of the coefficients of the two runs is not Π^{Δ} itself: by the definition of Δ_{k+1}, $\Pi^{\Delta}_{k+1} = \Pi^{\mathcal{G}}_{k+1} - \beta^A_{k+1}$, so that

$$\Delta\beta_{k+1} := \beta^B_{k+2} - \beta^A_{k+1} = \Pi^{\Delta}_{k+1} - (\Pi^{\mathcal{G}}_{k+1} - \beta^B_{k+2}) ,$$

and the bracket, the isotropic part of the moment defect, is at most $C_{\Pi} E_0 e^{-\delta_c M_1 L^{k+1}}$ by Proposition S.15(d). From $\delta_K = 0$, $\delta_j = \sum_{i=j+1}^{K} \Delta\beta_i$, and $\sum_{i>j} e^{-\delta_c M_1 L^i} \leq 2e^{-\delta_c M_1 L^{j+1}}$ for $\delta_c M_1 L \geq 1$. In the recursion the coupling difference enters with the factor $C_g \varepsilon^*_k g_k^2$, and $e^{-\delta_c M_1 L^{k+1}} \leq C\tilde{\theta}^{k+1}$, so that the new terms are sources of the form of e_{k+1}, with the constant C_e of (e) enlarged by $2C_g C_{\Pi} E_0 \sup_k \tilde{\theta}^{-k} e^{-\delta_c M_1 L^k}$.

(c). A term of the generation $k+1$ of the **R**-sector has two sources. The renormalisation of the large-field components at the step $k+1$ produces the logarithms of the quotients [12, (0.5)] with the bound [10, (2.31)], whose two-run differences are at most $c_{k+1}\mathcal{D}_{[k+1]} + c^B_{k+1}\mathcal{D}^*_{k+1} + c^V_{k+1}\mathcal{D}^V_{k+1} + \nu_{k+1}$ by (d) and Proposition S.14, with $c_j := C_R(100MR_j)^4 g_j^{\kappa_0}$, c^B_j, c^V_j of the same form, and ν_j the bounds of the fine scales and of the old histories, $\sum_{j\leq K} \nu_j \leq C_{\nu} w_K$. The fluctuation integral carries the frozen **R**-terms of the recent generations $j > k_1$ through the step by the s-interpolation [10, p. 271, ll. 1–30], k_1 the largest integer with $L^{-(k-k_1)} MR_{k_1} \leq L$ [10, p. 270, ll. 49–50], and its own **R**-output has the difference $2\varepsilon^*_k \sum_{j=k-N'_k+1}^{k} d^{\mathbf{R}}_j + c'_{k+1}(D_k + S_k)$, $c'_{k+1} := C_{\mathrm{KP}}(2R_1/E_0) N'_k g_k^{\kappa_0} \varepsilon^*_k$, where the memory satisfies $N'_k \leq N'(\hat{g}_k) = \log_L(MR(\hat{g}_k)) + 3$ by [10, (2.9), p. 256]. The window lemma, that $u_{i+1} \leq s_{i+1} + \eta_i \sum_{j=i-N'_i+1}^{i} u_j$ with $u, \eta \geq 0$ and $\sum_{i:\, i-N'_i<j\leq i} \eta_i \leq \frac{1}{2}$ imply $\sum_{j\leq k} u_j \leq 2\sum_{j\leq k} s_j$, which follows by summing over i and exchanging the sums, applies with $\eta_i = 2\varepsilon^*_i$ under the second condition of Paper, (C.2), because the i with $i - N'_i < j \leq i$ are at most $N'(g_j/(1+\beta_0))$ in number with $g_i \leq 2g_j$ by [10, (2.6)]. The terms $S_{m-1} \leq S_{j-1}$ inside the maxima feed S back into itself with the coefficients $c^S_j := 2(c_j + c^B_j + c^V_j + c'_j)$, and induction gives $S_k \leq B_k \exp(\sum_{j\leq k} c^S_j)$, which is Paper, (11.21), with $2c'_j D_{j-1} \leq 2c_j D_{j-1}$ for γ small. C_{Σ} is finite uniformly in K because $(100MR_j)^4$ is a power of $\log g_j^{-2}$, N'_j and $N_0^+(j)$ are double logarithms, and $g_j^{\kappa_0} \leq (\bar{g}^{-2} + b(K-j) - C)^{-\kappa_0/2}$ by Paper, (4.1).

(d). The two runs' integrands of the operation of a component differ, per point of its history at the scale m, through the frozen small-field exponents, bounded by $D_m \leq D^{\flat}_m$ since $L^{-\beta'} \leq q$, which holds because $(\beta' - 1/3)\log L \geq 1/12$ for $\beta' \geq 19/40$ and $L \geq 2$; through the carried **R**-differences, bounded by S_{m-1}; through the coupling, which enters the action as $C\varepsilon^2_m|\delta_m|$, the function [12, (1.6), p. 178] as $10|\delta_m|$ per variable, and the normalisation constants, the radial shells, and the denominators as at most $C|\delta_m|$ per point; through the depth terms e_m; and through the threshold shells and the large-field integrands, bounded by h_n per point of a history born at the scale n by Paper, Theorem 10.8, and carried by the term $m = n$ of the maximum. The weight $e^{-C_P(j-m)/2}$ is the memory of the large-field weight of Paper, Proposition 8.4.

(e). Let $u_i := \max(d^{\bullet}_i, d^{\sharp}_i)$, so that $d_i \leq u_i$. In (c) and (d) the differences of the regular terms are those of the two runs, bounded by $d_i \leq u_i$. Substituting (c) and (14.2) into the

two inequalities of (b) gives $u \le \mathcal{T}u + s$, with

$$
\begin{aligned}
(\mathcal{T}u)_{i+1} := \varepsilon^* \Big[& D_i[u] + C_S \sum_{j\le i} c_j^\sharp (\mathcal{D}'_{[j]}[u] + D_{j-1}[u]) + C_S \sum_{j\le i} c_j^{B\sharp} \mathcal{D}_j^{*\prime}[u] + C_S \sum_{j\le i} c_j^{V\sharp} \mathcal{D}_j^{V\prime}[u] \Big] \\
& + C_g \varepsilon^* g_i^2 C_\beta \sum_{l>i} u_l \ ,
\end{aligned}
\tag{14.5}
$$

positive, monotone, and positively homogeneous, and a source built from the e_i, the h_i, and $C_\nu w_K$, with $s_1 := 2$. Here the factor ϱ_i with which the **R**-sector enters the inequality for $d^\sharp$ has been paid by the **R**-terms themselves: for $j \le i$, $\varrho_i \le 1 + \hat{g}_j^{-2} \le \frac{5}{2} g_j^{-2}$, as in the proof of Lemma 3.5(b), so that $\varrho_i c_j \le c_j^\sharp := \frac{5}{2} C_R (100 M R_j)^4 g_j^{\kappa_0 - 2}$, and likewise for $c_j^{B\sharp}$ and $c_j^{V\sharp}$. In the source, the bounds ν_j of the fine scales are multiplied by $\varrho_i \le C_\varrho (K+1)$, $C_\varrho := 1 + \bar{g}^{-2} + b' + C_W$, which is the reason for the factor $K+1$ in the weight w_K of Paper, (11.23); C_ν contains C_ϱ. The estimates which follow are made with the numbers $c_j^\sharp$, $c_j^{B\sharp}$, $c_j^{V\sharp}$ in the place of c_j, c_j^B, c_j^V; they have the properties of the latter with $\kappa_0 - 2$ in the place of κ_0, and the conditions $\kappa_0 > 2$ and $\kappa_0 > 4$ of the sums below become $\kappa_0 > 4$ and $\kappa_0 > 6$. In the weighted norm: $\sum_{j \le i} L^{-\beta'(i-j)} m_j \le (1 - L^{-\beta'}/\tilde{\theta})^{-1} m_i$ since $\tilde{\theta} > L^{-\beta'}$; $\sum_{j\le i} q^{i-j} m_j \le (1 - L^{-1/3}/\tilde{\theta})^{-1} m_i$ since $q < L^{-1/3} < \tilde{\theta}$; $g_i^2 \sum_{l>i} m_l \le (2(1-\tilde{\theta})^{-2} + b^{-1}) m_{i+1}$, because $\sum_{l>i} l\tilde{\theta}^l \le 2(1-\tilde{\theta})^{-2}(i+1)\tilde{\theta}^{i+1}$ and $g_i^2 (K - i) \le b^{-1}$; $\sum_{j\le i} c_j m_j \le C_m m_{i+1}$, $C_m := C_\Sigma + C''_\mu$, the fine scales $j \le K/2$ paying with $c_j \le C_R \ell_K (\bar{g}^{-2} + bK/2 - C)^{-\kappa_0/2}$ and the coarse ones with $\tilde{\theta}^j$, which together give at most $C''_\mu \mu_K \le C''_\mu w_K$; the coupling part of $\mathcal{D}'$ gives C'_m in the same way, using $\sum_j c_j (K - j + 1) < \infty$ for $\kappa_0 > 4$; and the maxima are bounded by their values at the scale j, since $e^{-C_P/2} < \frac{1}{2} \le \tilde{\theta}$ and, for the maxima over $k_s/8 - 1 \le l \le j$, $l\tilde{\theta}^l \le C_\mu (k_s + 1) \tilde{\theta}^{k_s/8} \le C_\mu w_K$ with $C_\mu := \tilde{\theta}^{-1} \max(1, (e \log(1/\tilde{\theta}))^{-1})$, and $C_* := (1 - L^{-\beta'}/\tilde{\theta})^{-1} + 2C_\beta (2(1-\tilde{\theta})^{-2} + b^{-1} + 1) + 1$. Since $m_i \le \tilde{\theta}^{-1} m_{i+1}$, every component of $\mathcal{T}u$ is bounded by $\varepsilon^* C_{\mathcal{T}} \|u\|_m m_{i+1}$, and $C_{\mathcal{T}} \ge \tilde{\theta}^{-1}(1 - L^{-\beta'}/\tilde{\theta})^{-1} = (\tilde{\theta} - L^{-\beta'})^{-1} > (1 - L^{-\beta'})^{-1}$. Under the first condition of Paper, (C.2), $\|\mathcal{T}\|_m \le \frac{1}{4}$; the system is finite, so $\|u\|_m < \infty$ and $\|u\|_m \le \frac{4}{3} \|s\|_m \le 2C_s$, hence $d_i \le u_i \le 2C_s m_i$, with $C_s := \max(2/\tilde{\theta}, C_e + \varepsilon^* C_S ((C_e + C_h) C_m + C_e (C^B_{R,1}/C_R) C^N_\Sigma + C^B_V (C_e + C_h)(C^V_R/C_R) C_\Sigma + C_\nu))$. The depth sources were bounded in (b), in both settings, at the steps $k \ge k_{\text{src}}$, and at the earlier steps, whose number k_{src} does not depend on K, the outputs of both runs satisfy the bounds of their classes, so that the class norms of the differences are at most 2 there; since $\tilde{\theta} > L^{-1}$, $2 \le 2L^{k_{\text{src}}} \tilde{\theta}^{k+1}$ for $k + 1 \le k_{\text{src}}$, so $e_i \le C_e \tilde{\theta}^i$ at every i with $C_e \ge 2L^{k_{\text{src}}}$. The classical depth of Paper, Proposition 11.3, is not among these sources: it bounds the difference of the two actions at the minimisers of the stopping scale, where it enters the exponent with the bounded factor $(g^A_{K'})^{-2}$, and it is not used at the earlier scales, where this factor is not bounded. The bounds on S_i and $|\delta_i|$ follow from (c) and (b), and (11.25) with $C_a := 2C_s (1 - L^{-\beta'}/\tilde{\theta})^{-1} + C''_S + C_e + 4C_s C_\beta (1-\tilde{\theta})^{-2} C \sup_{g \le (1+\beta_0)\gamma} g^2 p_0(g)^2$.

(f). The deviation insertions are produced by the same fluctuation integral with the loop's words as further marked activities, with the same localisation and analyticity, so (a) and (b) apply with their class norms. A group born at the step $i+1$ has, besides its marked polymers, old vertices, the untouched old insertions merged into it and the occurrences of the reference gas of the step i from the expansion of Q_i, which enter as numbers outside the fluctuation integral and so carry no factor ε^*. Their differences are counted by the telescoped count of the two families, which have the same polymers, compatibilities, and Ursell coefficients, with one marked old vertex per term, and relative to the bound of the insertions this gives $\varepsilon_i^Q D_i^{\text{ins}}$ with $\varepsilon^Q(g) := 2L^4 C (M/\nu_e^2) K_0 K_\Delta (C'')^2 C'_S \rho(g) \mathrm{pl}(g)/\varepsilon_*$, $\rho(g) := C_\rho g^2 (1 + A_1 (\log g^{-2})^{p_0})^4$, and

$\mathrm{pl}(g) := 1 + (\log g^{-2})^{(q_1-q_0)_+}$, the constants of the inductive class of Paper, Section 6. The other sources are at most $C_{\mathrm{ins}} m_{i+1}$ by (e), and $(\varepsilon^* + \varepsilon_i^Q) D_i^{\mathrm{ins}} \le (\varepsilon^* + \varepsilon_i^Q) C_{\mathcal{T}} \|d^{\mathrm{ins}}\|_m m_{i+1} \le \frac{1}{2}\|d^{\mathrm{ins}}\|_m m_{i+1}$ under the first condition of (C.2) and (11.26), so $\|d^{\mathrm{ins}}\|_m \le 2C^*_{\mathrm{ins}}$ with $C^*_{\mathrm{ins}} := \max(C_{\mathrm{ins}}, 2/\tilde{\theta})$.

(g), (h): the relative differences of the constituents. The steps at which wraps and per-step constants are compared are the young steps $i \ge K'+1-\max(m_K^E, m_K^w)$, and $\max(m_K^E, m_K^w) \le \lfloor K'/4 \rfloor$ for $K \ge K_1$. The constituents of such a step are Bałaban's activities of the generations up to i and the marked activities of the loops, the terms $V(Y)$ which the $\mathbf{R}$-operation re-sums, the operations $\mathbf{T}_i'(X)$ of the components of the large-field region, and the polymer activities of the fluctuation integral. Relative to the bounds which the two runs have in common, their differences between the two runs are at most the following numbers. For the activities it is $r_i^E := C_r^E(i\tilde{\theta}^i + w_K)$ with $C_r^E := \max(2C_s(1-L^{-\beta'}/\tilde{\theta})^{-1} + C_S'', 4C_s, 4C^*_{\mathrm{ins}}(1-L^{-\beta'}/\tilde{\theta})^{-1})$, by Paper, (11.24), and (f), and at the young steps $i\tilde{\theta}^i \le (K'+1)\tilde{\theta}^{3K'/4}$. For the terms $V(Y)$ it is $r_K^V := 4C_h\tilde{\theta}^{K'/8} + C_V 2^{-K'/16} + 2\sup_i \mathcal{D}_i^{V\prime}$, the supremum over the young steps, by Proposition S.14(c), (d), and (d) above, with C_V the constant of the deep terms of Paper, Section 11. For the operation of a component born at a scale above $K' - \lfloor K'/2 \rfloor$ it is $3s_K$ in total variation, with s_K of Paper, (11.30): by Paper, Theorem 10.8, the difference is at most $3s_K\omega^s(X)$ times the larger of the two runs' bounds [13, (1.79)] of the total mass, and the factor $e^{-C_P P_i(X)}$ of (14.7) below pays $\omega^s(X) \le e^{C_P P_i(X)}$. For a polymer activity on X it is $3C_M s_K' n(X) e^{C_M s_K' n(X)}$, where $n(X) \le N^{\mathrm{pt}}(1 + d_i(X))$ is the number of its points and $N^{\mathrm{pt}} := 2\cdot 3^d M^d$, because a tour of a shortest tree of X has the length $2d_i(X)$ and each of its pieces of unit length meets at most 3^d cubes. We put

$$\begin{aligned} \rho_K^{\mathrm{LF}} &:= \max\Big(C_r^E[(K'+1)\tilde{\theta}^{3K'/4} + w_K],\ r_K^V,\ 3(1+eC_M N^{\mathrm{pt}})s_K\Big)\ , \\ \rho_K^E &:= \rho_K^{\mathrm{LF}} + C_\rho^w x_K^w\ , \end{aligned} \tag{14.6}$$

where x_K^w is the number of (g), which is defined through ρ_K^{LF} alone, and $C_\rho^w := 2c_o C_{\hat{\varepsilon}}$, with the constants of (i), bounds the dependence of the marked activities on the wraps of their step. By Paper, (11.24), (11.30), and (11.23), each of the three entries of ρ_K^{LF} is at most a constant times $(K+1)w_K + \ell_K L^{-\gamma_H K'/4}$. Hence ρ_K^{LF} and ρ_K^E tend to zero, and their products with any power of $\log(K+2)$ are summable in K, because $(K+1)w_K$ is at most a geometric sequence plus a constant times $K^{4-\kappa_0/2}\ell_K$ and $\kappa_0 \ge 13$. For $|\varsigma| \le 1/(2\rho_K^E)$ the interpolated constituents $x^\varsigma = x^A + \varsigma(x^B - x^A)$ are at most twice their bounds; for a polymer activity, with $\sigma := C_M N^{\mathrm{pt}} s_K' \le 1/(12e)$ and $1 + d \le e^d$, $|\varsigma|\, 3C_M s_K' n(X) e^{C_M s_K' n(X)} \le \frac{1}{2}e^{(1+\sigma)d_i(X)} \le \frac{1}{2}e^{\kappa_I d_i(X)}$ with $\kappa_I := \kappa/6 \ge 2$, so that it is within $2e^{\kappa_I d_i(X)}$ times its bound.

(g), (h): the old components. A component X of the large-field region of a young step i is *old* if it was born at a scale $n(X) \le K' - \lfloor K'/2 \rfloor$; the old components are not compared. The proof of Paper, (8.33), rests on Proposition S.11 and on Paper, (8.31), which hold at every step, and it gives, for every component of the step i,

$$\sup \mathbf{T}_i'(X)1 \le e^{-C_P P_i(X)} b(X)\ , \qquad b(X) := \exp(-2c_{\mathrm{LF}} p_0(g_a))\ , \tag{14.7}$$

with $P_i(X)$ the history volume of X up to the step i. The components which the step renormalises have at most 100^4 cubes, and $b(X)$ is the bound of their factors under which the expansion of the $\mathbf{R}$-part converges (Proposition S.10). An old component meets the large-field region at every scale from $n(X)$ to i, so that $P_i(X) \ge i - n(X) + 1 \ge K'/4$, and its mass is at most $e^{-C_P K'/4} b(X)$. We multiply the operation of every old component by a number $\mathfrak{z}$ with $|\mathfrak{z}| \le e^{C_P K'/4}$. Each of them still has the bound $b(X)$, so that at every such $\mathfrak{z}$ every activity of the gas satisfies the bound under which the expansion converges at $\mathfrak{z} = 1$. This is a bound on each activity of the complex family, component by component, and not on the sum of the weights of the terms; it is what makes the quantities of the step analytic in $\mathfrak{z}$ on the disc, with the bounds they have at $\mathfrak{z} = 1$. At $\mathfrak{z} = 0$ the step has no old components, and Cauchy's estimate on the disc bounds the difference between $\mathfrak{z} = 1$ and $\mathfrak{z} = 0$, in each run, by $2e^{-C_P K'/4}$ times the bound.

(g). The wrapping insertions of age m, born at the step $K' + 1 - m$, have total weighted mass at most $A_w L^m e^{-a_w L^m}$ in each run, because their mass decays in the smallest period of the torus at their scale, $N^{(j)}_{\text{min}} \geq c(T/u)L^{K'-j+j_q}$, and those of age at least m^w_K, the least m with $a_w L^m \geq 3\log(K+2)$, have total mass at most $\tau^w_K := 2A_w \max(L, 3L\log(K+2)/a_w)(K+2)^{-3}$, since $\sum_{m\geq m_0} L^m e^{-a_w L^m} \leq L^{m_0} e^{-a_w L^{m_0}}(1 + 1/(a_w(L-1)L^{m_0-1}))$. The younger ones are compared at their birth by the interpolation of (h), on the disc $|\varsigma| \leq 1/(2\rho^{\mathsf{LF}}_K)$ with ρ^{LF}_K of (14.6), which gives the relative difference $8\rho^{\mathsf{LF}}_K$ at the enlarged amplitudes, and are then carried, by evaluation at the new background and merging, with a difference which grows by at most $2L$ per step, since a wrap grows by at most L per step and the difference of a product replaces one factor at a time; over at most m^w_K steps this gives $x^w_K := 2\tau^w_K + (m^w_K + 1)(2L)^{m^w_K}[\bar{r}_K A_w S_w + 12\rho^{\mathsf{LF}}_K A^{w,2} S^{(2)}_w + 4e^{-C_P K'/4} A_w S_w]$, summable in K because $(2L)^{m^w_K}$ is a power of $\log K$ and $\bar{r}_K$, ρ^{LF}_K decay faster than any power of $\log K$ after summation. The weights $Q_{K'}(\mathfrak{B})$ are cluster activities of the reference gas of the stopping step, numbers given the labels, and the marked-cluster bound of (a) at the reference point gives $\sum_{\Gamma\ni W}|\varphi^B(\Gamma) - \varphi^A(\Gamma)| \leq 2C_{\text{pt}} C_X C_{\text{KP}} \bar{c}\,\theta(W)$, $\bar{c}$ the relative difference of the activities, which is x^Q_K after the weights of the families are summed with $\theta(W) \leq \nu\sigma(W)/16$.

(h). The per-step terms E_j of E_{const} of age m, among them the cross constants $E^{\mathbf{R}\times}_j$ of Proposition S.10(iv)(c), are at most $A_E L^{m/2} e^{-a_E L^{m/2}}$ by Paper, Section 11.6, "The loop constant"; those of age at least m^E_K are bounded by τ^E_K in each run as in (g). For the younger ones, every constituent x of the step is replaced by $x^\varsigma := x^A + \varsigma(x^B - x^A)$, after the operations of the old components have been multiplied by $\mathfrak{z}$ as above; for $|\varsigma| \leq 1/(2\rho^E_K)$, with ρ^E_K of (14.6), each constituent is at most twice its bound and the polymer activities are within $2e^{\kappa_I d(X)}$ times theirs, so the cluster expansion of the step converges at the enlarged amplitudes and the lowered decay $\kappa - \kappa_I$, and the per-step term is analytic in ς with the bound $A^{E,2} L^{m/2} e^{-a^{E,2} L^{m/2}}$. Cauchy's estimate on the disc of radius $1/(2\rho^E_K) \geq 2$ gives Paper, (11.29); Cauchy's estimate in $\mathfrak{z}$ on $|\mathfrak{z}| \leq e^{C_P K'/4}$ gives the cost $2e^{-C_P K'/4} A_E L^{m/2} e^{-a_E L^{m/2}}$ of the old components; and $\sum_{m\geq 1} \Lambda^m e^{-a\Lambda^m} \leq e^{-a}(1 + 6/a)$ for $\Lambda \geq 2$ sums the ages. Each run's constant is at most $B_E(\mathcal{C})$, which gives the second entry of the minimum.

(i). An insertion of the cross species carries no deviation bound and is not contracted (Paper, Section 6.5), so that the recursion of (f) does not apply to it. We compare it together with the wraps, because the two are coupled: a group with a member which wraps is a wrap, so that a difference of the cross species can pass into a wrap, and the wraps inside the groups of the reference gas change the cluster occurrences which every new group merges. We give the argument on the trivial term; k stands for K'.

(i), the ages. The insertions of the cross species born at the step $k - m$ have, at every later step and in each run, total mass at most $2^{m+1}\mathsf{A}^\times_{k-m}$, by Paper, (5.11). By the proof of Paper, Lemma 8.5, with $c := \nu_\times d_\times/M$ and $y := \kappa d_\times/(12MR_k)$, both at least 1, this is at most $A'_\times[(2L)^m e^{-cL^m} + (2L)^m e^{-y(L/2)^{m-1}}]$ for $m \geq 1$, with $A'_\times := 2P_k(\mathcal{C})\max(C_{\text{cr}}\bar{\mathfrak{a}}, C^{\mathbf{R}}_{\text{cr}})$. The wraps of age m have total weighted mass at most $t_m := A_w L^m e^{-a_w L^m}$ in each run, by (g). Let $m^\times_K$ be the least $m \geq 2$ with $\min(c, y)(L/2)^{m-1} \geq 4\log(K+2) + 2m\log(2L)$ and $a_w L^m \geq 4\log(K+2)$; it grows like $\log\log K$, so that $L^{m^\times_K}$ is at most a power of $\log(K+2)$. The first condition holds for every $m \geq m^\times_K$ once it holds at $m^\times_K$, and then each of the two terms of the cross species is at most $(2L)^{-m}(K+2)^{-4}$; hence the insertions of the cross species of

age at least $m_K^\times$ have total mass at most $\tau_K^\times := 2A'_\times(K+2)^{-4}$ in each run. For the wraps put $\tau_K^w := \sum_{m \geq m_K^\times} t_m$ and $\tau_K^{w,o} := \sum_{m \geq m_K^\times} 2^m t_m$; the ratio of consecutive terms of the second sum is $2Le^{-a_w L^m(L-1)} \leq \frac{1}{4}$, so that $\tau_K^w \leq \tau_K^{w,o} \leq 2A_w(2L)^{m_K^\times} e^{-a_w L^{m_K^\times}} \leq 2A_w(L^{m_K^\times})^2(K+2)^{-4}$. The insertions of these ages are not compared. By the product form (7.5) for the cross species, and since the words are multilinear in the wraps as in (g), the terms of a run's words which contain one of them have majorant at most $\tau_K^\times + \tau_K^w$ times the whole majorant, and dropping them in both runs changes the normalised dressed loop by at most $2(\tau_K^\times + \tau_K^w)\mathfrak{b}(\mathcal{C})$.

(i), the births. Let $i_0 := k - m_K^\times + 1$ be the first young step. An insertion of the cross species or a wrap born at a step $i \geq i_0$ is a function of the constituents of its step, of the carried insertions which it touches or holds, and of the cluster occurrences of the step $i-1$ which it merges. The production bounds Paper, (6.12), and (8.4) rest on the bounds of the constituents alone, and they hold at the enlarged amplitudes and the lowered decay of the interpolation of (h), with a total $A^{\times,2}(\mathcal{C})$ of the form of Paper, (8.38); for the wraps the corresponding total is $A^{w,2}S_w^{(2)}$ of (g), and we put $S_{\mathrm{cw}}^{(2)} := A^{\times,2}(\mathcal{C}) + A^{w,2}S_w^{(2)}$. With the old components removed as in (h), the two runs' insertions born at the step i, paired by shape and position, differ by at most $8\rho_K^{\mathrm{LF}}$ times their bound at the enlarged amplitudes, and each run's insertions born through an old component are at most $2e^{-C_P k/4}$ times their own bound. At every later step the evaluation at each run's background moves an insertion by at most $\bar{r}_K$ times its bound, as in (g).

(i), the carrying of a difference. The product form (7.5) and Proposition S.10(v) are multilinear in the carried insertions, and the difference of a product replaces one parent at a time, so that the increments of a step are doubled for a difference: a difference of the cross species grows by at most $(1+4\bar{c})(1+2y_i^{\mathbf{R}}) \leq 3(1+y_i^{\mathbf{R}})^2$ per step, and $\prod_i (1+y_i^{\mathbf{R}})^2 \leq 4$; a difference of a wrap grows by at most $2L$ per step, as in (g), with the squares of the factors by which the **R**-operations multiply a wrap (Proposition S.10(v)), whose product over the steps is at most 16 before squaring; and a difference of the cross species which enters a wrap does so with the factor $1+2y_i^{\mathbf{R}} \leq 3 \leq 2L$ and is carried as a wrap's from then on. Let $R_i \geq 1$ be the larger of the two factors of the **R**-operation at the step i, so that $\prod_i R_i \leq 1024$ over the young steps. A young insertion of the cross species which touches an old one, or holds it in a slot, carries the two runs' difference of the old ones, at most $2\tau_K^\times$, with the doubled factor $3(1+y_i^{\mathbf{R}})^2 \leq 12$; a young wrap takes in the old wraps and the old insertions of the cross species with the factor 3 each. With the sources

$$\begin{aligned}\beta_K^\times &:= (8\rho_K^{\mathrm{LF}} + \bar{r}_K)A^{\times,2}(\mathcal{C}) + 4e^{-C_P k/4}A^\times(\mathcal{C}) + 30\tau_K^\times \,, \\ \beta_K^w &:= (8\rho_K^{\mathrm{LF}} + \bar{r}_K)A^{w,2}S_w^{(2)} + 4e^{-C_P k/4}A_w S_w + 6\tau_K^w \,,\end{aligned}$$

and $\beta_K := \beta_K^\times + \beta_K^w$, each step adds at most β_K to the differences e_i of the young insertions of the two species, apart from the occurrences.

(i), the occurrences. A group born at the step i contains, from the expansion of the interactions (Paper, Section 7.1), the factor

$$\sum_m \frac{(-1)^m}{m!} \prod_{j=1}^m \Phi^\circ(\Gamma_j)$$

of the clusters Γ_j of the reference gas of the step $i-1$ which meet its members. The weight of a cluster is the Ursell function times the product of the activities $c_{\mathcal{G}}$ of its groups, which are groups on one loop, since a cross group has the activity 0; and $c_{\mathcal{G}}$ is multilinear in the old insertions of $\mathcal{G}$, the deviation insertions, which (f) compares, and the wraps. Let c^0 be the activities with the wraps of run A and the other constituents of run B, $\delta c := c^B - c^0$, and let

$$\hat{\varepsilon}_i := \sup_b \sum_{\mathcal{G}:\, b\in J_{\mathcal{G}}} |c_{\mathcal{G}}| e^{2a^\circ(\mathcal{G})}$$

be the root sum of Paper, (7.5), at the step i, $\hat{\varepsilon}^{(2)}$ its bound at the enlarged amplitudes, and $\delta\hat{\varepsilon}_i$ the same supremum with $|\delta c_{\mathcal{G}}|$ in place of $|c_{\mathcal{G}}|$. For γ_q small, $\hat{\varepsilon}^{(2)} \le \nu/(5184M)$. Along $c^0 + t\delta c$ with complex t, $|t| \le u_0 := \nu/(5184M\delta\hat{\varepsilon}_{i-1})$, the root sum is at most $\nu/(2592M)$, the condition of Paper, (7.7), so that the clusters which meet a member W sum to at most $(\nu/32)\sigma(W)$, the share of the reserve of W which pays the occurrences in the bound of the new group. The new insertion is linear in the factor of the occurrences, hence analytic in t, and if $\delta\hat{\varepsilon}_{i-1} \le \nu/(10368M)$, so that $u_0 \ge 2$, Cauchy's estimate bounds this part of its difference by $c_o\delta\hat{\varepsilon}_{i-1}$ times its bound at the enlarged amplitudes, $c_o := 10368M/\nu$. Over the births of a step this adds at most $c_o S^{(2)}_{\mathrm{cw}}\delta\hat{\varepsilon}_{i-1}$; the parts of the occurrences which come from the fresh constituents are interpolated with the birth and are contained in $8\rho^{\mathrm{LF}}_K$. For $\delta\hat{\varepsilon}_i$ itself we change first the wraps inside the groups, where the difference of a group through a touched or merged member is at most 3 times the member's and the weight e^{2a° with the shares of a group is at most a constant $C_{\hat{\varepsilon}}$ of ν, L, and M times the weight of the class, and then the occurrences inside the activities, by the same Cauchy estimate with $c_o\delta\hat{\varepsilon}_{i-1}\hat{\varepsilon}^{(2)} \le 2\delta\hat{\varepsilon}_{i-1}$:

$$\delta\hat{\varepsilon}_i \le 3C_{\hat{\varepsilon}}(2\tau^w_K + e_i) + 2\delta\hat{\varepsilon}_{i-1}\ . \tag{14.8}$$

(i), the deviation insertions. The count of (f) takes the occurrences merged into a deviation group with marks which are differences of deviation groups. The wraps in these occurrences are not among them, and by the Cauchy estimate above they add $o_{i+1} \le 2c_o(\delta\hat{\varepsilon}_i + C_{\hat{\varepsilon}}e_i)$ to the right side of (f) at $i+1$. The recursion of (f) is linear, so that $d^{\mathrm{ins}}_i \le d^{\mathrm{ins},0}_i + d^{\mathrm{ins},o}_i$ with $d^{\mathrm{ins},0}$ its solution without this source, $d^{\mathrm{ins},o}_{i+1} := (\varepsilon^* + \varepsilon^Q_i)F_i + o_{i+1}$, and $F_i := \sum_{j\le i} L^{-\beta'(i-j)} d^{\mathrm{ins},o}_j$. Since $(\varepsilon^* + \varepsilon^Q_i)C_{\mathcal{T}} \le \frac{1}{2}$ and $C_{\mathcal{T}} \ge (1 - L^{-\beta'})^{-1}$ by (e), $F_{i+1} = L^{-\beta'}F_i + d^{\mathrm{ins},o}_{i+1} \le \frac{1}{2}(1 + L^{-\beta'})F_i + o_{i+1}$. The births of the step i take the marked activities of the step $i-1$ among their constituents; their part from o is $x_i \le F_{i-1} + d^{\mathrm{ins},o}_i \le F_{i-1} + F_i$, and the interpolation is then made with $\rho^{\mathrm{LF}}_K + x_i$ in place of ρ^{LF}_K, which adds $8S^{(2)}_{\mathrm{cw}}x_i$ to the births.

(i), the joint recursion. Collecting the three parts, with the factor e^{cw} of the normalisation of the masses on the sources, and with $C_1 := 1 + 3C_{\hat{\varepsilon}}$ and $H_i := e_i + \delta\hat{\varepsilon}_i + F_i$,

$$e_i \le 2LR_ie_{i-1} + e^{cw}(c_oS^{(2)}_{\mathrm{cw}}\delta\hat{\varepsilon}_{i-1} + \beta_K + 8S^{(2)}_{\mathrm{cw}}x_i)\ ,$$
$$F_i \le F_{i-1} + 2c_oC_1H_{i-1}\ , \qquad x_i \le 2F_{i-1} + 2c_oC_1H_{i-1}\ ,$$

and with (14.8), $e_i + \delta\hat{\varepsilon}_i \le C_1e_i + 6C_{\hat{\varepsilon}}\tau^w_K + 2\delta\hat{\varepsilon}_{i-1}$. Since $2 \le 2C_1L$ and $R_i \ge 1$, these give

$$H_i \le \varpi R_iH_{i-1} + C_1e^{cw}\beta_K + 6C_{\hat{\varepsilon}}\tau^w_K\ ,$$
$$\varpi := C_1(2L + e^{cw}c_oS^{(2)}_{\mathrm{cw}}) + 1 + 16C_1e^{cw}S^{(2)}_{\mathrm{cw}}(1 + c_oC_1) + 2c_oC_1\ .$$

At the old steps $n < i_0$ the wraps are not compared. Removing them from the activities of either run changes the root sum by $\eta_n \le 3C_{\hat{\varepsilon}}W_n + 2\eta_{n-1}$, with $W_n \le \sum_{j\le n} t_{k-j}$ the mass of the wraps of the run, hence $\eta_n \le 3C_{\hat{\varepsilon}}\sum_{j\le n} 2^{n-j+1}t_{k-j} \le 6C_{\hat{\varepsilon}}\tau^{w,o}_K$; the same count with the wrap groups of the step gives the bound $16C_{\hat{\varepsilon}}\tau^{w,o}_K$ for the difference which the recursions take at $i_0 - 1$, and $o_{n+1} \le 32c_oC_{\hat{\varepsilon}}\tau^{w,o}_K$, $F_n \le 64c_oC_{\hat{\varepsilon}}(1 - L^{-\beta'})^{-1}\tau^{w,o}_K$. So $H_{i_0-1} \le B^H_K := C_{\hat{\varepsilon}}(16 + 64c_o(1 - L^{-\beta'})^{-1})\tau^{w,o}_K$, and over the at most $m^\times_K + 1$ young steps, with $\prod_i R_i \le 1024$ and $\varpi \ge 1$, $H_i \le D^{\mathrm{cw}}_K$ of (14.3). The Cauchy estimates at the step i need

$C_1 H_{i-1} \le \nu/(10368M)$; this holds by induction on i for $K \ge K_Q$, the least index from which $C_1 D^{\mathrm{cw}}_K \le \nu/(10368M)$, which is defined from the formula (14.3) and not from the bound. The number D^{cw}_K is summable in K: $\varpi^{m^{\times}_K+1} = \varpi(L^{m^{\times}_K})^{\log_L \varpi}$ is at most a power of $\log(K+2)$, and ρ^{LF}_K, $\bar{r}_K$, $e^{-C_P k/4}$, $\tau^{\times}_K$, τ^{w}_K, and $\tau^{w,o}_K$ times such a power are summable, as in (g).

(i), the conclusion. The young insertions of the cross species and the young wraps multiply each run's words through the product forms, so that their differences change the normalised dressed loop by at most $2(e^{D^{\mathrm{cw}}_K} - 1)\mathfrak{b}(\mathcal{C})$. For the weights $Q_{K'}(\mathfrak{B})$, Cauchy's estimate per cluster for the fresh constituents and the coefficient of the first order in u of the cluster series at the activities $\bar{c}+u|\delta c|$, which is at most its value at u_0 divided by u_0, give $\sum_{\Gamma \not\perp W} |\Phi^{\circ,B}(\Gamma) - \Phi^{\circ,A}(\Gamma)| \le q_K \sigma(W)$ with $q_K := 81M\hat{\varepsilon}^{(2)}(8\rho^{\mathrm{LF}}_K + 16(1+c_o C_1)D^{\mathrm{cw}}_K + 4e^{-C_P k/4}) + 162MD^{\mathrm{cw}}_K$; since $\sigma e^{-(\lambda-1/16)\nu\sigma} \le 32/(29e\nu)$ for $\lambda \ge \frac{31}{32}$, the weights change the normalised dressed loop by at most $x^{Q}_K \mathfrak{b}(\mathcal{C})$ with $x^{Q}_K := \exp(32 q_K M^{\times}/(29e\nu)) - 1$, $M^{\times}$ the total weighted mass in the exponent of $\mathfrak{b}(\mathcal{C})$, which replaces the x^{Q}_K of (g). The deviation insertions gain at most D^{cw}_K per unit of their class norm, since $F_i \le D^{\mathrm{cw}}_K$. Altogether $x^{\times}_K := 2(\tau^{\times}_K + \tau^{w}_K) + 2(e^{D^{\mathrm{cw}}_K} - 1) + x^{Q}_K + e^{M^{\times} D^{\mathrm{cw}}_K} - 1$, summable in K. On the young matched pairs of terms with large fields the same recursion runs, with the births compared from the first step of the pair on, and its result enters the comparison of the dressings through the numbers $s^{W,\times}_K$ and $s^{W,0,\times}_K$ of Proposition S.14(f). The cross constants $E^{\times}_j$ and $E^{\mathbf{R}\times}_j$ are per-step terms of the loop constant and are covered by (h). A cross group one of whose parents is attached belongs to the species of that parent (Paper, Definition 5.2(ii)) and is compared with it. The counts of the positions of a cross group are those of one loop with the total perimeter: two loops are at distance at least a_k at the scale k (Paper, Lemma 3.14), so that a cube of side less than a_k meets one loop. *q.e.d.*

15 The Gaussian Forms of Two Fine Lattices

At the step whose fluctuation field B lives on $\mathbb{T}_\ell$, $\ell := \varepsilon L^{j-1}$ (run A's step j, run B's step $j+1$), the Gaussian form of [10, (3.15)] is $-\frac{1}{2}\langle B, \Delta B\rangle$ with $\Delta = \Delta^{(j)}(U)$ of [9, (1.4)–(1.5)] on $\{\tilde{Q}B = 0\}$. By [9, p. 267, ll. 38–39], whose quadratic form is "equal to $-1/2\langle B', \Delta^{(k)} B'\rangle$, see the definition (3.156) [13] with the δ-function gauge fixing term replaced by the exponential one",

$$\Delta_U[B] = \langle B, (Q_m G_1 Q_m^*)^{-1} B\rangle - a\ell^{-2}\langle B, B\rangle - 2\langle H_1 h \tilde{C}^{(2)}(B), J\rangle + \mathbf{G}^{(2)}_{M^m U}(B) \,, \qquad (15.1)$$

with $G_1 = (\Delta_1 + DRD^* + a\ell^{-2} Q_m^* Q_m)^{-1}$ and $H_1 = G_1 Q_m^* (Q_m G_1 Q_m^*)^{-1}$ of [5, (3.128)–(3.129)], Q_m the linearised m-fold average, J the current of U, and $\mathbf{G}^{(2)}$ the quadratic part of the exponential axial gauge fixing [9, (2.5), (2.11)]; run A has $m = j-1$ and $U = U^A_j$ on $\mathbb{T}_\varepsilon$, run B $m = j$ and $U = U^B_{j+1}$ on $\mathbb{T}_{\varepsilon/L}$. Both are forms on fields on $\mathbb{T}_\ell$; we write $\delta\Delta_j := \Delta^B - \Delta^A$.

Proposition S.17 (The Gaussian Forms of Two Fine Lattices)**.** *Let* $0 < \varepsilon_0 < 1/125$, *let the constants satisfy the conditions of Paper, Appendix C, let* $\bar{g}$ *be small, and let* γ_H *be the exponent of Paper, (11.18). Then, with* $\delta_2 := \min(\delta_1', \frac{1}{2}\delta_1)$, δ_1 *and* δ_1' *the rates of [5, (3.132)] at real and at complex data,*

$$|\delta\Delta_j(y,y')| \le r_j B_0 \ell^{-2} \ell^{-d} e^{-\delta_2 d(y,y')} \,, \qquad r_j \le C_{\delta\Delta}[L^{-\gamma_H(j-1)} + L^{-(1-\varepsilon_0)j}] \,, \qquad (15.2)$$

with $C_{\delta\Delta}$ *independent of* j, K, ε, *and the torus; the Gaussian normalisations of the step differ relatively by at most* r_j *per variable.*

Proof. The Galerkin identity. At a real background G_1^{-1} is positive definite and Q_m is onto, under the hypotheses of [5, Theorem 3.12, p. 423]. At the complex backgrounds of the class of Proposition S.12(g) positivity has no meaning; there the operators G_1 and $(Q_m G_1 Q_m^*)^{-1}$ exist and are analytic (Lemma 17.10), and the identities below, which are algebraic consequences of $Q_m H_1 = 1$ and of the definitions of the two inverses, hold by analytic continuation, the word minimum being read as the bilinear expression which it equals at real points. At a real background the decomposition $A = H_1 B + A_0$ with $Q_m A_0 = 0$ is orthogonal for G_1^{-1}, so

$$P_U(B) := \langle B, (Q_m G_1 Q_m^*)^{-1} B\rangle - a\ell^{-2}\langle B, B\rangle = \min\{\langle A, (\Delta_1 + DRD^*)A\rangle \mid Q_m A = B\} \ , \tag{15.3}$$

attained at $A = H_1 B$, and, polarised, $\langle B', P_U B\rangle = \langle A', (\Delta_1 + DRD^*)H_1 B\rangle$ for every A' with $Q_m A' = B'$; this uses only $Q_m H_1 B = B$ and the two inverses, so it holds at complex backgrounds as a bilinear identity.

The alignment. Run A's form at its own background U_j^A is compared with its form at the aligned pair $(\tilde U_j, J_j')$, $\tilde U_j := M U_{j+1}^B$ and J_j' the intermediate current, the lift of run B's multiplier through run A's averaging. By Proposition S.13(b), $U_j^A = e^{i\varepsilon A'}\tilde U_j$ with $a_j|A'| + a_j^2|\nabla A'| \leq 4\rho_j$, $\rho_j := \frac{1}{2}C_G(\varepsilon_0)L^{-(1-\varepsilon_0)j}\varepsilon_1$, and the currents differ by $a_j^3|J(U_j^A) - J_j'| \leq x_J(\rho_j + (\varepsilon/a_j)\varepsilon_1)$, through the multipliers of the two constrained problems. The form is analytic in the pair on the class of radii α_1 for the field and α_J for the current, with the kernel bound $(B_0^{\mathrm{cx}} + a + C_{\mathbf{J}} + C_{\mathbf{G}})\ell^{-2}\ell^{-d}e^{-\delta_1' d(y,y')}$ there, so Cauchy's estimate along the straight path gives the difference $[16\rho_j/\alpha_1 + 8x_J(\rho_j + (\varepsilon/a_j)\varepsilon_1)/\alpha_J]$ times that bound.

The aligned comparison. At aligned backgrounds the constraints of run B are those of run A composed with the linearised one-step average, $Q^B = Q^A Q_1$, by the chain rule $d(M^j)(U^B) = d(M^{j-1})(MU^B) \circ dM(U^B)$ and the composition of the double bars [4, (90)–(91)], and the gauge-fixing terms of (15.1) are equal, since both are evaluated at the same block field $M^m U$. The current terms are not equal term by term, and their difference is given by the chain rule: the second-order part of run B's m-fold average is that of run A's $(m-1)$-fold average at the average of the field plus run A's linearised average of the second-order part $C_1^{(2)}$ of the one-step average, $d^2(M^{m-1} \circ M) = d^2 M^{m-1}[dM, dM] + dM^{m-1}[d^2 M]$. Paired with the common multiplier, the first part is run A's current term at the averaged fields, and the second is $2\langle J', C_1^{(2)}[\mathcal{A}'^B, \mathcal{A}^B]\rangle$, which is at most $2C_2'\varepsilon \sup|J'||\mathcal{A}'^B||\mathcal{A}^B|$ by the bound of $C_1^{(2)}$ in the proof of Proposition S.12(d), that is at most $C_2'\alpha_J(\varepsilon/\ell)$ times the bound of the form; this term is added to the defect β below. With $\mathcal{A}^X := H_1^X B$ and the bilinear defect $\beta(\mathcal{A}', \mathcal{A}) := \langle Q_1\mathcal{A}', G_A^{-1}Q_1\mathcal{A}\rangle - \langle \mathcal{A}', G_B^{-1}\mathcal{A}\rangle$, the Galerkin identity gives

$$\langle B', (P_B - P_A)B\rangle = -\beta(\mathcal{A}'^B, \mathcal{A}^B) + \langle E', G_A^{-1}E\rangle \ , \qquad E := Q_1\mathcal{A}^B - \mathcal{A}^A \in \ker Q^A \ , \tag{15.4}$$

and since E is the Green operator of $\ker Q^A$ applied to the source s with $\langle \psi, s\rangle = \beta(\mathcal{L}\psi, \mathcal{A}^B)$ for $\psi \in \ker Q^A$, $\mathcal{L}$ the lift of Proposition S.12(e), both terms are values of β. The defect β is the difference of the two runs' actions, penalties, and gauge-fixing terms on a field and its average, and it is bounded through the oscillation of the regular lifts $H_1 e_y$ over the cells of side 6ε: by [5, Theorem 3.12, (3.133), pp. 422–423] at the Hölder exponent $2\gamma_H$, which is admissible since $2\gamma_H < \frac{1}{3} < 1$, the lifts oscillate over a cell by at most $(\varepsilon/\ell)^{2\gamma_H}$ times their bound, and the lift $\mathcal{L}$ preserves these bounds up to the factor $C_{\mathcal{L}}'(2\gamma_H)$. So $|\langle B, (P_B - P_A)B\rangle| \leq r_\beta \sum_{y,y'}|B(y)||B(y')|B_0\ell^{-2}\ell^d e^{-\delta_1 d(y,y')/4}$ with $r_\beta := 2C_{\mathcal{L}}'(2\gamma_H)^2[C_\beta^{\mathcal{L}}(2\gamma_H) + C_{\mathrm{gf}}^{\mathcal{L}}(2\gamma_H)](\varepsilon/\ell)^{2\gamma_H}$, where $C_\beta^{\mathcal{L}}$ contains the constant $C_2'\alpha_J$ of the current term above, since $\varepsilon/\ell \leq (\varepsilon/\ell)^{2\gamma_H}$.

Interpolation. Since $P_B - P_A$ is symmetric, polarisation at unit vectors at y and y' gives $|(P_B - P_A)(y, y')| \leq 2r_\beta B_0\ell^{-2}\ell^{-d}$, without decay. The same difference is the difference of the two runs' $(Q_m G_1 Q_m^*)^{-1}$, the penalty terms cancelling, so [5, (3.132)] gives $|(P_B - P_A)(y, y')| \leq$

$2B_0\ell^{-2}\ell^{-d}e^{-\delta_1 d(y,y')}$. With $\min(u,v) \le (uv)^{1/2}$ and $\varepsilon/\ell = L^{-(j-1)}$,

$$|(P_B - P_A)(y,y')| \le 3C'_{\mathcal{L}}(2\gamma_H)[C^{\mathcal{L}}_{\beta}(2\gamma_H) + C^{\mathcal{L}}_{\mathrm{gf}}(2\gamma_H)]^{1/2} L^{-\gamma_H(j-1)} B_0 \ell^{-2}\ell^{-d} e^{-\delta_1 d(y,y')/2} \,. \tag{15.5}$$

Adding the alignment, with $\rho_j \le C_G(\varepsilon_0)L^{-(1-\varepsilon_0)j}\varepsilon_1$ and $\varepsilon/a_j = L^{-j}$, gives (15.2) with $C_{\delta\Delta} := C(\varepsilon_0,\gamma_H)\max(1,\varepsilon_1/\alpha_*)$, $\alpha_* := \min(\alpha_1,\alpha_J)$, and $C(\varepsilon_0,\gamma_H) := 3C'_{\mathcal{L}}(2\gamma_H)[C^{\mathcal{L}}_{\beta}(2\gamma_H) + C^{\mathcal{L}}_{\mathrm{gf}}(2\gamma_H)]^{1/2} + [16C_G(\varepsilon_0) + 8x_J(C_G(\varepsilon_0)+1)](B_0^{\mathrm{cx}} + a + C_{\mathbf{J}} + C_{\mathbf{G}})/B_0$. The normalisation of the Gaussian integral is $\det(C^*\Delta C)^{-1/2}$ up to factors common to the runs, and $|\log\det(1+X)| \le 2\|X\|_{\mathrm{tr}}$ for $\|X\| \le \frac{1}{2}$ with X local of size r_j per variable shows that the logarithms of the two normalisations differ by at most $2r_j$ per variable, so that the normalisations differ by a factor between e^{-2r_j} and e^{2r_j} for each variable; this is the sense of the last statement of the proposition. The exponent $\gamma_H = \beta\gamma_w/(1+\beta)$ of the paper is one admissible choice of the Hölder exponent here; it coincides with the exponent with which Paper, Theorem 9.5, carries the defect of the walk factors to the lifts H through the interpolation between the supremum and the Hölder seminorm of exponent β. *q.e.d.*

16 The Flat Values Are Real and Reflection Invariant

The flat values of the dressed loops at the reference data are quotients $N_j[W_\chi(C)](1;\ell_0)/N_j[1](1;\ell_0)$ (Paper, (12.9)), and the paper uses that the map $F \mapsto N_j[F](1;\ell_0)$ is real and invariant under the reflections θ_μ, $\mu \in \{0,\dots,3\}$, which replace the coordinate x_μ by $-x_\mu$ (Paper, Section 1.3). Both are statements about the construction as a whole, and the point is that every operation by which it is built has a real kernel and commutes with each θ_μ.

Proposition S.18 (The Flat Values). *Let $N_j[F]$ be the density obtained from $F\varrho_0$ by the operations of the first j steps on the torus T', and $N_j[F](1;\ell_0)$ the value of its term ℓ_0 at the block field $V_j = 1$. For every bounded measurable function F on the configurations of the finest lattice which is invariant under the gauge transformations, and every $\mu \in \{0,\dots,3\}$, $N_j[\bar{F}](1;\ell_0) = \overline{N_j[F](1;\ell_0)}$ and $N_j[F\circ\theta_\mu](1;\ell_0) = N_j[F](1;\ell_0)$, where $(F\circ\theta_\mu)(U) := F(\theta_\mu U)$ and $(\theta_\mu U)(x,y) := U(\theta_\mu x, \theta_\mu y)$.*

The functions to which the paper applies the proposition, $F = W_\chi(C)$ and $F = 1$, are gauge invariant, and so are $\bar{F}$ and $F\circ\theta_\mu$ with F. The invariance is used because every step fixes a gauge and returns from it: a gauge-invariant function is not changed by these two operations and remains a factor of the integrand at every step, whereas for a function which is not invariant the gauge-fixed kernels would have to be followed through the proof.

Proof. The operations of a step act on the integrand linearly: the integration over the fine variables with the δ-functions of the block averaging [9, (2.17)–(2.18), p. 269], the multiplication by the partition of unity into large- and small-field terms and by the test functions [10, (3.6)–(3.9), pp. 265–266], the change of variables to the fluctuation field with its gauge fixing, and, at the steps of an $\mathbf{R}$-operation, the multiplication by the renormalisation quotients and the integrations of [12, (0.4)–(0.5), p. 176] and [13, (1.70)–(1.72), pp. 377–378]. The density ϱ by which F is multiplied is not changed by F, so that $N_j[F](1;\ell_0) = \int F(U)k_j(U)\,\mathrm{d}U$ with a kernel k_j which is a product of Haar measures, δ-functions, test functions, Gaussian factors of real positive forms, renormalisation quotients, and the density at real configurations. Every one of these factors is real at real configurations: the test functions and characteristic functions by definition; the actions and the Gaussian forms because they are real quadratic or trace functions of unitary configurations; the renormalisation quotients because they are quotients of integrals of nonnegative functions [12, p. 176], [13, p. 380]; and the terms of the effective action because each localised term $\mathbf{E}(X, U)$ is obtained from real functions by Cauchy integrals over circles $|s| = e^{\kappa_1}$ in the decoupling parameters [11, (1.23), p. 7] and by cluster expansions, and a function f with $f(\bar{s}) = \overline{f(s)}$ has real Taylor coefficients at a real centre, so its Cauchy integrals over circles centred on the real axis are real. Hence k_j is real, and $N_j[\bar{F}] = \overline{N_j[F]}$ by the linearity of the integral.

Fix μ and write $\theta := \theta_\mu$. The reflection θ maps every lattice $\mathbb{T}_\eta$ of the construction to itself, since the lattices contain the origin, and maps every block $B(y)$, a cube of odd side L centred at y [9, (0.3)], to the block $B(\theta y)$; the partitions into M-cubes and MR_k-cubes are centred at the origin (Paper, Section 7.1), so θ permutes their cubes, and it maps the trivial term and the field $V_j = 1$ to themselves. It commutes with every operation, and we go through them in the order of a step.

(a) The Wilson action and the Haar measure are invariant under $U \mapsto \theta U$, since θ permutes the plaquettes and $\operatorname{Re}\operatorname{tr}$ is invariant under conjugation and inversion.

(b) The averages are defined through the families of all shortest contours from the centre of a block, in every order of the coordinate runs, so as to be "symmetric with respect to lattice Euclidean transformations" [9, p. 252, ll. 34–40]; θ maps the family of (y, x) onto that of $(\theta y, \theta x)$. The designated bond $b_0(c)$ of a coarse bond c is the fine bond contained in c which joins the blocks of its two end points [9, p. 267, ll. 19–20], so that $\theta b_0(c) = b_0(\theta c)$ up to orientation.

(c) The first gauge fixing is the exponential one built from the averaged contour variables, introduced "to preserve the Euclidean invariance" [10, (1.5)–(1.7), p. 247].

(d) At the steps $k \geq 2$ the fluctuation fields are written first in the axial gauge of [6, (1.19), p. 79], whose contours $\Gamma_{y,x}$ start at the centre y of a block and change the coordinates in a fixed order [2, (1.7), p. 18], and then in the Landau gauge, defined by the orthogonal projection onto $\Delta_{U_0} N(Q'(U_0))$ [6, (1.27), p. 80]. The reflection θ negates one coordinate about the centre of each block and keeps the order in which the coordinates are changed, so that $\theta\Gamma_{y,x} = \Gamma_{\theta y, \theta x}$; it therefore maps the axial conditions, the restricted gauge transformations [6, (1.29), p. 81], whose averages use the same contours, and the Landau condition to those of the image data, and the fluctuation integral [9, (2.17)–(2.18), p. 269] with them.

(e) At the first step the flat value is the quotient of two integrals over the free fluctuation variables of [1, Section 4], built from the blocks and contour families of [9, Section 0], the nonlinear average [9, (0.4)], the designated bonds, the exponential gauge fixing [10, (1.7)], a product of bond cutoffs in a norm invariant under $X \mapsto -X$, the Haar density, which is invariant under inversion, and a Jacobian; the orthogonal involution $B \mapsto \theta B$ of the fluctuation variables, with the sign of B reversed on the bonds whose image is reversed, maps each of these to itself, and the substitution $B \mapsto \theta B$ gives the invariance at $j = 1$.

(f) The test functions, soft or characteristic, are functions of plaquette deviations, of relative fields $|V(b) V_{\square'}(b)^{-1} - 1|$, and of fluctuation fields $|A(b)|$ over the cubes of the partitions [10, (3.2), (3.3), (3.16), pp. 265–268], in norms invariant under conjugation, inversion, and sign.

(g) The $\mathbf{R}$-operation averages its axial gauge-fixing graphs over the 2^d reflections in subsets of coordinates and the $d!$ permutations of coordinates [12, (1.100), p. 201, ll. 34–38], [13, p. 379], an average which is invariant under θ, and its cubes and characteristic functions are defined from the partitions and the fields.

So $k_j \circ \theta = k_j$, and the change of variables $U \mapsto \theta U$, which preserves the product of Haar measures, gives $N_j[F \circ \theta](1; \ell_0) = N_j[F](1; \ell_0)$. *q.e.d.*

17 Lemmata on Bałaban's Papers

The proofs above use a number of statements about Bałaban's own objects which he states without the calculation, states with a count that fails in a degenerate case, or proves for real arguments where the paper needs complex ones. They are collected here, ordered by the paper of Bałaban's that they concern; each is stated in his setting, and each is used in Part I where its label is cited.

17.1 The Small-Field Renormalisation Group

Bałaban bounds the number of cubes of a localisation domain by $3 \cdot 2^3$ times its linear size [11, (1.28), p. 8; p. 9, l. 23; (2.30), p. 18], a bound which fails for a single cube, where $d_k(Y) = 0$ and the number is 1; he excludes this case from the sum [11, (1.26)] "as it follows from our inductive construction" [11, p. 8, l. 8]. The following count holds for every domain.

Lemma 17.1 (The Cubes of a Localisation Domain)**.** *Let* $X \in \mathbf{D}_j$*, a union of a connected finite family of closed cubes of* π_j *of side* M *[9, p. 257, ll. 2–11], let* $N(X) := M^{-4}|X|$ *be the number of its cubes, and let* $d_j(X)$ *be the length, divided by* M*, of a shortest tree graph contained in* X *and meeting all its cubes [9, p. 257, ll. 14–16], measured in any norm* $|\cdot|_p$*,* $p \in [1, \infty]$*. Then*

$$N(X) \le 2^8(2 + d_j(X)) \le 2^9(1 + d_j(X)) \, , \qquad \operatorname{diam} X \le M(2d_j(X) + 4) \, , \tag{17.1}$$

with $\operatorname{diam} X$ *in the Euclidean norm. For* $\kappa \ge 2^9(6\log 2{+}1)$ *and* $\square \in \pi_j$*,* $\sum_{X \in \mathbf{D}_j, X \supset \square} e^{-\kappa d_j(X)} \le 2^{6 \cdot 2^{10}} + 1$*, which is [11, (1.26)] with an absolute constant, and [11, Lemmata 1–3, (2.41)] hold with* (17.1) *in place of Bałaban's count, with larger absolute constants.*

Proof. Let T be a tree graph of Bałaban's class for X, $\ell_p(T)$ the sum of the $|\cdot|_p$-lengths of its arcs, and $\ell := \ell_\infty(T) \le \ell_p(T)$. Doubling every arc gives a connected multigraph with all degrees even, so an Euler tour traverses it, and parametrised by $|\cdot|_\infty$-arclength it is a map $\gamma \colon [0, 2\ell] \to T$ onto T with $|\gamma(s) - \gamma(s')|_\infty \le |s - s'|$. With $n := \lceil \ell/M \rceil$ and $s_i := \min(2iM, 2\ell)$, every $s \in [0, 2\ell]$ lies within M of some s_i, so every point of T lies within $|\cdot|_\infty$-distance M of one of the $n + 1 \le \ell/M + 2$ points $\gamma(s_i)$. The corners of the cubes of π_j form a translate of $M\mathbb{Z}^4$, so a box $\prod_\nu [p_\nu - M, p_\nu + M]$ meets at most $4^4 = 2^8$ cubes, since in each coordinate the corner must lie in an interval of length $3M$. Every cube of X contains a point of T, which lies in one of these boxes, so $N(X) \le 2^8(2 + \ell_p(T)/M)$, and the infimum over T gives the first bound. Two points of X lie in cubes containing points y, y' of T with $|y - y'| \le 2|y - y'|_\infty \le 2\ell_p(T)$, and a cube has Euclidean diameter $2M$ in $d = 4$, which gives the second. For the sum, a family of N cubes connected through common walls and containing $\square$ is visited by a closed walk of $2(N-1)$ steps, each one of 8, so there are at most $8^{2(N-1)}$ of them; with $d_j(X) \ge N/2^9$ for $N \ge 2^{10}$, the sum is at most $\sum_{N < 2^{10}} 2^{6(N-1)} + \sum_{N \ge 1} e^{-N} < 2^{6 \cdot 2^{10}} + 1$. In Bałaban's uses the count enters through $N(Y) \le 2^9 e^{\delta\kappa d_k(Y)}$ for $\delta\kappa \ge 1$, and through factors $e^{cN(Z)} e^{-c' d(Z)}$ with $c' \ge 2^9 c$, each of which (17.1) bounds by $e^{2^9 c}$; these factors join the absolute constants $O(M^q)$ of [11, (1.29)], C_1 of [11, (1.36)], and C_3 of [11, p. 20]. *q.e.d.*

Bałaban's Lemmata 1 and 2 of [11, pp. 9–11] are stated for the real fluctuation field. The paper needs them for complex fields, and for a complex parameter in place of the coupling.

Lemma 17.2 (Bałaban's Lemmata 1–2 at Complex Fields)**.** *Let* α_0*,* α_1*, and* ε_1 *be small, depending on d, L, M, and Bałaban's constants. The function* $\mathbf{V}'_k(Y, \mathbf{U}, \mathbf{J}, B)$ *of [11, Lemma 1, p. 9] is holomorphic in* $(\mathbf{U}, \mathbf{J}, B)$ *on [11, (1.34)] with B complex,* $\mathfrak{g}^c$*-valued,* $|B| < \varepsilon_1 g_k^{-1}$ *on Y, and satisfies [11, (1.36)] there; the function* $\mathbf{V}_k(Y, \mathbf{U}, \mathbf{J}, B)$ *of [11, Lemma 2, p. 11] is holomorphic there and has the decomposition [11, (1.42)] with [11, (1.43)] and [11, (1.36)] for* $\mathbf{V}''_k$*. The same holds with a complex z in place of* g_k *and* z^{-2} *in place of* g_k^{-2} *where* $|z||B| < \varepsilon_1$*, also at* $z = 0$*, with* $|\tilde{D}(z\mathcal{P}B)| \leq 4c_{\tilde{C}}|z\mathcal{P}B|^2$ *and* $|B'| \leq C'_1|z||B|$*.*

Proof. Bałaban constructs the maps of [11, pp. 3–11] for pairs $(\mathbf{U}, \mathbf{J})$ satisfying the conditions (i)–(iii) of [9, p. 262, ll. 16–38], and states that each term depends only on the data on Y and is analytic on [11, (1.34)] [11, p. 7, ll. 39–42; p. 9, ll. 38–43]; this passage is used as printed. Writing $\mathbf{U} = e^{i\xi A'}U$ with U G-valued, holomorphy in $\mathbf{U}$ is holomorphy in A', and we follow the construction through the four places where B enters.

The fluctuation field enters first through $B' = g_k\mathcal{P}B - h\tilde{D}(g_k\mathcal{P}B)$ [11, (1.19), p. 6], [9, (3.2), p. 270], with $\mathcal{P}$ linear and $\tilde{D}$ the solution of $\tilde{D}(Z) = \tilde{C}(Z - h\tilde{D}(Z))$ [9, p. 267, l. 26], $\tilde{C}$ being the part of order at least two of $(1/i)\log[M(e^{iB'}V^{(k)})M(V^{(k)})^{-1}]$. At a coarse bond c this function is a fixed function of B' and of $V^{(k)}$ on the bonds with both ends in $B(c_-) \cup B(c_+)$, analytic near $(0, W)$ for every flat G-valued W [9, p. 253, ll. 14–16]. The configurations within a distance $\mathfrak{t}_0$ of the flat ones form a compact set which contains $V^{(k)}$ once α_0 and α_1 are small, by [4, (53), p. 26; Proposition 6, p. 43], so the function is holomorphic and bounded on a polydisc of fixed radius $r_{\tilde{C}}$ in B', uniformly in the configuration, and Schwarz's lemma gives $|\tilde{C}(\phi)| \leq c_{\tilde{C}}|\phi|^2$ there, with $c_{\tilde{C}}$ and $|h| \leq 2L^{d-1}$ depending on d, L, G, and the average only. On the bonds of c the equation for $\tilde{D}(c)$ involves $\tilde{D}$ only at c, and for complex Z with $|Z| < \varepsilon \leq \varepsilon_{\tilde{D}}$ the map $\mathfrak{x} \mapsto \tilde{C}(Z - h(c)\mathfrak{x})(c)$ is a contraction of $\{|\mathfrak{x}| < 4c_{\tilde{C}}\varepsilon^2\}$ with Lipschitz constant $\frac{1}{2}$, by Cauchy's estimate on the derivative of $\tilde{C}$. Its fixed point is a uniform limit of holomorphic maps, hence holomorphic in (Z, A'), satisfies $|\tilde{D}(Z)| \leq 4c_{\tilde{C}}|Z|^2$, and agrees with Bałaban's solution at real Z by uniqueness. So B' is holomorphic in (B, A') with $|B'| \leq C'_1 g_k|B|$, which is [11, (1.20)] at complex B.

The field B' enters next through $\mathbf{H}_k(s(Y_0), B')$ of [11, (1.17)], which Bałaban states to be analytic in $(s(Y_0), B')$ for $|B'| < \varepsilon_3$ [11, p. 6, ll. 39–41] and in $(\mathbf{U}, \mathbf{J})$ [11, p. 6, ll. 44–45]. At fixed $s(Y_0)$ it is thus separately holomorphic on an open set of finitely many complex variables, hence jointly holomorphic by Hartogs's theorem, and with $\varepsilon_3 := 2C'_1\varepsilon_1$ its composition with B' is holomorphic in $(B, A', \mathbf{J})$ and satisfies [11, (1.21)] at complex B.

The localisation of [11, pp. 7–11] then applies integrals over $[0, 1]$ in the decoupling parameters, Cauchy integrals on fixed circles in the parameters $\sigma(\Delta)$, and sums with uniform bounds, all of which preserve holomorphy. Two Cauchy integrals, [11, (1.22), p. 7] and [9, (3.15), p. 273, ll. 9–11], are taken on circles whose radii depend on $|B|$, which is not holomorphic in B. Such an integral $(2\pi i)^{-1}\oint \lambda^{-2}f(\lambda)\,d\lambda$ is however $f'(0)$, the same on every circle inside the disc of holomorphy, and we may replace each circle by a fixed one which lies inside that disc for every B of the domain: $1/|t_\square| = 16B_0C'_1e^{16\kappa_1}\alpha_2^{-1}\varepsilon_1$ for the first, under $8B_0C'_1e^{16\kappa_1}\varepsilon_1 \leq \frac{1}{4}\alpha_2$, and $|t_\square| = \frac{1}{3}\alpha_2(4B_0C'_1e^{16\kappa_1}m_X\varepsilon_1)^{-1}$ for the second, with m_X the factor of [9, (3.9)]. The bounds are then Bałaban's with $2\varepsilon_1$ in place of $g_k|B|$, the form [11, (1.29), p. 8] itself uses, and the factor 2 enters the absolute constant C_1 of [11, (1.36)]. The terms obtained by the transformations of [9, Sect. 4] are polynomials in local fields built from $\mathbf{H}_k(B')$, whose coefficients do not involve the fluctuation field [9, pp. 281–290], so they are holomorphic as well.

The terms of $\mathbf{P}^{(k)}$ in Lemma 2 depend on B also directly, through $Z = g_k\mathcal{P}B$ and $\tilde{D}(Z)$ [9, p. 268, ll. 9–13]. The series $\operatorname{Tr}\log(I - h(\delta\tilde{D}/\delta Z)(Z))$ converges, since Cauchy's estimate gives $\|h(\delta\tilde{D}/\delta Z)(Z)\| \leq \frac{1}{2}$; the logarithm of the Haar density at B' is $\sum_b \operatorname{Tr}\log(I + E_b)$ with $\|E_b\| \leq e^{2|B'(b)|} - 1 \leq \frac{1}{2}$ when $2C_1'\varepsilon_1 \leq \log\frac{3}{2}$; and the remaining terms are of order at least three in Z, so that $g_k^{-2}F_3(Z) = \langle \mathcal{P}B, (\int_0^1(1-t)\partial^2 F_3(tZ)\,dt)\mathcal{P}B\rangle$ is a quadratic form in B with a holomorphic operator. These terms enter $\mathbf{V}_k''$ with [11, (1.36)] or $Q(Y, B)$ with [11, (1.43)], at larger absolute constants, which Bałaban allows [11, p. 11, ll. 5–6]. Every step uses g_k only through $g_k|B| < \varepsilon_1$ and the prefactor g_k^{-2}, which gives the statement with a complex z; at $z = 0$ the form $z^{-2}F_3(z\mathcal{P}B)$ is holomorphic by the same Taylor formula. Two inputs are read from Bałaban's text rather than printed there: the bound $g_k|B^R| < 2\varepsilon_1$ for the gauge-transformed field of [9, p. 276, ll. 11–14], which he says has "almost the same bounds", and the condition [9, (3.14)] for the argument of the second term of [9, (3.21)], which he treats "in exactly the same way" [9, p. 274, ll. 30–33]. The identities [11, (1.33), (1.41)] are used at real B only. *q.e.d.*

17.2 The Variational Problems

Bałaban states the decay [7, (189), p. 308] of the second derivative of the potential V of [7, (80)] with the remark "We do not perform these calculations here, … let us formulate a final result only" [7, p. 308, ll. 28–30], and derives [7, (190)] from it.

Lemma 17.3 (The Decay of the Second Derivative of the Potential)**.** *Let* $d = 4$, $L \geq 5$, *let* U_0 *be as in [7, Sects. B–D], and let* $\frac{1}{8}\delta_0 RM \geq 4\log L + 1$, $RM \geq 4dL$, *and [3, (2.59)] at* $\alpha = 1/128$ *hold. There are* $\bar{\varepsilon} > 0$ *and* C_{189}, *constants of* d *and* L, *such that for complex* A' *with* $|A'|_\infty := \max(|A'|_{(-1)}, |\nabla A'|_{(-2)}) < \varepsilon_3 \leq \bar{\varepsilon}$, *for* $\mathfrak{A}$ *supported in* $\tilde{\Delta}(y')$, $y' \in \Lambda_{j'}$, *and for a bond* $b \in \Delta(y)$, $y \in \Lambda_j$,

$$(L^j\eta)^3\Big|\Big(\frac{\delta^2 V}{\delta A'^2}(A')\mathfrak{A}\Big)(b)\Big| \leq C_{189}\varepsilon_3 e^{-\delta_0 d(y,y')/4}|\mathfrak{A}|_\infty\,, \tag{17.2}$$

with d *the distance of [3, (2.46), p. 231].*

Proof. We write every object in the dimensionless units of its layer, $a_j := L^j\eta$: a field X at a bond b as $a_{j(b)}X(b)$, its gradient as $a_{j(b)}^2\nabla X(b)$, and a source as $a_{j(b)}^3 F(b)$, with $y(b)$ the block of the layer of b which contains an endpoint of b. A linear map is called a link of rate r and constant κ if its kernel in these units satisfies $|k(u,v)| \leq \kappa L^{|j(u)-j(v)|}e^{-rd(y(u),y(v))}$, and a local link with constant κ if $|k(u,v)| \leq \kappa$ and $k(u,v) = 0$ unless $d(y(u), y(v)) \leq \rho_0 := 3dL$.

The first task is to identify the links. By [5, (3.133)], as used in [7, (46), (86)], H and ∇H are links of rate δ_0; by [5, (3.132)], which Lemma 17.8 proves, so is $(QGQ^*)^{-1}$; by [7, (71), (73)], $(I + \mathfrak{R})^{-1}$, $\mathfrak{D}(A')$, and $\mathfrak{D}(A')^*$ are links of rate $\frac{1}{2}\delta_0$. The averages Q, Q^* are local, and so are the vertex functions $\mathcal{V}(X)(c,b) := L^j\eta(\delta C_j/\delta A(b))(L^j\eta X, c)$ of [7, (68), (70)], which depend on X only near c and are bounded by [7, (72)]. Locality is a statement about the distance [3, (2.46), p. 231], and it rests on the following fact: if $RM \geq 4dL$, $y \in \Lambda_j$, and a block w contains a site within ℓ^∞-distance $2a_j$ of $\Delta(y)$, then $d(y,w) \leq 3dL$, since a contour from y to w through at most $2d$ bonds of the coarser of the two layers and $d(L-1)$ bonds of the finer one is admissible for [3, (2.46)], the blocks it passes being, under $RM \geq 4dL$ and [3, (2.2)], blocks of the layers whose bonds it uses.

Chains of links are then controlled as follows. Under $\frac{1}{8}\delta_0 RM \geq 4\log L + 1$ a change of layer costs $L^{|j(u)-j(v)|} \leq Le^{(\delta_0/32)d(y(u),y(v))}$, since adjacent layers are separated by more than RM, and [3, (2.58)–(2.59)] at $\alpha = \frac{1}{128}$ give $\sum_v \nu(v)e^{-(\delta_0/128)d(y(u),y(v))} \leq 2dc_{\text{tot}}(\frac{1}{128})$. A link of rate at least $\frac{1}{2}\delta_0$ is therefore bounded by κLe^{-rd} for every $r \leq \frac{15}{32}\delta_0$, and a local link by $\kappa e^{\delta_0\rho_0}e^{-rd}$, so the triangle inequality [3, (2.54)] shows that a chain of n such links applied to an input g

with $|g(v)| \leq \gamma e^{-rd(y(v),y')}$, $\frac{\delta_0}{128} \leq r \leq \frac{15}{32}\delta_0$, is bounded by $\gamma \prod_i (c_C \kappa_i) e^{-(r-\delta_0/128)d(y(u_0),y')}$ with $c_C := 2dLe^{\delta_0\rho_0} c_{\mathrm{tot}}(\frac{1}{128})$.

We now differentiate. The identities $QH = \ell^{-1}$ and $RD^*H = 0$ of [7, (45)] hold for every A', so with $A^\sharp := A' - HD(A')$ the representation [7, (80)] of V has the gradient [7, (85), (88)–(93)], and we differentiate it along $\mathfrak{A}$. We write ℓ^{-1} for the multiplication by $(L^{j(\cdot)}\eta)^{-1}$, $S_a := (QGQ^*)^{-1} - a$, so that $\Delta_\pi H = Q^* S_a \ell^{-1}$ on the fields in the Landau gauge and $H^*\Delta_\pi H = \ell^{-1} S_a \ell^{-1}$ by [7, (87)], $\mathcal{N}_0 := \partial V_0$ for the local vertex of [7, (90)–(93)], a function of the field and of its covariant derivative on the plaquettes at a bond, and $\delta_{\mathfrak{A}}$ for the derivative along $\mathfrak{A}$. The fixed point $\mathfrak{D} = \mathcal{V}(A^\sharp)(I - H\mathfrak{D})$ of [7, (67)] has the derivative $\delta_{\mathfrak{A}}\mathfrak{D} = (I + \mathfrak{R})^{-1}(\delta_{\mathfrak{A}^\sharp}\mathcal{V})(A^\sharp)(I - H\mathfrak{D})$, by [7, (68)], with $\mathfrak{A}^\sharp := \mathfrak{A} - H\mathfrak{D}\mathfrak{A}$ the derivative of $A^\sharp$, and likewise for $\mathfrak{D}_2$. The gradient is the sum of $-\mathfrak{D}_2^* H^* J$ [7, (85)], of $-Q^* S_a \ell^{-1} D - \mathfrak{D}^* \ell^{-1} S_a Q A'$ [7, (88)], of $\mathfrak{D}^* \ell^{-1} S_a \ell^{-1} D$ [7, (89)], and of $(I - \mathfrak{D}^* H^*)\mathcal{N}_0(A^\sharp)$ [7, (90)–(93)], and its derivative is the sum of the nine terms

$$\begin{gathered} -(\delta_{\mathfrak{A}}\mathfrak{D}_2)^* H^* J, \qquad -Q^* S_a \ell^{-1} \mathfrak{D}\mathfrak{A}, \qquad -(\delta_{\mathfrak{A}}\mathfrak{D})^* \ell^{-1} S_a Q A', \qquad -\mathfrak{D}^* \ell^{-1} S_a Q \mathfrak{A}, \\ (\delta_{\mathfrak{A}}\mathfrak{D})^* \ell^{-1} S_a \ell^{-1} D, \qquad \mathfrak{D}^* \ell^{-1} S_a \ell^{-1} \mathfrak{D}\mathfrak{A}, \\ (\delta_{\mathfrak{A}^\sharp}\mathcal{N}_0)(A^\sharp), \qquad -\mathfrak{D}^* H^* (\delta_{\mathfrak{A}^\sharp}\mathcal{N}_0)(A^\sharp), \qquad -(\delta_{\mathfrak{A}}\mathfrak{D})^* H^* \mathcal{N}_0(A^\sharp) \,. \end{gathered}$$

In each of them the operators H^*, S_a, $\mathfrak{D}^*$, $(I + \mathfrak{R})^{-1}$ and their adjoints are links, Q, Q^*, ℓ^{-1}, and the vertex functions are local, and no term contains more than four links: the longest are the fifth and the ninth, with $(I + \mathfrak{R})^{-1}$, $\mathfrak{D}^*$, H^* from $\delta_{\mathfrak{A}}\mathfrak{D}$ and one further link. Each term is thus a chain of at most four links applied either to $\mathfrak{D}\mathfrak{A}$ or $Q\mathfrak{A}$, which decay at the rate $r_0 := \frac{15}{32}\delta_0$ away from y' by [7, (73)], or to the derivative of a vertex function along $\mathfrak{A}^\sharp := \mathfrak{A} - H\mathfrak{D}\mathfrak{A}$, which Cauchy's formula on a disc of radius $\bar{\varepsilon}/s_{\mathfrak{A}^\sharp}$ bounds by $\bar{\varepsilon}^{-1}$ times the bound of [7, (72)], times $s_{\mathfrak{A}^\sharp}$, which decays at the rate $r_0 - \frac{\delta_0}{128}$. Two applications of the chain bound bring the rate down to $r_2 := \frac{29}{64}\delta_0 > \frac{1}{4}\delta_0$, and summing the nine terms gives $a_{j(b)}^3 |((\delta^2 V/\delta A'^2)(A')\mathfrak{A})(b)| \leq C_T e^{-\delta_0 d(y(b),y')/4} |\mathfrak{A}|_\infty$ for every complex A' with $|A'|_\infty < \bar{\varepsilon}$, with C_T a constant of d and L.

The factor ε_3 comes from the order of V. By [7, (55), (56)] and [7, p. 282, l. 27], V is of order at least three in A' and analytic for $|A'|_\infty < a_3$ [7, p. 290, l. 32], so $f(\zeta) := a_{j(b)}^3((\delta^2 V/\delta A'^2)(\zeta A')\mathfrak{A})(b)$ is analytic for $|\zeta| < \bar{\varepsilon}/|A'|_\infty$ and vanishes at $\zeta = 0$, and Schwarz's lemma on that disc gives $|f(1)| \leq C_T e^{-\delta_0 d(y(b),y')/4} |\mathfrak{A}|_\infty |A'|_\infty / \bar{\varepsilon}$. Finally $b \in \Delta(y)$ has an endpoint in the block of y, so $j \leq j(b)$ and $(L^j \eta)^3 \leq a_{j(b)}^3$, and $y(b)$ has a site within η of $\Delta(y)$, so $d(y(b), y') \geq d(y, y') - \rho_0$ by the fact above and [3, (2.54)]; this gives (17.2) with $C_{189} := e^{\delta_0 \rho_0/4} C_T / \bar{\varepsilon}$. *q.e.d.*

The same chains of links, with one factor of the field kept at its point, give a kernel form of the uniform bound [7, (97)–(98)] on the gradient $\mathcal{N} := \delta V/\delta A'$ of the potential. For a block y of the layer j let $\Delta(y)$ be the set of bonds of the layer j whose lower endpoint lies in y, or whose upper endpoint does when the lower one lies in the layer $j - 1$, and $\|A\|_y := \sup_{b \in \Delta(y)} (a_j |A(b)| + a_j^2 |\nabla A(b)|)$.

Lemma 17.4 (The Gradient of the Potential as a Kernel)**.** *Under the hypotheses of Lemma 17.3 there is a constant C_N of d, L, and Bałaban's constants such that for complex A' with $|A'|_\infty \leq \varepsilon_3 \leq \frac{1}{4}a_3$ and every bond $b \in \Delta(y)$,*

$$a_{j(y)}^3 |\mathcal{N}(A')(b)| \leq C_N \varepsilon_3 \sum_{y''} e^{-\delta_0 d(y,y'')/8} \|A'\|_{y''} \,. \tag{17.3}$$

Proof. We begin with the vertices. For a block bond c of the layer j let $s_c(X)$ be the maximum of $a_j|X(b)|$ over the bonds of $B^j(c_-) \cup B^j(c_+)$. By [4, Proposition 4, p. 38, l. 48–p. 39, l. 10], $C_j(a_jX, c)$ is analytic for $s_c(X) < c_4$ with $|C_j(a_jX, c)| \le C_2 s_c(X)^2$, as in [7, (44)], and by [4, Proposition 5, (157), p. 42], rescaled as in [7, (72)], the vertex $\mathcal{V}(X)(c, b)$ satisfies $|\mathcal{V}(X)(c, b)| \le C_3 a_j^{1-d} s_c(X)$ for $s_c(X) < r_v$. Cauchy's formula on the circles $|\tau| = R$ for $\tau \mapsto C_j(\tau a_j X, c)$ and $\tau \mapsto \mathcal{V}(\tau X)$, and the maximum principle for $(\mathcal{V}(\tau X) - \tau\mathcal{V}^{(1)}(X))/\tau^2$, give, with $\mathcal{V}^{(1)}$ the linear part and $C_v := \max(C_2, C_3, 2C_3/r_v)$, $|\mathcal{V}^{(1)}(X)(c, b)| \le C_3 a_j^{1-d} s_c(X)$ for every X and $|(\mathcal{V} - \mathcal{V}^{(1)})(X)(c, b)| \le C_v a_j^{1-d} s_c(X)^2$ for $s_c(X) < r_v$. The plaquette vertices of [7, (90)] obey [7, (40)] with the power lowered by one [7, p. 291, ll. 58–59], and the commutator vertices of [7, (91)–(96)] are of order two, so in each vertex one factor is the local size of the field and the others are at most $2\varepsilon_3$. A local size at c is at most $L \max\|X\|_{y'}$, or $L^2 \max\|X\|_{y'}$ for the vertices with differences, over the blocks y' meeting the bonds of the vertex, since adjacent layers differ by L in a_j.

We next bound D, HD, and $\mathfrak{D}_2 := \mathfrak{D} - \mathcal{V}^{(1)}$. Since $D(0) = 0$ and $D(A') = \int_0^1 \langle \mathfrak{D}(tA'), A' \rangle \, dt$ [7, (63)–(64)], the bound [7, (73)] and $\sum_{b \in \Delta(y)} \eta^d a_j^{1-d} |X(b)| \le \frac{3}{2} d \|X\|_y$ give $|D(A', c)| \le \frac{3}{2} d O(1) C_3 \varepsilon_3 \sum_y e^{-\delta_0 d(c_-, y)/2} \|A'\|_y$. With [5, (3.133)] for H, as in [7, (69), (86)], the change of layer by [3, (2.60)], which gives $a_{j'}/a_j \le L e^{(\delta_0/32) d(y, y')}$ under $\frac{1}{8}\delta_0 RM \ge 4 \log L + 1$, and the triangle inequality [3, (2.54)],

$$\|HD(A')\|_y \le C_{HD} \varepsilon_3 \sum_{y''} e^{-\delta_0 d(y, y'')/4} \|A'\|_{y''} \,, \qquad C_{HD} := 3 d^2 L B_0 O(1) C_3 c_{\mathrm{tot}}(\tfrac{1}{128}) \,,$$

and the same with $1 + C_{HD}\varepsilon_3$ for $A^\sharp = A' - HD(A')$. By [7, (70)], $\mathfrak{D}(A') = (I + \mathfrak{R})^{-1} \mathcal{V}(A^\sharp)$, so

$$\mathfrak{D}_2 = (I + \mathfrak{R})^{-1} (\mathcal{V} - \mathcal{V}^{(1)})(A^\sharp) - (I + \mathfrak{R})^{-1} \mathcal{V}^{(1)}(HD(A')) - (I + \mathfrak{R})^{-1} \mathfrak{R} \mathcal{V}^{(1)}(A') \,,$$

and the bounds on the vertices, [7, (69), (71)], and the locality fact of the proof of Lemma 17.3 give $|\mathfrak{D}_2(A'; c, b)| \le C_{\mathfrak{D}} \varepsilon_3 a_j^{1-d} e^{-\delta_0 d(c_-, y)/4} \sum_{y''} e^{-\delta_0 d(y, y'')/4} \|A'\|_{y''}$ for $b \in \Delta(y)$.

Finally we assemble the chains. By [7, (80)], $\mathcal{N}$ is the sum of [7, (85), (88), (89), (90)] and of the derivatives of the first expression of [7, (39)] after the substitution [7, (47)], and in [7, (88), (89)] the identities $\Delta_\pi H = Q^*[(QGQ^*)^{-1} - a^V](L^{j(\cdot)}\eta)^{-1}$, $DRD^*H = 0$, and $H^*Q^* = (L^{j(\cdot)}\eta)^{-1}$ of [7, (45), (87)] reduce it to the five terms listed in the proof of Lemma 17.3: $-\mathfrak{D}_2^* H^* J$, $-Q^* S_a \ell^{-1} D$, $-\mathfrak{D}^* \ell^{-1} S_a Q A'$, $\mathfrak{D}^* \ell^{-1} S_a \ell^{-1} D$, and $(I - \mathfrak{D}^* H^*) \mathcal{N}_0(A^\sharp)$, where $D(A') = \int_0^1 \langle \mathfrak{D}(tA'), A' \rangle \, \mathrm{d}t$ contains one factor $\mathfrak{D}$ and $A^\sharp = A' - HD(A')$ the further factor H. Counting in each term the factors H, H^*, $(DH)^*$, Q, Q^*, S_a, $\mathfrak{D}^*$, $\mathfrak{D}$, $\mathfrak{D}_2$, and the vertex, the longest is $\mathfrak{D}^* H^* \mathcal{N}_0(A' - HD(A'))$, with $\mathfrak{D}^*$, H^*, the vertex, H, and $\mathfrak{D}$; so every term is a chain of at most six links. In each chain one link carries the local size, namely $\mathfrak{D}_2$ in [7, (85)], $D(A')$ or QA' in [7, (88), (89)], and the vertex in [7, (90)–(96)], and one link carries the factor ε_3, by the order of the vertex, by the bounds above, or by [7, (73)] for $\mathfrak{D}^*(A')$; the current is bounded by [7, (86)]. The links decay at the rates δ_0, $\frac{1}{2}\delta_0$, or $\frac{1}{4}\delta_0$, a change of layer costs $\frac{\delta_0}{32}$ of the rate of its link, and the local links span distances at most $3dL$, so every link decays at least at the rate $\frac{7}{32}\delta_0$; the sums over the at most five intermediate blocks cost $c_{\mathrm{tot}}(\frac{1}{128})^5$ by [3, (2.58)–(2.59)], and the chain decays from y to y'' at the rate $\frac{7}{32}\delta_0 - \frac{\delta_0}{128} = \frac{27}{128}\delta_0 > \frac{1}{8}\delta_0$, which gives (17.3). *q.e.d.*

Bałaban's regularity statement [7, Theorem 1, p. 279] holds on the cubes of a class: a cube $\Box$ of side $2ML^j\eta$, M a multiple of R_1M_1, which is a union of big blocks of the L^{-j}-lattice and whose concentric cube $\tilde{\Box}$ of side $(2M + 4R_1M_1)L^j\eta$ lies in $B^j(\Lambda_j) \cup B^{j+1}(\Lambda_{j+1}) = \Omega_j \setminus \Omega_{j+2}$ but not in $B^{j+1}(\Lambda_{j+1})$ [7, p. 279, ll. 2–4]; his proof works with the cubes which meet Ω_j but not Ω_{j+1}, with $\tilde{\Box} \subset B^{j-1}(\Lambda_{j-1}) \cup B^{j}(\Lambda_j)$ [7, p. 300, ll. 28–37]. The paper needs every point of a layer in such cubes.

Lemma 17.5 (The Layers Are Covered by the Cubes of the Class). *Let $\{\Omega_j\}$ be a sequence as in [7, (1)–(4), p. 277], $a_j := L^j\eta$, $M' := R_1M_1$, $L \geq 2$, and let every layer $\Omega_j \setminus \Omega_{j+1}$ below the top one have ℓ^∞-width at least $12LM'$ in the units a_j. Let $2M' \leq M(\varepsilon_1)$, where $M(\varepsilon_1)$ denotes the largest size M of the cubes of the class at which [7, Theorem 1] holds with the parameter ε_1; in the paper this is one of the conditions of Paper, Appendix C. Let x be a point of the layer j, and if $j = 0$ let its ℓ^∞-distance from Ω_0^c be at least $4M'a_0$. Then:*

- *(a) x lies in a cube of the class of Theorem 1 of the index j and side $2M'a_j$, or of the index $j-1$ and side $4LM'a_{j-1}$;*
- *(b) x lies in a cube of the class of the proof of the index j and side $2M'a_j$, or of the index $j+1$ and side $4M'a_{j+1}$, and in the second case the bounds [7, (9)–(10)] on it are at most those of the index j with M'.*

Proof. The big blocks of the L^{-j}-lattice have side M_1a_j [7, p. 277, l. 12], and M_1 divides M', so for M a multiple of M' the cubes of side $2Ma_j$ on the grid of big blocks are unions of big blocks, and for every point x and every choice of a side in each coordinate one of them contains x in its last big block on the chosen sides; it extends from x at least $(2M - M')a_j$ towards the other side. The concentric cube $\tilde{\Box}$ of such a cube $\Box \ni x$ lies within ℓ^∞-distance less than $(2M + 2M')a_j$ of x. Let $\delta(x)$ and $\delta^+(x)$ be the ℓ^∞-distances of x from Ω_j^c and from Ω_{j+1} in the units a_j, with $\delta^+(x) := \infty$ in the top layer.

For (a), suppose first that $\delta(x) \geq 4M'$, and take $\Box \ni x$ of side $2M'a_j$. Then $\tilde{\Box}$ lies within $4M'$ of x, so $\tilde{\Box} \subset \Omega_j$; the set Ω_{j+2} lies at distance at least $12LM'$ from Ω_{j+1}^c in the units a_{j+1}, that is at least $12L^2M' > 4M'$ in the units a_j, and $x \notin \Omega_{j+1}$, so $\tilde{\Box}$ misses Ω_{j+2}; and $\tilde{\Box} \not\subset \Omega_{j+1}$ because $x \in \tilde{\Box}$. Suppose next that $\delta(x) < 4M'$ and $j \geq 1$, let $y \in \Omega_j^c$ with $|x-y|_\infty = \delta(x)$, and take the cube of side $4LM'a_{j-1}$ which contains x in its last big block on the sides away from y. In each coordinate it extends from x towards y at least $(4L-1)M'$ and $\tilde{\Box}$ a further $2M'$, together more than $4LM'$ in the units a_{j-1}, that is more than $\delta(x)$ in the units a_j, so $y \in \tilde{\Box}$ and $\tilde{\Box} \not\subset B^j(\Lambda_j)$. The cube $\tilde{\Box}$ lies within $(4L+2)M' < 12LM'$ of $x \in \Omega_j$ in the units a_{j-1}, so $\tilde{\Box} \subset \Omega_{j-1}$ by the width of the layer $j-1$; and if the layer j is not the top one, x lies at distance more than $12LM' - 4M'$ from Ω_{j+1} in the units a_j, more than the reach $(4 + 2/L)M'$ of $\tilde{\Box}$, so $\tilde{\Box}$ misses Ω_{j+1}.

For (b), suppose first that $\delta^+(x) \geq 4M'$, and take $\Box \ni x$ of side $2M'a_j$, which meets Ω_j at x. Its $\tilde{\Box}$ lies within $4M'$ of x, so it misses Ω_{j+1}, and for $j \geq 1$ it lies in Ω_{j-1} since Ω_{j-1}^c is at distance at least $12M' > 4M'$ from Ω_j in the units a_j; for $j = 0$ the hypothesis on x gives $\tilde{\Box} \subset \Omega_0$. Suppose next that $\delta^+(x) < 4M'$, let $y \in \Omega_{j+1}$ with $|x-y|_\infty = \delta^+(x)$, less than $2M'$ in the units a_{j+1}, and take the cube of side $4M'a_{j+1}$ which contains x in its last big block on the sides away from y. It extends at least $3M' > 2M'$ from x towards y, so it meets Ω_{j+1}, and $\tilde{\Box}$ lies within $6M'$ of x in the units a_{j+1}; the set Ω_{j+2} lies at distance at least $12LM' > 6M'$ from x in these units, and Ω_j^c at distance more than $(12 - 4/L)M' \geq 10M'$, so $\tilde{\Box} \subset \Omega_j \setminus \Omega_{j+2}$. Finally $2M' \leq M(\varepsilon_1)$ by hypothesis, so that the cube is of the class, and the bounds [7, (9)–(10)] at the index $j+1$ with $2M'$ are $2B_3M'\varepsilon_1 a_{j+1}^{-p} = 2L^{-p}B_3M'\varepsilon_1 a_j^{-p}$ with $p \geq 1$, and likewise for the Hölder term, which is at most the bound of the index j with M' since $L \geq 2$. *q.e.d.*

At the stopping scale the paper needs the plaquettes of the stopping-scale field bounded by the action of its background. Bałaban states no such bound; the closest is [8, (69)–(71),

p. 273], which needs the regularity of the fine field and gives one averaged plaquette at a time.

Lemma 17.6 (The Action Bounds the Plaquettes of the Stopping Scale)**.** *Let* V *be a field of the stopping scale* K' *in the support of the density of the term with no large-field region, so that no test function [10, (2.17)] of the step* K' *vanishes, and let* $U := U_{K'}(V)$ *be the global minimiser of [10, (2.12)] on* $\mathbb{T}_\eta$, $\eta = L^{-K'}$, *so that* $M^{K'}U = V$ *with* M *the average of [9, (0.4)]; let* $d = 4$ *and* $G \subset U(N)$. *If* $\varepsilon_{K'}$ *is small, depending on* d, L, *and* G *only, then*

$$\sum_{p'} |V(\partial p') - 1|^2 \le 8N\,A(U)\,, \tag{17.4}$$

the sum over the plaquettes of the lattice of the step K' *and* A *the Wilson action of [7, (5), p. 278]. If moreover* $A(U) \le Tg_{K'}^2$, *every plaquette of* V *lies within* $2(2NT)^{1/2}g_{K'}$ *of* 1, *and if* $\varepsilon^{\mathrm{act}} := 3(2NT)^{1/2}g_{K'} \le \min(a_1, 2\varepsilon_{K'})$ *and* $2B_3\varepsilon_{K'} \le a_0$, *then* $|U(\partial p) - 1| < B_3\varepsilon^{\mathrm{act}}\eta^2$ *on* $\mathbb{T}_\eta$.

Proof. For $0 \le n \le K'$ let $U_n^{\mathrm{av}} := M^n U$, $E_n := \sum_p |U_n^{\mathrm{av}}(\partial p) - 1|^2$ over the positively oriented plaquettes of the level n, and $\sigma_n := \max_p |U_n^{\mathrm{av}}(\partial p) - 1| \le E_n^{1/2}$, with the operator norm of [4, (19)].

We first show that one averaging step does not increase the plaquette energy beyond a factor $1 + C^{\mathrm{av}}\sigma_n$, with $C^{\mathrm{av}} := (2d(d-1))^{1/2}L^4C_0$, when σ_n is small. Write $a_p := |U_n^{\mathrm{av}}(\partial p) - 1|$. For a plaquette p' of the level $n+1$ with first corner y_0, let $\Delta(p')$ be the union of the four blocks of its corners and $\alpha(p') := \max_{p \subset \Delta(p')} a_p$. The second line of [4, (50), p. 25], which holds for the average [9, (0.4)] by the identity [4, (48)] and is local in $\Delta(p')$, gives $|U_{n+1}^{\mathrm{av}}(\partial p') - 1| \le m(p') + C_0L^4\alpha(p')^2$, with $m(p') := \sum_{x \in B(y_0)} L^{-d} \sum_{p \subset (p')_x} a_p$ and $(p')_x$ the translate of p' by $x - y_0$, a square of L^2 plaquettes. The weights L^{-d} sum to 1, so the Cauchy–Schwarz inequality gives $m(p')^2 \le L^{2-d} \sum_{x \in B(y_0)} \sum_{p \subset (p')_x} a_p^2$; in a fixed plane the pairs (p', x) correspond to the fine points x, and a plaquette p lies in $(p')_x$ for exactly L^2 of them, so that $\sum_{p'} m(p')^2 \le L^{4-d}E_n = E_n$ in $d = 4$. A plaquette lies in $\Delta(p')$ for at most four p' in each of the $\frac{1}{2}d(d-1)$ planes, so $\sum_{p'} \alpha(p')^4 \le \sigma_n^2 \cdot 2d(d-1)E_n$, and Minkowski's inequality gives $E_{n+1}^{1/2} \le (1 + C^{\mathrm{av}}\sigma_n)E_n^{1/2}$.

We next bound σ_n. The test functions of the construction are soft, and on a cube of the test functions of the step K' the support of the test function bounds the plaquettes of the localised minimiser by $(1 + \kappa_s)\varepsilon_{K'}\eta^2$, with the soft width $\kappa_s = \beta/30$ of Paper, Appendix A, and not by $\varepsilon_{K'}\eta^2$ (Proposition S.8(b)). By [4, Proposition 2], which is local in the corner blocks, the plaquettes of the averaged configuration exceed this bound, in the units of the step K', by at most a term of the second order in it, so that $|V(\partial p') - 1| < 2\varepsilon_{K'}$ on every plaquette of the step K' once $\varepsilon_{K'}$ is small, because $1 + \kappa_s < 2$. For the determining set of the step K' the data of [7, (7)] are these plaquettes [7, p. 278, ll. 39–40], so [7, Theorem 1, p. 279] with $\varepsilon_1 = 2\varepsilon_{K'} \le a_1$ gives $|U(\partial p) - 1| < 2B_3\varepsilon_{K'}\eta^2$ on $\mathbb{T}_\eta$, and [4, (53)] bounds the plaquettes of U_n^{av} by $\alpha_0L^{2n}\eta^2 + 2C_0(\alpha_0L^{2n}\eta^2)^2 < 2\alpha_0L^{2(n-K')}$, $\alpha_0 := 2B_3\varepsilon_{K'}$, so $\sigma_n < 4B_3\varepsilon_{K'}L^{2(n-K')}$. Hence $\sum_{n<K'} C^{\mathrm{av}}\sigma_n \le 2C^{\mathrm{av}}B_3\varepsilon_{K'} \le \log 2$, and $E_{K'}^{1/2} \le 2E_0^{1/2}$.

Finally $E_0 \le 2N\,A(U)$, since $|X| \le \sqrt{N}\|X\|$ [4, (20), p. 21] and $\|U(\partial p) - 1\|^2 = 2[1 - \operatorname{Re}\operatorname{tr} U(\partial p)]$ with the normalised trace [4, (17)], and $E_{K'}$ is the left side of (17.4); a single term is at most the sum, which gives the second claim. For the third, [7, (7)] holds with $\varepsilon_1 = \varepsilon^{\mathrm{act}} \le a_1$, so [7, Theorem 1] gives a minimal orbit in [7, (2)] with $\varepsilon_0 = B_3\varepsilon^{\mathrm{act}}$, which is the unique critical orbit in [7, (6)] with $\varepsilon_0 = 2B_3\varepsilon_{K'} \le a_0$ because $B_3\varepsilon^{\mathrm{act}} \le 2B_3\varepsilon_{K'}$ [7, p. 279, l. 15]; U lies in that space, so it lies on that orbit. *q.e.d.*

17.3 Propagators in a Background Field

Bałaban's theorems on the propagators [5, Sects. A–D] are proved for the averages of [4] with corner blocks, at G-valued configurations, on sequences whose bottom domain Ω_0 is a union of big blocks near Ω_1. The paper uses them for the dressed average built from the average [9, (0.4)] with centred blocks, whose one step is described in the proof of Lemma 4.20, on sequences with $\Omega_0 = \mathbb{T}$, at complex configurations, and on the joins of the determining sets at a smaller block size. Throughout, a sequence $\{\Omega_j\}$ satisfies [3, (2.1)–(2.4), p. 224], d is the distance [3, (2.46)], and $\Delta(y)$ is the block of y.

Lemma 17.7 (The Averaging Inputs for the Dressed Average)**.** *Let G be a compact Lie group, $\{\Omega_j\}$ a sequence with $\Omega_0 = \mathbb{T}$, U a G-valued configuration of the class [5, (3.35)–(3.36)], and $U'U$ of the class [5, (3.37)–(3.38)]. In the definitions [5, (3.13)–(3.19)] replace the averages of [4] by the dressed average, so that $Q_j(U)$ is the linear part of its j-fold composite, $C_j(U;\cdot)$ the nonlinear part, and $Q'_j(U)$ the composite of [5, (3.19)] with the contour variables [9, (0.11)]. For $M\alpha_0$ and α_1 below thresholds depending on d, L, and G:*

- *(a) Q_j and Q'_j are local and gauge covariant as in [5, (3.32); p. 414, ll. 13–15];*
- *(b) the analogues of [4, (143), (149)] and of [5, (3.58)–(3.59), (3.79)–(3.81)] hold, analytically on [5, (3.37)], with constants depending on d, L, and G;*
- *(c) [5, Theorems 3.1–3.13, (3.131)–(3.137), (3.147)–(3.153)] hold for these operators as stated there, with constants depending on d, L, and G.*

Proof. By the proof of Lemma 4.20, one step of the dressed average from the level l to the level $l{+}1$, at a background with bond logarithms x and fluctuation logarithms y on the $n(d, L)$ bonds of $B(c_-) \cup B(c_+)$, has the logarithm $\Psi(x, y)$, with one analytic Ψ on a polydisc $P(r_0)$, $r_0 = r_0(d, L, G)$, independent of l, j, k, and c, and with $\Psi(x, 0) = 0$; let $K_0 := \sup_{P(r_0)}|\Psi|$. Write the normalised step as $Q^{(l+1)}(a)Y + C^{(l+1)}(a, Y)$ and let Q'' be the average with weight L^{-d} on every bond of $B(c_-) \cup B(c_+)$, $Q''_{m\leftarrow l}$ its composite. Cauchy's estimates on $P(r_0)$ give, with $c_1 := 16nK_0L^d/r_0^2$ and $c_2 := 8K_0/r_0^2$,

$$|Q^{(l+1)}(a) - Q^{(l+1)}(a')| \le c_1 L^{l-1}\eta|a - a'|Q'' \ , \qquad |C^{(l+1)}(a, Y)| \le c_2 L^{l-1}\eta|Y|^2 \ ,$$

and the analogous bound on $dC^{(l+1)}$, the analogues of [4, (139), (148)]. Two counts of the supports, $Q''_{m\leftarrow i}Q''_{i\leftarrow l} \le 2dQ''_{m\leftarrow l}$ and $Q^{(l+1)}Q''_{l\leftarrow i} \le 2L^{-1}Q''_{l+1\leftarrow i}$, give $Q_{m\leftarrow l} \le Q''_{m\leftarrow l}$ by induction.

At the base 1 every background is 1, and the normalised logarithms of the composite at a field X with $|X| \le \rho(L^j\eta)^{-1}$ grow by at most a factor $1 + c_2L^{l-1}\eta|X^{(l)}|$ per step, so they stay below $2|X|$ once $\exp(2c_2\rho/(L(L-1))) \le 2$; the raw bond variables are $v_l(b_-)e^{iL^l\eta X_b^{(l)}}v_l(b_+)^{-1}$ by the identity [4, (97)] for the chain, with dressing factors v_l whose logarithms are means over contours of at most $d(L-1)/2$ bonds, so the raw logarithms are at most $c_\beta L^l\eta|X|$, $c_\beta := 3(1 + d)$. At a base $W_0 = e^{i\eta X_0}$ with background logarithms a_l, the chain rule and $\Psi(x, 0) = 0$ give $Q_{l+1}(W_0) = Q^{(l+1)}(a_l)Q_l(W_0)$, and induction with the two counts gives $|Q_l(W_0)Z| \le Q_l|Z| + TL^l\eta|X_0|Q''_l|Z|$, $T := 4dc_1c_\beta/(L^2 - 2)$, the analogue of [4, (143)]. The background logarithms are analytic in X_0, so Cauchy's estimate and the telescoping of [5, (3.80)] give $|(Q_j(W_0) - Q_j(W'_0))Z| \le T'L^j\eta|X_0 - X'_0|Q''_j|Z|$, $T' := 32d^2c_1c_\beta/(L(L-1))$, which is [5, (3.79)–(3.81)]. For the nonlinear part, $C_{l+1}(X) = Q^{(l+1)}(a_l)C_l(X) + C^{(l+1)}(a_l, X^{(l)})$, and induction gives $|dC_l(X)[\delta X]| \le SL^l\eta|X|Q''_l|\delta X|$, $S := 12dc_1/(L^2 - 3)$, which in the normalisation of [4, (150)] is [4, (149)] with $C_3 = S$. The transports of Q'_j are the composites

$W^{(j)}(y,x)$ of the contour variables along the chain of blocks from x to y, with $|\log W^{(j)}(y,x)| \leq dc_\beta L^j\eta|X|$ and the Lipschitz bound in X by Cauchy's estimate, which is [5, (3.58)–(3.59)]. This proves (b).

For (a), under a G-valued u, with $U_b'^u := u(b_-)U_b'u(b_-)^{-1}$, the factors of the one-step dressed average [5, p. 418, l. 6] transform so that the step is conjugated by $u(c_-)$, by the point covariance of the raw averages and of the contour variables [9, (0.6)]; by induction the composite is conjugated by $u(c_-)$, which is [5, (3.32)] for Q_j, and $W^{(j)}(y,x) \mapsto u(y)W^{(j)}(y,x)u(x)^{-1}$ gives it for Q_j'. The blocks being centred, every point of $B^j(c_-) \cup B^j(c_+)$ lies within $(1+d/2)L^j\eta \leq RML^{j-1}\eta$ of c_-, so these sets lie in Ω_{j-1} by [3, (2.2)], which gives locality.

For (c), Bałaban uses the averages only through the definitions [5, (3.13)–(3.19); p. 393, l. 46–p. 394, l. 2], the covariance [5, (3.32)–(3.34)], the identities $QD = \bar{D}Q'$ and [5, (3.113)–(3.115)], the kernels and [5, (3.102)] on [5, p. 414, ll. 13–22], and the expansions [5, (3.55)–(3.59), (3.78)–(3.81)], all of which hold by (a) and (b). At $U = 1$ each step is the straight average with centred blocks, and since L is odd the fine translation by $\frac{1}{2}(L^k-1)\eta(1,\dots,1)$ maps the centred blocks of every level $l \leq k$ onto the corner blocks of [4, (2), p. 17] at once, so it carries the sequences and the operators at $U = 1$ to Bałaban's. Finally, with $\Omega_0 = \mathbb{T}$ and $R > 20dL$, a union of big blocks of the L^{-m}-lattice of side between $10ML^m\eta$ and $20ML^m\eta$ which meets $B^m(\Lambda_m)$ has ℓ^1-diameter less than $RML^{m-1}\eta$, so by [3, (2.2)] it meets neither Ω_{m+2} nor Ω_{m-1}^c, nor both Ω_m^c and Ω_{m+1}; it is therefore a cube of the class [5, (3.35)] of the index m or $m-1$, which proves [5, p. 408, l. 35–p. 409, l. 2] for $\Omega_0 = \mathbb{T}$ with the constant $12L$. *q.e.d.*

Bałaban states the kernel bound [5, (3.132)] on the inverses of the block forms with the remark "We will not repeat these considerations here, let us write only bounds" [5, p. 422, ll. 15–19].

Lemma 17.8 (The Inverses of the Block Forms). *Let U be G-valued and satisfy [5, (3.35)–(3.36)] for a sequence with $\Omega_0 = \mathbb{T}$, with Bałaban's averages or with the dressed average, and let $G^\sharp$ be the operator G of [5, (3.122)] or G_1 of [5, (3.128)]. There are M_2, a_2, C, and $\delta_1 > 0$, depending on d, L, and G, such that for $M \geq M_2$ and $M\alpha_0 \leq a_2$, and for block bonds c, c' with $c_- \in \Lambda_j$, $c'_- \in \Lambda_{j'}$,*

$$|(QG^\sharp Q^*)^{-1}(c,c')| \leq C(L^j\eta)^{-2}(L^{j'}\eta)^{-d}e^{-\delta_1 d(c,c')}\,. \tag{17.5}$$

Proof. We prove the bound first for $G^\sharp = G$ and then for G_1; the inputs for G_1 use [5, (3.137)], which rests on [5, (3.133)] for H, which Bałaban derives from (17.5) for G [5, p. 422, ll. 21–22], so no input uses the bound under proof. On the space of $\mathfrak{g}$-valued functions on the block bonds with $\langle B,B'\rangle = \sum_j\sum_{c\in\Lambda_j}(L^j\eta)^dB(c)\cdot B'(c)$, let W be the multiplication by $(L^j\eta)^2$, $G_C := (\Delta + DRD^* + Q^*aQ)^{-1}$, $X^\sharp := QG^\sharp Q^*$, and $X_C := QG_CQ^*$. Two sums are used throughout: under $\frac{1}{4}rRM \geq 2d\log c_0 + 1 + p\log L$, $c_0 := \sum_{z\in\mathbb{Z}}e^{-r|z|/2}$, one has $\sup_y\sum_{y'}L^{p|j-j'|}e^{-rd(y,y')} \leq 12L^pc_0^d$, by the proof of [3, (2.58)–(2.59)] with $c_0^{2d}L^p$ for c_0^{2d}; and, by Schur's test, a kernel bounded by $\kappa L^{q(j-j')}(L^{j'}\eta)^{-d}e^{-rd(c,c')}$ has norm at most $12d\kappa L^{d+|q|}c_0^d$.

Step 1: decay of $X^\sharp$. By [5, (3.42)] for G_C [5, Theorem 3.3] and for $G^\sharp - G_C$, the sum of the terms $n \geq 1$ of [5, (3.130)] or [5, (3.138)], by Lemma 17.9 with the current of order $M\alpha_0$, and by the locality of Q and Q^*, $|X^\sharp(c,c')| \leq C_X(L^j\eta)^2(L^{j'}\eta)^{-d}e^{-\delta_0d(c,c')}$ and $|(X^\sharp - X_C)(c,c')| \leq O(1)M\alpha_0(L^j\eta)^2(L^{j'}\eta)^{-d}e^{-\delta_0d(c,c')/2}$.

Step 2: local sequences. For a cube $\square$ of [5, p. 408] of the scale j let $\Omega_n(\square) := \square^{(n)} \cap \Omega_n$ with the cubes $\square^{(n)}$ of [5, p. 408, ll. 30–32]. Since $\tilde{\square}^5$ meets only two consecutive layers, this is a sequence with R_0M_0 in place of RM, its blocks and kernels agree with the global ones on $\tilde\square^2$, and $\Omega_0(\square) \subset \tilde\square^5$.

Step 3: a local lower bound. At $U = 1$, [3, Proposition 2.7, p. 249] and Schur's test give $\|W^{1/2}X_\square(1)^{-1}W^{1/2}\| \le C_4$, and $X_\square(1)$ is positive since $G_\square(1)$ is, so $W^{-1/2}X_\square(1)W^{-1/2} \ge C_4^{-1}$. At U the local gauge of [5, Corollaries 3.5–3.6] gives $U^u = e^{i\eta A}$ with $\alpha_1 = O(1)M\alpha_0$, the forms at U and U^u are orthogonally conjugate, and the expansions [5, (3.80)–(3.81), (3.84)–(3.85)] bound $W^{-1/2}(X_\square(U^u) - X_\square(1))W^{-1/2}$ by $O(1)M\alpha_0$. Hence $X_\square(U) \ge \gamma_\ell W$, $\gamma_\ell := \frac{1}{2}C_4^{-1}$, for $M\alpha_0$ small.

Step 4: a global lower bound for X_C. Since $G_C > 0$, $\langle v, G_Cv\rangle \ge 2\langle v, A\rangle - \langle A, G_C^{-1}A\rangle$ for every A; take $v := Q^*B$ and $A := t\hat{G}v$ with $\hat{G} := \sum_\square h_\square G_\square h_\square$, the partition $\sum_\square h_\square^2 = 1$ of [5, p. 408, l. 23]. By [5, (3.102)], $h_\square Q^*B = Q^*(\bar{h}_\square B) + e_\square$ with $|e_\square| \le O(1)LM^{-1}|B|$ locally, so Step 3 and the decay of $G_\square$ give $\langle v, \hat{G}v\rangle \ge (\gamma_\ell - O(1)M^{-1})\langle B, WB\rangle$, each block bond lying in at most $3 \cdot 9^d$ of the cubes $\tilde\square^3$. The decay of $G_\square$, $\nabla G_\square$, and Q bounds $\langle \hat{G}v, G_C^{-1}\hat{G}v\rangle \le C_A\langle B, WB\rangle$, and $t := \gamma_\ell/(2C_A)$ gives $X_C \ge \gamma_C W$, $\gamma_C := \gamma_\ell^2/(4C_A)$, for M large.

Step 5. By Step 1 and Schur's test, $X^\sharp \ge (\gamma_C - O(1)M\alpha_0)W \ge \frac{1}{2}\gamma_C W =: \gamma W$.

Step 6: Combes–Thomas. $T := W^{-1/2}X^\sharp W^{-1/2}$ is symmetric with $T \ge \gamma$ and has the kernel bound of Step 1 with $L^{j-j'}$. For $\rho(c) := d(c, c_\star)$, which is 1-Lipschitz for d, and $0 < \varepsilon \le \frac{1}{4}\delta_0$, $T_\varepsilon := e^{\varepsilon\rho}Te^{-\varepsilon\rho}$ satisfies $\|T_\varepsilon - T\| \le \varepsilon C_s$ by $|e^x - 1| \le |x|e^{|x|}$ and Schur's test, so for $\varepsilon := \min(\frac{1}{4}\delta_0, \gamma/(2C_s))$ we get $\|T_\varepsilon^{-1}\| \le 2/\gamma$, hence $|T^{-1}(c, c')| \le (2/\gamma)(L^j\eta)^{-d/2}(L^{j'}\eta)^{-d/2}e^{-\varepsilon d(c,c')}$ with $c_\star := c'$. Then $(X^\sharp)^{-1} = W^{-1/2}T^{-1}W^{-1/2}$, and $d(c, c') > RM(|j - j'| - 1)$ absorbs $L^{|d/2-1||j-j'|}$ into $L^{d/2}e^{\varepsilon d(c,c')/2}$ once $RM \ge 2|d/2 - 1|\log L/\varepsilon$, which gives (17.5) with $C := 2L^{d/2}/\gamma$ and $\delta_1 := \frac{1}{2}\varepsilon$. *q.e.d.*

The series [5, (3.130), (3.138)], by which Bałaban defines G and G_1 from G_C, are stated at G-valued configurations. The paper needs them at complex configurations and with an added perturbation without derivatives.

Lemma 17.9 (The Series for the Propagators at Complex Configurations)**.** *Let* $\mathbf{U} = U'U$ *with* U *G-valued of the class [5, (3.35)] and* A' *in [5, (3.37)], on a sequence with* $\Omega_0 = \mathbb{T}$*, with Bałaban's averages or with the dressed average, let* $\mathbf{J}$ *be* $\mathfrak{g}^c$*-valued with* $|\mathbf{J}(b)| < \gamma(L^j\eta)^{-3}$ *when* b *has an endpoint in* Ω_j*, and let* G'*,* R*,* $P := I - R$*, and* G_C *be the operators of [5, Theorem 3.4] at* $\mathbf{U}$*. Let* Δ'_π *be the form [5, (3.120)] with* $\mathbf{J}$ *in place of* J*,* $\Delta_\pi^{(2)}$ *the form [5, (3.135)] with a symmetric operator* $\Delta^{(2)}$ *analytic in* $(A', \mathbf{J})$ *satisfying [5, (3.137)] with a constant* κ_Δ*,* X_0 *a local operator of order* -2 *with constant* κ_X*, analytic in* $(A', \mathbf{J})$*, and* X *one of* X_0*,* $\Delta'_\pi + X_0$*, and* $\Delta'_\pi + \Delta_\pi^{(2)} + X_0$*. There is* $\mathfrak{a}_{\mathrm{ser}} > 0$*, depending on* d*,* L*, and* G*, such that for* $b_X := \gamma + \kappa_X + \kappa_\Delta \le \mathfrak{a}_{\mathrm{ser}}$*,* RM *large, and* α_1 *small, the series* $G_X := \sum_{n\ge0} G_C(XG_C)^n$ *converges,* $G_X(G_C^{-1} - X) = (G_C^{-1} - X)G_X = I$*, and* $G_X - G_C$ *satisfies the inequalities [5, (3.42)–(3.47)] with* $O(1)b_X$ *for the constants and* $\frac{3}{8}\delta_0$ *for* δ_0*, except the covariant-Laplacian inequality of [5, (3.42)], which holds when* $X = X_0$*, and the fourth inequality of [5, (3.47)]. Moreover* G_X *is analytic in* $(A', \mathbf{J})$ *and has a walk expansion whose terms depend on the configuration on the unions of the domains of their factors and whose absolute values obey the same bounds. The case* $X = \Delta'_\pi$ *is* G *of [5, (3.130)], and* $X = \Delta'_\pi + \Delta_\pi^{(2)}$ *is* G_1 *of [5, (3.138)].*

Proof. Call an operator K of type (s, r, κ) if $|K\lambda|_y \le \kappa(L^{j_y}\eta)^s e^{-rd(y,y')}|\lambda|$ for λ supported in $\Delta(y')$. A product of operators of types (s_m, r, κ_m) has type $(\sigma_n, r - s_0, (12L^{\sigma^*}c_0^d)^{n-1}\prod\kappa_m)$, with σ_n the total order and σ^* a bound on the orders of its final segments, under $\frac{1}{4}s_0RM \ge$

$2d \log c_0 + 1 + \sigma^* \log L$: insert the blocks between the factors, use Abel summation for the powers of $L^j \eta$ and the triangle inequality for d, and sum over the blocks one at a time with the first sum of the proof of Lemma 17.8.

By [5, (3.120)], with $S := G' R D^*$, the evaluations $E_\pm$ at the endpoints of a bond, the transport $R_\bullet$, and the multiplication m by the current, Δ'_π is a sum $\sum_{p,q \in \{0,1\}} D^p N_{pq} D^{*q}$ of six products, in each of which exactly one factor carries $E_\pm$; by [5, (3.135)], $\Delta^{(2)}_\pi = (I - (DS)^T) \Delta^{(2)} (I - DS)$ has the same form, and X_0 adds to N_{00}. By [5, Theorem 3.4] and [5, (3.49)], G' has type $(2, \delta_0, B_0)$, DG' and $G'D^*$ type $(1, \delta_0, B_0)$, and P, DP, PD^* types $(0, \frac{1}{2}\delta_0, O(1))$ and $(-1, \frac{1}{2}\delta_0, O(1))$; m, X_0, and $\Delta^{(2)}$ are local of orders -3, -2, and -2 with constants $c_{\mathfrak{g}}\gamma$, κ_X, and κ_Δ. Attaching the derivative of an E end to its factor and expanding $R = I - P$, each product has type $(\tilde{p} + \tilde{q} - 2, \frac{15}{32}\delta_0, O(1) b_X)$, by the composition rule with $s_0 = \delta_0/64$ and $\sigma^* = 7$, where $\tilde{p}$, $\tilde{q}$ record whether its ends are derivatives. A term of $G_C (X G_C)^n$ is then a product of units $D^{*\tilde{q}_{i-1}} G_C D^{\tilde{p}_i} T_i$. The units with $\tilde{p}_i = \tilde{q}_{i-1} = 1$ need an argument of their own, because $D^* G_C D$ has no bound of the type above: [5, (3.44)] bounds it at a point only on a field with a Hölder norm, $|(D^* G_C D u)(x)| \le B'_0(\varepsilon) e^{-\delta_0 d(y,y')} (\|u\|^{\xi'}_\varepsilon + |u|)$ for u supported in $\tilde{\Delta}(y')$. The Hölder norm is supplied by the factor to the right of the derivative. In every product with $\tilde{p} = 1$ the factor next to D is $R G' D^*$, as [5, (3.120), (3.135)] show, so that $T_i = R G' D^* T'_i$ with T'_i of a type (s', r, κ'); the operator X_0 occurs with $\tilde{p} = 0$ only. For a bounded λ supported in $\Delta(y')$ and a function ζ on $\tilde{\Delta}(y)$, [5, (3.43)] at $\beta = \frac{1}{2}$ gives $\|\zeta G' \nabla^* \lambda\|_{1/2} \le B_0(\frac{1}{2}) (L^{j_y}\eta)^{1/2} (\|\zeta\|^{\xi}_{1/2} + |\zeta|) e^{-\delta_0 d(y,y')} |\lambda|$, and the kernel of $P G' D^*$ has first differences bounded by [5, (3.42), (3.49)], so that $P G' D^* \lambda$ is Lipschitz, with a Hölder quotient of exponent $\frac{1}{2}$ at the scale of its block at most its bound. Hence the field $u := T_i \lambda$ satisfies

$$|u|_y + (L^{j_y}\eta)^{1/2} [u]_{1/2;y} \le C_{\mathrm{H}} \kappa' (L^{j_y}\eta)^{s'+1} e^{-\frac{15}{32}\delta_0 d(y,y')} |\lambda| \,,$$

with C_{H} a multiple of $B_0(\frac{1}{2})$ and of the constants of [5, (3.49)], where $[u]_{1/2;y}$ is the Hölder quotient over $\tilde{\Delta}(y)$. With a partition of unity $\sum_{y''} \zeta_{y''} = 1$ by functions supported in the cubes $\tilde{\Delta}(y'')$ with $(L^{j_{y''}}\eta)^{1/2} [\zeta_{y''}]_{1/2} \le c_\zeta$, the product rule $[\zeta u]_{1/2} \le |\zeta| [u]_{1/2} + [\zeta]_{1/2} |u|$ and [5, (3.44)] at $\varepsilon = \frac{1}{2}$, applied to each $\zeta_{y''} u$, give

$$|D^* G_C D u|_y \le (1 + c_\zeta) B'_0(\tfrac{1}{2}) \sum_{y''} e^{-\delta_0 d(y,y'')} (|u|_{y''} + (L^{j_{y''}}\eta)^{1/2} [u]_{1/2;y''}) \,,$$

and the sum over y'' is carried out as in the composition rule. So each unit has type $(\tilde{q}_i - \tilde{q}_{i-1}, \frac{13}{32}\delta_0, O(1) b_X)$, and the term has type $(2, \frac{3}{8}\delta_0, B_0 (C_{\mathrm{ser}} b_X)^n / 44^n)$, there being at most 44^n terms. The other inequalities of [5, Theorem 3.3] change only the end units, and the L^2 bounds follow in the same way from [5, (3.46)], Schur's test for P and its relatives, and the Riesz–Thorin theorem for the local factors.

For $C_{\mathrm{ser}} b_X \le \frac{1}{2}$ the sum of the terms $n \ge 1$ is at most $2 B_0 C_{\mathrm{ser}} b_X$ in each norm, so the series converges; on the finite-dimensional space of fields, $(G_C^{-1} - X) \sum_{n \le N} G_C (X G_C)^n = I - (X G_C)^{N+1} \to I$, and likewise on the other side. Each term is a finite product of operators analytic in $(A', \mathbf{J})$, and the convergence is uniform, so G_X is analytic. Replacing G_C, G', and P by their walk expansions [5, (3.90), (3.94), (3.98)–(3.99), (3.107)–(3.108)], every estimate holds for the sums of absolute values of the terms, since only the bounds on the factors were used, at the rate $\frac{1}{2}\delta_0$ which the walk expansions have. *q.e.d.*

Lemma 17.10 (The Propagators at Complex Configurations)**.** *In the setting of Lemma 17.9 with the dressed average, let* G*,* H*,* G_1*,* H_1 *be given by [5, (3.130), (3.126), (3.138), (3.129)] at* $(\mathbf{U}, \mathbf{J})$*, with* $\Delta^{(2)}$ *by [5, (3.136)] and* $\mathfrak{G} := G_1 - G_1 D R D^* G_1 - G_1 Q^* (Q G_1 Q^*)^{-1} Q G_1$*, the* *formula of [5, (3.153), p. 426]. There are* M_3 *and* $a_{\mathrm{cx}} > 0$*, depending on* d*,* L*, and* G*, such that for* $M \geq M_3$ *and* $M\alpha_0 + \alpha_1 + \gamma \leq a_{\mathrm{cx}}$*:*

(a) *[5, Corollaries 3.6, 3.8, Theorems 3.7–3.10] hold at* $\mathbf{U}$ *as stated there at* U*, with terms analytic in* A' *and depending on* $\mathbf{U}$ *as stated, and sums equal to the operators of [5, Theorem 3.4];*

(b) G *and* G_1 *satisfy the bounds of Lemma 17.9,* $\mathfrak{G}$ *satisfies them with* $\frac{1}{2}\delta_1'$ *for* δ_0 *except the bounds of* $\mathfrak{G}\nabla^*\lambda$ *for* λ *only bounded,* $(QGQ^*)^{-1}$ *and* $(QG_1Q^*)^{-1}$ *satisfy* (17.5) *with* $\delta_1' := \min(\delta_{\mathrm{cx}}, \frac{1}{2}\delta_0)/8$ *for* δ_1*, and* H*,* H_1 *satisfy [5, (3.133)] with* $\frac{1}{2}\delta_1'$ *for* $\frac{1}{2}\delta_1$*, all with walk expansions and analytic in* $(A', \mathbf{J})$*;*

(c) *when* $\mathbb{T}^{(j)}$ *has side at least* $20M$*, (a) and (b) hold on the domains (i)–(iii) of [9, p. 262] at the scale* j*, with every* $\Omega_n = \mathbb{T}$*, and on a domain* Y *for the terms of the expansions localised in* Y*;*

(d) *on these domains RG I's operator* $C^{(k)}$ *with the exponential gauge fixing and the resolvents* $(xI + \mathcal{P}^*\Delta^{(k)}\mathcal{P})^{-1}$*,* $0 \leq x \leq \gamma_1$*, of [11, (2.7)] are analytic, decay at a rate* $\varrho > 0$*, and have convergent expansions, uniformly in* x*.*

Proof. Local gauge. On a class cube $\square'$ of the index i let u be the G-valued gauge with $U^u = e^{i\eta A}$, and $x_b := \mathsf{Ad}(u(b_-))A'(b)$. With $\mathsf{bch}(x,y) := (1/i)\log(e^{ix}e^{iy})$, analytic near 0 with $|\partial\, \mathsf{bch}_2(x,y)| \leq c_{\mathsf{bch}}(|x|+|y|)$, the field $A'' := \eta^{-1}\mathsf{bch}(\eta x, \eta A)$ satisfies $\mathbf{U}^u = e^{i\eta A''}$, $|A''| < 2(O(1)M\alpha_0 + \alpha_1)(L^i\eta)^{-1}$, and the same bound for $|\nabla A''|$ with $(L^i\eta)^{-2}$, because $|\mathsf{Ad}(e^{iz}) - 1| \leq c_G|z|$. So $e^{i\eta A''}$ lies in [5, (3.37)] at $U = 1$ with $\alpha_1'' := 2L^4(O(1)M\alpha_0 + \alpha_1)$, [5, Corollary 3.5] gives [5, Theorems 3.1–3.3] there, and the covariance, which preserves every norm, carries them to $\mathbf{U}$.

Section C at $\mathbf{U}$. The proofs of [5, Theorems 3.7–3.10, Corollary 3.8] (pp. 408–416) take from the configuration only the localised operators of [5, Corollary 3.6], the identities [5, (3.88), (3.95), (3.101)–(3.105)], which hold at every G^c-valued configuration, bounds on the kernels of Q_j and on the transports, which Lemma 17.7 gives at complex configurations, the domain-difference bounds on local sequences, and the smallness and locality of Δ'; positivity and symmetry enter only in [5, Theorem 3.11]. So (a) holds.

The operators of Section D. $\Delta_\pi'(\mathbf{U}, \mathbf{J})$ is analytic and linear in $\mathbf{J}$, and Lemma 17.9 with $b_X = \gamma$ gives G and G_1. At $(\mathbf{U}, 0)$ we have $G = G_C$, and Bałaban builds the expansion of X_0^{-1}, $X_0 := Q(\mathbf{U})G_CQ^*(\mathbf{U})$, as he builds that of $(Q'G'^2Q'^*)^{-1}$ [5, (3.95)–(3.97), p. 411, l. 5–p. 412, l. 30], "in the same way" [5, p. 422, ll. 15–16]; these steps use only [5, Theorem 3.10] for G_C, the local operators of [5, Corollary 3.6], the local bound (17.5) for the inverses $C_\square$ of the local block forms, and bounds on kernels summed over walks. The local inverse at $\mathbf{U}$ follows from the local gauge: $T^{\mathrm{loc}}(\tau) := W^{-1/2}X_\square(e^{i\eta\tau A''})W^{-1/2}$ is analytic and bounded by a constant C_T for $|\tau| \leq \tau_0 := a_{\mathrm{cx}}/\alpha_1''$, the radius at which $e^{i\eta\tau A''}$ leaves the domain, where $\alpha_1'' \leq \frac{1}{2}a_{\mathrm{cx}}$ is the size of A''; it is symmetric with $T^{\mathrm{loc}}(0) \geq C_4^{-1}$ at the configuration 1. Cauchy's estimate on the disc of radius $\tau_0 \geq 2$ bounds $T^{\mathrm{loc}}(1) - T^{\mathrm{loc}}(0)$ by $C_T/(\tau_0 - 1) \leq 2C_T\alpha_1''/a_{\mathrm{cx}}$, which is small with α_1'' because the radius grows as α_1'' decreases, and Step 6 of the proof of Lemma 17.8, with $T^{\mathrm{loc}}(0)^{-1}$ in place of T^{-1}, gives (17.5) for $C_\square$ at $\mathbf{U}$. Hence X_0^{-1} has an expansion with kernels of type $(-2, \delta_{\mathrm{cx}})$, $0 < \delta_{\mathrm{cx}} \leq \delta_1$. At $(\mathbf{U}, \mathbf{J})$ the form is $X := Q(\mathbf{U})G(\mathbf{U}, \mathbf{J})Q^*(\mathbf{U})$, and $X - X_0 = Q(G - G_C)Q^*$; by Lemma 17.9 with $b_X = \gamma$ the difference $G - G_C$ satisfies the bounds of G_C with the constant $O(1)\gamma$, so that $X - X_0$ has kernels of type $(2, \frac{3}{8}\delta_0)$ with constant $O(1)\gamma$. With the weighted norm $m_r(E) := \sup_c \sum_{c'} \sum_\omega \mu(c')|E_\omega(c, c')|e^{rd(c,c')}$, which satisfies $m_r(E_1E_2) \leq m_r(E_1)m_r(E_2)$, the Neumann series $X^{-1} = X_0^{-1}\sum_n(-(X - X_0)X_0^{-1})^n$

converges for $r := \min(\delta_{\rm cx}, \frac{1}{2}\delta_0)/4$ and gives (17.5) with $\delta_1' = \frac{1}{2}r$ and the expansion. Then $H = Gf$ with $f := Q^*(QGQ^*)^{-1}$, and the composition rule gives [5, (3.133)] with $\frac{1}{2}\delta_1'$; G_1, H_1, and $\mathfrak{G}$ follow in the same way. This proves (b).

The domains of RG I. On the domains (i)–(iii) at the scale j take $\eta := \xi$, $k := j$, and $\Omega_n := \mathbb{T}$; the class cubes have side at most $20M$, so [9, (1.12)] on the cubes of side $20M$ gives [5, (3.35)] with $O(1)LMB\alpha_0$, [9, (1.13)] is [5, (3.37)], and [9, (1.14)] bounds the current, which gives (c).

The resolvents. At the real point $(U, 0)$ the form $\mathcal{P}^*\Delta^{(k)}\mathcal{P}$ is bounded below by $c_1 > 0$ at $U = 1$, uniformly in k; a partition of unity $\sum_z \psi_z^2 = 1$ by cubes of side 2ℓ, $\ell \leq 6M$, with $|\psi_z(y) - \psi_z(y')| \leq c_d|y - y'|/\ell$, and the local gauge reduce the lower bound at U to that at 1 up to errors of order ℓ^{-1} and $M\alpha_0$, so $xI + \mathcal{P}^*\Delta^{(k)}\mathcal{P} \geq \frac{1}{2}c_1$ at $(U, 0)$; the local inverses on the cubes and a Combes–Thomas argument as in Step 6 of the proof of Lemma 17.8 give the decay and the expansion at $(U, 0)$, and the Neumann series in the differences of $(QG_1Q^*)^{-1}$, H_1, $\mathbf{J}$, $\tilde{C}^{(2)}$, $\mathbf{G}^{(2)}$, and $\mathcal{P}$ between $(\mathbf{U}, \mathbf{J})$ and $(U, 0)$, all of order $O(1)LMB\alpha_0 + \alpha_1 + \gamma_0$, gives (d). *q.e.d.*

The determining sets of the paper have layers whose widths are a fixed multiple of the block size M_1 on which Bałaban's earlier theorems hold [10, p. 245, ll. 36–37], and at that block size a localisation cube of [5] may meet three layers. The theorems are therefore applied at a smaller block size.

Lemma 17.11 (The Theorems at the Block Size of the Joins)**.** *Let M' be a power of L with $M' \geq \max(M_1^{\rm PBG}, LR_0M_0, R_1M_1^{\rm VAR})$, the thresholds of [5, Theorems 3.1–3.3, p. 397], of the local sequences [5, p. 408, ll. 26–28], and of [7, p. 277, l. 17], and let $K_{M'}$ be a power of L with $K_{M'} \geq 12d+1$ and $K_{M'} \geq R_{\rm thr}(d, L)/L+1$, $R_{\rm thr}$ the largest threshold on R alone in [3], [5], and [7] at the block size M'; put $M_1 := K_{M'}M'$ and $R' := LK_{M'}-1 \geq 12dL$. Let every layer below the top one of a sequence be at least LM_1 wide in its own units. Then [3, (2.2)] holds at the block size M' with R', [7, (1)] holds with $R'R_1$ blocks of $M_1^{\rm VAR}$ after R_1 is raised to $M'/M_1^{\rm VAR}$, every localisation domain of [5] meets at most the layers $j-1$, j, $j+1$, and [5, Theorems 3.1–3.3], its expansions, and [7, Theorem 1] hold within their printed hypotheses with constants of d and L. The condition [3, (2.59)] at $\alpha = \frac{1}{128}$ together with $\frac{1}{8}\delta_*LM_1 \geq 4\log L + 1$ holds for $M_1 \geq 2M_J(d, L)$, with $M_J(d, L) := \max\{8(4\log L + 1), 512(2d\log(1024/\delta_*) + 1)\}/(\delta_*L)$ and δ_* the least of the rates of [3, (2.43)], of [5, Theorem 3.1], and of [5, p. 423, l. 39].*

Proof. The constants of Bałaban's theorems "do not depend on the sequence $\{\Omega_j\}$ …if the conditions (2.1), (2.2) are satisfied" [5, p. 399, l. 2], and in [3] the condition [3, (2.2)] enters through the length more than RM of each crossing of a layer [3, (2.57), p. 233], through [3, Lemma 2.1, p. 234], and through the factors $e^{-c\delta_0 RM\max(|j-j'|-1,0)}$ of [3, pp. 235, 237]. Since M' divides M_1 and M, the domains are unions of M'-blocks, and a layer at least $LM_1 > R'M'$ wide satisfies [3, (2.2)] with R' at the block size M'; $R'M' \geq \frac{1}{2}LM_1$ since $LK_{M'} \geq 2$. The printed conditions on R, M, R_1, and M_1 are lower bounds on RM and R_1M_1, met since $R'M' \geq \frac{1}{2}LM_1$ and $R_1M_1 = M'$, among them [7, (163), p. 304]; separate lower bounds on R, met by the choice of $K_{M'}$; divisibility conditions, met since M' is a multiple of R_0M_0 and the cube parameters M', $2M'$, $2LM'$ of Lemma 17.5 are multiples of $R_1M_1 = M'$; ratios, such as $L^{-1}RM > 2dR_1M_1$ of [6, p. 98, l. 18], which reads $R' > 2dL$, and $R > 4dLR_1$ for $\tilde{\Box} \subset \Omega_{j-1}$ in [7, p. 300, l. 36], which reads $R' > 4dL$; and upper bounds tied to M, which are smallness conditions on α_0 and ε.

The reach of Bałaban's cubes is decided by the largest one, $\tilde{\Box}^9$ of side $20M'L^j\eta$, which bounds the enlargements $\tilde{X}^5$ of the localisation domains of [5, Theorems 3.9–3.10] [5, p. 413, ll. 19–20; p. 416, ll. 7–8] and the boundary conditions of [5, p. 415, ll. 9–10]. The centre of $\square$ is a corner of a big block in Ω_j, so $\tilde{\square}^9$ reaches at most $10dM'L^j\eta$ in ℓ^1 beyond $\partial\Omega_j$, while the layer Λ_{j-1} is at least $K_{M'}M'L^j\eta$ wide and $K_{M'} \geq 10d+1$; so every localisation domain meets at most the layers $j-1$, j, $j+1$, as on Bałaban's own sequences. The regularity [5, (3.35)] at the cube size M' follows from the plaquette regularity in an axial gauge by the argument of [6, p. 98], under $L^{-1}RM > 2dR_1M_1$. Finally, the rate of [3, (2.43)] comes from the big blocks of its reference [3, p. 230, ll. 40–43], whose size is a constant of that proof, so δ_* does not depend on M_1; since $c_0(\frac{1}{256}) < 1024/\delta_*$, the condition [3, (2.59)] at $\alpha = \frac{1}{128}$, that is $\delta_*LM_1/512 > 2d\log c_0(\frac{1}{256}) + 1$, and $\frac{1}{8}\delta_*LM_1 \geq 4\log L + 1$ hold for $M_1 \geq M_J(d,L)$, and the factor 2 accounts for $R'M' \geq \frac{1}{2}LM_1$. *q.e.d.*

17.4 The Landau Gauge

Bałaban's construction of the Landau gauge [6] is carried out for the averages of [4], and the variational paper takes from it one step, the passage off the Landau slice [7, p. 296, l. 26–p. 297, l. 13; p. 302, l. 30], through the generalisation [6, Theorem 8, p. 101], whose proof is the remark that "inspecting the proofs of the theorems and propositions we can see easily that they work in this more general situation almost without any changes" [6, p. 101, ll. 9–10].

Lemma 17.12 (The Landau Gauge for the Dressed Average)**.** *Let G be a compact Lie group, U_0 and $U'U_0$ be G-valued as in [6, (1.33)–(1.35)], on a sequence with $\Omega_0 = \mathbb{T}$, and replace the averages of [4] by the dressed average: the raw average by [9, (0.4)] iterated over the levels as in [4, (43)], and every contour variable $W(\Gamma_{y,x})$ by the variable $W(y,x)$ of [9, (0.11)]. Then [6, Lemma 1] holds with a constant C_{L1} in place of $4d^2$, [6, (1.65)] with $C_{(1.65)} := 3C_{L1}$ in place of $11d^2$, and [6, Propositions 3, 5–7, Theorems 2, 4] and [4, Propositions 1, 2, 8–10, (213)–(214)] hold as stated, with constants depending on d, L, and G. Moreover, the conclusion of [7, p. 297, ll. 9–13; p. 302, l. 30], that the critical configuration of the action on the Landau slice is critical along every variation $\delta A'$ with $Q\delta A' = 0$, holds without [6, Theorem 8], for Bałaban's averages and for the dressed average.*

Proof. Covariance and size. By [9, (0.6)], $W^u(y,x) = u(y)W(y,x)u(x)^{-1}$, so the relative variables $(R_{0,y}W')(y,x) := (W'W_0)(y,x)W_0(y,x)^{-1}$ transform as in [4, p. 27] with (y,x) in place of $\Gamma_{y,x}$. A loop formed by two contours of $\mathbf{G}(y,x)$, by two contours and a bond of $B(y)$, or by a loop of [9, (0.4)], lies in a box which the plaquette bound β controls, and contracting it by plaquette moves gives $|W_0(\ell) - 1| \leq A_dL^2\beta$, $A_d := 3d^2$, since $|\prod g_i - 1| \leq \sum|g_i - 1|$ for unitary g_i.

Propositions 1–2 and Lemma 1. With these bounds in place of the contour bounds of [4], the second line of [4, (50), p. 25] holds for [9, (0.4)] with a constant C_0 by the identity [4, (48)], which gives [4, Propositions 1–2, (53)]; and [6, Lemma 1], whose proof estimates the relative variables of one level through such loops, holds with C_{L1}. Iterating it over the levels with the plaquette bound $2\alpha_0L^{-2(k-j)}$ of Proposition 2 and the coefficient 1 of α_1 gives $|\tilde{U}'^j - 1| < 2C_{L1}\alpha_0\sum_{m\geq 0}L^{-2m} + \alpha_1 < C_{(1.65)}\alpha_0 + \alpha_1$, which is [6, (1.65)].

Section F of [4]. In [4, (168)–(214)] the transports are automorphisms, the contour identities hold for each pair of contours and hence for their means, and the loop factors are within $8A_dL^2\alpha_0$ of 1, so the errors are Bałaban's plus $O(L^3\alpha_0\alpha'_4)$, of his form with a constant depending on L; hence [4, Propositions 8–10, (213)–(214)] hold, with the averaged gauge transformations composed by [4, (78)–(80)] with $V_0(y,x)$ in place of $V_0(\Gamma_{y,x})$.

The sections of [6]. Bałaban uses the averages only through the axial gauge [6, (1.19)] relative to U_0, Lemma 1, the Landau gauge [6, (1.27)] with the projection $R(U_0)$ of [5] for the dressed Q', the identities [6, (1.29)–(1.31)], [4, Proposition 4] in the form [6, (1.56)], the operators $G(U_0)$, $H(U_0)$, G', H' with their bounds, and the bounds above; the first are covariant by the first paragraph, the operators satisfy the bounds of [5] by Lemma 17.7, and every other step is his own argument on these objects. In [6, (1.71)] the bond logarithms of the dressed composite are $L^n\eta(Q_n(U_0)A + C_n(U_0; A))$, bounded by $2\kappa_G^2 B_1(\alpha_0 + \alpha_1)L^{n-j}$ by the proof of Lemma 17.7, so [6, (1.71)–(1.76)] hold with $4d\kappa_G^2 B_1$, $16d\kappa_G^2 B_1$, $32d\kappa_G^2 B_1$ for $4dB_1$, $16dB_1$, $32dB_1$, and the induction of [6, Theorem 4] closes because $B_1 = 5dLB_0$ does not depend on α_3, α_4. In [6, Proposition 6], Lemma 1 iterated as above gives [6, (1.130), (1.133)] with $(C_{(1.65)} + 5dM)L^2\alpha_0$ for $6dL^2M\alpha_0$.

The step off the Landau slice. Let A_1 be the critical configuration of $F(A) := A(e^{i\eta A}U_0)$ on $S := \{A \mid Q_j(U_0, \eta A) = B \text{ on } \Lambda_j\}$ with $RD^*A_1 = 0$, and u_1 the gauge transformation of [7, (124)]. Let E be the derivative at 0 of $C'(u_1, \cdot)$ of [4, (213)]; Cauchy's formula with [4, (214)] gives $|EH'X| \le \frac{1}{2}|X|$, and since $Q'H' = I$, the map $Q' + E$ is onto, so $N_{u_1} := \{\lambda \mid Q'(u_1, \lambda) = 0\}$ is a manifold near 0 with tangent space $\ker(Q' + E)$. This space is not $\ker Q'$, and we parametrise it by $\ker Q'$: for $\nu \in \ker Q'$ put

$$\omega(\nu) := \nu - H'(I + EH')^{-1}E\nu \; .$$

Since $Q'\nu = 0$ and $Q'H' = I$, $(Q' + E)\omega(\nu) = E\nu - (I + EH')(I + EH')^{-1}E\nu = 0$. The map $\nu \mapsto \omega(\nu)$ is injective, because $\omega(\nu) = 0$ gives $0 = Q'\nu = (I + EH')^{-1}E\nu$ and then $\nu = 0$, and it is onto $\ker(Q' + E)$, because Q' and $Q' + E$ are onto the same space and their kernels have the same dimension. Moreover $|\omega(\nu) - \nu| \le 2\|H'\||E\nu|$, which is at most a constant times $\varepsilon_2|\nu|$ by [4, (214)]. For ω in $\ker(Q' + E)$, the gauge orbit $A(s) := (1/i\eta)\log U_1^{e^{-i\lambda_s}}$ along a curve in N_{u_1} stays in S by [6, (1.30)–(1.31)] and keeps F fixed, so $T_\omega := \dot{A}(0)$ satisfies $dQ_{A_1}T_\omega = 0$ and $dF_{A_1}T_\omega = 0$. Next, RD^* is onto the range of R along these directions: for $h = Rh$ we look for $\nu \in \ker Q'$ with $RD^*T_{\omega(\nu)} = h$. Let $Z\nu := RD^*T_{\omega(\nu)} - \Delta\nu$; it consists of the error terms of $RD^*T_\nu - \Delta\nu$ and of the term $RD^*T_{\omega(\nu)-\nu}$ of the parametrisation, linear in ν because T is linear, and both are bounded by $C_{11}\varepsilon_2\|\nu\|$, the second by the bound of $\omega(\nu) - \nu$ above. The map $\nu \mapsto G'h - G'Z\nu$ takes values in $\ker Q'$ because $Q'G'R = 0$ [6, p. 93, l. 15] and contracts for $B_0'C_{11}\varepsilon_2 \le \frac{1}{2}$, and its fixed point ν gives $RD^*T_{\omega(\nu)} = h$ with $\omega := \omega(\nu) \in \ker(Q' + E)$, using $R(\Delta + Q'^*aQ')H' = 0$. So $A \mapsto (Q(U_0, \eta A), RD^*A)$ has onto derivative at A_1, by [6, (1.56)] and $RD^*H(U_0) = 0$, and $S_0 := S \cap \{RD^*A = 0\}$ is a manifold near A_1 on whose tangent space dF_{A_1} vanishes. For δ with $dQ_{A_1}\delta = 0$, choose ω with $RD^*T_\omega = RD^*\delta$; then $\delta - T_\omega$ is tangent to S_0, and $dF_{A_1}\delta = dF_{A_1}(\delta - T_\omega) + dF_{A_1}T_\omega = 0$. The transformation [7, (47)] maps the variations with $Q\delta A' = 0$ into $\ker dQ_{A_1}$, which is the conclusion; at [7, p. 302, l. 30], $U_0 = 1$ and the same argument applies to the minimum of [7, (157)]. *q.e.d.*

17.5 The Averaging Operations

Two properties of Bałaban's averaging operations [4] are used in a form he does not state: a bound on the adjoint of the linearised one-step average at a background, and the gauge transformation [4, (106), p. 33] at a complex base.

Lemma 17.13 (The Adjoint of the One-Step Average). *Let W be a real configuration on $\mathbb{T}_\eta$ with plaquette bound κ_W, $\bar{W} := MW$ with M the average [9, (0.4)], $Q_1(W) := dM(W)$ the linearisation of one step at W along $e^{i\eta A}W$, and ν a field on the bonds of $\mathbb{T}_{L\eta}$, with the pairings $\langle A, B\rangle = \eta^d \sum_b$ and $\langle \nu, \mu\rangle = (L\eta)^d \sum_c$. Let $\rho_{\mathfrak{g}} := \sup\{r \le \frac{1}{2} \mid$ every $g \in G$ with $|g - 1| < r$ is e^{iP} with $P \in \mathfrak{g}$, $|P| \le 1\}$ and $c_{\mathrm{av}} := \max(4d, 1/(8\rho_{\mathfrak{g}}))$. If $c_{\mathrm{av}}L^2\kappa_W \le \frac{1}{8}$, then*

$$\sup|Q_1(W)^*\nu| \le (1+c_Q L^{d+1}\kappa_W)\sup|\nu| + L^{d-1}\cdot L\eta\sup|D_{\bar{W}}^*\nu| \; , \qquad c_Q := 14(d+1)c_{\mathrm{av}} \; . \tag{17.6}$$

Proof. At $W = 1$ the linearisation of [9, (0.4)] averages A over the paths $\gamma_{c,x}^{\Gamma,\Gamma'} = \Gamma \cup [x, x'] \cup (-\Gamma')$, $x \in B(c_-)$, $\Gamma \in \mathbf{G}(c_-, x)$, $\Gamma' \in \mathbf{G}(c_+, x')$, with weights $L^{-d}|\mathbf{G}(c_-, x)|^{-2}$ and the factor L^{-1} per fine bond. At a real W, with Y_c the mean of the logarithms of the holonomies $h_{c,x}^{\Gamma,\Gamma'}$ of the loops $\gamma_{c,x}^{\Gamma,\Gamma'} \cup (-c)$, the derivative of $e^{iY_c}W(c)$ is $\mathsf{dexp}_{Y_c}(i\,dY_c) + \mathsf{Ad}(e^{iY_c})(dW(c)W(c)^{-1})$, and an occurrence of a bond b in a path contributes the adjoint action of the transport from c_- to b_-.

Each loop is contracted to the trivial loop by at most dL^2 plaquette moves, each multiplying the holonomy by a conjugated plaquette variable, so $|h_{c,x}^{\Gamma,\Gamma'} - 1| \leq dL^2\kappa_W$ and $|Y_c| \leq 2dL^2\kappa_W$; since $\kappa_W \leq \rho_{\mathfrak{g}}/L^2$, every move multiplies by e^{iQ} with $Q \in \mathfrak{g}$, $|Q| \leq 2\kappa_W$, and the path of partial products stays within $\frac{1}{16}$ of 1, along which the set where the principal logarithm lies in $i\mathfrak{g}$ is open and closed, so $Y_c \in \mathfrak{g}$. Hence $\mathsf{Ad}(h)$, $\mathsf{Ad}(e^{iY_c})$, dexp_{Y_c}, and dlog_h are within $c_{\text{av}}L^2\kappa_W =: \theta$ of the identity, and so are their adjoints, because $\mathsf{ad}(iZ)$ is antisymmetric for $Z \in \mathfrak{g}$.

Replacing these four operators by the identity, the adjoint splits into three groups. A bond b lies on the straight segment $[x, x']$ for exactly L points x, each of total weight L^{-1}, with isometric transports, which gives at most $\sup|\nu|$. A bond with both endpoints in a block $B(y)$ lies on the first contour of the paths of the d bonds leaving y and on the reversed last contour of the paths of the d bonds entering y, each with weight $L^{-1}|\mathbf{G}(y, \hat{x})|^{-1}$. At a curved W these contours do not carry $\nu(c)$ to b by one transport: the transports from y to b_- along two contours through b differ by the adjoint action of the holonomy of a closed contour inside $B(y)$, which is contracted by at most dL^2 plaquette moves and is therefore within θ of the identity, as above. We replace all of them by the transport along one fixed contour, which is a fifth replacement of an operator by one within θ of it. After it, all the path terms carry $\nu(c)$ to b by the same transport, the entering ones after $\mathsf{Ad}(\bar{W}(c))^{-1}$ and with the opposite sign; their sum is $-L\eta D^*_{\bar{W}}\nu(y)$ times $L^{-1}N_b$ with $N_b \leq L^d$, which gives the second term of (17.6). The deviations from the five replaced operators change each path term by at most $(1+\theta)^5 - 1 \leq 7\theta$ times its weight, since $\theta \leq \frac{1}{8}$, the path terms having total weight at most $1 + 2dL^{d-1}$, and the term of the outer factor $dW(c)W(c)^{-1}$, which vanishes at $W = 1$ because the weights sum to 1, is at most $5\theta L^{d-1}\sup|\nu|$. In all the deviations are at most $14(d+1)\theta L^{d-1}\sup|\nu| = c_Q L^{d+1}\kappa_W \sup|\nu|$. *q.e.d.*

Lemma 17.14 (The Gauge Transformation at a Complex Base)**.** *Let U be G-valued with the plaquette condition [4, (109)], $\mathbf{U} = e^{i\eta A'}U$ and $W = e^{i\eta X}\mathbf{U}$ with A' and X $\mathfrak{g}^c$-valued and bounded by ε_A and ε_X in the form of [6, (1.69)], $|\cdot| < \varepsilon(L^j\eta)^{-1}$ and $|\nabla^\eta\cdot| < \varepsilon(L^j\eta)^{-2}$ on Ω_j. For $\varepsilon_A + \varepsilon_X$ small, the gauge transformation $a_{\mathbf{U}}(W)$ near 1 with $W^a \in \mathrm{Ax}(\mathbf{U})$ and averaged transformation 1 on the sets Λ_j, which is [4, (106)] at a G-valued base, exists, is unique, and satisfies $|\log a_{\mathbf{U}}(W)| \leq 72d(\varepsilon_A + \varepsilon_X)$.*

Proof. Bałaban takes the base G-valued [6, (1.33), p. 82], [4, p. 34, ll. 4–9], and his bound [6, (1.70)] uses that conjugation by the transports of the base is unitary. Three facts extend to G^c-valued g near 1 by the identity theorem, both sides being analytic and equal at real arguments: the covariance $(\overline{V^g}^j)_c = g(c_-)(\bar{V}^j)_c g(c_+)^{-1}$ [6, p. 78, l. 14]; the description of the axial gauge [6, (1.19)–(1.20), p. 79] relative to a base B as the configurations with the same averaged transports along the axial contours as B, the product of [6, (1.19)] telescoping; and the averaged gauge transformation N_B of [4, (78)–(80), p. 30]. So $\mathrm{Ax}(B)$ and N_B depend on B only through its transports along the axial contours, [4, (106)] continued to a complex base is a_B, and $a_B(V)$ is unique: two solutions differ by an h with $h(x) = \mathsf{Ad}(B(\Gamma^{(j)}_{y,x})^{-1})h(y)$, for which $N_B(hv)(y) = h(y)N_B(v)(y)$, so the normalisation forces $h = 1$. The same uniqueness gives $a_{B^g}(V^g) = g a_B(V) g^{-1}$.

Let $\tilde{u} := a_U(\mathbf{U})$ and $\hat{u} := a_U(W)$, gauges at the G-valued base U. Since $\mathbf{U}^{\tilde{u}} \in \mathrm{Ax}(U)$ has the transports of U, $a_{\mathbf{U}^{\tilde{u}}} = a_U$, and the covariance gives $a_{\mathbf{U}}(W) = \tilde{u}^{-1} a_U(W^{\tilde{u}})\tilde{u}$; as $W^{\tilde{u}} = (W^{\hat{u}})^{\tilde{u}\hat{u}^{-1}}$ with $W^{\hat{u}} \in \mathrm{Ax}(U)$, the uniqueness argument gives $a_{\mathbf{U}}(W) = \tilde{u}^{-1}h'\hat{u}$ with $h'(x) = \mathrm{Ad}(U(\Gamma^{(j)}_{y,x})^{-1})h'(y)$, $h'(y) = N_U(\hat{u}\tilde{u}^{-1})(y)^{-1}$, and $|\log h'(x)| = |\log h'(y)|$ since U is unitary.

The gauges $\tilde{u}$ and $\hat{u}$ are [4, (106)] at the G-valued base U for the perturbations $e^{i\eta A'}$ and $e^{i\eta\hat{H}}$, $\eta\hat{H} := (1/i)\log(e^{i\eta X}e^{i\eta A'})$, and by the Baker–Campbell–Hausdorff formula $\hat{H}$ satisfies the bounds at $\varepsilon_W := (\varepsilon_A + \varepsilon_X)(1 + O(\varepsilon_A + \varepsilon_X))$. At a point of the layer j, [4, (106)] is a product over the levels $n < j$ of brackets conjugated by transports of U, each the averaged contour times the inverse of its normalisation [4, (105)]; both exponents of the n-th bracket are at most $t_n < 4d\varepsilon L^{n+1-j}$, $\varepsilon := \varepsilon_A + \varepsilon_W$, by [6, (1.71)], which prints L^{n+l-j} where the factor $1/(1 - L^{-1})$ of [6, (1.72)] requires L^{n+1-j}. The exponents therefore sum to at most $16d\varepsilon \le \frac{1}{16}$ for $256d\varepsilon \le 1$, and $|\log\tilde{u}| \le 32d\varepsilon_A$, $|\log\hat{u}| \le 32d\varepsilon_W$. For h', the averaged gauges of $\tilde{u}$ and $\hat{u}$ are 1 at the points of Λ_j and have oscillations at most $4t_n$ at the level n; the averaging step $M[w] = w(y)\exp(\sum_x L^{-d}\log(w(y)^{-1}w(x)))$ satisfies $|\log M[w] - \mathrm{avg}_x \log w(x)| \le C_0 s \max_x|\log w(x) - \log w(y)|$ for logarithms at most $s \le s_0$, by Schwarz's lemma in each argument, and the defect $r_n := \log N^n_U(\hat{u}\tilde{u}^{-1}) - \log(N^n_U(\hat{u})N^n_U(\tilde{u})^{-1})$ satisfies $\sup|r_{n+1}| \le (1 + 2C_0 s)\sup|r_n| + C_1(s + t_n)t_n$, so $|\log h'| = |r_j(y)| \le C_2 d^2\varepsilon^2 \le d\varepsilon$. Hence $|\log a_{\mathbf{U}}(W)| \le \frac{33}{32}(32d\varepsilon_A + d\varepsilon + 32d\varepsilon_W) \le 35d\varepsilon \le 72d(\varepsilon_A + \varepsilon_X)$ for $\varepsilon_A + \varepsilon_X$ small. *q.e.d.*

17.6 The Inductive Description of the Densities

Bałaban writes the expectation of a localised observable pinned at a bond b as a sum of terms $\mathbf{E}_0(\Lambda_{k+1}, X, b)$ over $X \ni b$ [10, (3.44)–(3.47), p. 277, l. 31–p. 278, l. 31], and bounds them by [10, (2.42)] with B_0 replaced by $O(p_0^3(g_k))$ [10, p. 278, ll. 32–33], at the rate κ. The improved decay of the newly created terms [10, p. 262, ll. 24–29] needs the larger rate which the construction in fact gives.

Lemma 17.15 (The Rate of the Pinned Expansion). *Let* $\mathfrak{a}_H := (1 - 8\delta)\frac{1}{2}L\kappa$ *and* $\mathfrak{a}_E := (1 - 10\delta)\frac{1}{2}L\kappa$ *be the rates of [11, (2.38), (2.41)],* $\eta_0 := 2^9(6\log 2 + 1)$, *and* $K_{\mathbf{D}} := 2^{6\cdot 2^{10}} + 1$. *Assume that the factors* $H'(Z_0)$ *of [10, (3.44)] for the domains* $Z_0 \ni b$ *satisfy* $|H'(Z_0)| \le B'e^{-\mathfrak{a}_H d_{k+1}(Z_0)}$, *that the exponents* $\tilde{\mathbf{E}}(\Lambda_{k+1}, Y)$ *of [10, (3.46)] satisfy* $|\tilde{\mathbf{E}}(\Lambda_{k+1}, Y)| \le c_{\mathrm{cl}}e^{-\mathfrak{a}_E d_{k+1}(Y)}$ *with* $c_{\mathrm{cl}} = O(1)C_3\varepsilon_1 \le \frac{1}{2}$, *and that* $\mathfrak{a}_E \ge \eta_0$, $\delta L\kappa \ge 1$, *and* $c_\Sigma := 9 \cdot 2^{10}c_{\mathrm{cl}}e^{5\mathfrak{a}_E}K_{\mathbf{D}} \le 1$. *Let*

$$\mathbf{E}_0(\Lambda_{k+1}, X, b) := \sum_{Z_0 \ni b} H'(Z_0) \sum_{\mathcal{Y}'} \prod_{Y \in \mathcal{Y}'} (e^{-\tilde{\mathbf{E}}(\Lambda_{k+1}, Y)} - 1) , \tag{17.7}$$

the inner sum over the finite sets $\mathcal{Y}'$ *of domains of [10, (3.46)] with* $Z_0 \cup \bigcup\mathcal{Y}' = X$, *be the Mayer expansion of the exponential of [10, (3.46)] resummed by* X *[10, p. 278, ll. 13–15]. Then* $|\mathbf{E}_0(\Lambda_{k+1}, X, b)| \le 2^6 K_{\mathbf{D}} B'e^{-(\mathfrak{a}_E - \eta_0)d_{k+1}(X)}$ *for every* $X \ni b$.

Proof. We first bound one term. Fix Z_0 and a family $\mathcal{Y}'$ with $Z_0 \cup \bigcup\mathcal{Y}' = X$. Each $Y \in \mathcal{Y}'$ has a cube equal to, or sharing a wall with, a cube of Z_0, and a segment between points of these two cubes lies in their union and has length at most $5M$, so shortest trees of the members joined by such segments give $d_{k+1}(X) \le d_{k+1}(Z_0) + \sum_{Y \in \mathcal{Y}'}(d_{k+1}(Y) + 5)$. Let $\mathfrak{a}_P := \mathfrak{a}_E - \eta_0 \ge 0$, so that $\mathfrak{a}_H = \mathfrak{a}_P + \delta L\kappa + \eta_0$ and $\mathfrak{a}_E = \mathfrak{a}_P + \eta_0$. Since $|e^{-u} - 1| \le |u|e^{|u|} \le 2|u|$ for $|u| \le \frac{1}{2}$, the hypotheses bound the term by

$$B'e^{-\mathfrak{a}_H d_{k+1}(Z_0)} \prod_{Y \in \mathcal{Y}'} 2c_{\mathrm{cl}}e^{-\mathfrak{a}_E d_{k+1}(Y)} \le B'e^{-\mathfrak{a}_P d_{k+1}(X)}e^{-(\delta L\kappa + \eta_0)d_{k+1}(Z_0)} \prod_{Y \in \mathcal{Y}'} u_Y ,$$

with $u_Y := 2c_{\mathrm{cl}}e^{5\mathfrak{a}_P}e^{-\eta_0 d_{k+1}(Y)}$, by the inequality for $d_{k+1}(X)$. We next sum over $\mathcal{Y}'$. Dropping the condition on the union, $\sum_{\mathcal{Y}'}\prod_{Y\in\mathcal{Y}'}u_Y \le \exp\sum_Y u_Y$, the sum over all domains Y of [10, (3.46)] for Z_0. Each of them contains one of the at most $9N(Z_0) \le 9\cdot 2^9(1+d_{k+1}(Z_0))$ cubes which equal or share a wall with a cube of Z_0, by Lemma 17.1, and the second part of that lemma with η_0 for κ bounds the sum over the Y containing a given cube, so that $\sum_Y u_Y \le c_\Sigma(1+d_{k+1}(Z_0))$, using $\mathfrak{a}_P \le \mathfrak{a}_E$. Finally a bond lies in at most 2^4 cubes of π_{k+1}, and since $\delta L\kappa \ge 1 \ge c_\Sigma$,

$$\sum_{Z_0\ni b} e^{c_\Sigma}e^{-(\delta L\kappa+\eta_0-c_\Sigma)d_{k+1}(Z_0)} \le e\sum_{\square\ni b}\sum_{Z_0\supset\square} e^{-\eta_0 d_{k+1}(Z_0)} \le 2^4 eK_{\mathbf{D}} \le 2^6 K_{\mathbf{D}}\ ,$$

by Lemma 17.1 again, which gives the bound. *q.e.d.*

Bałaban requires of the profile φ_j of the coupling function [10, (2.24)] only $\varphi_j \in C_0^\infty(\Lambda_j)$ and $\varphi_j = 1$ on $\Lambda_j^{\sim -1}$ [10, p. 259, ll. 20–21], of ζ_0 [12, (1.100)] only that it lie in $C_0^\infty(Z)$ and change from 0 to 1 near $\partial Z^{\sim -2}$ [12, p. 201, ll. 28–29], and of ζ_1 [13, (1.70)] only that it lie in $C_0^\infty(Z''_{h+1})$ and change from 0 to 1 "on a layer of the width 2 at the boundary, in L^{-h}-scale" [13, p. 360, ll. 34–35]. The paper needs their gradients bounded in the units of the layer.

Lemma 17.16 (The Profiles)**.** *Let* $\mathfrak{j} \ge 0$ *be smooth, of integral* 1*, supported in the Euclidean unit ball,* $\mathfrak{j}_r(z) := r^{-d}\mathfrak{j}(z/r)$*, and* $a_j := L^j\eta$*. Let* s_j^φ *be the least side of the partition cubes of* Λ_j*,* S_j^φ *the set of points at* ℓ^∞*-distance at least* $\frac{1}{2}s_j^\varphi$ *from* Λ_j^c*, and* $\varphi_j := 1_{S_j^\varphi} * \mathfrak{j}_{s_j^\varphi/4}$*; let* $\zeta_1 := 1_{S^{\zeta_1}} * \mathfrak{j}_{a_h/2}$ *with* S^{ζ_1} *the points at* ℓ^∞*-distance at least* a_h *from* $(Z''_{h+1})^c$*, and* $\zeta_0 := 1_{S^{\zeta_0}} * \mathfrak{j}_{a_k/2}$ *with* S^{ζ_0} *the points at* ℓ^∞*-distance at most* $\frac{1}{2}a_k$ *from* $Z^{\sim -2}$*. Then these profiles have Bałaban's properties, and, with* $j(x)$ *the layer of* x*,*

$$a_{j(x)}|\nabla\varphi_j(x)| \le c_{2.24} := 4\|\nabla\mathfrak{j}\|_1\ , \qquad a_{j(x)}|\nabla\zeta_0(x)|,\ a_{j(x)}|\nabla\zeta_1(x)| \le c_\zeta := 2\|\nabla\mathfrak{j}\|_1\ ; \tag{17.8}$$

for $j \le k-1$ *the gradient of* φ_j *vanishes off the layer* j*, also after* **R***-operations.*

Proof. For $F \subset \mathbb{R}^d$ and $r > 0$ the function $f := 1_F * \mathfrak{j}_r$ is smooth, $0 \le f \le 1$, $|\nabla f| \le \|\nabla\mathfrak{j}\|_1/r$, and $f(x) = 1$ if the ℓ^∞-ball of radius r about x lies in F, $f(x) = 0$ if it misses F, the Euclidean ball lying in the ℓ^∞ one; ∇f vanishes where f is 0 or 1. Hence $\varphi_j = 0$ within $\frac{1}{4}s_j^\varphi$ of Λ_j^c and $\varphi_j = 1$ beyond $\frac{3}{4}s_j^\varphi$, which contains $\Lambda_j^{\sim -1}$ because a point of Λ_j within s_j^φ of Λ_j^c lies in a cube touching Λ_j^c; and $|\nabla\varphi_j| \le 4\|\nabla\mathfrak{j}\|_1/s_j^\varphi$, supported in the strip between $\frac{1}{4}s_j^\varphi$ and $\frac{3}{4}s_j^\varphi$. For $j \le k-1$ this strip lies in the layer j if $\mathsf{dist}(\Omega_{j+1}, \Lambda_j^c) \ge s_j^\varphi$, since $\Lambda_j \subset \Omega_j$ [10, (2.1)], and then $a_{j(x)} = a_j \le s_j^\varphi/M$ there, which gives the first bound. The distance bound holds for every sequence of domains of the expansion, by induction over the times: a renormalisation step puts Ω_{j+1} at distance at least $8MR_{j+1}a_{j+1} \ge s_j^\varphi$ from Λ_j^c, by Lemma 17.17 and $R_j \le LR_{j+1}$, which follows from $R_j \le (L+1)R_{j+1}$ [10, (2.9), p. 256] because the R_j are powers of L [10, (2.5)]; an **R**-operation on Z replaces Z_j by Z''_j for $h+1 \le j \le k$ [12, (1.10)–(1.12), p. 179], which keeps the bound within Z by the widths of Lemma 17.17, keeps it across ∂Z because the points of Ω_k^c on either side of ∂Z are at least $2MR_k a_k \ge s_j^\varphi$ from the other side, and does not increase s_j^φ. For ζ_1, the same properties of the mollification give $\zeta_1 \in C_0^\infty(Z''_{h+1})$, changing from 0 to 1 on the layer of width $2a_h$, with $|\nabla\zeta_1| \le 2\|\nabla\mathfrak{j}\|_1/a_h$, and $a_{j(x)} \le a_h$ on Z''_{h+1} because $\Omega''_{h+1} = (Z''_{h+1})^c$ on Z [12, (1.12)]. For ζ_0, $Z^{\sim -2}$ lies at distance at least $MR_k a_k \ge 2a_k$ from Z^c, Z being a union of cubes of side $MR_k a_k$, so $\zeta_0 \in C_0^\infty(Z)$, $\zeta_0 = 1$ on $Z^{\sim -2}$, and $a_{j(x)}|\nabla\zeta_0| \le a_k \cdot 2\|\nabla\mathfrak{j}\|_1/a_k$. *q.e.d.*

The large-field paper and the paper use widths of the layers of the determining sets that Bałaban's papers use but do not state. For a step k let $\mathbf{B}''_k$ be the determining set after the

R-operations, $h = k - N$, and $\mathbf{B}_1$, $\mathbf{B}_2$, $\mathbf{B}_{1.70}$ the joins of [13, p. 371] and [13, (1.70)], formed by [10, (2.14), p. 257] within Z.

Lemma 17.17 (The Widths of the Layers)**.** *Every layer of* $\mathbf{B}''_k$ *below the top one is at least* M *wide in its own units; on* Z *the layer* j *is at least* $3MR_{j+1}a_{j+1}$ *wide for* $h \le j \le k - N_0 - 1$*, the layer* $k - N_0$ *at least* $LMR_{k-N_0+1}a_{k-N_0}$*, and the layers above it one cube of side* Ma_{j+1} *wide; and* $\mathsf{dist}(\Omega_{j+1}, \Lambda_j^c) \ge 8MR_{j+1}a_{j+1}$ *after a renormalisation step. Every layer of* $\mathbf{B}_1$, $\mathbf{B}_2$*, and* $\mathbf{B}_{1.70}$ *below the top one is at least* LM_1 *wide in its own units, the hypotheses of [10, (2.14)] hold for the three joins, and the top layers of* $\mathbf{B}_2$ *and* $\mathbf{B}_{1.70}$ *are crossed by no contour of [3, (2.47)]. Consequently the counts [3, (2.57)–(2.61)] hold on these sets with* RM *replaced by* LM_1 *under the condition (J-W) of Lemma 17.11.*

Proof. The step layers. The boundaries of Ω_j and Λ_j are at least $2MR_j$ apart [10, p. 256, ll. 21–24], and $\Omega_{j+1} \subset \Lambda_j \subset \Omega_j$ [10, (2.1)]. By [10, (3.5), p. 265], the step $j \to j+1$ puts Ω_{j+1} three layers of cubes of side $MR_{j+1}a_{j+1}$ inside $B^{j+1}(P^1_{j+1})$, which lies five further layers inside the complement of the union of the cubes meeting Λ_j^c [10, p. 264, ll. 52–55; p. 265, ll. 16–22], so $\mathsf{dist}(\Omega_{j+1}, \Lambda_j^c) \ge 8MR_{j+1}a_{j+1}$.

On Z*.* By [12, (1.2), p. 178], between the scales h and k only one-step operations without large-field characteristic functions occur, so the Ω_{j+1} on Z are the step domains. By [12, (1.10)–(1.12), p. 179] the layer j on Z is $\Omega_j^{\sim 5} \backslash \Omega_{j+1}^{\sim 5}$ for $h+1 \le j \le k-N_0-1$, $\Omega_{k-N_0}^{\sim 5} \backslash \Omega_{k-N_0+1}^{\sim 7}$ for $j = k - N_0$, and the layers above are separated "by one layer of M-cubes" of the scale $j+1$ [12, p. 179, ll. 21–24]; the enlargements move $\partial\Omega_{j+1}$ outwards by at most 5 or 7 cubes, which leaves the stated widths, each at least $LM > LM_1$ since $M > M_2 = L^2 M_1$ [10, p. 245, ll. 35–38]. An **R**-operation changes the domains only on its component and only enlarges them there. Across ∂Z, since $\Lambda_k^c \supseteq (\Omega_k^c)^{\sim 2}$ [10, (3.20), p. 269], a point of $\Omega_k^c \cap Z^c$ and its two-layer neighbourhood lie in one component of Λ_k^c other than Z, so the cross pairs are at least $2MR_k a_k \ge 2ML^{k-j}a_j$ apart; by induction over the times every layer of $\mathbf{B}''_k$ is at least M wide.

The joins. The collar $\mathbf{B}_{j_0}(\Omega)$ has $\mathsf{dist}(\Omega_n, \Omega_{n-1}^c) \ge L^n \xi M_1$ [10, p. 256, ll. 56–61], so its layers are LM_1 wide, and its strip $\Omega \backslash \Omega^{\sim -2}$ lies within $2M_1$ of Ω^c in ℓ^∞. The joins are formed within Z, where Z''_{h+1} lies at distance at least $5LMR_{h+1}$ from Z^c in the units a_h, since a path from Z to Z^c meets $\Lambda_k \subset \Omega_{h+1}$; their boundaries lie inside layers of $\mathbf{B}''_k$ at distance at least $LMR_{h+1} > 2M_1$ from the boundaries contained in Ω, which is the margin of [10, (2.14)] [10, p. 257, ll. 10–11], and they are unions of M_1-cubes since M, M_1, and the R_j are powers of L and the partitions are compatible [10, p. 245, ll. 38–40]. The domains of a join are $(\Omega''_j \cap \Omega^{\sim -2}) \cup (\Omega_j \cap S)$, and in each of the four cases (collar to collar, collar to Ω^c, $\mathbf{B}''_k$ to $\mathbf{B}''_k$, and across the strip) the distance from the domain $j+1$ to the complement of the domain j is at least LM_1 in the units a_j, the last case because $M/L - 2dM_1 > (L - 2d)M_1 \ge M_1$. The top layer j_0 of $\mathbf{B}_2$ and $\mathbf{B}_{1.70}$ is crossed by no contour, since a crossing in a layer joins the surfaces Σ_{j_0} and Σ_{j_0+1} [3, (2.47), p. 231] and these joins have no Σ_{j_0+1}. Finally Bałaban's count [3, (2.58)] uses [3, (2.2)] only through the length more than RM of each crossing [3, (2.57), p. 233], and [3, (2.60)] follows from [3, (2.48), (2.57)], so these counts hold with RM replaced by any common lower bound of the widths of the crossed layers. *q.e.d.*

Bałaban states the conditions [10, (2.35), (2.37), p. 261, ll. 17–21] on the configurations derived from a critical one by localising its determining set, as conditions of the class [12, p. 190, ll. 20–31], and does not derive them from the conditions [10, (2.34), (2.36)] on the critical configuration itself. Let $\mathbf{B}$ be the determining set of a sequence $\mathbb{T}_\eta = \Omega_0 \supset \cdots \supset \Omega_k$ of the step k, $X \in \mathbf{D}_m$ with $X \subset \Omega_{m+1}^c$ [12, (1.63), p. 189], $1 \le p \le k$, $\mathbf{B}^X$ the determining set of $U_{p,X} = U(\mathbf{B}_p(X) \cup \{\Gamma_i\}_{i<p}, \cdot)$ [10, p. 261, l. 10], and $\hat{X}$ the graph of the bonds with an endpoint in X.

Lemma 17.18 (The Derived Configurations)**.** *(a) If y is a point of $\mathbf{B}^X$ of index n, then $B^n(y) \subset \Omega_n$ and $B^n(y)$ lies in a block of $\mathbf{B}$ of index at least n; and if two configurations agree off X and have the same data on $\mathbf{B}^X$, they have the same data on $\mathbf{B}$.*

(b) Real data. *For $\varepsilon_1 \le \varepsilon_1^*$, a constant of d, L, and G, and a G-valued U which minimises the action on the space of $\mathbf{B}$ [7, (6)] with parameter $B_3\varepsilon_1$, such as $U_{\mathbf{B}}(V)$, the minimal configuration of $\mathbf{B}^X$ at the data W of U is $U_{p,X}(W) = (U\lceil_X)^w$ for a G-valued gauge transformation w on $\hat{X}$ with $w = 1$ at the points of $\mathbf{B}^X$ and of $\hat{X} \setminus X$.*

(c) Complex data. *For data $V = e^{iB}V_0$ with B in a convex ball $\mathcal{V}$ about 0, the minimal configuration $F(V)$ of $\mathbf{B}^X$ in the axial gauge at the data of the Landau representative $U_L(V)$ of $U_{\mathbf{B}}(V)$ satisfies $F(V) = U_L(V)^{w(V)}$ on $\hat{X}$, with $w(V)$ invertible-valued, analytic in B, equal to 1 at the points of $\mathbf{B}^X$ and of $\hat{X} \setminus X$, and G-valued at $\mathfrak{g}$-valued B near 0.*

(d) Plaquettes and currents. *On $\mathcal{V}$, at every plaquette P with bonds in $\hat{X}$ and base point x, $\partial F(P) = R(w(x))\partial U_L(P)$, and at every bond b whose plaquettes have their bonds in $\hat{X}$, with initial point x, $J(F)(b) = R(w(x))J(U_L)(b)$, with $J(Y) := D_Y^{\eta *}\eta^{-2}\pi \operatorname{Im}\partial Y$; hence $|\partial F(P) - 1| \le \Theta(x)|\partial U_L(P) - 1|$ and $|J(F)(b)| \le \Theta(x)|J(U_L)(b)|$ with $\Theta(x) := |w(x)||w(x)^{-1}|$.*

(e) The factor Θ. *$\Theta(x) \le e^{d(t(B)+(24\sqrt{N}+1)t^X)}$ at every $x \in \hat{X}$, where $t(B)$ and t^X bound the Landau fields of $U_L(V)$ and of $F(V)$ relative to the real configurations in the units of the layers.*

Proof. (a). The cubes $\Delta(y)$, $y \in \mathbb{T}^{(n)}$, form nested divisions, and a coordinate of a point of $\mathbb{T}^{(i)}$ is an odd multiple of $\frac{1}{2}L^i\eta$, while the faces of a cube of side $L^l\eta$ lie at multiples of $L^l\eta$. So for $n \le l$ and a union A of cubes of side $L^l\eta$, a cube Δ of side $L^n\eta$ either lies in A or has no lattice point in A; every block $b \subset \Omega_n$ of side $L^n\eta$ then lies in the block of $\mathbf{B}$ of the largest index $h \ge n$ with $b \subset \Omega_h$. By the joining [10, (2.14)] [10, p. 257, ll. 7–26], the points of $\mathbf{B}^X$ outside Ω_p are points of $\{\Gamma_i\}_{i<p}$ in $X \setminus X^{\sim -2}$ or of $\mathbf{B}_q(X)$ in the strip, and inside Ω_p points of $\mathbf{B}_p(X)$, whose blocks lie in the Ω_n by the margin of [10, (2.14)], which gives the first claim. A datum of index n depends only on the bonds with both endpoints in $\Delta(c_-) \cup \Delta(c_+)$, by the locality of the averages [9, (0.4)], and the first claim places these cubes in blocks of $\mathbf{B}$ or outside X, which gives the second.

(b). The data of $\mathbf{B}^X$ are functions of the configuration on $\hat{X}$, the problem of $\mathbf{B}^X$ is that of [7, (5)–(6)] for its sequence, and its sequence satisfies Bałaban's standing conditions at the block size of Lemma 17.11. The restriction $U\lceil_X$ satisfies [7, (2)] for this sequence with $\varepsilon_0^X := \max\{2(d-1), 2L^2B_3\}B_3\varepsilon_1$, its data satisfy [7, (7)] with $\varepsilon_1^X := 2L^2B_3\varepsilon_1 \le a_1$ by [4, Proposition 2] for the dressed average (Lemma 17.12), and it is critical for the restricted problem: a variation of $U\lceil_X$ preserving the data of $\mathbf{B}^X$ and vanishing at the bonds leaving X extends by U to a variation preserving the data of $\mathbf{B}$, by (a), and the action of $\mathbf{B}$ is the action of $\mathbf{B}^X$ plus terms off X. By [7, Theorem 1] the critical orbit in the space of $\mathbf{B}^X$ is unique, so $U\lceil_X$ lies on the minimal orbit, $U_{p,X}(W) = (U\lceil_X)^w$, and $w = 1$ at the points of $\mathbf{B}^X$ and off X by the invariance of the space under the transformations equal to 1 there [7, p. 278, ll. 16–19, 32–33].

(c). Fix a point y_0 of $\mathbf{B}^X$ and put $w(x) := F(\gamma)^{-1}U_L(\gamma)$ for a path γ in $\hat{X}$ from y_0 to x. The holonomies of F and U_L along every path between points of $\mathbf{B}^X$ agree: at real data by (b), and on $\mathcal{V}$ because both sides are analytic in B and agree on the real ball. So w does not

depend on the path, $F = U_L^w$ on $\hat{X}$, w is analytic, and $w = 1$ at the points of $\mathbf{B}^X$; off X, F and U_L agree.

(d). For every u with invertible values, $(\partial U^u)(p_{\mu\nu}(x)) = R(u(x))(\partial U)(p_{\mu\nu}(x))$, the product around the plaquette telescoping [6, p. 77, l. 48], and $R(w)1 = 1$; the current is a covariant derivative of the projected plaquette variables, which transforms in the same way.

(e). At real data near V_0 the configuration F is the axial representative of the orbit of its Landau form $F_L = e^{i\eta\mathcal{H}^X} F_0$, so $F = F_L^{v^X}$ with v^X the gauge transformation [4, (106)] in the form for the dressed average, and on $\mathcal{V}$ by analyticity. At a point x of a block $B^n(y)$ of $\mathbf{B}^X$, $n \geq 1$, the factor $l < n$ of [4, (106)] is a G-valued conjugate of a bracket whose logarithm is at most $dL\mathfrak{a}_l$, $\mathfrak{a}_l := 2\sqrt{N}t^X L^{l-n}$, by the size of the composed averages of $e^{i\eta\mathcal{H}^X}$ and the smallness of the plaquettes of F_0, so $|v^X(x)^{\pm 1}| \leq e^{6d\sqrt{N}t^X}$. A path γ from y to x monotone in every coordinate has at most $\frac{1}{2}d(L^n - 1)$ bonds in $B^n(y)$, on which $|\eta\mathcal{H}^X| \leq t^X L^{-n}$ and the Landau field of U_L is at most $t(B)L^{-n}$, so $|F_L(\gamma)^{\pm 1}| \leq e^{dt^X/2}$ and $|U_L(\gamma)^{\pm 1}| \leq e^{dt(B)/2}$, the real configurations being unitary. Since $w(y) = 1$, $w(x) = v^X(x)F_L(\gamma)^{-1}v^X(y)^{-1}U_L(\gamma)$, and both $w(x)$ and $w(x)^{-1}$ have norm at most $e^{12d\sqrt{N}t^X + d(t(B)+t^X)/2}$, whose square is the bound on $\Theta(x)$. *q.e.d.*

17.7 The Renormalisation Transformations with Large Fields

At a step of the **R**-operation the letters of the loops and the carried insertions are functions of the configuration before the operation, which depend on the fields of the class components it renormalises; the ratio of gases of Proposition S.10 needs them localised over these components. Bałaban's localisation expansion [13, pp. 372–376], which he applies to the terms of the action, applies to them.

We use the notation of Section 8: a step k of an **R**-operation, a term of [13, (1.72), p. 379], the union Z of its class components, the sets K^+, and the homes. The configuration U^{old} before the operation is Bałaban's [13, (1.61)] with the left factors $\exp i\eta\mathbf{H}_k''(g_k B)$ and the preliminary deformations of [12, p. 191, ll. 39–41]. Cubes are those of π_k, $d_\square$ is the ℓ^∞ distance of their indices, and $S^{[n]}$ is the set of cubes at $d_\square \leq n$ from S.

Lemma 17.19 (The Localisation of the Letters)**.** *Let F be one of the following functions of U^{old}: a letter $\pi(T_i(U^{\mathrm{old}}))$ of a bond i of a loop which lies in no set K^+; the tensor product F_h of the letters of the bonds of the loops in $\bigcup_{K \in h} K^+$, for a home h, with the projective norm; or a carried insertion. Let $\|F\|$ be its bound on the class, $X(F)$ the union of the cubes which meet the bonds on which F depends, $\hat{T}_0$ the union of $X(F)^{[1]}$, of the components meeting $X(F)$, and of the sets K^+ of the components of the home of F whose sets K^+ meet these bonds, and $\hat{T}$ the union of $\hat{T}_0$ and of the components at $d_\square \leq R_k - 3$ from $\hat{T}_0$; the other components of Z are the* units *of F. For g_k small there is a decomposition $F(U^{\mathrm{old}}) = \sum_{\mathcal{K}} F^{\mathcal{K}}$ over the finite sets $\mathcal{K}$ of units, each $F^{\mathcal{K}}$ a finite sum of pieces, with the following properties.*

- *(i)* No home. *If F has no home, $F^{\emptyset}$ is F at the renormalised configuration U^R.*
- *(ii)* Locality. *Each piece of $F^{\mathcal{K}}$ is a function of the external data and of the data of the components of Z in its domain, which are the components in $\hat{T}$ and the units of $\mathcal{K}$; in particular $F^{\mathcal{K}}$ depends among the units only on those in $\mathcal{K}$.*
- *(iii)* Covariance. *Each piece transforms under a gauge transformation u of the external configuration as F does: by $\pi(u)$ at the end sites of its bonds and intervals, and by* $\mathrm{Ad}(u)$ *on the fluctuation fields.*

(iv) Bounds. *For every $\mathcal{K}$ the pieces of $F^{\mathcal{K}}$ have moduli summing to at most $\|F\| e^{\varepsilon_F} C_R^{|\mathcal{K}|} e^{-\frac{63}{64}\ell_c(F,\mathcal{K})}$, where $C_R := 2e^{32\sqrt{d}}$, $\ell_c(F,\mathcal{K})$ is M^{-1} times the least total length of a finite union Γ of segments such that every component of $\Gamma \cup \hat{T} \cup \bigcup \mathcal{K}$ meets $\hat{T}$, and $\varepsilon_F \leq |X(F)| e^{-R_k/2}$ for a carried insertion and $\varepsilon_F \leq |X(F)| R_k^{-d/2}$ otherwise. For a unit K_0, with $\rho(A,B) := \max(d_\Box(A,B) - 1, 0)$, $c_{\text{dec}} := 128 \cdot 204/63$, and a number $x'_F \leq |X(F)| e^{-63R_k/512}$,*

$$\sum_{\mathcal{K} \ni K_0} C_R^{|\mathcal{K}|} e^{-\frac{63}{64}\ell_c(F,\mathcal{K})} \leq 2(e^{x'_F} - 1) e^{-\rho(\hat{T},K_0)/c_{\text{dec}}} \,. \tag{17.9}$$

(v) The factor per component. *In the ratio of gases of Proposition S.10(iii), the pieces of the letters, of the home tensors, and of the carried insertions add at most a factor e^{a_k} per first-class component, with $a_k \leq \frac{11}{10}$.*

Proof. The construction is Bałaban's localisation expansion [13, p. 372, ll. 12–28; p. 373, l. 13–p. 376, l. 13], applied to F in place of a term of the action; we describe its steps and the place where each property is decided.

The reference and the interpolation. The reference configuration U^0 is Bałaban's term with no further domains, $Y'_2 = Y'_3 = Y'_4 = \emptyset$ [13, p. 373, ll. 31–35; p. 374, ll. 29–31], for the domain made of $X(F)^{[3]}$ and the components meeting $X(F)^{[2]}$: it has no left factors, and it depends on the configuration on this domain only. Call a fine bond far if its sup-distance from the points of Z at distance at least $2MR_k$ from Z^c is at least $M(2R_k - 1)$. On the far bonds of $X(F)^{[2]}$ we interpolate between U^0 and U^{old} by complex parameters $z(\Delta)$, one for each cube Δ of the set X_o of the cubes of $X(F)^{[1]}$ outside Z, through a C^2 partition of unity subordinate to the cubes, and the fundamental theorem of calculus in each parameter, as at [11, (1.9)], gives $F = \sum_{I \subset X_o} F_I$, with F_I the integral over $[0,1]^I$ of the I-fold derivative.

The four operations. The configuration U^{old} depends on the data of the components through four families of functions: the left factors, and the functions $\mathbf{H}''_k$, $\mathbf{H}_{\mathbf{B}_1}$, and $\mathbf{H}_k$ of [13, (1.61)–(1.63)]. Their sources lie in the points of Z at distance at least $2MR_k$ from Z^c, and their operators have walk expansions decaying at the rate $\frac{1}{2}\delta_0$ in the distance of [3, (2.46)] at the complex parameters, by [7, (190)] with the bounds of [11, (1.11), p. 5] and [5, Theorem 3.13]. For each family in turn we multiply every walk term by the parameters $s(\Delta)$ of the cubes it meets, on the cubes away from the sources, and expand by [11, (1.9)], as Bałaban does for his operations [13, p. 373, ll. 31–35; p. 374, ll. 8–11, ll. 26–29]. The parameters do not spoil the decay of the walks. A maximal set of points of a walk's path of length ℓ at mutual ℓ^∞-distance more than M has at most $1 + 2\ell/M$ points, since the balls of radius $\frac{1}{2}M$ about them hold disjoint parts of the path of length at least $\frac{1}{2}M$ each, and every cube of side M which the domain of the walk term meets lies within $(2 \cdot 5^d + 26)M$ of one of them, by the size of the domains of [5, p. 408, ll. 20–24]; so the term carries at most $N_\sigma(1 + 2\ell/M)$ factors $s(\Delta)$, $N_\sigma := (4 \cdot 5^d + 54)^d$, and at $|s(\Delta)| \leq e^{\kappa_1}$ these cost at most $e^{N_\sigma \kappa_1} e^{2N_\sigma \kappa_1 \ell/M}$, which half of the decay of the walk absorbs once M is large compared with $N_\sigma \kappa_1$. The separation argument of [11, p. 4, ll. 17–30] uses only that the kernels vanish across the cubes with $s = 0$ and that the equations, the sources, and the local functions are local, so it applies: at fixed parameters a walk term which meets a cube with $s = 0$ vanishes, and the value of a family on a set depends only on the sources in the components, of the cubes with $s \neq 0$, which meet that set. Resumming the cubes adjacent to the already reached region in reverse leaves, at each of the four stages, a sum over *cores*, the connected sets of parameters which reach beyond that region. A piece is determined by I and its cores at the four stages;

its reached set $\mathcal{K}$ is the set of units which its region meets, and $F^{\mathcal{K}}$ is the sum of the pieces with reached set $\mathcal{K}$.

Locality and no home. Every walk which contributes to a piece lies in its domain together with every source it reaches, and a term depends on the field restricted to its domain [13, p. 374, ll. 29–33]; adding or removing a unit at $d_\square \geq 3$ from the region of a piece changes neither the region nor the piece. This gives (ii). If F has no home, a component which meets $X(F)$ would lie in the home, so no component meets $\hat{T}$; a piece with $\mathcal{K} = \emptyset$ then has no source in its region, the families vanish there because their functions vanish at the zero argument, $|\mathbf{H}(s, B')| \leq 4B_0 e^{16\kappa_1}|B'|$ [11, (1.18)], and the sum over I of the remaining values is F at the configuration with these functions zero, which is Bałaban's term "taken at the background field" [13, p. 373, ll. 5–6], that is U^R. This gives (i).

Covariance. The configurations entering the pieces are minimisers on determining sets, which transform by a fine gauge transformation when the block data do, since the gauge transformation maps the minimal orbit of [10, (2.12)] onto the minimal orbit. The parameters s multiply walk terms of the operators Δ_U^η, $Q_j'(U)$, $G'(U)$, $\Delta_a(U)$, and $G(U)$, which transform by conjugation with the multiplication operator $R(u)$ [5, (3.28)–(3.34), pp. 395–396], and so do their restrictions to domains and the inverses of these. Every walk term is therefore covariant, and so is the unique solution of the fixed-point equations [11, (1.3)–(1.4), pp. 5–6], since its conjugate solves the equations at U^u. The derivatives in the parameters at fixed u keep the law, which gives (iii).

Bounds. Cauchy's formula on the circles $|s(\Delta)| = e^{\kappa_1}$ bounds each s-derivative by $e^{-(\kappa_1-1)}$ times the supremum, and on the circles $|z(\Delta)| = R_z := e^{(\delta_0/128)MR_k}$ each z-derivative by $2/R_z$ times it, so a piece is at most $\|F\|(2/R_z)^{|I|}\prod_c e^{-(\kappa_1-1)|c|}$, the product over its cores; the parameters stay in the class, at whose points $|F| \leq \|F\|$. Under $\kappa_1 - 1 \geq 4L$ and $\frac{1}{2}(\kappa_1 - 1) \geq 2L\kappa$ [11, p. 20, l. 17], the factor $e^{-(\kappa_1-1)|c|}$ pays $e^{-2\sqrt{d}|c|}$, the count $9^{d|c|}$ of the connected sets of $|c|$ cubes through a root, and a small factor $e^{-(\kappa_1-1)w/4}$ for a core of at least w cubes. The cores and the edges joining them to $\hat{T}$ and to the units form a connected graph, and joining the centres of the cubes along a spanning tree gives $\ell_c(F, \mathcal{K}) \leq \sum_c 2\sqrt{d}|c| + \sqrt{d}(\lambda_T + 2\lambda_K)|\mathcal{K}|$, with $\sqrt{d}(\lambda_T + 2\lambda_K) \leq R_k/128 + 31\sqrt{d}$; since the units lie at $d_\square \geq R_k - 2$ from $\hat{T}$ and from one another, $\ell_c(F, \mathcal{K}) \geq \frac{1}{2}|\mathcal{K}|(R_k - 3)$, so $e^{-2\sqrt{d}\sum_c|c|} \leq e^{-\frac{63}{64}\ell_c} e^{(31\sqrt{d}+3/128)|\mathcal{K}|}$. The roots of the cores lie in neighbourhoods of $\hat{T}$, of I, and of the units of $\mathcal{K}$, of polynomial size in R_k; summing over them, over $I \subset X_o$ with $(1 + 2e/R_z)^{|X_o|} \leq e^{2e3^d|X(F)|/R_z}$, and using that the small factors decrease as a power of g_k while the counts grow as a power of $\log g_k^{-2}$, gives the bound of (iv) with the stated ε_F, since $R_z \geq L^{(12d+1)R_k}$ and $R_k \geq (\log g_k^{-2})^2$ [10, (2.5), p. 255]. For (17.9) we bound $e^{-\frac{63}{64}\ell_c}$ by the sum over the trees on $\{\hat{T}\} \cup \mathcal{K}$ of $\prod e^{-\frac{63}{128}\rho}$ over their edges, whose least total ρ-length is at most $2\ell_c$; a path from $\hat{T}$ to K_0 through j units has ρ-length at least $\rho(\hat{T}, K_0) - 100R_k j$ and at least $(j+1)(R_k - 3)$, hence at least $\rho(\hat{T}, K_0)/102$ as $R_k \geq 303$, and the tree sum, from the leaves inward, gives (17.9) for g_k small.

The factor per component. A first-class component X pays its constant factors and its positions from the factor $e^{-p_0(g_k)}$ of [13, (1.89)], extracted before the sum as on [13, p. 389, ll. 27–28], and each piece with a first-class unit is charged to the first such unit in a fixed order and bounded there by (17.9). A home tensor F_h has at most $4(104MR_k + 1)|h|$ bonds and $|X(F_h)| \leq 2^{d+2} \cdot 105R_k|h|$, so for g_k small its pieces cost at most $e^{|h|/10}(1 + \sum_{X'} 160\theta_F e^{-\rho(\hat{T}_h, X')/c_{\mathrm{dec}}})$, with $\theta_F := 2^{d+2} \cdot 105R_k e^{-63R_k/512}$, and each component of h carries the share $\frac{1}{10}$ of the first factor. What reaches X from the far letters i, for which $\hat{T}$ is the set $Y_1(i) \subset X(i)^{[1]}$ and $|X(i)| \leq 2^d$, is at most $8e2^d e^{-63R_k/512}\sum_i e^{-\rho(Y_1(i), X)/c_{\mathrm{dec}}}$. To bound the sum we count the bonds of the loops near X without any assumption on the shape or the number of the loops beyond their being vertex-disjoint: a site then lies on at most one loop and carries two of its bonds, and the bonds with $\rho(Y_1(i), X) < m$ lie in a box of side $100MR_k + 2(m+1)M + 1 \leq 105(m+1)MR_k$ about X, so that they number at most $2 \cdot 105^d(m+1)^d(MR_k)^d$. With $\sum_{m\geq 0}(m+2)^d e^{-m/c_{\text{dec}}} \leq 2^d d!(c_{\text{dec}}+1)^{d+1}$ the sum is at most $C_{\text{box}}(MR_k)^d$, $C_{\text{box}} := 2 \cdot 210^d d!(c_{\text{dec}}+1)^{d+1}$, a polynomial in R_k, and $8e2^d C_{\text{box}}(MR_k)^d e^{-63R_k/512} \leq \frac{1}{4}$ for g_k small, because R_k grows as g_k decreases. The carried insertions reach X in the same way, each with its own bound in the place of the bound 1 of a letter and with $|X(F)| \leq 2^d(1+\sigma(W))$ for the insertion of a group W; the weight $e^{\lambda_k \nu \sigma(W)}$ of Paper, Definition 5.2(ii), pays this size. The count concerns the insertions of the two species which the class bounds through weighted sums: for the deviation species the sum rooted at one bond is the first term of Paper, (5.9), with the weights, and for the cross species the sum over all bonds is bounded by Paper, (5.11); both are at most the root mass $\varepsilon_* \leq 1$ of the class for γ_q small, so that their contribution is at most that of the far letters, again at most $\frac{1}{4}$. The insertions attached at an earlier step j to a component Z of a large-field region are not bounded in this way, their norms are not small per bond, and they do not enter the count. They are normalised beforehand: by Proposition S.9(ii) the products of the insertions of the species of (j, Z), summed over the multilinear expansion of the words, are at most $\exp(2A_j(Z))$, and this factor has been paid by the part of the large-field factor of Z which Proposition S.11 sets aside; after this payment the words with attached insertions are bounded as the words without them, and it is to these that the bounds of the letters are applied. And what reaches X from the other homes is at most $320\theta_F 105^d(c_{\text{dec}}+1) \leq \frac{1}{2}$, since distinct homes have distinct components at mutual distance at least MR_k. Hence $a_k \leq \frac{1}{10} + \frac{1}{2} + \frac{1}{2} = \frac{11}{10}$. *q.e.d.*

The ratio of gases also contains the polymers of Bałaban's gas which do not meet the loops, and the bound of Proposition S.10(iii) on the change of $\log \Xi$ needs their sum through a cube of side MR_k. Bałaban proves the convergence of his expansion through the bound [13, (1.97), p. 389] on the relative linear size $d_{k,\cup Y_i}$ of [13, (1.67), p. 376], which gives no decay inside the components of the second class, so a count is needed which does not lose a factor of the size of these components. The next lemma provides it, together with the factor which the second-class components bring to the induction of [13, (1.80)]. For a domain X' we write $d_{k,\cup Y}(X')$ for $d_{k,Z}(X')$ of [13, (1.67)] with $Z = \bigcup_i Y_i$, that is M^{-1} times the length of a shortest tree graph contained in X' which meets every cube of $X' \setminus \bigcup_i Y_i$, and $d_k(X')$ for the case $Z = \emptyset$.

Lemma 17.20 (The Second-Class Components)**.** *Let k be an $\mathbf{R}$-step, fix a term of [13, (1.72), p. 379] with the second-class components $Y_1, \ldots, Y_m$, let $F(X')$ be the activities of [13, (1.91), p. 388], and let $F_q(X')$ be the part of [13, (1.91)] with q first-class components. Let κ be large depending on d and β, α small depending on d, κ, and β, and γ small. The domains below are the connected unions of cubes which contain or are disjoint from each Y_i.*

(i) The distance. $d_k(X') \leq d_{k,\cup Y}(X') + d\sum_{Y_i \subset X'}(\#\pi_k(Y_i) + 1)$ *for every domain X'.*

(ii) The parts. $\sum_q |F_q(X')|$ *obeys [13, (1.97)], and a domain with* $F_0(X') \neq 0$ *contains some Y_i and has $c_1 = \alpha^{1/3}$.*

(iii) The roots. *Each Y_i has a cube $r_i \not\subset Y_i$ with a face on Y_i such that $r_i \subset X' \setminus \bigcup_j Y_j$ for every domain $X' \supset Y_i$ with $F(X') \neq 0$. With $a(X') := \frac{1}{2}(\#\pi_k(X' \setminus \bigcup_j Y_j) + \#\{j \mid Y_j \subset X'\})$, there is a constant C_Ψ depending on d, κ, and β such that for every cube $c \not\subset \bigcup_j Y_j$*

$$\sum_{X' \supset c} e^{a(X')} e^{-\frac{1}{2}(1+\beta)\kappa d_{k,\cup Y}(X')} \leq C_\Psi \,, \tag{17.10}$$

the sum over the domains $X' \supset c$ *with* $F(X') \neq 0$*, and the same bound holds for the sum over the domains* $X' \supset Y_i$ *with* $F(X') \neq 0$*.*

(iv) A cube of side MR_k. *For every cube* Q *of side* MR_k*,*

$$\sum_{X' \cap Q \neq \emptyset} e^{a(X')}|F(X')| \leq 6^d C_\Psi(\alpha^{1/3} + e^{-p_0(g_k)}) + e^{-p_0(g_k)/2} \,. \tag{17.11}$$

(v) The small-domain branch. *On the support of* $\chi_k^c(Y_i^{\sim -6})$ *some plaquette* $p' \subset Y_i$ *has* $|V_k(\partial p') - 1| \geq (O(1)B_3M^2)^{-1}\varepsilon_k$*, and the Wilson action of the term gives the factor* $\exp(-p_0(g_k))$ *of the branch of [13, p. 385, l. 45–p. 386, l. 2], one factor of the product* $\prod'$ *of [13, (1.79), p. 383].*

Proof. (i). A shortest tree for $d_{k,\cup Y}(X')$ meets every cube of $X' \setminus \bigcup_i Y_i$. If $X' \neq Y_i$, the cubes of X' with a face on Y_i do not lie in Z, because Y_i is a component of Z, so they lie in $X' \setminus \bigcup_j Y_j$ and the tree meets them. For each $Y_i \subset X'$ we add the segments of length M which join the centres of adjacent cubes along a spanning tree of its $\#\pi_k(Y_i)$ cubes, and a path of length at most $M + \frac{1}{2}dM \leq dM$ from the centre of a cube of Y_i through the centre of an adjacent cube off Z to the tree. The result is a connected graph in X' which meets every cube of X', of length at most $Md_{k,\cup Y}(X') + M\sum_{Y_i \subset X'}(\#\pi_k(Y_i) - 1 + d)$, and a spanning tree of it is no longer. If X' is one Y_i, the spanning tree of its cubes alone gives the bound.

(ii). Replacing $F(X')$ by its parts $F_q(X')$, as activities on the same domain, leaves the gas (8.1) and its cluster expansion unchanged in sum, since two parts on one domain are incompatible. The bounds [13, (1.92)–(1.95), pp. 388–389] bound each $F_q(X')$ by the term q of the right side of [13, (1.95)], and [13, (1.96)] sums these terms over q to the bound [13, (1.97)]; so $\sum_q |F_q(X')|$ obeys it. A domain disjoint from $\bigcup_i Y_i$ has $q \geq 1$ [13, p. 389, ll. 27–28], so a domain with a part $q = 0$ meets $\bigcup_i Y_i$, hence contains some Y_i, and its c_1 is $\alpha^{1/3}$ [13, p. 390, l. 2].

(iii), the root. Fix a cube $r_i \not\subset Y_i$ with a face on Y_i. The set Z is a union of cubes of side MR_k [13, p. 380, ll. 47–49; p. 384, l. 24], each a union of cubes of π_k. The cube of side MR_k which contains r_i touches Y_i and is not contained in Z, since Y_i is a component of Z [13, p. 378, ll. 20–21]; so it lies in the layer which the operation $\sim$ adds [10, p. 245], and r_i lies in the component of $Z^\sim$ through Y_i. A domain $Y \in \mathbf{D}$ of a term of [13, (1.91)] which meets Y_i contains this component [13, p. 377, ll. 5–6], and with it r_i. A term of [13, (1.91)] determines the localisation domain $\bigcup_{Y \in \mathbf{D}} Y \cup \bigcup_h X_{j_h}$ [13, p. 387, l. 42], which equals X' [13, p. 388, l. 14]. If $X' \supset Y_i$, some domain $Y \in \mathbf{D}$ of the term meets Y_i: a cube of Z which touches a first-class component X_{j_h} lies in X_{j_h}, so that neither X_{j_h} nor $X_{j_h}^\sim$ [13, p. 389, ll. 9–10] contains a cube of Y_i. Hence $r_i \subset X'$, and $r_i \subset X' \setminus \bigcup_j Y_j$ because $r_i \not\subset Z$.

(iii), the count. Let $X' \supset c$, or $X' \supset Y_i$ and $c := r_i$, with $n \leq d_{k,\cup Y}(X') < n + 1$. The tree of [13, (1.67)] meets c, every cube of $X' \setminus \bigcup_j Y_j$, and, by the root, the cube r_j of every $Y_j \subset X'$; it has length less than $M(n+1)$, so the cubes which it meets form a connected set $S \ni c$ of at most $C_d(n+1)$ cubes, with C_d depending on d. The domain X' is the union of the cubes of S which lie in $X' \setminus \bigcup_j Y_j$ and of the components $Y_j \subset X'$, each of which has a face on its root $r_j \in S$, and at most $2d|S|$ components have a face on a cube of S. So X' is determined by S, a subset of S, and a set of at most $2d|S|$ components, and since $a(X') \leq \frac{1}{2}(1 + 2d)|S|$, the left side of (17.10) is at most $\sum_{n \geq 0} e^{C'_d(n+1)} e^{-\frac{1}{2}(1+\beta)\kappa n} =: C_\Psi$, with C'_d depending on d, which is finite once $\frac{1}{2}(1+\beta)\kappa > C'_d$.

(iv). The cube Q lies in one Y_l or outside $\bigcup_j Y_j$, since Z is a union of cubes of side MR_k. A domain X' which meets Q either is disjoint from $\bigcup_j Y_j$ or contains some Y_j.

In the first case $c_1 = e^{-p_0(g_k)}$ [13, (1.97)], $Q \not\subset \bigcup_j Y_j$, and summing (17.10) over the R_k^d cubes c of Q gives at most $R_k^d C_\Psi e^{-p_0(g_k)} \leq e^{-p_0(g_k)/2}$, because $R_k^d < p_0(g_k)$ by the condition $-\frac{1}{4}p_0(g_k) + 1 + 2\kappa(100R_k)^d \leq 0$ of [13, p. 389, ll. 16–18] and γ is small. In the second case we sum X' at the root r_l of a component $Y_l \subset X'$. The tree of X' meets a cube of $X' \cap Q$ outside $\bigcup_j Y_j$, or r_l itself when $Q \subset Y_l$, and the root r_l, each point of which lies within the ℓ^∞-distance M of Y_l; so $d_{k,\cup Y}(X') \geq mR_k - 1 \geq m - 1$ for every $Y_l \subset X'$ with $\mathsf{dist}_\infty(Y_l, Q) \geq mMR_k$. At most $(2m+3)^d$ components Y_l have $\mathsf{dist}_\infty(Y_l, Q) < (m+1)MR_k$, each containing its own cube of side MR_k in the box of side $(2m+3)MR_k$ about Q. We split $|F(X')| \leq c_1 e^{-(1+\beta)\kappa d_{k,\cup Y}(X')}$ into two halves: one half gives $e^{-2d(m-1)_+}$ for $\frac{1}{2}(1+\beta)\kappa \geq 2d$, and the other half, summed by (iii) at r_l with $c_1 \leq \alpha^{1/3} + e^{-p_0(g_k)}$, gives $C_\Psi(\alpha^{1/3} + e^{-p_0(g_k)})$. Finally

$$\sum_{m \geq 0} (2m+3)^d e^{-2d(m-1)_+} \leq 6^d \tag{17.12}$$

for $d \geq 2$. For $m \geq 2$ the term is $((2m+3)e^{-2(m-1)})^d$, whose base divided by 6 is $\frac{7}{6}e^{-2} < \frac{1}{6}$ at $m = 2$ and is multiplied by $\frac{2m+5}{2m+3}e^{-2} \leq \frac{9}{7}e^{-2} < \frac{1}{5}$ from m to $m+1$; so the left side of (17.12) divided by 6^d is at most $2^{-d} + (\frac{5}{6})^d + \frac{25}{24}6^{-d}$, which decreases in d and equals $(34 + \frac{25}{24})/36 < 1$ at $d = 2$.

(v). On the support of $\chi_k^c(Y_i^{\sim -6}) = 1 - \chi_k(Y_i^{\sim -6})$ [13, p. 378, l. 6, l. 21] some cube $\square \subset Y_i^{\sim -6}$ has a plaquette $p \subset \square^\sim$ with $|U_{k,\square}(V_k, \partial p) - 1| \geq \varepsilon_k \eta^2$ [10, (2.17), p. 257], the failed soft test function giving this bound at its threshold by Proposition S.2(b). If every plaquette of V_k on the neighbourhood of $\square$ which determines $U_{k,\square}$, which lies in Y_i by the margin ~ -6, were less than $(O(1)B_3M^2)^{-1}\varepsilon_k$, the bound of [7, Theorem 1, p. 279] would give $|\partial U_{k,\square} - 1| < \varepsilon_k \eta^2$, as Bałaban uses it for $U_{k,Z}$ at [12, p. 193, l. 20, ll. 28–29]. So some $p' \subset Y_i$ has $|V_k(\partial p') - 1| \geq (O(1)B_3M^2)^{-1}\varepsilon_k$, and from this condition the Wilson action gives, as for the function $1 - \chi_{k,\Lambda}$ [12, p. 193, ll. 30–31], the factor $\exp(-\frac{1}{4}g_k^{-2}(O(1)B_3M^2)^{-1}\varepsilon_k^2) \leq \exp(-A_1^2 p_0^2(g_k))$ [13, p. 381, ll. 29–31], which is at most $\exp(-p_0(g_k))$ for g_k small [13, p. 383, ll. 27–28]. The estimate uses only this condition on V_k in Y_i and the Wilson action of the term of Y_i, not the operation which introduced the function; so Y_i carries one factor of $\prod'$, which is a product over components [13, p. 383, ll. 40, 44], and the factor is part of $\mathbf{T}'_k(Y_i)1$. *q.e.d.*

The removal of the test function $\chi_{k,\Lambda}$ on the components of the first class is asserted without proof: “This is clear for $\chi_{k,\Lambda}$” [13, p. 378, l. 10]. The test function compares the global minimiser $U_{k,Z}$ of [12, (1.74)] with the threshold $2\varepsilon_k\eta^2$, while the test functions passed on the first class bound the local minimisers $U_{k,\square}$ of [10, (2.16)] by $(1+\kappa_s)\varepsilon_k\eta^2$; so the two minimisers have to be compared, and the margin between the two thresholds has to absorb the difference.

Lemma 17.21 (The Test Function $\chi_{k,\Lambda}$ on the First Class). *Let $\kappa_s \leq \frac{1}{4}$, let γ be small, and let X be a component of the first class of a term of an* **R***-step k. On the support of the soft test function $\chi_k(\Omega_k^{\sim 4})$ of [12, (1.71)] and of the soft small-field function on X of the decomposition of [13, p. 378, l. 6], every plaquette p of $\Omega_k^c \cap X$ satisfies*

$$|U_{k,Z}(\partial p) - 1| \leq (1 + \kappa_s + 2\tau + 9\tau^2)\varepsilon_k\eta^2 < 2\varepsilon_k\eta^2 \ , \tag{17.13}$$

with $U_{k,Z}$ the minimiser of [12, (1.74)], B_3 and δ the constants of [10, (3.7), p. 266], and

$$\tau := \frac{\pi}{2}B_3(22d^2B_3 + 4(d-1)MR_k)(1+\kappa_s)e^{-2\delta M_2 R_k} \leq \frac{1}{20} \ . \tag{17.14}$$

So the condition of [12, (1.75), p. 193] holds on X at V_k itself, and $\chi_{k,\Lambda}$ equals one on the first class.

Proof. The geometry. By [10, (3.20), p. 269], which builds Λ_k at the step to the scale k, $\Lambda_k \subset \Omega_k^{\sim -2}$, and the modification of [10, p. 269] only enlarges Λ_k^c. So every plaquette p of $\Omega_k^c \cap X$ is at distance at least $2MR_k$ from Λ_k, and hence from X^c, whose points lie in Λ_k or in other components of Λ_k^c, separated from X by Λ_k [10, p. 264]. The cube $\square$ of side LM_2R_k of [10, (2.17)] which contains p lies in $\Omega_k^{\sim 4}$ or in $(\Omega_k^{\sim 4})^c \cap X$, the partitions being compatible [10, p. 245], and in either case one of the two soft test functions bounds it, $\sup_{p' \subset \square^\sim} |U_{k,\square}(\partial p') - 1| < (1+\kappa_s)\varepsilon_k \eta^2$, by Proposition S.2(b). Since $M \geq (dL+4)M_2$ [10, p. 245], the set $\square^{\sim 4}$ lies within MR_k of p, inside $Z^{\sim -2}$ and so inside the deepest domain of $\mathbb{B}_k(Z)$, which is made of k-blocks [10, p. 256].

The data. By [10, (1.3)] and the averages of [9, (0.4), p. 253], the data $M^{\cdot}(Q_k^{s*}V_k)$ are 1 on the bonds inside the k-blocks and $V_k(c)$ on those joining the k-blocks of a bond c, so their plaquettes are 1 or plaquettes of V_k. On the k-plaquettes of $\square$, $V_k = M^k(U_{k,\square})$, the constraint of the problem of $\square^{\sim 4}$ on its deepest domain, which contains $\square^\sim$ [10, p. 256]. With $a := (1+\kappa_s)\varepsilon_k$, [4, Proposition 2], applied as at [6, p. 87, l. 35] with $L^k\eta = 1$, gives $|\partial V_k - 1| < a + 2C_0a^2 < 2a$ for γ small. So [7, (7)] holds for the problems of Z and of $\square^{\sim 4}$ with $\varepsilon_1 = 2(1+\kappa_s)\varepsilon_k$, and [7, Theorem 1, (8), p. 279] gives on $\square^{\sim 4}$

$$|U_{k,Z}(\partial p) - 1| < \alpha_0\eta^2 , \qquad \alpha_0 := 2B_3(1+\kappa_s)\varepsilon_k . \tag{17.15}$$

The global minimiser solves the local problem. A variation of $U_{k,Z}$ which keeps the data of $\mathbb{B}_k(\square^{\sim 4})$ is supported in the first domain Ω_1 of $\square^{\sim 4}$, and it keeps the data of $\mathbb{B}_k(Z)$: each one-step average reads only the bonds with both end-points in the blocks of the two ends of its bond [9, (0.4), p. 253], so the k-block averages on the bonds which meet $\square^{\sim 4}$ are compositions of the averages of $\mathbb{B}_k(\square^{\sim 4})$ and of $U_{k,Z}$ off $\square^{\sim 4}$; the other k-block averages read k-blocks whose bonds miss Ω_1 [10, p. 256]; and the data of the levels $j < k$ lie within $2M_1$ of Z^c, at distance at least $MR_k - 2M_1$ from $\square^{\sim 4}$. Such a variation changes the action by the same amount in both problems, so $U_{k,Z}$ is a local minimum of the problem of $\square^{\sim 4}$ at its own data; its regularity $(1+2B_3)\alpha_0$ lies within that problem's space for γ small, so it is that problem's minimiser by the uniqueness of the critical orbit [7, Theorem 1, p. 279], [10, p. 256]. By [7, Proposition 9, p. 309], $U_{k,\square} = (\exp(i\eta\mathbf{H}_{k,\square}(B))U_{k,Z})^{u^{-1}}$ for a gauge transformation u, with

$$B := \frac{1}{i}\log[M^{\cdot}(Q_k^{s*}V_k)(M^{\cdot}(U_{k,Z}))^{-1}] , \tag{17.16}$$

once B lies in the ball [7, (172)].

The size of B. Off the boundary layer of $\square^{\sim 4}$ of width $2M_1$ both data equal V_k, so $B = 0$ there. On the layer we compute after a gauge transformation w constant on the k-blocks, equal to the complete axial gauge of V_k on the k-blocks of $\square^{\sim 4}$ and of the layer of k-blocks around it, and to 1 elsewhere; the minimisers are covariant [4, (11), p. 19], B is conjugated without change of its norm, and $U_{k,Z}^w$ is taken in the axial gauge [6, (1.15), p. 78] of the k-block centres, which keeps the data. This region is a cube of side at most MR_k, so each bond of V_k^w on it is a product of at most $(d-1)MR_k$ transported plaquettes of V_k, each a k-plaquette of a cube of side LM_2R_k contained in X, and $|V_k^w - 1| < \alpha_1 := 2(d-1)MR_k(1+\kappa_s)\varepsilon_k$. The iteration of [6, Lemma 1] at [6, p. 87, ll. 35–39], which is local by [6, p. 80], applied with $U_0 = 1$ and $U' = U_{k,Z}^w$, gives $|M^j(U_{k,Z}^w) - 1| < 11d^2\alpha_0 + \alpha_1$ there for γ small. Since $|ab^{-1} - 1| = |a - b|$ for unitary a, b, $|e^{iB} - 1| < 11d^2\alpha_0 + 2\alpha_1$ on the layer, and

$$|B| \leq \frac{\pi}{2}(11d^2\alpha_0 + 2\alpha_1) , \tag{17.17}$$

which lies in the ball [7, (172)] for γ small, both data satisfying [7, (7)] at $11d^2\alpha_0 + 2\alpha_1$.

The decay. The layer is at distance at least $3M_2R_k - 2M_1 \geq 2M_2R_k$ from $\square^\sim$. Since $\mathbf{H}_{k,\square}(0) = 0$, we have $\mathbf{H}_{k,\square}(B) = \int_0^1 (\delta\mathbf{H}_{k,\square}/\delta B)(tB)B\,\mathrm{d}t$, each tB lying in the ball, and [7, (190), p. 308] bounds the derivative and its gradient at the points of $\square^\sim$ with the decay $\exp(-\frac{1}{8}\delta_0 d(y,y'))$ in the distance to the sources y'. Summed over the layer as in [10, (3.7), p. 266], this gives, by (17.17),

$$\sup_{\square^\sim}|\mathbf{H}_{k,\square}| + \sup_{\square^\sim}|\nabla\mathbf{H}_{k,\square}| \leq B_3 e^{-2\delta M_2 R_k}\sup|B| \leq \tau\varepsilon_k\ . \tag{17.18}$$

Since $R_k e^{-2\delta M_2 R_k} \to 0$ as $R_k \to \infty$ and $R_k \geq (\log g_k^{-2})^r$, we have $\tau \leq \frac{1}{20}$ for γ small.

The plaquette. The identity [6, (1.43)] with $U_0 := U^w_{k,Z}$ and $U_1 := \exp(i\eta\mathbf{H}_{k,\square})$ reads $W - 1 = (1+P)XQ + E_2$, with $X := U_0(\partial p) - 1$ and $W := (U_1U_0)(\partial p)$, where $1+P$ and Q are products of factors $U_1(b)$ and of transports of U_0, hence unitary; so $|X| \leq |W-1| + |E_2|$. By [6, (1.22), p. 79], E_2 is the product of the four factors $e^{iA_i} := U_1(b)^{\pm 1}$ of p, two of them conjugated by U_0, minus one, with $|A_i| \leq s := \tau\varepsilon_k\eta$ by (17.18). Its linear part $i\sum_i \pm A_i$ is $i\eta$ times two U_0-covariant differences of $\mathbf{H}_{k,\square}$, of norm at most $2\tau\varepsilon_k\eta^2$; the four remainders $e^{iA_i} - 1 - iA_i$ are at most $\frac{1}{2}s^2$ each; and the products of two, three, and four factors $e^{iA_i} - 1$ are at most $6s^2$, $4s^3$, and s^4 in all. Hence, by $\varepsilon_k, \eta \leq 1$ and $s \leq \frac{1}{20}$,

$$|E_2| \leq 2\tau\varepsilon_k\eta^2 + 8s^2 + 4s^3 + s^4 \leq (2\tau + 9\tau^2)\varepsilon_k\eta^2\ . \tag{17.19}$$

The configuration U_1U_0 is a gauge transform of $U_{k,\square}$, so $|W-1| = |U_{k,\square}(\partial p) - 1| < (1+\kappa_s)\varepsilon_k\eta^2$, and $|X| = |U_{k,Z}(\partial p) - 1|$. This is the first inequality of (17.13), and the second follows from $1 + \kappa_s + 2\tau + 9\tau^2 \leq 1 + \frac{1}{4} + \frac{1}{10} + \frac{9}{400} < 2$. *q.e.d.*

The last lemma of this section concerns the gauge in which the background is small. Bałaban's Theorem 1 of [7] gives, on a cube of the class of that theorem, a gauge in which the background is bounded by the size of the cube times the bound ε_1 of the plaquettes of its data on the whole determining set, and [10, (3.17), p. 268] localises the background on the cubes $\square'$ of the decomposition [10, (3.16)]. The paper needs a bound in which the plaquettes near one cube stand in the place of ε_1, because the deviation bound of an insertion of a dressed loop is a bound by the curvature of the background near that insertion. Such a bound holds on every cube $\mathcal{Q}$ which lies well inside the small-field region, and it is obtained by running [10, (3.17)] at $\mathcal{Q}$ in place of $\square'$. The difference from Bałaban's use of that identity is the distance over which the correction $\mathbf{H}$ decays: at $\square'$ it is of the order LM_2R_{k+1} and grows as the coupling decreases, while at $\mathcal{Q}$ it is a fixed multiple of M, so that the smallness of the correction becomes a condition on M.

We fix a step $k+1 \geq 2$ and a labelled term, with the domains Ω_j, the block field V_{k+1} in the support of the test functions of the term, and the background U_{k+1}. We write $\xi := L^{-(k+1)}$ for the spacing of the fine lattice in the units of the scale $k+1$, in which all lengths of this lemma are measured, and $\mathcal{Q}^{\sim n}$ for a union $\mathcal{Q}$ of cubes of π_{k+1} widened by n layers of such cubes. Let a_0, a_1, B_3, and R_1M_1 be the constants of [7, Theorem 1, p. 279], c_2 the constant of [4, Proposition 2, p. 26], and let

$$C_H := \sup_y \sum_{y'} K(y,y') \tag{17.20}$$

be the supremum, over the points y of a cube $\mathcal{Q}^{\sim 2}$, of the sum of the kernel K of the bounds [7, (190), p. 308], with its constant $O(1)$, over the points y' of the boundary layer of width $2LM_1$ of $\mathcal{Q}^{\sim 4}$. The kernel decays as $\exp(-\frac{1}{8}\delta_0 d(y,y'))$ and the distance from $\mathcal{Q}^{\sim 2}$ to that layer is at least $2M - 2LM_1$, so that the sum over the layer converges and C_H is bounded by a constant of d, L, M, and M_1 which does not depend on k or on the size of $\mathcal{Q}$ and tends to zero as $M \to \infty$ at fixed L and M_1. Finally let $c_F \geq 1$ be a constant such that the operator $\mathbf{F}(1, A)$ of [9, (3.11), p. 272], "a local operator depending on U, ∂U, $\mathbf{A}$, $\nabla_U^\xi \mathbf{A}$ only", satisfies $|\mathbf{F}(1, A)| \leq c_F \xi (|A| + |\nabla^\xi A|)^2$ for $\xi|A|, \xi|\nabla^\xi A| \leq 1$; such a constant exists because $\mathbf{F}(1, A)$ is the part of the second and higher orders in A of the left side of [9, (3.11)] at $U = 1$, by the expansion [7, (22), (23), p. 281] of $e^{i\xi A}$. Similarly let $c_B \geq 1$ be an absolute constant such that, for fields A' and $\mathbf{H}$ on a set of bonds with $\xi|A'|, \xi|\mathbf{H}| \leq 1$, the field r defined by $e^{i\xi\mathbf{H}} e^{i\xi A'} = e^{i\xi(A' + \mathbf{H} + \xi r)}$, and the differences between the covariant operators $\nabla_{e^{i\xi A'}}$, $\partial^*\partial$, Δ with respect to $e^{i\xi A'}$ and the plain ones, applied to $\mathbf{H}$, are bounded together with the first and second differences of ξr by $c_B \xi a' h$, where a' and h bound A' and $\mathbf{H}$ with their differences up to the second order; it exists by the Baker–Campbell–Hausdorff formula and by the formulae [6, (1.43)–(1.54)] for the covariant operators, which [9, p. 272] uses for the same purpose.

Lemma 17.22 (A Gauge Bounded by the Local Curvature)**.** *Suppose that* M *is a multiple of* $R_1 M_1$ *with* $M \geq 2LM_1 + 2R_1 M_1$, *that* $22d^2 C_H \leq 1$, *and that* $a_1 \leq \min(1/(2c_B), 1/(18 c_F B_3 R_1 M_1))$. *Let* $\mathcal{Q}$ *be a cube which is a union of cubes of* π_{k+1}, *with* $\mathcal{Q}^{\sim 5} \subset \Omega_{k+1}$, *let*

$$E := \max\{\xi^{-2} |U_{k+1}(\partial p) - 1| \,|\, p \textit{ a plaquette of the fine lattice in } \mathcal{Q}^{\sim 5}\} , \tag{17.21}$$

and let $M_{\mathcal{Q}}$ *be half the side of* $\mathcal{Q}^{\sim 2}$, *rounded up to a multiple of* $R_1 M_1$. *If*

$$2E \leq \min(a_1, c_2) \qquad \textit{and} \qquad 2M_{\mathcal{Q}} E \leq R_1 M_1 a_1 , \tag{17.22}$$

then there is a G*-valued gauge transformation* u, *defined on a neighbourhood of* $\mathcal{Q}^{\sim 2}$, *such that* $U_{k+1}^u = e^{i\xi A}$ *on* $\mathcal{Q}^{\sim 2}$ *with*

$$|A|,\ |\nabla^\xi A|,\ |\partial^{\xi *} \partial^\xi A|,\ |\Delta^\xi A| \leq 3 B_3 M_{\mathcal{Q}} E , \qquad |R(u) J_{k+1}| \leq 4 B_3 M_{\mathcal{Q}} E , \tag{17.23}$$

where J_{k+1} *is the current [9, (1.8), p. 261] of* U_{k+1} *in the normalisation of [9, (3.12), p. 272].*

Proof. Step 1: the local problem at $\mathcal{Q}$. For a cube $\square'$ of side $L^2 M_2 R_{k+1}$ Bałaban defines the minimal configuration $U_{k+1, \square'}$ on the determining set $\mathbf{B}_{k+1}(\square'^{\sim 4})$, with the data $Q_{k+1}^{s*} V_{k+1}$ and its averages [10, (2.16), p. 257; (3.4), p. 265]: the fine lattice of $\square'^{\sim 4}$ with the constraints of the scale $k+1$, and shells of the scales $k, \ldots, 0$ in a boundary layer of width $2LM_1$ at the boundary of $\square'^{\sim 4}$ [10, p. 268]. The construction uses the cube only through this boundary layer. The set $\mathcal{Q}^{\sim 4}$ is a union of cubes of side M, hence of blocks of side M_1, since M is a multiple of $R_1 M_1$; so the determining set $\mathbf{B}_{k+1}(\mathcal{Q}^{\sim 4})$ and its data $Q_{k+1}^{s*} V_{k+1}|_{\mathcal{Q}^{\sim 4}}$ are defined in the same way, they are of the class of [7, (1)–(3), p. 277], and we write $U_{k+1, \mathcal{Q}}$ for the minimal configuration. The two layers $\mathcal{Q}^{\sim 4} \setminus \mathcal{Q}^{\sim 2}$ have the width $2M \geq 2LM_1 + 2R_1 M_1$, so that they contain the shells and, beyond them, the margin $2R_1 M_1$ which the cube $\widetilde{\mathcal{Q}^{\sim 2}}$ of [7, p. 279] requires.

Step 2: the identity [10, (3.17)] at $\mathcal{Q}$. We claim that on $\mathcal{Q}^{\sim 2}$

$$\begin{aligned} U_{k+1} &= (e^{i\xi \mathbf{H}_{\mathcal{Q}}} U_{k+1, \mathcal{Q}})^{u_{\mathcal{Q}}^{-1}} , \qquad \mathbf{H}_{\mathcal{Q}} := \mathbf{H}_{k+1, \mathcal{Q}}(B_{\mathcal{Q}}) , \\ B_{\mathcal{Q}} &:= \frac{1}{i} \log[M^{\cdot}(U_{k+1})(M^{\cdot}(Q_{k+1}^{s*} V_{k+1}))^{-1}] , \end{aligned} \tag{17.24}$$

with a gauge transformation $u_{\mathcal{Q}}$ and the function $\mathbf{H}_{k+1, \mathcal{Q}}$ of [7, Proposition 9, p. 309] for the determining set of Step 1. The first line of [10, (3.17)] says that the restriction of U_{k+1} is the minimal configuration of the local problem whose data are the averages of U_{k+1} itself. For $\mathcal{Q}^{\sim 4}$ this is the localisation identity of the proof of Proposition S.8, Step 5(2), with $\mathcal{Q}^{\sim 4}$ in the place of $\Box'^{\sim 4}$: since $\mathcal{Q}^{\sim 4} \subset \Omega_{k+1} \subset \Omega_n$ for every $n \leq k+1$, the set $\mathbf{B}_{\mathrm{mix}}$ of (6.4) is $\mathbf{B}_{k+1}(\mathcal{Q}^{\sim 4})$; a configuration on $\mathcal{Q}^{\sim 4}$ with the same averages on this set, glued to U_{k+1} outside, satisfies all the constraints of U_{k+1}, so that $U_{k+1}|_{\mathcal{Q}^{\sim 4}}$ is critical for the local problem; and it lies in the space [7, (6), p. 278], in which the critical orbit is unique by [7, Theorem 1], "This orbit is a unique critical orbit in the space (6) if $B_3\varepsilon_1 \leq \varepsilon_0$ and $\varepsilon_0 \leq a_0$", a statement with no condition on the size of the cube. The second line is [7, Proposition 9] for the local problem, with $V_0 := M^{\cdot}(Q^{s*}_{k+1}V_{k+1})$ and $V' := M^{\cdot}(U_{k+1})V_0^{-1}$: "The function $U_k(V'V_0)U_k(V_0)^{-1}$ transformed to the Landau gauge is, by the definition, equal to $\exp i\eta\mathscr{H}(B)$, where $B = 1/i \log V'$".

Step 3: the argument $B_{\mathcal{Q}}$. For the cube $\Box'$ Bałaban finds that "the argument of the function $\mathbf{H}_{k+1,\Box'}$ is bounded by $44d^2B_3\varepsilon_{k+1}$, and has a support in a boundary layer of the width $2LM_1$ at the boundary of $\Box'^{\sim 4}$" [10, p. 268], from the bound $2B_3\varepsilon_{k+1}(L^{-1}\eta)^2$ on the plaquettes of U_{k+1}; the factor between the two bounds is $22d^2$. The support is that of the shells, for $\mathcal{Q}$ as for $\Box'$, because on the fine region the data are $V_{k+1} = M^{k+1}(U_{k+1})$ [10, p. 267]. The bound is obtained "by an application of the inequality (1.65) from [14], or rather by an application of the reasoning leading to that inequality" [10, p. 266], that is, by [6, Lemma 1, (1.24)–(1.25), p. 79] applied block by block as in [6, (1.65), p. 87]. At a bond b of a shell of the scale j it compares $M^j(U_{k+1})(b)$ with the lift at b of $M^{k+1}(U_{k+1})$; the average $M^j(U)(b)$ depends on U only in the blocks of the ends of b, "the result is local in the sense that if $p = \langle x, y, z, w\rangle$, then it is enough to assume (52) for $p \subset B^k(x) \cup B^k(y) \cup B^k(z) \cup B^k(w)$" [4, Proposition 2, p. 26], and the lift depends only on the block field at b. The comparison therefore uses the plaquettes of U_{k+1} in the blocks within one block of b, which for b in the boundary layer of $\mathcal{Q}^{\sim 4}$ lie in $\mathcal{Q}^{\sim 5}$, and it gives

$$|B_{\mathcal{Q}}(b)| \leq 22d^2E\ , \tag{17.25}$$

with $B_{\mathcal{Q}}$ supported in the boundary layer of width $2LM_1$ of $\mathcal{Q}^{\sim 4}$.

Step 4: the correction $\mathbf{H}_{\mathcal{Q}}$. By [7, Proposition 9] the functional derivative of $\mathscr{H}$ satisfies the inequalities [7, (190)], which bound the function, its gradient, and the operators $\partial^*\partial$ and Δ applied to it through the kernel K, and $\mathscr{H}(0) = 0$ because $B = 0$ means $V' = 1$. Integrating the derivative along the segment from 0 to $B_{\mathcal{Q}}$ and using (17.25) and (17.20), we find on $\mathcal{Q}^{\sim 2}$

$$|\mathbf{H}_{\mathcal{Q}}|,\ |\nabla\mathbf{H}_{\mathcal{Q}}|,\ |\partial^*\partial\mathbf{H}_{\mathcal{Q}}|,\ |\Delta\mathbf{H}_{\mathcal{Q}}| \leq C_H \cdot 22d^2E \leq E\ , \tag{17.26}$$

the last inequality being the hypothesis $22d^2C_H \leq 1$. Bałaban's bound for $\Box'$ has the factor $\exp(-\delta LM_2R_{k+1})$ in the place of C_H [10, p. 268], which is not available here, the distance to the support of the argument being $2M - 2LM_1$.

Step 5: Theorem 1 of [7] for $U_{k+1,\mathcal{Q}}$. The data of the local problem on the constraint lattice of the fine region are $V_{k+1} = M^{k+1}(U_{k+1})$, and by [4, Proposition 2], used with $\alpha_0 := E \leq c_2$ on the blocks of the corners of a plaquette p' of that lattice in $\mathcal{Q}^{\sim 4}$, which lie in $\mathcal{Q}^{\sim 5}$, we have $|(\partial V_{k+1})(p') - 1| < 2E$ [4, (54), p. 26]. On the shells the data are the lift $Q^{s*}_{k+1}V_{k+1}$ and its averages; the lift [10, (1.3), p. 246] equals 1 inside the blocks and the block field across them, so that its plaquettes are 1 or plaquettes of the block field, and its averages inherit the bound. Hence the data satisfy [7, (7), p. 278] with $\varepsilon_1 := 2E \leq a_1$; this is the local form of the remark that $U_{k+1,\Box'}$ "satisfies the regularity condition as above, only with the coefficient 1 in front of ε_{k+1}" [10, p. 267]. The cube $\mathcal{Q}^{\sim 2}$ has the side $2M_{\mathcal{Q}}$ up to the rounding, which the margin of Step 1 absorbs, it is in the class of [7, p. 279] by Step 1, and $M_{\mathcal{Q}} \leq M(\varepsilon_1) = R_1M_1a_1/(2E)$ by the second part of (17.22). So [7, (9), (10), p. 279] give a gauge transformation v on a neighbourhood of $\mathcal{Q}^{\sim 2}$ with $U^{v^{-1}}_{k+1,\mathcal{Q}} = e^{i\xi A'}$ on $\mathcal{Q}^{\sim 2}$ and

$$|A'|,\ |\nabla^{\xi}A'|,\ |\partial^{\xi *}\partial^{\xi}A'|,\ |\Delta^{\xi}A'| < 2B_3M_{\mathcal{Q}}E\ . \tag{17.27}$$

Step 6: the composition and the current. For two configurations and a gauge transformation w, $(U_1U_2)^w(b) = (w(b_-)U_1(b)w(b_-)^{-1})\, U_2^w(b)$ by [10, (1.17), p. 250]. Applying $u_{\mathcal{Q}}$ to (17.24) and then v^{-1}, we obtain on $\mathcal{Q}^{\sim 2}$

$$U_{k+1}^{v^{-1}u_{\mathcal{Q}}}(b) = e^{i\xi\, \mathrm{Ad}(v^{-1}(b_-))\mathbf{H}_{\mathcal{Q}}(b)} e^{i\xi A'(b)} = e^{i\xi A(b)} \,, \qquad A = A' + \mathrm{Ad}(v^{-1})\mathbf{H}_{\mathcal{Q}} + \xi r \,,$$

with the field r of the definition of c_B, since $\xi|A'|, \xi|\mathbf{H}_{\mathcal{Q}}| \leq 1$; we put $u := v^{-1}u_{\mathcal{Q}}$. The bounds (17.26) are bounds on $\mathbf{H}_{\mathcal{Q}}$ and on the covariant operators with respect to $U_{k+1,\mathcal{Q}}$ applied to it, and in the gauge v^{-1} these are the covariant operators with respect to $e^{i\xi A'}$ applied to $\mathrm{Ad}(v^{-1})\mathbf{H}_{\mathcal{Q}}$. By the definition of c_B, with $a' = 2B_3M_{\mathcal{Q}}E$ from (17.27) and $h = E$ from (17.26), each of the four quantities of (17.23) is therefore at most

$$2B_3M_{\mathcal{Q}}E + E + 2c_B\xi B_3M_{\mathcal{Q}}E^2 \leq 3B_3M_{\mathcal{Q}}E \,.$$

Indeed $E \leq \frac{1}{2}B_3M_{\mathcal{Q}}E$ because $B_3 \geq 1$ and $M_{\mathcal{Q}} \geq 2$, and $2c_B\xi E \leq c_Ba_1 \leq \frac{1}{2}$ because $E \leq \frac{1}{2}a_1$ and $\xi \leq 1$. The condition on a_1 is admissible: Theorem 1 of [7] holds with every smaller a_1, the class of its cubes shrinking with it. For the current, J_{k+1} is the expression [9, (1.8)] of U_{k+1} up to the factor of [9, (3.12)], and it is covariant, so that $R(u)J_{k+1}$ is the current of $U_{k+1}^u = e^{i\xi A}$, which by [9, (3.11)] at $U = 1$ equals $\partial^{\xi *}\partial^{\xi}A + \mathbf{F}(1, A)$. Therefore

$$|R(u)J_{k+1}| \leq 3B_3M_{\mathcal{Q}}E + c_F\xi(6B_3M_{\mathcal{Q}}E)^2 \leq 4B_3M_{\mathcal{Q}}E$$

once $36c_F\xi B_3M_{\mathcal{Q}}E \leq 1$, which follows from $2M_{\mathcal{Q}}E \leq R_1M_1a_1$, $\xi \leq 1$, and $a_1 \leq 1/(18c_FB_3R_1M_1)$. The same three inequalities give $\xi|A|, \xi|\nabla^{\xi}A| \leq 3B_3M_{\mathcal{Q}}E \leq 1$ and $\xi|A'| \leq 1$, and $\xi|\mathbf{H}_{\mathcal{Q}}| \leq E \leq 1$, which are the conditions under which c_F and c_B were defined. *q.e.d.*

When the hypothesis (17.22) fails, E or $M_{\mathcal{Q}}E$ is bounded below by a constant which does not depend on the coupling, and the paper then uses no gauge: a bound by the curvature holds trivially when the curvature is of order one in these units.

18 The Activities of the Reference Gases and First-Order Bounds for Their Cluster Weights

The interactions $Q_k(\mathfrak{B})$ and the local terms of the exponent $\mathcal{E}_k$ of a dressed loop are numbers on each labelled term (Paper, Definition 5.2(iv)): they are cluster terms of the reference gases of the steps, and they depend on the run only through the activities of these gases. On the trivial term Proposition S.16(i) compares the activities of the two runs and, through them, the weights. On a term with large fields there is a difficulty which the trivial term does not have. The insertions attached to a large-field region are compared between the two runs only in total, over the history of their component, because they are carried as functions which are never summed over the loop (Paper, Section 6.4); the difference of two such insertions has a part which is small relative to their bound and a part, the difference of the old wrapping insertions which they have taken in, which is small only in its total. Neither statement is a bound on the difference of a single activity, and a cluster series whose activities contained attached insertions could not be compared term by term. This section contains the two facts on which a comparison of the weights on such terms rests: the bounds which pass from a change of the activities to the change of the cluster weights, and the statement that the activities contain no attached insertion.

18.1 First-Order Bounds for a Reference Gas

We begin with estimates for the cluster series of a hard-core gas, which bound the change of the cluster weights under a change of the activities by the size of that change. They use the criterion of Kotecký and Preiss in the form in which Paper, Lemma 7.3, uses it, and nothing else. We fix a step, a labelled term, and the groups of the reference gas of that step, with the compatibility relation of Paper, Section 7.1, the sizes $\sigma(\mathcal{G})$, and the weights $a^\circ(\mathcal{G}) = (\nu/32)\sigma(\mathcal{G})$ of Paper, Definition 7.2; for a family x of complex numbers indexed by the groups, $\Phi^\circ(\Gamma; x) := \phi^{\mathrm{U}}(\Gamma) \prod_{\mathcal{G}\in\Gamma} x_{\mathcal{G}}$ is the weight of a cluster Γ, the product taken with the multiplicities. A *root* is a traversal of the step or a cube of its partition π, and a group is *at* a root if the root is a traversal in its arc or a cube which meets one of its domains or one of the clusters merged into it. For a family $x \geq 0$ put

$$\|x\|_\circ := \sup_r \sum_{\mathcal{G} \text{ at } r} x_{\mathcal{G}}\, e^{2a^\circ(\mathcal{G})} , \tag{18.1}$$

the supremum over the roots. An *object* W is a finite set of roots, $n(W)$ in number, and a cluster *meets* W if one of its groups is at a root of W. By the count in the proof of Paper, Lemma 7.3, the groups which are incompatible with a group $\mathcal{G}$ are at one of at most $81M\sigma(\mathcal{G})$ roots, the traversals of the arc of $\mathcal{G}$ and the cubes adjacent or equal to the cubes of its domains and of its merged clusters; we write $W(\mathcal{G})$ for this object, so that $n(W(\mathcal{G})) \leq 81M\sigma(\mathcal{G})$ and every cluster which contains $\mathcal{G}$ meets $W(\mathcal{G})$.

Lemma 18.1 (First-Order Bounds). *Let c and c' be two families of complex activities, $\bar{c}$ a family of numbers with $|c_{\mathcal{G}}| \leq \bar{c}_{\mathcal{G}}$ and $|c'_{\mathcal{G}}| \leq \bar{c}_{\mathcal{G}}$ for every group, and $\delta c := c' - c$. Suppose that*

$$\|\bar{c}\|_\circ \leq \frac{\nu}{5184M} . \tag{18.2}$$

(a) For every object W,

$$\sum_{\Gamma \text{ meets } W} |\Phi^\circ(\Gamma; c') - \Phi^\circ(\Gamma; c)|\, e^{a^\circ(\Gamma)} \leq 2n(W)\|\delta c\|_\circ , \tag{18.3}$$

and in particular the left side is at most $162M\sigma(\mathcal{G})\|\delta c\|_\circ$ for $W = W(\mathcal{G})$.

(b) Summed over all clusters,

$$\sum_\Gamma |\Phi^\circ(\Gamma; c') - \Phi^\circ(\Gamma; c)| \leq 162M \sum_{\mathcal{G}} |\delta c_{\mathcal{G}}|\, \sigma(\mathcal{G})\, e^{2a^\circ(\mathcal{G})} . \tag{18.4}$$

(c) Let c_0 and c'_0 be two further families with $|c_{0,\mathcal{G}}| \leq \bar{c}_{\mathcal{G}}$ and $|c'_{0,\mathcal{G}}| \leq \bar{c}_{\mathcal{G}}$, and put $d := c - c_0$, $d' := c' - c'_0$, $\bar{d}_{\mathcal{G}} := \max(|d_{\mathcal{G}}|, |d'_{\mathcal{G}}|)$, and $\bar{\delta}_{\mathcal{G}} := \max(|c'_{\mathcal{G}} - c_{\mathcal{G}}|, |c'_{0,\mathcal{G}} - c_{0,\mathcal{G}}|)$. Then

$$\begin{aligned} &\sum_\Gamma \Big| [\Phi^\circ(\Gamma; c') - \Phi^\circ(\Gamma; c'_0)] - [\Phi^\circ(\Gamma; c) - \Phi^\circ(\Gamma; c_0)] \Big| \\ &\qquad \leq 162M \sum_{\mathcal{G}} |d'_{\mathcal{G}} - d_{\mathcal{G}}|\, \sigma(\mathcal{G})\, e^{2a^\circ(\mathcal{G})} + C_{\mathrm{II}} \|\bar{\delta}\|_\circ \sum_{\mathcal{G}} \bar{d}_{\mathcal{G}}\, \sigma(\mathcal{G})\, e^{2a^\circ(\mathcal{G})} , \end{aligned} \tag{18.5}$$

with $C_{\mathrm{II}} := 2^9 \cdot 3^8 M^2/\nu$.

Proof. We first recall the bound which the criterion gives. Let $x \geq 0$ be a family with $\|x\|_\circ \leq \nu/(2592M)$. The groups incompatible with a group $\mathcal{G}$ are at the roots of $W(\mathcal{G})$, so that

$$\sum_{\mathcal{G}' \not\sim \mathcal{G}} x_{\mathcal{G}'}\, e^{2a^\circ(\mathcal{G}')} \leq 81M\sigma(\mathcal{G})\|x\|_\circ \leq \frac{\nu}{32}\sigma(\mathcal{G}) = a^\circ(\mathcal{G}) ,$$

which is the condition (7.7) of the paper for the activities x. As in the proof of Paper, Lemma 7.3, [14, Theorem 5.4] with the functions $a = a^\circ$ and $d = a^\circ$ then bounds the clusters which are incompatible with an external polymer by the sum of $x_{\mathcal{G}} e^{2a^\circ(\mathcal{G})}$ over the groups incompatible with it; with an object W as the external polymer, incompatible with the groups at its roots,

$$\sum_{\Gamma \text{ meets } W} |\Phi^\circ(\Gamma; x)|\, e^{a^\circ(\Gamma)} \le \sum_{r \in W} \sum_{\mathcal{G} \text{ at } r} x_{\mathcal{G}}\, e^{2a^\circ(\mathcal{G})} \le n(W) \|x\|_\circ \,. \tag{18.6}$$

(a). If $\|\delta c\|_\circ = 0$ there is nothing to prove, so let it be positive. For $t \in [0,1]$ the family $c^t := c + t\delta c = (1-t)c + tc'$ satisfies $|c^t_{\mathcal{G}}| \le \bar{c}_{\mathcal{G}}$. The weight of a cluster is a monomial in the activities, and differentiating it along c^t replaces one factor at a time, so that, with $m_\Gamma(\mathcal{G})$ the multiplicity of $\mathcal{G}$ in Γ and $\Gamma \setminus \mathcal{G}$ the multiset Γ with one copy of $\mathcal{G}$ removed,

$$|\Phi^\circ(\Gamma; c') - \Phi^\circ(\Gamma; c)| \le |\phi^{\mathrm{U}}(\Gamma)| \sum_{\mathcal{G} \in \Gamma} m_\Gamma(\mathcal{G}) |\delta c_{\mathcal{G}}| \prod_{\mathcal{G}' \in \Gamma \setminus \mathcal{G}} \bar{c}_{\mathcal{G}'} \,, \tag{18.7}$$

the sum over the distinct groups of Γ. The right side is the coefficient of u in the polynomial $p_\Gamma(u) := |\phi^{\mathrm{U}}(\Gamma)| \prod_{\mathcal{G} \in \Gamma} (\bar{c}_{\mathcal{G}} + u|\delta c_{\mathcal{G}}|)$. The coefficients of p_Γ are nonnegative, so that for every $u_0 > 0$ this coefficient is at most $p_\Gamma(u_0)/u_0$, and $p_\Gamma(u_0) = |\Phi^\circ(\Gamma; x)|$ for the family $x := \bar{c} + u_0|\delta c|$. We take $u_0 := \nu/(5184M\|\delta c\|_\circ)$; then $\|x\|_\circ \le \|\bar{c}\|_\circ + u_0\|\delta c\|_\circ \le \nu/(2592M)$ by (18.2), and (18.6) gives

$$\sum_{\Gamma \text{ meets } W} |\Phi^\circ(\Gamma; c') - \Phi^\circ(\Gamma; c)|\, e^{a^\circ(\Gamma)} \le \frac{1}{u_0}\, n(W)\, \frac{\nu}{2592M} = 2n(W)\|\delta c\|_\circ \,.$$

(b). Fix a group $\mathcal{G}_0$ and let $D_{\mathcal{G}_0} := \sum_\Gamma |\phi^{\mathrm{U}}(\Gamma)| m_\Gamma(\mathcal{G}_0) \prod_{\mathcal{G}' \in \Gamma \setminus \mathcal{G}_0} \bar{c}_{\mathcal{G}'}$, the sum over the clusters which contain $\mathcal{G}_0$. This is the coefficient of u in the sum of $|\Phi^\circ(\Gamma; x^u)|$ over these clusters, for the family $x^u := \bar{c} + u\mathbf{1}_{\mathcal{G}_0}$, in which only the activity of $\mathcal{G}_0$ is changed. The group $\mathcal{G}_0$ contributes $ue^{2a^\circ(\mathcal{G}_0)}$ to the sums (18.1) at its roots and nothing at the others, so that with $u_0 := \nu e^{-2a^\circ(\mathcal{G}_0)}/(5184M)$ we have $\|x^{u_0}\|_\circ \le \nu/(2592M)$. Every cluster which contains $\mathcal{G}_0$ meets $W(\mathcal{G}_0)$, and (18.6) gives, as in (a),

$$D_{\mathcal{G}_0} \le \frac{1}{u_0}\, 81M\sigma(\mathcal{G}_0)\, \frac{\nu}{2592M} = 162M\sigma(\mathcal{G}_0)\, e^{2a^\circ(\mathcal{G}_0)} \,. \tag{18.8}$$

Summing (18.7) over all clusters and exchanging the sums gives $\sum_\Gamma |\Phi^\circ(\Gamma; c') - \Phi^\circ(\Gamma; c)| \le \sum_{\mathcal{G}} |\delta c_{\mathcal{G}}| D_{\mathcal{G}}$, which with (18.8) is (18.4).

(c). For $s, t \in [0,1]$ put $c^t := (1-t)c + tc'$, $c^t_0 := (1-t)c_0 + tc'_0$, $d^t := c^t - c^t_0 = (1-t)d + td'$, and $F_\Gamma(s,t) := \Phi^\circ(\Gamma; c^t_0 + sd^t)$. The expression inside the modulus on the left of (18.5) is $F_\Gamma(1,1) - F_\Gamma(0,1) - F_\Gamma(1,0) + F_\Gamma(0,0) = \int_0^1 \int_0^1 \partial_s \partial_t F_\Gamma \,\mathrm{d}s\,\mathrm{d}t$. The family $c^t_0 + sd^t = (1-s)c^t_0 + sc^t$ is a convex combination of the four families and is bounded by $\bar{c}$; its derivative in s is d^t, with $|d^t_{\mathcal{G}}| \le \bar{d}_{\mathcal{G}}$, its derivative in t is $(1-s)(c'_0 - c_0) + s(c' - c)$, bounded by $\bar{\delta}_{\mathcal{G}}$, and its mixed derivative is $d' - d$. Differentiating the monomial twice therefore gives

$$\begin{aligned} |\partial_s \partial_t F_\Gamma| \le{}& |\phi^{\mathrm{U}}(\Gamma)| \sum_{\mathcal{G} \in \Gamma} m_\Gamma(\mathcal{G}) |d'_{\mathcal{G}} - d_{\mathcal{G}}| \prod_{\mathcal{G}'' \in \Gamma \setminus \mathcal{G}} \bar{c}_{\mathcal{G}''} \\ &+ |\phi^{\mathrm{U}}(\Gamma)| \sum_{(\mathcal{G}, \mathcal{G}')} \bar{d}_{\mathcal{G}}\, \bar{\delta}_{\mathcal{G}'} \prod_{\mathcal{G}'' \in \Gamma \setminus \{\mathcal{G}, \mathcal{G}'\}} \bar{c}_{\mathcal{G}''} \,, \end{aligned}$$

the second sum over the ordered pairs of distinct entries of the multiset. Summed over the clusters, the first term is bounded as in (b), which gives the first term of (18.5). In the second we fix the group $\mathcal{G}_0$ of the first entry: the sum over the clusters and over the second entry is the coefficient of uv in the sum of $|\Phi^\circ(\Gamma; x^{u,v})|$ over the clusters which contain $\mathcal{G}_0$, for $x^{u,v} := \bar{c} + u\mathbf{1}_{\mathcal{G}_0} + v\bar{\delta}$, a polynomial in u and v with nonnegative coefficients. With $u_0 := \nu e^{-2a^\circ(\mathcal{G}_0)}/(10368M)$ and $v_0 := \nu/(10368M\|\bar{\delta}\|_\circ)$ we have $\|x^{u_0,v_0}\|_\circ \le \nu/(2592M)$, and (18.6) bounds this coefficient by

$$\frac{1}{u_0 v_0}\, 81M\sigma(\mathcal{G}_0)\,\frac{\nu}{2592M} = \frac{10368^2 M^2}{32\nu}\,\sigma(\mathcal{G}_0)\, e^{2a^\circ(\mathcal{G}_0)}\|\bar{\delta}\|_\circ\,,$$

and $10368^2/32 = 2^9 \cdot 3^8$. Multiplying by $\bar{d}_{\mathcal{G}_0}$ and summing over $\mathcal{G}_0$ gives the second term.
q.e.d.

The bound (a) is the estimate by which the proof of Proposition S.16(i) passes from the activities to the cluster occurrences, and (b), (c) are its forms for a sum over all clusters and for the difference of two such sums, which the local terms require: the local term of a step on a labelled term ℓ is the difference of the logarithms of the reference gases of ℓ and of the trivial term (Proposition S.9(iii)), so that its difference between two runs is a second difference of the kind bounded in (c), with d and d' supported on the groups whose activities on ℓ differ from those on the trivial term.

18.2 The Activities Contain No Attached Insertion

At a step of the step map the insertion $Y_{\mathcal{G}}$ of a new group is written as $Y^{\mathrm{d}}_{\mathcal{G}} + c_{\mathcal{G}}\,\mathrm{id}$, the words are expanded in the two terms, and the scalars $c_{\mathcal{G}}$ are resummed into the reference gas, whose cluster series gives the interactions and the local terms of the new step (Paper, Section 7.1). The expansion and the resummation are identities for every choice of the numbers $c_{\mathcal{G}}$, one for each group: they use only that $c_{\mathcal{G}}\,\mathrm{id}$ commutes with every factor of a word and that the compatibility of two groups does not depend on these numbers. The choice matters for the bounds alone. For a group of the deviation species $c_{\mathcal{G}}$ has to be the value of $Y_{\mathcal{G}}$ at the reference data, since the deviation bound of $Y^{\mathrm{d}}_{\mathcal{G}}$ rests on its vanishing at pure-gauge data (Paper, Section 6.2). Call a group *attached* if it is attached to a large-field region of its step or contains a part of an insertion of an attached species; by Section 7 such a group belongs to the species of that insertion, and the name does not depend on how a group with parents of several species is counted. For the cross groups and for the attached groups the paper takes $c_{\mathcal{G}} := 0$, so that these groups are carried whole and are never split. We recall that a group which contains an insertion with intervals on two loops has intervals on two loops and is a cross group.

No bound of the class rests on the subtraction of a scalar from an attached group. The proof of Proposition S.9(i) bounds the attached insertions of a step by the second term of Paper, (5.9); that of Proposition S.9(ii) uses of a carried attached insertion its supremum on the subdomain on which its bound is stated, in the evaluation at the new background, in Cauchy's estimate for the touched parts, and in the bound of the merged parts; and none of the three uses a value at the reference data. In Proposition S.9(iii) the choice is used: the statement that a cluster which meets neither the large-field region nor the cutoffs on its boundary layer has the same weight on ℓ and on the trivial term holds because the groups which exist on ℓ alone, those which contain a touched or a merged part of an attached insertion and which reach into the small-field region with it, have the activity 0.

Lemma 18.2 (The Activities of a Step of the Step Map)**.** *Fix a labelled term and a step $n \to n+1$ of the step map, and let $\mathcal{G}$ be a group of this step on one loop which is not attached. Its activity $c_{\mathcal{G}}$ is the value at the reference data of the term of a function which is multilinear* *in the insertions of the members of $\mathcal{G}$, each of which has its intervals on that loop, is not attached, and is therefore of the deviation species or of the wrapping species, and linear in the factor*

$$\sum_{m\geq 0} \frac{(-1)^m}{m!} \prod_{j=1}^{m} \Phi^\circ(\Gamma_j) \tag{18.9}$$

of the clusters Γ_j of the reference gas of the step n which meet its members, and whose other constituents are those of the step $n+1$ on the domains of $\mathcal{G}$. The clusters in (18.9) *consist of groups of the step n on one loop which are not attached. Consequently, by induction on the steps of the step map, no activity of a reference gas depends on an attached insertion or on an insertion with intervals on two loops.*

Proof. A member of $\mathcal{G}$ is an insertion of an earlier step which the step touches or merges into $\mathcal{G}$. If a member were attached, $\mathcal{G}$ would contain a part of an insertion of an attached species and would be attached; if a member had intervals on two loops, so would $\mathcal{G}$. Hence every member is an insertion on the loop of $\mathcal{G}$ which is not attached, and the species of such an insertion are the deviation species and the wrapping species (Paper, Definition 5.2(ii)). The insertion of $\mathcal{G}$ is the integral over the fluctuation field of the step of a product, in the order of the loop, of the slots and transports of the step, of the insertions of the members, and of the factors of the clusters of the step n which are merged into $\mathcal{G}$ when the interactions of the step n are expanded (Paper, Section 7.1); it is therefore multilinear in the insertions of the members and linear in the factor (18.9), which is the term of the exponential series of these clusters, and $c_{\mathcal{G}}$ is its value at the reference data. A cluster of the reference gas of the step n has a nonzero weight only if each of its groups has a nonzero activity, and then each of them is a group on one loop which is not attached. The last statement follows by induction: the activities of the step $n+1$ depend on the insertions of members which are neither attached nor on two loops, on the constituents of the step, and on the activities of the step n, and an insertion which is not attached contains, by the definition of an attached group, no part of an attached insertion. *q.e.d.*

At an **R**-step the scalars are the activities $\mathfrak{a}_g$ of the eligible groups of Proposition S.10(iv), and here no choice has to be made, because the construction of the polymers already keeps the data of the components which survive the step out of every activity.

Lemma 18.3 (The Activities of an **R**-Step)**.** *Fix an* **R***-step k and a term as in Proposition S.10. The activity $F(P)$ of a polymer which does not meet the ensemble involves no letter of a loop. The tensor T_P of a polymer of an eligible group g and the scalar activity $\mathfrak{a}_g$ are functions of the operations $\mathbf{T}'_k(X)$ of the first-class components X of the polymers of g, of the factors $V(Y)$ of [13, (1.91)], and of the pieces of the letters in the slots; every one of these pieces is a function of the external data and of the data of first-class components of its polymer. In particular none of them is a piece of an insertion whose set $\hat{T}$ contains a second-class or an unchanged component, and none depends on a unit of these two classes. Consequently the numbers $E_k^{\mathbf{R}}(C;\ell_0)$, $E_k^{\mathbf{R}\times}$, $E_k^{\mathbf{R},\partial}(\ell)$, and $\Psi(S)$ depend on the insertions of the term only through the pieces, in the slots of eligible groups and at first-class components, of insertions on one loop.*

Proof. A polymer is a connected union of class components, of localisation domains, and of the domains of the pieces of the letters which it holds, and by Lemma 17.19(ii) the domain of a piece of $F^{\mathcal{K}}$ contains the components of $\hat{T}$ and the units of $\mathcal{K}$, of which the piece is a function. A group is eligible only if none of its polymers meets a second-class or an unchanged component (Proposition S.10(iv)). A polymer meets the ensemble if its activity carries a slot (Section 8), so that the activity of a polymer which does not meet it is the expression [13, (1.91)] without letters. The polymers of an eligible group contain first-class components only, and the pieces in their slots have only such components in their domains. The tensor T_P is the expression [13, (1.91)] with these pieces in loop order, in which the operations are local, "All the expressions in the above integral operation are localized in the domain X", and "the terms $V(Y, U_k)$ depend on the sequence $\{\Omega_j^c, Z_j\}$ restricted to the components of Z contained in Y" [13, p. 379]; and $\mathfrak{a}_g$ is the contraction of the tensors T_P, $P \in g$, with the transports 1 of the flat reference. This is the statement for T_P and $\mathfrak{a}_g$. An insertion one of whose components in $\hat{T}$ is of the second class or unchanged has no piece without that component in its domain, and a piece with a unit of these classes has that unit in its domain, so neither occurs in an eligible group; this agrees with the proof of Proposition S.10(iii), by which the letters in the slots of the second-class and of the unchanged components, and the pieces all of whose units are of these classes, enter no activity. A group which holds, in a slot, an insertion with intervals on two loops has slots on two loops and is a cross group, with $\mathfrak{a}_g = 0$. The four numbers are cluster terms of the gas $\Xi^\flat$, whose activities are the $F(P)$ and the $\mathfrak{a}_g$, and of the gas Ξ, which gives the last statement. *q.e.d.*

The members of a group, the insertions of the term whose intervals lie in its span and in none of its slots, are not part of its scalar activity: by the proof of Proposition S.10(v), D_g is the contraction with the members on their intervals, less $\mathfrak{a}_g$ times the small-field word with the same members.

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